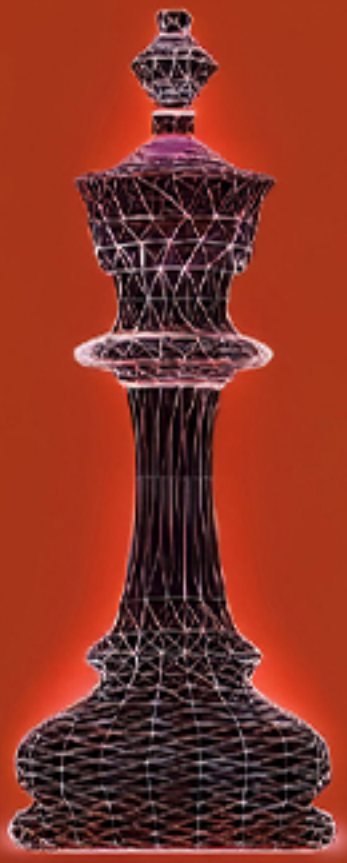
A Course in Reinforcement Learning

2nd Edition

Dimitri P. Bertsekas



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A Course in Reinforcement Learning (2nd Edition)

by

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Dimitri Bertsekas studied Mechanical and Electrical Engineering at the National Technical University of Athens, Greece, and obtained his Ph.D. in system science from the Massachusetts Institute of Technology. He has held faculty positions with the Engineering-Economic Systems Department, Stanford University, and the Electrical Engineering Department of the University of Illinois, Urbana. From 1979 to 2019 he was a professor at the Electrical Engineering and Computer Science Department of the Massachusetts Institute of Technology (M.I.T.), where he continues to hold the title of McAfee Professor of Engineering. In 2019, he joined the School of Computing and Augmented Intelligence at the Arizona State University, Tempe, AZ, as Fulton Professor of Computational Decision Making.

Professor Bertsekas' teaching and research have spanned several fields, including deterministic optimization, dynamic programming and stochastic control, large-scale and distributed computation, artificial intelligence, and data communication networks. He has authored or coauthored numerous research papers and twenty books, several of which are currently used as textbooks in MIT classes, including "Dynamic Programming and Optimal Control," "Data Networks," "Introduction to Probability," and "Nonlinear Programming." At ASU, he has been focusing on teaching and research in reinforcement learning, and he has written several textbooks and research monographs in this field since 2019.

Professor Bertsekas was awarded the INFORMS 1997 Prize for Research Excellence in the Interface Between Operations Research and Computer Science for his book "Neuro-Dynamic Programming" (co-authored with John Tsitsiklis), the 2001 AACC John R. Ragazzini Education Award, the 2009 INFORMS Expository Writing Award, the 2014 AACC Richard Bellman Heritage Award, the 2014 INFORMS Khachiyan Prize for Life-Time Accomplishments in Optimization, the 2015 MOS/SIAM George B. Dantzig Prize, and the 2022 IEEE Control Systems Award. In 2018 he shared with his coauthor, John Tsitsiklis, the 2018 INFORMS John von Neumann Theory Prize for the contributions of the research monographs "Parallel and Distributed Computation" and "Neuro-Dynamic Programming." Professor Bertsekas was elected in 2001 to the United States National Academy of Engineering for "pioneering contributions to fundamental research, practice and education of optimization/control theory."

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CONTENTS

1. Exact and Approximate Dynamic Programming

1.1. AlphaZero, Off-Line Training, and On-Line Play . . . . . p. 4
1.2. Deterministic Dynamic Programming . . . . . p. 10
1.2.1. Finite Horizon Problem Formulation . . . . . p. 10
1.2.2. The Dynamic Programming Algorithm . . . . . p. 14
1.2.3. Approximation in Value Space and Rollout . . . . . p. 22
1.3. Stochastic Exact and Approximate Dynamic Programming . . p. 28
1.3.1. Finite Horizon Problems . . . . . p. 28
1.3.2. Approximation in Value Space for Stochastic DP . . . p. 34
1.3.3. Approximation in Policy Space . . . . . p. 38
1.3.4. Training of Cost Function and Policy . . . . . Approximations . . . . . p. 41
1.4. Infinite Horizon Problems - An Overview . . . . . p. 42
1.4.1. Infinite Horizon Methodology . . . . . p. 46
1.4.2. Approximation in Value Space - Infinite Horizon . . . p. 49
1.4.3. Understanding Approximation in Value Space . . . . . p. 55
1.5. Newton's Method - Linear Quadratic Problems . . . . . p. 56
1.5.1. Visualizing Approximation in Value Space - Region of . . . Stability . . . . . p. 62
1.5.2. Rollout and Policy Iteration . . . . . p. 70
1.5.3. Local and Global Error Bounds for Approximation in . . . Value Space . . . . . p. 74
1.5.4. How Approximation in Value Space Can Fail and . . . . . What to Do About It . . . . . p. 78
1.6. Examples, Reformulations, and Simplifications . . . . . p. 79
1.6.1. A Few Words About Modeling . . . . . p. 79
1.6.2. Problems with a Termination State . . . . . p. 83
1.6.3. General Discrete Optimization Problems . . . . . p. 85
1.6.4. General Finite to Infinite Horizon Reformulation . . . p. 89
1.6.5. State Augmentation, Time Delays, Forecasts, and . . . . . Uncontrollable State Components . . . . . p. 91
1.6.6. Partial State Information and Belief States . . . . . p. 98
1.6.7. Multiagent Problems and Multiagent Rollout . . . . p. 102
1.6.8. Problems with Unknown or Changing Model - . . . . . Adaptive Control . . . . . p. 107
1.6.9. Model Predictive Control . . . . . p. 118
1.7. Reinforcement Learning and Decision/Control . . . . . p. 131
1.7.1. Differences in Terminology . . . . . p. 132
1.7.2. Differences in Notation . . . . . p. 133
1.7.3. Relations Between DP and RL . . . . . p. 135

1.7.4. Synergy Between Large Language Models and RL . p. 136
1.7.5. Machine Learning and Optimization . . . . . p. 136
1.7.6. Mathematics in Machine Learning and Optimization p. 139
1.8. Notes, Sources, and Exercises . . . . . p. 140

2. Approximation in Value Space - Rollout Algorithms

2.1. Deterministic Finite Horizon Problems . . . . . p. 166
2.2. Approximation in Value Space - Deterministic Problems . . p. 173
2.3. Rollout Algorithms for Discrete Optimization . . . . . p. 177
2.3.1. Cost Improvement with Rollout - Sequential Consistency, . .
Sequential Improvement . . . . . p. 183
2.3.2. The Fortified Rollout Algorithm . . . . . p. 189
2.3.3. Using Multiple Base Heuristics - Parallel Rollout . . . p. 192
2.3.4. Using Multiple Rollout Trajectories - General Rollout . p. 194
2.3.5. Simplified Rollout Algorithms . . . . . p. 195
2.3.6. Truncated Rollout with Terminal Cost Approximation . p. 196
2.3.7. Rollout with an Expert - Model-Free Rollout . . . . . p. 197
2.3.8. Using a World Model for Rollout . . . . . p. 202
2.3.9. Local Search with Rollout for Discrete Optimization . . p. 204
2.3.10. Inference in n-Grams, Transformers, HMMs, and . . . . . .
Markov Chains . . . . . p. 208
2.4. Approximation in Value Space with Multistep Lookahead . p. 221
2.4.1. Iterative Deepening Using Forward Dynamic . . . . . .
Programming . . . . . p. 227
2.4.2. Incremental Multistep Rollout . . . . . p. 229
2.5. Constrained Forms of Rollout Algorithms . . . . . p. 232
2.5.1. Constrained Rollout for Discrete Optimization and Integer . .
Programming . . . . . p. 245
2.6. Small Stage Costs and Long Horizon - Continuous-Time . . . . .
Rollout . . . . . p. 249
2.7. Approximation in Value Space - Stochastic Problems . . . p. 257
2.7.1. Simplified Rollout and Policy Iteration . . . . . p. 261
2.7.2. Certainty Equivalence Approximations . . . . . p. 262
2.7.3. Simulation-Based Implementation of the Rollout . . . . . .
Algorithm . . . . . p. 265
2.7.4. Variance Reduction in Rollout - Comparing Advantages p. 267
2.7.5. Monte Carlo Tree Search . . . . . p. 270
2.7.6. Randomized Policy Improvement by Monte Carlo . . . . . .
Tree Search . . . . . p. 274
2.8. Rollout for Infinite-Spaces Problems - Optimization . . . . . .
Heuristics . . . . . p. 274
2.8.1. Rollout for Infinite-Spaces Deterministic Problems . . . p. 275
2.8.2. Rollout Based on Stochastic Programming . . . . . p. 279
2.8.3. Stochastic Rollout with Certainty Equivalence . . . . . p. 281

2.9. Multiagent Rollout . . . . . p. 283
2.9.1. Asynchronous and Autonomous Multiagent Rollout . . p. 294
2.10. Rollout for Bayesian Optimization and Sequential Estimation . . . . . p. 297
2.11. Adaptive Control by Rollout with a POMDP Formulation . p. 309
2.12. Minimax Control and Reinforcement Learning . . . . . p. 317
2.12.1. Exact Dynamic Programming for Minimax Problems . p. 318
2.12.2. Minimax Approximation in Value Space and Rollout . p. 321
2.12.3. Combined Approximation in Value and Policy Space for Minimax Control . . . . . p. 327
2.12.4. A Meta Algorithm for Computer Chess Based on Reinforcement Learning . . . . . p. 328
2.12.5. Combined Approximation in Value and Policy Space for Sequential Noncooperative Games . . . . . p. 332
2.13. Notes, Sources, and Exercises . . . . . p. 334

3. Learning Values and Policies . . . . . .

3.1. Parametric Approximation Architectures . . . . . p. 351
3.1.1. Cost Function Approximation . . . . . p. 352
3.1.2. Feature-Based Architectures . . . . . p. 353
3.1.3. Training of Linear and Nonlinear Architectures . . . p. 364
3.2. Neural Networks . . . . . p. 372
3.2.1. Training of Neural Networks . . . . . p. 377
3.2.2. Multilayer and Deep Neural Networks . . . . . p. 378
3.3. Learning Cost Functions in Approximate DP . . . . . p. 380
3.3.1. Fitted Value Iteration . . . . . p. 380
3.3.2. Q-Factor Parametric Approximation - Model-Free Implementation . . . . . p. 382
3.3.3. Parametric Approximation in Infinite Horizon Problems - Approximate Policy Iteration . . . . . p. 384
3.3.4. Optimistic Policy Iteration with Parametric Q-Factor Approximation - SARSA and DQN . . . . . p. 388
3.3.5. Approximate Policy Iteration for Infinite Horizon POMDP . . . . . p. 390
3.3.6. Advantage Updating - Approximating Q-Factor Differences . . . . . p. 395
3.3.7. Differential Training of Cost Differences for Rollout . p. 397
3.4. Learning a Policy in Approximate DP . . . . . p. 399
3.4.1. The Use of Classifiers for Approximation in Policy Space . . . . . p. 400
3.4.2. Policy Iteration with Value and Policy Networks . . p. 404
3.4.3. Why Use On-Line Play and Not Just Train a Policy Network to Emulate the Lookahead Minimization? . p. 406
3.5. Policy Gradient and Related Methods . . . . . p. 408

3.5.1. Gradient Methods for Parametric Cost Optimization . . . p. 409
3.5.2. Policy Gradient-Like Methods . . . p. 412
3.5.3. Scaling and Proximal Policy Optimization Methods . . . p. 417
3.5.4. Random Direction Methods . . . p. 424
3.5.5. Random Search and Cross-Entropy Methods . . . p. 426
3.5.6. Refining and Retraining Parametric Policies . . . p. 428
3.6. Aggregation . . . p. 429
3.6.1. Aggregation with Representative States . . . p. 430
3.6.2. Continuous Control Space Discretization . . . p. 437
3.6.3. Continuous State Space - POMDP Discretization . . . p. 438
3.6.4. General Aggregation . . . p. 440
3.6.5. Types of Aggregation and Error Bounds . . . p. 443
3.6.6. Aggregation Using Features . . . p. 445
3.6.7. Biased Aggregation . . . p. 446
3.6.8. Asynchronous Distributed Aggregation . . . p. 449
3.7. Notes, Sources, and Exercises . . . p. 451

References . . . p. 463

Preface

It is a capital mistake to theorize before one has data.

Sherlock Holmes

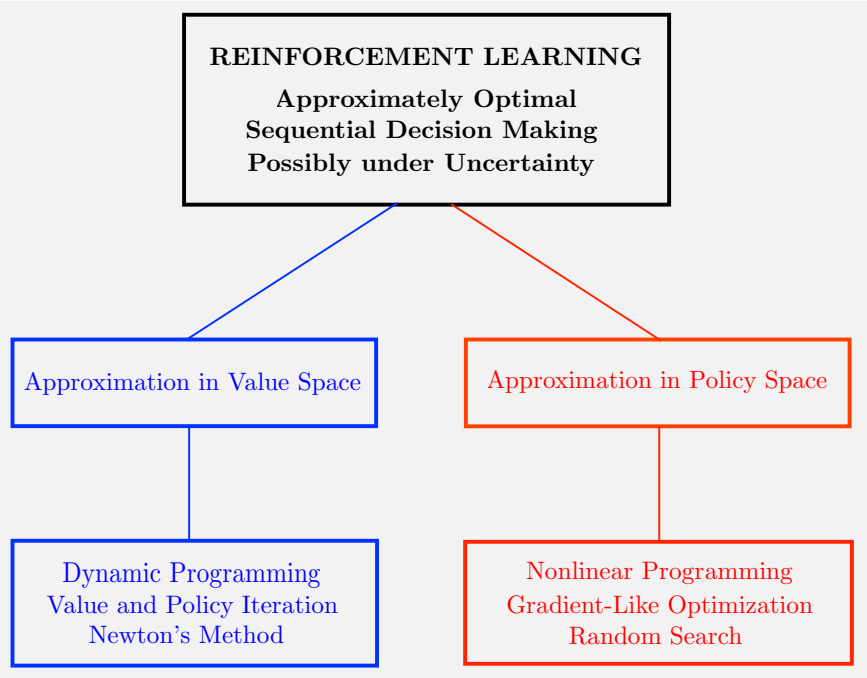
We are surrounded by data, but starved for insights.†

This book is based on lecture notes for the ASU research-oriented course on Reinforcement Learning (RL) that I have offered in each of the years from 2019 to 2025, as the field was rapidly evolving. The purpose of the book is to give an overview of the RL methodology, with a particular focus on problems of optimal and suboptimal control, as well as discrete optimization. More broadly, the book aims to provide a framework for structured thinking about RL and its connections to decision and control, which is grounded in mathematics, but is not dominated by it.

Generally, RL can be viewed as the art and science of approximately optimal sequential decision making for large and difficult problems, often in the presence of imprecisely known and changing environment conditions.‡ Dynamic Programming (DP), a well-established optimization methodology,

† Attributed to Jay Baer.

‡ The name "reinforcement learning" evolved from the concept of "reinforcement" in behavioral psychology. It refers to learning through interaction with the environment, shaped by rewards and penalties. The meaning of the name



deals with the same problem as RL, and forms its theoretical foundation. This is unlikely to change in the future, despite the current rapid pace of innovation. In fact, there are strong connections between sequential decision making and the new wave of technological progress, generative artificial intelligence, transformers, GPT applications, and natural language processing, as we will aim to show in this book.

Approximations in Value and Policy Space

In DP there are two principal objects to compute: the *optimal value function*, which provides the optimal cost that can be attained starting from any given initial state, and the *optimal policy*, which provides the optimal decision to apply at any given state and time. Unfortunately, the exact application of DP runs into formidable computational challenges, commonly known as the *curse of dimensionality*. To address these, RL aims to approximate the optimal value function and policy, by using manageable off-line and/or on-line computation, which often involves neural networks.

Thus there are two major methodological approaches in RL: *approximation in value space*, where we approximate in some way the optimal value function, and *approximation in policy space*, where we construct a

has significantly evolved since its initial popularization in the 1980s.

suboptimal policy by optimizing over a restricted set of policies; see the figure.† These two approaches can sometimes be combined to exploit the strengths of both. Generally, approximation in value space aligns more closely to the central DP ideas of value and policy iteration, which in turn connect closely to methods for solution of Bellman's equation using Newton's method.

On the other hand, approximation in policy space often relies on nonlinear programming ideas, and particularly gradient-like descent. In this connection it is worth noting two contexts:

(a) *Approximation of a target policy*, whereby we want to approximate a given policy. The policy is assumed to be accessible for sampling, and it can be used to obtain sample pairs, each consisting of a state and the control generated by the policy at that state. Many such sample pairs are collected and they are used to train a policy approximation with some form of supervised learning algorithm. Methods of this type are described in Section 3.4, and to some extent, in Sections 3.1-3.3 and 3.6.

(b) *Direct optimization over a parametric class of policies*. Here there is no target policy to approximate. Instead, we aim to approximate an optimal policy directly, through optimization of the cost function over a class of policies that depend on a parameter. This optimization is done with some kind of gradient-like algorithm (sometimes called a *policy gradient method*) or some form of random search. Methods of this type are described in Section 3.5.

From our point of view, RL consists of *all DP-like methods that aim to solve suboptimally problems formulated in terms of sequential decisions, as long as they involve approximation, either in value space or in policy space (or in both)*.‡ This definition is very broad and applies among others, to deterministic, stochastic, and minimax problem contexts not ordinarily viewed in terms of sequential decisions, such as integer programming and tuning of LLMs (see Chapter 2).

Let us finally note that while the connections between approximation in value and policy space are a recurring theme throughout our narrative, our presentation focuses primarily on approximation in value space. How-

† The terms "approximation in value space" and "approximation in policy space" were introduced by Bellman [Bel57], and are consistent with the name "neuro-dynamic programming" of the Bertsekas and Tsitsiklis book [BeT96] (the marriage of dynamic programming and neural network technology).

‡ The name "reinforcement learning" describes well approximation in policy space and its gradient-like methodology: reinforcing/improving parametrically defined policies can be effected by changing parameters along their gradients. It describes less well approximation in value space, because reinforcing good value functions does not make much sense from a methodological point of view.

ever, the book is structured to allow instructors to go into approximation in policy space in greater depth, using any of a number of available sources.

Approximation in Value Space - Off-Line Training and On-Line Play

An important part of our development of the approximation in value space approach is a new conceptual framework, which aims to bridge the gaps between the artificial intelligence, control theory, and operations research views of our subject. This framework, the focus of the author's recent monograph "Lessons from AlphaZero for Optimal, Model-Predictive, and Adaptive Control," [Ber22a], centers on approximate forms of DP that are inspired by some of the major successes of RL involving games. Primary examples are the recent (2017) AlphaZero program (which plays chess), and the similarly structured and earlier (1990s) TD-Gammon program (which plays backgammon).

Our framework is couched in two general algorithms that are designed largely independently of each other and operate in synergy through the powerful mechanism of Newton's method, applied to solution of the fundamental Bellman equation of DP. We call these the *off-line training* and the *on-line play* algorithms. In the AlphaZero and TD-Gammon game contexts, the off-line training algorithm is the method used to teach the program how to evaluate positions and to generate good moves at any given position, while the on-line play algorithm is the method used to play in real time against human or computer opponents.

Our synergistic view of off-line training and on-line play is motivated by some striking empirical observations. In particular, both AlphaZero and TD-Gammon were trained off-line extensively using neural networks and an approximate version of the fundamental DP algorithm of policy iteration. Yet the AlphaZero player that was obtained off-line is not used directly during on-line play (it is too inaccurate due to approximation errors that are inherent in off-line neural network training). Instead, a separate on-line player is used to select moves, based on multistep lookahead minimization and a terminal position evaluator that was trained using a deep neural network, simultaneously with the off-line player. The on-line player performs a form of policy improvement, which is not degraded by neural network approximations. As a result, it greatly improves the performance of the off-line player.

Similarly, TD-Gammon performs on-line a policy improvement step using one-step or two-step lookahead minimization, which is not degraded by neural network approximations. To this end, it uses an off-line neural network-trained terminal position evaluator, and importantly it also extends its on-line lookahead by rollout (simulation with the one-step lookahead player that is based on the position evaluator). Thus in summary:

(a) The on-line player of AlphaZero plays much better than its extensively trained off-line player. This is due to the beneficial effect of exact policy improvement with long lookahead minimization, which corrects for the inevitable imperfections of the neural network-trained off-line player, and position evaluator/terminal cost approximation.

(b) The TD-Gammon player that uses long rollout plays much better than TD-Gammon without rollout. This is due to the beneficial effect of the rollout, which serves as a supplement and substitute for long lookahead minimization.

An important lesson from AlphaZero and TD-Gammon is that the performance of an off-line trained policy can be greatly improved by on-line approximation in value space, with long lookahead (involving minimization or rollout with the off-line policy, or both), and terminal cost approximation that is obtained off-line. This performance enhancement is often dramatic and is due to a simple fact, which is couched in algorithmic mathematics and is a focal point of our course: *approximation in value space with one-step lookahead minimization amounts to a step of Newton's method for solving Bellman's equation, while the starting point for the Newton step is based on the results of off-line training, and may be enhanced by longer lookahead minimization and on-line rollout*. Indeed the major determinant of the quality of the on-line policy is the Newton step that is performed on-line, while off-line training plays a secondary role by comparison.

An additional benefit of policy improvement by approximation in value space, not observed in the context of games (which have stable rules and environment), is that it works well with changing problem parameters and on-line replanning, similar to the methodology of indirect adaptive control. In particular, the Bellman equation is perturbed due to the parameter changes, but approximation in value space still operates as a Newton step. An essential requirement here is that a system model is estimated on-line through some identification method, and is used during the one-step or multistep lookahead minimization process. This may be a mathematical model represented by equations and probability distributions, or a computer model that involves simulation and/or neural network software.

In this book, we will describe the basic RL methodologies, and we will aim to explain (often with visualization) their effectiveness (or lack thereof) through the lens of the off-line and on-line synergy. In the process, we will bring out the strong connections between the artificial intelligence view of RL, the control theory view of sequential decision making, and the operations research view of discrete optimization. Moreover, we will describe a broad variety of algorithms that can be used for on-line play.

In particular, we will aim to show that the methodology of approximation in value space and rollout applies very broadly to deterministic and stochastic optimal control problems, involving both discrete and continuous search spaces, as well as finite and infinite horizon. We will also show that

our conceptual framework can be effectively integrated with other important methodologies such as multiagent systems and decentralized control, discrete and Bayesian optimization, two-person antagonistic games, and heuristic algorithms for discrete optimization.

Reinforcement Learning and Model Predictive Control

Significantly, the synergy between off-line training and on-line play also underlies Model Predictive Control (MPC), a major control system design methodology that has been extensively developed since the 1980s.† One of the major themes of this book is that the RL approximation in value space approach, as practiced by the artificial intelligence community, and MPC, as practiced by the decision and control community, *can be viewed as essentially identical methodologies at a mathematical and conceptual level*.‡

This is particularly true in the case where the system model (either mathematical or simulated or trained neural network) is known and not changing over time. Problems with an unknown, partially known, or time-varying model belong to the realm of adaptive control; see Section 1.6.8. Important methods here include PID control, direct adaptive control, and indirect adaptive control. The latter method (also known as "dual control" or "combined identification and control") is essentially MPC, augmented with an online system identification algorithm that continuously updates the system model using sensor measurements or other information. Note that approximation in policy space methods are ill-suited for adaptive control, since online retraining of policies is prohibitively sample-intensive.

The similarity of MPC and RL will become evident from the discussion of Section 1.1. In particular, both fields are founded on DP, and both involve an on-line search algorithm and an off-line training process. Their differences are largely a matter of emphasis: RL has its roots in theories of game playing and learning, and places far more emphasis on off-line training, and discrete state and action spaces, while MPC has its roots in predictive on-line optimization and places far more emphasis in on-line play, and continuous state and control spaces. Other differences,

† "Model predictive control" is an abbreviation for the far more descriptive term "model-based predictive control." Here, the word "model" refers to either a mathematical model or a computer software model of a discrete-time dynamic system that evolves under the influence of control. The term "predictive" generally refers to taking into account the system's future, while applying control in the present.

‡ This view, while somewhat controversial within the AI community, has found support outside the decision and control community, notably in public lectures and interviews by Yann LeCun, a pioneer in deep learning and co-recipient of the 2018 Turing Award. His statement is "Abandon RL in favor of MPC." Our related but somewhat different view, reflected in this book, is that "MPC is part of RL, and conversely, the most reliable part of RL is MPC."

which can also be traced to the respective origins of the two fields, are that RL includes a strong focus on approximation in policy space (which has been of peripheral interest in MPC), while MPC includes a strong focus on closed-loop system stability issues (which are hardly ever discussed in RL).

Supporting Literature

In this book, we will deemphasize mathematical proofs, and focus instead on visualizations and intuitive (but still rigorous) explanations. However, there is considerable related analysis, which supports our narrative, and can be found within a broad range of sources, which we describe next.

The author's approximate DP/RL books

[1] Bertsekas, D. P., 2019. Reinforcement Learning and Optimal Control, Athena Scientific, Belmont, MA.

[2] Bertsekas, D. P., 2020. Rollout, Policy Iteration, and Distributed Reinforcement Learning, Athena Scientific, Belmont, MA (can be downloaded from the author's website).

provide a far more detailed discussion of MPC, adaptive control, discrete optimization, and distributed computation topics, than the present book. Moreover, some other popular methods, such as temporal difference algorithms and Q-learning, are discussed in the books [1] and [2], but not in the present book.

The author's two-volume DP book

[3] Bertsekas, D. P., 2017. Dynamic Programming and Optimal Control, Vol. I, 4th Edition, Athena Scientific, Belmont, MA.

[4] Bertsekas, D. P., 2012. Dynamic Programming and Optimal Control, Vol. II, 4th Edition, Athena Scientific, Belmont, MA.

is an extensive source on the modeling and mathematical aspects of finite and infinite horizon DP. Modeling aspects and finite horizon problems are the principal focus of [3], while the mathematical aspects of infinite horizon problems are the principal focus of [4]. Both books [3] and [4] provide substantial accounts of approximate DP/RL methods. Thus these books are good entry points for a research-oriented reader that wishes to go into DP and its connections to RL more deeply.

Two of the author's recent research monographs are also highly relevant to our narrative:

[5] Bertsekas, D. P., 2022. Abstract Dynamic Programming, 3rd Ed., Athena Scientific, Belmont, MA (can be downloaded from the author's website).

This monograph focuses on the analytical aspects of abstract DP on which the Newton-based methodology is couched, and may serve as a mathe-

matical supplement to the present book. It also provides some supportive mathematical foundation for the more visually oriented monograph

[6] Bertsekas, D. P., 2022. Lessons from AlphaZero for Optimal, Model Predictive, and Adaptive Control, Athena Scientific, Belmont, MA (can be downloaded from the author's website).

This monograph relies on intuitive visualization to convey the main ideas of our off-line training/on-line play/Newton's method conceptual framework for approximation in value space. It also focuses on model predictive and adaptive control, and associated issues of stability. Like our Section 1.5 of the present book, it is visually oriented, but goes into greater detail on various special cases, including pathological exceptions.

All of the above books are available as ebooks as well as in print form; see the Athena Scientific website. A lot of the material of the present book is adapted from these books. However, the books themselves collectively provide a far more detailed and mathematically rigorous presentation.

An extensive overview of the connections and applications of the conceptual framework of this book with model predictive and adaptive control is given in the paper

[7] Bertsekas, D. P., 2024. "Model Predictive Control, and Reinforcement Learning: A Unified Framework Based on Dynamic Programming," published in Proc. IFAC NMPC (can be downloaded from the author's website; a videolecture is available on-line).

Together with Chapter 1 of this book, it can be used as a starting point for a course in modern control system design, particularly a course that aims to touch upon the synergies of the subject with modern trends in artificial intelligence.

The present book can also be fruitfully supplemented by the extensive textbook and research monograph literature on RL. This literature is summarized in Section 1.8, and includes several accounts of RL that are based on alternative viewpoints of artificial intelligence, control theory, and operations research.

Structure of the Book - Course Adaptations

A key structural feature of this book is its modular organization, with a view towards flexibility, so it can be easily modified to accommodate changes in course content. In particular, the book is divided in two parts:

(1) A *foundational platform*, which consists of Chapter 1. It provides a selective overview of the approximate DP/RL landscape, and a starting point for a more detailed in-class development of other RL topics, whose choice can be at the instructor's discretion.

(2) An *in-depth coverage*, which consists of Chapters 2 and 3. It offers a deeper focus on specific methodologies. In particular, Chapter 2 de-

scribes primarily the deterministic and stochastic rollout techniques, as well as some related approximation in value methods. Chapter 3 addresses the use of neural networks and other approximation architectures for off-line training.

This modular structure allows for customization based on a course's focus. For example, instructors can use the foundational platform of Chapter 1 to build either more mathematically-oriented or less formal courses, depending on the needs of their class.

Let us also mention that the book contains more material than can be reasonably covered in class in one semester. This provides some flexibility to an instructor regarding the choice of material to present.

Videolectures, Slides, and Other Resources

The present book and my RL books above were developed while teaching several versions of my course at ASU. Videolectures and slides from this course, as well as links to overview videolectures by the author are available from my website

http://web.mit.edu/dimitrib/www/RLbook.html

and provide a good supplement and companion resource to this book.

The website

https://space.bilibili.com/2036999141

contains much Chinese-language material, including research papers and videolectures. A translation of the author's 2019 "Reinforcement Learning and Optimal Control" book, translated into Chinese by Yuchao Li, has been published by Tsinghua Press.

An incisive review by Antonio Montano that focuses on the relations of the present book with alternative RL approaches and sources can be found at

https://4m4.it/collections/fkposts/bertsekas-a-course-in-reinforcement-learning/index.html

Thanks and Appreciation

The hospitable and stimulating environment at ASU contributed much to shaping my course during the period 2019-2025. I am thankful to several colleagues and students, and to my teaching assistants, Sushmita Bhattacharya, Sahil Badyal, and Jamison Weber, for their comments and suggestions.

I am especially thankful to Yuchao Li for extensive interactions, collaborative research, and proofreading support since 2019. Moreover, Yuchao has shared with me the lectures of the 2025 offering of the course, and has

undertaken to translate much of the course material into Chinese; see the website

https://space.bilibili.com/2036999141

Dimitri P. Bertsekas

Winter 2026

1

Exact and Approximate Dynamic Programming

Contents

1.1. AlphaZero, Off-Line Training, and On-Line Play . . . . p. 4
1.2. Deterministic Dynamic Programming . . . . . p. 10
1.2.1. Finite Horizon Problem Formulation . . . . . p. 10
1.2.2. The Dynamic Programming Algorithm . . . . . p. 14
1.2.3. Approximation in Value Space and Rollout . . . . p. 22
1.3. Stochastic Exact and Approximate Dynamic Programming p. 28
1.3.1. Finite Horizon Problems . . . . . p. 28
1.3.2. Approximation in Value Space for Stochastic DP . p. 34
1.3.3. Approximation in Policy Space . . . . . p. 38
1.3.4. Training of Cost Function and Policy Approximations . . . . . p. 41
1.4. Infinite Horizon Problems - An Overview . . . . . p. 42
1.4.1. Infinite Horizon Methodology . . . . . p. 46
1.4.2. Approximation in Value Space - Infinite Horizon . p. 49
1.4.3. Understanding Approximation in Value Space . . . p. 55
1.5. Newton's Method - Linear Quadratic Problems . . . . . p. 56
1.5.1. Visualizing Approximation in Value Space - Region of Stability . . . . . p. 62
1.5.2. Rollout and Policy Iteration . . . . . p. 70
1.5.3. Local and Global Error Bounds for Approximation in Value Space . . . . . p. 74
1.5.4. How Approximation in Value Space Can Fail and What to Do About It . . . . . p. 78

1.6. Examples, Reformulations, and Simplifications . . . . . p. 79
1.6.1. A Few Words About Modeling . . . . . p. 79
1.6.2. Problems with a Termination State . . . . . p. 83
1.6.3. General Discrete Optimization Problems . . . . . p. 85
1.6.4. General Finite to Infinite Horizon Reformulation . p. 89
1.6.5. State Augmentation, Time Delays, Forecasts, and . . . Uncontrollable State Components . . . . . p. 91
1.6.6. Partial State Information and Belief States . . . . p. 98
1.6.7. Multiagent Problems and Multiagent Rollout . . p. 102
1.6.8. Problems with Unknown or Changing Model - . . . . Adaptive Control . . . . . p. 107
1.6.9. Model Predictive Control . . . . . p. 118
1.7. Reinforcement Learning and Decision/Control . . . . p. 131
1.7.1. Differences in Terminology . . . . . p. 132
1.7.2. Differences in Notation . . . . . p. 133
1.7.3. Relations Between DP and RL . . . . . p. 135
1.7.4. Synergy Between Large Language Models and RL p. 136
1.7.5. Machine Learning and Optimization . . . . . p. 136
1.7.6. Mathematics in Machine Learning and Optimization p. 139
1.8. Notes, Sources, and Exercises . . . . . p. 140

This chapter has multiple purposes:

(a) *To provide an overview of the exact dynamic programming (DP) methodology, with a focus on suboptimal solution methods*. We will first discuss finite horizon problems, which involve a finite sequence of successive decisions, and are thus conceptually and analytically simpler. We will consider separately deterministic and stochastic finite horizon problems (Sections 1.2 and 1.3, respectively). The reason is that deterministic problems are simpler and have some favorable characteristics, which allow the application of a broader variety of methods. Significantly, they include challenging discrete and combinatorial optimization problems, which can be fruitfully addressed with some of the reinforcement learning (RL) methods that are the main subject of the book. We will also discuss somewhat briefly the more intricate infinite horizon methodology (Section 1.4), and refer to the author's DP textbooks [Ber12], [Ber17a], the RL books [Ber19a], [Ber20a], and the neuro-dynamic programming monograph [BeT96] for a fuller presentation.

(b) *To summarize the principal RL methodologies, with primary emphasis on approximation in value space*. This is the approximate DP-based architecture that underlies the AlphaZero, AlphaGo, TD-Gammon and other related programs, as well as the Model Predictive Control (MPC) methodology, one of the principal control system design methods. We will also argue later (Chapter 2) that approximation in value space provides the entry point for the use of RL methods for solving discrete optimization and integer programming problems.

(c) *To explain the major principles of approximation in value space, and its division into the off-line training and the on-line play algorithms*. A key idea here is the connection of these two algorithms through the algorithmic methodology of Newton's method for solving the problem's Bellman equation. This viewpoint, recently developed in the author's "Rollout and Policy Iteration ..." book [Ber20a] and the visually oriented "Lessons from AlphaZero ..." monograph [Ber22a], underlies the entire course and is discussed for the simple, intuitive, and important class of linear quadratic problems in Section 1.5.

(d) *To give an overview of the range of problem types where our RL methods apply, and to explain some of their major algorithmic ideas (Section 1.6)*. Included here are partial state observation problems (POMDP), multiagent problems, and problems with unknown model parameters, which can be addressed with adaptive control methods.

This chapter will also discuss selectively some major algorithms for approximate DP and RL, including rollout and policy iteration. A broader discussion of DP/RL may be found in the RL books [Ber19a], [Ber20a], the DP textbooks [Ber12], [Ber17a], the neuro-dynamic programming mono-

graph [BeT96], alongside other textbooks referenced in Section 1.8.

The book reflects the author's decision/control and operations research orientation, which has in turn guided the choices of terminology, notation, and mathematical style throughout. On the other hand, RL methods have been developed within the artificial intelligence (AI) community, as well as the decision/control and operations research communities. Despite similarities in the mathematical structures of the problems that these communities address, there are notable differences in terminology, notation, and culture, which can be confusing to researchers entering the field. We have thus provided in Section 1.7 a glossary and an orientation to assist the reader in navigating the full range of the DP/RL literature.

1.1 ALPHAZERO, OFF-LINE TRAINING, AND ON-LINE PLAY

One of the most exciting recent success stories in RL is the development of the AlphaGo and AlphaZero programs by DeepMind Inc; see [SHM16], [SHS17], [SSS17]. AlphaZero plays chess, Go, and other games, and is an improvement in terms of performance and generality over AlphaGo, which plays the game of Go only. Both programs play better than all competitor computer programs available in 2022, and much better than all humans. These programs are remarkable in several other ways. In particular, they have learned how to play without human instruction, just data generated by playing against themselves. Moreover, they learned how to play very quickly. In fact, AlphaZero learned how to play chess better than all humans and computer programs within hours (with the help of awesome parallel computation power, it must be said).

Perhaps the most impressive aspect of AlphaZero/chess is that its play is not just better, but it is also very different from human play in terms of long-term strategic vision. Remarkably, AlphaZero has discovered new ways to play a game that has been studied intensively by humans for hundreds of years. Still, for all of its impressive success and brilliant implementation, AlphaZero is couched in well-established theory and methodology, which is the subject of the present book, and is portable to far broader realms of engineering, economics, and other fields. This is the methodology of DP, policy iteration, limited lookahead, rollout, and approximation in value space.

It is also worth noting that the principles of the AlphaZero design have much in common with the work of Tesauro [Tes94], [Tes95], [TeG96] on computer backgammon. Tesauro's programs stimulated much interest in RL in the middle 1990s, and exhibit similarly different and better play than human backgammon players. A related impressive program for the (one-player) game of Tetris, also based on the method of policy iteration, is described by Scherrer et al. [SGG15], who relied on several earlier works, including those of Tsitsiklis and Van Roy [TsV96], and Bertsekas and Ioffe

[BeI96]. For a better understanding of the connections of AlphaZero and AlphaGo Zero with Tesauro's programs and the concepts developed here, the "Methods" section of the paper [SSS17] is recommended.

To understand the overall structure of AlphaZero and its connection to our DP/RL methodology, it is useful to divide its design into two parts: *off-line training*, which is an algorithm that learns how to evaluate chess positions, and how to steer itself towards good positions with a default/base chess player, and *on-line play*, which is an algorithm that generates good moves in real time against a human or computer opponent, using the training it went through off-line. We will next briefly describe these algorithms, and relate them to DP concepts and principles.

Off-Line Training and Policy Iteration

This is the part of the program that learns how to play through off-line self-training, and is illustrated in Fig. 1.1.1. The algorithm generates a sequence of *chess players* and *position evaluators*. A chess player assigns "probabilities" to all legal moves in a given position, representing the likelihood of each move being chosen. A position evaluator assigns a numerical score to a given position, predicting the player's chances of winning from that position. The chess player and the position evaluator are represented by two neural networks, a *policy network* and a *value network*, which accept a chess position and generate a set of move probabilities and a position evaluation, respectively.†

In more traditional DP terminology, a position is the state of the game, a position evaluator is a cost function that gives (an estimate of) the optimal cost-to-go at a given state, and the chess player is a randomized policy for selecting actions/controls at a given state.‡

The overall training algorithm is a form of *policy iteration*, a classical DP algorithm that will be of primary interest to us in this book. Starting from a given player, it repeatedly generates (approximately) improved players, and settles on a final player that is judged empirically to be "best"

† Here the neural networks play the role of *function approximators*; see Chapter 3. By viewing a player as a function that assigns move probabilities to a position, and a position evaluator as a function that assigns a numerical score to a position, the policy and value networks provide approximations to these functions based on training with data (training algorithms for neural networks and other approximation architectures are also discussed in the RL books [Ber19a], [Ber20a], and the neuro-dynamic programming book [BeT96]).

‡ One more complication is that chess and Go are two-player games, while most of our development will involve single-player optimization. However, DP theory extends to two-player games, although we will not focus on this extension. Alternately, we can consider training a game program to play against a known fixed opponent; this is a one-player setting, discussed further in Section 2.12.

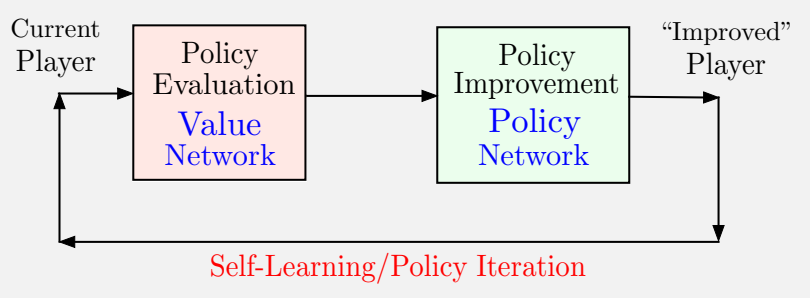


Figure 1.1.1 Illustration of the AlphaZero training algorithm. It generates a sequence of position evaluators and chess players. The position evaluator and the chess player are represented by two neural networks, a value network and a policy network, which accept a chess position and generate a position evaluation and a set of move probabilities, respectively.

out of all the players generated.† Policy iteration may be separated conceptually in two stages (see Fig. 1.1.1).

(a) *Policy evaluation*: Given the current player and a chess position, the outcome of a game played out from the position provides a single data point. Many data points are collected and used to train a value network, which then serves as the position evaluator for the player.

(b) *Policy improvement*: Given the current player and its position evaluator, trial move sequences are selected and evaluated for the rest of the game starting from many positions. An improved player is then generated by adjusting the move probabilities of the current player towards the trial moves that have yielded the best results. In AlphaZero this is done with a complicated algorithm called *Monte Carlo Tree Search*. However, policy improvement can also be done more simply. For example, all possible move sequences from a given position could be tried, extending forward a few moves, with the terminal

† Quoting from the paper [SSS17]: "The AlphaGo Zero selfplay algorithm can similarly be understood as an approximate policy iteration scheme in which MCTS is used for both policy improvement and policy evaluation. Policy improvement starts with a neural network policy, executes an MCTS based on that policy's recommendations, and then projects the (much stronger) search policy back into the function space of the neural network. Policy evaluation is applied to the (much stronger) search policy: the outcomes of selfplay games are also projected back into the function space of the neural network. These projection steps are achieved by training the neural network parameters to match the search probabilities and selfplay game outcome respectively." Note, however, that a two-person game player, trained through selfplay, may fail against a particular human or computer player that can exploit training vulnerabilities. This is a theoretical but rare possibility; see our discussion in Section 2.12.

position evaluated by the player's position evaluator. The move evaluations obtained in this way are used to nudge the move probabilities of the current player towards more successful moves, thereby obtaining data that is used to train a policy network that represents the new player.

Tesauro's TD-Gammon algorithm [Tes94] is similarly based on approximate policy iteration, but uses a different methodology for approximate policy evaluation, based on the TD(λ) algorithm. Unlike AlphaZero, TD-Gammon does not employ a policy network or MCTS. Instead, it relies solely on a value network, to replicate the functionality of a policy network by generating moves on-line via a one-step or two-step lookahead minimization. For a detailed description, see Section 8.6 of the neuro-dynamic programming book [BeT96].

On-Line Play and Approximation in Value Space - Rollout

Suppose that a "final" player has been obtained through the AlphaZero off-line training process just described. This player could, in principle, play chess against any human or computer opponent by generating move probabilities using its off-line-trained policy network. At any position, the player would simply select the move with the highest probability from the policy network. While this approach would allow the player to make decisions quickly, it would not be strong enough to defeat highly skilled human opponents. AlphaZero's extraordinary strength arises only when the off-line-trained player and position evaluator are embedded into another algorithm, which we refer to as the on-line player (see Fig. 1.1.2). At a given position, it generates a lookahead tree of all possible multiple move and countermove sequences, up to a given depth. It then evaluates the effect of the remaining moves by using the position evaluator of the off-line obtained value network.

The architecture of the final version of Tesauro's TD-Gammon program [TeG96] is similar to that of AlphaZero, and uses an off-line neural network-trained terminal position evaluator; see Fig. 1.1.3. However, it also includes a middle portion, called "truncated rollout," which involves running a player for a few moves, before using the position evaluator. In effect, rollout can be viewed as an economical way to extend the length of the lookahead. In the published version of AlphaZero/chess [SHS17], there is also a rollout portion, but it is rather rudimentary; the first portion (multistep lookahead) is quite long and efficiently implemented, so that a sophisticated rollout portion is not essential. Rollout plays a significant role in AlphaGo [SHM16], and is critically important in Tesauro's backgammon program, where long multistep lookahead is infeasible due to the rapid expansion of the lookahead tree.

Architectures that are similar to those of AlphaZero and TD-Gammon will be generically referred to as *approximation in value space* in this book.

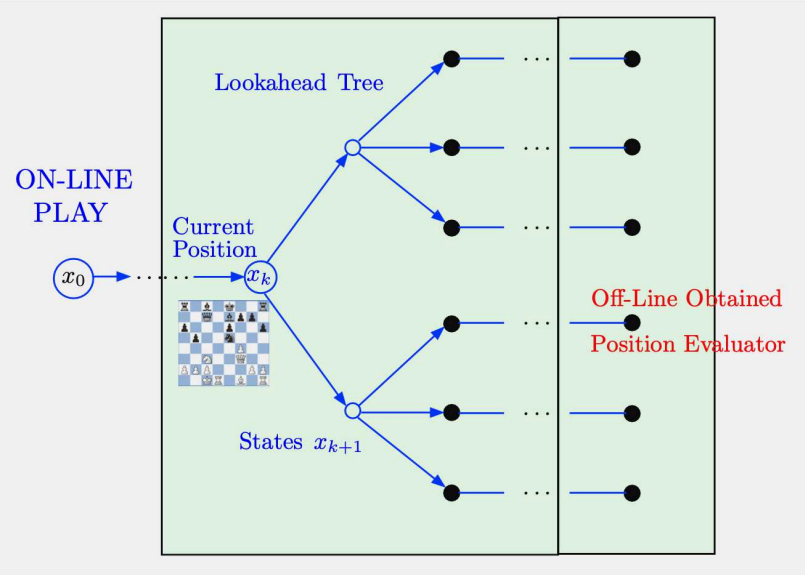


Figure 1.1.2 Illustration of the on-line player of AlphaZero and many other computer chess programs. At a given position, it generates a lookahead tree of multiple moves up to some depth, and evaluates the effect of the remaining moves by using the position evaluator of the off-line player. There are many implementation details that we do not discuss here; for example the lookahead is selective, because some lookahead paths are pruned, by using a form of Monte Carlo tree search. Note that the off-line-trained neural network of AlphaZero produces both a position evaluator and a playing policy. However, the neural network-trained policy is not used directly for on-line play.

Architectures of this type are also known as *approximate dynamic programming*, or *neuro-dynamic programming*, and will be central for our purposes.†

Among other settings, approximation in value space is used in control system design, particularly in *model predictive control* (MPC), which is described in Section 1.6.9. There, the number of steps in lookahead minimization is called the *control interval*, while the total number of steps in

† The names "approximate dynamic programming" and "neuro-dynamic programming" are often used as synonyms to RL. However, RL is generally thought to also subsume the methodology of approximation in policy space, which involves search for optimal parameters within a parametrized set of policies. The search is done with methods that are largely unrelated to DP, such as for example stochastic gradient or random search methods. Approximation in policy space may be used off-line to design a policy that can be used for on-line rollout. It will be discussed in Sections 3.4 and 3.5. A more detailed discussion, consistent with the terminology used here, can be found in Chapter 5 of the RL book [Ber19a].

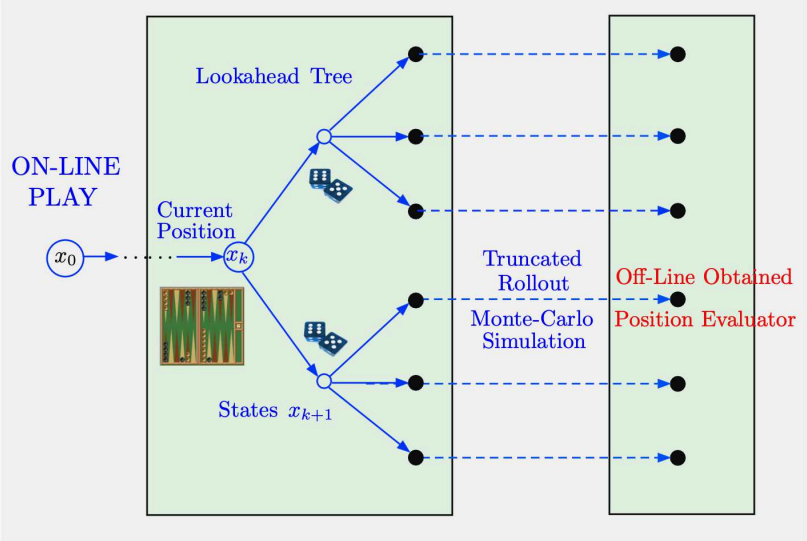


Figure 1.1.3 Illustration of the architecture of TD-Gammon with truncated rollout [TeG96]. It uses a relatively short lookahead minimization followed by rollout and terminal position evaluation (i.e., game simulation with the one-step lookahead player; the game is truncated after a number of moves, with a position evaluation at the end). Note that backgammon involves stochastic uncertainty, and its state is the pair of current board position and dice roll. For this reason, rollout requires Monte-Carlo simulation, and is quite time-consuming. As a result, backgammon programs that play under restrictive time controls, use limited or highly truncated forms of rollout.

lookahead minimization and truncated rollout is called the *prediction interval*; see e.g., Magni et al. [MDM01].† The benefit of truncated rollout in providing an economical substitute for longer lookahead minimization is well known in MPC.

It should be noted that the preceding description of AlphaZero and related approximation in value space architectures is somewhat simplified. We will be discussing refinements and details as the book progresses. However, DP ideas with cost function approximations, similar to the on-line player illustrated in Fig. 1.1.2, will be central to the book. Moreover, the algorithmic division between off-line training and on-line policy implementation will be conceptually very important for our discussions.

We finally note that in approximation in value space the off-line training and the on-line play algorithms can often be decoupled and independently designed. For example the off-line training portion may be simple, such as using a known policy for rollout without truncation, or without

† The Matlab toolbox for MPC design explicitly allows the user to set these two intervals.

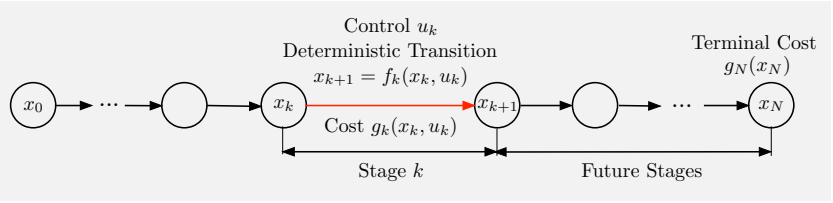


Figure 1.2.1 Illustration of a deterministic N-stage optimal control problem. Starting from state x_k, the next state under control u_k is generated nonrandomly, according to

$$x_{k+1} = f_k(x_k, u_k),$$

and a stage cost $g_k(x_k, u_k)$ is incurred.

terminal cost approximation. Conversely, a sophisticated process may be used for off-line training of a terminal cost function approximation, which is used immediately following one-step or multistep lookahead in a value space approximation scheme.

1.2 DETERMINISTIC DYNAMIC PROGRAMMING

In all DP problems, the central object is a discrete-time dynamic system that generates a sequence of states under the influence of actions or controls. The system may evolve deterministically or randomly (under the additional influence of a random disturbance).

1.2.1 Finite Horizon Problem Formulation

In finite horizon problems the system evolves over a finite number N of time steps (also called stages). The state and control at time k of the system will be generally denoted by x_k and u_k, respectively. In deterministic systems, x_{k+1} is generated nonrandomly, i.e., it is determined solely by x_k and u_k; see Fig. 1.2.1. Thus, a deterministic DP problem involves a system of the form

$$x_{k+1} = f_k(x_k, u_k), \qquad k = 0, 1, \ldots, N-1, \tag{1.1}$$

where

k is the time index,

x_k is the state of the system, an element of some space,

u_k is the control or decision variable, to be selected at time k from some given set $U_k(x_k)$ that depends on x_k,

f_k is a function of (x_k, u_k) that describes the mechanism by which the state is updated from time k to time $k+1$,

N is the horizon, i.e., the number of times control is applied.

In the case of a finite number of states, the system function f_k may be represented by a table that gives the next state x_{k+1} for each possible value of the pair (x_k, u_k). Otherwise a mathematical expression or a computer implementation is necessary to represent f_k.

The set of all possible x_k is called the *state space* at time k. It can be any set and may depend on k. Similarly, the set of all possible u_k is called the *control space* at time k. Again it can be any set and may depend on k. Similarly, the system function f_k can be arbitrary and may depend on k.†

The problem also involves a cost function that is additive in the sense that the cost incurred at time k, denoted by $g_k(x_k, u_k)$, accumulates over time. Formally, g_k is a function of (x_k, u_k) that takes scalar values, and may depend on k. For a given initial state x_0, the total cost of a control sequence $\{u_0, \ldots, u_{N-1}\}$ is

$$J(x_0; u_0, \ldots, u_{N-1}) = g_N(x_N) + \sum_{k=0}^{N-1} g_k(x_k, u_k), \tag{1.2}$$

where $g_N(x_N)$ is a terminal cost incurred at the end of the process. This is a well-defined scalar, since the control sequence $\{u_0, \ldots, u_{N-1}\}$ together with x_0 determines exactly the state sequence $\{x_1, \ldots, x_N\}$ via the system equation (1.1). We want to minimize the cost (1.2) over all sequences $\{u_0, \ldots, u_{N-1}\}$ that satisfy the control constraints, thereby obtaining the optimal value as a function of x_0:‡

$$J^*(x_0) = \min_{\substack{u_k \in U_k(x_k) \\ k=0,\ldots,N-1}} J(x_0; u_0, \ldots, u_{N-1}). \tag{1.3}$$

† This generality is one of the great strengths of the DP methodology and guides the exposition style of this book, and the author's other DP works. By allowing general state and control spaces (discrete, continuous, or mixtures thereof), and a k-dependent choice of these spaces, we can focus attention on the truly essential algorithmic aspects of the DP approach, exclude extraneous assumptions and constraints from our model, and avoid duplication of analysis.

The generality of our DP model is also partly responsible for our choice of notation. In the AI and operations research communities, finite-state models, often referred to as Markovian Decision Problems (MDP), are common and use a transition probability notation (see Section 1.7.2). Unfortunately, this notation is not well suited for deterministic models, and also for continuous-spaces models, both of which are important for the purposes of this book. For the latter models, it involves transition probability distributions over continuous spaces, and leads to mathematics that is far more complex as well as less intuitive than that based on the use of the system function (1.1).

‡ Here and later we write "min" (rather than "inf") even if we are not sure that the minimum is attained; similarly we write "max" (rather than "sup") even if we are not sure that the maximum is attained.

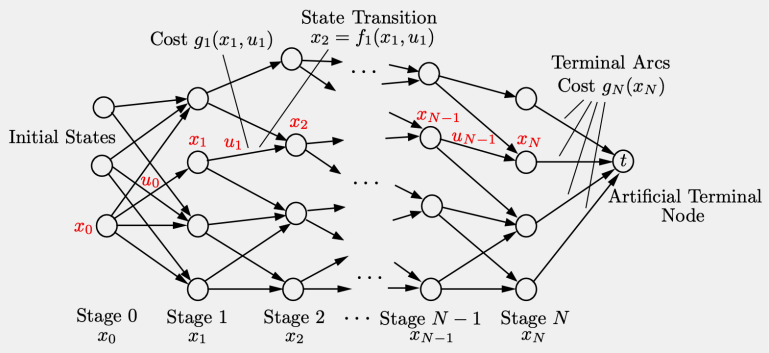


Figure 1.2.2 Transition graph for a deterministic finite-state system. Nodes correspond to states x_k. Arcs correspond to state-control pairs (x_k, u_k). An arc (x_k, u_k) has start and end nodes x_k and $x_{k+1} = f_k(x_k, u_k)$, respectively. We view the cost $g_k(x_k, u_k)$ of the transition as the length of this arc. The problem is equivalent to finding a shortest path from initial nodes of stage 0 to the terminal node t.

Discrete Optimal Control Problems

There are many situations where the state and control spaces are naturally discrete and consist of a finite number of elements. Such problems are often conveniently described with an acyclic graph specifying for each state x_k the possible transitions to next states x_{k+1}. The nodes of the graph correspond to states x_k and the arcs of the graph correspond to state-control pairs (x_k, u_k). Each arc with start node x_k corresponds to a choice of a single control $u_k \in U_k(x_k)$ and has as end node the next state $f_k(x_k, u_k)$. The cost of an arc (x_k, u_k) is defined as $g_k(x_k, u_k)$; see Fig. 1.2.2. To handle the final stage, an artificial terminal node t is added. Each state x_N at stage N is connected to the terminal node t with an arc having cost $g_N(x_N)$.

Note that control sequences $\{u_0, \ldots, u_{N-1}\}$ correspond to paths originating at the initial state (a node at stage 0) and terminating at one of the nodes corresponding to the final stage N. If we view the cost of an arc as its length, we see that *a deterministic finite-state finite-horizon problem is equivalent to finding a minimum-length (or shortest) path from the initial nodes of the graph (stage 0) to the terminal node t.* Here, by the length of a path we mean the sum of the lengths of its arcs.†

Generally, combinatorial optimization problems can be formulated as deterministic finite-state finite-horizon optimal control problems. The idea

† It turns out also that any shortest path problem (with a possibly nonacyclic graph) can be reformulated as a finite-state deterministic optimal control problem. See [Ber17a], Section 2.1, and [Ber91], [Ber98] for extensive accounts of shortest path methods, which connect with our discussion here.

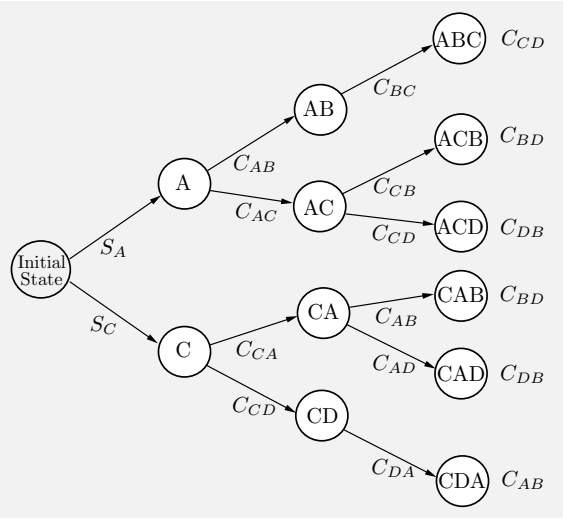


Figure 1.2.3 The transition graph of the deterministic scheduling problem of Example 1.2.1. Each arc of the graph corresponds to a decision leading from some state (the start node of the arc) to some other state (the end node of the arc). The corresponding cost is shown next to the arc. The cost of the last operation is shown as a terminal cost next to the terminal nodes of the graph.

is to break down the solution into components, which can be computed sequentially. The following is an illustrative example.

Example 1.2.1 (A Deterministic Scheduling Problem)

Suppose that to produce a certain product, four operations must be performed on a given machine. The operations are denoted by A, B, C, and D. We assume that operation B can be performed only after operation A has been performed, and operation D can be performed only after operation C has been performed. (Thus the sequence CDAB is allowable but the sequence CDBA is not.) The setup cost C_{mn} for passing from any operation m to any other operation n is given. There is also an initial startup cost S_A or S_C for starting with operation A or C, respectively (cf. Fig. 1.2.3). The cost of a sequence is the sum of the setup costs associated with it; for example, the operation sequence ACDB has cost $S_A + C_{AC} + C_{CD} + C_{DB}$.

We can view this problem as a sequence of three decisions, namely the choice of the first three operations to be performed (the last operation is determined from the preceding three). It is appropriate to consider as state the set of operations already performed, the initial state being an artificial state corresponding to the beginning of the decision process. The possible state transitions corresponding to the possible states and decisions for this problem are shown in Fig. 1.2.3. Here the problem is deterministic, i.e., at a given state, each choice of control leads to a uniquely determined state. For example, at state AC the decision to perform operation D leads to state ACD

with certainty, and has cost C_{CD}. Thus the problem can be conveniently represented with the transition graph of Fig. 1.2.3 (which in turn is a special case of the graph of Fig. 1.2.2). The optimal solution corresponds to the path that starts at the initial state and ends at some state at the terminal time and has minimum sum of arc costs plus the terminal cost.

1.2.2 The Dynamic Programming Algorithm

In this section we will state the DP algorithm and formally justify it. Generally, DP is used to solve a problem of sequential decision making over N stages, by breaking it down to a sequence of simpler single-stage problems.

In particular, the algorithm aims to find a sequence of optimal controls $u_0^*, \ldots, u_{N-1}^*$ by generating a corresponding sequence of optimal cost functions $J_0^*, \ldots, J_{N-1}^*$. It starts with J_N^* equal to the terminal cost function g_N and computes the next function J_{N-1}^*, by solving a single stage decision problem whose optimization variable is u_{N-1}. It then uses J_{N-1}^* to compute J_{N-2}^*, and proceeds similarly to compute all the remaining cost functions $J_{N-3}^*, \ldots, J_0^*$.

The algorithm rests on a simple idea, the *principle of optimality*, which roughly states the following; see Fig. 1.2.4.

Principle of Optimality

Let $\{u_0^*, \ldots, u_{N-1}^*\}$ be an optimal control sequence, which together with x_0 determines the corresponding state sequence $\{x_1^*, \ldots, x_N^*\}$ via the system equation (1.1). Consider the subproblem whereby we start at x_k^* at time k and wish to minimize the "cost-to-go" from time k to time N,

$$g_k(x_k^*, u_k) + \sum_{m=k+1}^{N-1} g_m(x_m, u_m) + g_N(x_N),$$

over $\{u_k, \ldots, u_{N-1}\}$ with $u_m \in U_m(x_m)$, $m = k, \ldots, N-1$. Then the truncated optimal control sequence $\{u_k^*, \ldots, u_{N-1}^*\}$ is optimal for this subproblem.

The subproblem referred to above is called the *tail subproblem* that starts at x_k^*. Stated succinctly, the principle of optimality says that *the tail of an optimal sequence is optimal for the tail subproblem*. Its intuitive justification is simple. If the truncated control sequence $\{u_k^*, \ldots, u_{N-1}^*\}$ were not optimal as stated, we would be able to reduce the cost further by switching to an optimal sequence for the subproblem once we reach x_k^* (since the preceding choices of controls, $u_0^*, \ldots, u_{k-1}^*$, do not restrict our future choices).

For an auto travel analogy, suppose that the fastest route from Phoenix to Boston passes through St Louis. The principle of optimality translates

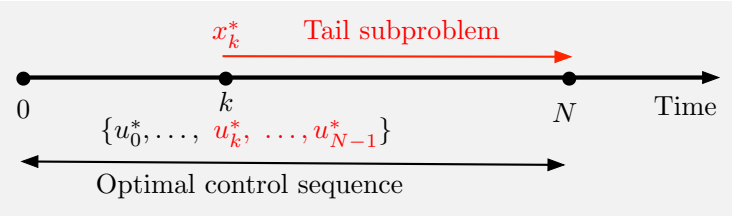


Figure 1.2.4 Schematic illustration of the principle of optimality. The tail $\{u_k^*, \ldots, u_{N-1}^*\}$ of an optimal sequence $\{u_0^*, \ldots, u_{N-1}^*\}$ is optimal for the tail subproblem that starts at the state x_k^* of the optimal state trajectory.

to the obvious fact that the St Louis to Boston portion of the route is also the fastest route for a trip that starts from St Louis and ends in Boston.†

The principle of optimality suggests that the optimal cost function can be constructed in piecemeal fashion going backwards: first compute the optimal cost function for the "tail subproblem" involving the last stage, then solve the "tail subproblem" involving the last two stages, and continue in this manner until the optimal cost function for the entire problem is constructed.

The DP algorithm is based on this idea: it proceeds sequentially by *solving all the tail subproblems of a given time length, using the solution of the tail subproblems of shorter time length*. We illustrate the algorithm with the scheduling problem of Example 1.2.1. The calculations are simple but tedious, and may be skipped without loss of continuity. However, they may be worth going over by a reader who has no prior experience in the use of DP.

Example 1.2.1 (Scheduling Problem - Continued)

Let us consider the scheduling Example 1.2.1, and let us apply the principle of optimality to calculate the optimal schedule. We have to schedule optimally the four operations A, B, C, and D. There is a cost for a transition between two operations, and the numerical values of the transition costs are shown in Fig. 1.2.5 next to the corresponding arcs.

According to the principle of optimality, the "tail" portion of an optimal schedule must be optimal. For example, suppose that the optimal schedule is CABD. Then, having scheduled first C and then A, it must be optimal to complete the schedule with BD rather than with DB. With this in mind, we solve all possible tail subproblems of length two, then all tail subproblems of length three, and finally the original problem that has length four (the subproblems of length one are of course trivial because there is only one operation

† In the words of Bellman [Bel57]: "An optimal trajectory has the property that at an intermediate point, no matter how it was reached, the rest of the trajectory must coincide with an optimal trajectory as computed from this intermediate point as the starting point."

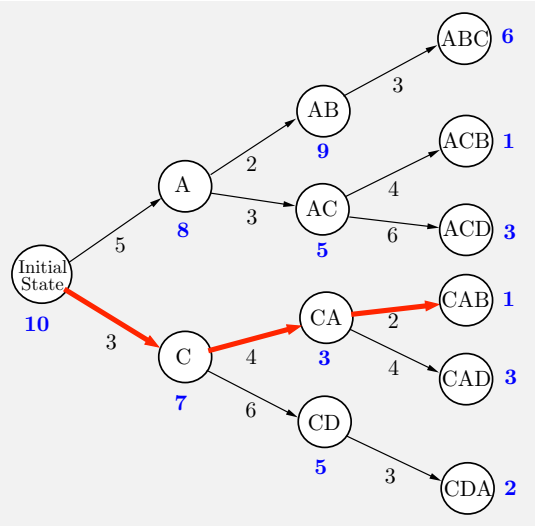


Figure 1.2.5 Transition graph of the deterministic scheduling problem, with the cost of each decision shown next to the corresponding arc. Next to each node/state we show the cost to optimally complete the schedule starting from that state. This is the optimal cost of the corresponding tail subproblem (cf. the principle of optimality). The optimal cost for the original problem is equal to 10, as shown next to the initial state. The optimal schedule corresponds to the thick-line arcs.

that is as yet unscheduled). As we will see shortly, the tail subproblems of length $k+1$ are easily solved once we have solved the tail subproblems of length k, and this is the essence of the DP technique.

Tail Subproblems of Length 2: These subproblems are the ones that involve two unscheduled operations and correspond to the states AB, AC, CA, and CD (see Fig. 1.2.5).

State AB: Here it is only possible to schedule operation C as the next operation, so the optimal cost of this subproblem is 9 (the cost of scheduling C after B, which is 3, plus the cost of scheduling D after C, which is 6).

State AC: Here the possibilities are to (a) schedule operation B and then D, which has cost 5, or (b) schedule operation D and then B, which has cost 9. The first possibility is optimal, and the corresponding cost of the tail subproblem is 5, as shown next to node AC in Fig. 1.2.5.

State CA: Here the possibilities are to (a) schedule operation B and then D, which has cost 3, or (b) schedule operation D and then B, which has cost 7. The first possibility is optimal, and the corresponding cost of the tail subproblem is 3, as shown next to node CA in Fig. 1.2.5.

State CD: Here it is only possible to schedule operation A as the next operation, so the optimal cost of this subproblem is 5.

Tail Subproblems of Length 3: These subproblems can now be solved using the optimal costs of the subproblems of length 2.

State A: Here the possibilities are to (a) schedule next operation B (cost 2) and then solve optimally the corresponding subproblem of length 2 (cost 9, as computed earlier), a total cost of 11, or (b) schedule next operation C (cost 3) and then solve optimally the corresponding subproblem of length 2 (cost 5, as computed earlier), a total cost of 8. The second possibility is optimal, and the corresponding cost of the tail subproblem is 8, as shown next to node A in Fig. 1.2.5.

State C: Here the possibilities are to (a) schedule next operation A (cost 4) and then solve optimally the corresponding subproblem of length 2 (cost 3, as computed earlier), a total cost of 7, or (b) schedule next operation D (cost 6) and then solve optimally the corresponding subproblem of length 2 (cost 5, as computed earlier), a total cost of 11. The first possibility is optimal, and the corresponding cost of the tail subproblem is 7, as shown next to node C in Fig. 1.2.5.

Original Problem of Length 4: The possibilities here are (a) start with operation A (cost 5) and then solve optimally the corresponding subproblem of length 3 (cost 8, as computed earlier), a total cost of 13, or (b) start with operation C (cost 3) and then solve optimally the corresponding subproblem of length 3 (cost 7, as computed earlier), a total cost of 10. The second possibility is optimal, and the corresponding optimal cost is 10, as shown next to the initial state node in Fig. 1.2.5.

Note that having computed the optimal cost of the original problem through the solution of all the tail subproblems, we can construct the optimal schedule: we begin at the initial node and proceed forward, each time choosing the optimal operation, i.e., the one that starts the optimal schedule for the corresponding tail subproblem. In this way, by inspection of the graph and the computational results of Fig. 1.2.5, we determine that CABD is the optimal schedule.

Finding an Optimal Control Sequence by DP

We now state the DP algorithm for deterministic finite horizon problems by translating into mathematical terms the heuristic argument underlying the principle of optimality. The algorithm constructs functions

$$J_N^*(x_N), J_{N-1}^*(x_{N-1}), \ldots, J_0^*(x_0),$$

sequentially, starting from J_N^*, and proceeding backwards to J_{N-1}^*, J_{N-2}^*, etc. We will show that the value $J_k^*(x_k)$ represents the optimal cost of the tail subproblem that starts at state x_k at time k.

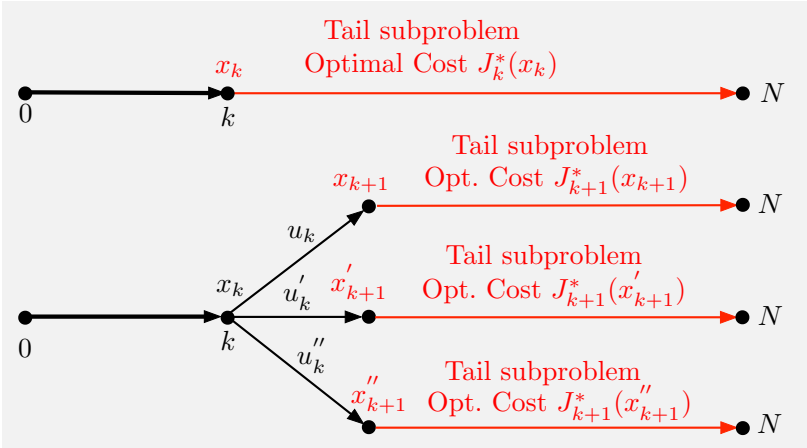


Figure 1.2.6 Illustration of the DP algorithm. The tail subproblem that starts at x_k at time k minimizes over $\{u_k, \ldots, u_{N-1}\}$ the "cost-to-go" from k to N,

$$g_k(x_k, u_k) + \sum_{m=k+1}^{N-1} g_m(x_m, u_m) + g_N(x_N).$$

To solve it, we choose u_k to minimize the (1st stage cost + Optimal tail problem cost) or

$$J_k^*(x_k) = \min_{u_k \in U_k(x_k)} \Big[g_k(x_k, u_k) + J_{k+1}^*\big(f_k(x_k, u_k)\big) \Big].$$

DP Algorithm for Deterministic Finite Horizon Problems

Start with

$$J_N^*(x_N) = g_N(x_N), \qquad \text{for all } x_N, \tag{1.4}$$

and for $k = 0, \ldots, N-1$, let

$$J_k^*(x_k) = \min_{u_k \in U_k(x_k)} \Big[g_k(x_k, u_k) + J_{k+1}^*\big(f_k(x_k, u_k)\big) \Big], \qquad \text{for all } x_k. \tag{1.5}$$

The DP algorithm together with the construction of the functions $J_k^*(x_k)$ are illustrated in Fig. 1.2.6. Note that at stage k, the calculation in Eq. (1.5) must be done for all states x_k before proceeding to stage $k-1$. The key fact about the DP algorithm is that for every initial state x_0, the number $J_0^*(x_0)$ obtained at the last step, is equal to the optimal cost $J^*(x_0)$. Indeed, a more general fact can be shown, namely that for all

$k = 0, 1, \ldots, N-1$, and all states x_k at time k, we have

$$J_k^*(x_k) = \min_{\substack{u_m \in U_m(x_m) \\ m=k,\ldots,N-1}} J(x_k; u_k, \ldots, u_{N-1}), \tag{1.6}$$

where $J(x_k; u_k, \ldots, u_{N-1})$ is the cost generated by starting at x_k and using subsequent controls $u_k, \ldots, u_{N-1}$:

$$J(x_k; u_k, \ldots, u_{N-1}) = g_N(x_N) + \sum_{t=k}^{N-1} g_t(x_t, u_t). \tag{1.7}$$

Thus, $J_k^*(x_k)$ is the optimal cost for an $(N-k)$-stage tail subproblem that starts at state x_k and time k, and ends at time N.† Based on the interpretation (1.6) of $J_k^*(x_k)$, we call it the *optimal cost-to-go* from state x_k at stage k, and refer to J_k^* as the *optimal cost-to-go function* or *optimal cost function* at time k. In maximization problems the DP algorithm (1.5) is written with maximization in place of minimization, and then J_k^* is referred to as the *optimal value function* at time k.

Once the functions $J_0^*, \ldots, J_N^*$ have been obtained, we can use a forward algorithm to construct an optimal control sequence $\{u_0^*, \ldots, u_{N-1}^*\}$ and corresponding state trajectory $\{x_1^*, \ldots, x_N^*\}$ for the given initial state x_0.

---

† We can prove this by induction. The assertion holds for $k = N$ in view of the initial condition

$$J_N^*(x_N) = g_N(x_N).$$

To show that it holds for all k, we use Eqs. (1.6) and (1.7) to write

$$\begin{aligned} J_k^*(x_k) &= \min_{\substack{u_t \in U_t(x_t) \\ t=k,\ldots,N-1}} \left[g_N(x_N) + \sum_{t=k}^{N-1} g_t(x_t, u_t) \right] \\ &= \min_{u_k \in U_k(x_k)} \Bigg[g_k(x_k, u_k) \\ &\qquad + \min_{\substack{u_t \in U_t(x_t) \\ t=k+1,\ldots,N-1}} \left[g_N(x_N) + \sum_{t=k+1}^{N-1} g_t(x_t, u_t) \right] \Bigg] \\ &= \min_{u_k \in U_k(x_k)} \Big[g_k(x_k, u_k) + J_{k+1}^*\big(f_k(x_k, u_k)\big) \Big], \end{aligned}$$

where for the last equality we use the induction hypothesis. A subtle mathematical point here is that, through the minimization operation, the cost-to-go functions J_k^* may take the value $-\infty$ for some x_k. Still the preceding induction argument is valid even if this is so.

Construction of Optimal Control Sequence $\{u_0^*, \ldots, u_{N-1}^*\}$

Set

$$u_0^* \in \arg \min_{u_0 \in U_0(x_0)} \Big[g_0(x_0, u_0) + J_1^*\big(f_0(x_0, u_0)\big) \Big],$$

and

$$x_1^* = f_0(x_0, u_0^*).$$

Sequentially, going forward, for $k = 1, 2, \ldots, N-1$, set

$$u_k^* \in \arg \min_{u_k \in U_k(x_k^*)} \Big[g_k(x_k^*, u_k) + J_{k+1}^*\big(f_k(x_k^*, u_k)\big) \Big], \qquad (1.8)$$

and

$$x_{k+1}^* = f_k(x_k^*, u_k^*).$$

The same algorithm can be used to find an optimal control sequence for any tail subproblem. Figure 1.2.5 traces the calculations of the DP algorithm for the scheduling Example 1.2.1. The numbers next to the nodes give the corresponding cost-to-go values, and the thick-line arcs give the construction of the optimal control sequence using the preceding algorithm.

The following example deals with the classical traveling salesman problem involving N cities. Here, the number of states grows exponentially with N, and so does the corresponding amount of computation for exact DP. We will show later that with rollout, we can solve the problem approximately with computation that grows polynomially with N.

Example 1.2.2 (Traveling Salesman Problem)

Here we are given N cities and the travel time between each pair of cities. We wish to find a minimum-time travel that visits each of the cities exactly once and returns to the start city. To convert this problem to a DP problem, we form a graph whose nodes are the sequences of k distinct cities, where $k = 1, \ldots, N$. The k-city sequences correspond to the states of the kth stage. The initial state x_0 consists of some city, taken as the start (city A in the example of Fig. 1.2.7). A k-city node/state leads to a $(k+1)$-city node/state by adding a new city at a cost equal to the travel time between the last two of the $k+1$ cities; see Fig. 1.2.7. Each sequence of N cities is connected to an artificial terminal node t with an arc of cost equal to the travel time from the last city of the sequence to the starting city, thus completing the transformation to a DP problem.

The optimal costs-to-go from each node to the terminal state can be obtained by the DP algorithm and are shown next to the nodes. Note, however, that the number of nodes grows exponentially with the number of cities N. This makes the DP solution intractable for large N. As a result, large traveling salesman and related scheduling problems are typically not addressed with

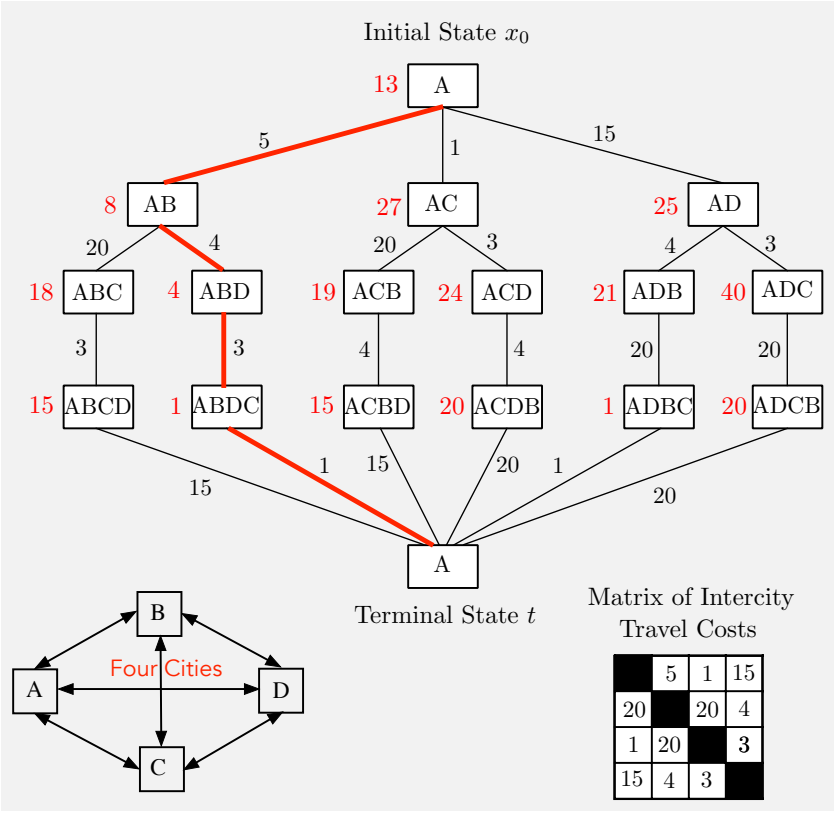


Figure 1.2.7 The DP formulation of the traveling salesman problem of Example 1.2.2. The travel times between the four cities A, B, C, and D are shown in the matrix at the bottom. We form a graph whose nodes are the k-city sequences and correspond to the states of the kth stage, assuming that A is the starting city. The transition costs/travel times are shown next to the arcs. The optimal costs-to-go are generated by DP starting from the terminal state and going backwards towards the initial state, and are shown next to the nodes. There is a unique optimal sequence here (ABDCA), and it is marked with thick lines. The optimal sequence can be obtained by forward minimization [cf. Eq. (1.8)], starting from the initial state x_0.

exact DP, but rather with approximation methods. Some of these methods are based on DP and will be discussed later.

Q-Factors and Q-Learning

An alternative (and equivalent) form of the DP algorithm (1.5), uses the optimal cost-to-go functions J_k^* indirectly. In particular, it generates the

optimal Q-factors, defined for all pairs (x_k, u_k) and k by

$$Q_k^*(x_k, u_k) = g_k(x_k, u_k) + J_{k+1}^*\big(f_k(x_k, u_k)\big). \tag{1.9}$$

Thus the optimal Q-factors are simply the expressions that are minimized in the right-hand side of the DP equation (1.5).†

Note that the optimal cost function J_k^* can be recovered from the optimal Q-factor Q_k^* by means of the minimization

$$J_k^*(x_k) = \min_{u_k \in U_k(x_k)} Q_k^*(x_k, u_k). \tag{1.10}$$

Moreover, the DP algorithm (1.5) can be written in an essentially equivalent form that involves Q-factors only [cf. Eqs. (1.9)-(1.10)]:

$$Q_k^*(x_k, u_k) = g_k(x_k, u_k) + \min_{u_{k+1} \in U_{k+1}(f_k(x_k, u_k))} Q_{k+1}^*\big(f_k(x_k, u_k), u_{k+1}\big).$$

Exact and approximate forms of this and other related algorithms, including counterparts for stochastic optimal control problems, comprise an important class of RL methods known as *Q-learning*.

1.2.3 Approximation in Value Space and Rollout

The forward optimal control sequence construction of Eq. (1.8) is possible only after we have computed $J_k^*(x_k)$ by DP for all x_k and k. Unfortunately, in practice this is often prohibitively time-consuming, because the number of possible x_k and k can be very large. However, a similar forward algorithmic process can be used if the optimal cost-to-go functions J_k^* are replaced by some approximations $\tilde{J}_k$. This is the basis for an idea that is central in RL: *approximation in value space*.‡ It constructs a suboptimal solution $\{\tilde{u}_0, \ldots, \tilde{u}_{N-1}\}$ in place of the optimal $\{u_0^*, \ldots, u_{N-1}^*\}$, based on using $\tilde{J}_k$ in place of J_k^* in the DP algorithm (1.8).

† The term "Q-factor" has been used in the books [BeT96], [Ber19a], [Ber20a] and is adopted here as well. Another term used is "action value" (at a given state). The terms "state-action value" and "Q-value" are also common in the literature. The name "Q-factor" originated in reference to the notation used in an influential Ph.D. thesis [Wat89] that proposed the use of Q-factors in RL.

‡ Approximation in value space (sometimes called "search" or "tree search" in the AI literature) is a simple idea that has been used quite extensively for deterministic problems, well before the development of the modern RL methodology. For example it conceptually underlies the widely used A^* method for computing approximate solutions to large scale shortest path problems. For a view of A^* that is consistent with our approximate DP framework, the reader may consult the author's DP book [Ber17a].

Approximation in Value Space - Use of $\tilde{J}_k$ in Place of J_k^*

Start with

$$\tilde{u}_0 \in \arg\min_{u_0 \in U_0(x_0)} \Big[g_0(x_0, u_0) + \tilde{J}_1\big(f_0(x_0, u_0)\big) \Big],$$

and set

$$\tilde{x}_1 = f_0(x_0, \tilde{u}_0).$$

Sequentially, going forward, for $k = 1, 2, \ldots, N-1$, set

$$\tilde{u}_k \in \arg\min_{u_k \in U_k(\tilde{x}_k)} \Big[g_k(\tilde{x}_k, u_k) + \tilde{J}_{k+1}\big(f_k(\tilde{x}_k, u_k)\big) \Big], \tag{1.11}$$

and

$$\tilde{x}_{k+1} = f_k(\tilde{x}_k, \tilde{u}_k).$$

In approximation in value space the calculation of the suboptimal sequence $\{\tilde{u}_0, \ldots, \tilde{u}_{N-1}\}$ is done by going forward (no backward calculation is needed once the approximate cost-to-go functions $\tilde{J}_k$ are available). This is similar to the calculation of the optimal sequence $\{u_0^*, \ldots, u_{N-1}^*\}$, and is independent of how the functions $\tilde{J}_k$ are computed. The motivation for approximation in value space for stochastic DP problems is vastly reduced computation relative to the exact DP algorithm (once $\tilde{J}_k$ have been obtained): the minimization (1.11) needs to be performed only for the N states $x_0, \tilde{x}_1, \ldots, \tilde{x}_{N-1}$ that are encountered during the on-line control of the system, and not for every state within the potentially enormous state space, as is the case for exact DP.

The algorithm (1.11) is said to involve a *one-step lookahead minimization*, since it solves a one-stage DP problem for each k. In what follows we will also discuss the possibility of *multistep lookahead*, which involves the solution of an ℓ-step DP problem, where ℓ is an integer, $1 < \ell < N - k$, with a terminal cost function approximation $\tilde{J}_{k+\ell}$. Multistep lookahead typically (but not always) provides better performance over one-step lookahead in RL approximation schemes. For example in AlphaZero chess, long multistep lookahead is critical for good on-line performance. The intuitive reason is that with ℓ stages being treated "exactly" (by optimization), the effect of the approximation error

$$\tilde{J}_{k+\ell} - J_{k+\ell}^*$$

tends to become less significant as ℓ increases. However, the solution of the multistep lookahead optimization problem, instead of the one-step lookahead counterpart of Eq. (1.11), becomes more time consuming.

Rollout with a Base Heuristic for Deterministic Problems

A major issue in value space approximation is the construction of suitable approximate cost-to-go functions $\tilde{J}_k$. This can be done in many different ways, giving rise to some of the principal RL methods. For example, $\tilde{J}_k$ may be constructed with a sophisticated off-line training method, as discussed in Section 1.1. Alternatively, $\tilde{J}_k$ may be obtained on-line with *rollout*, which will be discussed in detail in this book. In rollout, the approximate values $\tilde{J}_k(x_k)$ are obtained when needed by running a heuristic control scheme, called *base heuristic* or *base policy*, for a suitably large number of stages, starting from the state x_k, and accumulating the costs incurred at these stages.

The major theoretical property of rollout is *cost improvement*: the cost obtained by rollout using some base heuristic is less or equal to the corresponding cost of the base heuristic. This is true for any starting state, provided the base heuristic satisfies some simple conditions, which will be discussed in Chapter 2.†

There are also several variants of rollout, including versions involving multiple heuristics, combinations with other forms of approximation in value space methods, multistep lookahead, and stochastic uncertainty. We will discuss such variants later. For the moment we will focus on a deterministic DP problem with a finite number of controls. Given a state x_k at time k, this algorithm considers all the tail subproblems that start at every possible next state x_{k+1}, and solves them suboptimally by using some algorithm, referred to as base heuristic.

Thus when at x_k, rollout generates on-line the next states x_{k+1} that correspond to all $u_k \in U_k(x_k)$, and uses the base heuristic to compute the sequence of states $\{x_{k+1}, \ldots, x_N\}$ and controls $\{u_{k+1}, \ldots, u_{N-1}\}$ such that

$$x_{t+1} = f_t(x_t, u_t), \qquad t = k, \ldots, N-1,$$

and the corresponding cost

$$H_{k+1}(x_{k+1}) = g_{k+1}(x_{k+1}, u_{k+1}) + \cdots + g_{N-1}(x_{N-1}, u_{N-1}) + g_N(x_N).$$

The rollout algorithm then applies the control that minimizes over $u_k \in U_k(x_k)$ the tail cost expression for stages k to N:

$$g_k(x_k, u_k) + H_{k+1}(x_{k+1}).$$

† For an intuitive justification of the cost improvement mechanism, note that the rollout control $\tilde{u}_k$ is calculated from Eq. (1.11) to attain the minimum over u_k over the sum of two terms: the first stage cost $g_k(\tilde{x}_k, u_k)$ plus the cost of the remaining stages ($k+1$ to N) using the heuristic controls. Thus rollout involves a first stage optimization (rather than just using the base heuristic), which accounts for the cost improvement. This reasoning also explains why multistep lookahead tends to provide better performance than one-step lookahead in rollout schemes.

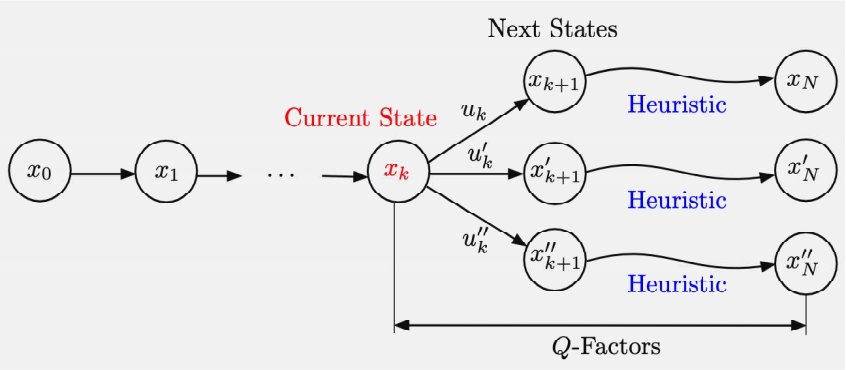


Figure 1.2.8 Schematic illustration of rollout with one-step lookahead for a deterministic problem. At state x_k, for every pair (x_k, u_k), $u_k \in U_k(x_k)$, the base heuristic generates an approximate Q-factor

$$\tilde{Q}_k(x_k, u_k) = g_k(x_k, u_k) + H_{k+1}\big(f_k(x_k, u_k)\big),$$

and selects the control $\tilde{\mu}_k(x_k)$ with minimal Q-factor.

Equivalently, and more succinctly, the rollout algorithm applies at state x_k the control $\tilde{\mu}_k(x_k)$ given by the minimization

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} \tilde{Q}_k(x_k, u_k), \tag{1.12}$$

where $\tilde{Q}_k(x_k, u_k)$ is the approximate Q-factor defined by

$$\tilde{Q}_k(x_k, u_k) = g_k(x_k, u_k) + H_{k+1}\big(f_k(x_k, u_k)\big); \tag{1.13}$$

see Fig. 1.2.8.

Note that the rollout algorithm requires running the base heuristic for a number of times that is bounded by Nn, where n is an upper bound on the number of control choices available at each state. Thus if n is small relative to N, it requires computation equal to a small multiple of N times the computation time for a single application of the base heuristic. Similarly, if n is bounded by a polynomial in N, the ratio of the rollout algorithm computation time to the base heuristic computation time is a polynomial in N.

Example 1.2.3 (Traveling Salesman Problem - Continued)

Let us consider the traveling salesman problem of Example 1.2.2, whereby a salesman wants to find a minimum cost tour that visits each of N given cities $c = 0, \ldots, N-1$ exactly once and returns to the city he started from. With each pair of distinct cities c, c', we associate a traversal cost $g(c, c')$. Note

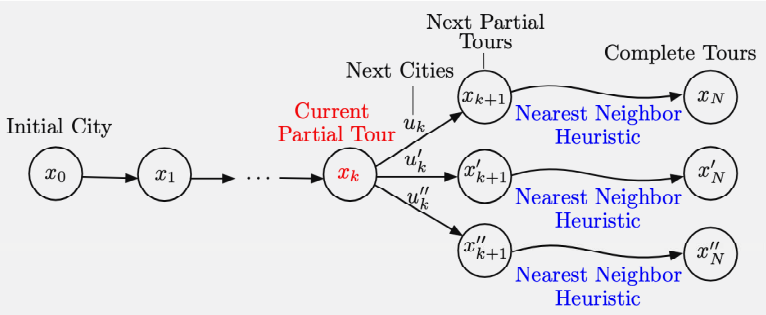


Figure 1.2.9 Rollout with the nearest neighbor heuristic for the traveling salesman problem of Example 1.2.3. The initial state x_0 consists of a single city. The final state x_N is a complete tour of N cities, containing each city exactly once.

that we assume that we can go directly from every city to every other city. There is no loss of generality in doing so because we can assign a very high cost $g(c, c')$ to any pair of cities (c, c') that is precluded from participation in the solution. The problem is to find a visit order that goes through each city exactly once and whose sum of costs is minimum.

There are many heuristic approaches for solving the traveling salesman problem. For illustration purposes, let us focus on the simple *nearest neighbor* heuristic, which starts with a partial tour, i.e., an ordered collection of distinct cities, and constructs a sequence of partial tours, adding to each partial tour a new city that does not close a cycle and minimizes the cost of the enlargement. In particular, given a sequence $\{c_0, c_1, \ldots, c_k\}$ (with $k < N-1$) consisting of distinct cities, the nearest neighbor heuristic adds a city c_{k+1} that minimizes $g(c_k, c_{k+1})$ over all cities $c_{k+1} \neq c_0, \ldots, c_k$, thereby forming the sequence $\{c_0, c_1, \ldots, c_k, c_{k+1}\}$. Continuing in this manner, the heuristic eventually forms a sequence of N cities, $\{c_0, c_1, \ldots, c_{N-1}\}$, thus yielding a complete tour with cost

$$g(c_0, c_1) + \cdots + g(c_{N-2}, c_{N-1}) + g(c_{N-1}, c_0). \tag{1.14}$$

We can formulate the traveling salesman problem as a DP problem as we discussed in Example 1.2.2. We choose a starting city, say c_0, as the initial state x_0. Each state x_k corresponds to a partial tour $(c_0, c_1, \ldots, c_k)$ consisting of distinct cities. The states x_{k+1}, next to x_k, are sequences of the form $(c_0, c_1, \ldots, c_k, c_{k+1})$ that correspond to adding one more unvisited city $c_{k+1} \neq c_0, c_1, \ldots, c_k$ (thus the unvisited cities are the feasible controls at a given partial tour/state). The terminal states x_N are the complete tours of the form $(c_0, c_1, \ldots, c_{N-1}, c_0)$, and the cost of the corresponding sequence of city choices is the cost of the corresponding complete tour given by Eq. (1.14). Note that the number of states at stage k increases exponentially with k, and so does the computation required to solve the problem by exact DP.

Let us now use as a base heuristic the nearest neighbor method. The corresponding rollout algorithm operates as follows: After $k < N-1$ iterations, we have a state x_k, i.e., a sequence $\{c_0, \ldots, c_k\}$ consisting of distinct cities. At the next iteration, we add one more city by running the

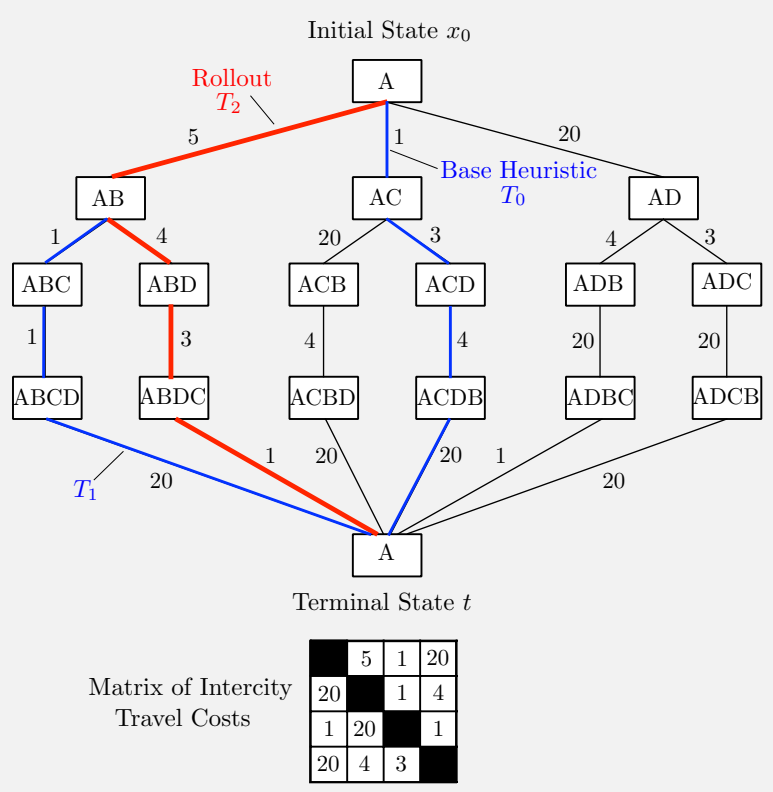


Figure 1.2.10 Rollout with the nearest neighbor base heuristic, applied to a traveling salesman problem. At city A, the nearest neighbor heuristic generates the tour ACDBA (labeled T_0). At city A, the rollout algorithm compares the tours ABCDA, ACDBA, and ADCBA, finds ABCDA (labeled T_1) to have the least cost, and moves to city B. At AB, the rollout algorithm compares the tours ABCDA and ABDCA, finds ABDCA (labeled T_2) to have the least cost, and moves to city D. The rollout algorithm then moves to cities C and A (it has no other choice). The final tour T_2 generated by rollout turns out to be optimal in this example, while the tour T_0 generated by the base heuristic is suboptimal. This is suggestive of a general result: the rollout algorithm for deterministic problems generates a sequence of solutions of decreasing cost under some conditions on the base heuristic that we will discuss in Chapter 2, and which are satisfied by the nearest neighbor heuristic.

nearest neighbor heuristic starting from each of the sequences of the form $\{c_0, \ldots, c_k, c\}$ where $c \neq c_0, \ldots, c_k$. We then select as next city c_{k+1} the city c that yielded the minimum cost tour under the nearest neighbor heuristic; see Fig. 1.2.9. The overall computation for the rollout solution is bounded by a polynomial in N, and is much smaller than the exact DP computation. Figure 1.2.10 provides an example where the nearest neighbor heuristic and the corresponding rollout algorithm are compared; see also Exercise 1.1.

1.3 STOCHASTIC EXACT AND APPROXIMATE DYNAMIC PROGRAMMING

We will now extend the DP algorithm and our discussion of approximation in value space to problems that involve stochastic uncertainty in their system equation and cost function. We will first discuss the finite horizon case, and the extension of the ideas underlying the principle of optimality and approximation in value space schemes. We will then consider the infinite horizon version of the problem, and provide an overview of the underlying theory and algorithmic methodology.

1.3.1 Finite Horizon Problems

The stochastic optimal control problem differs from its deterministic counterpart primarily in the nature of the discrete-time dynamic system that governs the evolution of the state x_k. This system includes a random "disturbance" w_k with a probability distribution $P_k(\cdot \mid x_k, u_k)$ that may depend explicitly on x_k and u_k, but not on values of prior disturbances $w_{k-1}, \ldots, w_0$. The system has the form

$$x_{k+1} = f_k(x_k, u_k, w_k), \qquad k = 0, 1, \ldots, N-1,$$

where, as earlier, x_k is an element of some state space, the control u_k is an element of some control space.† The cost per stage is denoted by $g_k(x_k, u_k, w_k)$ and also depends on the random disturbance w_k; see Fig. 1.3.1. The control u_k is constrained to take values in a given subset $U_k(x_k)$, which depends on the current state x_k.

Given an initial state x_0 and a policy $\pi = \{\mu_0, \ldots, \mu_{N-1}\}$, the future states x_k and disturbances w_k are random variables with distributions defined through the system equation

$$x_{k+1} = f_k\big(x_k, \mu_k(x_k), w_k\big), \qquad k = 0, 1, \ldots, N-1,$$

† The discrete equation format and corresponding x-u-w notation is standard in the optimal control literature. For finite-state stochastic problems, also called *Markovian Decision Problems* (MDP), the system is often represented conveniently in terms of control-dependent transition probabilities. A common notation in the RL literature is $p(s, a, s')$ for transition probability from s to s' under action a. This type of notation is not well suited for deterministic problems, which involve no probabilistic structure at all and are of major interest in this book. The transition probability notation is also cumbersome for problems with a continuous state space; see Sections 1.7.1 and 1.7.2 for further discussion. The reader should note, however, that mathematically the system equation and transition probabilities are equivalent, and any analysis that can be done in one notational system can be translated to the other notational system.

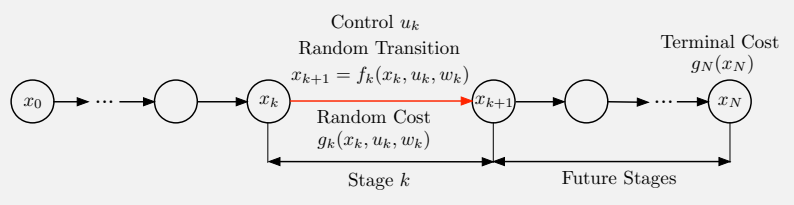


Figure 1.3.1 Illustration of an N-stage stochastic optimal control problem. Starting from state x_k, the next state under control u_k is generated randomly, according to $x_{k+1} = f_k(x_k, u_k, w_k)$, where w_k is the random disturbance, and a random stage cost $g_k(x_k, u_k, w_k)$ is incurred.

and the given distributions $P_k(\cdot \mid x_k, u_k)$. Thus, for given functions g_k, $k = 0, 1, \ldots, N$, the expected cost of π starting at x_0 is

$$J_\pi(x_0) = \mathop{E}_{\substack{w_k \\ k=0,\ldots,N-1}} \left\{ g_N(x_N) + \sum_{k=0}^{N-1} g_k\big(x_k, \mu_k(x_k), w_k\big) \right\},$$

where the expected value operation $E\{\cdot\}$ is taken with respect to the joint distribution of all the random variables w_k and x_k.† An optimal policy π^* is one that minimizes this cost; i.e.,

$$J_{\pi^*}(x_0) = \min_{\pi \in \Pi} J_\pi(x_0),$$

where Π is the set of all policies.

An important difference from the deterministic case is that we optimize not over control sequences $\{u_0, \ldots, u_{N-1}\}$ [cf. Eq. (1.3)], but rather over *policies* (also called *closed-loop control laws*, or *feedback policies*) that consist of a sequence of functions

$$\pi = \{\mu_0, \ldots, \mu_{N-1}\},$$

where μ_k maps states x_k into controls $u_k = \mu_k(x_k)$, and satisfies the control constraints, i.e., is such that $\mu_k(x_k) \in U_k(x_k)$ for all x_k. Policies are more general objects than control sequences, and in the presence of stochastic uncertainty, they can result in improved cost, since they allow choices of controls u_k that incorporate knowledge of the state x_k. Without this knowledge, the controller cannot adapt appropriately to unexpected values of the state, and as a result the cost can be adversely affected. This is a fundamental distinction between deterministic and stochastic optimal control problems.

† We assume an introductory probability background on the part of the reader. For an account that is consistent with our use of probability in this book, see the textbook by Bertsekas and Tsitsiklis [BeT08].

The optimal cost depends on x_0 and is denoted by $J^*(x_0)$; i.e.,

$$J^*(x_0) = \min_{\pi \in \Pi} J_\pi(x_0).$$

We view J^* as a function that assigns to each initial state x_0 the optimal cost $J^*(x_0)$, and call it the *optimal cost function* or *optimal value function*.

Stochastic Dynamic Programming

The DP algorithm for the stochastic finite horizon optimal control problem has a similar form to its deterministic version, and shares several of its major characteristics:

(a) Using tail subproblems to break down the minimization over multiple stages to single stage minimizations.

(b) Generating backwards for all k and x_k the values $J_k^*(x_k)$, which give the optimal cost-to-go starting from state x_k at stage k.

(c) Obtaining an optimal policy by minimization in the DP equations.

(d) A structure that is suitable for approximation in value space, whereby we replace J_k^* by approximations $\tilde{J}_k$, and obtain a suboptimal policy by the corresponding minimization.

DP Algorithm for Stochastic Finite Horizon Problems

Start with

$$J_N^*(x_N) = g_N(x_N),$$

and for $k = 0, \ldots, N-1$, let

$$J_k^*(x_k) = \min_{u_k \in U_k(x_k)} E_{w_k}\Big\{ g_k(x_k, u_k, w_k) + J_{k+1}^*\big(f_k(x_k, u_k, w_k)\big)\Big\}. \tag{1.15}$$

For each x_k and k, define $\mu_k^*(x_k) = u_k^*$ where u_k^* attains the minimum in the right side of this equation. Then, the policy $\pi^* = \{\mu_0^*, \ldots, \mu_{N-1}^*\}$ is optimal.

The key fact is that starting from any initial state x_0, the optimal cost is equal to the number $J_0^*(x_0)$, obtained at the last step of the above DP algorithm. This can be proved by induction similar to the deterministic case; we will omit the proof (which incidentally involves some mathematical fine points; see the discussion of Section 1.3 in the textbook [Ber17a]).

Simultaneously with the off-line computation of the optimal cost-to-go functions $J_0^*, \ldots, J_N^*$, we can compute and store an optimal policy $\pi^* = \{\mu_0^*, \ldots, \mu_{N-1}^*\}$ by minimization in Eq. (1.15). We can then use this

policy on-line to retrieve from memory and apply the control $\mu_k^*(x_k)$ once we reach state x_k. The alternative is to forego the storage of the policy π^* and to calculate the control $\mu_k^*(x_k)$ by executing the minimization (1.15) on-line.

There are a few favorable cases where the optimal cost-to-go functions J_k^* and the optimal policies μ_k^* can be computed analytically using the stochastic DP algorithm. A prominent such case involves a linear system and a quadratic cost function, which is a fundamental problem in control theory. We illustrate the scalar version of this problem next. The analysis can be generalized to multidimensional systems (see optimal control textbooks such as [Ber17a]).

Example 1.3.1 (Linear Quadratic Optimal Control)

Here the system is linear,

$$x_{k+1} = ax_k + bu_k + w_k, \qquad k = 0, \ldots, N-1,$$

and the state and control are scalars. Moreover, the disturbance w_k has zero mean and given variance σ^2. The cost is quadratic of the form:

$$qx_N^2 + \sum_{k=0}^{N-1} (qx_k^2 + ru_k^2),$$

where q and r are known positive weighting parameters. We assume no constraints on x_k and u_k (in reality such problems include constraints, but it is common to neglect the constraints initially, and check whether they are seriously violated later).

As an illustration, consider a vehicle that moves on a straight-line road under the influence of a force u_k and without friction. Our objective is to maintain the vehicle's velocity at a constant level $\bar{v}$ (as in an oversimplified cruise control system). The velocity v_k at time k, after time discretization of its Newtonian dynamics and addition of stochastic noise, evolves according to

$$v_{k+1} = v_k + bu_k + w_k, \tag{1.16}$$

where w_k is a stochastic disturbance. By introducing $x_k = v_k - \bar{v}$, the deviation between the vehicle's velocity v_k at time k from the desired level $\bar{v}$, we obtain the system equation

$$x_{k+1} = x_k + bu_k + w_k.$$

Here the coefficient b relates to a number of problem characteristics including the weight of the vehicle and the road conditions. The cost function expresses our desire to keep x_k near zero with relatively little force.

We will apply the DP algorithm, and derive the optimal cost-to-go functions J_k^* and optimal policy. We have

$$J_N^*(x_N) = qx_N^2,$$

and by applying Eq. (1.15), we obtain

$$\begin{aligned} J^*_{N-1}(x_{N-1}) &= \min_{u_{N-1}} E\Big\{qx^2_{N-1} + ru^2_{N-1} + J^*_N(ax_{N-1} + bu_{N-1} + w_{N-1})\Big\} \\ &= \min_{u_{N-1}} E\Big\{qx^2_{N-1} + ru^2_{N-1} + q(ax_{N-1} + bu_{N-1} + w_{N-1})^2\Big\} \\ &= \min_{u_{N-1}} \Big[qx^2_{N-1} + ru^2_{N-1} + q(ax_{N-1} + bu_{N-1})^2 \\ &\qquad + 2qE\{w_{N-1}\}(ax_{N-1} + bu_{N-1}) + qE\{w^2_{N-1}\}\Big], \end{aligned}$$

and finally, using the assumptions $E\{w_{N-1}\} = 0$, $E\{w^2_{N-1}\} = \sigma^2$, and bringing out of the minimization the terms that do not depend on u_{N-1},

$$J^*_{N-1}(x_{N-1}) = qx^2_{N-1} + q\sigma^2 + \min_{u_{N-1}} \Big[ru^2_{N-1} + q(ax_{N-1} + bu_{N-1})^2\Big]. \quad (1.17)$$

The expression minimized over u_{N-1} in the preceding equation is convex quadratic in u_{N-1}, so by setting to zero its derivative with respect to u_{N-1},

$$0 = 2ru_{N-1} + 2qb(ax_{N-1} + bu_{N-1}),$$

we obtain the optimal policy for the last stage:

$$\mu^*_{N-1}(x_{N-1}) = -\frac{abq}{r + b^2q}\, x_{N-1}.$$

Substituting this expression into Eq. (1.17), we obtain with a straightforward calculation

$$J^*_{N-1}(x_{N-1}) = K_{N-1}x^2_{N-1} + q\sigma^2,$$

where

$$K_{N-1} = \frac{a^2rq}{r + b^2q} + q.$$

We can now continue the DP algorithm to obtain J^*_{N-2} from J^*_{N-1}. An important observation is that J^*_{N-1} is quadratic (plus an inconsequential constant term), so with a similar calculation we can derive μ^*_{N-2} and J^*_{N-2} in closed form, as a linear and a quadratic (plus constant) function of x_{N-2}, respectively. This process can be continued going backwards, and it can be verified by induction that for all k, we obtain the optimal policy and optimal cost-to-go function in the form

$$\mu^*_k(x_k) = L_kx_k, \qquad k = 0, 1, \ldots, N-1,$$

$$J^*_k(x_k) = K_kx^2_k + \sigma^2 \sum_{t=k}^{N-1} K_{t+1}, \qquad k = 0, 1, \ldots, N-1,$$

where

$$L_k = -\frac{abK_{k+1}}{r + b^2K_{k+1}}, \qquad k = 0, 1, \ldots, N-1, \quad (1.18)$$

and the sequence $\{K_k\}$ is generated backwards by the equation

$$K_k = \frac{a^2 r K_{k+1}}{r + b^2 K_{k+1}} + q, \qquad k = 0, 1, \ldots, N-1, \tag{1.19}$$

starting from the terminal condition $K_N = q$.

The process by which we obtained an analytical solution in this example is noteworthy. A little thought while tracing the steps of the algorithm will convince the reader that what simplifies the solution is the quadratic nature of the cost and the linearity of the system equation. Indeed, it can be shown in generality that when the system is linear and the cost is quadratic, the optimal policy and cost-to-go function are given by closed-form expressions, even for multi-dimensional linear systems (see [Ber17a], Section 3.1). The optimal policy is a linear function of the state, and the optimal cost function is a quadratic in the state plus a constant.

Another remarkable feature of this example, which can also be extended to multi-dimensional systems, is that the optimal policy does not depend on the variance of w_k, and remains unaffected when w_k is replaced by its mean (which is zero in our example). This is known as *certainty equivalence*, and occurs in several types of problems involving a linear system and a quadratic cost; see [Ber17a], Sections 3.1 and 4.2. For example it holds even when w_k has nonzero mean. For other problems, certainty equivalence can be used as a basis for problem approximation, e.g., assume that certainty equivalence holds (i.e., replace stochastic quantities by some typical values, such as their expected values) and apply exact DP to the resulting deterministic optimal control problem. This is an important part of the RL methodology, which we will discuss later in this chapter, and in more detail in Chapter 2.

Note that the linear quadratic type of problem illustrated in the preceding example is exceptional in that it admits an elegant analytical solution. Most DP problems encountered in practice require a computational solution.

Q-Factors and Q-Learning for Stochastic Problems

Similar to the case of deterministic problems [cf. Eq. (1.9)], we can define optimal Q-factors for a stochastic problem, as the expressions that are minimized in the right-hand side of the stochastic DP equation (1.15). They are given by

$$Q_k^*(x_k, u_k) = E_{w_k}\Big\{ g_k(x_k, u_k, w_k) + J_{k+1}^*\big(f_k(x_k, u_k, w_k)\big)\Big\}. \tag{1.20}$$

The optimal cost-to-go functions J_k^* can be recovered from the optimal Q-factors Q_k^* by means of

$$J_k^*(x_k) = \min_{u_k \in U_k(x_k)} Q_k^*(x_k, u_k),$$

and the DP algorithm can be written in terms of Q-factors as

$$Q_k^*(x_k,u_k) = E_{w_k}\Big\{g_k(x_k,u_k,w_k) + \min_{u_{k+1}\in U_{k+1}(f_k(x_k,u_k,w_k))} Q_{k+1}^*\big(f_k(x_k,u_k,w_k),u_{k+1}\big)\Big\}.$$

We will later be interested in approximate Q-factors, where J_{k+1}^* in Eq. (1.20) is replaced by an approximation $\tilde{J}_{k+1}$. Again, the Q-factor corresponding to a state-control pair (x_k,u_k) is the sum of the expected first stage cost using (x_k,u_k), plus the expected cost of the remaining stages starting from the next state as estimated by the function $\tilde{J}_{k+1}$.

1.3.2 Approximation in Value Space for Stochastic DP

Generally the computation of the optimal cost-to-go functions J_k^* can be very time-consuming or impossible. One of the principal RL methods to deal with this difficulty is approximation in value space. Here approximations $\tilde{J}_k$ are used in place of J_k^*, similar to the deterministic case; cf. Eqs. (1.8) and (1.11).

Approximation in Value Space - Use of $\tilde{J}_k$ in Place of J_k^*

At any state x_k encountered at stage k, set

$$\tilde{\mu}_k(x_k) \in \arg\min_{u_k\in U_k(x_k)} E_{w_k}\Big\{g_k(x_k,u_k,w_k) + \tilde{J}_{k+1}\big(f_k(x_k,u_k,w_k)\big)\Big\}. \tag{1.21}$$

The one-step lookahead minimization (1.21) needs to be performed only for the N states $x_0,\ldots,x_{N-1}$ that are encountered during the on-line control of the system. By contrast, exact DP requires that this type of minimization be done for every state and stage.

The Three Approximations

When designing approximation in value space schemes, one may consider several interesting simplification ideas, which are aimed at alleviating the computational overhead. Aside from cost function approximation (use $\tilde{J}_{k+1}$ in place of J_{k+1}^*), there are other possibilities. One of them is to simplify the lookahead minimization over $u_k \in U_k(x_k)$ [cf. Eq. (1.15)] by replacing $U_k(x_k)$ with a suitably chosen subset of controls that are viewed as most promising based on some heuristic criterion.

In Section 1.6.7, we will discuss a related idea for control space simplification for the multiagent case where the control consists of multiple

components, $u_k = (u_k^1, \ldots, u_k^m)$. Then, a sequence of m single component minimizations can be used instead, with potentially enormous computational savings resulting.

Another type of simplification relates to approximations in the computation of the expected value in Eq. (1.21) by using limited Monte Carlo simulation. The Monte Carlo Tree Search method, which will be discussed in Chapter 2, Section 2.7.5, is one possibility of this type.

Still another type of expected value simplification is based on the *certainty equivalence approach*, to be discussed in more detail in Chapter 2, Section 2.7.2. In this approach, at stage k, we replace the future random variables $w_{k+1}, \ldots, w_{k+m}$ by some deterministic values $\overline{w}_{k+1}, \ldots, \overline{w}_{k+m}$, such as their expected values. We may also view this as a form of problem approximation, whereby for the purpose of computing $\tilde{J}_{k+1}(x_{k+1})$, we "pretend" that the problem is deterministic, with the future random quantities replaced by deterministic typical values. This is one of the most effective techniques to make approximation in value space for stochastic problems computationally tractable, particularly when it is combined with multistep lookahead minimization, as we will discuss later.

Figure 1.3.2 illustrates the three approximations involved in approximation in value space for stochastic problems: *cost-to-go approximation, simplified minimization,* and *expected value approximation*. They may be designed largely independently of each other, and may be implemented with a variety of methods. Much of the discussion in this book will revolve around different ways to organize these three approximations for both cases of one-step and multistep lookahead.

As indicated in Fig. 1.3.2, an important approach for cost-to-go approximation is *problem approximation*, whereby the functions $\tilde{J}_{k+1}$ in Eq. (1.21) are obtained as the optimal or nearly optimal cost functions of a simplified optimization problem, which is more convenient for computation. Simplifications may include exploiting decomposable structure, ignoring various types of uncertainties, and reducing the size of the state space. Several types of problem approximation approaches are discussed in the author's RL book [Ber19a]. A major approach is *aggregation*, which will be discussed in Section 3.6. In this book, problem approximation will not receive much attention, despite the fact that it can often be combined very effectively with the approximation in value space methodology that is our main focus.

Another important approach for on-line cost-to-go approximation is rollout, which we discuss next. This is similar to the rollout approach for deterministic problems, discussed in Section 1.2.

Rollout for Stochastic Problems - Truncated Rollout

In the rollout approach, we select $\tilde{J}_{k+1}$ in Eq. (1.21) to be the cost function of a suitable base policy (perhaps with some approximation). Note that

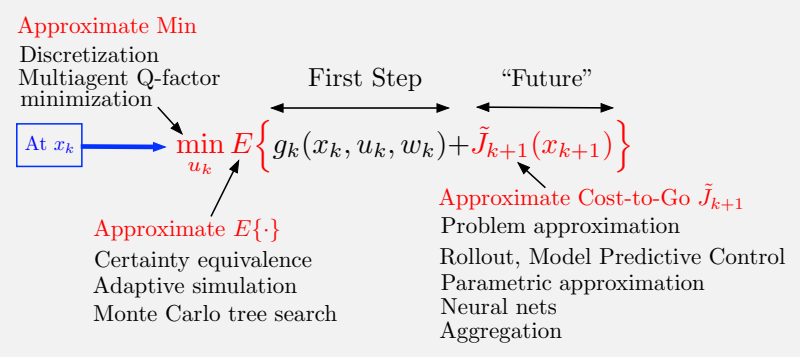


Figure 1.3.2 Schematic illustration of approximation in value space for stochastic problems, and the three approximations involved in its design. Typically the approximations can be designed independently of each other, and with a variety of approaches. There are also multistep lookahead versions of approximation in value space, which will be discussed later.

any policy can be used on-line as base policy, including policies obtained by a sophisticated off-line procedure, using for example neural networks and training data. The rollout algorithm has a cost improvement property, whereby it yields an improved cost relative to its underlying base policy. We will discuss this property and some conditions under which it is guaranteed to hold in Chapter 2.

A major variant of rollout is *truncated rollout*, which combines the use of one-step optimization, simulation of the base policy for a certain number of steps m, and then adds an approximate cost $\tilde{J}_{k+m+1}(x_{k+m+1})$ to the cost of the simulation, which depends on the state x_{k+m+1} obtained at the end of the rollout. Note that if one foregoes the use of a base policy (i.e., $m = 0$), one recovers as a special case the general approximation in value space scheme (1.21); see Fig. 1.3.3. Thus rollout provides an extra layer of lookahead to the one-step minimization, but this lookahead need not extend to the end of the horizon.

Note also that versions of truncated rollout with multistep lookahead minimization are possible. They will be discussed later. The terminal cost approximation is necessary in infinite horizon problems, since an infinite number of stages of the base policy rollout is impossible. However, even for finite horizon problems it may be necessary and/or beneficial to artificially truncate the rollout horizon. Generally, a large combined number of multistep lookahead minimization and rollout steps is likely to be beneficial.

Cost Versus Q-Factor Approximations - Robustness and On-Line Replanning

Similar to the deterministic case, Q-learning involves the calculation of either the optimal Q-factors (1.20) or approximations $\tilde{Q}_k(x_k, u_k)$. The

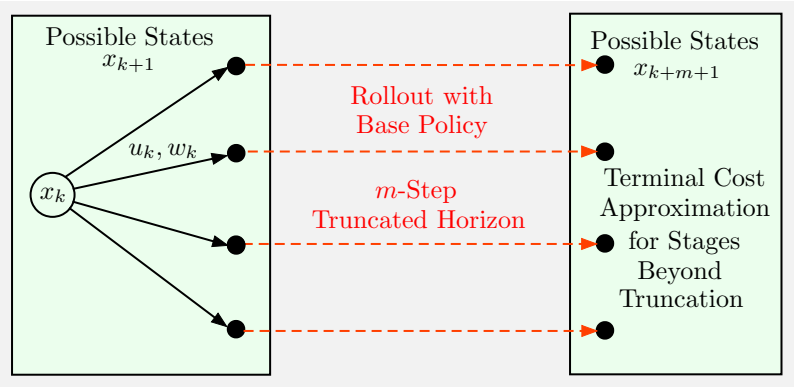


Figure 1.3.3 Schematic illustration of truncated rollout. One-step lookahead is followed by simulation of the base policy for m steps, and an approximate cost $\tilde{J}_{k+m+1}(x_{k+m+1})$ is added to the cost of the simulation, which depends on the state x_{k+m+1} obtained at the end of the rollout. If the base policy simulation is omitted (i.e., $m = 0$), one recovers the general approximation in value space scheme (1.21). Truncated rollout with multistep lookahead is also possible and is discussed in some detail in Chapter 2.

approximate Q-factors may be obtained using approximation in value space schemes, and can be used to obtain approximately optimal policies through the Q-factor minimization

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} \tilde{Q}_k(x_k, u_k). \tag{1.22}$$

Since it is possible to implement approximation in value space by using cost function approximations [cf. Eq. (1.21)] or by using Q-factor approximations [cf. Eq. (1.22)], the question arises which one to use in a given practical situation. One important consideration is the facility of obtaining suitable cost or Q-factor approximations. This depends largely on the problem and also on the availability of data on which the approximations can be based. However, there are some other major considerations.

In particular, the cost function approximation scheme

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} E_{w_k} \Big\{ g_k(x_k, u_k, w_k) + \tilde{J}_{k+1}\big(f_k(x_k, u_k, w_k)\big) \Big\}, \tag{1.23}$$

has an important disadvantage: *the expected value above needs to be computed on-line for all $u_k \in U_k(x_k)$, and this may involve substantial computation*. It also has an important advantage in situations where the system function f_k, the cost per stage g_k, or the control constraint set $U_k(x_k)$ can change as the system is operating. Assuming that the new f_k, g_k, or $U_k(x_k)$ become known to the controller at time k, *on-line replanning may be used, and this may improve substantially the robustness of the approximation in*

value space scheme. By comparison, the Q-factor function approximation scheme (1.22) does not allow for on-line replanning. On the other hand, for problems where there is no need for on-line replanning, the Q-factor approximation scheme may not require the on-line computation of expected values and may allow a much faster on-line computation of the minimizing control $\tilde{\mu}_k(x_k)$ via Eq. (1.22).

One more disadvantage of using Q-factors will emerge later, as we discuss the synergy between off-line training and on-line play based on Newton's method; see Section 1.5. In particular, we will interpret the cost function of the lookahead minimization policy $\{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$ as the result of one step of Newton's method for solving the Bellman equation that underlies the DP problem, starting from the terminal cost function approximations $\{\tilde{J}_1, \ldots, \tilde{J}_N\}$. This synergy tends to be negatively affected when Q-factor (rather than cost) approximations are used.

1.3.3 Approximation in Policy Space

The major alternative to approximation in value space is *approximation in policy space*, whereby we select the policy from a suitably restricted class of policies, usually a parametric class of some form. In particular, we can introduce a parametric family of policies (or approximation architecture, as we will call it in Chapter 3),

$$\tilde{\mu}_k(x_k, r_k), \qquad k = 0, \ldots, N-1,$$

where r_k is a parameter, and then estimate the parameters r_k using some type of training process or optimization; cf. Fig. 1.3.4.

In this section and throughout this book, *we focus on selecting a policy off-line*, possibly through training with off-line-collected data. There are algorithms that aim to improve parametric policies by using data that is collected on-line, but this subject is beyond our scope (see also the relevant discussion on policy gradient methods in Chapter 3).

Neural networks, described in Chapter 3, are often used to generate the parametric class of policies, in which case r_k is the vector of weights/parameters of the neural network. In Chapter 3, we will also discuss methods for obtaining the training data required for obtaining the parameters r_k, and we will consider several other classes of approximation architectures.

A general scheme for parametric approximation in policy space is to somehow obtain a training set, consisting of a large number of sample state-control pairs

$$(x_k^s, u_k^s), \qquad s = 1, \ldots, q,$$

such that for each s, u_k^s is a "good" control at state x_k^s. We can then choose the parameter r_k by solving the least squares/regression problem

$$\min_{r_k} \sum_{s=1}^{q} \left\| u_k^s - \tilde{\mu}_k(x_k^s, r_k) \right\|^2 \tag{1.24}$$

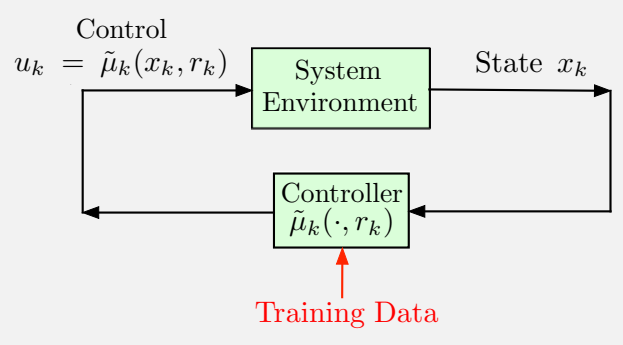


Figure 1.3.4 Schematic illustration of parametric approximation in policy space. A policy

$$\tilde{\mu}_k(x_k, r_k), \qquad k = 0, 1, \ldots, N-1,$$

from a parametric class is computed off-line based on data, and it is used to generate the control $u_k = \tilde{\mu}_k(x_k, r_k)$ on-line, when at state x_k.

(possibly modified to add regularization).† In particular, we may determine u_k^s using a human or a software "expert" that can choose "near-optimal" controls at given states, so $\tilde{\mu}_k$ is trained to match the behavior of the expert. Methods of this type are commonly referred to as *supervised learning* in AI.

An important approach for generating the training set (x_k^s, u_k^s), $s = 1, \ldots, q$, for the least squares training problem (1.24) is based on approximation in value space. In particular, we may use a one-step lookahead minimization of the form

$$u_k^s \in \arg\min_{u \in U_k(x_k^s)} E\Big\{ g_k(x_k^s, u, w_k) + \tilde{J}_{k+1}\big(f_k(x_k^s, u, w_k)\big) \Big\},$$

† Here $\|\cdot\|$ denotes the standard quadratic Euclidean norm. It is implicitly assumed here (and in similar situations later) that the controls are members of a Euclidean space (i.e., the space of finite-dimensional vectors with real-valued components) so that the distance between two controls can be measured by their normed difference (randomized controls, i.e., probabilities that a particular action will be used, fall in this category). Regression problems of this type arise in the training of *parametric classifiers* based on data, including the use of neural networks (see Section 3.4). Assuming a finite control space, the classifier is trained using the data $\big(x_k^s, u_k^s\big)$, $s = 1, \ldots, q$, which are viewed as state-category pairs, and then a state x_k is classified as being of "category" $\tilde{\mu}_k(x_k, r_k)$. Parametric approximation architectures, and their training through the use of classification and regression techniques are described in Chapter 3. An important modification is to use *regularized regression* where a quadratic regularization term is added to the least squares objective. This term is a positive multiple of the squared deviation $\|r - \hat{r}\|^2$ of r from some initial guess $\hat{r}$.

where $\tilde{J}_{k+1}$ is a suitable (separately obtained) approximation in value space. Alternatively, we may use an approximate Q-factor based minimization

$$u_k^s \in \arg \min_{u_k \in U_k(x_k^s)} \tilde{Q}_k(x_k^s, u_k),$$

where $\tilde{Q}_k$ is a (separately obtained) Q-factor approximation. We may view this as *approximation in policy space built on top of approximation in value space*.

There is a significant advantage of the least squares training procedure of Eq. (1.24), and more generally approximation in policy space: once the parametrized policy $\tilde{\mu}_k$ is obtained, the computation of controls

$$u_k = \tilde{\mu}_k(x_k, r_k), \qquad k = 0, \ldots, N-1,$$

during on-line operation of the system is often much easier compared with the lookahead minimization (1.23). For this reason, one of the major uses of approximation in policy space is to provide an *approximate implementation of a known policy* (no matter how obtained) for the purpose of convenient on-line use. On the negative side, such an implementation is less well suited for on-line replanning.

Model-Free Approximation in Policy Space

There are also alternative optimization-based approaches for policy space approximation. The main idea is that once we use a vector $(r_0, r_1, \ldots, r_{N-1})$ to parametrize the policies π, the expected cost $J_\pi(x_0)$ is parametrized as well, and can be viewed as a function of $(r_0, r_1, \ldots, r_{N-1})$. We can then optimize this cost by using a gradient-like or random search method. This is a widely used approach for optimization in policy space, which will be discussed somewhat briefly in this book (see Section 3.5, and the RL book [Ber19a], Section 5.7).

An interesting feature of this approach is that in principle it does not require a mathematical model of the system and the cost function; a computer simulator (or availability of the real system for experimentation) suffices instead. This is sometimes called a *model-free implementation*. The advisability of implementations of this type, particularly when they rely exclusively on simulation (i.e., without the use of any prior mathematical model knowledge), is a hotly debated and much contested issue; see for example the review paper by Alamir [Ala22].†

† The term "model-free" can be confusing. In reality, *there is always a model in DP/RL problem formulations*. It is just a question of whether it is a mathematical model (i.e., based on equations), or a computer model (i.e., based on computer simulation or a trained neural network), or a hybrid model (i.e., one that relies both on mathematical equations and computer software).

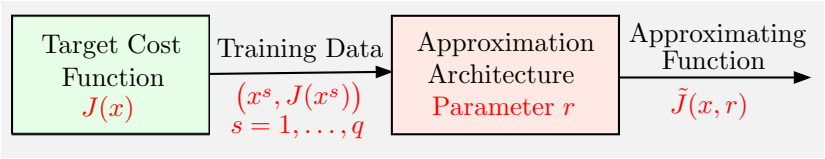


Figure 1.3.5 The general structure for parametric cost approximation. We approximate the target cost function $J(x)$ with a member from a parametric class $\tilde{J}(x, r)$ that depends on a parameter vector r. We use training data $\big(x^s, J(x^s)\big)$, $s = 1, \ldots, q$, and a form of optimization that aims to find a parameter $\hat{r}$ that "minimizes" the size of the errors $J(x^s) - \tilde{J}(x^s, \hat{r})$, $s = 1, \ldots, q$.

We finally note an important conceptual difference between approximation in value space and approximation in policy space. The former is primarily an on-line method (with off-line training used optionally to construct cost function approximations for one-step or multistep lookahead). The latter is primarily an off-line training method (which may be used without modification for on-line play or optionally to provide a policy for on-line rollout).

1.3.4 Training of Cost Function and Policy Approximations

When it comes to off-line constructed approximations, a major approach is based on the use of parametric approximation. Feature-based architectures and neural networks are very useful within our RL context, and will be discussed in Chapter 3, together with methods that can be used for training them.†

A general structure for parametric cost function approximation is illustrated in Fig. 1.3.5. We have a target function $J(x)$ that we want to

† The principal role of neural networks within the context of this book is to provide the means for approximating various target functions from input-output data. This includes cost functions and Q-factors of given policies, and optimal cost-to-go functions and Q-factors; in this case the neural network is referred to as a *value network* (sometimes the alternative term *critic network* is also used). In other cases the neural network represents a policy viewed as a function from state to control, in which case it is called a *policy network* (the alternative term *actor network* is also used). The training methods for constructing the cost function, Q-factor, and policy approximations from data are mostly based on optimization and regression, and will be reviewed in Chapter 3. Further DP-oriented discussions are found in many sources, including the RL books [Ber19a], [Ber20a], and the neuro-dynamic programming book [BeT96]. Machine learning books, including those describing at length neural network architectures and training are also recommended; see e.g., the recent book by Bishop and Bishop [BiB24], and the references quoted therein.

approximate with a member of a parametric class of functions $\tilde{J}(x,r)$ that depend on a parameter vector r (to simplify, we drop the time index, using J in place of J_k). To this end, we collect training data $\big(x^s, J(x^s)\big)$, $s = 1,\ldots,q$, which we use to determine a parameter $\hat{r}$ that leads to a good "fit" between the data $J(x^s)$ and the predictions $\tilde{J}(x^s,\hat{r})$ of the parametrized function. This is usually done through some form of optimization that aims to minimize in some sense the size of the errors $J(x^s) - \tilde{J}(x^s,\hat{r})$.

The methodological ideas for parametric cost approximation can also be used for approximation of a target policy μ with a policy from a parametric class $\tilde{\mu}(x,r)$. The training data may be obtained, for example, from rollout control calculations, thus enabling the construction of both value and policy networks that can be combined for use in a perpetual rollout scheme. However, there is an important difference: the approximate cost values $\tilde{J}(x,r)$ are real numbers, whereas the approximate policy values $\tilde{\mu}(x,r)$ are elements of a control space U. Thus if U consists of m-dimensional vectors, $\tilde{\mu}(x,r)$ consists of m numerical components. In this case the parametric approximation problems for cost functions and for policies are fairly similar, and both involve continuous space approximations.

On the other hand, the case where the control space is finite, $U = \{u^1,\ldots,u^m\}$ is markedly different. In this case, for any x, $\tilde{\mu}(x,r)$ consists of one of the m possible controls $u^1,\ldots,u^m$. This ushers a connection with traditional classification schemes, whereby objects x are classified as belonging to one of the categories $u^1,\ldots,u^m$, so that $\mu(x)$ *defines the category of x, and can be viewed as a classifier*. Some of the most prominent classification schemes actually produce randomized outcomes, i.e., x is associated with a probability distribution

$$\{\tilde{\mu}(u^1,r),\ldots,\tilde{\mu}(u^m,r)\}, \tag{1.25}$$

which is a randomized policy in our policy approximation context; see Fig. 1.3.6. This is usually done for reasons of algorithmic convenience, since many optimization methods, including least squares regression, require that the optimization variables are continuous. In this case, the randomized policy (1.25) can be converted to a nonrandomized policy using a maximization operation: associate x with the control of maximum probability (cf. Fig. 1.3.6),

$$\tilde{\mu}(x,r) \in \arg\max_{i=1,\ldots,m} \tilde{\mu}^i(x,r). \tag{1.26}$$

The use of classification methods for approximation in policy space will be discussed in Chapter 3 (Section 3.4).

1.4 INFINITE HORIZON PROBLEMS - AN OVERVIEW

We will now provide an outline of infinite horizon stochastic DP with an emphasis on its aspects that relate to our RL/approximation methods. We

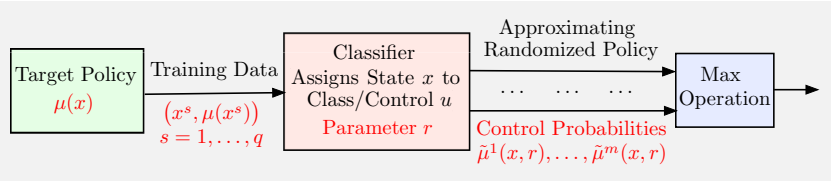


Figure 1.3.6 A general structure for parametric policy approximation for the case where the control space is finite, $U = \{u^1, \ldots, u^m\}$, and its relation to a classification scheme. It produces a randomized policy of the form (1.25), which is converted to a nonrandomized policy through the maximization operation (1.26).

will deal primarily with infinite horizon stochastic problems, where we aim to minimize the total cost over an infinite number of stages, given by

$$J_\pi(x_0) = \lim_{N\to\infty} \mathop{E}_{\substack{w_k \\ k=0,1,\ldots}} \left\{ \sum_{k=0}^{N-1} \alpha^k g\big(x_k, \mu_k(x_k), w_k\big) \right\}; \tag{1.27}$$

see Fig. 1.4.1. Here, $J_\pi(x_0)$ denotes the cost associated with an initial state x_0 and a policy $\pi = \{\mu_0, \mu_1, \ldots\}$, and α is a scalar in the interval $(0, 1]$. The functions g and f that define the cost per stage and the system equation

$$x_{k+1} = f(x_k, u_k, w_k),$$

do not change from one stage to the next. The stochastic disturbances, $w_0, w_1, \ldots$, have a common probability distribution $P(\cdot \mid x_k, u_k)$.

When α is strictly less than 1, it has the meaning of a *discount factor*, and its effect is that future costs matter to us less than the same costs incurred at the present time. Among others, a discount factor guarantees that the limit defining $J_\pi(x_0)$ exists and is finite (assuming that the range of values of the stage cost g is bounded). This is a nice mathematical property that makes discounted problems analytically and algorithmically tractable.

Thus, by definition, the infinite horizon cost of a policy is the limit of its finite horizon costs as the horizon tends to infinity. The three types of problems that we will focus on are:

(a) *Stochastic shortest path problems* (SSP for short). Here, $\alpha = 1$ but there is a special cost-free termination state; once the system reaches that state it remains there at no further cost. In some types of problems, the termination state may represent a goal state that we are trying to reach at minimum cost, while in others it may be a state that we are trying to avoid for as long as possible. We will mostly assume a problem structure such that termination is inevitable under all policies. Thus the horizon is in effect finite, but its length is random and may be affected by the policy being used. A significantly

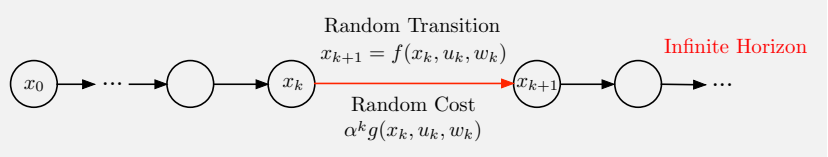


Figure 1.4.1 Illustration of an infinite horizon problem. The system and cost per stage are stationary, except for the use of a discount factor α. If $\alpha = 1$, there is typically a special cost-free termination state that we aim to reach.

more complicated type of SSP problems, which we will discuss selectively, arises when termination can be guaranteed only for a subset of policies, which includes all optimal policies. Some common types of SSP belong to this category, including deterministic shortest path problems that involve graphs with cycles.

(b) *Discounted problems*. Here, $\alpha < 1$ and there need not be a termination state. However, we will see that a discounted problem with a finite number of states can be readily converted to an SSP problem. This can be done by introducing an artificial termination state to which the system moves with probability $1 - \alpha$ at every state and stage, thus making termination inevitable. As a result, algorithms and analysis for SSP problems can be easily adapted to discounted problems; the DP textbook [Ber17a] provides a detailed account of this conversion, and an accessible introduction to discounted and SSP problems with a finite number of states.

(c) *Deterministic nonnegative cost problems*. Here, the disturbance w_k takes a single known value. Equivalently, there is no disturbance in the system equation and the cost expression, which now take the form

$$x_{k+1} = f(x_k, u_k), \qquad k = 0, 1, \ldots, \tag{1.28}$$

and

$$J_\pi(x_0) = \lim_{N\to\infty} \sum_{k=0}^{N-1} \alpha^k g\big(x_k, \mu_k(x_k)\big). \tag{1.29}$$

We assume further that there is a cost-free and absorbing termination state t, and that we have

$$g(x, u) \geq 0, \qquad \text{for all } x \neq t,\ u \in U(x), \tag{1.30}$$

and $g(t, u) = 0$ for all $u \in U(t)$. This type of structure expresses the objective to reach or approach t at minimum cost, a classical control problem. An extensive analysis of the undiscounted version of this problem was given in the author's paper [Ber17b].

Discounted stochastic problems with a finite number of states [also referred to as *discounted MDP (abbreviation for Markovian Decision Problem)*] are very common in the DP/RL literature, particularly because of

their benign analytical and computational nature. Moreover, there is a widespread belief that discounted MDP can be used as a universal model, i.e., that in practice any other kind of problem (e.g., undiscounted problems with a termination state and/or a continuous state space) can be painlessly converted to a discounted MDP with a discount factor that is close enough to 1. This is questionable, however, for a number of reasons:

(a) Deterministic models are common as well as natural in many practical contexts (including discrete optimization/integer programming problems), so to convert them to MDP does not make sense.

(b) The conversion of a continuous-state problem to a finite-state problem through some kind of discretization involves mathematical subtleties that can lead to serious practical/algorithmic complications. In particular, the character of the optimal solution may be seriously distorted by converting to a discounted MDP through some form of discretization, regardless of how fine the discretization is.

(c) For some practical shortest path contexts it is essential that the termination state is ultimately reached. However, when a discount factor α is introduced in such a problem, the character of the problem may be fundamentally altered. In particular, the threshold for an appropriate value of α may be very close to 1 and may be unknown in practice. For a simple example consider a shortest path problem with states 1 and 2 plus a termination state t. From state 1 we can go to state 2 at cost 0, from state 2 we can go to either state 1 at a small cost $\epsilon > 0$ or to the termination state at a substantial cost $C > 0$. The optimal policy over an infinite horizon is to go from 1 to 2 and from 2 to t. Suppose now that we approximate the problem by introducing a discount factor $\alpha \in (0, 1)$. Then it can be shown that if $\alpha < 1 - \epsilon/C$, it is optimal to move indefinitely around the cycle $1 \to 2 \to 1 \to 2$ and never reach t, while for $\alpha > 1 - \epsilon/C$ the shortest path $2 \to 1 \to t$ will be obtained. Thus the solution of the discounted problem varies discontinuously with α: it changes radically at some threshold, which in general may be unknown.

An important class of problems that we will consider in some detail in this book is finite-state deterministic problems with a large number of states. Finite horizon versions of these problems include challenging discrete optimization problems, whose exact solution is practically impossible. An important fact to keep in mind is that we can transform such problems to infinite horizon SSP problems with a termination state at the end of the horizon, so that the conceptual framework of the present section applies. The approximate solution of discrete optimization problems by RL methods, and particularly by rollout, will be considered in Chapter 2, and has been discussed at length in the books [Ber19a] and [Ber20a].

1.4.1 Infinite Horizon Methodology

There are several analytical and computational issues regarding our infinite horizon problems. Many of them revolve around the relation between the optimal cost function J^* of the infinite horizon problem and the optimal cost functions of the corresponding N-stage problems.

In particular, let $J_N(x)$ denote the optimal cost of the problem involving N stages, initial state x, cost per stage $g(x,u,w)$, and zero terminal cost. This cost is generated after N iterations of the algorithm

$$J_{k+1}(x) = \min_{u\in U(x)} E_w\Big\{g(x,u,w) + \alpha J_k\big(f(x,u,w)\big)\Big\}, \qquad k = 0,1,\ldots, \tag{1.31}$$

starting from $J_0(x) \equiv 0$.† The algorithm (1.31) is known as the *value iteration* algorithm (VI for short). Since the infinite horizon cost of a given policy is, by definition, the limit of the corresponding N-stage costs as $N \to \infty$, it is natural to speculate that:

(a) The optimal infinite horizon cost is the limit of the corresponding N-stage optimal costs as $N \to \infty$; i.e.,

$$J^*(x) = \lim_{N\to\infty} J_N(x) \tag{1.32}$$

for all states x.

(b) The following equation should hold for all states x,

$$J^*(x) = \min_{u\in U(x)} E_w\Big\{g(x,u,w) + \alpha J^*\big(f(x,u,w)\big)\Big\}. \tag{1.33}$$

This is obtained by taking the limit as $N \to \infty$ in the VI algorithm (1.31) using Eq. (1.32). The preceding equation, called *Bellman's equation*, is really a system of equations (one equation per state x), which has as solution the optimal costs-to-go of all the states.

(c) If $\mu(x)$ attains the minimum in the right-hand side of the Bellman equation (1.33) for each x, then the policy $\{\mu,\mu,\ldots\}$ should be optimal. This type of policy is called *stationary*, and for simplicity it is denoted by μ.

---

† This is just the finite horizon DP algorithm of Section 1.3.1, except that we have reversed the time indexing to suit our infinite horizon context. In particular, consider the N-stages problem and let $V_{N-k}(x)$ be the optimal cost-to-go starting at x with k stages to go, and with terminal cost equal to 0. Applying DP, we have for all x,

$$V_{N-k}(x) = \min_{u\in U(x)} E_w\Big\{\alpha^{N-k} g(x,u,w) + V_{N-k+1}\big(f(x,u,w)\big)\Big\}, \quad V_N(x) = 0.$$

By defining $J_k(x) = V_{N-k}(x)/\alpha^{N-k}$, we obtain the VI algorithm (1.31).

(d) The cost function J_μ of a stationary policy μ satisfies

$$J_\mu(x) = E_w\Big\{g\big(x,\mu(x),w\big) + \alpha J_\mu\big(f(x,\mu(x),w)\big)\Big\}, \qquad \text{for all } x. \tag{1.34}$$

We can view this as just the Bellman equation (1.33) for a different problem, where for each x, the control constraint set $U(x)$ consists of just one control, namely $\mu(x)$. Moreover, we expect that J_μ is obtained in the limit by the VI algorithm:

$$J_\mu(x) = \lim_{N\to\infty} J_{\mu,N}(x), \qquad \text{for all } x,$$

where $J_{\mu,N}$ is the N-stage cost function of μ generated by

$$J_{\mu,k+1}(x) = E_w\Big\{g\big(x,\mu(x),w\big) + \alpha J_{\mu,k}\big(f(x,\mu(x),w)\big)\Big\}, \tag{1.35}$$

starting from $J_{\mu,0}(x) \equiv 0$ or some other initial condition; cf. Eqs. (1.31)-(1.32).

All four of the preceding results can be shown to hold for finite-state discounted problems, and also for finite-state SSP problems under reasonable assumptions. The results also hold for infinite-state discounted problems, provided the cost per stage function g is bounded over the set of possible values of (x,u,w), in which case we additionally can show that J^* is the unique solution of Bellman's equation. The VI algorithm is also valid under these conditions, in the sense that $J_k \to J^*$, even if the initial function J_0 is nonzero. The motivation for a different choice of J_0 is faster convergence to J^*; generally the convergence is faster as J_0 is chosen closer to J^*. The associated mathematical proofs can be found in several sources, e.g., [Ber12], Chapter 1, or [Ber19a], Chapter 4.†

It is important to note that for infinite horizon problems, there are additional important algorithms that are amenable to approximation in value space. Approximate policy iteration, Q-learning, temporal difference methods, linear programming, and their variants are some of these; see the RL books [Ber19a], [Ber20a]. For this reason, in the infinite horizon case, there is a richer set of algorithmic options for approximation in value space, despite the fact that the associated mathematical theory is more complex. In this book, we will only discuss approximate forms and variations of the policy iteration algorithm, which we describe next.

† For undiscounted problems and discounted problems with unbounded cost per stage, we may still adopt the four preceding results as a working hypothesis. However, we should also be aware that exceptional behavior is possible under unfavorable circumstances, including nonuniqueness of solution of Bellman's equation, and nonconvergence of the VI algorithm to J^* from some initial conditions; see the books [Ber12], [Ber22b].

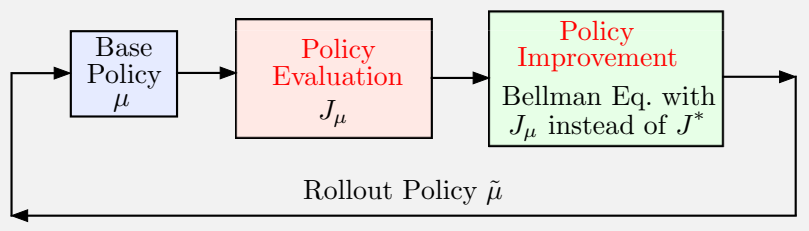


Figure 1.4.2 Schematic illustration of PI as repeated rollout. It generates a sequence of policies, with each policy μ in the sequence being the base policy that generates the next policy $\tilde{\mu}$ in the sequence as the corresponding rollout policy. This rollout policy is used as the base policy in the subsequent iteration.

Policy Iteration

A major infinite horizon algorithm is *policy iteration* (PI for short). We will argue that PI, together with its variations, forms the foundation for self-learning in RL, i.e., learning from data that is self-generated (from the system itself as it operates) rather than from data supplied from an external source. Figure 1.4.2 describes the method as repeated rollout, and indicates that each of its iterations consists of two phases:

(a) *Policy evaluation*, which computes the cost function J_μ of the current (or base) policy μ. One possibility is to solve the corresponding Bellman equation

$$J_\mu(x) = E_w\Big\{g\big(x,\mu(x),w\big) + \alpha J_\mu\big(f(x,\mu(x),w)\big)\Big\}, \qquad \text{for all } x,$$

cf. Eq. (1.34). However, the value $J_\mu(x)$ for any x can also be computed by Monte Carlo simulation, by averaging over many randomly generated trajectories the cost of the policy starting from x.

(b) *Policy improvement*, which computes the "improved" (or rollout) policy $\tilde{\mu}$ using the one-step lookahead minimization

$$\tilde{\mu}(x) \in \arg\min_{u\in U(x)} E_w\Big\{g(x,u,w) + \alpha J_\mu\big(f(x,u,w)\big)\Big\}, \qquad \text{for all } x.$$

We call $\tilde{\mu}$ "improved policy" because we can generally prove that

$$J_{\tilde{\mu}}(x) \leq J_\mu(x), \qquad \text{for all } x.$$

This cost improvement property will be shown in Chapter 2, Section 2.7, and can be used to show that PI produces an optimal policy in a finite number of iterations under favorable conditions (for example for finite-state discounted problems; see the DP books [Ber12], [Ber17a], or the RL book [Ber19a]).

The rollout algorithm in its pure form is just *a single iteration of the PI algorithm*. It starts from a given base policy μ and produces the rollout policy $\tilde{\mu}$. It may be viewed as approximation in value space with one-step lookahead that uses J_μ as terminal cost function approximation. It has the advantage that it can be applied on-line by computing the needed values of $J_\mu(x)$ by simulation. By contrast, approximate forms of PI for challenging problems, involving for example neural network training, can only be implemented off-line.

1.4.2 Approximation in Value Space - Infinite Horizon

The approximation in value space approach that we discussed in connection with finite horizon problems can be extended in a natural way to infinite horizon problems. Here in place of J^*, we use an approximation $\tilde{J}$, and generate at any state x, a control $\tilde{\mu}(x)$ by the *one-step lookahead minimization*

$$\tilde{\mu}(x) \in \arg\min_{u \in U(x)} E\Big\{ g(x,u,w) + \alpha \tilde{J}\big(f(x,u,w)\big) \Big\}. \tag{1.36}$$

This minimization yields a stationary policy $\{\tilde{\mu}, \tilde{\mu}, \ldots\}$, with cost function denoted $J_{\tilde{\mu}}$ [i.e., $J_{\tilde{\mu}}(x)$ is the total infinite horizon discounted cost obtained when using $\tilde{\mu}$ starting at state x]; see Fig. 1.4.3. Note that when $\tilde{J} = J^*$, the one-step lookahead policy attains the minimum in the Bellman equation (1.33) and is expected to be optimal. This suggests that one should try to use $\tilde{J}$ as close as possible to J^*, which is generally true as we will argue later.

Naturally an important goal to strive for is that $J_{\tilde{\mu}}$ is close to J^* in some sense. However, for classical control problems, which involve steering and maintaining the state near a desired reference state (e.g., problems with a cost-free and absorbing terminal state, and positive cost for all other states), *stability of $\tilde{\mu}$ may be a principal objective*. In this book, we will discuss stability issues primarily for this one class of problems, and *we will consider the policy $\tilde{\mu}$ to be stable if $J_{\tilde{\mu}}$ is real-valued*, i.e.,

$$J_{\tilde{\mu}}(x) < \infty, \qquad \text{for all states } x.$$

Selecting $\tilde{J}$ so that $\tilde{\mu}$ is stable is a question of major interest for some application contexts, such as model predictive and adaptive control, and will be discussed in the next section within the limited context of linear quadratic problems.

ℓ-Step Lookahead

An important extension of one-step lookahead minimization is *ℓ-step lookahead*, whereby at a state x_k we minimize the cost of the first $\ell > 1$ stages

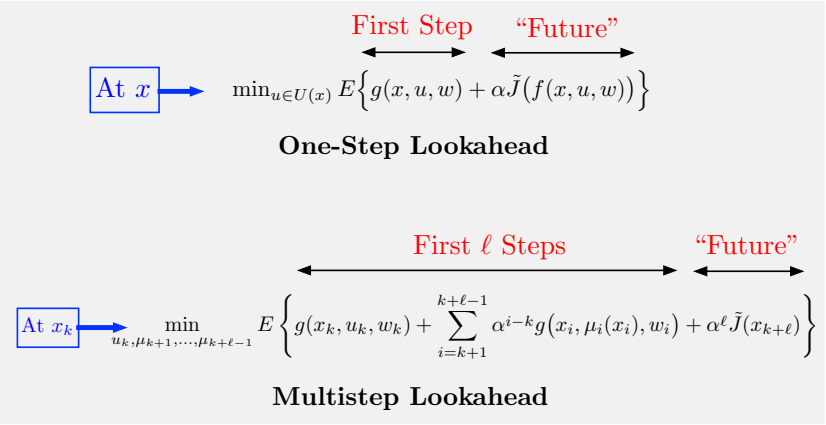


Figure 1.4.3 Schematic illustration of approximation in value space with one-step and ℓ-step lookahead minimization for infinite horizon problems. In the former case, the minimization yields at state x a control $\tilde{u}$, which defines the one-step lookahead policy $\tilde{\mu}$ via

$$\tilde{\mu}(x) = \tilde{u}.$$

In the latter case, the minimization yields a control $\tilde{u}_k$ and policies $\tilde{\mu}_{k+1}, \ldots, \tilde{\mu}_{k+\ell-1}$. The control $\tilde{u}_k$ is applied at x_k while the remaining sequence $\tilde{\mu}_{k+1}, \ldots, \tilde{\mu}_{k+\ell-1}$ is discarded. The control $\tilde{u}_k$ defines the ℓ-step lookahead policy $\tilde{\mu}$.

with the future costs approximated by a function $\tilde{J}$ (see the bottom half of Fig. 1.4.3).† This minimization yields a control $\tilde{u}_k$ and a sequence $\tilde{\mu}_{k+1}, \ldots, \tilde{\mu}_{k+\ell-1}$. The control $\tilde{u}_k$ is applied at x_k, and defines the ℓ-step lookahead policy $\tilde{\mu}$ via $\tilde{\mu}(x_k) = \tilde{u}_k$, while $\tilde{\mu}_{k+1}, \ldots, \tilde{\mu}_{k+\ell-1}$ are discarded. Actually, we may view ℓ-step lookahead minimization as the special case of its one-step counterpart where the lookahead function is the optimal cost function of an $(\ell-1)$-stage DP problem with a terminal cost $\tilde{J}(x_{k+\ell})$ on the state $x_{k+\ell}$ obtained after $\ell - 1$ stages.

The motivation for ℓ-step lookahead minimization is that *by increasing the value of ℓ, we may require a less accurate approximation $\tilde{J}$ to obtain good performance*. Otherwise expressed, for the same quality of cost function approximation, better performance may be obtained as ℓ becomes larger. This will be explained visually later, using the formalism of Newton's method in Section 1.5. In particular, for AlphaZero chess, long multistep lookahead is critical for good on-line performance. Another motivation for multistep lookahead is to *enhance the stability properties of the gener-*

† On-line play with multistep lookahead minimization (and possibly truncated rollout) is referred to by a number of different names in the RL literature, such as *on-line search*, *predictive learning*, *learning from prediction*, etc; in the model predictive control literature the combined interval of lookahead minimization and truncated rollout is referred to as the *prediction interval*.

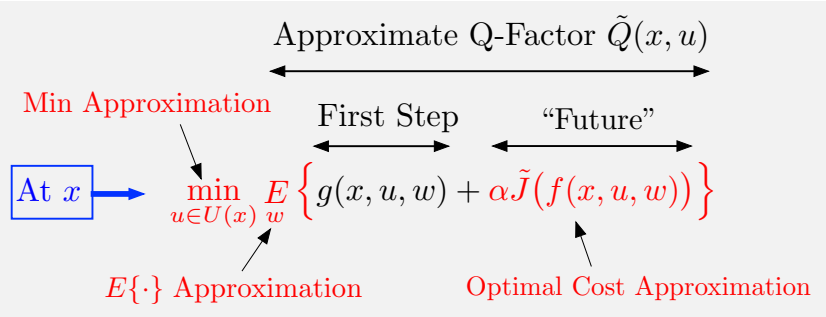


Figure 1.4.4 Approximation in value space with one-step lookahead for infinite horizon problems. There are three potential areas of approximation, which can be considered independently of each other: optimal cost approximation, expected value approximation, and minimization approximation.

ated on-line policy, as we will discuss later in Section 1.5. On the other hand, solving the multistep lookahead minimization problem, instead of the one-step lookahead counterpart of Eq. (1.36), is more time consuming.

The Three Approximations: Optimal Cost, Expected Value, and Lookahead Minimization Approximations

There are three potential areas of approximation for infinite horizon problems: optimal cost approximation, expected value approximation, and minimization approximation; cf. Fig. 1.4.4. They are similar to their finite horizon counterparts that we discussed in Section 1.3.2. In particular, we have potentially:

(a) A *terminal cost approximation* $\tilde{J}$ *of the optimal cost function* J^*: A major advantage of the infinite horizon context is that only one approximate cost function $\tilde{J}$ is needed, rather than the N functions $\tilde{J}_1,\ldots,\tilde{J}_N$ of the N-step horizon case.

(b) An *approximation of the expected value operation*: This operation can be very time consuming. It may be simplified in various ways. For example, some of the random quantities $w_k, w_{k+1},\ldots,w_{k+\ell-1}$ appearing in the ℓ-step lookahead minimization may be replaced by deterministic quantities; this is another example of the *certainty equivalence approach*, which we discussed in Section 1.3.2.

(c) A *simplification of the minimization operation*: For example in multiagent problems the control consists of multiple components,

$$u=(u^1,\ldots,u^m),$$

with each component u^i chosen by a different agent/decision maker. In this case the size of the control space can be enormous, but it

can be simplified in ways that will be discussed later (e.g., choosing components sequentially, one-agent-at-a-time). This will form the core of our approach to multiagent problems; see Section 1.6.7 and Chapter 2, Section 2.9.

We will next describe briefly various approaches for selecting the terminal cost function approximation.

Constructing Terminal Cost Approximations for On-Line Play

A major issue in value space approximation is the construction of a suitable approximate cost function $\tilde{J}$. This can be done in many different ways, giving rise to some of the principal RL methods.

For example, $\tilde{J}$ may be constructed with sophisticated off-line training methods. Alternatively, the approximate values $\tilde{J}(x)$ may be obtained on-line as needed with truncated rollout, by running an off-line obtained policy for a suitably large number of steps, starting from x, and supplementing it with a suitable, perhaps primitive, terminal cost approximation.

For orientation purposes, let us describe briefly four broad types of approximation. We will return to these approaches later, and we also refer to the RL and approximate DP literature for more detailed discussions.

(a) *Off-line problem approximation*: Here the function $\tilde{J}$ is computed off-line as the optimal or nearly optimal cost function of a simplified optimization problem, which is more convenient for computation. Simplifications may include exploiting decomposable structure, reducing the size of the state space, neglecting some of the constraints, and ignoring various types of uncertainties. For example we may consider using as $\tilde{J}$ the cost function of a related deterministic problem, obtained through some form of certainty equivalence approximation, thus allowing computation of $\tilde{J}$ by gradient-based optimal control methods or shortest path-type methods.
A major type of problem approximation method is *aggregation*, described in Section 3.6, and in the books [Ber12], [Ber19a] and papers [Ber18a], [Ber18b]. Aggregation provides a systematic procedure to simplify a given problem. A principal example is to group states together into a relatively small number of subsets, called aggregate states. The optimal cost function of the simpler aggregate problem is computed by exact DP methods, possibly involving the use of simulation. This cost function is then used to provide an approximation $\tilde{J}$ to the optimal cost function J^* of the original problem, using some form of interpolation.

(b) *On-line simulation*: This possibility arises in rollout algorithms for stochastic problems, where we use Monte-Carlo simulation and some suboptimal policy μ (the base policy) to compute (whenever needed) values $\tilde{J}(x)$ that are exactly or approximately equal to $J_\mu(x)$. The

policy μ may be obtained by any method, e.g., one based on heuristic reasoning (such as in the case of the traveling salesman Example 1.2.3), or off-line training based on a more principled approach, such as approximate policy iteration or approximation in policy space. Note that while simulation is time-consuming, it is uniquely well-suited for the use of parallel computation. Moreover, it can be simplified through the use of certainty equivalence approximations.

(c) *On-line approximate optimization*. This approach involves the solution of a suitably constructed shorter horizon version of the problem, with a simple terminal cost approximation. It can be viewed as either approximation in value space with multistep lookahead, or as a form of rollout algorithm. It is often used in model predictive control (MPC).

(d) *Parametric cost approximation*, where $\tilde{J}$ is obtained from a given parametric class of functions $J(x, r)$, where r is a parameter vector, selected by a suitable algorithm. The parametric class typically involves prominent characteristics of x called *features*, which can be obtained either through insight into the problem at hand, or by using training data and some form of neural network (see Chapter 3).

Such methods include approximate forms of PI, as discussed in Section 1.1 in connection with chess and backgammon. The policy evaluation portion of the PI algorithm can be done by approximating the cost function of the current policy using an approximation architecture such as a neural network (see Chapter 3). It can also be done with stochastic iterative algorithms such as TD(λ), LSPE(λ), and LSTD(λ), which are described in the DP book [Ber12] and the RL book [Ber19a]. These methods are somewhat peripheral to our course, and will not be discussed at any length. We note, however, that approximate PI methods do not just yield a parametric approximate cost function $J(x, r)$, but also a suboptimal policy, which can be improved on-line by using (possibly truncated) rollout.

Aside from approximate PI, parametric approximate cost functions $J(x, r)$ may be obtained off-line with methods such as Q-learning, linear programming, and aggregation methods, which are also discussed in the books [Ber12] and [Ber19a].

Let us also mention that for problems with special structure, $\tilde{J}$ may be chosen so that the one-step lookahead minimization (1.36) is facilitated. In fact, under favorable circumstances, the lookahead minimization may be carried out in closed form. An example is when the system is nonlinear, but the control enters linearly in the system equation and quadratically in the cost function, while the terminal cost approximation is quadratic. Then the one-step lookahead minimization can be carried out analytically, because it involves a function that is quadratic in u.

From Off-Line Training to On-Line Play

Generally off-line training will produce either just a cost approximation (as in the case of TD-Gammon), or just a policy (as for example, by some approximation in policy space/policy gradient approach), or both (as in the case of AlphaZero). We have already discussed in this section one-step lookahead and multistep lookahead schemes to implement on-line approximation in value space using $\tilde{J}$; cf. Fig. 1.4.3. Let us now consider some additional possibilities, which involve the use of a policy μ that has been obtained off-line (possibly in addition to a terminal cost approximation). Here are some of the main possibilities:

(a) *Given a policy* μ that has been obtained off-line, we may use as terminal cost approximation $\tilde{J}$ the cost function J_μ of the policy. For the case of one-step lookahead, this requires a policy evaluation operation, and can be done on-line, by computing (possibly by simulation) just the values of

$$E\Big\{J_\mu\big(f(x_k,u_k,w_k)\big)\Big\}$$

that are needed [cf. Eq. (1.36)]. For the case of ℓ-step lookahead, the values

$$E\big\{J_\mu(x_{k+\ell})\big\}$$

for all states $x_{k+\ell}$ that are reachable in ℓ steps starting from x_k are needed. This is the simplest form of rollout, and only requires the off-line construction of the policy μ.

(b) *Given a terminal cost approximation* $\tilde{J}$ that has been obtained off-line, we may use it on-line to compute fast the controls of a corresponding one-step or multistep lookahead policy $\tilde{\mu}$ when needed. The policy $\tilde{\mu}$ can in turn be used for rollout as in (a) above. In a truncated variation of this scheme, we may also use $\tilde{J}$ to approximate the tail end of the rollout process (an example of this is the rollout-based TD-Gammon algorithm).

(c) *Given a policy* μ *and a terminal cost approximation* $\tilde{J}$, we may use them together in a truncated rollout scheme, whereby the tail end of the rollout with μ is approximated using the cost approximation $\tilde{J}$. This is similar to the truncated rollout scheme noted in (b) above, except that the policy μ is computed off-line rather than on-line using $\tilde{J}$ and one-step or multistep lookahead.

The preceding three possibilities are the principal ones for using the results of off-line training within on-line play schemes. Naturally, there are variations where additional information is computed off-line to facilitate and/or expedite the on-line play algorithm. As an example, in MPC, in addition to a terminal cost approximation, a target tube may need to be computed off-line in order to guarantee that some state constraints can

be satisfied on-line; see the discussion of MPC in Section 1.6.9. Other examples of this type will be noted in the context of specific applications.

Finally, let us note that while we have emphasized approximation in value space with cost function approximation, our discussion applies to Q-factor approximation, involving functions

$$\tilde{Q}(x,u) \approx E\Big\{g(x,u,w) + \alpha J^*\big(f(x,u,w)\big)\Big\}.$$

The corresponding one-step lookahead scheme has the form

$$\tilde{\mu}(x) \in \arg\min_{u\in U(x)} E\Big\{g(x,u,w)+\alpha \min_{u'\in U(f(x,u,w))} \tilde{Q}\big(f(x,u,w),u'\big)\Big\}; \quad (1.37)$$

cf. Eq. (1.36). The second term on the right in the above equation represents the cost function approximation

$$\tilde{J}\big(f(x,u,w)\big) = \min_{u'\in U(f(x,u,w))} \tilde{Q}\big(f(x,u,w),u'\big).$$

The use of Q-factors is common in the "model-free" case where a computer simulator is used to generate samples of w, and corresponding values of g and f. Then, having obtained $\tilde{Q}$ through off-line training, the one-step lookahead minimization in Eq. (1.37) must be performed on-line with the use of the simulator.

1.4.3 Understanding Approximation in Value Space

We will now discuss some of our aims as we try to get insight into the process of approximation in value space. Clearly, it makes sense to approximate J^* with a function $\tilde{J}$ that is as close as possible to J^*. However, we should also try to understand quantitatively the relation between $\tilde{J}$ and $J_{\tilde{\mu}}$, the cost function of the resulting one-step lookahead (or multistep lookahead) policy $\tilde{\mu}$. Interesting questions in this regard are the following:

(a) *How is the quality of the lookahead policy $\tilde{\mu}$ affected by the quality of the off-line training?* A related question is how much should we care about improving $\tilde{J}$ through a longer and more sophisticated training process, for a given approximation architecture? A fundamental fact that provides a lot of insight in this respect is that *$J_{\tilde{\mu}}$ is the result of a step of Newton's method that starts at $\tilde{J}$ and is applied to the Bellman Eq. (1.33).* This will be the focus of our discussion in the next section, and has been a major point in the narrative of the author's books, [Ber20a] and [Ber22a].

A related fact is that in approximation in value space with multistep lookahead, *$J_{\tilde{\mu}}$ is the result of a step of Newton's method that starts at the function obtained by applying multiple value iterations to $\tilde{J}$.*

(b) *How do simplifications in the multistep lookahead implementation affect $J_{\tilde{\mu}}$?* The Newton step interpretation of approximation in value space leads to an important insight into the special character of the initial step of the multistep lookahead. In particular, *it is only the first step that acts as the Newton step, and needs to be implemented with precision*. The subsequent steps are value iterations, which only serve to enhance the quality of the starting point of the Newton step, and hence *their precise implementation is not critical*.

This idea suggests that simplifications of the lookahead steps after the first can be implemented with relatively small (if any) performance loss for the multistep lookahead policy. Important examples of such simplifications are the use of certainty equivalence (Sections 1.6.9, 2.7.2, 2.8.3), and forms of pruning of the lookahead tree (Section 2.4). In practical terms, simplifications after the first step of the multistep lookahead can save a lot of on-line computation, which can be fruitfully invested in extending the length of the lookahead.

(c) *When is $\tilde{\mu}$ stable?* The question of stability is very important in many control applications where the objective is to keep the state near some reference point or trajectory. Indeed, in such applications, stability is the dominant concern, and optimality is secondary by comparison. Among others, here we are interested to characterize the set of terminal cost approximations $\tilde{J}$ that lead to a stable $\tilde{\mu}$.

(d) *How does the length of lookahead minimization or the length of the truncated rollout affect the stability and quality of the multistep lookahead policy $\tilde{\mu}$?* While it is generally true that the length of lookahead has a beneficial effect on quality, it turns out that it also has a beneficial effect on the stability properties of the multistep lookahead policy, and we are interested in the mechanism by which this occurs.

In what follows we will be keeping in mind these questions. In particular, in the next section, we will discuss them in the context of the simple and convenient linear quadratic problem. Our conclusions, however, hold within a far more general context with the aid of the abstract DP formalism; see the author's books [Ber20a] and [Ber22a] for a broader presentation and analysis, which address these questions in greater detail and generality.

1.5 NEWTON'S METHOD - LINEAR QUADRATIC PROBLEMS

We will now aim to understand the character of the Bellman equation, approximation in value space, and the VI and PI algorithms within the context of an important deterministic problem. This is the classical continuous-spaces problem where the system is linear, with no control constraints, and the cost function is nonnegative quadratic. While this prob-

lem can be solved analytically, it provides a uniquely insightful context for understanding visually the Bellman equation and its algorithmic solution, both exactly and approximately.

In its general form, the problem deals with the system

$$x_{k+1} = Ax_k + Bu_k,$$

where x_k and u_k are elements of the Euclidean spaces $\Re^n$ and $\Re^m$, respectively, A is an $n \times n$ matrix, and B is an $n \times m$ matrix. It is assumed that there are no control constraints. The cost per stage is quadratic of the form

$$g(x, u) = x'Qx + u'Ru,$$

where Q and R are positive definite symmetric matrices of dimensions $n \times n$ and $m \times m$, respectively (all finite-dimensional vectors in this work are viewed as column vectors, and a prime denotes transposition). The analysis of this problem is well known and is given with proofs in several control theory texts, including the author's DP books [Ber17a] and [Ber12].

In what follows, we will focus for simplicity only on the one-dimensional version of the problem, where the system has the form

$$x_{k+1} = ax_k + bu_k; \tag{1.38}$$

cf. Example 1.3.1. Here the state x_k and the control u_k are scalars, and the coefficients a and b are also scalars, with $b \neq 0$. The cost function is undiscounted and has the form

$$\sum_{k=0}^{\infty}(qx_k^2 + ru_k^2), \tag{1.39}$$

where q and r are positive scalars. The one-dimensional case allows a convenient and insightful analysis of the algorithmic issues that are central for our purposes. This analysis generalizes to multidimensional linear quadratic problems and beyond, but requires a more demanding mathematical treatment.

The Riccati Equation and its Justification

The analytical results for our problem may be obtained by taking the limit in the results derived in the finite horizon Example 1.3.1, as the horizon length tends to infinity. In particular, we can show that the optimal cost function is expected to be quadratic of the form

$$J^*(x) = K^*x^2, \tag{1.40}$$

where the scalar K^* solves the equation

$$K = F(K), \tag{1.41}$$

with F defined by

$$F(K) = \frac{a^2 rK}{r + b^2 K} + q. \tag{1.42}$$

This is the limiting form of Eq. (1.19).

Moreover, the optimal policy is linear of the form

$$\mu^*(x) = L^* x, \tag{1.43}$$

where L^* is the scalar given by

$$L^* = -\frac{abK^*}{r + b^2 K^*}. \tag{1.44}$$

To justify Eqs. (1.41)-(1.44), we show that J^* as given by Eq. (1.40), satisfies the Bellman equation

$$J(x) = \min_{u \in \Re} \big\{ qx^2 + ru^2 + J(ax + bu) \big\}, \tag{1.45}$$

and that $\mu^*(x)$, as given by Eqs. (1.43)-(1.44), attains the minimum above for every x when $J = J^*$. Indeed for any quadratic cost function $J(x) = Kx^2$ with $K \geq 0$, the minimization in Bellman's equation (1.45) is written as

$$\min_{u \in \Re} \big\{ qx^2 + ru^2 + K(ax + bu)^2 \big\}. \tag{1.46}$$

Thus it involves minimization of a positive definite quadratic in u and can be done analytically. By setting to 0 the derivative with respect to u of the expression in braces in Eq. (1.46), we obtain

$$0 = 2ru + 2bK(ax + bu),$$

so the minimizing control and corresponding policy are given by

$$\mu_K(x) = L_K x, \tag{1.47}$$

where

$$L_K = -\frac{abK}{r + b^2 K}. \tag{1.48}$$

By substituting this control, the minimized expression (1.46) takes the form

$$\Big(q + rL_K^2 + K(a + bL_K)^2 \Big) x^2.$$

After straightforward algebra, using Eq. (1.48) for L_K, it can be verified that this expression is written as $F(K)x^2$, with F given by Eq. (1.42). Thus when $J(x) = Kx^2$, the Bellman equation (1.45) takes the form

$$Kx^2 = F(K)x^2$$

or equivalently $K = F(K)$ [cf. Eq. (1.41)].

In conclusion, when restricted to quadratic functions $J(x) = Kx^2$ with $K \geq 0$, the Bellman equation (1.45) is equivalent to the equation

$$K = F(K) = \frac{a^2 rK}{r + b^2 K} + q. \tag{1.49}$$

We refer to this equation as the *Riccati equation*† and to the function F as the *Riccati operator*.‡ Moreover, the policy corresponding to K^*, as per Eqs. (1.47)-(1.48), attains the minimum in Bellman's equation, and is given by Eqs. (1.43)-(1.44).

The Riccati equation can be visualized and solved graphically as illustrated in Fig. 1.5.1. As shown in the figure, the quadratic coefficient K^* that corresponds to the optimal cost function J^* [cf. Eq. (1.40)] is the unique solution of the Riccati equation $K = F(K)$ within the nonnegative real line.

The Riccati Equation for a Stable Linear Policy

We can also characterize the cost function of a policy μ that is linear of the form $\mu(x) = Lx$, and is also stable, in the sense that the scalar L satisfies $|a + bL| < 1$, so that the corresponding closed-loop system

$$x_{k+1} = (a + bL)x_k$$

is stable (its state x_k converges to 0 as $k \to \infty$). In particular, we can show that its cost function has the form

$$J_\mu(x) = K_L x^2,$$

† This is an algebraic form of the Riccati differential equation, which was invented in its one-dimensional form by count Jacopo Riccati in the 1700s, and has played an important role in control theory. It has been studied extensively in its differential and difference matrix versions; see the paper collection by Bittanti, Laub, and Willems [BLW91], which also includes a historical account by Bittanti [Bit91] of Riccati's remarkable life and accomplishments.

‡ The Riccati operator is a special case of the *Bellman operator*, denoted by T, which transforms a function J into the right side of Bellman's equation:

$$(TJ)(x) = \min_{u \in U(x)} E_w \Big\{ g(x,u,w) + \alpha J\big(f(x,u,w)\big) \Big\}, \qquad \text{for all } x.$$

Thus the Bellman operator T transforms a function J of x into another function TJ also of x. Bellman operators allow a succinct abstract description of the problem's data, and are fundamental in the theory of abstract DP (see the author's monographs [Ber22a] and [Ber22b]). We may view the Riccati operator as the restriction of the Bellman operator to the subspace of quadratic functions of x.

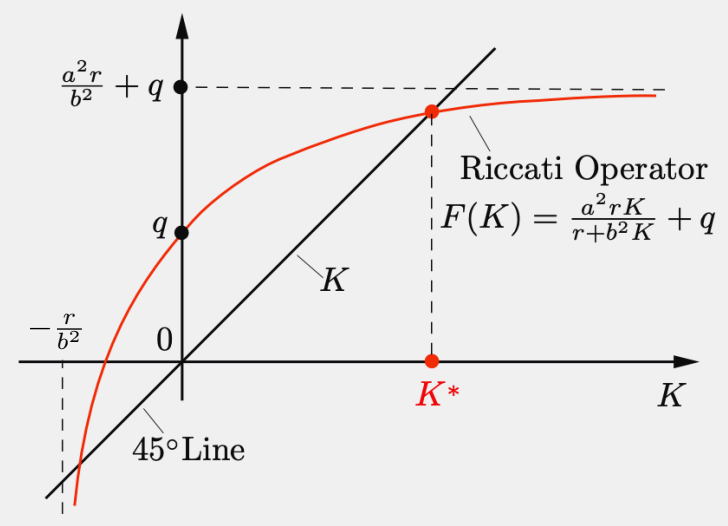


Figure 1.5.1 Graphical construction of the solutions of the Riccati equation (1.41)-(1.42) for the linear quadratic problem. The optimal cost function is $J^*(x) = K^*x^2$, where the scalar K^* solves the fixed point equation $K = F(K)$, with F being the Riccati operator given by

$$F(K) = \frac{a^2rK}{r + b^2K} + q.$$

Note that F is concave and monotonically increasing in the interval $(-r/b^2, \infty)$ and "flattens out" as $K \to \infty$, as shown in the figure. The quadratic Riccati equation $K = F(K)$ also has another solution, denoted by $\bar{K}$, which is negative and therefore of no interest.

where K_L solves the equation

$$K = F_L(K), \tag{1.50}$$

with F_L defined by

$$F_L(K) = (a + bL)^2K + q + rL^2. \tag{1.51}$$

This equation is called the *Riccati equation for the stable policy* $\mu(x) = Lx$. It is illustrated in Fig. 1.5.2, and it is linear, with linear coefficient $(a+bL)^2$ that is strictly less than 1. Hence the line that represents the graph of F_L intersects the 45-degree line at a unique point, which defines the quadratic cost coefficient K_L.

The Riccati equation (1.50)-(1.51) for $\mu(x) = Lx$ may be justified by verifying that it is in fact the Bellman equation for μ,

$$J(x) = (q + rL^2)x^2 + J\big((a + bL)x\big),$$

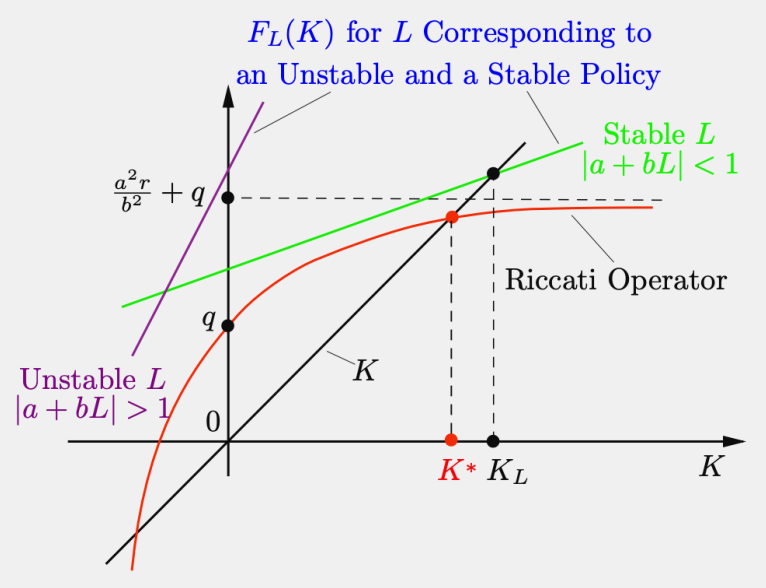


Figure 1.5.2 Illustration of the construction of the cost function of a linear policy $\mu(x) = Lx$, which is stable, i.e., $|a + bL| < 1$. The cost function $J_\mu(x)$ has the form

$$J_\mu(x) = K_L x^2,$$

with K_L obtained as the unique solution of the linear equation $K = F_L(K)$, where

$$F_L(K) = (a + bL)^2 K + q + rL^2,$$

is the Riccati equation operator corresponding to $\mu(x) = Lx$. If μ is not stable, i.e., $|a + bL| \geq 1$, we have $J_\mu(x) = \infty$ for all $x \neq 0$, but the equation has $K = F_L(K)$ still has a solution that is of no interest within our context.

[cf. Eq. (1.34)], restricted to quadratic functions of the form $J(x) = Kx^2$.

We note, however, that $J_\mu(x) = K_L x^2$ is the solution of the Riccati equation (1.50)-(1.51) only when $\mu(x) = Lx$ is stable. If μ is not stable, i.e., $|a + bL| \geq 1$, then (since $q > 0$ and $r > 0$) we have $J_\mu(x) = \infty$ for all $x \neq 0$. Then, the Riccati equation (1.50)-(1.51) is still defined, but its solution is negative and is of no interest within our context.

Value Iteration

The VI algorithm for our linear quadratic problem is given by

$$J_{k+1}(x) = \min_{u \in \Re} \big\{ qx^2 + ru^2 + J_k(ax + bu) \big\}.$$

When J_k is quadratic of the form $J_k(x) = K_k x^2$ with $K_k \geq 0$, it can be seen that the VI iterate J_{k+1} is also quadratic of the form $J_{k+1}(x) = K_{k+1} x^2$,

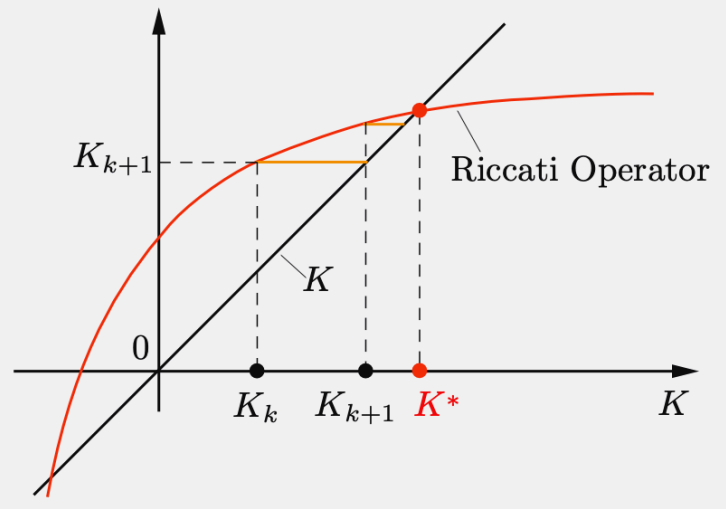


Figure 1.5.3 Graphical illustration of value iteration for the linear quadratic problem. It has the form $K_{k+1} = F(K_k)$, where F is the Riccati operator,

$$F(K) = \frac{a^2 rK}{r + b^2 K} + q.$$

The algorithm converges to K^* starting from any $K_0 \geq 0$.

where

$$K_{k+1} = F(K_k),$$

with F being the Riccati operator of Eq. (1.49). The algorithm is illustrated in Fig. 1.5.3. As can be seen from the figure, when starting from any $K_0 \geq 0$, the algorithm generates a sequence $\{K_k\}$ of nonnegative scalars that converges to K^*.

1.5.1 Visualizing Approximation in Value Space - Region of Stability

The use of Riccati equations allows insightful visualization of approximation in value space. This visualization, although specialized to linear quadratic problems, is consistent with related visualizations for more general infinite horizon problems; this is a recurring theme in what follows. In particular, in the books [Ber20a] and [Ber22a], Bellman operators, which define the Bellman equations, are used in place of Riccati operators, which define the Riccati equations.

In summary, we will aim to show that:

(a) Approximation in value space with one-step lookahead can be viewed as a Newton step for solving the Bellman equation, and maps the

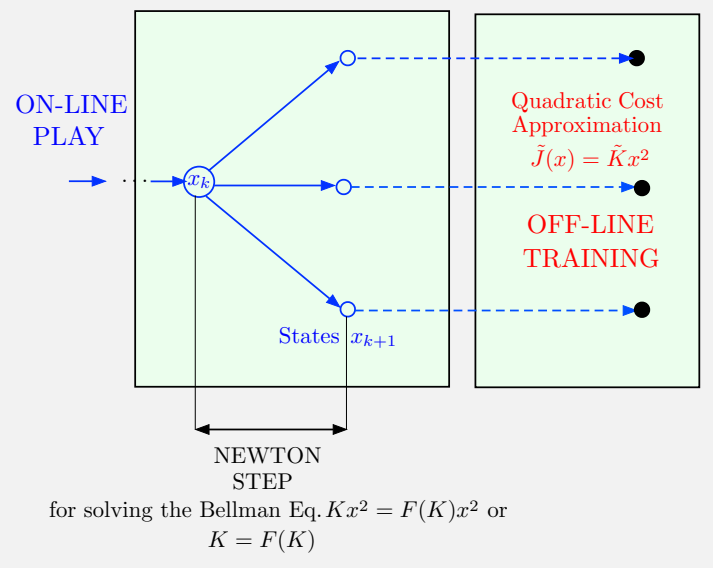


Figure 1.5.4 Illustration of the interpretation of approximation in value space with one-step lookahead as a Newton step that maps $\tilde{J}$ to the cost function $J_{\tilde{\mu}}$ of the one-step lookahead policy.

terminal cost function approximation $\tilde{J}$ to the cost function $J_{\tilde{\mu}}$ of the one-step lookahead policy; see Fig. 1.5.4.

(b) Approximation in value space with multistep lookahead and truncated rollout can be viewed as a Newton step for solving the Bellman equation, and maps the result of multiple VI iterations starting with the terminal cost function approximation $\tilde{J}$ to the cost function $J_{\tilde{\mu}}$ of the multistep lookahead policy; see Fig. 1.5.5.

Our derivation will be given for the one-dimensional linear quadratic problem, but *applies far more generally*. The reason is that the Bellman equation is valid universally in DP, and the corresponding Bellman operator has a concavity property that is well-suited for the application of Newton's method; see the books [Ber20a] and [Ber22a], where the connection of approximation in value space with Newton's method was first developed in detail.

Let us consider one-step lookahead minimization with any terminal cost function approximation of the form $\tilde{J}(x) = Kx^2$, where $K \geq 0$. We have derived the one-step lookahead policy $\mu_K(x)$ in Eqs. (1.47)-(1.48), by minimizing the right side of Bellman's equation when $J(x) = Kx^2$:

$$\min_{u \in \Re} \{qx^2 + ru^2 + K(ax + bu)^2\}.$$

We can break this minimization into a sequence of two minimizations as

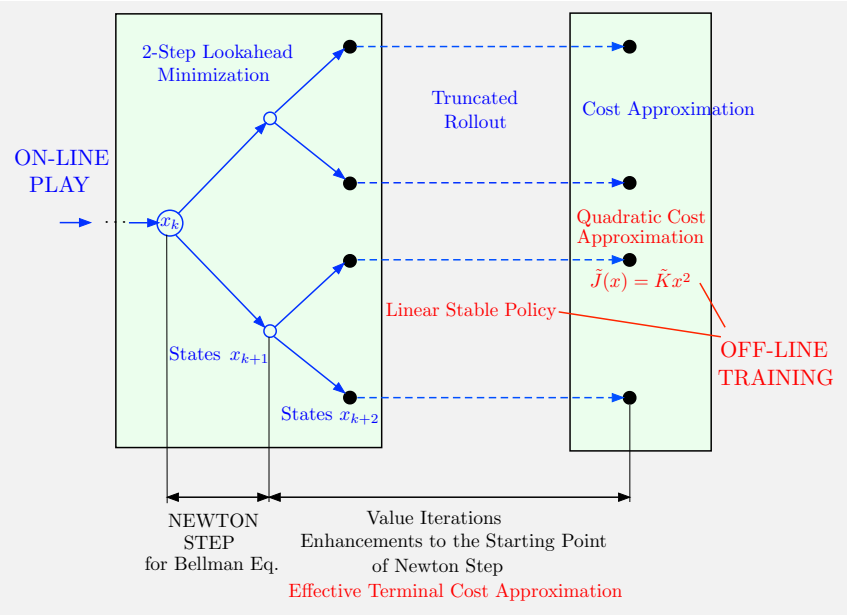


Figure 1.5.5 Illustration of the interpretation of approximation in value space with multistep lookahead and truncated rollout as a Newton step, which maps the result of multiple VI iterations starting with the terminal cost function approximation $\tilde{J}$ to the cost function $J_{\tilde{\mu}}$ of the multistep lookahead policy.

follows:

$$F(K)x^2 = \min_{L\in\Re} \min_{u=Lx} \left\{qx^2+ru^2+K(ax+bu)^2\right\} = \min_{L\in\Re} \left\{q+rL^2+K(a+bL)^2\right\}x^2.$$

From this equation, it follows that

$$F(K) = \min_{L\in\Re} F_L(K), \tag{1.52}$$

where the function $F_L(K)$ is defined by

$$F_L(K) = (a+bL)^2K + q + rL^2. \tag{1.53}$$

Figure 1.5.6 illustrates the relation (1.52)-(1.53), and shows how the graph of *the Riccati operator F can be obtained as the lower envelope of the linear operators F_L*, as L ranges over the real numbers.

One-Step Lookahead Minimization and Newton's Method

Let us now fix the terminal cost function approximation to some $\tilde{K}x^2$, where $\tilde{K} \geq 0$, and consider the corresponding one-step lookahead policy,

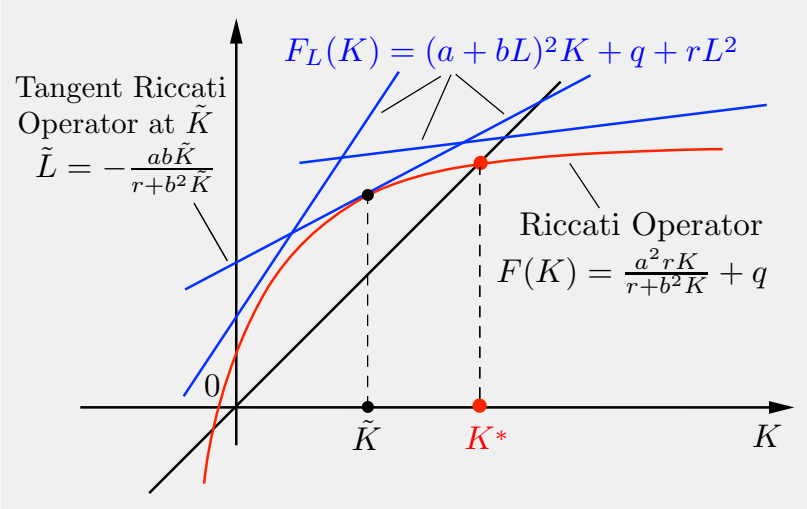


Figure 1.5.6 Illustration of how the graph of the Riccati operator F can be obtained as the lower envelope of the linear operators

$$F_L(K) = (a+bL)^2K + q + rL^2,$$

as L ranges over the real numbers. We have

$$F(K) = \min_{L\in\Re} F_L(K);$$

cf. Eq. (1.52). Moreover, for any fixed $\tilde{K}$, the scalar $\tilde{L}$ that attains the minimum is given by

$$\tilde{L} = -\frac{ab\tilde{K}}{r+b^2\tilde{K}}$$

[cf. Eq. (1.48)], and is such that the line corresponding to the graph of $F_{\tilde{L}}$ is tangent to the graph of F at $\tilde{K}$, as shown in the figure.

which we will denote by $\tilde{\mu}$. Figure 1.5.7 illustrates the corresponding linear function $F_{\tilde{L}}$, and shows that its graph is a tangent line to the graph of F at the point $\tilde{K}$ [cf. Fig. 1.5.6 and Eq. (1.53)].

Thus the function $F_{\tilde{L}}$ can be viewed as a linearization of F at the point $\tilde{K}$, and defines a linearized problem: to find a solution of the equation

$$K = F_{\tilde{L}}(K) = q + r\tilde{L}^2 + K(a+b\tilde{L})^2.$$

The important point now is that *the solution of this equation, denoted $K_{\tilde{L}}$, is the same as the one obtained from a single iteration of Newton's method for solving the Riccati equation, starting from the point $\tilde{K}$*. This is illustrated in Fig. 1.5.7, and is also justified analytically in Exercise 1.7.

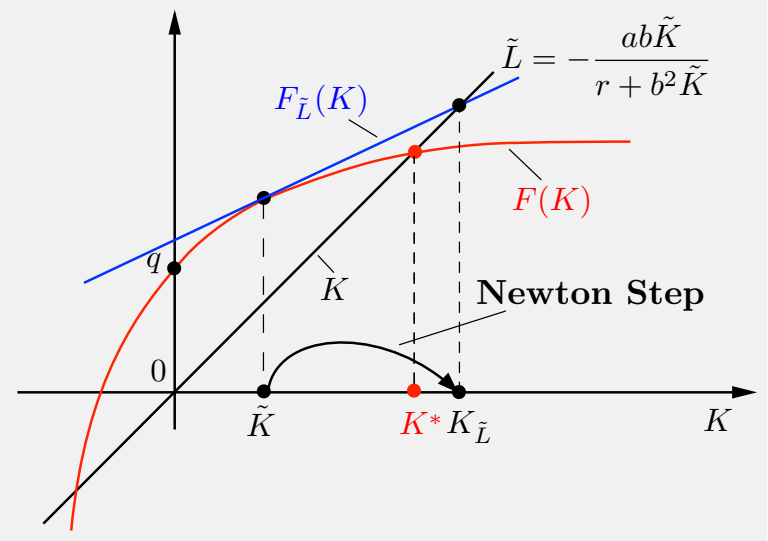


Figure 1.5.7 Illustration of approximation in value space with one-step lookahead for the linear quadratic problem. Given a terminal cost approximation $\tilde{J}(x) = \tilde{K}x^2$, we compute the corresponding linear policy $\tilde{\mu}(x) = \tilde{L}x$, where

$$\tilde{L} = -\frac{ab\tilde{K}}{r + b^2\tilde{K}},$$

and the corresponding cost function $K_{\tilde{L}}x^2$, using the Newton step shown.

To explain this connection, we note that the classical form of Newton's method for solving a fixed point problem of the form $y = T(y)$, where y is an n-dimensional vector, operates as follows: At the current iterate y_k, we linearize T and find the solution y_{k+1} of the corresponding linear fixed point problem. Assuming T is differentiable, the linearization is obtained by using a first order Taylor expansion:

$$y_{k+1} = T(y_k) + \frac{\partial T(y_k)}{\partial y}(y_{k+1} - y_k),$$

where $\partial T(y_k)/\partial y$ is the $n \times n$ Jacobian matrix of T evaluated at the vector y_k, as indicated in Fig. 1.5.7.

The most commonly given convergence rate property of Newton's method is *quadratic convergence*. It states that near the solution y^*, we have

$$\|y_{k+1} - y^*\| = O\big(\|y_k - y^*\|^2\big),$$

where $\|\cdot\|$ is the Euclidean norm, and holds assuming the Jacobian matrix exists and is Lipschitz continuous (see [Ber16], Section 1.4). There are extensions of Newton's method that are based on solving a linearized

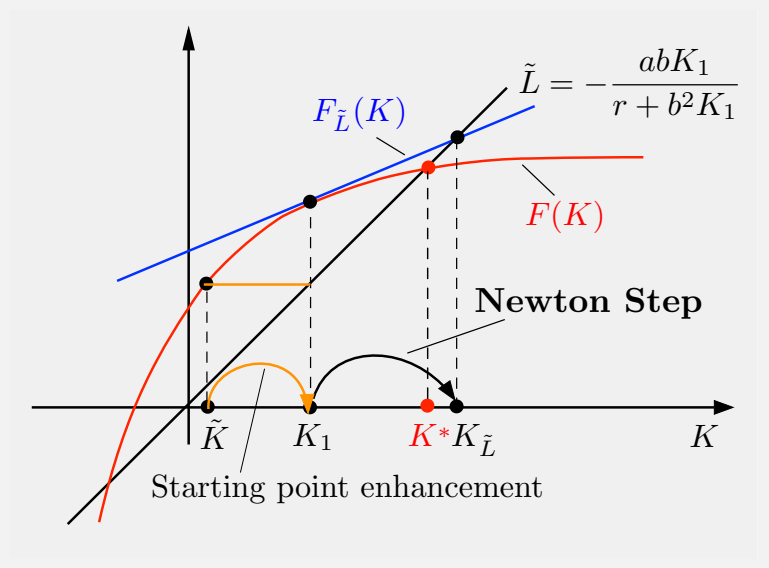


Figure 1.5.8 Illustration of approximation in value space with two-step lookahead for the linear quadratic problem. Starting with a terminal cost approximation $\tilde{J}(x) = \tilde{K}x^2$, we obtain K_1 using a single value iteration, thereby enhancing the starting point of the Newton step. We then compute the corresponding linear policy $\tilde{\mu}(x) = \tilde{L}x$, where

$$\tilde{L} = -\frac{abK_1}{r + b^2K_1}$$

and the corresponding cost function $K_{\tilde{L}}x^2$, using the Newton step shown. The figure shows that for any $K \geq 0$, the corresponding ℓ-step lookahead policy will be stable for all ℓ larger than some threshold.

system at the current iterate, but relax the differentiability requirement to piecewise differentiability, and/or component concavity, while maintaining the either a quadratic or a similarly fast superlinear convergence property of the method; see the monograph [Ber22a] (Appendix A) and the paper [Ber22c], which also provide a convergence analysis.

Note also that if the one-step lookahead policy is stable, i.e., $|a+b\tilde{L}| < 1$, then $K_{\tilde{L}}$ is the quadratic cost coefficient of its cost function, i.e.,

$$J_{\tilde{\mu}}(x) = K_{\tilde{L}}x^2.$$

The reason is that $J_{\tilde{\mu}}$ solves the Bellman equation for policy $\tilde{\mu}$. On the other hand, if $\tilde{\mu}$ is not stable, then in view of the positive definite quadratic cost per stage, we have $J_{\tilde{\mu}}(x) = \infty$ for all $x \neq 0$.

Multistep Lookahead

In the case of ℓ-step lookahead minimization, a similar Newton step inter-

Riccati Equation Formulas for One-Dimensional Problems

Riccati equation for minimization [cf. Eqs. (1.41) and (1.42)]

$$K = F(K), \qquad F(K) = \frac{a^2 rK}{r + b^2 K} + q.$$

Riccati equation for a linear policy $\mu(x) = Lx$

$$K = F_L(K), \qquad F_L(K) = (a + bL)^2 K + q + rL^2.$$

Cost coefficient K_L of a stable linear policy $\mu(x) = Lx$

$$K_L = \frac{q + rL^2}{1 - (a + bL)^2}.$$

Linear coefficient L_K of the one-step lookahead linear policy μ_K for K in the region of stability [cf. Eq. (1.48)]

$$L_K = \arg\min_L F_L(K) = -\frac{abK}{r + b^2 K}.$$

Quadratic cost coefficient $\tilde{K}$ of a one-step lookahead linear policy μ_K for K in the region of stability

Obtained as the solution of the linearized Riccati equation

$$\tilde{K} = F_{L_K}(\tilde{K}),$$

or equivalently by a Newton iteration starting from K.

pretation is possible. Instead of linearizing F at $\tilde{K}$, we linearize at

$$K_{\ell-1} = F^{\ell-1}(\tilde{K}),$$

i.e., the result of $\ell - 1$ successive applications of F starting with $\tilde{K}$. Each application of F corresponds to a value iteration. Thus *the effective starting point for the Newton step is* $F^{\ell-1}(\tilde{K})$. Figure 1.5.8 depicts the case $\ell = 2$.

Region of Stability

It is also useful to define the *region of stability* as the set of $K \geq 0$ such

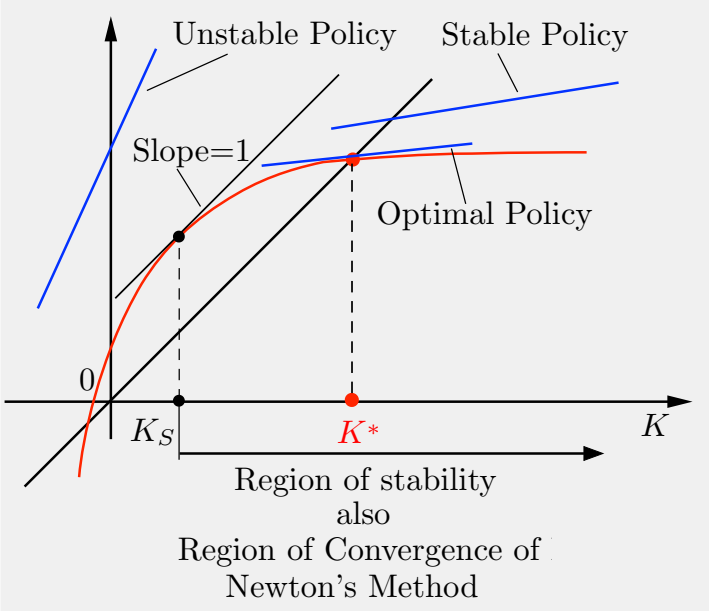


Figure 1.5.9 Illustration of the region of stability, i.e., the set of $K \geq 0$ such that the one-step lookahead policy μ_K is stable. This is also the set of initial conditions for which Newton's method converges to K^* asymptotically.

that

$$|a + bL_K| < 1,$$

where L_K is the linear coefficient of the one-step lookahead policy corresponding to K; cf. Eq. (1.48). The region of stability may also be viewed as *the region of convergence of Newton's method*. It is the set of starting points K for which Newton's method, applied to the Riccati equation $F = F(K)$, converges to K^* asymptotically, and with a quadratic convergence rate (asymptotically as $K \to K^*$). Note that for our one-dimensional problem, the region of stability is the interval (K_S, ∞) that is characterized by the single point K_S where F has derivative equal to 1; see Fig. 1.5.9.

For multidimensional problems, the region of stability may not be characterized as easily. Still, however, it is generally true that *the region of stability is enlarged as the length of the lookahead increases*.

Indeed, with increased lookahead, the effective starting point

$$F^{\ell-1}(\tilde{K})$$

is pushed more and more within the region of stability. In particular, *for any given $K \geq 0$, the corresponding ℓ-step lookahead policy will be stable for all ℓ larger than some threshold*; see Fig. 1.5.8. The book [Ber22a], Section 3.3, contains a broader discussion of the region of stability and the role of multistep lookahead in enhancing it; see also Exercises 1.8 and 1.9.

Newton Step Interpretation of Approximation in Value Space in General Infinite Horizon Problems

The interpretation of approximation in value space as a Newton step, and related notions of stability that we have discussed in this section admit a broad generalization to the infinite horizon problems that we consider in this book and beyond. The key fact in this respect is that our DP problem formulation allows arbitrary state and control spaces, both discrete and continuous, and can be extended even further to general abstract models with a DP structure; see the abstract DP book [Ber22b].

Within this context, the Riccati operator is replaced by an abstract Bellman operator, and valuable insight can be obtained from graphical interpretations of the Bellman equation, the VI and PI algorithms, one-step and multistep approximation in value space, the region of stability, and exceptional behavior; see the book [Ber22a] for an extensive discussion. Naturally, the graphical interpretations and visualizations are limited to one dimension. However, they provide insight, and motivate conjectures and mathematical analysis, much of which is given in the book [Ber20a].

The Importance of the First Step in Multistep Lookahead

The Newton step interpretation of approximation in value space leads to an important insight into the special character of the initial step in ℓ-step lookahead implementations. In particular, *it is only the first step that acts as the Newton step*, and needs to be implemented with precision; cf. Fig. 1.5.5. The subsequent $\ell-1$ steps are a sequence of value iterations starting with $\tilde{J}$, and only serve to enhance the quality of the starting point of the Newton step. As a result, *their precise implementation is not critical*, a major point in the narrative of the author's book [Ber22a].

This idea suggests that we can simplify (within reason) the lookahead steps after the first with small (if any) performance loss for the multistep lookahead policy. An important example of such a simplification is the use of certainty equivalence, which will be discussed later in various contexts (Sections 1.6.9, 2.7.2, 2.8.3). Other possibilities include the "pruning" of the lookahead tree after the first step; see Section 2.4. In practical terms, simplifications after the first step of the multistep lookahead can save a lot of on-line computation, which can be fruitfully invested in extending the length of the lookahead. This insight is supported by substantial computational experimentation, starting with the paper by Bertsekas and Castañon [BeC99], which verified the beneficial effect of using certainty equivalence after the first step.

1.5.2 Rollout and Policy Iteration

We will now consider the rollout algorithm for the linear quadratic problem, starting from a linear stable base policy μ. It generates the rollout policy

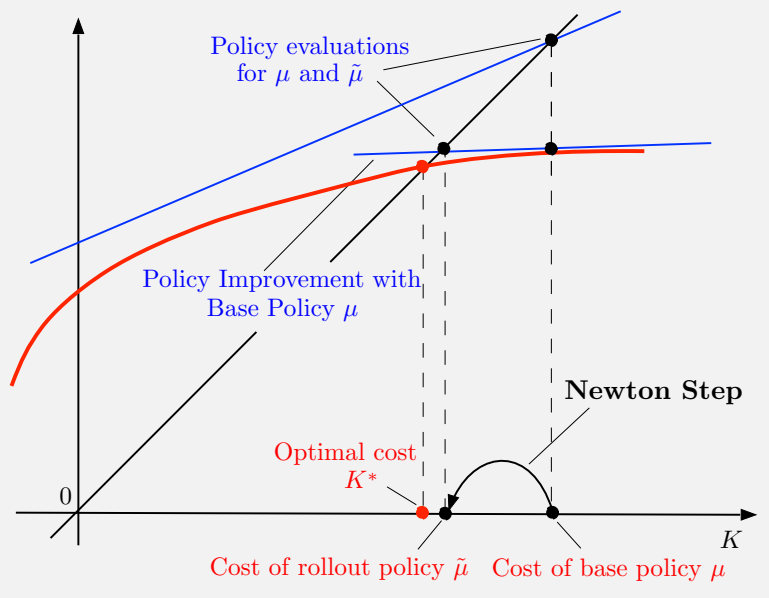


Figure 1.5.10 Illustration of the rollout algorithm for the linear quadratic problem. Starting from a linear stable base policy μ, it generates a stable rollout policy $\tilde{\mu}$. The quadratic cost coefficient of $\tilde{\mu}$ is obtained from the quadratic cost coefficient of μ with a Newton step for solving the Riccati equation.

$\tilde{\mu}$ by using a policy improvement operation, which by definition, yields the one-step lookahead policy that corresponds to terminal cost approximation $\tilde{J} = J_\mu$. Figure 1.5.10 illustrates the rollout algorithm. It can be seen from the figure that the rollout policy is in fact an improved policy, in the sense that $J_{\tilde{\mu}}(x) \leq J_\mu(x)$ for all x. Among others, this implies that the rollout policy is stable, since μ is assumed stable so that $J_\mu(x) < \infty$ for all x.

Since the rollout policy is a one-step lookahead policy, it can also be described using the formulas that we developed earlier in this section. In particular, let the base policy have the form

$$\mu^0(x) = L_0 x,$$

where L_0 is a scalar. We require that the base policy must be stable, i.e., $|a + bL_0| < 1$. From our earlier calculations, we have that the cost function of μ^0 is

$$J_{\mu^0}(x) = K_0 x^2, \tag{1.54}$$

where

$$K_0 = \frac{q + rL_0^2}{1 - (a + bL_0)^2}. \tag{1.55}$$

Moreover, the rollout policy μ^1 has the form $\mu^1(x) = L_1 x$, where

$$L_1 = -\frac{abK_0}{r + b^2 K_0}; \tag{1.56}$$

cf. Eqs. (1.47)-(1.48).

The PI algorithm is simply the repeated application of nontruncated rollout, and generates a sequence of stable linear policies $\{\mu^k\}$. By replicating our earlier calculations, we see that the policies have the form

$$\mu^k(x) = L_k x, \qquad k = 0, 1, \ldots,$$

where L_k is generated by the iteration

$$L_{k+1} = -\frac{abK_k}{r + b^2 K_k},$$

with K_k given by

$$K_k = \frac{q + rL_k^2}{1 - (a + bL_k)^2},$$

[cf. Eqs. (1.55)-(1.56)].

The corresponding cost function sequence is

$$J_{\mu^k}(x) = K_k x^2;$$

cf. Eq. (1.54). Part of the classical linear quadratic theory is that J_{μ^k} converges to the optimal cost function J^*, while the generated sequence of linear policies $\{\mu^k\}$, where $\mu^k(x) = L_k x$, converges to the optimal policy, assuming that the initial policy is linear and stable. The convergence rate of the sequence $\{K_k\}$ is quadratic, as indicated earlier. This result was proved by Kleinman [Kle68] for the continuous-time multidimensional version of the linear quadratic problem, and it was extended later to more general problems; see the references given in the books [Ber20a] and [Ber22a] (Kleinman gives credit to Bellman and Kalaba [BeK65] for the one-dimensional version of his results).

Truncated Rollout

An m-step truncated rollout scheme with a stable linear base policy $\mu(x) = Lx$, one-step lookahead minimization, and terminal cost approximation $\tilde{J}(x) = \tilde{K}x^2$ is geometrically interpreted as in Fig. 1.5.11. The truncated rollout policy $\tilde{\mu}$ is obtained by starting at $\tilde{K}$, executing m VI steps using μ, followed by a Newton step for solving the Riccati equation.

We mentioned some interesting performance issues in our discussion of truncated rollout in Section 1.1. In particular we noted that:

(a) Lookahead by rollout may be an economic substitute for lookahead by minimization: it may achieve a similar performance for the truncated rollout policy at a much reduced computational cost.

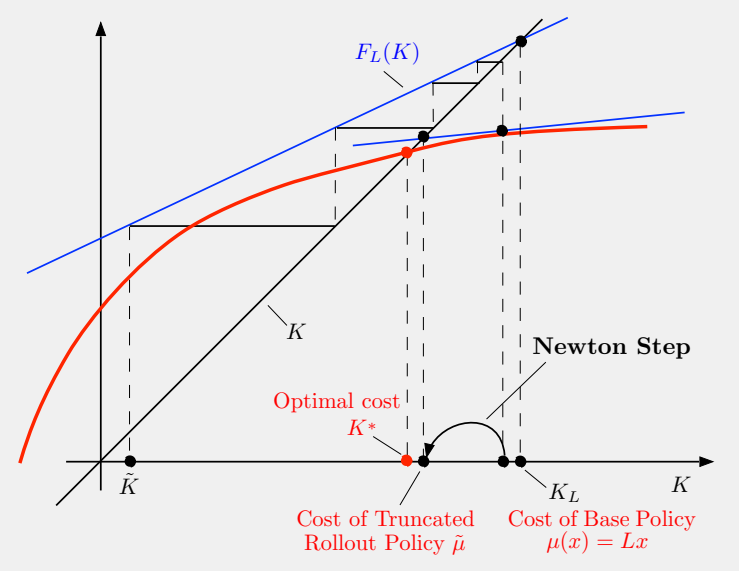


Figure 1.5.11 Illustration of truncated rollout with one-step lookahead minimization, a stable base policy $\mu(x) = Lx$, and terminal cost approximation $\tilde{K}$ for the linear quadratic problem. In this figure the number of rollout steps is $m = 4$.

(b) Lookahead by rollout with a stable policy has a beneficial effect on the stability properties of the lookahead policy.

These statements are difficult to establish analytically in some generality. However, they can be intuitively understood in the context with our one-dimensional linear quadratic problem, using geometrical constructions like the one in Fig. 1.5.11. They are also consistent with the results of computational experimentation. We refer to the monograph [Ber22a] for further discussion.

Double Newton Step - Rollout on Top of Approximation in Value Space

Given a cost function approximation $\tilde{K}$ that defines the policy $\tilde{\mu}(x) = L_{\tilde{K}}x$, it is possible to consider rollout that uses $\tilde{\mu}$ as a base policy. This can be viewed as rollout built on top of approximation in value space, and is referred to as a *double Newton step*; it is a Newton step that starts from the result of the Newton step that starts from $\tilde{K}$ (see Fig. 1.5.12). The double Newton step is much more powerful than approximation in value space with two-step lookahead starting from $\tilde{K}$, which amounts to a value iteration followed by a Newton step. Moreover, the idea of a double Newton step extends to general infinite horizon problems.

Note that it is also possible to consider variants of rollout on top of approximation in value space, such as truncated, simplified, and multi-

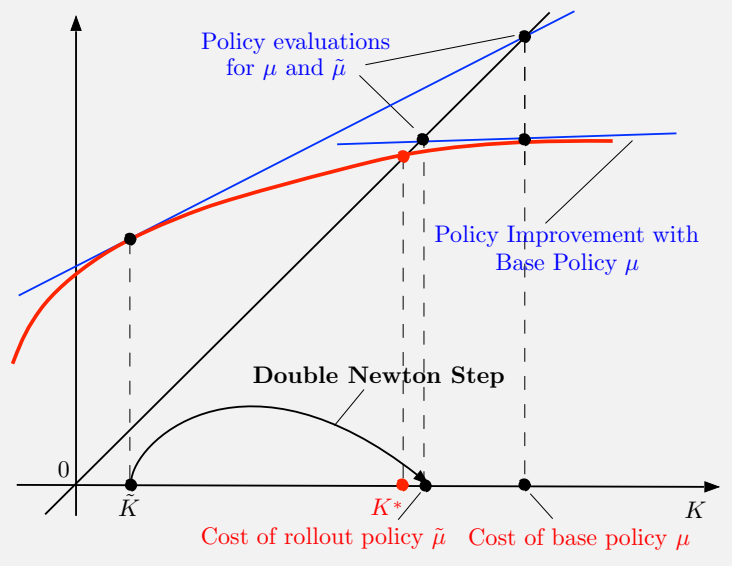


Figure 1.5.12 Illustration of a double Newton step. Here, a base policy μ is obtained by one-step lookahead using cost function approximation $\tilde{J}(x) = \tilde{K}x^2$. The cost function of the corresponding rollout policy $\tilde{\mu}$ is obtained with two successive Newton steps starting from $\tilde{K}$.

step lookahead versions. An important example of the truncated version is the 1996 TD-Gammon architecture [TeG96], where the base policy is obtained through approximation in value space with a terminal cost function approximation that is computed off-line using a neural network.

1.5.3 Local and Global Error Bounds for Approximation in Value Space

In approximation in value space, an important analytical issue is to quantify the level of suboptimality of the one-step or multistep lookahead policy obtained. It is thus important to understand the character of the critical mapping between the approximation error $\tilde{J} - J^*$ and the performance error $J_{\tilde{\mu}} - J^*$, where as earlier, $J_{\tilde{\mu}}$ is the cost function of the lookahead policy $\tilde{\mu}$ and J^* is the optimal cost function.

There is a classical one-step lookahead error bound for the case of an α-discounted problem with finite state space X, which has the form

$$\|J_{\tilde{\mu}} - J^*\| \leq \frac{2\alpha}{1-\alpha}\|\tilde{J} - J^*\|; \tag{1.57}$$

where $\|\cdot\|$ denotes the maximum norm,

$$\|J_{\tilde{\mu}} - J^*\| = \max_{x \in X} \big|J_{\tilde{\mu}}(x) - J^*(x)\big|, \qquad \|\tilde{J} - J^*\| = \max_{x \in X} \big|\tilde{J}(x) - J^*(x)\big|;$$

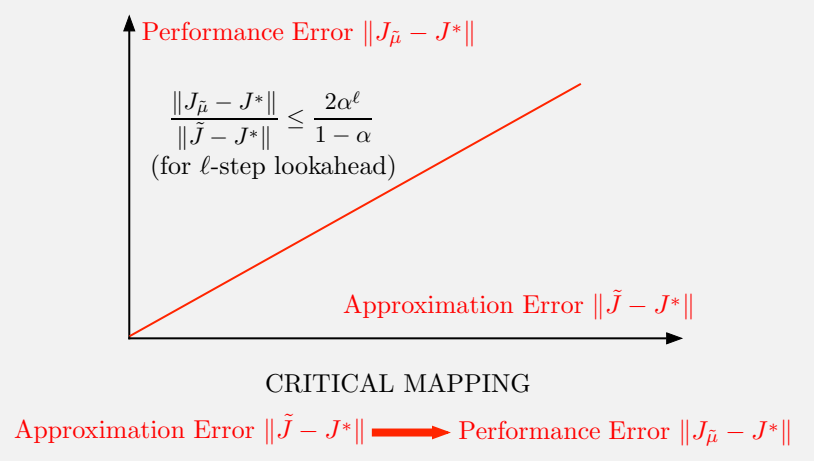


Figure 1.5.13 Illustration of the linear error bound (1.58) for ℓ-step lookahead approximation in value space. For $\ell = 1$, we obtain the one-step bound (1.57).

see e.g., [Ber19a], Prop. 5.1.1. The bound (1.57) predicts a linear relation between the size of the approximation error $\|\tilde{J} - J^*\|$ and the performance error $\|J_{\tilde{\mu}} - J^*\|$. For a generalization, we may view ℓ-step lookahead as one-step lookahead with a terminal cost function $T^{\ell-1}\tilde{J}$, i.e., $\tilde{J}$ transformed by $\ell - 1$ value iterations. We then obtain the ℓ-step bound

$$\|J_{\tilde{\mu}} - J^*\| \leq \frac{2\alpha^\ell}{1-\alpha}\|\tilde{J} - J^*\|. \tag{1.58}$$

The linear bounds (1.57)-(1.58) are illustrated in Fig. 1.5.13, and apply beyond the α-discounted case, to problems where the Bellman equation involves a contraction mapping over a subset of functions; see the RL book [Ber19a], Section 5.9.1, or the abstract DP book [Ber22b], Section 2.2.

Unfortunately, *the linear error bounds are very conservative, and do not reflect practical reality, even qualitatively so.* The main reason is that they are *global* error bounds, i.e., they hold for all $\tilde{J}$, even the worst possible. In practice, $\tilde{J}$ is often chosen sufficiently close to J^*, so that the error $J_{\tilde{\mu}} - J^*$ behaves consistently with the superlinear convergence rate of the Newton step that starts at $\tilde{J}$. In other words, for $\tilde{J}$ relatively close to J^*, we have the *local* estimate

$$\|J_{\tilde{\mu}} - J^*\| = o\big(\|\tilde{J} - J^*\|\big), \tag{1.59}$$

illustrated in Fig. 1.5.14.

A salient characteristic of this superlinear relation is that the performance error rises rapidly outside the region of superlinear convergence of Newton's method. Note that small improvements in the quality of $\tilde{J}$ (e.g., better sampling methods, improved confidence intervals, and the like, without changing the approximation architecture) have *little effect, both inside and outside the region of convergence*.

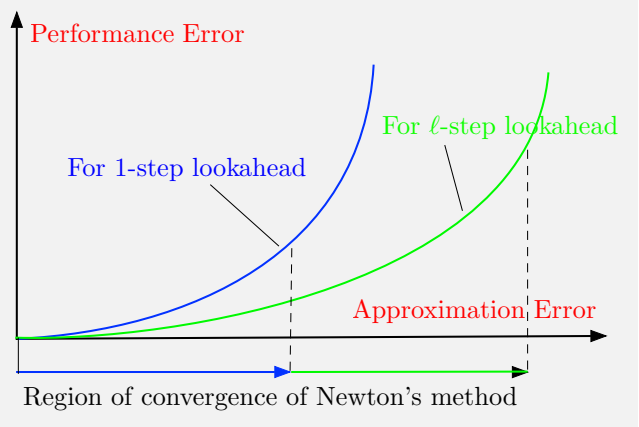


Figure 1.5.14 Schematic illustration of the correct superlinear error bound (1.59) for the case of ℓ-step lookahead approximation in value space scheme. The performance error rises rapidly outside the region of convergence of Newton's method [the illustration in the figure is not realistic; in fact the region of convergence is not bounded as it contains lines of the form γe, where γ is a scalar and e is the unit vector (all components equal to 1)]. Note that this region expands as the size of lookahead ℓ increases. Furthermore, with long enough lookahead ℓ, the ℓ-step lookahead policy $\tilde{\mu}$ can be shown to be *exactly optimal* for many problems of interest; this is a theoretical result, which holds for α-discounted finite-state problems, among others, and has been known since the 60s-70s (Prop. 2.3.1 of [Ber22a] proves a general form of this result that applies beyond discounted problems).

In practical terms, *there is often a huge difference, both quantitative and qualitative, between the linear error bounds (1.57)-(1.58) and the superlinear error bound (1.59)*. Moreover, the linear bounds, despite their popularity in academia, often misdirect academic research and confuse practitioners.† Note that as we have mentioned earlier, the qualitative performance behavior predicted in Fig. 1.5.14 holds very broadly in ap-

† A study by Laidlaw, Russell, and Dragan [LRD23] has assessed the practical performance of popular methods on a set of 155 problems, and found wide disparities relative to theoretical predictions. Quoting from this paper: "we find that prior bounds do not correlate well with when deep RL succeeds vs. fails."

The study goes on to assert the importance of long multistep lookahead (the size of ℓ) in stabilizing the performance of approximation in value space schemes. It also confirms computationally a known theoretical result, namely that with long enough lookahead ℓ, the ℓ-step lookahead policy $\tilde{\mu}$ is exactly optimal (but the required length of ℓ depends on the approximation error $\tilde{J} - J^*$). This fact has been known since the 60s-70s for α-discounted finite-state problems. A generalization of this result is given as Prop. 2.3.1 of the abstract DP book [Ber22b]; see also Section A.4 of the book [Ber22a], which discusses the convergence of Newton's method for systems of equations that involve nondifferentiable mappings (such as the Bellman operator).

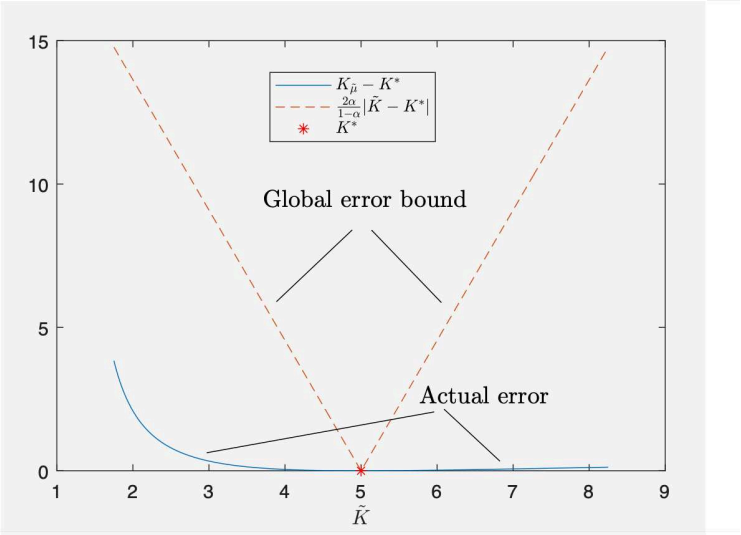


Figure 1.5.15 Illustration of the global error bound for the one-step lookahead error $K_{\tilde{\mu}} - K^*$ as a function of $\tilde{K}$, compared with the true error obtained by one step of Newton's method starting from $\tilde{K}$; cf. Example 1.5.1.

The problem data are $a = 2$, $b = 2$, $q = 1$, and $r = 5$. With these numerical values, we have $K^* = 5$ and the region of stability is (S, ∞) with $S = 1.25$. The modulus of contraction α used in the figure is computed at $\overline{S} = S + 0.5$. Depending on the chosen value of $\overline{S}$, α can be arbitrarily close to 1, but decreases as $\overline{S}$ increases. Note that the error $K_{\tilde{\mu}} - K^*$ is much smaller when $\tilde{K}$ is larger than K^* than when it is lower, because the slope of F diminishes as K increases. This is not reflected by the global error bound.

proximation in value space, because it relies on notions of abstract DP that apply very generally, for arbitrary state spaces, control spaces, and other problem characteristics; see the abstract DP book [Ber22a].

We illustrate the failure of the linear error bound (1.57) to predict the performance of the one-step lookahead policy with an example given in Fig. 1.5.15.

Example 1.5.1 (Global and Actual Error Bounds for a Linear Quadratic Problem)

Consider the one-dimensional linear quadratic problem, involving the system $x_{k+1} = ax_k + bu_k$, and the cost per stage $qx_k^2 + ru_k^2$. We will consider one-step lookahead, and a quadratic cost function approximation

$$\tilde{J}(x) = \tilde{K}x^2,$$

with $\tilde{K}$ within the region of stability, which is some interval of the form (S, ∞). The Riccati operator is

$$F(K) = \frac{a^2 rK}{r + b^2 K} + q,$$

and the one-step lookahead policy $\tilde{\mu}$ has cost function

$$J_{\tilde{\mu}}(x) = K_{\tilde{\mu}} x^2,$$

where $K_{\tilde{\mu}}$ is obtained by applying one step of Newton's method for solving the Riccati equation $K = F(K)$, starting at $K = \tilde{K}$.

Let S be the boundary of the region of stability, i.e., the value of K at which the derivative of F with respect to K is equal to 1:

$$\frac{\partial F(K)}{\partial K}\Big|_{K=S} = 1.$$

Then the Riccati operator F is a contraction within any interval $[\overline{S}, \infty)$ with $\overline{S} > S$, with a contraction modulus α that depends on $\overline{S}$. In particular, α is given by

$$\alpha = \frac{\partial F(K)}{\partial K}\Big|_{K=\overline{S}}$$

and satisfies $0 < \alpha < 1$ because $\overline{S} > S$, and the derivative of F is positive and monotonically decreasing to 0 as K increases to ∞.

The error bound (1.57) can be rederived for the case of quadratic functions and can be rewritten in terms of quadratic cost coefficients as

$$K_{\tilde{\mu}} - K^* \leq \frac{2\alpha}{1-\alpha}|\tilde{K} - K^*|, \tag{1.60}$$

where $K_{\tilde{\mu}}$ is the quadratic cost coefficient of the lookahead policy $\tilde{\mu}$ [and also the result of a Newton step for solving the fixed point Riccati equation $F = F(K)$ starting from $\tilde{K}$]. A plot of $(K_{\tilde{\mu}} - K^*)$ as a function of $\tilde{K}$, compared with the bound on the right side of this equation is shown in Fig. 1.5.15. It can be seen that $(K_{\tilde{\mu}} - K^*)$ exhibits the qualitative behavior of Newton's method, which is very different than the bound (1.60). An interesting fact is that the bound (1.60) depends on α, which in turn depends on how close $\tilde{K}$ is to the boundary S of the region of stability, while the local behavior of Newton's method is independent of S.

1.5.4 How Approximation in Value Space Can Fail and What to Do About It

Let us finally discuss the most common way that approximation in value space can fail. Consider the case where the terminal cost approximation $\tilde{J}$ is obtained through training with data of an approximation architecture such as a neural network (e.g., as in AlphaZero and TD-Gammon). Then there are three components that determine the approximation error $\tilde{J} - J^*$:

(a) The *power of the architecture*, which roughly speaking is a measure of the *representation error*, i.e., the error that would be obtained even if infinite data were available and were used optimally to obtain $\tilde{J}$.

(b) The *error degradation due to the limited availability of training data*.

(c) The additional *error degradation due to imperfections in the training methodology*.

Thus *if the architecture is not powerful enough to bring $\tilde{J} - J^*$ within the region of convergence of Newton's method, approximation in value space with one-step lookahead will likely fail, no matter how much data is collected and how effective the associated training method is*.

In this case, there are two potential practical remedies:

(1) Use a more powerful architecture/neural network for representing $\tilde{J}$.

(2) Extend the combined length of the lookahead minimization and truncated rollout in order to bring the effective value of $\tilde{J}$ within the region of convergence of Newton's method.

The first remedy typically requires a deep neural network or transformer, which uses more weights and requires more expensive training (see Chapter 3).† The second remedy requires longer on-line computation and/or simulation, which may run up against some practical real-time implementation constraint. Parallel computation and sophisticated multistep lookahead implementation methods may help to mitigate these requirements (see Chapter 2).

1.6 EXAMPLES, REFORMULATIONS, AND SIMPLIFICATIONS

In this section we provide a few examples that illustrate problem formulation techniques, exact and approximate solution methods, and adaptations of the basic DP algorithm to various contexts. We refer to DP textbooks for extensive additional discussions of modeling and problem formulation techniques (see e.g., the many examples that can be found in the author's DP and RL textbooks [Ber12], [Ber17a], [Ber19a], [Ber20a], as well as in the neuro-dynamic programming book [BeT96]).

An important fact to keep in mind is that there are many ways to model a given practical problem in terms of DP, and that there is no unique choice for state and control variables. This will be brought out by the examples in this section, and is facilitated by the generality of DP: its basic algorithmic principles apply for arbitrary state, control, and disturbance spaces, and system and cost functions.

1.6.1 A Few Words About Modeling

In practice, optimization problems seldom come neatly packaged as mathematical problems that can be solved by DP/RL or some other methodology.

† For a recent example of implementation of a grandmaster-level chess program with *one-step lookahead* and a huge-size (270M parameter) neural network position evaluator, see Ruoss et al. [RDM24].

Generally, a practical problem is a prime candidate for a DP formulation if it involves multiple sequential decisions, which are separated by feedback, i.e., by observations that can be used to enhance the effectiveness of future decisions.

However, there are other types of problems that can be fruitfully formulated by DP. These include the entire class of deterministic problems, where there is no information to be collected: all the information needed in a deterministic problem is either known or can be predicted from the problem data that is available at time 0 (see, e.g., the traveling salesman Example 1.2.3). Moreover, for deterministic problems there is a plethora of non-DP methods, such as linear, nonlinear, and integer programming, random and nonrandom search, discrete optimization heuristics, etc. Still, however, the use of RL methods for deterministic optimization is a major subject in this book, which will be discussed in Chapter 2. We will argue there that rollout and its variations, when suitably applied, can improve substantially on the performance of other heuristic or suboptimal methods, however derived. Moreover, we will see that often for discrete optimization problems the DP sequential structure is introduced artificially, with the aim to facilitate the use of approximate DP/RL methods.

There are also problems that fit quite well into the sequential structure of DP, but can be fruitfully addressed by RL methods that do not have a fundamental connection with DP. An important case in point is *policy gradient* and *policy search* methods, which will be considered somewhat briefly in Chapter 3. Here the policy of the problem is parametrized by a set of parameters, so that the cost of the policy becomes a function of these parameters, and can be optimized by non-DP methods such as gradient or random search-based suboptimal approaches. This generally relates to the approximation in policy space approach, which we have discussed in Section 1.3.3 and we will discuss further in Section 3.4; see also Section 5.7 of the RL book [Ber19a].

A Practical Guide

As a guide for formulating optimal control problems in a manner that is suitable for a DP solution the following two-stage process is suggested:

(a) Identify the controls/decisions u_k and the times k at which these controls are applied. Usually this step is fairly straightforward. However, in some cases there may be some choices to make. For example in deterministic problems, where the objective is to select an optimal sequence of controls $\{u_0, \ldots, u_{N-1}\}$, one may lump multiple controls to be chosen together, e.g., view the pair (u_0, u_1) as a single choice. This is usually not possible in stochastic problems, where distinct decisions are differentiated by the information/feedback available when making them.

(b) Select the states x_k. The basic guideline here is that x_k should encompass *all the information that is relevant for future optimization*, i.e., the information that is known to the controller at time k and can be used with advantage in choosing u_k. In effect, at time k *the state x_k should separate the past from the future*, in the sense that anything that has happened in the past (states, controls, and disturbances from stages prior to stage k) is irrelevant to the choices of future controls as long we know x_k. Sometimes this is described by saying that the state should have a "Markov property" to express an analogy with states of Markov chains, where (by definition) the conditional probability distribution of future states depends on the past history of the chain only through the present state.

The control and state selection may also have to be refined or specialized in order to enhance the application of known results and algorithms. This includes the choice of a finite or an infinite horizon, and the availability of good base policies or heuristics in the context of rollout.

Note that there may be multiple possibilities for selecting the states, because information may be packaged in several different ways that are equally useful from the point of view of control. It may thus be worth considering alternative ways to choose the states; for example try to use states that minimize the dimensionality of the state space. For a trivial example that illustrates the point, if a quantity x_k qualifies as state, then (x_{k-1}, x_k) also qualifies as state, since (x_{k-1}, x_k) contains all the information contained within x_k that can be useful to the controller when selecting u_k. However, using (x_{k-1}, x_k) in place of x_k, gains nothing in terms of optimal cost while complicating the DP algorithm that would have to be executed over a larger space.

The concept of a *sufficient statistic*, which refers to a quantity that summarizes all the essential content of the information available to the controller, may be useful in providing alternative descriptions of the state space. An important paradigm is problems involving *partial* or *imperfect* state information, where x_k evolves over time but is not fully accessible for measurement (for example, x_k may be the position/velocity vector of a moving vehicle, but we may obtain measurements of just the position). If I_k is the collection of all measurements and controls up to time k (the *information vector*), it is correct to use I_k as state in a reformulated DP problem that involves perfect state observation. However, a better alternative may be to use as state the conditional probability distribution

$$P_k(x_k \mid I_k),$$

called *belief state*, which (as it turns out) subsumes all the information that is useful for the purposes of choosing a control. On the other hand, the belief state $P_k(x_k \mid I_k)$ is an infinite-dimensional object, whereas I_k may be finite dimensional, so the best choice may be problem-dependent.

Still, in either case, the stochastic DP algorithm applies, with the sufficient statistic [whether I_k or $P_k(x_k \mid I_k)$] playing the role of the state.

A Few Words about the Choice of an RL Method

An attractive aspect of the current RL methodology, inherited by the generality of our DP formulation, is that it can address a very broad range of challenging problems, deterministic as well as stochastic, discrete as well as continuous, etc. However, in the practical application of RL one has to contend with limited theoretical guarantees. In particular, several of the RL methods that have been successful in practice have less than solid performance properties, and may not work on a given problem, even one of the type for which they are designed.

This is a reflection of the state of the art in the field: *there are no methods that are guaranteed to work for all or even most DP problems*. However, there are enough methods to try on a given problem with a reasonable chance of success in the end (after some heuristic and problem specific tuning). For this reason, it is important to develop insight into the inner workings of various methods, as a means of selecting the proper type of methodology to try on a given problem.†

A related consideration is the context within which a method is applied. In particular, is it a single problem that is being addressed, such as chess that has fixed rules and a fixed initial condition, or is it a family of related problems that must be periodically be solved with small variations in its data or its initial conditions? Also, are the problem data fixed or may they change over time as the system is being controlled?

A related consideration is the context within which a method is applied. In particular, is it a single problem that is being addressed, such as chess that has fixed rules and a fixed initial condition, or is it a family of related problems that must be periodically be solved with large or small variations in its data or its initial conditions? Also, if the problem data may change, how predictably and how fast do they change?

Generally, convenient but relatively unreliable methods, which can be tuned to the problem at hand, may be tried with a reasonable chance of success if a single problem is addressed. Similarly, RL methods that require extensive tuning of parameters, including ones that involve approximation in policy space and the use of neural networks, may be well suited for a stable problem environment and a single problem solution. However, they are not well suited for problems with a variable environment and/or real-

† Aside from insight and intuition, it is also important to have a foundational understanding of the analytical principles of the field and of the mechanisms underlying the central computational methods. The role of the theory in this respect is to structure mathematically the methodology, guide the art, and delineate the sound from the flawed ideas.

time changes of model parameters. For such problems, RL methods based on approximation in value space and on-line play, possibly involving on-line replanning, are much better suited.

Note also that even when on-line replanning is not needed, on-line play may improve substantially the performance of off-line trained policies, so we may wish to use it in conjunction with off-line training. This is due to the Newton step that is implicit in one-step or multistep lookahead minimization, cf. our discussion of the AlphaZero and TD-Gammon architectures in Section 1.1. Of course the computational requirements of an on-line play method may be substantial and have to be taken into account when assessing its suitability for a particular application. In this connection, deterministic problems are better suited than stochastic problems for on-line play. Moreover, methods that are well-suited for parallel computation, and/or involve the use of certainty equivalence approximations are generally better suited for a stochastic control environment.

1.6.2 Problems with a Termination State

Many DP problems of interest involve a *termination state*, i.e., a state t that is cost-free and absorbing in the sense that for all k,

$$g_k(t, u_k, w_k) = 0, \quad f_k(t, u_k, w_k) = t, \qquad \text{for all } w_k \text{ and } u_k \in U_k(t).$$

Thus the control process essentially terminates upon reaching t, even if this happens before the end of the horizon. One may reach t by choice if a special stopping decision is available, or by means of a random transition from another state. Problems involving games, such as chess, Go, backgammon, and others include a termination state (the end of the game) and have played an important role in the development of the RL methodology.†

Generally, when it is known that an optimal policy will reach the termination state with certainty within at most some given number of stages N, the DP problem can be formulated as an N-stage horizon problem, with a very large termination cost for the nontermination states.‡ The reason is that even if the termination state t is reached at a time $k < N$, we can extend our stay at t for an additional $N - k$ stages at no additional cost, so the optimal policy will still be optimal, since it will not incur the large termination cost at the end of the horizon.

† Games often involve two players/decision makers, in which case they can be addressed by suitably modified exact or approximate DP algorithms. The DP algorithm that we have discussed in this chapter involves a single decision maker, but can be used to find an optimal policy for one player against a fixed and known policy of the other player.

‡ When an upper bound on the number of stages to termination is not known, the problem may be formulated as an infinite horizon problem of the stochastic shortest path problem.

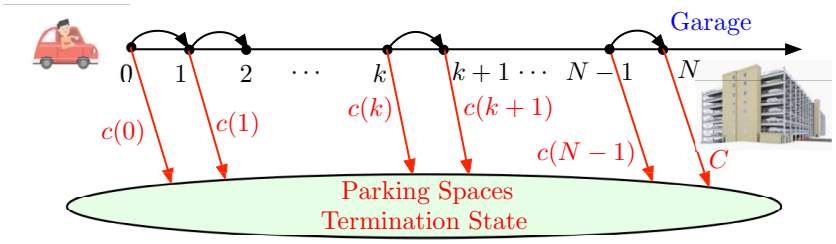


Figure 1.6.1 Cost structure of the parking problem. The driver may park at space $k = 0, 1, \ldots, N-1$ at cost $c(k)$, if the space is free, or continue to the next space $k+1$ at no cost. At space N (the garage) the driver must park at cost C.

Example 1.6.1 (Parking)

A driver is looking for inexpensive parking on the way to his destination. The parking area contains N spaces, numbered $0, \ldots, N-1$, and a garage following space $N-1$. The driver starts at space 0 and traverses the parking spaces sequentially, i.e., from space k he goes next to space $k+1$, etc. Each parking space k costs $c(k)$ and is free with probability $p(k)$ independently of whether other parking spaces are free or not. If the driver reaches the last parking space $N-1$ and does not park there, he must park at the garage, which costs C. The driver can observe whether a parking space is free only when he reaches it, and then, if it is free, he makes a decision to park in that space or not to park and check the next space. The problem is to find the minimum expected cost parking policy.

We formulate the problem as a DP problem with N stages, corresponding to the parking spaces, and an artificial termination state t that corresponds to having parked; see Fig. 1.6.1. At each stage $k = 1, \ldots, N-1$, we have three states: the artificial termination state t, and the two states F and $\overline{F}$, corresponding to space k being free or taken, respectively. At stage 0, we have only two states, F and $\overline{F}$, and at the final stage there is only one state, the termination state t. The decision/control is to park or continue at state F [there is no choice at states $\overline{F}$ and state t]. From location k, the termination state t is reached at cost $c(k)$ when a parking decision is made (assuming location k is free). Otherwise, the driver continues to the next state at no cost. At stage N, the driver must park at cost C.

Let us now derive the form of the DP algorithm, denoting:

$J_k^*(F)$: The optimal cost-to-go upon arrival at a space k that is free.

$J_k^*(\overline{F})$: The optimal cost-to-go upon arrival at a space k that is taken.

$J_k^*(t)$: The cost-to-go of the "parked"/termination state t.

The DP algorithm for $k = 0, \ldots, N-1$ takes the form

$$J_k^*(F) = \begin{cases} \min\big[c(k),\, p(k+1)J_{k+1}^*(F) + \big(1-p(k+1)\big)J_{k+1}^*(\overline{F})\big] & \text{if } k < N-1, \\ \min\big[c(N-1),\, C\big] & \text{if } k = N-1, \end{cases}$$

$$J_k^*(\overline{F}) = \begin{cases} p(k+1)J_{k+1}^*(F) + \big(1-p(k+1)\big)J_{k+1}^*(\overline{F}) & \text{if } k < N-1, \\ C & \text{if } k = N-1, \end{cases}$$

for the states other than the termination state t, while for t we have

$$J_k^*(t) = 0, \qquad k = 1, \ldots, N.$$

The minimization above corresponds to the two choices (park or not park) at the states F that correspond to a free parking space.

While this algorithm is easily executed, it can be written in a simpler and equivalent form. This can be done by introducing the scalars

$$\hat{J}_k = p(k)J_k^*(F) + \big(1-p(k)\big)J_k^*(\overline{F}), \qquad k = 0, \ldots, N-1,$$

which can be viewed as the optimal expected cost-to-go upon arriving at space k but *before verifying its free or taken status*. Indeed, from the preceding DP algorithm, we have

$$\hat{J}_{N-1} = p(N-1)\min\big[c(N-1),\, C\big] + \big(1-p(N-1)\big)C,$$

$$\hat{J}_k = p(k)\min\big[c(k),\, \hat{J}_{k+1}\big] + \big(1-p(k)\big)\hat{J}_{k+1}, \qquad k = 0, \ldots, N-2.$$

From this algorithm we can also obtain the optimal parking policy:

Park at space $k = 0, \ldots, N-1$ if it is free and we have $c(k) \leq \hat{J}_{k+1}$.

This is an example of DP simplification that occurs when the state involves components that are not affected by the choice of control, and will be addressed in the next section.

1.6.3 General Discrete Optimization Problems

Discrete deterministic optimization problems, including challenging combinatorial problems, can be typically formulated as DP problems by breaking down each feasible solution into a sequence of decisions/controls, similar to the scheduling Example 1.2.1, and the traveling salesman Examples 1.2.2 and 1.2.3. This formulation often leads to an intractable exact DP computation because of an exponential explosion of the number of states as time progresses. However, a reformulation to a discrete optimal control problem brings to bear approximate DP methods, such as rollout and others, to be discussed later, which can deal with the exponentially increasing size of the state space.

Let us now extend the ideas of the examples just noted to the general discrete optimization problem:

$$\begin{aligned} &\text{minimize} \quad G(u) \\ &\text{subject to} \quad u \in U, \end{aligned} \tag{1.61}$$

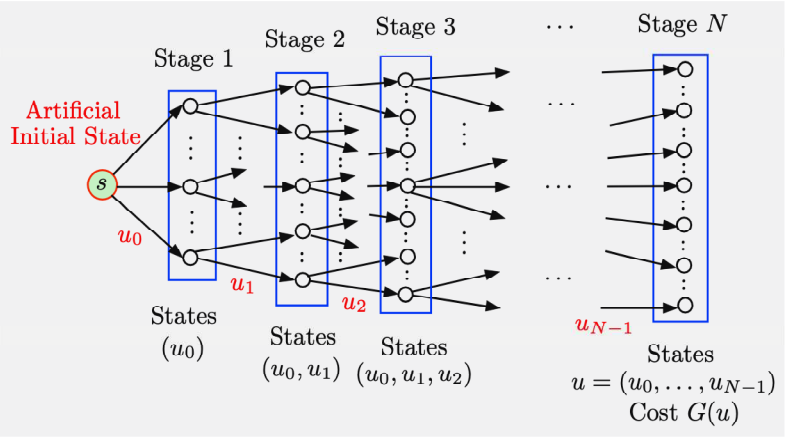


Figure 1.6.2 Formulation of a discrete optimization problem as a DP problem with N stages. There is a cost $G(u)$ only at the terminal stage on the arc connecting an N-solution $u = (u_0, \ldots, u_{N-1})$ upon reaching the terminal state. Note that there is only one incoming arc at each node.

where U is a finite set of feasible solutions and $G(u)$ is a cost function.

We assume that each solution u has N components; i.e., it has the form

$$u = (u_0, \ldots, u_{N-1}),$$

where N is a positive integer. We can then view the problem as a sequential decision problem, where the components $u_0, \ldots, u_{N-1}$ are selected one-at-a-time. A k-tuple $(u_0, \ldots, u_{k-1})$ consisting of the first k components of a solution is called a *k-solution*. We associate k-solutions with the kth stage of the finite horizon discrete optimal control problem shown in Fig. 1.6.2. In particular, for $k = 1, \ldots, N$, we view as the states of the kth stage all the k-tuples $(u_0, \ldots, u_{k-1})$. For stage $k = 0, \ldots, N-1$, we view u_k as the control. The initial state is an artificial state denoted s. From this state, by applying u_0, we may move to any "state" (u_0), with u_0 belonging to the set

$$U_0 = \big\{\tilde{u}_0 \mid \text{there exists a solution of the form } (\tilde{u}_0, \tilde{u}_1, \ldots, \tilde{u}_{N-1}) \in U\big\}. \tag{1.62}$$

Thus U_0 is the set of choices of u_0 that are consistent with feasibility.

More generally, from a state $(u_0, \ldots, u_{k-1})$, we may move to any state of the form $(u_0, \ldots, u_{k-1}, u_k)$, upon choosing a control u_k that belongs to the set

$$U_k(u_0, \ldots, u_{k-1}) = \big\{u_k \mid \text{for some } \overline{u}_{k+1}, \ldots, \overline{u}_{N-1} \text{ we have } (u_0, \ldots, u_{k-1}, u_k, \overline{u}_{k+1}, \ldots, \overline{u}_{N-1}) \in U\big\}. \tag{1.63}$$

These are the choices of u_k that are consistent with the preceding choices $u_0, \ldots, u_{k-1}$, and are also consistent with feasibility [we do not exclude the

possibility that the set (1.63) is empty]. The last stage corresponds to the N-solutions $u = (u_0, \ldots, u_{N-1})$, and the terminal cost is $G(u)$; see Fig. 1.6.2. All other transitions in this DP problem formulation have cost 0.

Let $J_k^*(u_0, \ldots, u_{k-1})$ denote the optimal cost starting from the k-solution $(u_0, \ldots, u_{k-1})$, i.e., the optimal cost of the problem over solutions whose first k components are constrained to be equal to $u_0, \ldots, u_{k-1}$. The DP algorithm is described by the equation

$$J_k^*(u_0, \ldots, u_{k-1}) = \min_{u_k \in U_k(u_0, \ldots, u_{k-1})} J_{k+1}^*(u_0, \ldots, u_{k-1}, u_k),$$

with the terminal condition

$$J_N^*(u_0, \ldots, u_{N-1}) = G(u_0, \ldots, u_{N-1}).$$

This algorithm executes backwards in time: starting with the known function $J_N^* = G$, we compute J_{N-1}^*, then J_{N-2}^*, and so on up to computing J_0^*. An optimal solution $(u_0^*, \ldots, u_{N-1}^*)$ is then constructed by going forward through the algorithm

$$u_k^* \in \arg \min_{u_k \in U_k(u_0^*, \ldots, u_{k-1}^*)} J_{k+1}^*(u_0^*, \ldots, u_{k-1}^*, u_k), \quad k = 0, \ldots, N-1, \tag{1.64}$$

where U_0 is given by Eq. (1.62), and U_k is given by Eq. (1.63): first compute u_0^*, then u_1^*, and so on up to u_{N-1}^*; cf. Eq. (1.8).

Of course here the number of states typically grows exponentially with N, but we can use the DP minimization (1.64) as a starting point for approximation methods. For example we may try to use approximation in value space, whereby we replace J_{k+1}^* with some suboptimal $\tilde{J}_{k+1}$ in Eq. (1.64). One possibility is to use as

$$\tilde{J}_{k+1}(u_0^*, \ldots, u_{k-1}^*, u_k),$$

the cost generated by a heuristic method that solves the problem suboptimally with the values of the first $k+1$ decision components fixed at $u_0^*, \ldots, u_{k-1}^*, u_k$. This is the *rollout algorithm*, which turns out to be a very simple and effective approach for approximate combinatorial optimization.

Let us finally note that while we have used a general cost function G and constraint set U in our discrete optimization model of this section, in many problems G and/or U may have a special (e.g., additive) structure, which is consistent with a sequential decision making process and may be computationally exploited. The traveling salesman Example 1.2.2 is a case in point, where G consists of the sum of N components (the intercity travel costs), one per stage. Our next example deals with a problem of great current interest.

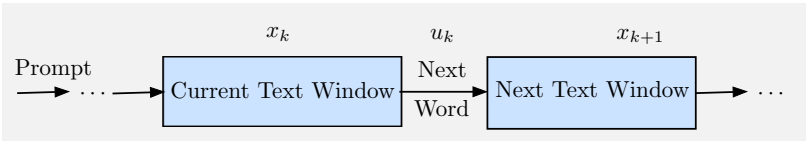


Figure 1.6.3 Schematic visualization of an LLM problem based on N-grams.

Example 1.6.2 (A Large Language Model Based on N-Grams)

Let us consider an N-gram model, whereby a text string consisting of N words is transformed into another string of N words by adding a word at the front of the string and deleting the word at the back of the string. We view the text strings as states of a dynamic system affected by the added word choice, which we view as the control. We denote by x_k the string obtained at time k, and by u_k the word added at time k. We assume that u_k is chosen from a given set $U(x_k)$. Thus we have a controlled dynamic system, which is deterministic and is described by an equation of the form

$$x_{k+1} = f(x_k, u_k),$$

where f specifies the operation of adding u_k at the front of x_k and removing the word at the back of x_k. The initial string x_0 is assumed given.

If we have a cost function G by which to evaluate a text string, we can pose a DP problem with either a finite or an infinite horizon. For example if the string evolution terminates after exactly N steps, we obtain the finite horizon problem of minimizing the function $G(x_N)$ of the final text string x_N. In this case, x_N is obtained after we have a chance to change successively all the words of the initial string x_0, subject to the constraints $u_k \in U(x_k)$.

Another possibility is to introduce a termination action, whereby addition/deletion of words is optionally stopped at some time and the final text string x is obtained with cost $G(x)$. In such a problem formulation, we may also include an additive stage cost that depends on u. This is an infinite horizon formulation that involves an additional termination state t in the manner of Section 1.6.2.

Note that in both the finite and the infinite horizon formulation of the problem, the initial string x_0 may include a "prompt," which may be subject to optimization through some kind of "prompt engineering." Depending on the context, this may include the use of another optimization or heuristic algorithm, perhaps unrelated to DP, which searches for a favorable prompt from within a given set of choices.

Interesting policies for the preceding problem formulation may be provided by a neural network, such as a Generative Pre-trained Transformer (GPT). In our terms, the GPT can be viewed simply as a policy that generates next words. This policy may be either deterministic, i.e., $u_k = \mu(x_k)$ for some function μ, or it may be a "randomized" policy, which generates u_k according to a probability distribution that depends on x_k. Our DP formulation can also form the basis for policy improvement algorithms such as rollout, which aim to improve the quality of the output generated by the GPT. Another, more ambitious, possibility is to consider an approximate,

neural network-based, policy iteration/self-training scheme, such as the ones discussed earlier, based on the AlphaZero/TD-Gammon architecture. Such a scheme generates a sequence of GPTs, with each GPT trained with data provided by the preceding GPT, a form of self-learning in the spirit of the AlphaZero and TD-Gammon policy iteration algorithms, cf. Section 1.1.

It is also possible to provide an infinite horizon formulation of the general discrete optimization problem

$$\begin{aligned} &\text{minimize} \quad G(u) \\ &\text{subject to} \quad u \in U, \end{aligned} \tag{1.65}$$

where U is a finite set of feasible solutions, $G(u)$ is a cost function, and u consists of N components, $u = (u_0, \ldots, u_{N-1})$; cf. Eq. (1.61). To this end, we introduce a termination state t that the system enters after N steps. At step k, the component u_k is selected subject to $u_k \in U_k(u_0, \ldots, u_{k-1}$, where the constraint set $U_k(u_0, \ldots, u_{k-1}$ is given by Eq. (1.63). This is a special case of a general finite to infinite horizon stochastic DP problem reformulation, which we describe in the next section.

1.6.4 General Finite to Infinite Horizon Reformulation

There is a conceptually important reformulation that transforms a finite horizon problem, possibly involving a nonstationary system and cost per stage, to an equivalent infinite horizon problem. It is based on introducing an expanded state space, which is the union of the state spaces of the finite horizon problem plus an artificial cost-free termination state that the system moves into at the end of the horizon. This reformulation is of great conceptual value, as it provides a mechanism to bring to bear ideas that can be most conveniently understood within an infinite horizon context. For example, it helps to understand the synergy of off-line training and on-line play based on Newton's method, and the related insights that explain the good performance of rollout algorithms in practice.

To define the reformulation, let us consider the N-stage horizon stochastic problem of Section 1.3.1, whose system has the form

$$x_{k+1} = f_k(x_k, u_k, w_k), \qquad k = 0, \ldots, N-1, \tag{1.66}$$

and let us denote by X_k, $k = 0, \ldots, N$, and U_k, $k = 0, \ldots, N-1$, the corresponding state spaces and control spaces, respectively. We introduce an artificial termination state t, and we consider an infinite horizon problem with state and control spaces X and U given by

$$X = \left(\cup_{k=0}^{N} X_k\right) \cup \{t\}, \qquad U = \cup_{k=0}^{N-1} U_k; \tag{1.67}$$

see Fig. 1.6.4.

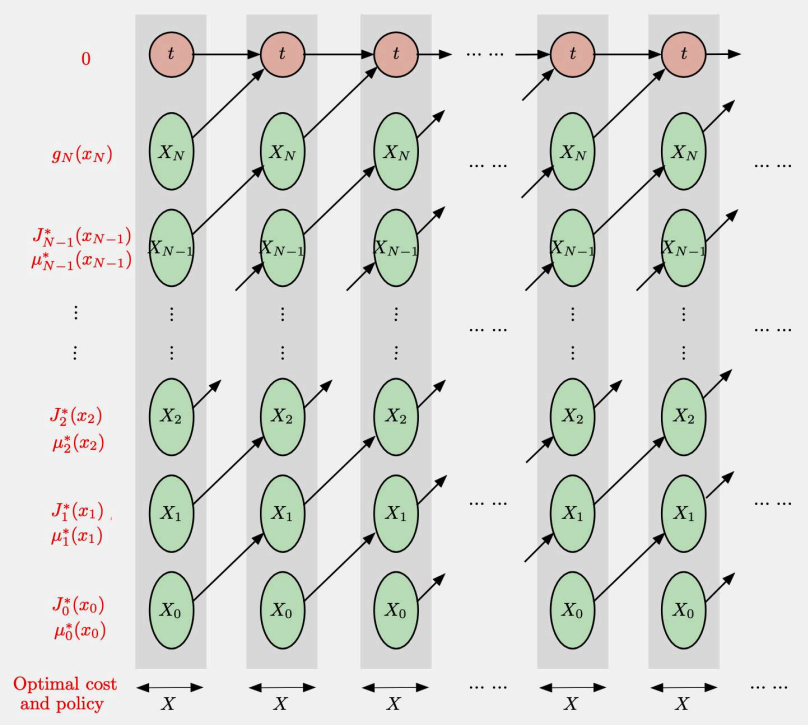


Figure 1.6.4 Illustration of the infinite horizon equivalent of a finite horizon problem. The state space is $X = \left(\cup_{k=0}^{N} X_k\right) \cup \{t\}$, and the control space is $U = \cup_{k=0}^{N-1} U_k$. Transitions from states $x_k \in X_k$ lead to states in $x_{k+1} \in X_{k+1}$ according to the system equation $x_{k+1} = f_k(x_k, u_k, w_k)$, and they are stochastic when they involve the random disturbance w_k. The transition from states $x_N \in X_N$ lead deterministically to the termination state at cost $g_N(x_N)$. The termination state t is cost-free and absorbing.

The infinite horizon optimal cost $J^*(x_k)$ and optimal policy $\mu^*(x_k)$ at state $x_k \in X_k$ of the infinite horizon problem are equal to optimal cost-to-go $J^*_k(x_k)$ and optimal policy $\mu^*_k(x_k)$ of the finite horizon problem.

The system equation and the control constraints of this problem are also reformulated so that states in X_k, $k = 0, \ldots, N-1$, are mapped to states in X_{k+1}, according to Eq. (1.66), while states $x_N \in X_N$ are mapped to the termination state t at cost $g_N(x_N)$. Upon reaching t, the state stays at t at no cost. Thus the policies of the infinite horizon problem map states $x_k \in X_k$ to controls in $U_k(x_k) \subset U_k$, and consist of functions $\mu_k(x_k)$ that are policies of the finite horizon problem. Moreover, the Bellman equation for the infinite horizon problem is identical to the DP algorithm for the finite horizon problem.

It can be seen that the optimal cost and optimal control, $J^*(x_k)$ and $\mu^*(x_k)$, at a state $x_k \in X_k$ in the infinite horizon problem are equal to the optimal cost-to-go $J^*_k(x_k)$ and optimal control $\mu^*_k(x_k)$ of the original finite

horizon problem, respectively; cf. Fig. 1.6.4. Moreover approximation in value space and rollout in the finite horizon problem translate to infinite horizon counterparts, and can be understood as Newton steps for solving the Bellman equation of the infinite horizon problem (or equivalently the DP algorithm of the finite horizon problem).

In summary, finite horizon problems can be viewed as infinite horizon problems with a special structure that involves a termination state t, and the state and control spaces of Eq. (1.67), as illustrated in Fig. 1.6.4. The Bellman equation of the infinite horizon problem coincides with the DP algorithm of the finite horizon problem. The PI algorithm for the infinite horizon problem can be translated directly to a PI algorithm for the finite horizon problem, involving repeated policy evaluations and policy improvements. Finally, the Newton step interpretations for approximation in value space and rollout schemes for the infinite horizon problem have straightforward analogs for finite horizon problems, and explain the powerful cost improvement mechanism that underlies the rollout algorithm and its variations.

1.6.5 State Augmentation, Time Delays, Forecasts, and Uncontrollable State Components

In practice, we are often faced with situations where some of the assumptions of our stochastic optimal control problem formulation are violated. For example, the disturbances may involve a complex probabilistic description that may create correlations that extend across stages, or the system equation may include dependences on controls applied in earlier stages, which affect the state with some delay.

Generally, in such cases the problem can be reformulated into our DP problem format through a technique, which is called *state augmentation* because it typically involves the enlargement of the state space. The general intuitive guideline in state augmentation is to *include in the enlarged state at time k all the information that is known to the controller at time k and can be used with advantage in selecting u_k*. State augmentation allows the treatment of time delays in the effects of control on future states, correlated disturbances, forecasts of probability distributions of future disturbances, and many other complications. We note, however, that state augmentation often comes at a price: the reformulated problem may have a very complex state space. We provide some examples.

Time Delays

In some applications the system state x_{k+1} depends not only on the preceding state x_k and control u_k, but also on earlier states and controls. Such situations can be handled by expanding the state to include an appropriate number of earlier states and controls.

As an example, assume that there is at most a single stage delay in the state and control; i.e., the system equation has the form

$$x_{k+1} = f_k(x_k, x_{k-1}, u_k, u_{k-1}, w_k), \qquad k = 1, \ldots, N-1, \tag{1.68}$$

$$x_1 = f_0(x_0, u_0, w_0).$$

If we introduce additional state variables y_k and s_k, and we make the identifications $y_k = x_{k-1}$, $z_k = u_{k-1}$, the system equation (1.68) yields

$$\begin{pmatrix} x_{k+1} \\ y_{k+1} \\ z_{k+1} \end{pmatrix} = \begin{pmatrix} f_k(x_k, y_k, u_k, z_k, w_k) \\ x_k \\ u_k \end{pmatrix}. \tag{1.69}$$

By defining $\tilde{x}_k = (x_k, y_k, z_k)$ as the new state, we have

$$\tilde{x}_{k+1} = \tilde{f}_k(\tilde{x}_k, u_k, w_k),$$

where the system function $\tilde{f}_k$ is defined from Eq. (1.69).

By using the preceding equation as the system equation and by expressing the cost function in terms of the new state, the problem is reduced to a problem without time delays. Naturally, the control u_k should now depend on the new state $\tilde{x}_k$, or equivalently a policy should consist of functions μ_k of the current state x_k, as well as the preceding state x_{k-1} and the preceding control u_{k-1}.

When the DP algorithm for the reformulated problem is translated in terms of the variables of the original problem, it takes the form

$$J_N^*(x_N) = g_N(x_N),$$

$$\begin{aligned} &J_{N-1}^*(x_{N-1}, x_{N-2}, u_{N-2}) \\ &\quad = \min_{u_{N-1} \in U_{N-1}(x_{N-1})} E_{w_{N-1}} \Big\{ g_{N-1}(x_{N-1}, u_{N-1}, w_{N-1}) \\ &\qquad\qquad + J_N^*\big(f_{N-1}(x_{N-1}, x_{N-2}, u_{N-1}, u_{N-2}, w_{N-1})\big) \Big\}, \end{aligned}$$

$$\begin{aligned} J_k^*(x_k, x_{k-1}, u_{k-1}) = \min_{u_k \in U_k(x_k)} E_{w_k} \Big\{ & g_k(x_k, u_k, w_k) \\ & + J_{k+1}^*\big(f_k(x_k, x_{k-1}, u_k, u_{k-1}, w_k), x_k, u_k\big) \Big\}, \quad k = 1, \ldots, N-2, \end{aligned}$$

$$J_0^*(x_0) = \min_{u_0 \in U_0(x_0)} E_{w_0} \Big\{ g_0(x_0, u_0, w_0) + J_1^*\big(f_0(x_0, u_0, w_0), x_0, u_0\big) \Big\}.$$

Similar reformulations are possible when time delays appear in the cost or the control constraints; for example, in the case where the cost is

$$E\left\{ g_N(x_N, x_{N-1}) + g_0(x_0, u_0, w_0) + \sum_{k=1}^{N-1} g_k(x_k, x_{k-1}, u_k, w_k) \right\}.$$

The extreme case of time delays in the cost arises in the nonadditive form

$$E\big\{g_N(x_N, x_{N-1}, \ldots, x_0, u_{N-1}, \ldots, u_0, w_{N-1}, \ldots, w_0)\big\}.$$

Then, the problem can be reduced to the standard problem format, by using as augmented state

$$\tilde{x}_k = \big(x_k, x_{k-1}, \ldots, x_0, u_{k-1}, \ldots, u_0, w_{k-1}, \ldots, w_0\big)$$

and $E\big\{g_N(\tilde{x}_N)\big\}$ as reformulated cost. Policies consist of functions μ_k of the present and past states $x_k, \ldots, x_0$, the past controls $u_{k-1}, \ldots, u_0$, and the past disturbances $w_{k-1}, \ldots, w_0$. Naturally, we must assume that the past disturbances are known to the controller. Otherwise, we are faced with a problem where the state is imprecisely known to the controller, which will be discussed in the next section.

Forecasts

Consider a situation where at time k the controller has access to a forecast y_k that results in a reassessment of the probability distribution of the subsequent disturbance w_k and, possibly, future disturbances. For example, y_k may be an exact prediction of w_k or an exact prediction that the probability distribution of w_k is a specific one out of a finite collection of distributions. Forecasts of interest in practice are, for example, probabilistic predictions on the state of the weather, the interest rate for money, and the demand for inventory. Generally, forecasts can be handled by introducing additional state variables corresponding to the information that the forecasts provide. We will illustrate the process with a simple example.

Assume that at the beginning of each stage k, the controller receives an accurate prediction that the next disturbance w_k will be selected according to a particular probability distribution out of a given collection of distributions $\{P_1, \ldots, P_m\}$; i.e., if the forecast is i, then w_k is selected according to P_i. The a priori probability that the forecast will be i is denoted by p_i and is given.

The forecasting process can be represented by means of the equation

$$y_{k+1} = \xi_k,$$

where y_{k+1} can take the values $1, \ldots, m$, corresponding to the m possible forecasts, and ξ_k is a random variable taking the value i with probability p_i. The interpretation here is that when ξ_k takes the value i, then w_{k+1} will occur according to the distribution P_i.

By combining the system equation with the forecast equation $y_{k+1} = \xi_k$, we obtain an augmented system given by

$$\begin{pmatrix} x_{k+1} \\ y_{k+1} \end{pmatrix} = \begin{pmatrix} f_k(x_k, u_k, w_k) \\ \xi_k \end{pmatrix}.$$

The new state and disturbance are

$$\tilde{x}_k = (x_k, y_k), \qquad \tilde{w}_k = (w_k, \xi_k).$$

The probability distribution of $\tilde{w}_k$ is determined by the distributions P_i and the probabilities p_i, and depends explicitly on $\tilde{x}_k$ (via y_k) but not on the prior disturbances.

Thus, by suitable reformulation of the cost, the problem can be cast as a stochastic DP problem. Note that the control applied depends on both the current state and the current forecast. The DP algorithm takes the form

$$J_N^*(x_N, y_N) = g_N(x_N),$$

$$J_k^*(x_k, y_k) = \min_{u_k \in U_k(x_k)} E_{w_k} \Big\{ g_k(x_k, u_k, w_k) + \sum_{i=1}^{m} p_i J_{k+1}^*\big(f_k(x_k, u_k, w_k), i\big) \mid y_k \Big\}, \tag{1.70}$$

where y_k may take the values $1, \ldots, m$, and the expectation over w_k is taken with respect to the distribution P_{y_k}.

Note that the preceding formulation admits several extensions. One example is the case where forecasts can be influenced by the control action (e.g., pay extra for a more accurate forecast), and may involve several future disturbances. However, the price for these extensions is increased complexity of the corresponding DP algorithm.

Problems with Uncontrollable State Components

In many problems of interest the natural state of the problem consists of several components, some of which cannot be affected by the choice of control. In such cases the DP algorithm can be simplified considerably, and be executed over the controllable components of the state.

As an example, let the state of the system be a composite (x_k, y_k) of two components x_k and y_k. The evolution of the main component, x_k, is affected by the control u_k according to the equation

$$x_{k+1} = f_k(x_k, y_k, u_k, w_k),$$

where the distribution $P_k(w_k \mid x_k, y_k, u_k)$ is given. The evolution of the other component, y_k, is governed by a given conditional distribution $P_k(y_k \mid x_k)$ and cannot be affected by the control, except indirectly through x_k. One is tempted to view y_k as a disturbance, but there is a difference: y_k is observed by the controller before applying u_k, while w_k occurs after u_k is applied, and indeed w_k may probabilistically depend on u_k.

It turns out that we can formulate a DP algorithm that is executed over the controllable component of the state, with the dependence on the uncontrollable component being "averaged out" (see also the parking Example 1.6.1). In particular, let $J_k^*(x_k, y_k)$ denote the optimal cost-to-go at stage k and state (x_k, y_k), and define

$$\hat{J}_k(x_k) = E_{y_k}\big\{J_k^*(x_k, y_k) \mid x_k\big\}.$$

Note that the preceding expression can be interpreted as an "average cost-to-go" at x_k (averaged over the values of the uncontrollable component y_k). Then, similar to the parking Example 1.6.1, a DP algorithm that generates $\hat{J}_k(x_k)$ can be obtained, and has the following form:

$$\hat{J}_k(x_k) = E_{y_k}\Bigg\{\min_{u_k \in U_k(x_k, y_k)} E_{w_k}\Big\{g_k(x_k, y_k, u_k, w_k) + \hat{J}_{k+1}\big(f_k(x_k, y_k, u_k, w_k)\big) \mid x_k, y_k, u_k\Big\}\Big| x_k\Bigg\}. \tag{1.71}$$

This is a consequence of the calculation

$$\begin{aligned}
\hat{J}_k(x_k) &= E_{y_k}\big\{J_k^*(x_k, y_k) \mid x_k\big\} \\
&= E_{y_k}\Bigg\{\min_{u_k \in U_k(x_k, y_k)} E_{w_k, x_{k+1}, y_{k+1}}\big\{g_k(x_k, y_k, u_k, w_k) \\
&\qquad + J_{k+1}^*(x_{k+1}, y_{k+1}) \mid x_k, y_k, u_k\big\} \mid x_k\Bigg\} \\
&= E_{y_k}\Bigg\{\min_{u_k \in U_k(x_k, y_k)} E_{w_k, x_{k+1}}\Big\{g_k(x_k, y_k, u_k, w_k) \\
&\qquad + E_{y_{k+1}}\big\{J_{k+1}^*(x_{k+1}, y_{k+1}) \mid x_{k+1}\big\} \mid x_k, y_k, u_k\Big\} \mid x_k\Bigg\}.
\end{aligned}$$

Note that the minimization in the right-hand side of the preceding equation must still be performed for all values of the full state (x_k, y_k) in order to yield an optimal control law as a function of (x_k, y_k). Nonetheless, the equivalent DP algorithm (1.71) has the advantage that it is executed over a significantly reduced state space. Later, when we consider approximation in value space, we will find that it is often more convenient to approximate $\hat{J}_k(x_k)$ than to approximate $J_k^*(x_k, y_k)$; see the following discussions of forecasts and of the game of tetris.

As an example, consider the augmented state resulting from the incorporation of forecasts, as described earlier. Then, the forecast y_k represents an uncontrolled state component, so that the DP algorithm can be simplified as in Eq. (1.71). In particular, assume that the forecast y_k can take values $i = 1, \ldots, m$ with probability p_i. Then, by defining

$$\hat{J}_k(x_k) = \sum_{i=1}^{m} p_i J_k^*(x_k, i), \qquad k = 0, 1, \ldots, N-1,$$

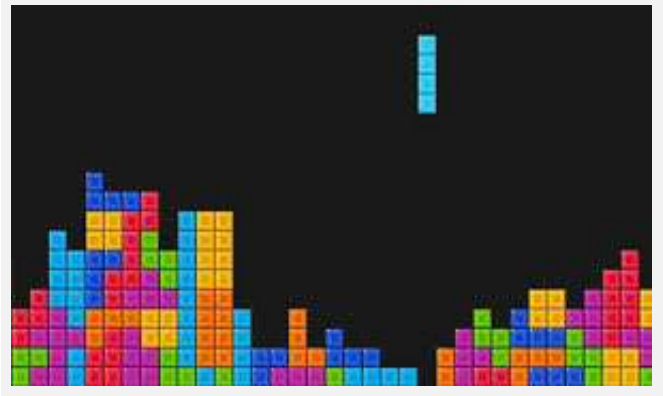


Figure 1.6.5 Illustration of a tetris board.

and $\hat{J}_N(x_N) = g_N(x_N)$, we have, using Eq. (1.70),

$$\hat{J}_k(x_k) = \sum_{i=1}^{m} p_i \min_{u_k \in U_k(x_k)} E_{w_k}\Big\{ g_k(x_k, u_k, w_k) + \hat{J}_{k+1}\big(f_k(x_k, u_k, w_k)\big) \mid y_k = i \Big\},$$

which is executed over the space of x_k rather than x_k *and* y_k. Note that this is a simpler algorithm to approximate than the one of Eq. (1.70).

Uncontrollable state components often occur in arrival systems, such as queueing, where action must be taken in response to a random event (such as a customer arrival) that cannot be influenced by the choice of control. Then the state of the arrival system must be augmented to include the random event, but the DP algorithm can be executed over a smaller space, as per Eq. (1.71). Here is an example of this type.

Example 1.6.3 (Tetris)

Tetris is a popular video game played on a two-dimensional grid. Each square in the grid can be full or empty, making up a "wall of bricks" with "holes" and a "jagged top" (see Fig. 1.6.5). The squares fill up as blocks of different shapes fall from the top of the grid and are added to the top of the wall. As a given block falls, the player can move horizontally and rotate the block in all possible ways, subject to the constraints imposed by the sides of the grid and the top of the wall. The falling blocks are generated independently according to some probability distribution, defined over a finite set of standard shapes. The game starts with an empty grid and ends when a square in the top row becomes full and the top of the wall reaches the top of the grid. When a row of full squares is created, this row is removed, the bricks lying above this row move one row downward, and the player scores a point. The player's objective is to maximize the score attained (total number of rows removed) up to termination of the game, whichever occurs first.

We can model the problem of finding an optimal tetris playing strategy as a finite horizon stochastic DP problem, with very long horizon. The state consists of two components:

(1) The board position, i.e., a binary description of the full/empty status of each square, denoted by x.

(2) The shape of the current falling block, denoted by y.

The control, denoted by u, is the horizontal positioning and rotation applied to the falling block. There is also an additional termination state which is cost-free. Once the state reaches the termination state, it stays there with no change in score. Moreover there is a very large amount added to the score when the end of the horizon is reached without the game having terminated.

The shape y is generated according to a probability distribution $p(y)$, independently of the control, so it can be viewed as an uncontrollable state component. The DP algorithm (1.71) is executed over the space of board positions x and has the intuitive form

$$\hat{J}_k(x) = \sum_y p(y) \max_u \Big[g(x,y,u) + \hat{J}_{k+1}\big(f(x,y,u)\big) \Big], \qquad \text{for all } x, \qquad (1.72)$$

where

$g(x,y,u)$ is the number of points scored (rows removed),

$f(x,y,u)$ is the next board position (or termination state),

when the state is (x,y) and control u is applied, respectively. The DP algorithm (1.72) assumes a finite horizon formulation of the problem.

Alternatively, we may consider an undiscounted infinite horizon formulation, involving a termination state (i.e., a stochastic shortest path problem). The "reduced" form of Bellman's equation, which corresponds to the DP algorithm (1.72), has the form

$$\hat{J}(x) = \sum_y p(y) \max_u \Big[g(x,y,u) + \hat{J}\big(f(x,y,u)\big) \Big], \qquad \text{for all } x.$$

The value $\hat{J}(x)$ can be interpreted as an "average score" at x (averaged over the values of the uncontrollable block shapes y).

Finally, let us note that despite the simplification achieved by eliminating the uncontrollable portion of the state, the number of states x is still enormous, and the problem can only be addressed by suboptimal methods.†

† Tetris is widely considered to be an interesting and challenging stochastic testbed for RL algorithms, and has received a lot of attention over a period spanning 20 years (1995-2015), starting with the paper [TsV96], the subsequent paper [BeI96] and the neuro-dynamic programming book [BeT96], and ending with the papers [GGS13], [SGG15], which contain many references to related works in the intervening years. All of these works are based on approximation in value space and various forms of approximate policy iteration.

1.6.6 Partial State Information and Belief States

We have assumed so far that the controller has access to the exact value of the current state x_k, so a policy consists of a sequence of functions of x_k. However, in many practical settings, this assumption is unrealistic because some components of the state may be inaccessible for observation, the sensors used for measuring them may be inaccurate, or the cost of measuring them more accurately may be prohibitive.

Often in such situations, the controller has access to only some of the components of the current state, and the corresponding observations may also be corrupted by stochastic uncertainty. For example in three-dimensional motion problems, the state may consist of the six-tuple of position and velocity components, but the observations may consist of noise-corrupted radar measurements of the three position components. This gives rise to problems of *partial* or *imperfect* state information, which have received a lot of attention in the optimization and AI literature (see e.g., [Ber17a], [RuN16]; these problems are also popularly referred to with the acronym POMDP for *partially observed Markovian Decision problem*).

Generally, solving a POMDP exactly is typically intractable, even though there are DP algorithms for doing so. Thus in practice, POMDP are solved approximately, except under very special circumstances.

Despite their inherent computational difficulty, it turns out that conceptually, partial state information problems are no different than the perfect state information problems we have been addressing so far. In fact by various reformulations, we can reduce a partial state information problem to one with perfect state information, which involves a different and more complicated state, called a *sufficient statistic*. Once this is done, we can state an exact DP algorithm that is defined over the space of the sufficient statistic. Roughly speaking, a sufficient statistic is a quantity that summarizes the content of the information available up to k for the purposes of optimal control. This statement can be made more precise, but we will not elaborate further in this book; see e.g., the DP textbook [Ber17a].

A common sufficient statistic is the *belief state*, which we will denote by b_k. It is the probability distribution of x_k given all the observations that have been obtained by the controller and all the controls applied by the controller up to time k, and it can serve as "state" in an appropriate DP algorithm. The belief state can in principle be computed and updated by a variety of methods that are based on Bayes' rule, such as *Kalman filtering* (see e.g., [AnM79], [KuV86], [Kri16], [ChC17]) and *particle filtering* (see e.g., [GSS93], [DoJ09], [Can16], [Kri16]).

Example 1.6.4 (Bidirectional Parking)

Let us consider a more complex version of the parking problem of Example 1.6.1. As in that example, a driver is looking for inexpensive parking on the

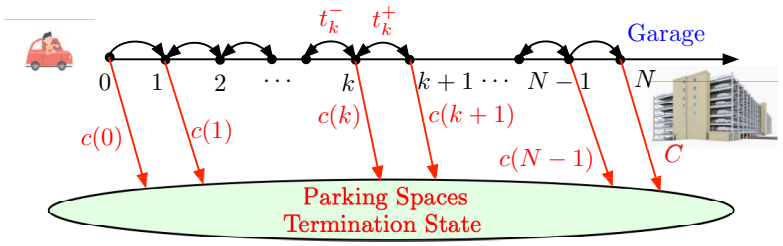


Figure 1.6.6 Cost structure and transitions of the bidirectional parking problem. The driver may park at space $k = 0, 1, \ldots, N-1$ at cost $c(k)$, if the space is free, can move to $k-1$ at cost t_k^- or can move to $k+1$ at cost t_k^+. At space N (the garage) the driver must park at cost C.

way to his destination, along a line of N parking spaces with a garage at the end. The difference is that the driver can move in either direction, rather than just forward towards the garage. In particular, at space i, the driver can park at cost $c(i)$ if i is free, can move to $i-1$ at a cost t_i^- or can move to $i+1$ at a cost t_i^+. Moreover, the driver records and remembers the free/taken status of the spaces previously visited and may return to any of these spaces; see Fig. 1.6.6.

We assume that the probability $p(i)$ of a space i being free changes over time, i.e., a space found free (or taken) at a given visit may get taken (or become free, respectively) by the time of the next visit. The initial probabilities $p(i)$, before visiting any spaces, are known, and the mechanism by which these probabilities change over time is also known to the driver. As an example, we may assume that at each time stage, $p(i)$ increases by a certain known factor with some probability ξ and decreases by another known factor with the complementary probability $1-\xi$.

Here the belief state is the vector of current probabilities

$$\big(p(0), \ldots, p(N-1)\big),$$

and it can be updated with a simple algorithm at each time based on the new observation: the free/taken status of the space visited at that time.

We can use the belief state as the basis of an exact DP algorithm for computing an optimal policy. This algorithm is typically intractable computationally, but it is conceptually useful, and it can form the starting point for approximations. It has the form

$$J_k^*(b_k) = \min_{u_k \in U_k} \Big[\hat{g}_k(b_k, u_k) + E_{z_{k+1}} \Big\{ J_{k+1}^*\big(F_k(b_k, u_k, z_{k+1})\big) \;\Big|\; b_k, u_k \Big\} \Big], \tag{1.73}$$

where:

$J_k^*(b_k)$ denotes the optimal cost-to-go starting from belief state b_k at stage k.

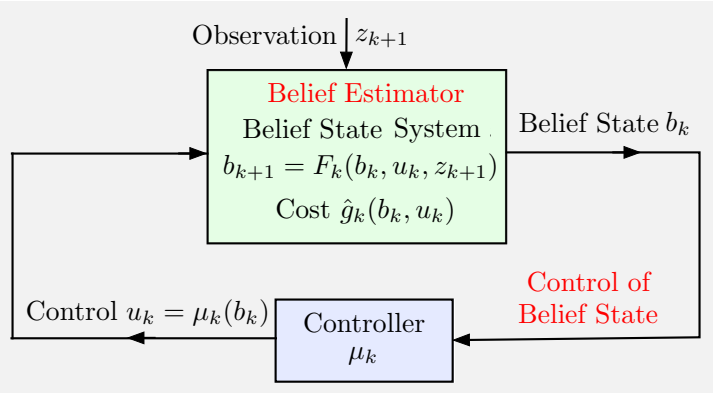


Figure 1.6.7 Schematic illustration of the view of an imperfect state information problem as one of perfect state information, whose state is the belief state b_k, i.e., the conditional probability distribution of x_k given all the observations up to time k. The observation z_{k+1} plays the role of the stochastic disturbance. The function F_k is a sequential estimator that updates the current belief state b_k.

U_k is the control constraint set at time k (since the state x_k is unknown at stage k, U_k must be independent of x_k).

$\hat{g}_k(b_k, u_k)$ denotes the expected stage cost of stage k. It is calculated as the expected value of the stage cost $g_k(x_k, u_k, w_k)$, with the joint distribution of (x_k, w_k) determined by the belief state b_k and the distribution of w_k.

$F_k(b_k, u_k, z_{k+1})$ denotes the belief state at the next stage, given that the current belief state is b_k, control u_k is applied, and observation z_{k+1} is received following the application of u_k:

$$b_{k+1} = F_k(b_k, u_k, z_{k+1}). \tag{1.74}$$

This is the system equation for a perfect state information problem with state b_k, control u_k, "disturbance" z_{k+1}, and cost per stage $\hat{g}_k(b_k, u_k)$. The function F_k is viewed as a sequential *belief estimator*, which updates the current belief state b_k based on the new observation z_{k+1}. It is given by either an explicit formula or an algorithm (such as Kalman filtering or particle filtering) that is based on the probability distribution of z_k and the use of Bayes' rule.

The expected value $E_{z_{k+1}}\{\cdot \mid b_k, u_k\}$ is taken with respect to the distribution of z_{k+1}, given b_k and u_k. Note that z_{k+1} is random, and its distribution depends on x_k and u_k, so the expected value

$$E_{z_{k+1}}\Big\{ J^*_{k+1}\big(F_k(b_k, u_k, z_{k+1})\big) \mid b_k, u_k \Big\}$$

in Eq. (1.73) is a function of b_k and u_k.

The algorithm (1.73) is just the ordinary DP algorithm for the perfect state information problem shown in Fig. 1.6.7. It involves the system/belief estimator (1.74) and the cost per stage $\hat{g}_k(b_k, u_k)$. Note that since b_k takes values in a continuous space, the algorithm (1.73) will typically require an approximate implementation, using approximation in value space methods.

We refer to the textbook [Ber17a], Chapter 4, for a detailed derivation of the DP algorithm (1.73), and to the monograph [BeS78] for a mathematical treatment that applies to infinite-dimensional state and disturbance spaces as well.

An Alternative DP Algorithm for POMDP

The DP algorithm (1.73) is not the only one that can be used for POMDP. There is also an exact DP algorithm that operates in the space of information vectors I_k, defined by

$$I_k = \{z_0, u_0, \ldots, z_{k-1}, u_{k-1}, z_k\},$$

where z_k is the observation received at time k. This is another sufficient statistic, and hence an alternative to the belief state b_k. In particular, we can view I_k as a state of the POMDP, which evolves over time according to the equation

$$I_{k+1} = (I_k, z_{k+1}, u_k).$$

Denoting by $J_k^*(I_k)$ the optimal cost starting at information vector I_k at time k, the DP algorithm takes the form

$$J_k^*(I_k) = \min_{u_k \in U_k(x_k)} E_{w_k, z_{k+1}} \Big\{ g_k(x_k, u_k, w_k) + J_{k+1}^*\big(I_k, z_{k+1}, u_k\big) \mid I_k, u_k \Big\}, \tag{1.75}$$

for $k = 0, \ldots, N-1$, with $J_N^*(I_N) = E\big\{g_N(x_N) \mid I_N\big\}$; see e.g., the DP textbook [Ber17a], Section 4.1.

A drawback of the preceding approach is that the information vector I_k is growing in size over time, thereby leading to a nonstationary system even in the case of an infinite horizon problem with a stationary system and cost function. This difficulty can be remedied in an approximation scheme that uses a finite history of the system (a fixed number of most recent observations) as state, thereby working effectively with a stationary finite-state system; see the paper by White and Scherrer [WhS94]. In particular, this approach is used in large language models such as ChatGPT and DeepSeek.

Finite-memory approximations for POMDP can be viewed within the context of feature-based approximation architectures, as we will discuss in Chapter 3 (see Example 3.1.6). Moreover, the finite-history scheme can be

generalized through the concept of a *finite-state controller*; see the paper by Yu and Bertsekas [YuB08], which also addresses the issue of convergence of the approximation error to zero as the size of the finite-history or finite-state controller is increased.

1.6.7 Multiagent Problems and Multiagent Rollout

In this book, we will view a multiagent system as a collection of decision making entities, called *agents*, which aim to optimally achieve a common goal.† The agents accomplish this by collecting and exchanging information, and otherwise interacting with each other. The agents can be software programs or physical entities such as robots, and they may have different capabilities.

Among the generic challenges of efficient implementation of multiagent systems, one may note issues of limited communication and lack of fully shared information, due to factors such as limited bandwidth, noisy channels, and lack of synchronization. Another important generic issue is that as the number of agents increases, the size of the set of possible joint decisions of the agents increases exponentially, thereby complicating control selection by lookahead minimization. In this section, we will focus on ways to resolve this latter difficulty for problems where the agents fully share information, and in Section 2.9 we will address some of the challenges of problems where the agents may have some autonomy, and act without fully coordinating with each other.

For a mathematical formulation, let us consider the discounted infinite horizon problem and a special structure of the control space, whereby the control u consists of m components, $u = (u^1, \ldots, u^m)$, with a separable control constraint structure $u^\ell \in U^\ell(x)$, $\ell = 1, \ldots, m$. Thus the control constraint set is the Cartesian product

$$U(x) = U^1(x) \times \cdots \times U^m(x), \tag{1.76}$$

where the sets $U^\ell(x)$ are given. This structure arises in applications involving distributed decision making by multiple agents; see Fig. 1.6.8.

In particular, we will view each component u^ℓ, $\ell = 1, \ldots, m$, as being chosen from within $U^\ell(x)$ by a separate "agent" (a decision making entity). For the sake of the following discussion, we assume that each set $U^\ell(x)$ is finite. Then the one-step lookahead minimization of the standard rollout scheme with base policy μ is given by

$$\tilde{u} \in \arg\min_{u \in U(x)} E_w\Big\{g(x,u,w) + \alpha J_\mu\big(f(x,u,w)\big)\Big\}, \tag{1.77}$$

† In a more general version of a multiagent system, which is outside our scope, the agents may have different goals, and act in their own self-interest.

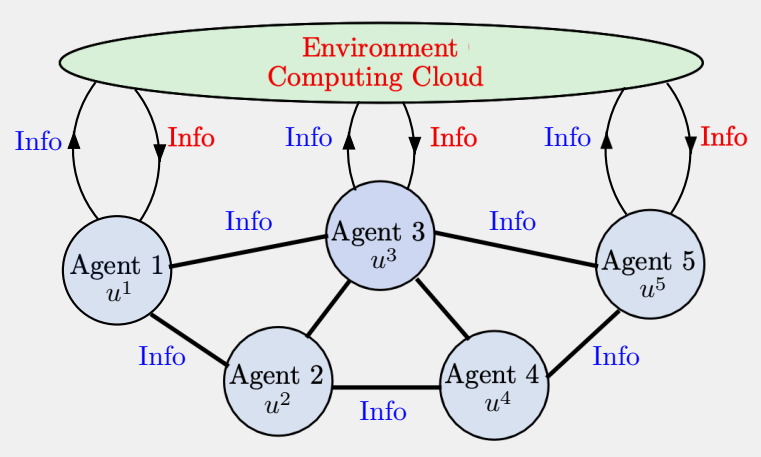


Figure 1.6.8 Schematic illustration of a multiagent problem. There are multiple "agents," and each agent $\ell = 1, \ldots, m$ controls its own decision variable u^ℓ. At each stage, agents exchange new information and also exchange information with the "environment," and then select their decision variables for the stage.

and involves as many as n^m Q-factors, where n is the maximum number of elements of the sets $U^\ell(x)$ [so that n^m is an upper bound to the number of controls in $U(x)$, in view of its Cartesian product structure (1.76)]. Thus the standard rollout algorithm requires an exponential [order $O(n^m)$] number of Q-factor computations per stage, which can be overwhelming even for moderate values of m.

This potentially large computational overhead motivates a far more computationally efficient rollout algorithm, whereby the one-step lookahead minimization (1.77) is replaced by a sequence of m successive minimizations, *one-agent-at-a-time*, with the results incorporated into the subsequent minimizations. In particular, given a base policy $\mu = (\mu^1, \ldots, \mu^m)$, we perform at state x the sequence of minimizations

$$\tilde{\mu}^1(x) \in \arg\min_{u^1 \in U^1(x)} E_w\Big\{g\big(x, u^1, \mu^2(x), \ldots, \mu^m(x), w\big) + \alpha J_\mu\big(f(x, u^1, \mu^2(x), \ldots, \mu^m(x), w)\big)\Big\},$$

$$\tilde{\mu}^2(x) \in \arg\min_{u^2 \in U^2(x)} E_w\Big\{g\big(x, \tilde{\mu}^1(x), u^2, \mu^3(x) \ldots, \mu^m(x), w\big) + \alpha J_\mu\big(f(x, \tilde{\mu}^1(x), u^2, \mu^3(x), \ldots, \mu^m(x), w)\big)\Big\},$$

$$\cdots \qquad \cdots \qquad \cdots \qquad \cdots$$

$$\tilde{\mu}^m(x) \in \arg\min_{u^m \in U^m(x)} E_w\Big\{g\big(x, \tilde{\mu}^1(x), \tilde{\mu}^2(x), \ldots, \tilde{\mu}^{m-1}(x), u^m, w\big) + \alpha J_\mu\big(f(x, \tilde{\mu}^1(x), \tilde{\mu}^2(x), \ldots, \tilde{\mu}^{m-1}(x), u^m, w)\big)\Big\}.$$

Thus *each agent component u^ℓ is obtained by a minimization with the preceding agent components $u^1, \ldots, u^{\ell-1}$ fixed at the previously computed values of the rollout policy, and the following agent components $u^{\ell+1}, \ldots, u^m$ fixed at the values given by the base policy*. This algorithm requires order $O(nm)$ Q-factor computations per stage, a potentially huge computational saving over the order $O(n^m)$ computations required by standard rollout.

A key idea here is that the computational requirements of the rollout one-step minimization (1.77) are proportional to the number of controls in the set $U(x_k)$ and are independent of the size of the state space. This motivates a reformulation of the problem, first suggested in the neurodynamic programming book [BeT96], Section 6.1.4, whereby *control space complexity is traded off with state space complexity*, by "unfolding" the control u_k into its m components, which are applied *one agent-at-a-time* rather than all-agents-at-once.

In particular, we can reformulate the problem by breaking down the collective decision u_k into m individual component decisions, thereby reducing the complexity of the control space while increasing the complexity of the state space. The potential advantage is that *the extra state space complexity does not affect the computational requirements of some RL algorithms, including rollout*.

To this end, we introduce a modified but equivalent problem, involving one-at-a-time agent control selection. At a state x, we break down the control u into the sequence of the m controls $u^1, u^2, \ldots, u^m$, and between x and the next state $\bar{x} = f(x, u, w)$, we introduce artificial intermediate "states" $(x, u^1), (x, u^1, u^2), \ldots, (x, u^1, \ldots, u^{m-1})$, and corresponding transitions. The choice of the last control component u^m at "state" $(x, u^1, \ldots, u^{m-1})$ marks the transition to the next state $\bar{x} = f(x, u, w)$, while incurring cost $g(x, u, w)$; see Fig. 1.6.9.

It is evident that this reformulated problem is equivalent to the original, since any control choice that is possible in one problem is also possible in the other problem, while the cost structure of the two problems is the same. In particular, every policy $\mu = (\mu^1, \ldots, \mu^m)$ of the original problem, including a base policy in the context of rollout, is admissible for the reformulated problem, and has the same cost function for the original as well as the reformulated problem.

The motivation for the reformulated problem is that the control space is simplified at the expense of introducing $m-1$ additional layers of states, and the corresponding $m-1$ cost-to-go functions

$$J^1(x, u^1), J^2(x, u^1, u^2), \ldots, J^{m-1}(x, u^1, \ldots, u^{m-1}).$$

The increase in size of the state space does not adversely affect the operation of rollout, since the Q-factor minimization (1.77) is performed for just one state at each stage.

The major fact that can be proved about multiagent rollout (see Section 2.9 and the end-of-chapter references) is that it *achieves cost improve-*

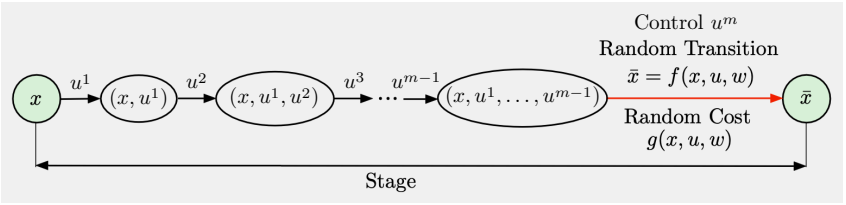


Figure 1.6.9 Equivalent formulation of the stochastic optimal control problem for the case where the control u consists of m components $u^1, u^2, \ldots, u^m$:

$$u = (u^1, \ldots, u^m) \in U(x) = U^1(x) \times \cdots \times U^m(x).$$

The figure depicts the kth stage transitions. Starting from state x, we generate the intermediate states

$$(x, u^1), (x, u^1, u^2), \ldots, (x, u^1, \ldots, u^{m-1}),$$

using the respective controls $u^1, \ldots, u^{m-1}$. The final control u^m leads from $(x, u^1, \ldots, u^{m-1})$ to $\bar{x} = f(x, u, w)$, and the random cost $g(x, u, w)$ is incurred.

ment:

$$J_{\tilde{\mu}}(x) \leq J_\mu(x), \qquad \text{for all } x,$$

where $J_\mu(x)$ is the cost function of the base policy $\mu = (\mu^1, \ldots, \mu^m)$, and $J_{\tilde{\mu}}(x)$ is the cost function of the rollout policy $\tilde{\mu} = (\tilde{\mu}^1, \ldots, \tilde{\mu}^m)$, starting from state x. Furthermore, this cost improvement property can be extended to multiagent PI schemes that involve one-agent-at-a-time policy improvement operations, and have sound convergence properties. Moreover, multiagent rollout becomes the starting point for related PI schemes that are well suited for distributed operation in contexts involving multiple autonomous decision makers; see Section 2.9, the book [Ber20a], the papers [Ber20b] and [BKB23], and the tutorial survey [Ber21a].

Example 1.6.5 (Spiders and Flies)

This example is representative of a broad range of practical problems such as multirobot service systems involving delivery, maintenance and repair, search and rescue, firefighting, etc. Here there are m spiders and several flies moving on a 2-dimensional grid; cf. Fig. 1.6.10. The objective is for the spiders to catch all the flies as fast as possible.

During a stage, each fly moves to some other position according to a given state-dependent probability distribution. Each spider learns the current state (the vector of spiders and fly locations) at the beginning of each stage, and either moves to a neighboring location or stays where it is. Thus each spider has as many as 5 choices at each stage. The control is $u = (u^1, \ldots, u^m)$, where u^ℓ is the choice of the ℓth spider, so there are about 5^m possible values of u.

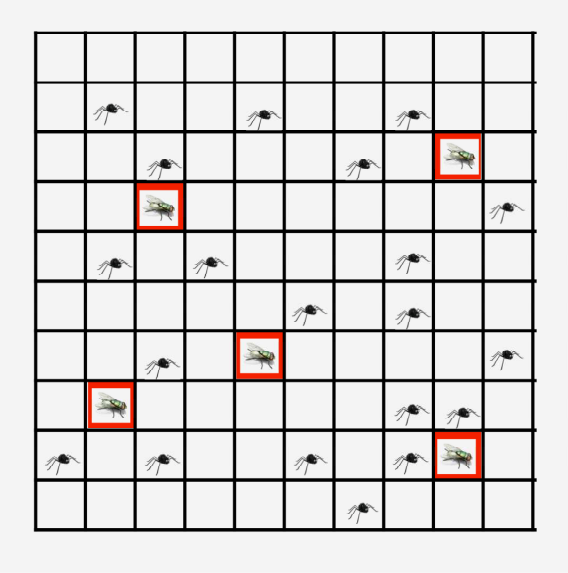


Figure 1.6.10 Illustration of a 2-dimensional spiders-and-fly problem with 20 spiders and 5 flies (cf. Example 1.6.5). The flies moves randomly, regardless of the position of the spiders. During a stage, each spider moves to a neighboring location or stays where it is, so there are 5 moves per spider (except for spiders at the edges of the grid). The total number of possible joint spiders moves is a little less than 5^{20}.

To apply multiagent rollout, we need a base policy. A simple possibility is to use the policy that directs each spider to move on the path of minimum distance to the closest fly position. According to the multiagent rollout formalism, the spiders choose their moves one-at-time in the order from 1 to m, taking into account the current positions of the flies and the earlier moves of other spiders, and assuming that future moves will be chosen according to the base policy, which is a tractable computation.

In particular, at the beginning at the typical stage, spider 1 selects its best move (out of the no more than 5 possible moves), assuming the other spiders $2, \ldots, m$ will move towards their closest surviving fly during the current stage, and all spiders will move towards their closest surviving fly during the following stages, up to the time where no surviving flies remain. Spider 1 then broadcasts its selected move to all other spiders. Then spider 2 selects its move taking into account the move already chosen by spider 1, and assuming that spiders $3, \ldots, m$ will move towards their closest surviving fly during the current stage, and all spiders will move towards their closest surviving fly during the following stages, up to the time where no surviving flies remain. Spider 2 then broadcasts its choice to all other spiders. This process of one-spider-at-a-time move selection is repeated for the remaining spiders $3, \ldots, m$, marking the end of the stage.

Note that while standard rollout computes and compares 5^m Q-factors (actually a little less to take into account edge effects), multiagent rollout

computes and compares ≤ 5 moves per spider, for a total of less than $5m$. Despite this tremendous computational economy, experiments with this type of spiders and flies problems have shown that multiagent rollout achieves a comparable performance to the one of standard rollout.

1.6.8 Problems with Unknown or Changing Model - Adaptive Control

Our discussion so far dealt with problems with a known mathematical model, i.e., one where the system equation, cost function, control constraints, and probability distributions of disturbances are perfectly known. The mathematical model may be available through explicit mathematical formulas and assumptions, or through computer software, such as a trained neural network, or a simulator that can emulate mathematical operations involved in the model or Monte Carlo simulation for the calculation of expected values.

From our point of view, *it makes no difference whether the mathematical model is available through closed form mathematical expressions or through a computer model, such as a simulator or a neural network*: the methods that we discuss are valid in both cases, only their suitability for a given problem may be affected by the type of model.

In practice, however, it is common that the system parameters are either not known exactly or can change over time, and this introduces potentially enormous complications.† As an example consider our oversimplified cruise control system that we noted in Example 1.3.1 or its infinite horizon version. The state evolves according to

$$x_{k+1} = x_k + bu_k + w_k, \tag{1.78}$$

where x_k is the deviation $v_k - \bar{v}$ of the vehicle's velocity v_k from the nominal $\bar{v}$, u_k is the force that propels the car forward, and w_k is the disturbance. However, the coefficient b and the distribution of w_k change frequently, and cannot be modeled with any precision because they depend on unpredictable time-varying conditions, such as the slope and condition of the road, and the weight of the car (which is affected by the number of passengers). Moreover, the nominal velocity $\bar{v}$ is set by the driver, and when

† The difficulties of decision and control within a changing environment are often underestimated. Among others, they complicate the balance between offline training and on-line play, which we discussed in Section 1.1 in connection with AlphaZero. It is worth keeping in mind that as much as learning to play high quality chess is a great challenge, the rules of play are stable and do not change unpredictably in the middle of a game! Problems with changing system parameters can be far more challenging!

it changes it may affect the parameter b in the system equation, and other parameters.†

In this section, we will briefly review some of the most commonly used approaches for dealing with unknown models and/or parameters in optimal control theory and practice. We should note also that unknown problem environments are an integral part of the AI view of RL. In particular, to quote from the popular book by Sutton and Barto [SuB18], RL is viewed as "a computational approach to learning from interaction," and "learning from interaction with the environment is a foundational idea underlying nearly all theories of learning and intelligence."

The idea of learning from interaction with the environment is often connected with the idea of exploring the environment to identify its characteristics. In control theory this is often viewed as part of the *system identification* methodology, which aims to construct mathematical models of dynamic systems. The system identification process is often combined with the control process to deal with unknown or changing problem parameters, in a framework that is sometimes called *dual control*. This is one of the most challenging areas of stochastic optimal and suboptimal control, and has been studied intensively since the early 1960s.

Let us also note that modern advances in neural network and AI technologies have expanded significantly the range of the system identification methodology. As an example, we refer to Section 2.3.8, which provides a brief discussion of *world models*. These can be used as a computational surrogate for an exact model to provide predictions of the system's trajectories beyond the first step of lookahead minimization, in contexts of approximation in value space and rollout.

Robust and Adaptive Control

Given a controller design that has been obtained assuming a nominal DP problem model, one possibility is to simply ignore changes in problem parameters. We may then try to investigate the performance of the current design for a suitable range of problem parameter values, and ensure that it is adequate for the entire range. This is sometimes called a *robust controller design*. For example, consider the oversimplified cruise control system of Eq. (1.78) with a linear controller of the form $\mu(x) = Lx$ for some scalar L. Then we check the range of parameters b for which the current controller is stable (this is the interval of values b for which $|1 + bL| < 1$), and ensure that b remains within that range during the system's operation.

The more general class of methods where the controller is modified in response to problem parameter changes is part of a broad field known as

† Adaptive cruise control, which can also adapt the car's velocity based on its proximity to other cars, has been studied extensively and has been incorporated in several commercially sold car models.

adaptive control, i.e., control that adapts to changing parameters. This is a rich methodology that we will discuss only briefly in this book. For the moment, let us just mention a simple time-honored adaptive control approach for a broad class of continuous-state problems without a known mathematical or computer model, called *PID (Proportional-Integral-Derivative) control*, for which we refer to the control literature, including the books by Aström and Hagglund [AsH95], [AsH06], and the end-of-chapter references on adaptive control (also Section 5.7 of the RL textbook [Ber19a]).

In particular, PID control aims to maintain the output of a single-input single-output dynamic system around a set point or to follow a given trajectory, as the system parameters change within a relatively broad range. In its simplest form, the PID controller is parametrized by three scalar parameters, which may be determined by a variety of methods, some of them manual/heuristic. PID control is used widely and with success, although its range of application is mainly restricted to single-input, single-output continuous-state control systems.

Dealing with Unknown Parameters Through System Identification

In PID control, no attempt is made to maintain a mathematical model and to track unknown model parameters as they change. An alternative and apparently reasonable form of suboptimal control is to separate the control process into two phases, a *system identification phase* and a *control phase*. In the first phase the unknown parameters are estimated, while the control takes no account of the interim results of estimation. The final parameter estimates from the first phase are then used to implement an optimal or suboptimal policy in the second phase. This alternation of estimation and control phases may be repeated several times during any system run in order to take into account subsequent changes of the parameters. Moreover, it is not necessary to introduce a hard separation between the identification and the control phases. They may be going on simultaneously, with new parameter estimates being introduced into the control process, whenever this is thought to be desirable; see Fig. 1.6.11.

One drawback of this approach is that it is not always easy to determine when to terminate one phase and start the other. A second difficulty, of a more fundamental nature, is that the control process may make some of the unknown parameters invisible to the estimation process. This is known as the problem of *parameter identifiability*, which is discussed in the context of optimal control in several sources, including [BoV79] and [Kum83]; see also [Ber17a], Section 6.7.

Example 1.6.6 (Parameter Identifiability Under Closed-Loop Control)

For a simple example, consider the scalar system

$$x_{k+1} = ax_k + bu_k, \qquad k = 0, \ldots, N-1,$$

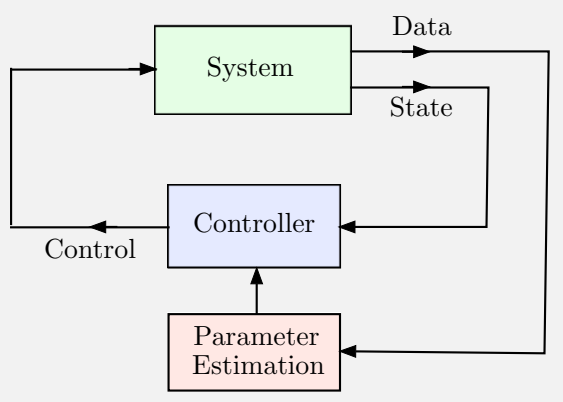


Figure 1.6.11 Schematic illustration of concurrent parameter estimation and system control. The system parameters are estimated on-line and the estimates are periodically passed on to the controller.

and the quadratic cost

$$\sum_{k=1}^{N}(x_k)^2.$$

Assuming perfect state information, if the parameters a and b are known, it can be seen that the optimal control law is

$$\mu_k^*(x_k) = -\frac{a}{b}x_k,$$

which sets all future states to 0. Assume now that the parameters a and b are unknown, and consider the two-phase method. During the first phase the control law

$$\tilde{\mu}_k(x_k) = \gamma x_k \tag{1.79}$$

is used (γ is some scalar; for example, $\gamma = -\frac{\overline{a}}{\overline{b}}$, where $\overline{a}$ and $\overline{b}$ are some a priori estimates of a and b, respectively). At the end of the first phase, the control law is changed to

$$\overline{\mu}_k(x_k) = -\frac{\hat{a}}{\hat{b}}x_k,$$

where $\hat{a}$ and $\hat{b}$ are the estimates obtained from the estimation process. However, with the control law (1.79), the closed-loop system is

$$x_{k+1} = (a + b\gamma)x_k,$$

so the estimation process can at best yield the value of $(a + b\gamma)$ but not the values of both a and b. In other words, the estimation process cannot discriminate between pairs of values (a_1, b_1) and (a_2, b_2) such that

$$a_1 + b_1\gamma = a_2 + b_2\gamma.$$

Therefore, a and b are not identifiable when feedback control of the form (1.79) is applied.

On-line parameter estimation algorithms, which address among others the issue of identifiability, have been discussed extensively in the control theory literature, but the corresponding methodology is complex and beyond our scope in this book. However, assuming that we can make the estimation phase work somehow, we are free to revise the controller using the newly estimated parameters in a variety of ways, in an on-line replanning process.

Unfortunately, there is still another difficulty with this type of on-line replanning: it may be hard to recompute an optimal or near-optimal policy on-line, using a newly identified system model. In particular, it may be impossible to use time-consuming methods that involve for example the training of a neural network or discrete/integer control constraints. A simpler possibility is to use rollout, which we discuss next.†

Adaptive Control by Rollout and On-Line Replanning

We will now consider an approach for dealing with unknown or changing parameters, which is based on on-line replanning. We have discussed this approach in the context of rollout and multiagent rollout, where we stressed the importance of fast on-line policy improvement.

Let us assume that some problem parameters change and the current controller becomes aware of the change "instantly" (i.e., very quickly before the next stage begins). The method by which the problem parameters are recalculated or become known is immaterial for the purposes of the following discussion. It may involve a limited form of parameter estimation, whereby the unknown parameters are "tracked" by data collection over a few time stages, with due attention paid to issues of parameter identifiability; or it may involve new features of the control environment, such as a changing number of servers and/or tasks in a service system (think of new spiders and/or flies appearing or disappearing unexpectedly in the spiders-and-flies Example 1.6.5).

† Another possibility is to deal with this difficulty by precomputation. In particular, assume that the set of problem parameters may take a known finite set of values (for example each set of parameter values may correspond to a distinct maneuver of a vehicle, motion of a robotic arm, flying regime of an aircraft, etc). Then we may precompute a separate controller for each of these values. Once the control scheme detects a change in problem parameters, it switches to the corresponding predesigned current controller. This is sometimes called a *multiple model control design* or *gain scheduling*, and has a long history of success in various settings over the years; see e.g., Athans et al., [ACD77].

We thus assume away/ignore issues of parameter estimation, and focus on revising the controller by on-line replanning based on the newly obtained parameters. This revision may be based on any suboptimal method, but rollout with the current policy used as the base policy is particularly attractive. Here the advantage of rollout is that it is simple and reliable. In particular, it does not require a complicated training procedure to revise the current policy, based for example on the use of neural networks or other approximation architectures, so *no new policy is explicitly computed in response to the parameter changes*. Instead the current policy is used as the base policy for rollout, and the available controls at the current state are compared by a one-step or mutistep minimization, with cost function approximation provided by the base policy (cf. Fig. 1.6.12).

Note that *over time the base policy may also be revised* (on the basis of an unspecified rationale), in which case the rollout policy will be revised both in response to the changed current policy and in response to the changing parameters. This is necessary in particular when the constraints of the problem change.

The principal requirement for using rollout in an adaptive control context is that the rollout control computation should be fast enough to be performed between stages. Note, however, that accelerated/truncated versions of rollout, as well as parallel computation, can be used to meet this time constraint.

The following example considers on-line replanning with the use of rollout in the context of the simple one-dimensional linear quadratic problem that we discussed earlier in this chapter. The purpose of the example is to illustrate analytically how rollout with a policy that is optimal for a nominal set of problem parameters works well when the parameters change from their nominal values. This property is not practically useful in linear quadratic problems because when the parameter change, it is possible to calculate the new optimal policy in closed form, but it is indicative of the performance robustness of rollout in other contexts. Generally, adaptive control by rollout and on-line replanning makes sense in situations where the calculation of the rollout controls for a given set of problem parameters is faster and/or more convenient than the calculation of the optimal controls for the same set of parameter values. These problems include cases involving nonlinear systems and/or difficult (e.g., integer) constraints.

Example 1.6.7 (On-Line Replanning for Linear Quadratic Problems Based on Rollout)

Consider the deterministic undiscounted infinite horizon linear quadratic problem. It involves the linear system

$$x_{k+1} = x_k + bu_k,$$

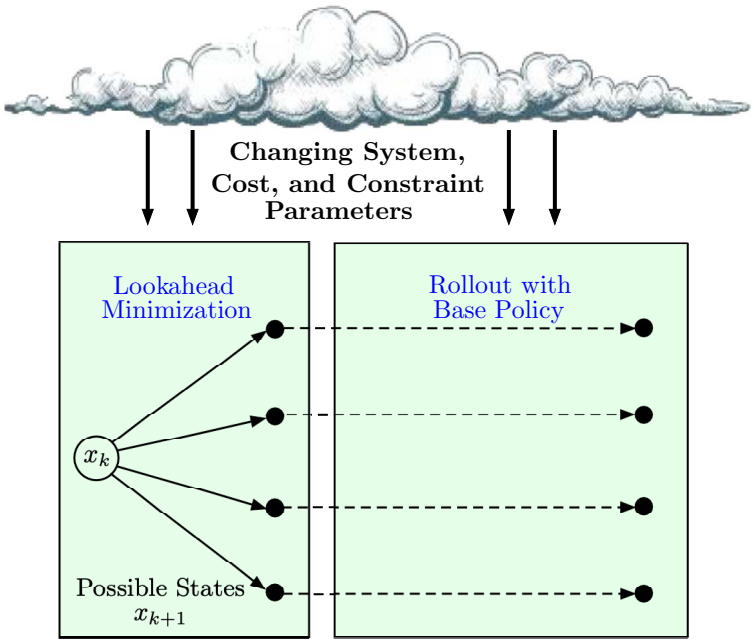


Figure 1.6.12 Schematic illustration of adaptive control by rollout. One-step lookahead is followed by simulation with the base policy, which stays fixed. The system, cost, and constraint parameters are changing over time, and the most recent values are incorporated into the lookahead minimization and rollout operations. For the discussion in this section, we may assume that all the changing parameter information is provided by some computation and sensor "cloud" that is beyond our control. The base policy may also be revised based on various criteria.

and the quadratic cost function

$$\lim_{N\to\infty}\sum_{k=0}^{N-1}(x_k^2+ru_k^2).$$

The optimal cost function is given by

$$J^*(x)=K^*x^2,$$

where K^* is the unique positive solution of the Riccati equation

$$K=\frac{rK}{r+b^2K}+1. \tag{1.80}$$

The optimal policy has the form

$$\mu^*(x)=L^*x, \tag{1.81}$$

where

$$L^* = -\frac{bK^*}{r + b^2K^*}. \tag{1.82}$$

As an example, consider the optimal policy that corresponds to the nominal problem parameters $b = 2$ and $r = 0.5$: this is the policy (1.81)-(1.82), with K obtained as the positive solution of the quadratic Riccati Eq. (1.80) for $b = 2$ and $r = 0.5$. In particular, we can verify that

$$K = \frac{2 + \sqrt{6}}{4}.$$

From Eq. (1.82) we then obtain

$$L = -\frac{2 + \sqrt{6}}{5 + 2\sqrt{6}}. \tag{1.83}$$

We will now consider changes of the values of b and r while keeping L constant, and we will compare the quadratic cost coefficient of the following three cost functions as b and r vary:

(a) The optimal cost function K^*x^2, where K^* is given by the positive solution of the Riccati Eq. (1.80).

(b) The cost function K_Lx^2 that corresponds to the base policy

$$\mu_L(x) = Lx,$$

where L is given by Eq. (1.83). From our earlier discussion, we have

$$K_L = \frac{1 + rL^2}{1 - (1 + bL)^2}.$$

(c) The cost function $\tilde{K}_Lx^2$ that corresponds to the rollout policy

$$\tilde{\mu}_L(x) = \tilde{L}x,$$

obtained by using the policy μ_L as base policy. Using the formulas given earlier, we have

$$\tilde{L} = -\frac{bK_L}{r + b^2K_L},$$

and

$$\tilde{K}_L = \frac{1 + r\tilde{L}^2}{1 - (1 + b\tilde{L})^2}.$$

Figure 1.6.13 shows the coefficients K^*, K_L, and $\tilde{K}_L$ for a range of values of r and b. We have

$$K^* \leq \tilde{K}_L \leq K_L.$$

The difference $K_L - K^*$ is indicative of the robustness of the policy μ_L, i.e., the performance loss incurred by ignoring the values of b and r, and continuing to use the policy μ_L, which is optimal for the nominal values $b = 2$ and $r = 0.5$, but suboptimal for other values of b and r. The difference $\tilde{K}_L - K^*$ is indicative of the performance loss due to using on-line replanning by rollout rather than using optimal replanning. Finally, the difference $K_L - \tilde{K}_L$ is indicative of the performance improvement due to on-line replanning using rollout rather than keeping the policy μ_L unchanged.

Note that Fig. 1.6.13 illustrates the behavior of the error ratio

$$\frac{\tilde{J} - J^*}{J - J^*},$$

where for a given initial state, $\tilde{J}$ is the rollout performance, J^* is the optimal performance, and J is the base policy performance. This ratio approaches 0 as $J - J^*$ becomes smaller because of the quadratic convergence rate of Newton's method that underlies the rollout algorithm.

Adaptive Control as POMDP

The preceding adaptive control formulation strictly separates the dual objective of estimation and control: first parameter identification and then controller reoptimization (either exact or rollout-based). In an alternative adaptive control formulation, the parameter estimation and the application of control are done simultaneously, and indeed part of the control effort may be directed towards improving the quality of future estimation. This alternative (and more principled) approach is based on a view of adaptive control as a partially observed Markovian decision problem (POMDP) with a special structure. We will see in Section 2.11 that this approach is well-suited for approximation in value space schemes, including forms of rollout.

To describe briefly the adaptive control reformulation as POMDP, we introduce a system whose state consists of two components:

(a) A perfectly observed component x_k that evolves over time according to a discrete-time equation.

(b) A component θ which is unobserved but stays constant, and is estimated through the perfect observations of the component x_k.

We view θ as a parameter in the system equation that governs the evolution of x_k. Thus we have

$$x_{k+1} = f_k(x_k, \theta, u_k, w_k), \tag{1.84}$$

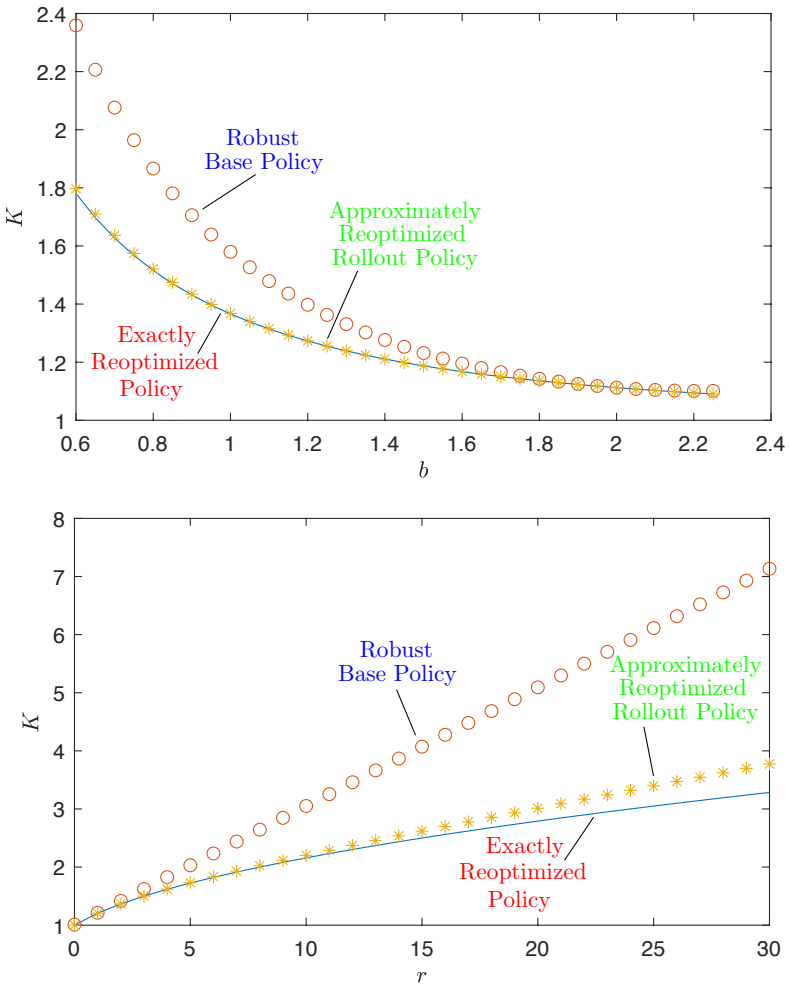


Figure 1.6.13 Illustration of adaptive control by rollout under changing problem parameters. The quadratic cost coefficients K^* (optimal, denoted by solid line), K_L (base policy, denoted by circles), and $\tilde{K}_L$ (rollout policy, denoted by asterisks) for the two cases where $r = 0.5$ and b varies, and $b = 2$ and r varies. The value of L is fixed at the value that is optimal for $b = 2$ and $r = 0.5$ [cf. Eq. (1.83)].

The rollout policy performance is very close to the one of the exactly reoptimized policy, while the base policy yields much worse performance. This is a consequence of the quadratic convergence rate of Newton's method that underlies rollout:

$$\lim_{J \to J^*} \frac{\tilde{J} - J^*}{J - J^*} = 0,$$

where for a given initial state, $\tilde{J}$ is the rollout performance, J^* is the optimal performance, and J is the base policy performance.

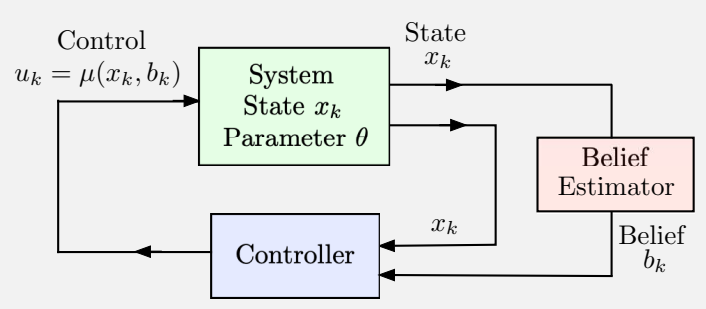


Figure 1.6.14 Schematic illustration of simultaneous control and belief estimation for the unknown system parameter θ. The control applied is a function of the current belief state (x_k, b_k), where b_k is the conditional probability distribution of θ given the observations accumulated up to time k (the current and past states $x_k, \ldots, x_0$, and the past controls $u_{k-1}, \ldots, u_0$).

where u_k is the control at time k, selected from a set $U_k(x_k)$, and w_k is a random disturbance with given probability distribution that depends on (x_k, θ, u_k). For convenience, we will assume that θ can take one of m known values $\theta^1, \ldots, \theta^m$.

The a priori probability distribution of θ is given and is updated based on the observed values of the state components x_k and the applied controls u_k. In particular, the information vector

$$I_k = \{x_0, \ldots, x_k, u_0, \ldots, u_{k-1}\}$$

is available at time k, and is used to compute the conditional probabilities

$$b_{k,i} = P\{\theta = \theta^i \mid I_k\}, \qquad i = 1, \ldots, m.$$

These probabilities form a vector

$$b_k = (b_{k,1}, \ldots, b_{k,m}),$$

which together with the perfectly observed state x_k, form the pair (x_k, b_k), which is the *belief state* of the POMDP at time k. The overall control scheme takes the form illustrated in Fig. 1.6.14.

As discussed in Section 1.6.6, an exact DP algorithm can be written for the equivalent POMDP, and this algorithm is suitable for the use of approximation in value space and rollout. We will describe this approach in some detail in Section 2.11. Related ideas will also be discussed in the context of Bayesian estimation and sequential estimation in Section 2.10.

Note that the case of a deterministic system

$$x_{k+1} = f_k(x_k, \theta, u_k),$$

is particularly interesting, because we can then typically expect that the true parameter θ^* will be identified in a finite number of stages. The reason is that at each stage k, we are receiving a noiseless observation relating to θ, namely the state x_k. Once the true parameter θ^* is identified, the problem becomes one of perfect state information.

1.6.9 Model Predictive Control

In this section, we will provide a brief summary of the model predictive control (MPC) methodology for control system design, with a view towards its connection with DP/RL, approximation in value space, and rollout schemes.† We will focus on the classical control problem: keeping the state of a deterministic system close to some reference point, taken here to be the origin of the state space (see Fig. 1.6.15). Another type of classical control problem is to keep the system close to a given trajectory (see Fig. 1.6.16). It can also be treated by forms of MPC, but will not be addressed in this book.

We discussed earlier the linear quadratic approach, whereby the system is represented by a linear model, the cost is quadratic in the state and the control, and there are no state and control constraints. The linear quadratic and other approaches based on state variable system representations and optimal control became popular, starting in the late 50s and early 60s. Unfortunately, however, the analytically convenient linear quadratic problem formulations are often not satisfactory. There are two main reasons for this:

(a) The system may be nonlinear, and it may be inappropriate to use for control purposes a model that is linearized around the desired point or trajectory. Moreover, some of the control variables may be naturally discrete, and this is incompatible with the linear system viewpoint.

(b) There may be control and/or state constraints, which are not handled adequately through quadratic penalty terms in the cost function. For example, the motion of a car may be constrained by the presence of obstacles and hardware limitations (see Fig. 1.6.16). The solution obtained from a linear quadratic model may not be suitable for such a problem, because quadratic penalties treat constraints "softly" and may produce trajectories that violate the constraints.

These inadequacies of the linear quadratic formulation have motivated MPC, which combines elements of several ideas that we have discussed so

† An extensive overview of the connections of the conceptual framework of this book with model predictive and adaptive control is given in the author's paper [Ber24]. The corresponding video is a good supplement to the present section and can be found at https://www.youtube.com/watch?v=UeVs0Op-Ui4 and a related video can also be found at https://www.youtube.com/watch?v=ZBRouvMat2Q

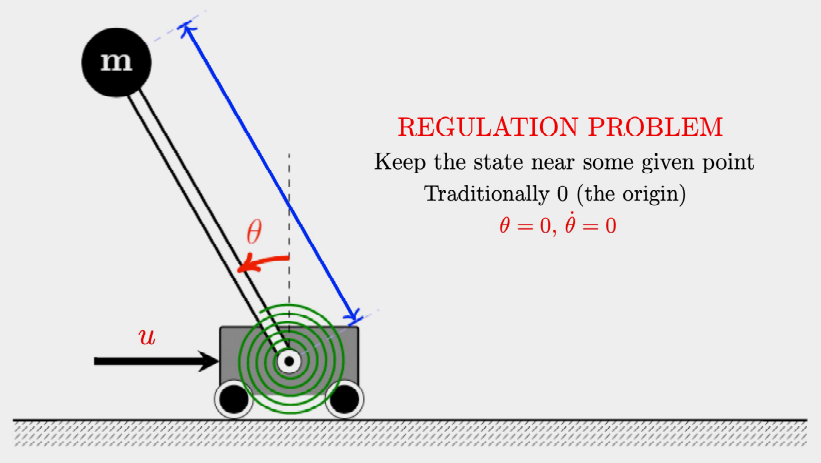


Figure 1.6.15 Illustration of a classical regulation problem, known as the "cart-pole problem" or "inverse pendulum problem." The state is the two-dimensional vector of angular position and angular velocity. We aim to keep the pole at the upright position (state equal to 0) by exerting horizontal force u on the cart.

far, such as multistep lookahead, rollout with a base policy, and certainty equivalence. Aside from dealing adequately with state and control constraints, MPC is well-suited for on-line replanning, like all approximation in value space methods.

Note that the ideas of MPC were developed independently of the approximate DP/RL methodology. However, the two fields are closely related, and there is much to be gained from understanding one field within the context of the other, as the subsequent development will aim to show. A major difference between MPC and finite-state stochastic control problems that are popular in the RL/AI literature is that in MPC the state and control spaces are continuous/infinite, such as for example in self-driving cars, the control of aircraft and drones, or the operation of chemical processes. At the same time, at a fundamental level, this difference turns out to be inconsequential, because the key underlying framework for approximation in value space, which is based on Newton's method, is valid for both discrete and continuous state and control spaces.

In this section, we will primarily focus on the undiscounted infinite horizon deterministic problem, which involves the system

$$x_{k+1} = f(x_k, u_k),$$

whose state x_k and control u_k are finite-dimensional vectors. The cost per stage is assumed nonnegative

$$g(x_k, u_k) \geq 0, \qquad \text{for all } (x_k, u_k),$$

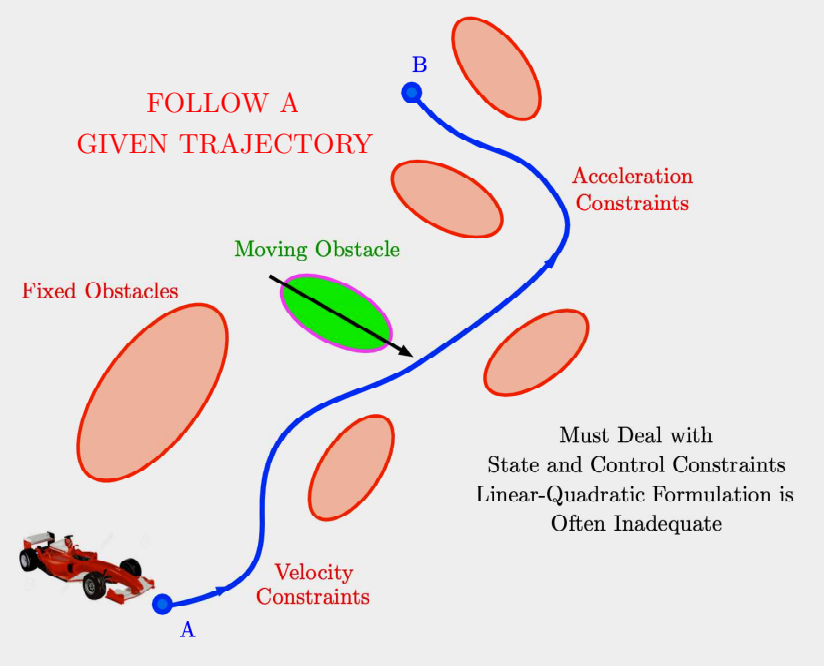


Figure 1.6.16 Illustration of constrained motion of a car from point A to point B. There are state (position/velocity) constraints, and control (acceleration) constraints. When there are mobile obstacles, the state constraints may change unpredictably, necessitating on-line replanning.

(e.g., a positive definite quadratic cost). There are control constraints $u_k \in U(x_k)$, and to simplify the following discussion, we will initially consider no state constraints. We assume that the system can be kept at the origin at zero cost, i.e.,

$$f(0, \overline{u}_k) = 0, \quad g(0, \overline{u}_k) = 0 \qquad \text{for some control } \overline{u}_k \in U(0).$$

For a given initial state x_0, we want to obtain a sequence $\{u_0, u_1, \ldots\}$ that satisfies the control constraints, while minimizing the total cost.

This is a classical problem in control system design, known as the *regulation problem*, where the aim is to keep the state of the system near the origin (or more generally some desired set point), in the face of disturbances and/or parameter changes. In an important variant of the problem, there are additional state constraints of the form $x_k \in X$, and there arises the issue of maintaining the state within X, not just at the present time but also in future times. We will address this issue later in this section.

The Classical Form of MPC - View as a Rollout Algorithm

We will first focus on a classical form of the MPC algorithm, discussed in

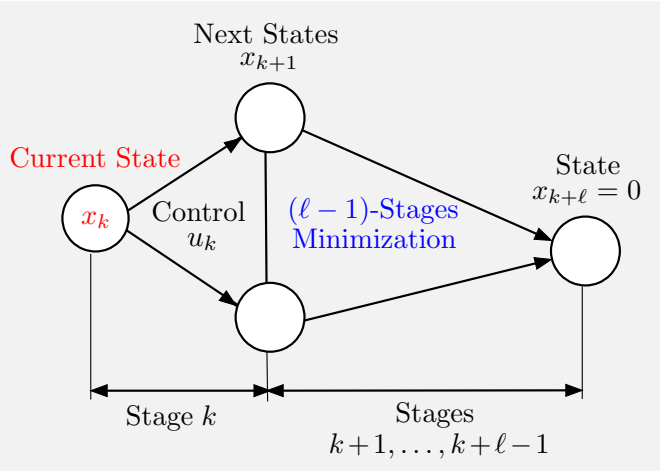


Figure 1.6.17 Illustration of the problem solved by a classical form of MPC at state x_k. We minimize the cost function over the next ℓ stages while imposing the requirement that $x_{k+\ell}=0$. We then apply the first control of the optimizing sequence. In the context of rollout, the minimization over u_k is the one-step lookahead, while the minimization over $u_{k+1},\ldots,u_{k+\ell-1}$ that drives $x_{k+\ell}$ to 0 is the base heuristic.

the form given here by Keerthi and Gilbert [KeG88], with a view towards stability analysis. In this algorithm, at each encountered state x_k, we apply a control $\tilde{u}_k$ that is computed as follows; see Fig. 1.6.17:

(a) We solve an ℓ-stage optimal control problem involving the same cost function and the requirement that the state after ℓ steps is driven to 0, i.e., $x_{k+\ell}=0$. This is the problem

$$\min_{u_t,\, t=k,\ldots,k+\ell-1} \sum_{t=k}^{k+\ell-1} g(x_t,u_t), \tag{1.85}$$

subject to the system equation constraints

$$x_{t+1}=f(x_t,u_t), \qquad t=k,\ldots,k+\ell-1, \tag{1.86}$$

the control constraints

$$u_t\in U(x_t), \qquad t=k,\ldots,k+\ell-1, \tag{1.87}$$

and the terminal state constraint

$$x_{k+\ell}=0. \tag{1.88}$$

Here ℓ is an integer with $\ell>1$, which is chosen in some largely empirical way.

(b) If $\{\tilde{u}_k, \ldots, \tilde{u}_{k+\ell-1}\}$ is the optimal control sequence of this problem, we apply $\tilde{u}_k$ and we discard the other controls $\tilde{u}_{k+1}, \ldots, \tilde{u}_{k+\ell-1}$.

(c) At the next stage, we repeat this process, once the next state x_{k+1} is revealed.

To make the connection of the preceding MPC algorithm with rollout, we note that *the one-step lookahead function $\tilde{J}$ implicitly used by MPC [cf. Eq. (1.85)] is the cost function of a certain stable base policy*. This is the policy that drives to 0 the state after $\ell - 1$ stages (*not ℓ stages*) and keeps the state at 0 thereafter, while observing the state and control constraints, and minimizing the associated $(\ell - 1)$-stages cost. This rollout view of MPC was first discussed in the author's papers [Ber05a], [Ber05b]. It is useful for making a connection with the approximate DP/RL, rollout, and its interpretation in terms of Newton's method. In particular, an important consequence is that *the MPC policy is stable*, since rollout with a stable base policy can be shown to yield a stable policy under very general conditions, as we have noted earlier for the special case of linear quadratic problems in Section 1.5; cf. Fig. 1.5.10.

Terminal Cost Approximation - Stability Issues

In a common variant of MPC, the requirement of driving the system state to 0 in ℓ steps in the ℓ-stage MPC problem (1.85), is replaced by a terminal cost $G(x_{k+\ell})$, which is positive everywhere except at 0. Thus at state x_k, we solve the problem

$$\min_{u_t,\, t=k,\ldots,k+\ell-1} \left[G(x_{k+\ell}) + \sum_{t=k}^{k+\ell-1} g(x_t, u_t) \right], \tag{1.89}$$

instead of problem (1.85) where we require that $x_{k+\ell} = 0$. This variant can be viewed as rollout with one-step lookahead, and a base policy, which at state x_{k+1} applies the first control $\tilde{u}_{k+1}$ of the sequence $\{\tilde{u}_{k+1}, \ldots, \tilde{u}_{k+\ell-1}\}$ that minimizes

$$G(x_{k+\ell}) + \sum_{t=k+1}^{k+\ell-1} g(x_t, u_t).$$

On the other hand, this MPC variant can also be viewed as approximation in value space with ℓ-step lookahead minimization and terminal cost approximation given by G. It can be interpreted in terms of a Newton step, as illustrated in Fig. 1.5.7 for the case of one-step lookahead, and in Fig. 1.5.8 for the case of multistep lookahead.

An important question is how to choose the terminal cost approximation so that the resulting MPC controller is stable. Our discussion of Section 1.5 on the region of stability of approximation in value space

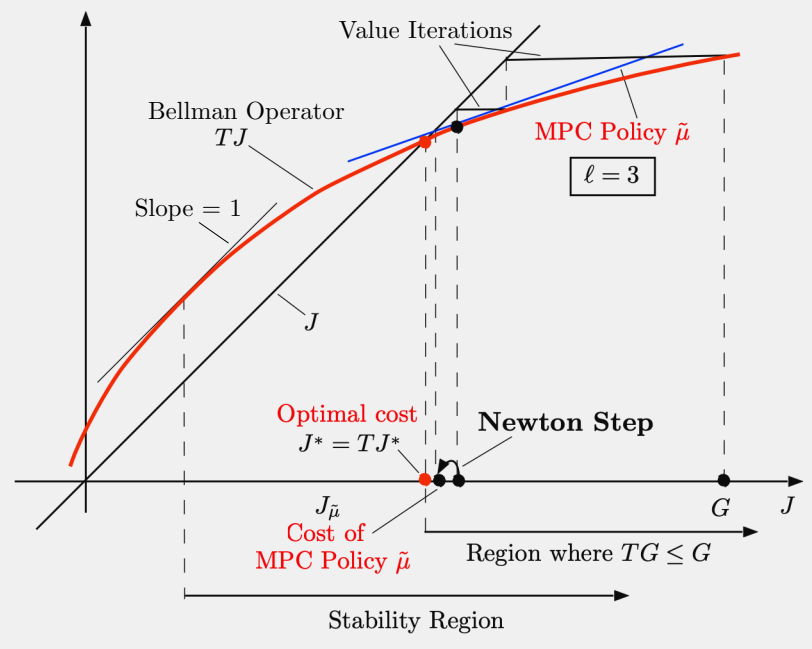


Figure 1.6.18 Illustration of the Bellman operator, defined by

$$(TJ)(x) = \min_{u \in U(x)} \Big\{ g(x,u) + J\big(f(x,u)\big) \Big\}, \qquad \text{for all } x.$$

The condition in (1.90) can be written compactly as $(TG)(x) \leq G(x)$ for all x. When satisfied by the terminal cost function G, it guarantees stability of the MPC policy $\tilde{\mu}$ with ℓ-step lookahead minimization. In this figure, $\ell = 3$.

schemes applies here. In particular, under the nonnegative cost assumption of this section, the MPC controller can be proved to be stable if a single value iteration (VI) starting from G produces a function that takes uniformly smaller values than G:

$$\min_{u \in U(x)} \Big\{ g(x,u) + G\big(f(x,u)\big) \Big\} \leq G(x), \qquad \text{for all } x. \tag{1.90}$$

This is also known as the *Lyapunov condition* in MPC. Figure 1.6.18 provides a graphical illustration, showing how this condition guarantees that successive iterates of value iteration, as implemented through multistep lookahead, lie within the region of stability, so that the policy produced by MPC is stable.

We expect that as the length ℓ of the lookahead minimization is increased, the stability properties of the MPC controller are improved. In particular, *given $G \geq 0$, the resulting MPC controller is likely to be stable*

for ℓ sufficiently large, since the VI algorithm ordinarily converges to J^*, which lies within the region of stability. Results of this type are known within the MPC framework under various conditions (see the papers by Mayne et al. [MRR00], Magni et al. [MDM01], the MPC book [RMD17], and the author's book [Ber20a], Section 3.1.2). Our discussion of stability in Section 1.5 is also relevant within this context; cf. Fig. 1.5.8.

In another variant of MPC, in addition to the terminal cost function approximation G, we use truncated rollout, which involves running some stable base policy μ for a number of steps m; see Fig. 1.6.19. This is quite similar to standard truncated rollout, except that the computational solution of the lookahead minimization problem (1.89) may become complicated when the control space is infinite. As discussed in Section 1.5, *increasing the length of the truncated rollout enlarges the region of stability of the MPC controller*. The reason is that by increasing this length, we push the start of the Newton step towards of the cost function J_μ of the stable policy, which lies within the region of stability. The base policy may also be used to address state constraints; see the papers by Rosolia and Borelli [RoB17], [RoB19], Li et al. [LJM21], and the discussions in the author's RL books [Ber20a], [Ber22a].

Finally, let us note that when faced with changing problem parameters, it is natural to consider on-line replanning as per our earlier adaptive control discussion. In this context, once new estimates of system and/or cost function parameters become available, MPC can readily adapt by introducing the new parameter estimates into the ℓ-stage optimization problem in (a) above. This is an important and often decisive advantage of MPC over approximation in policy space for problems with changing environments.

State Constraints and Invariant Sets

Our discussion so far has skirted a major issue in MPC, which is that there may be additional state constraints of the form $x_k \in X$, for all k, where X is some subset of the true state space. Indeed much of the original work on MPC was motivated by control problems with state constraints, imposed by the physics of the problem, which could not be handled effectively with the nice unconstrained framework of the linear quadratic problem that we discussed in Section 1.5.

To deal with additional state constraints of the form $x_k \in X$, where X is some subset of the state space, the MPC problem to be solved at the kth stage [cf. Eq. (1.89)] must be modified. Assuming that the current state x_k belongs to the constraint set X, the MPC problem should take the form

$$\min_{u_t,\, t=k,\ldots,k+\ell-1} \left[G(x_{k+\ell}) + \sum_{t=k}^{k+\ell-1} g(x_t, u_t) \right], \tag{1.91}$$

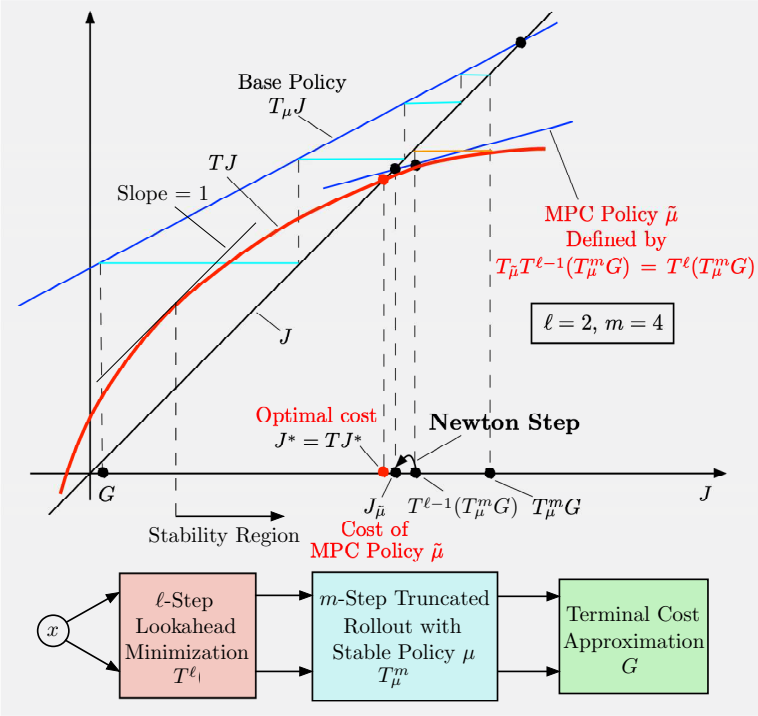


Figure 1.6.19 An MPC scheme with ℓ-step lookahead minimization, m-step truncated rollout with a stable base policy μ, and a terminal cost function approximation G, together with its interpretation as a Newton step. In this figure, $\ell = 2$ and $m = 4$. The truncated rollout with base policy μ consists of m value iterations, starting with the function G, and using the Bellman operator corresponding to μ, which is given by

$$(T_\mu J)(x) = g\big(x, \mu(x)\big) + J\Big(f\big(x, \mu(x)\big)\Big)$$

Thus, truncated rollout yields the function $T_\mu^m G$. Then $\ell - 1$ value iterations are applied to this function through the $(\ell - 1)$-step minimization, yielding the function

$$T^{\ell-1}(T_\mu^m G).$$

Finally, the Newton step is applied to this function to yield the cost function of the MPC policy $\tilde{\mu}$. As m increases, the starting point for the Newton step moves closer to J_μ, which lies within the region of stability.

subject to the control constraints

$$u_t \in U(x_t), \qquad t = k, \ldots, k + \ell - 1, \tag{1.92}$$

and the state constraints

$$x_t \in X, \qquad t = k + 1, \ldots, k + \ell. \tag{1.93}$$

The control $\tilde{u}_k$ thus obtained will generate a state

$$x_{k+1} = f(x_k, \tilde{u}_k)$$

that will belong to X, and similarly the entire state trajectory thus generated will satisfy the state constraint $x_t \in X$ for all t, assuming that the initial state does.

However, there is an important difficulty with the preceding MPC scheme, namely *there is no guarantee that the problem (1.91)-(1.93) has a feasible solution for all initial states $x_k \in X$*. Here is a simple example.

Example 1.6.8 (State Constraints in MPC)

Consider the scalar system

$$x_{k+1} = 2x_k + u_k,$$

with control constraint

$$|u_k| \leq 1,$$

and state constraints of the form $x_k \in X$, for all k, where

$$X = \big\{x_k \mid |x_k| \leq \beta\big\}. \tag{1.94}$$

Then if $\beta > 1$, the state constraint cannot be satisfied for all initial states $x_0 \in X$. In particular, if we take $x_0 = \beta$, then $2x_0 > 2$ and $x_1 = 2x_0 + u_0$ will satisfy $x_1 > x_0 = \beta$ for any value of u_0 with $|u_0| \leq 1$. Similarly the entire sequence of states $\{x_k\}$ generated by any set of feasible controls will satisfy

$$x_{k+1} > x_k \quad \text{for all } k, \qquad x_k \uparrow \infty.$$

The state constraint can be satisfied only for initial states x_0 in the set $\hat{X}$ given by

$$\hat{X} = \big\{x_k \mid |x_k| \leq 1\big\};$$

see Fig. 1.6.20, which also illustrates the trajectories generated by the MPC scheme of Eq. (1.89), which does not involve state constraints.

The preceding example illustrates a fundamental point in state-constrained MPC: *the state constraint set X must be invariant in the sense that starting from any one of its points x_k there must exist a control $u_k \in U(x_k)$ for which the next state $x_{k+1} = f(x_k, u_k)$ must belong to X*. Mathematically, X is invariant if

$$\text{for every } x \in X, \text{ there exists } u \in U(x) \text{ such that } f(x, u) \in X.$$

In particular, it can be seen that the set X of Eq. (1.94) is invariant if and only if $\beta \leq 1$.

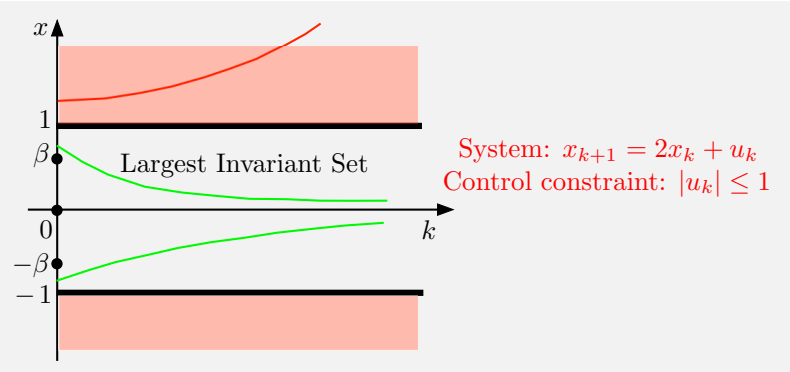


Figure 1.6.20 Illustration of invariance of a state constraint set X. Here the sets of the form $X = \{x_k \mid |x_k| \leq \beta\}$ are invariant for $\beta \leq 1$. For $\beta = 1$, we obtain the largest invariant set (the one that contains all other invariant sets). The figure shows some state trajectories produced by MPC. Note that starting with an initial condition x_0 with $|x_0| > 1$ (or $|x_0| < 1$) the closed-loop system obtained by MPC is unstable (or stable, respectively); cf. the red and green trajectories shown.

Given an MPC calculation of the form (1.91)-(1.93), we must make sure that the set X is invariant, or else it should be replaced by an invariant subset $\hat{X} \subset X$. Then the MPC calculation (1.91)-(1.93) will be feasible provided the initial state x_0 belongs to $\hat{X}$.

This brings up the question of how we compute an invariant subset of a given constraint set. This is typically an off-line calculation that cannot be performed during on-line play. It turns out that given X there exists a largest possible invariant subset of X, which can be computed in the limit with an algorithm that resembles value iteration. In particular, starting with $X_0 = X$, we obtain a nested sequence of subsets through the recursion

$$X_{k+1} = \{x \in X_k \mid f(x,u) \text{ belongs to } X_k \text{ for some } u \in U(x)\}, \quad k \geq 0. \tag{1.95}$$

Clearly, we have $X_{k+1} \subset X_k$ for all k, and under mild conditions it can be shown that the intersection set $\hat{X} = \cap_{k=0}^{\infty} X_k$, is the largest invariant subset of X; see the author's PhD thesis [Ber71] and subsequent paper [Ber72a], which introduced the concept of invariance and its use in satisfying state constraints in control over a finite and an infinite horizon.†

To illustrate, in the preceding Example 1.6.8, the sequence of value iterates (1.95) starting with the set $X_0 = \{x \mid |x| \leq \beta\}$, where $\beta > 1$, is given by

$$X_k = \{x \mid |x| \leq \beta_k\}, \quad \text{with } \beta_0 = \beta \text{ and } \beta_{k+1} = \tfrac{\beta_k + 1}{2} \text{ for all } k \geq 0.$$

† The term used in [Ber71] and [Ber72a] is *reachability of a target tube* $\{X, X, \ldots\}$, which is synonymous to invariance of X.

It can be seen that we have $\beta_{k+1} < \beta_k$ for all k and $\beta_k \downarrow 1$, so the intersection $\hat{X} = \cap_{k=0}^{\infty} X_k$ yields the largest invariant set $\hat{X} = \big\{x_k \mid |x_k| \leq 1\big\}$.

Suboptimal Invariant Subsets

Since computing the largest invariant subset of a constraint set X is often computationally intractable, one may consider using smaller invariant subsets of X. A relatively simple possibility is to compute an invariant subset $\hat{X}$ that corresponds to some nominal policy $\hat{\mu}$ [i.e., starting from any point $x \in \hat{X}$, the state $f\big(x, \hat{\mu}(x)\big)$ belongs to $\hat{X}$]. Such an invariant subset may be obtained or approximated with some form of simulation using the policy $\hat{\mu}$. Moreover, $\hat{\mu}$ can also be used for truncated rollout and also provide a terminal cost function approximation. An alternative for the case of a linear system driven by ellipsoid-bounded disturbances, is to construct an ellipsoidal invariant subset; the author's PhD thesis [Ber71] provided an algorithm for doing so.

For more broadly applicable possibilities, we refer to the MPC literature; see e.g., the book by Rawlings, Mayne, and Diehl [RMD17] (Chapter 3), and the surveys by Mayne [May14], and Houska, Muller, and Villanueva [HMV24], which give additional references. An important point is that the computation of an invariant subset of the given constraint set X must be done off-line, thus becoming part of the off-line training phase, along with the terminal cost function G.

To deal with state constraints in the context of a partially unknown or changing system model, combinations of MPC with robust control ideas have been suggested [see [RMD17] (Chapter 3), where robust tube-based MPC ideas are discussed]. An alternative is to replace the state constraints with penalty or barrier functions as part of the cost per stage. This approach has received attention, using what is known as *control barrier functions* and *control Lyapunov functions*. We refer to the several survey papers in the literature, and to the book by Xiao, Cassandras, and Belta [XCB23], for accounts of this research direction.

Stochastic MPC by Certainty Equivalence

We note that while we have focused on deterministic problems in this section, there are variants of MPC that include the treatment of uncertainty. The books and papers cited earlier contain several ideas along these lines; see e.g. the books by Kouvaritakis and Cannon [KoC16], Rawlings, Mayne, and Diehl [RMD17], and the survey by Mesbah [Mes16].

In this connection, it is also worth mentioning the *certainty equivalence approach* that we discussed briefly earlier. In particular, upon reaching state x_k we may perform the MPC calculations after replacing the uncertain quantities $w_{k+1}, w_{k+2}, \ldots$ with deterministic quantities $\overline{w}_{k+1}, \overline{w}_{k+2}, \ldots$, *while allowing for the stochastic character of the disturbance w_k of just the current stage k.* Note that only the first step of this

MPC calculation is stochastic. Thus the calculation needed per stage is not much more difficult than the one for deterministic problems, while still implementing a Newton step for solving the associated Bellman equation; see our earlier discussion, and also Section 2.5.3 of the RL book [Ber19a], Section 3.2 of the book [Ber22a], and the MPC overview paper [Ber24].

Data-Driven MPC

One of the principal limitations of the mainstream methodology of MPC is the need for a mathematical model. In many cases, such a model may not be available or may be difficult to derive. To deal with this situation, data-driven versions of MPC have been suggested, where sampled triplets, consisting of state, control, and next state, are used for prediction of system trajectories.

In this approach, the system is implicitly represented by a dataset of observed trajectories, rather than an explicit mathematical model, an approach pioneered by J. C. Willems; see Markovsky and Dorfler [MaD21] for a recent review. The data consists of many sample triplets (x_k, u_k, x_{k+1}), and the idea is to use this data to approximate the system dynamics and perform MPC. We can do this by fitting a parametric system equation model to the data, and predicting future system trajectories using this model. Among other approaches, we may use a neural network (possibly a transformer) or other type of system identification method for this purpose. Nonparametric models are also possible, based on statistical models, such as Bayesian process regression.

Once the data-approximated model has been constructed off-line, it can be used on-line to predict and optimize the future system trajectory over a finite prediction horizon, and to apply the first control in the optimal sequence at the current state. Naturally, the quality of the model has a major impact on the performance of the corresponding MPC policy. Some challenges here are the choice of state, and also the fact that the learned model may be inaccurate outside the training set. Moreover, major difficulties arise in an adaptive control environment, where the model must be modified on-line by using data. For some representative relevant papers, see [CLD19], [GrZ19], [HWM20], [KRW21], [CLR21], [BGH22], [CWA22], [GrZ22], [MHD23], and [BeA24].

MPC with Imperfect State Information

The MPC methodology, in the form we have described it here, requires full state feedback. When the exact state is not available, an estimate may be used instead. In classical control methodology, it is common to use a Kalman filtering algorithm or an observer to reconstruct the state. This is possible when the state is continuous and a system model is known. Also, in some noncontinuous state problems, it may be possible to use some inference method to obtain a state estimate, and use it in the MPC

algorithm as if it were the exact state. Otherwise, it is necessary to view the problem as one of imperfect state information, introduce belief states in place of the unobservable states, and adopt the corresponding POMDP-like methodology. This is, however, much more demanding computationally, both for training a terminal cost function approximation and for on-line computation of the MPC controls.

The alternative for problems with imperfect state information is a non-MPC/approximation in policy space approach, which may also have some serious drawbacks:

(a) The training of suitable policies can be complicated and unreliable due to local minima and other difficulties (see Chapter 3).

(b) In an adaptive control/changing model setting, it may be necessary to retrain policies on-line, which can be very sample-inefficient or practically impossible.

(c) Restricting the structure of the policy to be a function of just part of the available information (e.g., the current system output) places a limit on performance: we are not using the full information available.

(d) Fundamentally, MPC/approximation in value space has a generic advantage over approximation in policy space: the MPC policy is the result of a Newton step for solving Bellman's equation. Thus the error from optimality $J_{\tilde{\mu}} - J^*$ of the MPC policy $\tilde{\mu}$ is governed by a superlinear relation, as we have discussed, and is negligible if the terminal cost approximation error $\tilde{J} - J^*$ is small.

Model Mismatch and Disturbance Estimation in MPC

In practical implementations of MPC, an important source of performance degradation is model mismatch. Even when a reasonably accurate model is available to start with, unmodeled dynamics, parameter drift, or varying workloads, may cause the true evolution of the system to differ from the predictions used by the controller. If such mismatch is ignored, the closed-loop system may exhibit long-term state deviations from their desired targets or degraded constraint handling. For this reason, many practical MPC schemes for control system design incorporate some mechanism for estimating and compensating for the dominant model errors and other disturbances during real-time operation.

A viewpoint, emphasized in chemical process control (see the classical textbook by Stephanopoulos [Ste84] and its updated sequel [Ste25]), is that in many industrial settings, modeling errors can be significant, but can effectively be viewed as a constant (or slowly varying) disturbance. If the controller can infer this disturbance from available measurements, it can suitably compensate for it; this is an idea that stems from the so-called *internal model principle*, articulated by Francis and Wonham [FrW76]. The

simplest instance of this is the integral term in a PID control scheme (cf. Section 1.6.8), which acts as a dynamic estimator of the unknown average disturbance: by accumulating the output error, it gradually identifies the control action needed to make the measured output coincide with the desired one.

A similar strategy can be adopted in MPC formulations in a more general and systematic way. Rather than relying solely on integral action, one may introduce a small number of disturbance variables into the system model and allow an observer to update their values based on the mismatch between predicted and measured state evolution. Within our DP framework, this can be implemented through state augmentation (cf. Section 1.6.5). For a simple example, let the system $x_{k+1} = f(x_k, u_k)$ be replaced by

$$x_{k+1} = f(x_k, u_k) + w_k,$$

where w_k is a disturbance that represents a constant (or slowly varying) modeling error, so that

$$w_{k+1} = w_k.$$

We may then introduce w_k as an additional state variable and an augmented system with state $\tilde{x}_k = (x_k, w_k)$. If the state x_k is observed perfectly, the same is true for the augmented state $\tilde{x}_k$, since w_k can be computed exactly as the difference $x_k - f(x_{k-1}, u_{k-1})$.

A popular generalization of this approach, is to model the effect of the disturbance through a system equation of the form

$$x_{k+1} = f(x_k, u_k) + Ew_k,$$

where E is a known matrix, which models the effects of the components of w_k on different state variables/components of x_k. When the state is not measured exactly, the MPC controller is naturally combined with a Kalman filter or related estimator. This estimator updates both the state and disturbance estimates at each stage, essentially producing a continually refined model of the system. The MPC controller then applies its optimization step as if the estimates of state and disturbance were exact.

For many systems of interest in chemical process control, the combination of disturbance modeling and state estimation, outlined above, captures the main benefits of more elaborate adaptive or robust MPC techniques; see the book by Stephanopoulos [Ste25] noted earlier. For larger and more rapidly varying uncertainties, more sophisticated identification or robust formulations may be necessary; we refer to the MPC literature for further discussion, particularly the textbooks by Rawlings, Mayne, and Diehl [RMD17], and by Borrelli, Bemporad, and Morari [BBM17].

1.7 REINFORCEMENT LEARNING AND DECISION/CONTROL

The current state of RL has greatly benefited from the cross-fertilization of ideas from decision and control, and from AI; see Fig. 1.7.1. The strong con-

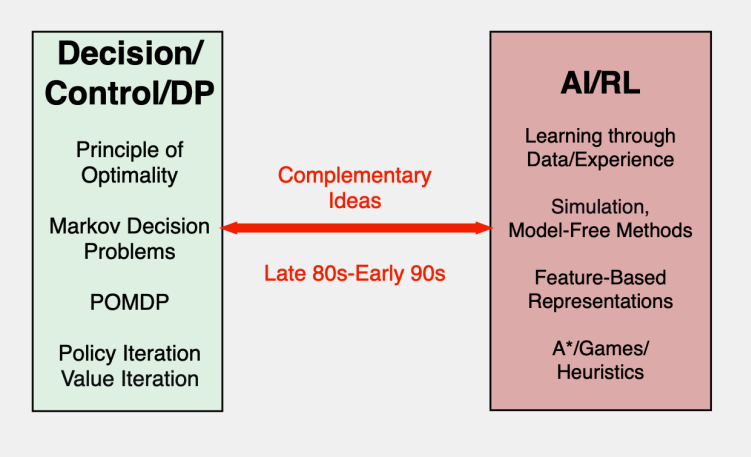


Figure 1.7.1 A schematic illustration of the synergy of ideas between decision and control on one hand, and AI on the other.

nections between these two fields are now widely recognized. Still, however, there are cultural differences, including the traditional reliance on mathematical analysis for the decision and control field, and the emphasis on challenging problem implementations in the AI field. Moreover, substantial differences in language and emphasis remain between RL-based discussions (where AI-related terminology is used) and DP-based discussions (where optimal control-related terminology is used).

1.7.1 Differences in Terminology

The terminology used in this book is standard in DP and optimal control, and in an effort to forestall confusion of readers that are accustomed to either the AI or the decision and control terminology, we provide a list of terms commonly used in AI/RL, and their optimal control counterparts.

(a) **Environment** = System.

(b) **Agent** = Decision maker or controller.

(c) **Action** = Decision or control.

(d) **Reward of a stage** = (Opposite of) Cost of a stage.

(e) **State value** = (Opposite of) Cost starting from a state.

(f) **Value (or reward) function** = (Opposite of) Cost function.

(g) **Maximizing the value function** = Minimizing the cost function.

(h) **Action (or state-action) value** = Q-factor (or Q-value) of a state-control pair. (Q-value is also used often in RL.)

(i) **Planning** = Solving a DP problem with a known mathematical model. (Often related to MPC and approximation in value space.)

(j) **Learning** = Solving a DP problem without using a known mathematical model. (This is the principal meaning of the term "learning" in AI/RL. Other meanings are also common.)

(k) **Self-learning** (or self-play in the context of games) = Solving a DP problem using some form of policy iteration.

(l) **Deep reinforcement learning** = Approximate DP using value and/or policy approximation with deep neural networks.

(m) **Prediction** = Policy evaluation.

(n) **Generalized policy iteration** = Optimistic policy iteration.

(o) **State abstraction** = State aggregation.

(p) **Temporal abstraction** = Time aggregation.

(q) **Learning a model** = System identification.

(r) **Episodic task or episode** = Finite-step system trajectory.

(s) **Continuing task** = Infinite-step system trajectory.

(t) **Experience replay** = Reuse of samples in a simulation process.

(u) **Bellman operator** = DP mapping or operator.

(v) **Backup** = Applying the DP operator at some state.

(w) **Sweep** = Applying the DP operator at all states.

(x) **Greedy policy with respect to a cost function** J = Minimizing policy in the DP expression defined by J.

(y) **Afterstate** = Post-decision state.

(z) **Ground truth** = Empirical evidence or information provided by direct observation.

Some of the preceding terms will be introduced in future chapters; see also the RL textbook [Ber19a]. The reader may then wish to return to this section as an aid in connecting with the relevant RL literature.

1.7.2 Differences in Notation

Unfortunately, the confusion caused by differing terminology has been further compounded by the use of inconsistent notations across various sources. This book adheres to the "standard" notation that emerged during the Bellman/Pontryagin optimal control era; see e.g., the books by Athans and Falb [AtF66], Bellman [Bel67], and Bryson and Ho [BrH75]. This notation is consistent with the author's other DP books and is the most appropriate for a unified treatment of the subject, which simultaneously addresses discrete and continuous spaces problems.

A summary of the most prominently used symbols in our notational system is as follows:

(a) x: state (also i for finite-state systems).

(b) u: control.

(c) w: stochastic disturbance.

(d) J: cost function.

(e) f: system function. For deterministic systems,

$$x_{k+1} = f(x_k, u_k)$$

and for stochastic systems,

$$x_{k+1} = f(x_k, u_k, w_k).$$

Also f_k in place of f for time-varying systems.

(f) g: cost per stage [$g(x, u)$ for deterministic systems, and $g(x, u, w)$ for stochastic systems; also g_k in place of g for time-varying systems].

(g) $p_{xy}(u)$: transition probability from state x to state y under control u in finite-state systems [also $p_{ij}(u)$].

(h) α: discount factor in discounted problems.

The x-u-J notation is standard in deterministic optimal control textbooks (e.g., the classical books [AtF66] and [BrH75], noted earlier, as well as the more recent books by Stengel [Ste94], Kirk [Kir04], and Liberzon [Lib11]). The symbols f (system function) and g (cost per stage) are also widely used in both early and later optimal control literature (unfortunately the more natural symbol "c" has not been used much in place of "g" for the cost per stage).

The notations i (state) and $p_{ij}(u)$ (transition probability) are common in the discrete-state Markov decision process (MDP) and operations research literature. Sometimes the alternative notation $p(j \mid i, u)$ is used for the transition probabilities.

In the AI literature, the focus is primarily on finite-state MDPs, particularly discounted and stochastic shortest path infinite horizon problems. The most commonly used notation is s for state, a for action, $r(s, a, s')$ for reward per stage, $p(s' \mid s, a)$ or $p(s, a, s')$ for transition probability from s to s' under action a, and γ for discount factor. While this notation is well-suited to finite-state problems, it is not ideal for continuous spaces models. The reason is that it requires the use of transition probability distributions defined over continuous spaces, and leads to more complex and less intuitive mathematics. Moreover, for deterministic problems, which lack a probabilistic component, the transition probability notation becomes cumbersome and unnecessary.

1.7.3 Relations Between DP and RL

When comparing the RL and DP methodologies, it is important to recognize that they are fundamentally connected by their shared focus on sequential decision making. Thus, any problem that can be addressed by DP can, in principle, also be addressed by RL, and vice versa.

One may argue that the RL algorithmic methodology is broader than that of DP. It includes the use of gradient descent and random search algorithms, simulation-based methods, statistical methods of sampling and performance evaluation, and neural network design and training ideas. However, methods of this type have also been considered in DP-related research and applications for many years, albeit less intensively.

In the AI view of RL, a machine learns through trial and error by interacting with an environment.† In practical terms, this is more or less the same as what DP aims to do, but in RL there is often an emphasis on the presence of uncertainty and exploration of the environment. In the decision, control, and optimization community, there is a lot of interest in using RL methods to address intractable problems, including deterministic discrete/integer optimization, which need not involve data collection, interaction with the environment, uncertainty, and learning (adaptive control is the only decision and control problem type, where uncertainty and exploration arise in a significant way).

In terms of applications, DP was originally developed in the 1950s and 1960s as part of the then emerging methodologies of operations research and optimal control. These methodologies are now mature and provide important tools and perspectives, as well as a rich variety of applications, such as robotics, autonomous transportation, and aerospace, which can benefit from the use of RL. Moreover, DP has been used in a broad range of applications in industrial engineering, operations research, economics, and finance, so these applications can also benefit from the use of RL methods and perspectives.

At the same time, RL and machine learning have ushered opportunities for the application of DP techniques in new domains, such as machine translation, image recognition, knowledge representation, database organization, large language models, and automated planning, where they can have a significant practical impact. We may also add that RL has brought into the field of sequential decision making a fresh and ambitious spirit that has made possible the solution of problems thought to be well outside the capabilities of DP. Indeed, before the connections between RL and DP were recognized, large dimensional problems, like those involving a Euclidean state space of even moderate dimension, or POMDP problems, were considered totally intractable with the DP methodology.

† A common description is that "the machine learns sequentially how to make decisions that maximize a reward signal, based on the feedback received from the environment."

1.7.4 Synergy Between Large Language Models and DP/RL

Can RL and large language models (LLMs) work synergistically? This is an important question, as these two AI paradigms operate with distinct methodologies, objectives, and capabilities. While RL focuses on sequential decision-making, LLMs specialize in natural language understanding and generation, including computer code.

In particular, RL is designed to optimize policies for sequential control tasks, excelling in applications such as robotics and resource allocation, where adaptive decision-making is essential. In contrast, LLMs process and generate human-like text, enabling them to perform tasks such as translation, summarization, sentiment analysis, and code generation. Moreover, by leveraging vast amounts of pre-trained knowledge, LLMs can generalize across diverse contexts.

Despite their differences, the capabilities of RL and LLMs are *complementary*, making them powerful tools when used in combination. Let us now summarize ways in which the synergy between RL and LLMs can manifest itself in practice.

The advent of pre-trained transformers, such as ChatGPT, has revolutionized natural language processing. These transformers can undergo further refinement through off-line training to specialize in specific tasks or mitigate undesirable biases. Notably, RL methodologies, particularly policy-space approximation techniques, have played a crucial role in this refinement process, as discussed in Section 3.5 of Chapter 3. Thus, RL has been instrumental in enhancing the capabilities of LLMs through off-line optimization techniques.

Conversely, LLMs serve as catalysts for RL by injecting domain knowledge, improving interpretability, and enabling more human-aligned training. By leveraging natural language input, LLMs facilitate the design of RL policies that are more transparent and adaptable. Additionally, LLMs support RL applications by assisting with mathematical formulation, algorithm selection, and code generation. This interplay between the two fields continues to evolve, driving new innovations in AI.

In conclusion, the interaction of RL and LLMs is not merely additive, it is multiplicative: RL equips LLMs with the ability to learn from interaction and feedback, while LLMs equip RL with contextual awareness, explainability, accessibility, and code generation capability. Together, they pave the way for enhanced applications, which align with human intent, and can also communicate and explain their reasoning.

1.7.5 Machine Learning and Optimization

Machine learning and optimization are closely intertwined fields, sharing similar mathematical models and computational algorithms.† However,

† Both fields are also closely connected to the field of statistical analysis.

they differ in their cultures and application contexts, so it is worth reflecting on their similarities and differences.

Machine learning can be broadly categorized into three main types of methods, all of which involve the collection and use of data in some form:

(a) *Supervised learning*: Here a dataset of many input-output pairs (also called labeled data) is collected. An optimization algorithm is used to create a parametrized function that fits well the data, as well as make accurate predictions on new, unseen data. Supervised learning problems are typically formulated as optimization problems, examples of which we will see in Chapter 3. A common algorithmic approach is to use a gradient-type algorithm to minimize a loss function that measures the difference between the actual outputs of the dataset and the predicted outputs of the parametrized model.

(b) *Unsupervised learning*: Here the dataset is "unlabeled" in the sense that the data are not separated into input and matching output pairs. Unsupervised learning algorithms aim to identify patterns or structures within the data, which is useful for tasks like clustering, dimensionality reduction, and density estimation. The objective is to extract meaningful insights from the data. Some unsupervised learning methods can be related to DP, but the connection is not strong. Generally speaking, unsupervised learning does not seem to align well with the types of sequential decision making applications of this book.

(c) *Reinforcement learning*: RL differs in an important way from supervised and unsupervised learning. *It does not use a dataset as a starting point*. Instead, it generates data on-line or off-line as dictated by the needs of the optimization algorithm it uses, be it multistep lookahead minimization, approximate policy iteration and rollout, or approximation in policy space.† A further complication in RL is that the generated data depends on the policy that is used to control the system. Ideally, the data should be generated using an optimal or near-optimal policy, but such a policy is unlikely to be available. We are thus forced to collect data using a sequence of (hopefully) improving policies, which is the essence of the approximate policy iteration algorithm of DP. This is a primary reason why this algorithm and its variations will be a focal point for our discussions in this book.

Another type of machine learning approach, which relates to DP/RL methods, is *semi-supervised learning*. It involves training a model using a dataset containing both labeled and unlabeled data. Here, some initial

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However, in this section, we will not focus on this connection, as it is less relevant to the content of this book.

† A variant of RL called *offline RL* or *batch RL*, starts from a historical dataset, and does not explore the environment to collect new data.

labeled data are sequentially augmented with unlabeled data, with the aim of constructing an "informative" data set that enhances machine learning tasks such as classification. This approach lies between supervised learning (which requires all data to be labeled) and unsupervised learning (which works with exclusively unlabeled data). Semi-supervised learning is related to the field of *active learning*, where DP-like methods are used to augment sequentially the labeled set; see e.g., the monograph by Zhu and Goldberg [ZhG22], the survey by Van Engelen and Hoos [VaH20], and the illustrative application papers by Marchesoni-Acland et al. [MMK23], and Bhusal, Miller, and Merkurjev [BMM24].

Optimization problems and algorithms on the other hand may or may not involve the collection and use of data. They involve data only in the context of special applications, most of which are related to machine learning. In theoretical terms, optimization problems are categorized in terms of their mathematical structure, which is the primary determinant of the suitability of particular types of methods for their solution. In particular, it is common to distinguish between *static optimization problems* and *dynamic optimization problems*. The latter problems involve sequential decision making, with feedback between decisions, while the former problems involve a single decision.

Stochastic problems with perfect or imperfect state observations are dynamic (unless they involve open-loop decision making without the use of any feedback), and they require the use of DP for their optimal solution. Deterministic problems can be formulated as static, but they can also be formulated as dynamic for reasons of algorithmic expediency. In this case, the decision making process is (sometimes artificially) broken down into stages, as is often done in this book for discrete optimization and other contexts.

Another important categorization of optimization problems is based on whether their search space is *discrete* or is *continuous*. Discrete problems include deterministic problems such as integer and combinatorial optimization problems, and can be addressed by formal methods of integer programming as well as by DP. These problems tend to be challenging, so they are often addressed (suboptimally) with the use of heuristics.

Continuous problems are usually addressed with very different methods, which are based on calculus and convexity, such as Lagrange multiplier theory and duality, and the computational machinery of linear, nonlinear, and convex programming. Some discrete problems, particularly those that involve graphs (such as matching, transportation, and transshipment problems), can be addressed using continuous spaces network optimization methods that rely on linear programming and duality. Hybrid problems, which combine discrete and continuous variables, usually require discrete optimization techniques, but can also benefit from convex duality methods, which are fundamentally continuous.

The DP methodology, generally speaking, applies to just about any

kind of optimization problem, deterministic or stochastic, static or dynamic, discrete or continuous, *as long as it is formulated as a sequential decision problem*, in the manner described in Sections 1.2-1.4. In terms of algorithmic structure, DP differs significantly from other optimization techniques, particularly those based on calculus and convexity. Notably, DP can handle both discrete and continuous problems and is not concerned with local minima, focusing instead on finding global minima.

Notice a qualitative difference between optimization and machine learning: *the former is mostly organized around mathematical structures and the analysis of the foundational issues of the corresponding algorithms, while the latter is mostly organized around how data is collected, used, and analyzed, often with a strong emphasis on statistical issues*. This is an important distinction, which affects profoundly the perspectives of researchers in the two fields.

1.7.6 Mathematics in Machine Learning and Optimization

Let us now discuss some differences between the research cultures of the optimization and machine learning fields, as they pertain to the use of mathematics. In optimization, the emphasis is often on general purpose methods that offer broad and mathematically rigorous performance guarantees, for a wide variety of problems. In particular, it is generally believed that a solid mathematical foundation for a given optimization methodology enhances its reliability and clarifies the boundaries of its applicability. Furthermore, it is recognized that formulating practical problems and matching them to the right algorithms is greatly enhanced by one's understanding of the mathematical structure of the underlying optimization methodology.

Machine learning research includes important lines of analysis with a strongly mathematical character, particularly relating to theoretical computer science, complexity theory, and statistical analysis. At the same time, in machine learning there are eminently useful algorithmic structures, such as neural networks, large language models, and image generative models, which are not well-understood mathematically and defy to a large extent mathematical analysis.† This can add to a perception that focusing on rigorous mathematics, as opposed to practical implementation, may be a low payoff investment in many practical machine learning contexts.

† As an illustration, the paper by He et al., "Deep Residual Learning for Image Recognition," published in Proc. of the IEEE Conf. on Computer Vision and Pattern Recognition, 2016, has been cited over 296,276 times as of November 2025, and contains only two equations. The famous neural network architecture paper by Vaswani et al., "Attention is all you Need," published in NIPS, 2017, which laid the foundation for GPT, has been cited over 205,945 times as of November 2025, and contains only six equations.

Moreover, the starting point in machine learning is often a specific dataset or a specialized type of training problem (e.g., language translation or image recognition). The priority is to find a method that works well for that specific dataset or problem, even if it is not generalizable to others. Thus specialized approximation architectures, implementation techniques, and heuristics, which perform well for the given problem and dataset type, may be perfectly acceptable in a machine learning context, even if they do not provide rigorous and generally applicable performance guarantees.

In conclusion, both optimization and machine learning involve mathematical models and rigorous analysis in important ways, and often overlap in the techniques and tools that they use, as well as in the practical applications that they address. However, depending on the problem at hand, there may be differences in the emphasis and priority placed on mathematical analysis, insight, and generality versus practical effectiveness and problem-specific efficiency. This can lead to some tension, as different fields may not fully appreciate each other's perspective.

1.8 NOTES, SOURCES, AND EXERCISES

We will now summarize this chapter and describe how it can be used flexibly as a foundation for a few different courses. We will also provide a selective overview of the DP and RL literature, and give a few exercises that have been used in ASU classes.

Chapter Summary

In this chapter, we have aimed to provide an overview of the approximate DP/RL landscape, which can serve as the foundation for a deeper in-class development of other RL topics. In particular, we have described in varying levels of depth the following:

(a) The algorithmic foundation of exact DP in all its major forms: deterministic and stochastic, discrete and continuous, finite and infinite horizon.

(b) Approximation in value space with one-step and multistep lookahead, the workhorse of RL, which underlies its major success stories, including AlphaZero. We contrasted approximation in value space with approximation in policy space, and discussed how the two may be combined.

(c) The important division between off-line training and on-line play in the context of approximation in value space. We highlighted how their synergy can be intuitively explained in terms of Newton's method.

(d) The fundamental methods of policy iteration and rollout, the former being primarily an off-line method, and the latter being primarily a less ambitious on-line method. Both methods and their variants bear

close relation to Newton's method and draw their effectiveness from this relation.

(e) Some major models with a broad range of applications, such as discrete optimization, POMDP, multiagent problems, adaptive control, and model predictive control. We delineated their principal characteristics and the major RL implementation issues within their contexts.

(f) The use of function approximation, which has been a recurring theme in our presentation. We have touched upon some of the principal schemes for approximation, e.g., neural networks and feature-based architectures.

One of the principal aims of this chapter was to provide a foundational platform for a range of RL courses that explore at a deeper level various algorithmic methodologies, such as:

(1) Rollout and policy iteration.

(2) Neural networks and other approximation architectures for off-line training.

(3) Aggregation, which can be used for cost function approximation in the context of approximation in value space.

(4) A broader discussion of sequential decision making in contexts involving changing system parameters, sequential estimation, and simultaneous system identification and control.

(5) Stochastic algorithms, such as temporal difference methods and Q-learning, which can be used for off-line policy evaluation in the context of approximate policy iteration.

(6) Sampling methods to collect data for off-line training in the context of cost and policy approximations.

(7) Statistical estimates and efficiency enhancements of various sampling methods used in simulation-based schemes. This includes confidence intervals and computational complexity estimates.

(8) On-line methods for specially structured contexts, including problems of the multi-armed bandit type.

(9) Simulation-based algorithms for approximation in policy space, including policy gradient and random search methods.

(10) A deeper exploration of control system design methodologies such as model predictive control and adaptive control, and their applications in robotics and automated transportation.

We have focused selectively on the methodologies (1)-(4), with a limited coverage of (9) in Section 3.5. In a different course, other choices from the above list may be made, based on the content of the present chapter.

Notes and Sources for Individual Sections

In the literature review that follows, we will focus primarily on textbooks, research monographs, and broad surveys, which supplement our discussions, present related viewpoints, and collectively provide a guide to the literature. Inevitably, our selection reflects a certain cultural bias and an overemphasis on sources that are familiar to the author and aligned in style with this book (including the author's own works). We acknowledge in advance that this may lead to omissions of research references that fall outside our own understanding and perspective on the field, and we apologize for any such exclusions.

Sections 1.1-1.4: Our discussion of exact DP in this chapter has been brief since our focus in this book will be on approximate DP and RL. For a more comprehensive treatment of finite-horizon exact DP and its applications to both discrete and continuous space problems, the author's DP textbook [Ber17a] provides an extensive overview, using notation and style consistent with this book. The books by Puterman [Put94] (written from an operations research perspective) and by the author [Ber12] (written from a decision and control perspective) provide detailed (but substantially different) treatments of infinite horizon finite-state stochastic DP problems. The book [Ber12] also covers continuous/infinite state and control spaces problems, including the linear quadratic problems that we have discussed for one-dimensional problems in this chapter. Continuous spaces problems present special analytical and computational challenges, which are at the forefront of research of the RL methodology. The author's 1976 DP textbook [Ber76] was the first to develop discrete-time DP within a general framework that allows arbitrary state, control, and disturbance spaces.

Some of the more complex mathematical aspects of exact DP were addressed in the monograph by Bertsekas and Shreve [BeS78], particularly the probabilistic/measure-theoretic issues associated with stochastic optimal control, including partial state information problems. This monograph provides an extensive treatment of these issues. The followup work by Huizhen Yu and the author [YuB15] addresses the special measurability issues that relate to policy iteration, and provides further analysis relating to the convergence of value iteration. The second volume of the author's DP book [Ber12], Appendix A, includes an accessible summary introduction of the measure-theoretic framework of the book [BeS78].† In the RL

† The rigorous mathematical theory of stochastic optimal control, including the development of an appropriate Borel space measure-theoretic framework, originated in the 60s, with the work of Blackwell [Bla65], [Bla67]. It relies on the theory of analytic sets of descriptive set theory, introduced in 1917 by M. Suslin, a young Russian mathematician, and further developed by his mentor N. Luzin. It culminated in the Bertsekas and Shreve monograph [BeS78], which provides the now "standard" framework, based on the formalism of Borel spaces,

literature, the mathematical difficulties around measurability are usually neglected (as they are in this book), and this is fine because they do not play an important role in practical applications. Moreover, measurability issues do not arise for problems involving finite or countably infinite state and control spaces. We note, however, that there are quite a few published works in RL as well as exact DP, which purport to address measurability issues with a mathematical narrative that is either confusing or plain incorrect.

The third edition of the author's abstract DP monograph [Ber22b], expands on the original 2013 first edition, and aims at a unified development of the core theory and algorithms of total cost sequential decision problems. It addresses simultaneously stochastic, minimax, game, risk-sensitive, and other DP problems, through the use of abstract DP operators (or Bellman operators as we call them here). The idea is to gain insight through abstraction. In particular, the structure of a DP model is encoded in its abstract Bellman operator, which serves as the "mathematical signature" of the model. Thus, characteristics of this operator (such as monotonicity and contraction) largely determine the analytical results and computational algorithms that can be applied to that model. Abstract DP ideas are also useful for visualizations and interpretations of RL methods using the Newton method formalism that we have discussed somewhat briefly in this book in the context of linear quadratic problems.

Approximation in value space, rollout, and policy iteration are the principal subjects of this book.† These are very powerful and general tech-

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lower semianalytic functions, and universally measurable policies. This development involves daunting mathematical complications, which stem, among others, from the observation that when a Borel measurable function $F(x, u)$, of the two variables x and u, is minimized with respect to u, the resulting function $G(x) = \min_u F(x, u)$ need not be Borel measurable (it belongs to the broader class of lower semianalytic functions); this key fact was the starting point of Suslin's analysis. Moreover, even if the minimum is attained by several policies μ, i.e., $G(x) = F\big(x, \mu(x)\big)$ for all x, it is possible that none of these μ is Borel measurable (however, there does exist a minimizing policy that belongs to the broader class of universally measurable policies). Thus, starting with a Borel measurability framework for cost functions and policies, we quickly get outside that framework when executing DP algorithms, such as value and policy iteration. The broader framework of universal measurability, introduced in [BeS78], is required to correct this deficiency, in the absence of additional (fairly strong) assumptions.

† The name "rollout" (also called "policy rollout") was introduced by Tesauro and Galperin [TeG96] in the context of rolling the dice in the game of backgammon. In Tesauro's proposal, a given backgammon position is evaluated by "rolling out" many games starting from that position to the end of the game. To quote from the paper [TeG96]: "In backgammon parlance, the expected value of a po-

niques: they can be applied to deterministic and stochastic problems, finite and infinite horizon problems, discrete and continuous spaces problems, and mixtures thereof. Moreover, rollout is reliable, easy to implement, and can be used in conjunction with on-line replanning. It is also compatible with new and exciting technologies such as transformer networks and large language models (see Section 2.3.7).

As we have noted, rollout with a given base policy is simply the first iteration of the policy iteration algorithm starting from the base policy. Truncated rollout can be interpreted as an "optimistic" form of a single policy iteration, whereby a policy is evaluated inexactly, by using a limited number of value iterations; see the books [Ber20a], [Ber22a].†

Policy iteration, which can be seen as repeated rollout, is more ambitious and challenging than rollout. It requires off-line training, possibly in conjunction with the use of neural networks. Together with its neural network and distributed implementations, it will be discussed in more detail later. Note that rollout does not require any off-line training, once the base policy is available; this is its principal advantage over policy iteration.

Section 1.5: There is a vast literature on linear quadratic problems. The connection of policy iteration with Newton's method within this context was first derived by Kleinman [Kle68], as part of his doctoral research at MIT, under the supervision of M. Athans. Kleinman's work focused on continuous-time linear quadratic problems (see Hewer [Hew71] for the discrete-time case). For followup work, which relates to approximate policy iteration, see Feitzinger, Hylla, and Sachs [FHS09], and Hylla [Hyl11].

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sition is known as the "equity" of the position, and estimating the equity by Monte-Carlo sampling is known as performing a "rollout." This involves playing the position out to completion many times with different random dice sequences, using a fixed policy to make move decisions for both sides."

† Truncated rollout was also proposed in the context of backgammon in the paper [TeG96]. To quote from this paper: "Using large multi-layer networks to do full rollouts is not feasible for real-time move decisions, since the large networks are at least a factor of 100 slower than the linear evaluators described previously. We have therefore investigated an alternative Monte-Carlo algorithm, using so-called "truncated rollouts." In this technique trials are not played out to completion, but instead only a few steps in the simulation are taken, and the neural net's equity estimate of the final position reached is used instead of the actual outcome. The truncated rollout algorithm requires much less CPU time, due to two factors: First, there are potentially many fewer steps per trial. Second, there is much less variance per trial, since only a few random steps are taken and a real-valued estimate is recorded, rather than many random steps and an integer final outcome. These two factors combine to give at least an order of magnitude speed-up compared to full rollouts, while still giving a large error reduction relative to the base player." Analysis and computational experience with truncated rollout since 1996 are consistent with the preceding assessment.

The general relation of approximation in value space with Newton's method, beyond policy iteration, and its connections with MPC and adaptive control was first presented in the author's book [Ber20a], the papers [Ber21b], [Ber22c], and in the book [Ber22a], which contains an extensive discussion. This relation provides the starting point for an in-depth understanding of the synergy between the off-line training and the on-line play components of the approximation in value space architecture, including the role of multistep lookahead in enhancing the starting point of the Newton step. The monograph [Ber22a] (Appendix A) also provides analysis of variants of Newton's method for nondifferentiable fixed point problems, such as the ones arising in Bellman's equation (which involves a nondifferentiable right-hand side in finite-control space problems, among others).

Note that in approximation in value space, we are applying Newton's method to the solution of a system of equations (the Bellman equation). This context has no connection with the "gradient descent" methods that are popular for the solution of special types of optimization problems in RL, arising for example in neural network training problems (see Chapter 3). In particular, there are no gradient descent methods that can be used for the solution of systems of equations such as the Bellman equation. There are, however, "first order" deterministic algorithms such as the Gauss-Seidel and Jacobi methods (and stochastic asynchronous extensions) that can be applied to the solution of systems of equations with special structure, including Bellman equations. Such methods include various Q-learning algorithms, which are discussed in the neuro-dynamic programming book by Bertsekas and Tsitsiklis [BeT89], as well as the recent book by Meyn [Mey22]. While these methods can be useful, they are much slower than Newton's method and have limited utility in the context of on-line play.

Section 1.6: Many applications of DP are discussed in the 1st volume of the author's DP book [Ber17a]. This book also covers a broad variety of state augmentation and problem reformulation techniques, including the mathematics of how problems with imperfect state information can be transformed to perfect state information problems. In Section 1.6 we have aimed to provide an overview, with an emphasis on the use of approximations. In what follows we provide some related historical notes.

Multiagent problems: This subject has a long history (Marschak [Mar55], Radner [Rad62], Witsenhausen [Wit68], [Wit71a], [Wit71b]), and was researched extensively in the 70s; see the review paper by Ho [Ho80] and the references cited there. The names used at that time were *team theory* and *decentralized control*. For a sampling of subsequent works in team theory and multiagent optimization, we refer to the papers by Krainak, Speyer, and Marcus [KLM82a], [KLM82b], and de Waal and van Schuppen [WaS00]. For more recent works, see Nayyar, Mahajan, and Teneketzis [NMT13], Nayyar and Teneketzis [NaT19], Li et al. [LTZ19], Qu and Li [QuL19], Gupta [Gup20], the book by Zoppoli, Sanguineti, Gnecco, and

Parisini [ZSG20], and the references quoted there. In addition to the aforementioned works, surveys of multiagent DP from an RL perspective were given by Busoniu, Babuska, and De Schutter [BBD08], [BBD10b].

The term "multiagent" has been used with various meanings in the literature. Some authors emphasize scenarios where agents lack common information when making their decisions, leading to sequential decision problems with "nonclassical information patterns." These problems are particularly complex because they cannot be solved using exact DP techniques. Other authors focus on situations where the agents are "weakly" coupled through the system equation, the cost function, or the constraints. They consider methods that exploit the weak coupling to address the problem with (suboptimal) decoupled computations.

Agent-by-agent minimization in multiagent approximation in value space and rollout was proposed and theoretically justified in the author's paper [Ber19c], which also discusses extensions to infinite horizon policy iteration algorithms, and explores connections with the concept of person-by-person optimality from team theory; see also the textbook [Ber20a], the papers [Ber19d], [Ber20b] for further elaboration. The papers by Bhattacharya et al. [BKB23], Di Gennaro et al. [DBF22], Di Gennaro et al. [DBP23], Macesker et al. [MPL23], and Weber et al. [WGP23], Garces et al. [GBB24], present computational studies with challenging problems, where several of the multiagent algorithmic ideas were adapted, tested, and validated. These papers consider large-scale multi-robot and vehicle routing problems, involving partial state information, and explore some of the attendant implementation issues, including autonomous multiagent rollout, through the use of policy neural networks and other precomputed signaling policies. They also compare the performance of these multiagent methods against alternative approaches, including those based on policy gradient techniques.

A different form of distributed computation and multiagent optimization, where each agent has a partial/local model of the system within part of the state space and relies on aggregate information from other agents for DP computations, is proposed in the author's DP book [Ber12], Section 6.5.4; see also Section 3.6.8 of the present book.

Adaptive control: The research on adaptive control has a long history and its literature is very extensive; see the books by Aström and Wittenmark [AsW94], Aström and Hagglund [AsH06], Bodson [Bod20], Goodwin and Sin [GoS84], Ioannou and Sun [IoS96], Jiang and Jiang [JiJ17], Krstic, Kanellakopoulos, and Kokotovic [KKK95], Kumar and Varaiya [KuV86], Liu, et al. [LWW17], Lavretsky and Wise [LaW13], Narendra and Annaswamy [NaA12], Sastry and Bodson [SaB11], Slotine and Li [SlL91], and Vrabie, Vamvoudakis, and Lewis [VVL13]. These books describe a vast array of methods spanning 60 years, and ranging from adaptive and PID model-free approaches, to simultaneous or sequential control and identifi-

cation, to time series models, to extremum-seeking methods, to simulation-based RL techniques, etc.

The ideas of PID control have been applied widely to adaptive and robust control contexts, and have a long history; see the books by Aström and Hagglund [AsH95], [AsH06], which provide many references. According to Wikipedia, "a formal control law for what we now call PID or three-term control was first developed using theoretical analysis, by Russian American engineer Nicolas Minorsky" in 1922 [Min22].

The DP framework for adaptive control was introduced in a series of papers by Feldbaum, starting in 1960 with [Fel60], under the name *dual control theory*. These papers emphasized the division of effort between system estimation and control, now more commonly referred to as the *exploration-exploitation tradeoff*. In the last paper of the series [Fel63], Feldbaum prophetically concluded as follows: "At the present time, the most important problem for the immediate future is the development of approximate solution methods for dual control theory problems, the formulation of sub-optimal strategies, the determination of the numerical value of risk in quasi-optimal systems and its comparison with the value of risk in existing systems."

The research on problems involving unknown models and using data for model identification simultaneously with control was rekindled with the advent of the AI side of RL and its focus on the active exploration of the environment. Here there is emphasis on "learning from interaction with the environment" [SuB18] through the use of (possibly hidden) Markov decision models, machine learning, and neural networks, in a wide array of methods that are under active development at present. This is more or less the same as the classical problems of dual and adaptive control that have been discussed in the control literature since the 60s.

The formulation of adaptive and dual control problems as POMDP (cf. Section 2.11) is classical. The use of rollout within this context was first suggested in the author's book [Ber22a], Section 6.7.

Model predictive control: The idea underlying MPC, on-line optimization with a truncated rolling horizon and a terminal cost function approximation, has arisen in several contexts, motivated by different types of applications. It has been part of the folklore of the control theory and operations research literature, dating to the 1960s and 1970s. Simultaneously, it was used in important chemical process control applications, where the name "model predictive control" (or "model-based predictive control") and the related name "dynamic matrix control" were introduced. The term "predictive" arises often in this path breaking literature, and generally refers to taking into account the system's future, while applying control in the present. Related ideas appeared independently in the computer science literature, in contexts of heuristic search, planning, and game playing. The history of the subject within the decision and control domain is recounted

in the paper by Morari [Mor25].

The literature on the theory and applications of MPC is voluminous. Some early widely cited papers are Richalet et al. [RRT78], Cutler and Ramaker [CuR80], Rouhani and Mehra [RoM82], Clarke, Mohtadi, and Tuffs [CMT87a], [CMT87b], Keerthi and Gilbert [KeG88], Mayne and Michalska [MaM88], Rawlings and Muske [RaM93], Mayne et al. [MRR00]. For early surveys, see Morari and Lee [MoL99], and Findeisen et al. [FIA03]. For a more recent review, which addresses issues of tube MPC and robustness among others, see Mayne [May14]. Textbooks on MPC include Maciejowski [Mac02], Goodwin, Seron, and De Dona [GSD06], Camacho and Bordons [CaB07], Kouvaritakis and Cannon [KoC16], Borrelli, Bemporad, and Morari [BBM17], Rawlings, Mayne, and Diehl [RMD17]. The textbook by Stephanopoulos [Ste25] focuses on applications of MPC and related algorithms in process and biological systems control. The connections between MPC and rollout were discussed in the author's paper [Ber05a]. For an extensive survey of the DP/Newton step framework of this chapter, as applied to MPC, see the author's paper [Ber24].

Reinforcement Learning Sources

The first DP/RL books were written in the 1990s, setting the tone for subsequent developments in the field. One in 1996 by Bertsekas and Tsitsiklis [BeT96], which reflects a decision, control, and optimization viewpoint, and another in 1998 by Sutton and Barto, which is culturally different and reflects an artificial intelligence viewpoint (a 2nd edition, [SuB18], was published in 2018). We refer to the former book and also to the author's DP textbooks [Ber12], [Ber17a] for a broader discussion of some of the topics of this book, including algorithmic convergence issues and additional DP models, such as those based on average cost and semi-Markov problem optimization. Note that both of these books deal with finite-state Markovian decision models and use a transition probability notation, as they do not address continuous spaces problems, which are one of the major focal points of this book.

More recent books are by Gosavi [Gos15] (a much expanded 2nd edition of his 2003 monograph), which emphasizes simulation-based optimization and RL algorithms, Cao [Cao07], which focuses on a sensitivity approach to simulation-based methods, Chang, Fu, Hu, and Marcus [CHF13] (a 2nd edition of their 2007 monograph), which emphasizes finite-horizon/multistep lookahead schemes and adaptive sampling, Busoniu, Babuska, De Schutter, and Ernst [BBD10a], which focuses on function approximation methods for continuous space systems and includes a discussion of random search methods, Szepesvari [Sze10], which is a short monograph that selectively treats some of the major RL algorithms such as temporal differences, armed bandit methods, and Q-learning, Powell [Pow11], which emphasizes resource allocation and operations research applications, Powell and Ryzhov [PoR12], which focuses on specialized topics

in learning and Bayesian optimization, Vrabie, Vamvoudakis, and Lewis [VVL13], which discusses neural network-based methods and on-line adaptive control, Kochenderfer et al. [KAC15], which selectively discusses applications and approximations in DP and the treatment of uncertainty, Jiang and Jiang [JiJ17], which addresses adaptive control and robustness issues within an approximate DP framework, Liu, Wei, Wang, Yang, and Li [LWW17], which deals with forms of adaptive dynamic programming, and topics in both RL and optimal control, and Zoppoli, Sanguineti, Gnecco, and Parisini [ZSG20], which addresses neural network approximations in optimal control as well as multiagent/team problems with nonclassical information patterns. The book by Meyn [Mey22] focuses on the connections of RL and optimal control, from a mathematical perspective, and treats stochastic problems and algorithms in more detail.

There are also several books that, while not exclusively focused on DP and/or RL, touch upon some of the topics of the present book. The book by Borkar [Bor08] is an advanced monograph that addresses rigorously many of the convergence issues of iterative stochastic algorithms in approximate DP, mainly using the so-called ODE approach. The book by Meyn [Mey07] is broader in its coverage, but discusses some of the popular approximate DP/RL algorithms. The book by Haykin [Hay08] discusses approximate DP in the broader context of neural network-related subjects. The book by Krishnamurthy [Kri16] focuses on partial state information problems, with a discussion of both exact DP, and approximate DP/RL methods. The textbooks by Kouvaritakis and Cannon [KoC16], Borrelli, Bemporad, and Morari [BBM17], and Rawlings, Mayne, and Diehl [RMD17] collectively provide a comprehensive view of the MPC methodology. The book by Lattimore and Szepesvari [LaS20] is focused on multiarmed bandit methods. The book by Brandimarte [Bra21] is a tutorial introduction to DP/RL that emphasizes operations research applications and includes MATLAB codes. The book by Hardt and Recht [HaR21] focuses on broader subjects of machine learning and covers selectively approximate DP and RL topics.

The present book is similar in style, terminology, and notation to the author's recent textbooks [Ber19a] (Reinforcement Learning and Optimal Control), [Ber20a] (Rollout and Policy Iteration), [Ber22a] (Lessons from AlphaZero), and the 3rd edition of the abstract DP monograph [Ber22b], which collectively provide a fairly comprehensive and more mathematical account of the subject. In particular, the book [Ber19a] includes a broader coverage of approximation in value space methods, including certainty equivalent control and aggregation methods. It also addresses approximation in policy space in greater detail than the present book. The book [Ber20a] focuses more closely on rollout, policy iteration, and multiagent problems, and introduced the connection of approximation in value space with Newton's method. The book [Ber22a] focuses primarily on this connection, relying on analysis first provided in the book [Ber20a] and the paper [Ber22c]. The abstract DP monograph [Ber22b] (a 3rd edition of

the original 2013 1st edition) is an advanced treatment of exact DP, which provides the mathematical framework of Bellman operators that are central for some of the Newton method visualizations presented in the present book and in the books [Ber20a], [Ber22a].

In addition to textbooks, there are many surveys and short research monographs relating to our subject, which are rapidly multiplying in number. Influential early surveys were written, from an AI viewpoint, by Barto, Bradtke, and Singh [BBS95] (which dealt with the methodologies of real-time DP and its antecedent, real-time heuristic search [Kor90], and the use of asynchronous DP ideas [Ber82a], [Ber83], [BeT89] within their context), and by Kaelbling, Littman, and Moore [KLM96] (which focused on general principles of RL).

Several overview papers in the volume by Si, Barto, Powell, and Wunsch [SBP04] describe some approximation methods that we will not be covering in much detail in this book: linear programming approaches (De Farias [DeF04]), large-scale resource allocation methods (Powell and Van Roy [PoV04]), and deterministic optimal control approaches (Si, Yang, and Liu [SYL04]). Updated accounts of these and other related topics are given in the survey collection by Lewis and Liu [LeL13].

Recent extended surveys and short monographs are Borkar [Bor08] (a methodological point of view that explores connections with other Monte Carlo schemes), Lewis and Vrabie [LeV09] (a control theory point of view), Szepesvari [Sze10] (which discusses approximation in value space from a RL point of view), Deisenroth, Neumann, and Peters [DNP11], and Grondman et al. [GBL12] (which focus on policy gradient methods), Browne et al. [BPW12] (which focuses on Monte Carlo Tree Search), Geffner and Bonet [GeB13], and Moerland et al. [MBP23] (which deal with methods related to search and automated planning), Schmidhuber [Sch15], Arulkumaran et al. [ADB17], Li [Li17], and Busoniu et al. [BDT18] (which deal with reinforcement learning schemes that are based on the use of deep neural networks), and the author's [Ber05a] (which focuses on rollout algorithms and MPC), [Ber11a] (which focuses on approximate policy iteration), [Ber18a] (which focuses on aggregation methods), [Ber20b] (which focuses on multiagent problems), and [Ber24] (which focuses on the relations between RL and MPC).

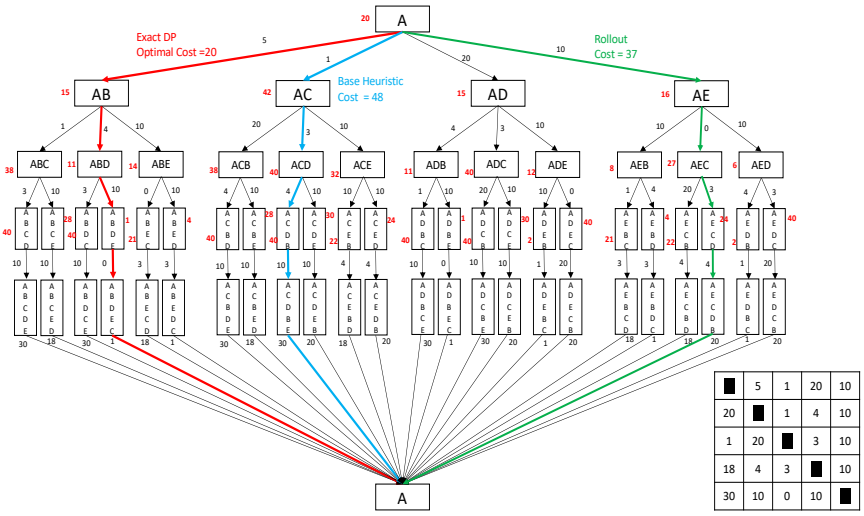

Figure 1.8.1 Solution of parts (a), (b), and (c) of Exercise 1.1. A 5-city traveling salesman problem illustration of rollout with the nearest neighbor base heuristic.

E X E R C I S E S

1.1 (Computational Exercise - Traveling Salesman Problem)

Consider a modified version of the four-city traveling salesman problem of Example 1.2.3, where there is a fifth city E. The intercity travel costs are shown in Fig. 1.8.1, which also gives the solutions to parts (a), (b), and (c).

(a) Use exact DP with starting city A to verify that the optimal tour is ABDECA with cost 20.

(b) Verify that the nearest neighbor heuristic starting with city A generates the tour ACDBEA with cost 48.

(c) Apply rollout with one-step lookahead minimization, using as base heuristic the nearest neighbor heuristic. Show that it generates the tour AECDBA with cost 37.
Illustration of the algorithm: At city A, the nearest neighbor heuristic generates the tour ACDBEA with cost 48, as per part (b). At city A, the rollout algorithm considers the four options of moving to cities B, C, D, E, or equivalently to states AB, AC, AD, AE, and it computes the nearest neighbor-generated tours corresponding to each of these states. These tours are ABCDEA with cost 49, ACDBEA with cost 48, ADCEBA with cost

63, and AECDBA with cost 37. The tour AECDBA has the least cost, so the rollout algorithm moves to city E or equivalently to state AE.

At AE, the rollout algorithm considers the three options of moving to cities B, C, D, or equivalently to states AEB, AEC, AED, and it computes the nearest neighbor-generated tours corresponding to each of these states. These tours are AEBCDA with cost 42, AECDBA with cost 37, AEDCBA with cost 63. The tour AECDBA has the least cost, so the rollout algorithm moves to city C or equivalently to state AEC.

At AEC, the rollout algorithm considers the two options of moving to cities B, D, and compares the nearest neighbor-generated tours corresponding to each of these. These tours are AECBDA with cost 52 and AECDBA with cost 37. The tour AECDBA has the least cost, so the rollout algorithm moves to city D or equivalently to state AECD. Then the rollout algorithm has only one option and generates the tour AECDBA with cost 37.

(d) Apply rollout with two-step lookahead minimization, using as base heuristic the nearest neighbor heuristic. This rollout algorithm operates as follows. For $k = 1, 2, 3$, it starts with a k-city partial tour, it generates every possible two-city addition to this tour, uses the nearest neighbor heuristic to complete the tour, and selects as next city to add to the k-city partial tour the city that corresponds to the best tour thus obtained (only one city is added to the current tour at each step of the algorithm, not two). Show that this algorithm generates the optimal tour.

(e) Estimate roughly the complexity of the computations in parts (b) and (c), assuming that we are given the graph of the N-city traveling salesman problem. *Answer*: The nearest neighbor heuristic that starts at city A performs $O(N)$ comparisons at each of N stages, so it requires $O(N^2)$ computation. The rollout algorithm at stage k runs the nearest neighbor heuristic $N - k$ times, so it must run the heuristic $O(N^2)$ times for a total computation of $O(N^4)$. Thus the rollout algorithm's complexity involves only a low order polynomial increase over the complexity of the base heuristic, something that is typical for practical discrete optimization problems. Note that even though this may represent a substantial increase in computation over the base heuristic, it is a potentially enormous improvement over the complexity of the exact DP algorithm.

1.2 (Computational Exercise - Linear Quadratic Problem)

In this problem we focus on the one-dimensional linear quadratic problem of Section 1.5 and the interpretation of approximation space as a Newton step for solving the Riccati equation. Consider the undiscounted linear quadratic problem with parameters $a = 2$, $b = 1$, $q = 1$, $r = 5$. For this problem:

(a) Plot and solve graphically the Riccati equation as in Fig. 1.5.1.

(b) Plot and solve graphically the Riccati equation corresponding to the linear policy $\mu(x) = -(3/2)x$.

(c) Plot graphically the numerical solution of the Riccati equation by value iteration as in Fig. 1.5.3, using a starting point $K_0 < K^*$ and a starting point $K_0 > K^*$.

(d) Interpret graphically approximation in value space with one-step, two-step, and three-step lookahead as a Newton step in the manner of Figs. 1.5.7 and 1.5.8. Use cost function approximations $\tilde{K}x^2$ with $\tilde{K} < K^*$ and $\tilde{K} > K^*$. What is the region of stability, i.e., the set of $\tilde{K}$ for which approximation in value space produces a stable policy under one-step, two-step, and three-step lookahead.

(e) Plot the performance error $|K_{\tilde{\mu}} - K^*|$ as a function of $|\tilde{K} - K^*|$ for one-step, two-step, and three-step lookahead approximation in value space.

(f) Plot graphical interpretations of rollout and truncated rollout in the manner of Figs. 1.5.10 and 1.5.11 using a stable starting linear policy of your choice.

1.3 (Computational Exercise - Spiders and Flies)

Consider the spiders and flies problem of Example 1.6.5 with two differences: the five flies stay still (rather than moving randomly), and there are only two spiders, both of which start at the fourth square from the right at the top row of the grid of Fig. 1.6.10. The base policy is to move each spider one square towards its nearest fly, with distance measured by the Manhattan metric, and with preference given to a horizontal direction over a vertical direction in case of a tie. Apply the multiagent rollout algorithm of Section 1.6.7, and compare its performance with the one of the ordinary rollout algorithm, and with the one of the base policy. This problem is also discussed in Section 2.9.

1.4 (Computational Exercise - Exercising an Option)

This exercise deals with a computational comparison of the optimal policy, a heuristic policy, and on-line approximation in value space using the heuristic policy, in the context of a problem that involves the timing of the sale of a stock.

An investor has the option to sell a given amount of stock at any one of N time periods. The initial price of the stock is an integer x_0. The price x_k, if it is positive and it is less than a given positive integer value $\bar{x}$, it evolves according to

$$x_{k+1} = \begin{cases} x_k + 1 & \text{with probability } p^+, \\ x_k & \text{with probability } 1 - p^+ - p^-, \\ x_k - 1 & \text{with probability } p^-, \end{cases}$$

where p^+ and p^- have known values with

$$0 < p^- \leq p^+, \qquad p^+ + p^- < 1.$$

If $x_k = 0$, then x_{k+1} moves to 1 with probability p^+, and stays unchanged at 0 with probability $1 - p^+$. If $x_k = \bar{x}$, then x_{k+1} moves to $\bar{x} - 1$ with probability p^-, and stays unchanged at $\bar{x}$ with probability $1 - p^-$.

At each period $k = 0, \ldots, N-1$ for which the stock has not yet been sold, the investor (with knowledge of the current price x_k), can either sell the stock at the current price x_k or postpone the sale for a future period. If the stock has not been sold at any of the periods $k = 0, \ldots, N-1$, it must be sold at period N at price x_N. The investor wants to maximize the expected value of the sale. For the following computations, use reasonable values of your choice for N, p^+, p^-, $\bar{x}$, and x_0 (you should choose x_0 between 0 and $\bar{x}$). You are encouraged to experiment with different sets of values. A set of values that you may try first is

$$N = 14, \quad x_0 = 3, \quad \bar{x} = 7, \quad p^+ = p^- = 0.25.$$

(a) Formulate the problem as a finite horizon DP problem by identifying the state, control, and disturbance spaces, the system equation, the cost function, and the probability distribution of the disturbance. Write the corresponding exact DP algorithm, and use it to compute the optimal policy and the optimal cost as a function of x_0.
Solution: The optimal reward-to-go is generated by the following DP algorithm:

$$J_N^*(x_N) = x_N, \tag{1.96}$$

and for $k = 0, \ldots, N-1$, if $x_k = 0$, then

$$J_k^*(0) = p^+ J_{k+1}^*(1) + (1 - p^+) J_{k+1}^*(0), \tag{1.97}$$

if $x_k = \bar{x}$, then

$$J_k^*(\bar{x}) = \bar{x}, \tag{1.98}$$

(since the price cannot go higher than $\bar{x}$, once at $\bar{x}$, but can go lower), and if $0 < x_k < \bar{x}$, then

$$J_k^*(x_k) = \max\left\{x_k,\, p^+ J_{k+1}^*(x_k{+}1) + (1{-}p^+{-}p^-) J_{k+1}^*(x_k) + p^- J_{k+1}^*(x_k{-}1)\right\}. \tag{1.99}$$

The optimal policy is to sell at $x_k = 1, \ldots, \bar{x}-1$, if x_k attains the maximum in the above equation, and not to sell otherwise. When $x_k = 0$, it is optimal not to sell, while when $x_k = \bar{x}$, it is optimal to sell.

The values of $J_k^*(x_k)$ and the optimal policy are tabulated as shown in Fig. 1.8.2. For this figure, all the calculations are done for the following special case:

$$N = 10, \quad x_0 = 2, \quad \bar{x} = 10, \quad p^+ = p^- = 0.25.$$

These values are also used for parts (b) and (c). However, you are asked to solve the problem for different values as noted earlier. Note that for the problem to have an interesting solution, the problem data must be chosen so that the problem's policies are materially affected by the presence of the upper and lower bounds on the price x_k. As an example consider the case where

$$N = 10, \quad x_0 = 20, \quad \bar{x} = 40, \quad p^+ = p^- = 0.25.$$

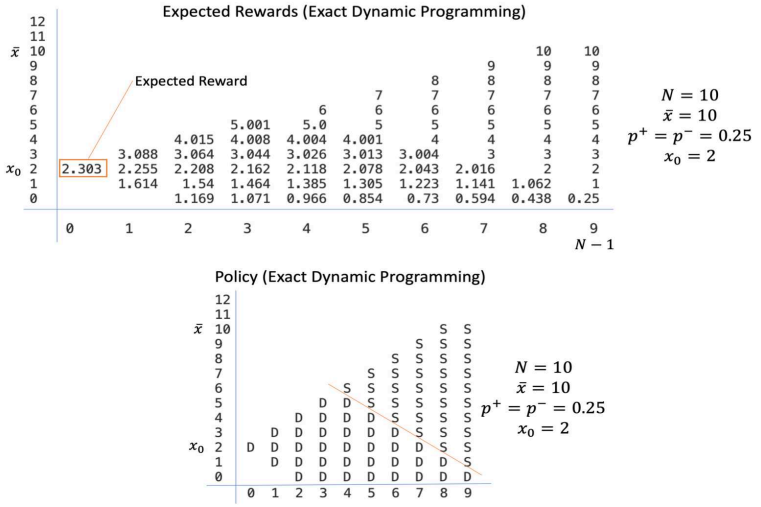

Figure 1.8.2 Table of values of optimal reward-to-go, obtained by exact DP, and corresponding optimal policy [cf. the algorithm (1.96)-(1.99). Only the states x_k that are reachable from x_0 at time k are considered (this is the state space for time k).

Then the bounds $0 \leq x_k$ and $x_k \leq \bar{x}$ never become "active," and it can be verified that the optimal expected reward is $J^*(x_0) = x_0$, while all policies are optimal and attain this optimal expected reward.

(b) Suppose the investor adopts a heuristic, referred to as base heuristic, whereby he/she sells the stock if its price is greater or equal to βx_0, where β is some number with $\beta > 1$. Write an exact DP algorithm to compute the expected value of the sale under this heuristic.
Solution: The reward-to-go for the base heuristic starting from state x_k, denoted $J_k^{x_k}(x_k)$, can be generated by the following (exact) DP algorithm. (Note here the use of superscript x_k in the quantities $J_n^{x_k}(x_n)$ computed by the algorithm. The reason is that the computed values $J_n^{x_k}(x_n)$ depend on x_k, which incidentally implies that base heuristic is not sequentially consistent, as defined later in Section 2.3.) The algorithm is given by

$$J_N^{x_k}(x_N) = x_N, \tag{1.100}$$

and for $n = k, \ldots, N-1$, if $0 < x_n < \beta x_k$, then

$$J_n^{x_k}(x_n) = p^+ J_{n+1}^{x_k}(x_n + 1) + (1 - p^+ - p^-) J_{n+1}^{x_k}(x_n) + p^- J_{n+1}^{x_k}(x_n - 1), \tag{1.101}$$

if $x_n = 0$, then

$$J_n^{x_k}(0) = p^+ J_{n+1}^{x_k}(1) + (1 - p^+) J_{n+1}^{x_k}(0), \tag{1.102}$$

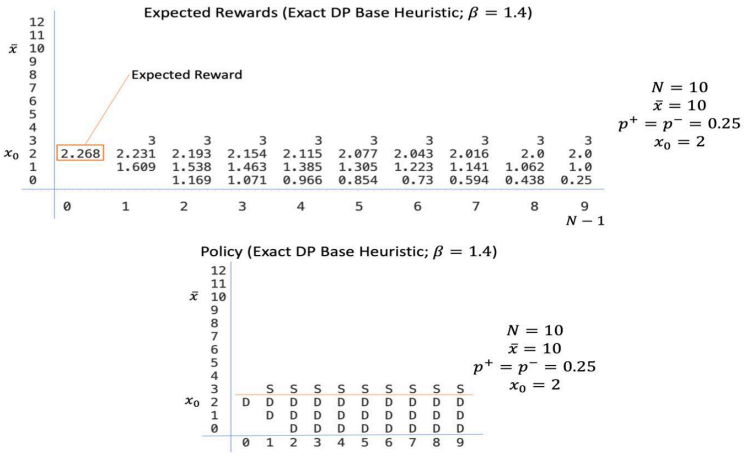

Figure 1.8.3 Table of rewards-to-go for the base policy with $\beta = 1.4$, starting from x_0 [cf. the algorithm (1.100)-(1.103) for $k = 0$].

and if $x_n \geq \beta x_k$, then

$$J_n^{x_k}(x_n) = x_n. \tag{1.103}$$

The values of $J_k^{x_k}(x_k)$ computed by this algorithm are shown in Fig. 1.8.3, together with the decisions applied by the base heuristic.

While the reward-to-go for the base heuristic starting from state x_k is very simple to compute for our problem, in order to apply the rollout algorithm only the values $J_{k+1}^{x_k+1}(x_k+1)$, $J_{k+1}^{x_k}(x_k)$, and $J_{k+1}^{x_k-1}(x_k-1)$ need to be calculated for each state x_k encountered during on-line operation. Moreover, the base heuristic's reward-to-go $J_k^{x_k}(x_k)$ can also be computed on-line by Monte Carlo simulation for the relevant states x_k. This would be the principal option in a more complicated problem where the exact DP algorithm is too time-consuming.

(c) Apply approximation in value space with one-step lookahead minimization and with function approximation that is based on the heuristic of part (b). In particular, use $\tilde{J}_N(x_N) = x_N$, and for $k = 1, \ldots, N-1$, use $\tilde{J}_k(x_k)$ that is equal to the expected value of the sale when starting at x_k and using the heuristic that sells the stock when its price exceeds βx_k. Use exact DP as well as Monte Carlo simulation to compute/approximate on-line the needed values $\tilde{J}_k(x_k)$. Compare the expected values of sale price computed with the optimal, heuristic, and approximation in value space methods.
Solution: The rollout policy $\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$ is determined by the base heuristic, where for every possible state x_k, and stage $k = 0, \ldots, N-1$, the rollout decision $\tilde{\mu}_k(x_k)$ is

$$\tilde{\mu}_k(x_k) = \text{sell at } x_k,$$

if

$$p^+ J_{k+1}^{x_k+1}(x_k+1) + (1-p^+-p^-)J_{k+1}^{x_k}(x_k) + p^- J_{k+1}^{x_k-1}(x_k-1) \leq x_k,$$

and

$$\tilde{\mu}_k(x_k) = \text{don't sell at } x_k,$$

otherwise. The sell or don't sell decision of the rollout algorithm is made on-line according to the preceding criterion, at each state x_k encountered during on-line operation.

Figure 1.8.4 shows the rollout policy, which is computed by the preceding equations using the rewards-to-go of the base heuristic $J_k^{x_k}(x_k)$, as given in Fig. 1.8.3. Once the rollout policy is computed, the corresponding reward function $\tilde{J}_k(x_k)$ can be calculated similar to the case of the base heuristic. Of course, during on-line operation, the rollout decision need only be computed for the states x_k encountered on-line.

The important observation when comparing Figs. 1.8.3 and 1.8.4 is that the rewards-to-go of the rollout policy are greater or equal to the ones for the base heuristic. In particular, starting from x_0, the rollout policy attains reward 2.269, and the base heuristic attains reward 2.268. The optimal policy attains reward 2.4. The rollout policy reward is slightly closer to the optimal than the base heuristic reward.

The rollout reward-to-go values shown in Fig. 1.8.4 are "exact," and correspond to the favorable case where the heuristic rewards needed at x_k, $J_{k+1}^{x_k+1}(x_k+1)$, $J_{k+1}^{x_k}(x_k)$, and $J_{k+1}^{x_k-1}(x_k-1)$, are computed exactly by DP or by infinite-sample Monte Carlo simulation.

When finite-sample Monte Carlo simulation is used to approximate the needed base heuristic rewards at state x_k, i.e., $J_{k+1}^{x_k+1}(x_k+1)$, $J_{k+1}^{x_k}(x_k)$, and $J_{k+1}^{x_k-1}(x_k-1)$, the performance of the rollout algorithm will be degraded. In particular, by using a computer program to implement rollout with Monte Carlo simulation, it can be shown that when $J_{k+1}^{x_k+1}(x_k+1)$, $J_{k+1}^{x_k}(x_k)$, and $J_{k+1}^{x_k-1}(x_k-1)$ are approximated using a 20-sample Monte-Carlo simulation per reward value, the rollout algorithm achieves reward 2.264 starting from x_0. This reward is evaluated by (almost exact) 400-sample Monte Carlo simulation of the rollout algorithm.

When $J_{k+1}^{x_k+1}(x_k+1)$, $J_{k+1}^{x_k}(x_k)$, and $J_{k+1}^{x_k-1}(x_k-1)$ were approximated using a 200-sample Monte-Carlo simulation per reward value, the rollout algorithm achieves reward 2.273 [as evaluated by (almost exact) 400-sample Monte Carlo simulation of the rollout algorithm]. Thus with 20-sample simulation, the rollout algorithm performs worse than the base heuristic starting from x_0. With the more accurate 200-sample simulation, the rollout algorithm performs better than the base heuristic starting from x_0, and performs nearly as well as the optimal policy (but still somewhat worse than in the case where exact values of the needed base heuristic rewards are used (based on an "infinite" number of Monte Carlo samples).

It is worth noting here that the heuristic is not a legitimate policy because at any state x_n it makes a decision that depends on the state x_k where it started. Thus the heuristic's decision at x_n depends not just on x_n, but also on the starting state x_k. However, the rollout algorithm is always an approximation in value space scheme with approximation reward $\tilde{J}_k(x_k)$ defined by the heuristic, and it provides a legitimate policy.

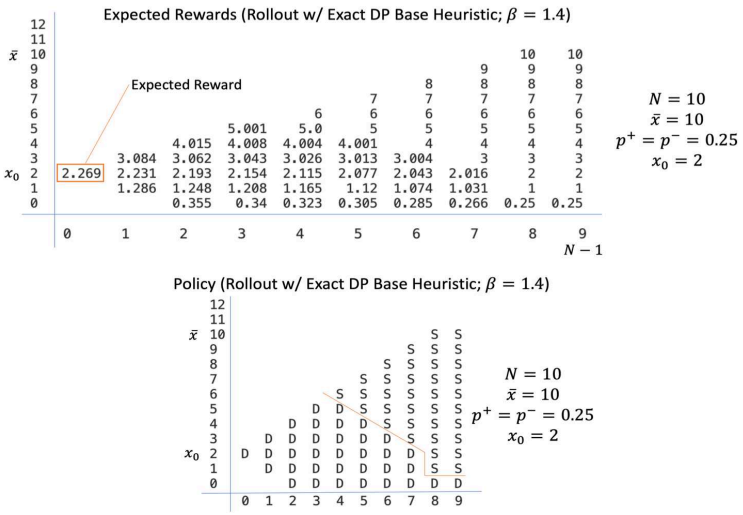

Figure 1.8.4 Table of values of reward-to-go and decisions applied by the rollout policy that corresponds to the base heuristic with $\beta = 1.4$.

(d) Repeat part (c) but with two-step instead of one-step lookahead minimization.

Answer: The implementation is very similar to the one-step lookahead case. The main difference is that at state x_k, the rollout algorithm needs to calculate the base heuristic reward values $J_{k+2}^{x_k+2}(x_k+2)$, $J_{k+2}^{x_k+1}(x_k+1)$, $J_{k+2}^{x_k}(x_k)$, $J_{k+2}^{x_k-1}(x_k-1)$, and $J_{k+2}^{x_k-2}(x_k-2)$. Thus the on-line Monte Carlo simulation work is accordingly increased. Generally the simulation work per stage of the rollout algorithm is proportional to $2\ell+1$, when ℓ-stage lookahead minimization is used, since the number of leaves at the end of the lookahead tree is $2\ell+1$.

1.5 (Computational Exercise - Linear Quadratic Problem)

In a more realistic version of the cruise control system of Example 1.3.1, the system has the form

$$x_{k+1} = ax_k + bu_k + w_k,$$

where the coefficient a satisfies $0 < a \leq 1$, and the disturbance w_k has zero mean and variance σ^2. The cost function has the form

$$(x_N - \bar{x}_N)^2 + \sum_{k=0}^{N-1} \big((x_k - \bar{x}_k)^2 + ru_k^2\big),$$

where $\bar{x}_0, \ldots, \bar{x}_N$ are given nonpositive target values (a velocity profile) that serve to adjust the vehicle's velocity, in order to maintain a safe distance from

the vehicle ahead, etc. In a practical setting, the velocity profile is recalculated by using on-line radar measurements.

Design an experiment to compare the performance of a fixed linear policy π, derived for a fixed nominal velocity profile, and the performance of the algorithm that uses on-line replanning, whereby the optimal policy π^* is recalculated each time the velocity profile changes. Compare with the performance of the rollout policy $\tilde{\pi}$ that uses π as the base policy and on-line replanning.

1.6 (Computational Exercise - Parking Problem)

In reference to Example 1.6.4, a driver aims to park at an inexpensive space on the way to his destination. There are L parking spaces available and a garage at the end. The driver can move in either direction. For example if he is in space i he can either move to $i-1$ with a cost $t-i$, or to $i+1$ with a cost $t+i$, or he can park at a cost $c(i)$ (if the parking space i is free). The only exception is when he arrives at the garage (indicated by index N) and he has to park there at a cost C. Moreover, after the driver visits a parking space he remembers its free/taken status and has an option to return to any parking space he has already visited. However, the driver must park within a given number of stages N, so that the problem has a finite horizon. The initial probability of space i being free is given, and the driver can observe the free/taken status of a parking only after he/she visits the space. Moreover, the free/taken status of a parking visited so far does not change over time.

Write a program to calculate the optimal solution using exact dynamic programming over a state space that is as small as possible. Try to experiment with different problem data, and try to visualize the optimal cost/policy with suitable graphical plots. Comment on the run-time as you increase the number of parking spots.

1.7 (Newton's Method for Solving the Riccati Equation)

The classical form of Newton's method applied to a scalar equation of the form $H(K) = 0$ takes the form

$$K_{k+1} = K_k - \left(\frac{\partial H(K_k)}{\partial K} \right)^{-1} H(K_k), \tag{1.104}$$

where $\frac{\partial H(K_k)}{\partial K}$ is the derivative of H, evaluated at the current iterate K_k. This exercise shows algebraically (rather than graphically), within the context of linear quadratic problems, that in approximation in value space with quadratic cost approximation, the cost function of the corresponding one-step lookahead policy is the result of a Newton step for solving the Riccati equation. To this end, we will apply Newton's method to the solution of the Riccati Eq. (1.42), which we write in the form $H(K) = 0$, where

$$H(K) = K - \frac{a^2 r K}{r + b^2 K} - q. \tag{1.105}$$

(a) Show that the operation that generates K_L starting from K is a Newton iteration of the form (1.104). In other words, show that for all K that lead to a stable one-step lookahead policy, we have

$$K_L = K - \left(\frac{\partial H(K)}{\partial K}\right)^{-1} H(K), \tag{1.106}$$

where

$$K_L = \frac{q + rL^2}{1 - (a + bL)^2} \tag{1.107}$$

is the quadratic cost coefficient of the one-step lookahead linear policy $\mu(x) = Lx$ corresponding to the cost function approximation $J(x) = Kx^2$:

$$L = -\frac{abK}{r + b^2K}. \tag{1.108}$$

Proof: Our approach for showing the Newton step formula (1.106) is to express each term in this formula in terms of L, and then show that the formula holds as an identity for all L. To this end, we first note from Eq. (1.108) that K can be expressed in terms of L as

$$K = -\frac{rL}{b(a + bL)}. \tag{1.109}$$

Furthermore, by using Eqs. (1.108) and (1.109), $H(K)$ as given in Eq. (1.105), can be expressed in terms of L as follows:

$$H(K) = -\frac{rL}{b(a + bL)} + \frac{arL}{b} - q. \tag{1.110}$$

Moreover, by differentiating the function H of Eq. (1.105), we obtain after a straightforward calculation

$$\frac{\partial H(K)}{\partial K} = 1 - \frac{a^2r^2}{(r + b^2K)^2} = 1 - (a + bL)^2, \tag{1.111}$$

where the second equation follows from Eq. (1.108). Having expressed all the terms in the Newton step formula (1.106) in terms of L through Eqs. (1.107), (1.109), (1.110), and (1.111), we can write this formula in terms of L only as

$$\frac{q + rL^2}{1 - (a + bL)^2} = -\frac{rL}{b(a + bL)} - \frac{1}{1 - (a + bL)^2}\left(-\frac{rL}{b(a + bL)} + \frac{arL}{b} - q\right),$$

or equivalently as

$$q + rL^2 = -\frac{rL\big(1 - (a + bL)^2\big)}{b(a + bL)} + \frac{rL}{b(a + bL)} - \frac{arL}{b} + q.$$

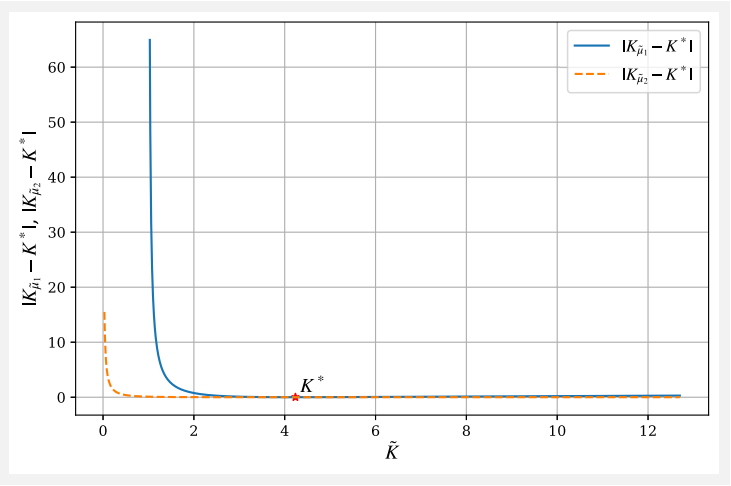

Figure 1.8.5 Illustration of the performance errors of the one-step and two-step lookahead policies as a function of $\tilde{K}$; cf. Exercise 1.8.

A straightforward calculation now shows that this equation holds as an identity for all L.

(b) What happens when K lies outside the region of stability?

(c) Show that in the case of ℓ-step lookahead, the analog of the quadratic convergence rate estimate has the form

$$|K_{\tilde{L}} - K^*| \leq c \left| F^{\ell-1}(\tilde{K}) - K^* \right|^2,$$

where $F^{\ell-1}(\tilde{K})$ is the result of the $(\ell-1)$-fold application of the mapping F to $\tilde{K}$. Thus a stronger bound for $|K_{\tilde{L}} - K^*|$ is obtained.

1.8 (Error Bounds and Region of Stability)

Consider a one-dimensional linear quadratic problem with problem data $a = 2$, $b = 1$, $q = 1$, $r = 1$.

(a) Plot the regions of stability for one-step, two-step, and four-step lookahead (cf. Fig. 1.5.9 in Section 1.5).

(b) Let $\tilde{\mu}_1$ and $\tilde{\mu}_2$ be the one-step and two-step lookahead policies for a given quadratic cost approximation coefficient $\tilde{K}$. Verify that the performance errors $|K_{\tilde{\mu}_1} - K^*|$ and $|K_{\tilde{\mu}_2} - K^*|$ as a function of $\tilde{K}$ are as shown in the plot of Fig. 1.8.5. Interpret the figure in terms of your results of part (a). Verify that longer lookahead expands the region of stability, and that the performance error increases sharply as $\tilde{K}$ approaches the boundary of the region of stability.

(c) Experiment with other problem data of your choice, including a range of values of a. Verify that for a system that is already stable ($|a| < 1$), the region of stability includes the entire nonnegative axis.

1.9 (Region of Stability and the Role of Multistep Lookahead)

In Section 1.5, we discussed the concept of the region of stability in the context of linear quadratic problems. The concept extends to far more general infinite horizon problems (see e.g., the book [Ber22a], Section 3.3). The idea is to call a stationary policy μ unstable if $J_\mu(x) = \infty$ for some states x, and call it stable otherwise. For $\ell \geq 1$, the ℓ-step region of stability is the set of cost function approximations $\tilde{J}$ for which the corresponding ℓ-step lookahead policy is stable.

Generally, the ℓ-step region of stability expands as ℓ increases. Note also that in finite-state discounted problems all policies are stable, so all $\tilde{J}$ belong to the region of stability. However, for SSP this is not so: there are policies, called *improper*, that do not terminate with positive probability for some initial states (this is very common, for example, in nonacyclic deterministic shortest path problems; see the books [Ber12] and [Ber22b] for extensive discussions). Such policies can be unstable. The following example illustrates difficult problems where the region of instability includes functions that are very close to J^*, even with large ℓ. This example involves small stage costs, a class of problems that pose challenges for approximation in value space; see Section 2.6.

Consider a deterministic shortest path problem with a single state 1, plus the termination state t. At state 1 we can either stay at that state at cost $\epsilon > 0$ or move to the state t at cost 1. Thus the optimal policy at state 1 is to move to t, the optimal cost $J^*(1) = 1$, and is the unique solution of Bellman's equation

$$J^*(1) = \min\big\{1,\, \epsilon + J^*(1)\big\}.$$

(In shortest path problems the optimal cost at t is 0 by assumption, and Bellman's equation involves only the costs of the states other than t.)

(a) Show that the one-step region of stability is the set of all $\tilde{J}(1) > 1 - \epsilon$. What happens in the case where $\tilde{J}(1) = 1 - \epsilon$? Show also that the ℓ-step region of stability is the set of all $\tilde{J}(1) > 1 - \ell\epsilon$. *Note*: The ℓ-step region of stability becomes arbitrarily large for sufficiently large ℓ. However, the boundary of the ℓ-step region of stability is arbitrarily close to $J^*(1)$ for sufficiently small ϵ.

(b) What happens in the case where there are additional states $i = 2, \ldots, n$, and for each of these states i there is the option to stay at i at cost ϵ or to move to $i - 1$ at cost 0? *Partial answer*: The one-step region of stability consists of all $\tilde{J} = \big(\tilde{J}(n), \ldots, \tilde{J}(1)\big)$ such that $\epsilon + \tilde{J}(i) > \tilde{J}(i-1)$ for all $i \geq 2$ and $\epsilon + \tilde{J}(1) > 1$.

2

Approximation in Value Space - Rollout Algorithms

Contents

2.1. Deterministic Finite Horizon Problems . . . . . p. 166
2.2. Approximation in Value Space - Deterministic Problems p. 173
2.3. Rollout Algorithms for Discrete Optimization . . . . . p. 177
2.3.1. Cost Improvement with Rollout - Sequential Consistency, Sequential Improvement . . . . . p. 183
2.3.2. The Fortified Rollout Algorithm . . . . . p. 189
2.3.3. Using Multiple Base Heuristics - Parallel Rollout . p. 192
2.3.4. Using Multiple Rollout Trajectories - General Rollout p. 194
2.3.5. Simplified Rollout Algorithms . . . . . p. 195
2.3.6. Truncated Rollout with Terminal Cost Approximation p. 196
2.3.7. Rollout with an Expert - Model-Free Rollout . . . p. 197
2.3.8. Using a World Model for Rollout . . . . . p. 202
2.3.9. Local Search with Rollout for Discrete Optimization p. 204
2.3.10. Inference in n-Grams, Transformers, HMMs, and Markov Chains . . . . . p. 208
2.4. Approximation in Value Space with Multistep Lookahead p. 221
2.4.1. Iterative Deepening Using Forward Dynamic Programming . . . . . p. 227
2.4.2. Incremental Multistep Rollout . . . . . p. 229
2.5. Constrained Forms of Rollout Algorithms . . . . . p. 232
2.5.1. Constrained Rollout for Discrete Optimization and Integer Programming . . . . . p. 245
2.6. Small Stage Costs and Long Horizon - Continuous-Time Rollout . . . . . p. 249

2.7. Approximation in Value Space - Stochastic Problems . p. 257
2.7.1. Simplified Rollout and Policy Iteration . . . . . p. 261
2.7.2. Certainty Equivalence Approximations . . . . . p. 262
2.7.3. Simulation-Based Implementation of the Rollout . . . . . Algorithm . . . . . p. 265
2.7.4. Variance Reduction in Rollout - Comparing Advantages p. 267
2.7.5. Monte Carlo Tree Search . . . . . p. 270
2.7.6. Randomized Policy Improvement by Monte Carlo . . . . Tree Search . . . . . p. 274
2.8. Rollout for Infinite-Spaces Problems - Optimization . . . . . Heuristics . . . . . p. 274
2.8.1. Rollout for Infinite-Spaces Deterministic Problems . p. 275
2.8.2. Rollout Based on Stochastic Programming . . . . p. 279
2.8.3. Stochastic Rollout with Certainty Equivalence . . . p. 281
2.9. Multiagent Rollout . . . . . p. 283
2.9.1. Asynchronous and Autonomous Multiagent Rollout p. 294
2.10. Rollout for Bayesian Optimization and Sequential . . . . . Estimation . . . . . p. 297
2.11. Adaptive Control by Rollout with a POMDP Formulation p. 309
2.12. Minimax Control and Reinforcement Learning . . . . p. 317
2.12.1. Exact Dynamic Programming for Minimax Problems p. 318
2.12.2. Minimax Approximation in Value Space and Rollout p. 321
2.12.3. Combined Approximation in Value and Policy Space . for Minimax Control . . . . . p. 327
2.12.4. A Meta Algorithm for Computer Chess Based on . Reinforcement Learning . . . . . p. 328
2.12.5. Combined Approximation in Value and Policy Space . for Sequential Noncooperative Games . . . . . p. 332
2.13. Notes, Sources, and Exercises . . . . . p. 334

In this chapter, we discuss various aspects of algorithms that are based on prediction of future costs starting from the current state, such as approximation in value space and rollout algorithms. We focus primarily on the case where the state and control spaces are finite. In Sections 2.1-2.6, we consider finite horizon deterministic problems, which in addition to arising often in practice, offer some important advantages in the context of RL. In particular, a finite horizon is well suited for the use of rollout, while the deterministic character of the problem eliminates the need for costly on-line Monte Carlo simulation.

An interesting aspect of our methodology for discrete deterministic problems is that it admits extensions that we have not discussed so far. The extensions include multistep lookahead variants, as well as variants that apply to constrained forms of DP, which involve constraints on the entire system trajectory, and also allow the use of heuristic algorithms that are more general than policies within the context of rollout. These variants rely on the problem's deterministic structure, and do not extend to stochastic problems.

Another interesting aspect of finite state deterministic problems is that they can serve as a framework for an important class of commonly encountered discrete optimization problems, including integer programming and combinatorial optimization problems such as scheduling, assignment, routing, etc. This brings to bear the methodology of approximation in value space, rollout, adaptive control, and MPC, and provides effective suboptimal solution methods for these problems.

In Sections 2.7-2.11, we consider various problems that involve stochastic uncertainty. In Section 2.12, we consider minimax problems that involve set membership uncertainty. The present chapter draws heavily on Chapters 2 and 3 of the book [Ber20a], and Chapter 6 of the book [Ber22a]. These books may be consulted for more details and additional examples.

While our focus in this chapter will be on finite horizon problems, our discussion applies to infinite horizon problems as well, because approximation in value space and rollout are essentially finite-stages algorithms, while the nature of the original problem horizon (be it finite or infinite) affects only the terminal cost function approximation. Thus in implementing one-step or multistep approximation in value space, it makes little difference whether the original problem has finite or infinite horizon. At the same time, for conceptual purposes, we can argue that finite horizon problems, even when they involve a nonstationary system and cost per stage, can be transformed to infinite horizon problems, by introducing an artificial cost-free termination state that the system moves into at the end of the horizon; see Sections 1.6.3 and 1.6.4. Through this transformation, the synergy of off-line training and on-line play based on Newton's method is brought to bear, and the insights that we discussed in Chapter 1 in the context of an infinite horizon apply and explain the good performance of our methods in practice.

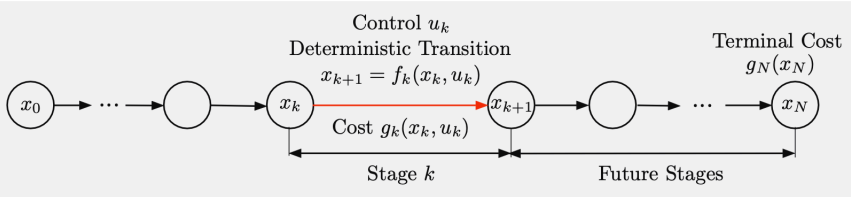

Figure 2.1.1 Illustration of a deterministic N-stage optimal control problem. Starting from state x_k, the next state under control u_k is generated nonrandomly, according to

$$x_{k+1} = f_k(x_k, u_k),$$

and a stage cost $g_k(x_k, u_k)$ is incurred.

2.1 DETERMINISTIC FINITE HORIZON PROBLEMS

We recall from Chapter 1, Section 1.2, that in deterministic finite horizon DP problems, the state is generated nonrandomly over N stages, through a system equation of the form

$$x_{k+1} = f_k(x_k, u_k), \qquad k = 0, 1, \ldots, N-1, \tag{2.1}$$

where k is the time index, and

x_k is the state of the system, an element of some state space X_k,

u_k is the control or decision variable, to be selected at time k from some given set $U_k(x_k)$, a subset of a control space U_k, which depends on x_k,

f_k is a function of (x_k, u_k) that describes the mechanism by which the state is updated from time k to time $k+1$.

The state space X_k and control space U_k are arbitrary sets and may depend on k. Similarly, the system function f_k can be arbitrary and may depend on k. The cost incurred at time k is denoted by $g_k(x_k, u_k)$, and the function g_k may depend on k. For a given initial state x_0, the total cost of a control sequence $\{u_0, \ldots, u_{N-1}\}$ is

$$J(x_0; u_0, \ldots, u_{N-1}) = g_N(x_N) + \sum_{k=0}^{N-1} g_k(x_k, u_k), \tag{2.2}$$

where $g_N(x_N)$ is a terminal cost incurred at the end of the process. This is a well-defined number, since the control sequence $\{u_0, \ldots, u_{N-1}\}$ together with x_0 determines exactly the state sequence $\{x_1, \ldots, x_N\}$ via the system equation (2.1); see Figure 2.1.1. We want to minimize the cost (2.2) over all sequences $\{u_0, \ldots, u_{N-1}\}$ that satisfy the control constraints, thereby obtaining the optimal value as a function of x_0

$$J^*(x_0) = \min_{\substack{u_k \in U_k(x_k) \\ k=0,\ldots,N-1}} J(x_0; u_0, \ldots, u_{N-1}).$$

Notice an important difference from the stochastic case: we optimize over sequences of controls $\{u_0, \ldots, u_{N-1}\}$, rather than over policies that consist of a sequence of functions $\pi = \{\mu_0, \ldots, \mu_{N-1}\}$, where μ_k maps states x_k into controls $u_k = \mu_k(x_k)$, and satisfies the control constraints $\mu_k(x_k) \in U_k(x_k)$ for all x_k. It is well-known that in the presence of stochastic uncertainty, policies are more effective than control sequences, and can result in improved cost. On the other hand, for deterministic problems, minimizing over control sequences yields the same optimal cost as over policies, since the cost of any policy starting from a given state determines with certainty the controls applied at that state and the future states, and hence can also be achieved by the corresponding control sequence. This point of view allows more general forms of rollout, which we will discuss in this chapter: instead of using a policy for rollout, we will allow the use of more general heuristics for choosing future controls.

The DP algorithm for finite horizon deterministic problems was derived in Section 1.2. It constructs functions

$$J_0^*(x_0), \ldots, J_{N-1}^*(x_{N-1}), J_N^*(x_N),$$

sequentially, starting from J_N^*, and proceeding backwards to J_{N-1}^*, J_{N-2}^*, etc. The value $J_k^*(x_k)$ will be viewed as the optimal cost of the tail subproblem that starts at state x_k at time k and ends at some state x_N.

DP Algorithm for Deterministic Finite Horizon Problems

Start with

$$J_N^*(x_N) = g_N(x_N), \qquad \text{for all } x_N, \tag{2.3}$$

and for $k = 0, \ldots, N-1$, let

$$J_k^*(x_k) = \min_{u_k \in U_k(x_k)} \Big[g_k(x_k, u_k) + J_{k+1}^*\big(f_k(x_k, u_k)\big) \Big], \qquad \text{for all } x_k. \tag{2.4}$$

The key fact about the DP algorithm is that for every initial state x_0, the number $J_0^*(x_0)$ obtained at the last step, is equal to the optimal cost $J^*(x_0)$. Indeed, a more general fact was shown in Section 1.2, namely that for all $k = 0, 1, \ldots, N-1$, and all states x_k at time k, we have

$$J_k^*(x_k) = \min_{\substack{u_m \in U_m(x_m) \\ m=k,\ldots,N-1}} J(x_k; u_k, \ldots, u_{N-1}), \tag{2.5}$$

where $J(x_k; u_k, \ldots, u_{N-1})$ is the cost generated by starting at x_k and using subsequent controls $u_k, \ldots, u_{N-1}$:

$$J(x_k; u_k, \ldots, u_{N-1}) = g_N(x_N) + \sum_{t=k}^{N-1} g_t(x_t, u_t).$$

Thus, $J_k^*(x_k)$ is the optimal cost for an $(N-k)$-stage tail subproblem that starts at state x_k and time k, and ends at time N. Based on this interpretation of $J_k^*(x_k)$, we have called it the *optimal cost-to-go* from state x_k at stage k, and refer to J_k^* as the *optimal cost-to-go function* or *optimal cost function* at time k.

We have also discussed in Section 1.2 the construction of an optimal control sequence. Once the functions $J_0^*, \ldots, J_N^*$ have been obtained, we can use a forward algorithm to construct an optimal control sequence $\{u_0^*, \ldots, u_{N-1}^*\}$ and state trajectory $\{x_1^*, \ldots, x_N^*\}$ for a given initial state x_0.

Construction of Optimal Control Sequence $\{u_0^*, \ldots, u_{N-1}^*\}$

Set

$$u_0^* \in \arg \min_{u_0 \in U_0(x_0)} \Big[g_0(x_0, u_0) + J_1^*\big(f_0(x_0, u_0)\big) \Big],$$

and

$$x_1^* = f_0(x_0, u_0^*).$$

Sequentially, going forward, for $k = 1, 2, \ldots, N-1$, set

$$u_k^* \in \arg \min_{u_k \in U_k(x_k^*)} \Big[g_k(x_k^*, u_k) + J_{k+1}^*\big(f_k(x_k^*, u_k)\big) \Big], \qquad (2.6)$$

and

$$x_{k+1}^* = f_k(x_k^*, u_k^*).$$

Note an interesting conceptual division of the optimal control sequence construction: there is *off-line training* to obtain J_k^* by precomputation [cf. the DP Eqs. (2.3)-(2.4)], which is followed by *on-line play* to obtain u_k^* [cf. Eq. (2.6)]. This is analogous to the two algorithmic processes described in Section 1.1 in connection with computer chess and backgammon.

Finite-State Deterministic Problems

For the first five sections of this chapter, we will consider the case where the state and control spaces are discrete and consist of a finite number of elements. As we have noted in Section 1.2, such problems can be described with an acyclic graph specifying for each state x_k the possible transitions to next states x_{k+1}. The nodes of the graph correspond to states x_k and the arcs of the graph correspond to state-control pairs (x_k, u_k). Each arc with start node x_k corresponds to a choice of a single control $u_k \in U_k(x_k)$ and has as end node the next state $f_k(x_k, u_k)$. The cost of an arc (x_k, u_k) is

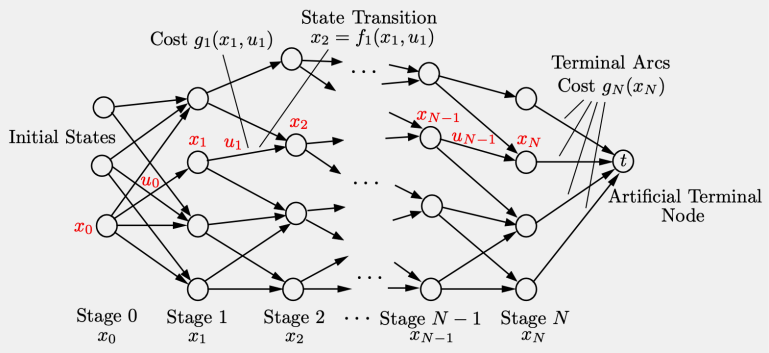

Figure 2.1.2 Illustration of a deterministic finite-state DP problem. Nodes correspond to states x_k. Arcs correspond to state-control pairs (x_k, u_k). An arc (x_k, u_k) has start and end nodes x_k and $x_{k+1} = f_k(x_k, u_k)$, respectively. The cost $g_k(x_k, u_k)$ of the transition is the length of this arc. An artificial terminal node t is connected with an arc of cost $g_N(x_N)$ with each state x_N. The problem is equivalent to finding a shortest path from initial nodes of stage 0 to node t.

defined as $g_k(x_k, u_k)$; see Fig. 2.1.2. To handle the final stage, an artificial terminal node t is added. Each state x_N at stage N is connected to the terminal node t with an arc having cost $g_N(x_N)$. The control sequences $\{u_0, \ldots, u_{N-1}\}$ correspond to paths originating at the initial state (a node at stage 0) and terminating at one of the nodes corresponding to the final stage N. With this description it can be seen that *a deterministic finite-state finite-horizon problem is equivalent to finding a minimum-length (or shortest) path from the initial nodes of the graph (stage 0) to the terminal node t*, as we have discussed in Section 1.2.

Shortest path problems arise in a great variety of application domains. While there are quite a few efficient polynomial algorithms for solving them, some practical shortest path problems are extraordinarily difficult because they involve an astronomically large number of nodes. For example, deterministic scheduling problems of the type discussed in Example 1.2.1 can be formulated as shortest path problems, but with a number of nodes that grows exponentially with the number of tasks. For such problems, neither exact DP nor any other shortest path algorithm can compute an exact optimal solution in practice. In what follows, we will aim to show that suboptimal solution methods, and rollout algorithms in particular, offer a viable alternative.

Many types of search problems involving games and puzzles also admit in principle exact solution by DP, but have to be solved by suboptimal methods in practice. The following is a characteristic example.

Example 2.1.1 (The Four Queens Problem)

Four queens must be placed on a 4×4 portion of a chessboard so that no queen can attack another. In other words, the placement must be such that

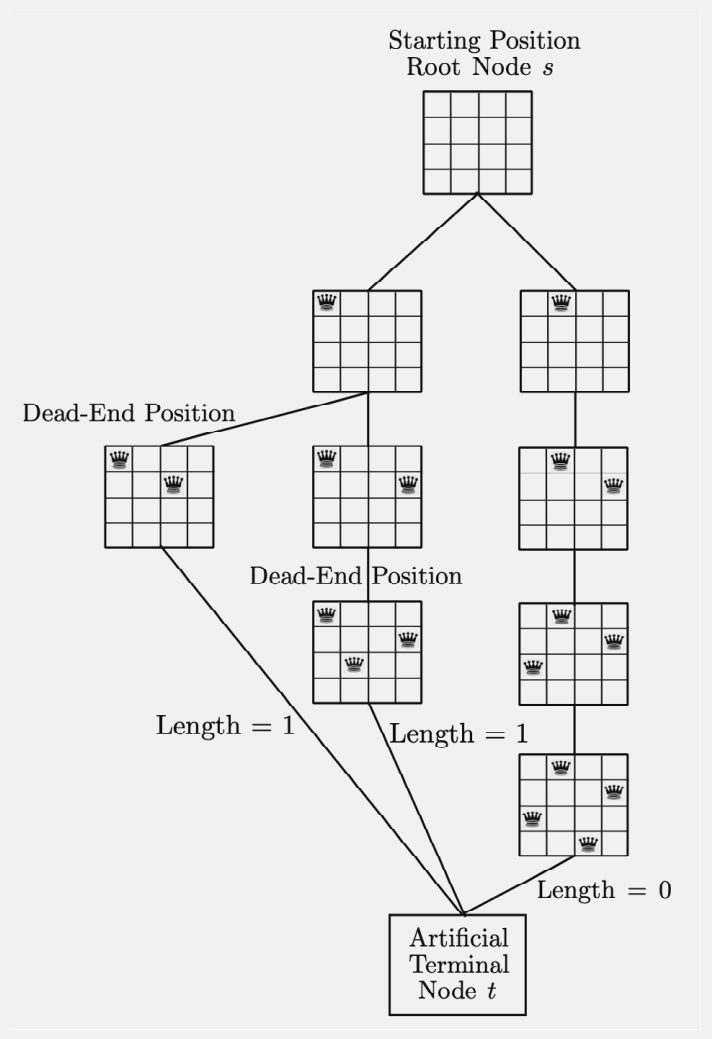

Figure 2.1.3 A finite horizon deterministic DP formulation of the four queens problem. Symmetric positions resulting from placing a queen in one of the rightmost squares in the top row have been ignored. Squares containing a queen have been darkened. All arcs have length zero except for those connecting dead-end positions to the artificial terminal node.

every row, column, or diagonal of the 4×4 board contains at most one queen. Equivalently, we can view the problem as a sequence of problems; first, placing a queen in one of the first two squares in the top row, then placing another queen in the second row so that it is not attacked by the first, and similarly placing the third and fourth queens. (It is sufficient to consider only the first two squares of the top row, since the other two squares lead to symmetric

positions; this is an example of a situation where we have a choice between several possible state spaces, but we select the one that is smallest.)

We can associate positions with nodes of an acyclic graph where the root node s corresponds to the position with no queens and the terminal nodes correspond to the positions where no additional queens can be placed without some queen attacking another. Let us connect each terminal position with an artificial terminal node t by means of an arc. Let us also assign to all arcs cost zero except for the artificial arcs connecting terminal positions with less than four queens with the artificial node t. These latter arcs are assigned a cost of 1 (see Fig. 2.1.3) to express the fact that they correspond to dead-end positions that cannot lead to a solution. Then, the four queens problem reduces to finding a minimal cost path from node s to node t, with an optimal sequence of queen placements corresponding to cost 0.

Note that once the states/nodes of the graph are enumerated, the problem is essentially solved. In this 4×4 problem the states are few and can be easily enumerated. However, we can think of similar problems with much larger state spaces. For example consider the problem of placing N queens on an $N \times N$ board without any queen attacking another. Even for moderate values of N, the state space for this problem can be extremely large (for $N = 8$ the number of possible placements with exactly one queen in each row is $8^8 = 16,777,216$). It can be shown that there exist solutions to the N queens problem for all $N \geq 4$ (for $N = 2$ and $N = 3$, clearly there is no solution). Moreover, effective (non-DP) search algorithms have been devised for its solution up to very large values of N.

The preceding example illustrates some of the difficulties of applying exact DP to discrete/combinatorial problems with the type of formulation that we have described. The state space typically becomes very large, particularly as k increases. In the preceding example, to start a backward DP algorithm, we need to consider all the possible terminal positions, which are too many when N is large. There is an alternative exact DP algorithm for deterministic problems, which proceeds forwards from the initial state. It is simply the backward DP algorithm applied to an equivalent shortest path problem, derived from the one of Fig. 2.1.2 by reversing the directions of all the arcs, and exchanging the roles of the origin and the destination. It will be discussed in Section 2.4; see also [Ber17a], Chapter 2. Still, however, this forward DP algorithm cannot overcome the difficulty with a very large state space.

Discrete Optimization Problems

A major class of deterministic problems that can be formulated as DP problems involves the minimization of a cost function $G(u)$ over all u within a constraint set U. For the purposes of this chapter, we assume that U is finite, although a similar DP formulation is also possible for the more general case of an infinite set U. In Section 1.6.3, we discussed the case

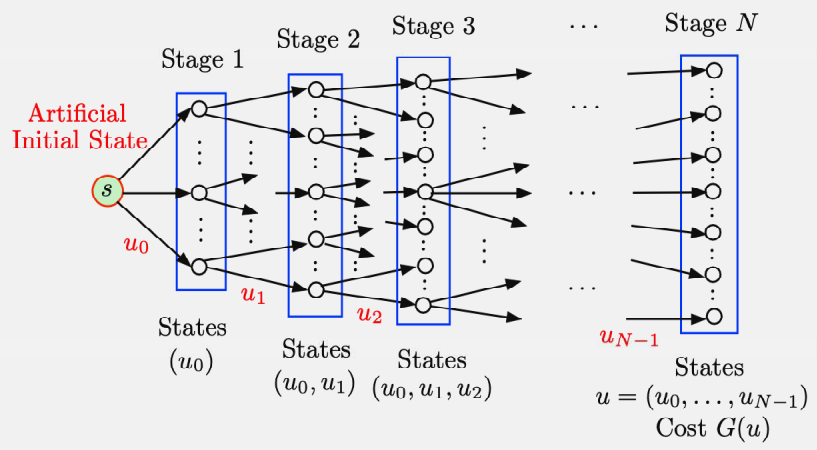

Figure 2.1.4 A DP formulation of the discrete optimization problem of minimizing a cost function $G(u)$ over all u within a finite set U, for the case where u consists of N components (cf. Section 1.6.3).

where u consists of N components,

$$u = (u_0, \ldots, u_{N-1}),$$

the system equation takes the simple form

$$x_{k+1} = (x_k, u_k),$$

and there is just a terminal cost $G(x_N) = G(u)$; see Fig. 1.6.2, repeated here for convenience as Fig. 2.1.4.

The following is a simple but challenging example.

Example 2.1.2 (Constraint Programming)

An interesting special case of the general optimization problem $\min_{u \in U} G(u)$, where $u = (u_0, \ldots, u_{N-1})$, is a *feasibility problem*, where $G(u) \equiv 0$, and the problem reduces to finding a value of u that satisfies the constraint. Generally, the structure of the constraint set U is encoded in a graph representing the problem such as the one of Fig. 2.1.4; cf. Section 1.6.3. This type of feasibility problem is also known as a *constraint programming problem*. The four queens problem (Example 2.1.1) provides an illustration. Another example is the breakthrough problem to be discussed in Section 2.3.

Constraint programming problems can also be transformed into equivalent unconstrained (or less constrained) problems by using problem-dependent penalty functions that eliminate constraints while quantifying the level of constraint violation. As an illustration, consider the case where the problem is to find a feasible solution of a system of constraints of the form

$$h_k(u_k, u_{k+1}) \leq 0, \qquad k = 0, \ldots, N-1,$$

$$u_k \in U_k, \qquad k = 0, \ldots, N-1.$$

This problem can be transformed into the equivalent DP problem of minimizing

$$\sum_{k=1}^{N} \max\big\{0, h_k(x_k, u_k)\big\},$$

subject to the system equation $x_{k+1} = u_k$, and the control constraints $u_k \in U_k$, $k = 0, \ldots, N-1$. Other penalty functions can also be used, such as a quadratic; see the author's nonlinear programming text [Ber16]. This approach is convenient, but it offers no guarantee that it can find a complete feasible solution $(u_0, \ldots, u_{N-1})$, even if one exists. It simply aims to minimize (suboptimally) a measure of the total constraint violation. However, in the process it may be able to find a complete feasible solution, or an infeasible solution that is adequate for practical purposes.

2.2 APPROXIMATION IN VALUE SPACE - DETERMINISTIC PROBLEMS

The forward optimal control sequence construction of Eq. (2.6) is possible only after we have computed $J_k^*(x_k)$ by DP for all x_k and k. Unfortunately, in practice this is often prohibitively time-consuming. However, a similar forward algorithmic process can be used if the optimal cost-to-go functions J_k^* are replaced by some approximations $\tilde{J}_k$. This is the idea of approximation in value space that we discussed in Section 1.2.3. It constructs a suboptimal solution $\{\tilde{u}_0, \ldots, \tilde{u}_{N-1}\}$ in place of the optimal $\{u_0^*, \ldots, u_{N-1}^*\}$, by using $\tilde{J}_k$ in place of J_k^* in the DP procedure (2.6).

Approximation in Value Space - Use of $\tilde{J}_k$ in Place of J_k^*

Start with

$$\tilde{u}_0 \in \arg\min_{u_0 \in U_0(x_0)} \Big[g_0(x_0, u_0) + \tilde{J}_1\big(f_0(x_0, u_0)\big)\Big],$$

and set

$$\tilde{x}_1 = f_0(x_0, \tilde{u}_0).$$

Sequentially, going forward, for $k = 1, 2, \ldots, N-1$, set

$$\tilde{u}_k \in \arg\min_{u_k \in U_k(\tilde{x}_k)} \Big[g_k(\tilde{x}_k, u_k) + \tilde{J}_{k+1}\big(f_k(\tilde{x}_k, u_k)\big)\Big], \qquad (2.7)$$

and

$$\tilde{x}_{k+1} = f_k(\tilde{x}_k, \tilde{u}_k).$$

The expression

$$\tilde{Q}_k(x_k, u_k) = g_k(x_k, u_k) + \tilde{J}_{k+1}\big(f_k(x_k, u_k)\big),$$

which is minimized in approximation in value space [cf. Eq. (2.7)] is known as the (approximate) *Q-factor of* (x_k, u_k). Note that the computation of the suboptimal control (2.7) can be done through the Q-factor minimization

$$\tilde{u}_k \in \arg\min_{u_k \in U_k(\tilde{x}_k)} \tilde{Q}_k(\tilde{x}_k, u_k).$$

This suggests the possibility of using approximate off-line trained Q-factors in place of cost functions in approximation in value space schemes. However, contrary to the cost approximation scheme (2.7) and its multistep counterparts, the performance may be degraded through the errors in the off-line training of the Q-factors (depending on how the training is done).

Exploiting Structure to Expedite the Lookahead Minimization

An important practical idea is to choose the cost function approximation $\tilde{J}_{k+1}$ in Eq. (2.7) in a way that exploits the problem's structure to expedite the computation of the one-step lookahead minimizing control $\tilde{u}_k$. A noteworthy example arises in multiagent problems with decomposable structure, where the use of separable function approximations $\tilde{J}_{k+1}$ leads to decomposition of the minimization of Eq. (2.7) (see the books [Ber19a], Section 2.3.1, and [Ber20a], Section 3.2.4; the author has first encountered this idea through his supervision of the MIT Ph.D. thesis of J. Kimemia [Kim82], [KGB82], which dealt with multi-machine flexible manufacturing). The following example, which is based on recent research by Musunuru et al. [MLW24], illustrates a similar idea but in a different context.

Example 2.2.1 (Multidimensional Assignment)

Let us consider multidimensional assignment problems, a class of combinatorial problems that have both a temporal and a spatial allocation structure. They arise in various settings including scheduling, resource allocation, and data association. The general idea is to group together tasks, or resources, or data points, so as to optimize some objective. An example is when we are given a set of m jobs, m persons, and m machines, and we want to select m non-overlapping (job, person, machine) triplets that correspond to minimum cost. Another example is data association problems, whereby we have collected data relating to the movement of some entities (e.g., persons, vehicles) sequentially over a number of time periods, and we want to group together data points that correspond to distinct entities for better inference purposes.

Mathematically, multidimensional assignment problems involve graphs consisting of $N+1$ subsets of nodes $\mathcal{N}_0, \mathcal{N}_1, \ldots, \mathcal{N}_N$, and referred to as *layers*. The arcs of the graphs are directed and are of the form (i, j), where i is a node in a layer $\mathcal{N}_k$, $k = 0, 1, \ldots, N-1$, and j is a node in the corresponding

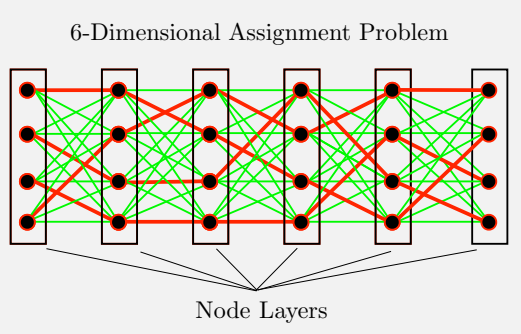

Figure 2.2.1 Illustration of the graph of an $(N+1)$-dimensional assignment problem (here $N = 5$). There are $N+1$ node layers each consisting of m nodes (here $m = 4$). Each grouping consists of $N+1$ nodes, one from each layer, and N corresponding arcs. An $(N+1)$-dimensional assignment consists of m node-disjoint groupings, where each node belongs to one and only one grouping (illustrated in the figure with thick lines). For each grouping, there is an associated cost that depends on the N-tuple of arcs comprising the grouping. The cost of an $(N+1)$-dimensional assignment is the sum of the costs of its m groupings. The difficulty here is that the cost of a grouping does not decompose into a sum of its N arc costs, so the problem cannot be solved by solving N decoupled 2-dimensional assignment problems (for a suboptimal approach based on enforced decoupling, see [Ber20a], Section 3.4.2).

next layer $\mathcal{N}_{k+1}$. Thus we have a directed graph with nodes arranged in $N+1$ layers, and arcs connecting the nodes of each layer to the nodes in their adjacent layers; see Fig. 2.2.1.

We assume that $N \geq 2$, so there are at least three layers. For simplicity, we also assume that each of the layers $\mathcal{N}_k$ contains the same number of nodes, say m, and that there is a unique arc connecting each node in a given layer with each of the nodes of the adjacent layers. We will present an approximation in value space approach that can be implemented by solving 2-dimensional assignment problems for which very fast algorithms exist, such as the Hungarian method and the auction algorithm (see e.g., [Ber98]).

We consider subsets of $N+1$ nodes, referred to as *groupings*, which consist of a single node from every layer. For each grouping, there is an associated cost, which depends on the N-tuple of arcs that comprise the grouping. A partition of the set of nodes into m disjoint groupings (so that each node belongs to one and only one grouping) is called an $(N+1)$-*dimensional assignment*. The cost of an $(N+1)$-dimensional assignment is the sum of the costs of its m groupings. The problem is to find an $(N+1)$-dimensional assignment of minimum cost. The exact solution of the problem is very difficult when the cost of a grouping does not decompose into the sum of costs of the N arcs of the grouping (in which case the problem decouples into N easily solvable 2-dimensional assignment problems).

We formulate the problem as an N-stage sequential decision problem where at the kth stage we select the assignment arcs

$$u_k = \big\{(i_n, j_n) \mid n = 1, \ldots, m\big\}, \qquad k = 0, 1, \ldots, N-1, \tag{2.8}$$

which connect the nodes i_n of layer $\mathcal{N}_k$ to nodes j_n of layer $\mathcal{N}_{k+1}$ on a one-to-one basis. The control constraint set U_k is the set of legitimate assignments, namely those involving exactly one incident arc per node $i_n \in \mathcal{N}_k$, and exactly one incident arc per node j_n of layer $\mathcal{N}_{k+1}$.

We assume a cost function that has the general form

$$G(u_0, \ldots, u_{N-1}).$$

We use as state x_k the partial solution up to time k,

$$x_k = (u_0, \ldots, u_{k-1}),$$

so the system equation takes the simple form

$$x_{k+1} = (x_k, u_k),$$

cf. Fig. 2.1.4 and Section 1.6.3.

The exact DP algorithm takes the form

$$J_N^*(x_N) = G(x_N) = G(u_0, \ldots, u_{N-1}),$$

and,

$$J_k^*(x_k) = \min_{u_k \in U_k} J_{k+1}^*(x_k, u_k), \qquad \text{for all } x_k,\ k = 0, \ldots, N-1,$$

cf. Eqs. (2.3) and (2.4).

Since this DP algorithm is intractable, we resort to approximation of J_{k+1}^* by a function $\tilde{J}_{k+1}$. The one-step lookahead minimization becomes

$$\min_{u_k \in U_k} \tilde{J}_{k+1}(x_k, u_k), \qquad \text{for all } x_k,$$

but is still formidable because its search space U_k is very large. However, it can be greatly simplified by using a cost function approximation $\tilde{J}_{k+1}$ with a special structure that is suitable for the use of fast 2-dimensional assignment algorithms. This cost function approximation has the form

$$\tilde{J}_{k+1}(x_k, u_k) = \sum_{(i,j) \in u_k} c_{k+1}^{ij}(x_k),$$

where $\{(i,j) \in u_k\}$ denotes the set of m arcs (i,j) that correspond to the 2-dimensional assignment u_k, cf. Eq. (2.8) [thus the dependence of the right-hand side on u_k comes through the choice of arcs (i,j) specified by u_k]. The arc costs $c_{k+1}^{ij}(x_k)$ in this equation must be calculated for every possible arc (i,j) that connects a node $i \in \mathcal{N}_k$ with a node $j \in \mathcal{N}_{k+1}$; see Fig. 2.2.2.

Note that the arc costs $c_{k+1}^{ij}(x_k)$ may depend in a complicated way on the previous assignment choices $x_k = (u_0, \ldots, u_{k-1})$, and future assignment choices, which may possibly be determined by a heuristic. For problems that involve tracking the movement of people or vehicles over time, the computation of $c_{k+1}^{ij}(x_k)$ may rely on sensor data and problem-dependent circumstances. For other problems, enforced decomposition methods may be useful; see [Ber20a], Section 3.4.2 and the survey by Emami et al. [EPE20]. For an implementation of the approach of the present example and computational results, see Musunuru et al. [MLW24].

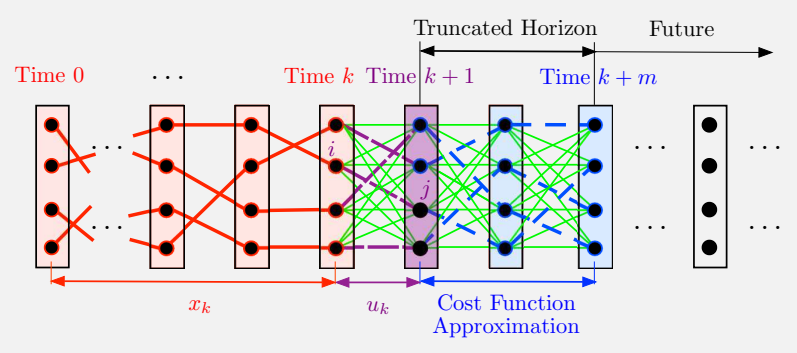

Figure 2.2.2 Illustration of a scheme for approximation in value space in a multiassignment problem. At time k, the next states have the form (x_k, u_k), where $x_k = (u_0, \ldots, u_{k-1})$ is the multiassignment trajectory generated up to time k. The scheme calculates the arc cost $c^{ij}_{k+1}(x_k)$ for each arc (i, j) connecting a node i of the time k layer with a node j of the time $k+1$ layer by extending heuristically the multiassignment trajectory along the "most likely" assignment arcs over some truncated horizon of m steps ($m = 2$ in the figure), and calculating $c^{ij}_{k+1}(x_k)$ as the "cost" of the multiassignment trajectory that starts at time 0, extends to time $k+m$ and passes through arc (i, j).

Once the arc costs $c^{ij}_{k+1}(x_k)$ have been calculated, the assignment u_k and corresponding multiassignment trajectory x_{k+1} are obtained by solving a 2-dimensional assignment problem, using one of the fast available algorithms.

Multistep Lookahead Minimization

The approximation in value space algorithm (2.7) involves a one-step lookahead minimization, since it solves a one-stage DP problem for each k. We may also consider *ℓ-step lookahead*, which involves the solution of an ℓ-step deterministic DP problem, where ℓ is an integer, $1 < \ell < N - k$, with a terminal cost function approximation $\tilde{J}_{k+\ell}$.

As we have noted in Chapter 1, multistep lookahead typically provides better performance over one-step lookahead in approximation in value space schemes. For example in AlphaZero chess, long multistep lookahead is critical for good on-line performance. On the negative side, the solution of the multistep lookahead minimization problem is more time consuming than its one-step lookahead counterpart. However, the deterministic character of the lookahead minimization problem and the fact that it is solved for the single initial state x_k at each time k helps to limit the growth of the lookahead tree and to keep the computation manageable. Moreover, one may try to approximate the solution of the multistep lookahead minimization problem (see Section 2.4).

2.3 ROLLOUT ALGORITHMS FOR DISCRETE OPTIMIZATION

The construction of suitable approximate cost-to-go functions $\tilde{J}_{k+1}$ for ap-

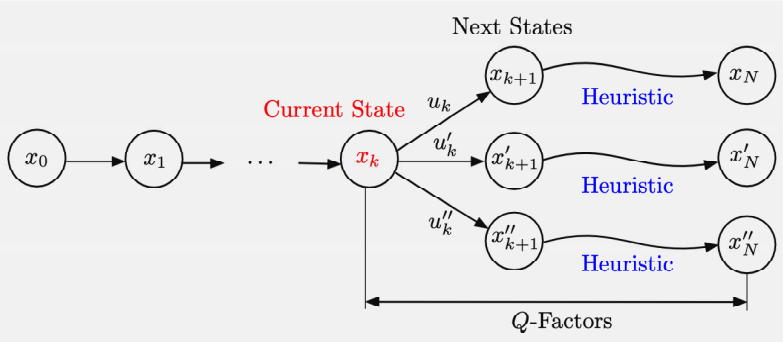

Figure 2.3.1 Schematic illustration of rollout with one-step lookahead. At state x_k, for every pair (x_k, u_k), $u_k \in U_k(x_k)$, the base heuristic generates a Q-factor

$$\tilde{Q}_k(x_k, u_k) = g_k(x_k, u_k) + H_{k+1}\big(f_k(x_k, u_k)\big),$$

and the rollout algorithm selects the control $\tilde{\mu}_k(x_k)$ with minimal Q-factor.

proximation in value space can be done in many different ways, including some of the principal RL methods. A method of particular interest for our course is *rollout*, whereby the approximate values $\tilde{J}_{k+1}(x_{k+1})$ in Eq. (2.7) are obtained when needed by running for each $u_k \in U_k(x_k)$ a heuristic control scheme, called *base heuristic*, for a suitably large number of steps, starting from $x_{k+1} = f_k(x_k, u_k)$.

The base heuristic can be any method, which starting from a state x_{k+1} generates a sequence of controls $u_{k+1}, \ldots, u_{N-1}$, the corresponding sequence of states $x_{k+2}, \ldots, x_N$, and the cost of the heuristic starting from x_{k+1}, which we will generically denote by $H_{k+1}(x_{k+1})$ in this chapter:

$$H_{k+1}(x_{k+1}) = g_{k+1}(x_{k+1}, u_{k+1}) + \cdots + g_{N-1}(x_{N-1}, u_{N-1}) + g_N(x_N).$$

This value of $H_{k+1}(x_{k+1})$ is the one used as the approximate cost $\tilde{J}_{k+1}(x_{k+1})$ in the corresponding approximation in value space scheme (2.7).

In this section, we will develop in more detail the theory of rollout with one-step lookahead minimization for deterministic problems, including the important issue of cost improvement. We will also illustrate several variants of the method, and we will consider questions of efficient implementation. We will then discuss examples of discrete optimization applications.

Let us consider a deterministic DP problem with a finite number of controls and a given initial state (so the number of states that can be reached from the initial state is also finite). We first focus on the pure form of rollout that uses one-step lookahead without truncation, and hence no terminal cost approximation. Given a state x_k at time k, this algorithm considers the tail subproblems that start at every possible next state x_{k+1}, and solves them suboptimally with the base heuristic.

Thus when at x_k, rollout generates on-line the next states x_{k+1} that correspond to all $u_k \in U_k(x_k)$, and uses the base heuristic to compute the sequence of states $\{x_{k+1}, \ldots, x_N\}$ and controls $\{u_{k+1}, \ldots, u_{N-1}\}$ such that

$$x_{t+1} = f_t(x_t, u_t), \qquad t = k+1, \ldots, N-1,$$

and the corresponding cost

$$H_{k+1}(x_{k+1}) = g_{k+1}(x_{k+1}, u_{k+1}) + \cdots + g_{N-1}(x_{N-1}, u_{N-1}) + g_N(x_N).$$

The rollout algorithm then applies the control that minimizes over $u_k \in U_k(x_k)$ the tail cost expression for stages k to N:

$$g_k(x_k, u_k) + H_{k+1}(x_{k+1}).$$

Equivalently, and more succinctly, the rollout algorithm applies at state x_k the control $\tilde{\mu}_k(x_k)$ given by the minimization

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} \tilde{Q}_k(x_k, u_k), \tag{2.9}$$

where $\tilde{Q}_k(x_k, u_k)$ is the approximate Q-factor defined by

$$\tilde{Q}_k(x_k, u_k) = g_k(x_k, u_k) + H_{k+1}\big(f_k(x_k, u_k)\big); \tag{2.10}$$

see Fig. 2.3.1. The rollout algorithm thus defines a suboptimal policy $\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$, referred to as the *rollout policy*, where for each x_k and k, $\tilde{\mu}_k(x_k)$ is the control produced by the Q-factor minimization (2.9).

Note that the rollout algorithm requires running the base heuristic a number of times that is bounded by Nn, where n is an upper bound on the number of control choices available at each state. Thus if n is small relative to N, the algorithm requires computation equal to a small multiple of N times the computation time for a single application of the base heuristic. Similarly, if n is bounded by a polynomial in N, the ratio of the rollout algorithm computation time to the base heuristic computation time is a polynomial in N.

In Section 1.2 we considered an example of rollout involving the traveling salesman problem and the nearest neighbor heuristic (cf. Examples 1.2.2 and 1.2.3). Let us consider another example, which involves a classical discrete optimization problem.

Example 2.3.1 (Multi-Vehicle Routing)

Consider m vehicles that move along the arcs of a given graph. Some of the nodes of the graph include a task to be performed by the vehicles. Each task will be performed only once, immediately after some vehicle reaches the corresponding node for the first time. We assume a horizon that is large

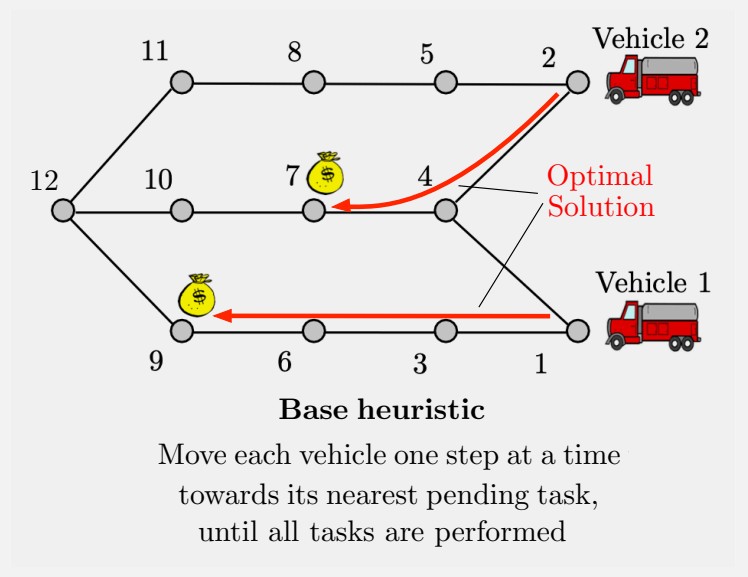

Figure 2.3.2 An instance of the vehicle routing problem of Example 2.3.1. The two vehicles aim to collectively perform the two tasks, at nodes 7 and 9, as fast as possible, by each moving to a neighboring node at each step. The optimal routes are shown.

enough to allow every task to be performed. The problem is to find a route for each vehicle so that the tasks are collectively performed by the vehicles in a minimum number of moves. To express this objective, we assume that for each move by a vehicle there is a cost of one unit. These costs are summed up to the point where all the tasks have been performed.

For a large number m of vehicles and a complicated graph, this is a nontrivial combinatorial problem. It can be approached by DP, like any discrete deterministic optimization problem, as we have discussed. In particular, we can view as state at a given stage the m-tuple of current positions of the vehicles together with the list of pending tasks. Unfortunately, however, the number of these states can be enormous (it increases exponentially with the number of tasks and the number of vehicles), so an exact DP solution is intractable.

This motivates an optimization in value space approach based on rollout. For this we need an easily implementable base heuristic that will solve suboptimally the problem starting from any state x_{k+1}, and will provide the cost approximation $\tilde{J}_{k+1}(x_{k+1})$ in Eq. (2.7). One possibility is based on the vehicles choosing their actions selfishly and without coordination, along shortest paths to their nearest pending task.

To illustrate, consider the two-vehicle problem of Fig. 2.3.2. The base heuristic is to move each vehicle one step at a time towards its nearest pending task, until all tasks have been performed.

The rollout algorithm will work as follows. At a given state x_k [involving

for example vehicle positions at the node pair $(1, 2)$ and tasks at nodes 7 and 9, as in Fig. 2.3.2], we consider all possible joint vehicle moves (the controls u_k at the state) resulting in the node pairs (3,5), (4,5), (3,4), (4,4), corresponding to the next states x_{k+1} [thus, as an example (3,5) corresponds to vehicle 1 moving from 1 to 3, and vehicle 2 moving from 2 to 5]. We then run the base heuristic starting from each of these node pairs, and accumulate the incurred costs up to the time when both tasks are completed. For example starting from the vehicle positions/next state (3,5), the heuristic will produce the following sequence of moves:

- Vehicles 1 and 2 move from (3,5) to (6,2).
- Vehicles 1 and 2 move from (6,2) to (9,4), and the task at 9 is performed.
- Vehicles 1 and 2 move from (9,4) to (12,7), and the task at 7 is performed.

The two tasks are thus performed in a total of 6 vehicle moves once the move to (3,5) has been made.

The process of running the heuristic is repeated from the other three vehicle position pairs/next states (4,5), (3,4) (4,4), and the heuristic cost (number of moves) is recorded. We then choose the next state that corresponds to minimum cost. In our case the joint move to state x_{k+1} that involves the pair $(3, 4)$ produces the sequence

- Vehicles 1 and 2 move from (3,4) to (6,7), and the task at 7 is performed.
- Vehicles 1 and 2 move from (6,7) to (9,4), and the task at 9 is performed.

and performs the two tasks in a total of 6 vehicle moves. It can be verified that it yields minimum first stage cost plus heuristic cost from the next state, as per Eq. (2.7). Thus, the rollout algorithm will choose to move the vehicles to state (3,4) from state (1,2). At that state the rollout process will be repeated, i.e., consider the possible next joint moves to the node pairs (6,7), (6,2), (6,1), (1,7), (1,2), (1,1), perform a heuristic calculation from each, compare, etc.

It can be verified that the rollout algorithm starting from the state (1,2) shown in Fig. 2.3.2 will attain the optimal cost (a total of 6 vehicle moves). It will perform much better than the heuristic, which starting from state (1,2), will move the two vehicles together to state (4,4), then to (7,7), then to (10,10), then to (12,12), and finally to (9,9), (a total of 10 vehicle moves). This is an instance of the cost improvement property of the rollout algorithm: it performs better than its base heuristic (under appropriate conditions).

Let us finally note that the computation required by the rollout algorithm increases exponentially with the number m of vehicles, since the number of m-tuples of moves at each stage increases exponentially with m. This is the type of problem where multiagent rollout can attain great computational savings; cf. Section 1.6.7, and Section 2.9.

Here is an example of a search problem, whose exact solution complexity grows exponentially with the problem size, but can be addressed with a greedy heuristic as well as with the corresponding rollout algorithm.

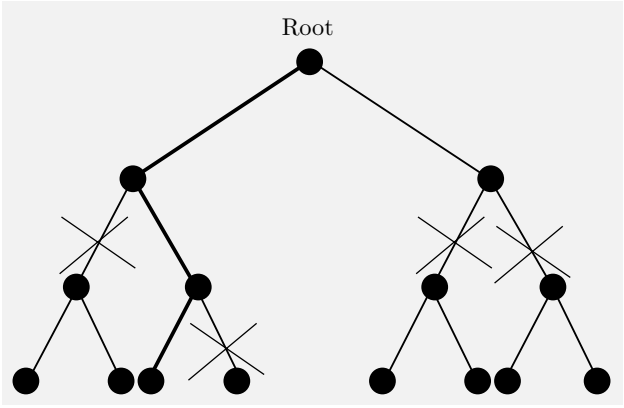

Figure 2.3.3 Binary tree for the breakthrough problem. Each arc is either free or is blocked (crossed out in the figure). The problem is to find a path from the root to one of the leaves, which is free (such as the one shown with thick lines). Note that the problem is related to the constraint programming problem discussed earlier in Section 2.1.

Example 2.3.2 (The Breakthrough Problem)

Consider a binary tree with N stages as shown in Fig. 2.3.3. Stage k of the tree has 2^k nodes, with the node of stage 0 called *root* and the nodes of stage N called *leaves*. There are two types of tree arcs: *free* and *blocked*. A free (or blocked) arc can (cannot, respectively) be traversed in the direction from the root to the leaves. The objective is to break through the graph with a sequence of free arcs (a free path) starting from the root, and ending at one of the leaves. (A variant of this problem is to introduce a positive cost $c > 0$ for traversing a blocked arc, and 0 cost for traversing a free arc.)

One may use DP to discover a free path (if one exists) by starting from the last stage and by proceeding backwards to the root node. The kth step of the algorithm determines for each node of stage $N - k$ whether there is a free path from that node to some leaf node, by using the results of the preceding step. The amount of calculation at the kth step is $O(2^{N-k})$. Adding the computations for the N stages, we see that the total amount of calculation is $O(N2^N)$, so it increases exponentially with the number of stages. For this reason it is interesting to consider heuristics requiring computation that is linear or polynomial in N, but may sometimes fail to determine a free path, even when a free path exists.

Thus, one may suboptimally use a *greedy* algorithm, which starts at the root node, selects a free outgoing arc (if one is available), and tries to construct a free path by adding successively nodes to the path. At the current node, if one of the outgoing arcs is free and the other is blocked, the greedy algorithm selects the free arc. Otherwise, it selects one of the two outgoing arcs according to some fixed rule that depends only on the current node (and not on the status of other arcs). Clearly, the greedy algorithm may fail to find a free path even if such a path exists, as can be seen from Fig. 2.3.3. On the other hand the amount of computation associated with the greedy

algorithm is $O(N)$, which is much faster than the $O(N2^N)$ computation of the DP algorithm. Thus we may view the greedy algorithm as a fast heuristic, which is suboptimal in the sense that there are problem instances where it fails while the DP algorithm succeeds.

One may also consider a rollout algorithm that uses the greedy algorithm as the base heuristic. There is an analysis that compares the probability of finding a breakthrough solution with the greedy and with the rollout algorithm for random instances of binary trees (each arc is independently free or blocked with given probability p). This analysis is given in Section 6.4 of the book [Ber17a], and shows that asymptotically, the rollout algorithm requires $O(N)$ times more computation, but has an $O(N)$ times larger probability of finding a free path than the greedy algorithm.

This tradeoff is qualitatively typical: the rollout algorithm achieves a substantial performance improvement over the base heuristic at the expense of extra computation that is equal to the computation time of the base heuristic times a factor that is a low-order polynomial of the problem size.

2.3.1 Cost Improvement with Rollout - Sequential Consistency, Sequential Improvement

The definition of the rollout algorithm leaves open the choice of the base heuristic. There are several types of suboptimal solution methods that can be used as base heuristics, such as greedy algorithms, local search, genetic algorithms, and others.

Intuitively, we expect that the rollout policy's performance is no worse than the one of the base heuristic: since rollout optimizes over the first control before applying the heuristic, it makes sense to conjecture that it performs better than applying the heuristic without the first control optimization. However, some special conditions must hold in order to guarantee this cost improvement property. We provide two such conditions, *sequential consistency* and *sequential improvement*, introduced in the paper by Bertsekas, Tsitsiklis, and Wu [BTW97], and we later show how to modify the algorithm to deal with the case where these conditions are not met.

Definition 2.3.1: We say that the base heuristic is *sequentially consistent* if it has the property that when it generates the sequence

$$\{x_k, u_k, x_{k+1}, u_{k+1}, \ldots, x_N\}$$

starting from state x_k, it also generates the sequence

$$\{x_{k+1}, u_{k+1}, \ldots, x_N\}$$

starting from state x_{k+1}.

In other words, the base heuristic is sequentially consistent if it "stays the course": when the starting state x_k is moved forward to the next state x_{k+1} of its state trajectory, the heuristic will not deviate from the remainder of the trajectory.

As an example, the reader may verify that the nearest neighbor heuristic described in the traveling salesman Example 1.2.3 and the heuristics used in the multi-vehicle routing Example 2.3.1 are sequentially consistent. Similar examples include the use of various types of greedy/myopic heuristics (Section 6.4 of the book [Ber17a] provides additional examples).† Generally most heuristics used in practice satisfy the sequential consistency condition at "most" states x_k. However, some heuristics of interest may violate this condition at some states.

A sequentially consistent base heuristic can be recognized by the fact that it will apply the same control u_k at a state x_k, no matter what position x_k occupies in a trajectory generated by the base heuristic. Thus *a base heuristic is sequentially consistent if and only if it defines a legitimate DP policy*. This is the policy that moves from x_k to the state x_{k+1} that lies on the state trajectory $\{x_k, x_{k+1}, \ldots, x_N\}$ that the base heuristic generates. Similarly the policy moves from x_n to the state x_{n+1} for $n = k+1, \ldots, N-1$.

We will now show that the rollout algorithm obtained with a sequentially consistent base heuristic has a fundamental cost improvement property: it yields no worse cost than the base heuristic. The amount of cost improvement cannot be easily quantified, but is determined by the performance of the Newton step associated with the rollout policy, so it can be very substantial; cf. the discussion of Chapter 1.

Proposition 2.3.1: (Cost Improvement Under Sequential Consistency) Consider the rollout policy $\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$ obtained with a sequentially consistent base heuristic, and let $J_{k,\tilde{\pi}}(x_k)$ denote the cost obtained with $\tilde{\pi}$ starting from x_k at time k. Then we have

$$J_{k,\tilde{\pi}}(x_k) \leq H_k(x_k), \qquad \text{for all } x_k \text{ and } k, \tag{2.11}$$

where $H_k(x_k)$ denotes the cost of the base heuristic starting from x_k.

† A subtle but important point relates to how one breaks ties while implementing greedy base heuristics. For sequential consistency, one must break ties in a consistent way at various states, i.e., using a fixed rule at each state encountered by the base heuristic. In particular, randomization among multiple controls, which are ranked as equal by the greedy optimization of the heuristic, violates sequential consistency, and can lead to serious degradation of the corresponding rollout algorithm's performance.

Proof: We prove the inequality (2.11) by induction. Clearly it holds for $k = N$, since

$$J_{N,\tilde{\pi}} = H_N = g_N.$$

Assume that it holds for index $k+1$. For any state x_k, let $\overline{u}_k$ be the control applied by the base heuristic at x_k. Then we have

$$\begin{aligned} J_{k,\tilde{\pi}}(x_k) &= g_k\big(x_k, \tilde{\mu}_k(x_k)\big) + J_{k+1,\tilde{\pi}}\Big(f_k\big(x_k, \tilde{\mu}_k(x_k)\big)\Big) \\ &\leq g_k\big(x_k, \tilde{\mu}_k(x_k)\big) + H_{k+1}\Big(f_k\big(x_k, \tilde{\mu}_k(x_k)\big)\Big) \\ &= \min_{u_k \in U_k(x_k)} \Big[g_k(x_k, u_k) + H_{k+1}\big(f_k(x_k, u_k)\big)\Big] \\ &\leq g_k\big(x_k, \overline{u}_k\big) + H_{k+1}\big(f_k(x_k, \overline{u}_k)\big) \\ &= H_k(x_k), \end{aligned} \tag{2.12}$$

where:

- The first equality is the DP equation for the rollout policy $\tilde{\pi}$.
- The first inequality holds by the induction hypothesis.
- The second equality holds by the definition of the rollout algorithm.
- The third equality is the DP equation for the policy that corresponds to the base heuristic (this is the step where we need sequential consistency).

This completes the proof of the cost improvement property (2.11). **Q.E.D.**

Sequential Improvement

We will next show that the rollout policy has no worse performance than its base heuristic under a condition that is weaker than sequential consistency. Let us recall that the rollout algorithm $\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$ is defined by the minimization

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} \tilde{Q}_k(x_k, u_k),$$

where $\tilde{Q}_k(x_k, u_k)$ is the approximate Q-factor defined by

$$\tilde{Q}_k(x_k, u_k) = g_k(x_k, u_k) + H_{k+1}\big(f_k(x_k, u_k)\big),$$

[cf. Eq. (2.10)], and $H_{k+1}\big(f_k(x_k, u_k)\big)$ denotes the cost of the trajectory of the base heuristic starting from state $f_k(x_k, u_k)$.

Definition 2.3.2: We say that the base heuristic is *sequentially improving* if for all x_k and k, we have

$$\min_{u_k \in U_k(x_k)} \tilde{Q}_k(x_k, u_k) \leq H_k(x_k). \tag{2.13}$$

In words, the sequential improvement property (2.13) states that

Minimal heuristic Q-factor at $x_k \leq$ Heuristic cost at x_k.

Note that *when the heuristic is sequentially consistent it is also sequentially improving*. This follows from the preceding relation, since for a sequentially consistent heuristic, the heuristic cost at x_k is equal to the Q-factor of the control $\overline{u}_k$ that the heuristic applies at x_k,

$$\tilde{Q}_k(x_k, \overline{u}_k) = g_k(x_k, \overline{u}_k) + H_{k+1}\big(f_k(x_k, \overline{u}_k)\big),$$

which is greater or equal to the minimal Q-factor at x_k. This implies Eq. (2.13). A sequentially improving heuristic yields policy improvement as the next proposition shows.

Proposition 2.3.2: (Cost Improvement Under Sequential Improvement) Consider the rollout policy $\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$ obtained with a sequentially improving base heuristic, and let $J_{k,\tilde{\pi}}(x_k)$ denote the cost obtained with $\tilde{\pi}$ starting from x_k at time k. Then

$$J_{k,\tilde{\pi}}(x_k) \leq H_k(x_k), \qquad \text{for all } x_k \text{ and } k,$$

where $H_k(x_k)$ denotes the cost of the base heuristic starting from x_k.

Proof: Follows from the calculation of Eq. (2.12), by replacing the last two steps (which rely on sequential consistency) with Eq. (2.13). **Q.E.D.**

Thus the rollout algorithm obtained with a sequentially improving base heuristic, will improve or at least will perform no worse than the base heuristic, from every starting state x_k. In fact *the algorithm has a monotonic improvement property, whereby it discovers a sequence of improved trajectories*. In particular, let us denote the trajectory generated by the base heuristic starting from x_0 by

$$T_0 = (x_0, u_0, \ldots, x_{N-1}, u_{N-1}, x_N),$$

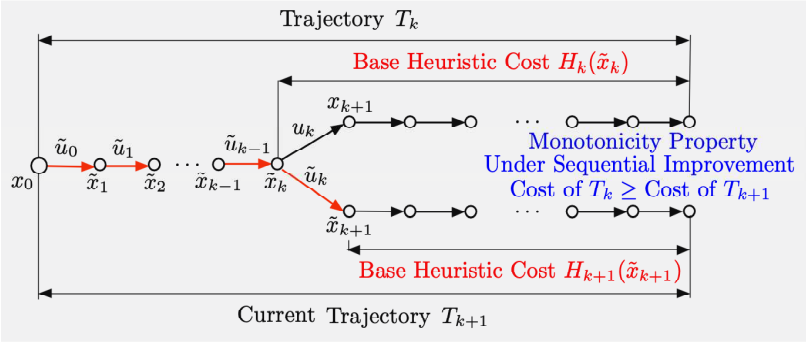

Figure 2.3.4 Proof of the monotonicity property (2.14). At $\tilde{x}_k$, the kth state generated by the rollout algorithm, we compare the "current" trajectory T_k whose cost is the sum of the cost of the current partial trajectory $(x_0, \tilde{u}_0, \tilde{x}_1, \tilde{u}_1, \ldots, \tilde{x}_k)$ and the cost $H_k(\tilde{x}_k)$ of the base heuristic starting from $\tilde{x}_k$, and the trajectory T_{k+1} whose cost is the sum of the cost of the partial rollout trajectory $(x_0, \tilde{u}_0, \tilde{x}_1, \tilde{u}_1, \ldots, \tilde{x}_k)$, and the Q-factor $\tilde{Q}_k(\tilde{x}_k, \tilde{u}_k)$ of the base heuristic starting from $(\tilde{x}_k, \tilde{u}_k)$. The sequential improvement condition guarantees that

$$H_k(\tilde{x}_k) \geq \tilde{Q}_k(\tilde{x}_k, \tilde{u}_k),$$

which implies that

$$\text{Cost of } T_k \geq \text{Cost of } T_{k+1}.$$

If strict inequality holds, the rollout algorithm will switch from T_k and follow T_{k+1}; cf. the traveling salesman Example 1.2.3.

and the final trajectory generated by the rollout algorithm starting from x_0 by

$$T_N = (x_0, \tilde{u}_0, \tilde{x}_1, \tilde{u}_1, \ldots, \tilde{x}_{N-1}, \tilde{u}_{N-1}, \tilde{x}_N).$$

Consider also the intermediate trajectories generated by the rollout algorithm given by

$$T_k = (x_0, \tilde{u}_0, \tilde{x}_1, \tilde{u}_1, \ldots, \tilde{x}_k, u_k, \ldots, x_{N-1}, u_{N-1}, x_N), \quad k = 1, \ldots, N-1,$$

where

$$(\tilde{x}_k, u_k, \ldots, x_{N-1}, u_{N-1}, x_N),$$

is the trajectory generated by the base heuristic starting from $\tilde{x}_k$. Then, by using the sequential improvement condition, it can be proved (see Fig. 2.3.4) that

$$\text{Cost of } T_0 \geq \cdots \geq \text{Cost of } T_k \geq \text{Cost of } T_{k+1} \geq \cdots \geq \text{Cost of } T_N. \quad (2.14)$$

Empirically, it has been observed that the cost improvement obtained by rollout with a sequentially improving heuristic is typically considerable and often dramatic. In particular, many case studies, dating to the middle

1990s, indicate consistently good performance of rollout; see the last section of this chapter for a bibliography. The DP textbook [Ber17a] provides some detailed worked-out examples (Chapter 6, Examples 6.4.2, 6.4.5, 6.4.6, and Exercises 6.11, 6.14, 6.15, 6.16). The price for the performance improvement is extra computation that is typically equal to the computation time of the base heuristic times a factor that is a low order polynomial of N. It is generally hard to quantify the amount of performance improvement, but the computational results obtained from the case studies are consistent with the Newton step interpretations that we discussed in Chapter 1.

The books [Ber19a] (Section 2.5.1) and [Ber20a] (Section 3.1) show that the sequential improvement condition is satisfied in the context of MPC, and is the underlying reason for the stability properties of the MPC scheme. On the other hand the base heuristic underlying the classical form of the MPC scheme is not sequentially consistent (see the preceding references).

Generally, the sequential improvement condition may not hold for a given base heuristic. This is not surprising since any heuristic (no matter how inconsistent or silly) is in principle admissible to use as base heuristic. Here is an example:

Example 2.3.3 (Sequential Improvement Violation)

Consider the 2-stage problem shown in Fig. 2.3.5, which involves two states at each of stages 1 and 2, and the controls shown. Suppose that the unique optimal trajectory is $(x_0, u_0^*, x_1^*, u_1^*, x_2^*)$, and that the base heuristic produces this optimal trajectory starting at x_0. The rollout algorithm chooses a control at x_0 as follows: it runs the base heuristic to construct a trajectory starting from x_1^* and $\tilde{x}_1$, with corresponding costs $H_1(x_1^*)$ and $H_1(\tilde{x}_1)$. If

$$g_0(x_0, u_0^*) + H_1(x_1^*) > g_0(x_0, \tilde{u}_0) + H_1(\tilde{x}_1), \tag{2.15}$$

the rollout algorithm rejects the optimal control u_0^* in favor of the alternative control $\tilde{u}_0$. The inequality above will occur if the base heuristic chooses $\bar{u}_1$ at x_1^* (there is nothing to prevent this from happening, since the base heuristic is arbitrary), and moreover the cost $g_1(x_1^*, \bar{u}_1) + g_2(\tilde{x}_2)$, which is equal to $H_1(x_1^*)$ is high enough.

Let us also verify that if the inequality (2.15) holds then the heuristic is not sequentially improving at x_0, i.e., that

$$H_0(x_0) < \min\big\{g_0(x_0, u_0^*) + H_1(x_1^*),\, g_0(x_0, \tilde{u}_0) + H_1(\tilde{x}_1)\big\}.$$

Indeed, this is true because $H_0(x_0)$ is the optimal cost

$$H_0(x_0) = g_0(x_0, u_0^*) + g_1(x_1^*, u_1^*) + g_2(x_2^*),$$

and must be smaller than both

$$g_0(x_0, u_0^*) + H_1(x_1^*),$$

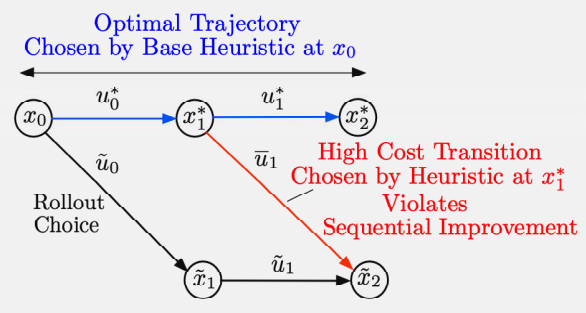

Figure 2.3.5 A 2-stage problem with states $x_1^*, \tilde{x}_1$ at stage 1, and states $x_2^*, \tilde{x}_2$ at stage 2. The controls and corresponding transitions are as shown in the figure. The rollout choice at the initial state x_0 is strictly suboptimal, while the base heuristic choice is optimal. The reason is that the base heuristic is not sequentially improving and makes the suboptimal choice $\overline{u}_1$ at x_1^*, but makes the different (optimal) choice u_1^* when run from x_0.

which is the cost of the trajectory $(x_0, u_0^*, x_1^*, \overline{u}_1, \tilde{x}_2)$, and

$$g_0(x_0, \tilde{u}_0) + H_1(\tilde{x}_1),$$

which is the cost of the trajectory $(x_0, \tilde{u}_0, \tilde{x}_1, \tilde{u}_1, \tilde{x}_2)$.

The preceding example and the monotonicity property (2.14) suggest a simple enhancement to the rollout algorithm, which detects when the sequential improvement condition is violated and takes corrective measures. In this algorithmic variant, called *fortified rollout*, we maintain the best trajectory obtained so far, and keep following that trajectory up to the point where we discover another trajectory that has improved cost (see the next section).

2.3.2 The Fortified Rollout Algorithm

In this section we describe a rollout variant that implicitly enforces the sequential improvement property. This variant, called the *fortified rollout algorithm*, starts at x_0, and generates step-by-step a sequence of states $\{x_0, x_1, \ldots, x_N\}$ and corresponding sequence of controls. Upon reaching state x_k we have the trajectory

$$\overline{P}_k = \{x_0, u_0, \ldots, u_{k-1}, x_k\}$$

that has been constructed by rollout, called *permanent trajectory*, and we also store a *tentative best trajectory*

$$\overline{T}_k = \{x_0, u_0, \ldots, u_{k-1}, x_k, \overline{u}_k, \overline{x}_{k+1}, \overline{u}_{k+1}, \ldots, \overline{u}_{N-1}, \overline{x}_N\}$$

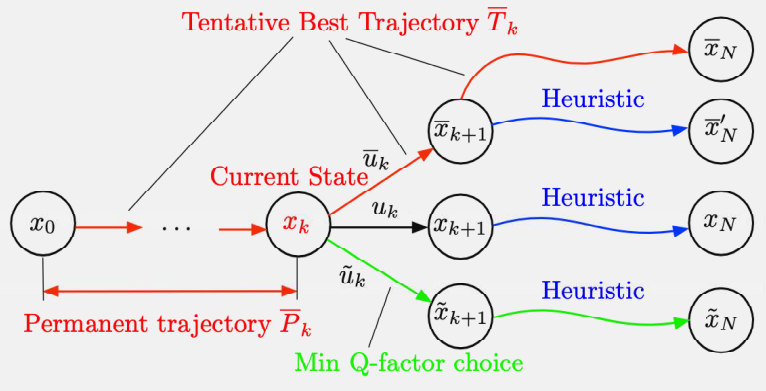

Figure 2.3.6 Schematic illustration of fortified rollout. After k steps, we have constructed the permanent trajectory

$$\overline{P}_k = \{x_0, u_0, \ldots, u_{k-1}, x_k\},$$

and the tentative best trajectory

$$\overline{T}_k = \{x_0, u_0, \ldots, u_{k-1}, x_k, \overline{u}_k, \overline{x}_{k+1}, \overline{u}_{k+1}, \ldots, \overline{u}_{N-1}, \overline{x}_N\},$$

the best end-to-end trajectory computed so far. We now run the rollout algorithm at x_k, i.e., we find the control $\tilde{u}_k$ that minimizes over u_k the sum of $g_k(x_k, u_k)$ plus the heuristic cost from the state $x_{k+1} = f_k(x_k, u_k)$, and the corresponding trajectory

$$\tilde{T}_k = \{x_0, u_0, \ldots, u_{k-1}, x_k, \tilde{u}_k, \tilde{x}_{k+1}, \tilde{u}_{k+1}, \ldots, \tilde{u}_{N-1}, \tilde{x}_N\}.$$

If the cost of the end-to-end trajectory $\tilde{T}_k$ is lower than the cost of $\overline{T}_k$, we add $(\tilde{u}_k, \tilde{x}_{k+1})$ to the permanent trajectory and set the tentative best trajectory to $\overline{T}_{k+1} = \tilde{T}_k$. Otherwise we add $(\overline{u}_k, \overline{x}_{k+1})$ to the permanent trajectory and keep the tentative best trajectory unchanged: $\overline{T}_{k+1} = \overline{T}_k$.

with corresponding cost

$$C(\overline{T}_k) = \sum_{t=0}^{k-1} g_t(x_t, u_t) + g_k(x_k, \overline{u}_k) + \sum_{t=k+1}^{N-1} g_t(\overline{x}_t, \overline{u}_t) + g_N(\overline{x}_N).$$

The tentative best trajectory $\overline{T}_k$ is the end-to-end trajectory that has minimum cost out of all end-to-end trajectories computed up to stage k of the algorithm. Initially, $\overline{T}_0$ is the trajectory generated by the base heuristic starting at the initial state x_0. The idea now is to *discard the suggestion of the rollout algorithm at every state x_k where it produces a trajectory that is inferior to $\overline{T}_k$, and use $\overline{T}_k$ instead* (see Fig. 2.3.6).

In particular, upon reaching state x_k, we run the rollout algorithm as earlier, i.e., for every $u_k \in U_k(x_k)$ and next state $x_{k+1} = f_k(x_k, u_k)$, we

run the base heuristic from x_{k+1}, and find the control $\tilde{u}_k$ that gives the best trajectory, denoted

$$\tilde{T}_k = \{x_0, u_0, \ldots, u_{k-1}, x_k, \tilde{u}_k, \tilde{x}_{k+1}, \tilde{u}_{k+1}, \ldots, \tilde{u}_{N-1}, \tilde{x}_N\}$$

with corresponding cost

$$C(\tilde{T}_k) = \sum_{t=0}^{k-1} g_t(x_t, u_t) + g_k(x_k, \tilde{u}_k) + \sum_{t=k+1}^{N-1} g_t(\tilde{x}_t, \tilde{u}_t) + g_N(\tilde{x}_N).$$

Whereas the ordinary rollout algorithm would choose control $\tilde{u}_k$ and move to $\tilde{x}_{k+1}$, the fortified algorithm compares $C(\overline{T}_k)$ and $C(\tilde{T}_k)$, and depending on which of the two is smaller, chooses $\overline{u}_k$ or $\tilde{u}_k$ and moves to $\overline{x}_{k+1}$ or to $\tilde{x}_{k+1}$, respectively. In particular, if

$$C(\overline{T}_k) \leq C(\tilde{T}_k),$$

the algorithm sets the next state and corresponding tentative best trajectory to

$$x_{k+1} = \overline{x}_{k+1}, \qquad \overline{T}_{k+1} = \overline{T}_k,$$

and if

$$C(\overline{T}_k) > C(\tilde{T}_k),$$

it sets the next state and corresponding tentative best trajectory to

$$x_{k+1} = \tilde{x}_{k+1}, \qquad \overline{T}_{k+1} = \tilde{T}_k.$$

In other words the fortified rollout at x_k follows the current tentative best trajectory $\overline{T}_k$ unless a lower cost trajectory $\tilde{T}_k$ is discovered by running the base heuristic from all possible next states x_{k+1}.† It follows that at every state the tentative best trajectory has no larger cost than the initial tentative best trajectory, which is the one produced by the base heuristic starting from x_0. Moreover, it can be seen that if the base heuristic is sequentially improving, the rollout algorithm and its fortified version coincide. Experimental evidence suggests that it is often important to use the fortified version if the base heuristic is not known to be sequentially improving. Fortunately, the fortified version involves hardly any additional computational cost.

As expected, when the base heuristic generates an optimal trajectory, the fortified rollout algorithm will also generate the same trajectory. This is illustrated by the following example.

† The base heuristic may also be run from a subset of the possible next states x_{k+1}, as in the case where a simplified version of rollout is used; cf. Section 2.3.4. Then fortified rollout will still guarantee a cost improvement property.

Example 2.3.4

Let us consider the application of the fortified rollout algorithm to the problem of Example 2.3.3 and see how it addresses the issue of cost improvement. The fortified rollout algorithm stores as initial tentative best trajectory the optimal trajectory $(x_0, u_0^*, x_1^*, u_1^*, x_2^*)$ generated by the base heuristic at x_0. Then, starting at x_0, it runs the heuristic from x_1^* and $\tilde{x}_1$, and (despite the fact that the ordinary rollout algorithm prefers going to $\tilde{x}_1$ rather than x_1^*) it discards the control $\tilde{u}_0$ in favor of u_0^*, which is dictated by the tentative best trajectory. It then sets the tentative best trajectory to $(x_0, u_0^*, x_1^*, u_1^*, x_2^*)$.

We finally note that the fortified rollout algorithm can be used in a different setting to restore and maintain the cost improvement property. Suppose in particular that the rollout minimization at each step is performed with approximations. For example the control u_k may have multiple independently constrained components, i.e.,

$$u_k = (u_k^1, \ldots, u_k^m), \qquad U_k(x_k) = U_k^1(x_k) \times \cdots \times U_k^m(x_k).$$

Then, to take advantage of distributed computation, it may be attractive to decompose the optimization over u_k in the rollout algorithm,

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} \Big[g_k(x_k, u_k) + H_{k+1}\big(f_k(x_k, u_k)\big) \Big],$$

into an (approximate) parallel optimization over the components u_k^i (or subgroups of these components). However, as a result of approximate optimization over u_k, the cost improvement property may be degraded, even if the sequential improvement assumption holds. In this case by maintaining the tentative best trajectory, starting with the one produced by the base heuristic at the initial condition, we can ensure that the fortified rollout algorithm, even with approximate minimization, will not produce an inferior solution to the one of the base heuristic.

2.3.3 Using Multiple Base Heuristics - Parallel Rollout

In many problems, several promising heuristics may be available. It is then possible to use all of these heuristics in the rollout framework. The idea is to construct a *superheuristic*, which selects the best out of the trajectories produced by the entire collection of heuristics. The superheuristic can then be used as the base heuristic for a rollout algorithm.†

† A related practically interesting possibility is to introduce a partition of the state space into subsets, and a collection of multiple heuristics that are specially tailored to the subsets. We may then select the appropriate heuristic to use on each subset of the partition. In fact one may use a collection of multiple heuristics tailored to each subset of the state space partition, and at each state, select out of all the heuristics that apply, the one that yields minimum cost.

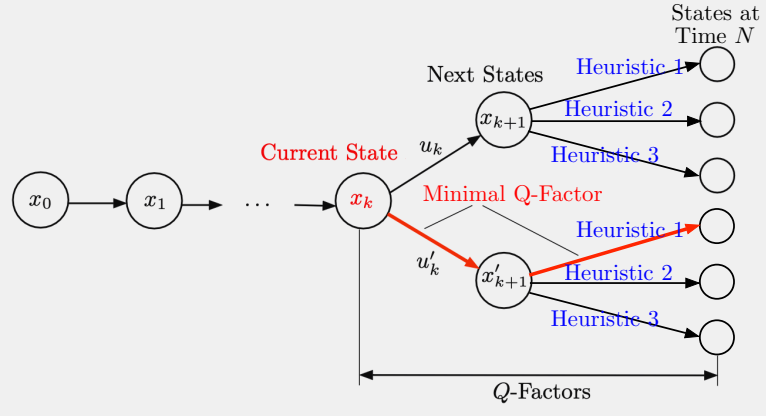

Figure 2.3.7 Schematic illustration of parallel rollout. From every possible next state x_{k+1}, we run each of the heuristics, we compute the Q-factor corresponding to each possible control u_k and heuristic, and select the control with minimal Q-factor. Thus the number of Q-factors that are compared is equal to $m \cdot h$, where m is the number of controls available at state x_k and h is the number of heuristics used in the parallel rollout. In the figure, there are six Q-factors to compare, and the best/minimal Q-factor corresponds to the pair (u'_k, Heuristic 1), as indicated, thus leading to selection of u'_k as the rollout control at x_k.

In particular, let us assume that we have m heuristics, and that the ℓth of these, given a state x_{k+1}, produces a trajectory

$$\tilde{T}^\ell_{k+1} = \{x_{k+1}, \tilde{u}^\ell_{k+1}, x_{k+2}, \ldots, \tilde{u}^\ell_{N-1}, \tilde{x}^\ell_N\},$$

and corresponding cost $C(\tilde{T}^\ell_{k+1})$. The superheuristic then produces at x_{k+1} the trajectory $\tilde{T}^\ell_{k+1}$ for which $C(\tilde{T}^\ell_{k+1})$ is minimum. The rollout algorithm selects at state x_k the control u_k that minimizes the minimal Q-factor:

$$\tilde{u}_k \in \arg \min_{u_k \in U_k(x_k)} \min_{\ell=1,\ldots,m} \tilde{Q}^\ell_k(x_k, u_k),$$

where

$$\tilde{Q}^\ell_k(x_k, u_k) = g_k(x_k, u_k) + C(\tilde{T}^\ell_{k+1})$$

is the cost of the trajectory $(x_k, u_k, \tilde{T}^\ell_{k+1})$. Note that the Q-factors of the different heuristics can be computed independently and in parallel. In view of this fact, the rollout scheme just described is sometimes referred to as *parallel rollout*; see Fig. 2.3.7.

An interesting property, which can be readily verified by using the definitions, is that *if all the heuristics are sequentially improving, the same is true for the superheuristic*, something that is also suggested by the proof of the cost improvement property of Fig. 2.3.4. Indeed, let us write the sequential improvement condition (2.13) for each of the base heuristics

$$\min_{u_k \in U_k(x_k)} \tilde{Q}^\ell_k(x_k, u_k) \leq H^\ell_k(x_k), \qquad \ell = 1, \ldots, m,$$

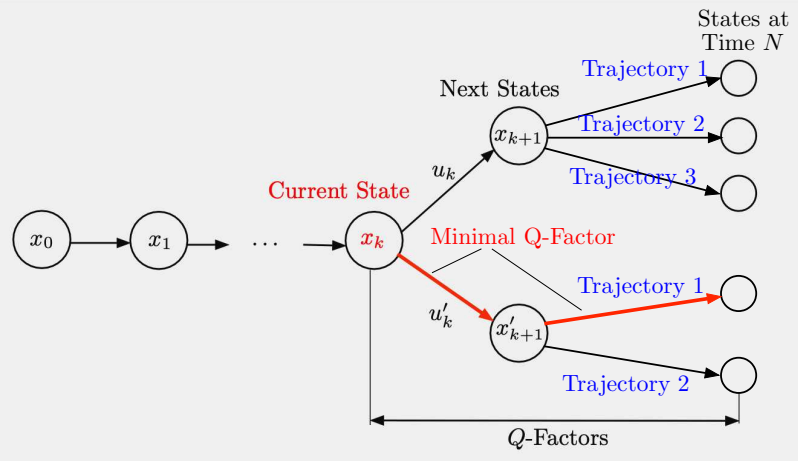

Figure 2.3.8 Schematic illustration of general rollout. From every possible next state x_{k+1}, we run a number of trajectories, we compute the Q-factor corresponding to all the possible (control, trajectory) pairs, and select the control with minimal Q-factor. The number of Q-factors compared is at most $m \cdot h$, where m is the number of controls available at state x_k and h is the maximum number of trajectories generated from each possible next state x_{k+1}. In the figure, there are five Q-factors to compare, and the best/minimal Q-factor corresponds to the pair (u'_k, Trajectory 1), as indicated, thus leading to selection of u'_k as the rollout control at x_k.

where $\tilde{Q}^\ell_k(x_k, u_k)$ and $H^\ell_k(x_k)$ are Q-factors and heuristic costs that correspond to the ℓth heuristic. Then by taking minimum over ℓ, we have

$$\min_{\ell=1,\ldots,m} \min_{u_k \in U_k(x_k)} \tilde{Q}^\ell_k(x_k, u_k) \leq \min_{\ell=1,\ldots,m} H^\ell_k(x_k),$$

for all x_k and k. By interchanging the order of the minimizations of the left side, we then obtain

$$\min_{u_k \in U_k(x_k)} \underbrace{\min_{\ell=1,\ldots,m} \tilde{Q}^\ell_k(x_k, u_k)}_{\text{Superheuristic Q-factor}} \leq \underbrace{\min_{\ell=1,\ldots,m} H^\ell_k(x_k)}_{\text{Superheuristic cost}},$$

which is precisely the sequential improvement condition (2.13) for the superheuristic.

2.3.4 Using Multiple Rollout Trajectories - General Rollout

A generalization of the parallel rollout scheme, called *general rollout*, involves the generation of multiple trajectories starting from each possible next state x_{k+1} (see Fig. 2.3.8). Here, similar to parallel rollout, for each possible x_{k+1}, we compute multiple Q-factors, but we do not require that

each of these Q-factors correspond to a distinct base heuristic. Instead the Q-factors may be associated with multiple trajectories, however generated, which start at x_{k+1} and end at the terminal time N.

Note that some of the trajectories that start from a state x_{k+1} may share some intermediate states from times $k+2$ to N, and they may also be randomly generated. As a result the general rollout framework is well suited for the use of large language models (LLM) as base heuristics. The reason is that the trajectories generated by LLM often involve randomizations and may also involve suboptimization over multiple trajectories. For example, LLM may consider multiple next word choices at selected states, and generate multiple trajectories corresponding to each of these choices.

Similar to parallel rollout, general rollout is conceptually no different from the basic rollout algorithm with a single superheuristic that involves optimization over multiple trajectories starting at any possible next state x_{k+1} (cf. Section 2.3.3). One possibility in general rollout is to carry out this optimization by DP, over a tree of trajectories that is rooted at x_{k+1}. We finally note an important point: with multiple trajectories, the verification of properties such as sequential consistency and sequential improvement, may become complicated. Thus the use of rollout fortification (cf. Section 2.3.2) may be essential to guarantee a cost improvement property.

2.3.5 Simplified Rollout Algorithms

We will now consider a rollout variant, called *simplified rollout*, which is motivated by problems where the control constraint set $U_k(x_k)$ is either infinite or finite but very large. Then the minimization

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} \tilde{Q}_k(x_k, u_k), \tag{2.16}$$

[cf. Eqs. (2.9) and (2.10)], may be unwieldy, since the number of Q-factors

$$\tilde{Q}_k(x_k, u_k) = g_k(x_k, u_k) + H_{k+1}\big(f_k(x_k, u_k)\big)$$

is accordingly infinite or large.

To remedy this situation, we may replace $U_k(x_k)$ with a smaller finite subset $\overline{U}_k(x_k)$:

$$\overline{U}_k(x_k) \subset U_k(x_k).$$

The rollout control $\tilde{\mu}_k(x_k)$ in this variant is one that attains the minimum of $\tilde{Q}_k(x_k, u_k)$ over $u_k \in \overline{U}_k(x_k)$:

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in \overline{U}_k(x_k)} \tilde{Q}_k(x_k, u_k). \tag{2.17}$$

An example is when $\overline{U}_k(x_k)$ results from discretization of an infinite set $U_k(x_k)$. Another possibility is when by using some preliminary approximate optimization, perhaps using a trained neural network or other heuristic method, we can identify a subset $\overline{U}_k(x_k)$ of promising controls, and to

save computation, we restrict attention to this subset. A related possibility is to generate $\overline{U}_k(x_k)$ by some iterative or random search method that explores intelligently the set $U_k(x_k)$ with the aim to minimize $\tilde{Q}_k(x_k, u_k)$ [cf. Eq. (2.16)].

It turns out that the proof of the cost improvement property of Prop. 2.3.2,

$$J_{k,\tilde{\pi}}(x_k) \leq H_k(x_k), \qquad \text{for all } x_k \text{ and } k,$$

goes through if the following modified sequential improvement property holds:

$$\min_{u_k \in \overline{U}_k(x_k)} \tilde{Q}_k(x_k, u_k) \leq H_k(x_k). \tag{2.18}$$

This can be seen by verifying that Eq. (2.18) is sufficient to guarantee that the monotone improvement Eq. (2.14) is satisfied. The condition (2.18) is very simple to satisfy if the base heuristic is sequentially consistent, in which case the control $\overline{u}_k$ selected by the base heuristic satisfies

$$\tilde{Q}_k(x_k, \overline{u}_k) = H_k(x_k).$$

In particular, for the property (2.18) to hold, it is sufficient that $\overline{U}_k(x_k)$ contains the base heuristic choice $\overline{u}_k$.

The idea of replacing the minimization (2.16) by the simpler minimization (2.17) can be extended. In particular, by working through the preceding argument, it can be seen that *any policy*

$$\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$$

such that $\tilde{\mu}_k(x_k)$ *satisfies the condition*

$$\tilde{Q}_k\big(x_k, \tilde{\mu}_k(x_k)\big) \leq H_k(x_k),$$

for all x_k *and* k*, guarantees the modified sequential improvement property (2.18), and hence also the cost improvement property.* A prominent example of such an algorithm arises in the multiagent case where u has m components, $u = (u^1, \ldots, u^m)$, and the minimization over $U_k^1(x_k) \times \cdots \times U_k^m(x_k)$ is replaced by a sequence of single component minimizations, one-component-at-a-time; cf. Section 1.6.7. Of course in the multiagent case, the one-component-at-a-time implementation has an additional favorable property: it can be viewed as rollout (without simplification) for a modified but equivalent DP problem (see Section 1.6.7).

2.3.6 Truncated Rollout with Terminal Cost Approximation

An important variation of rollout algorithms is *truncated rollout* with terminal cost approximation. Here the rollout trajectories are obtained by running the base policy from the leaf nodes of the lookahead tree, but they

are truncated after a given number of steps, while a terminal cost approximation is added to the heuristic cost to compensate for the resulting error. This is important for problems with a large number of stages, and it is also essential for infinite horizon problems where the rollout trajectories have infinite length.

One possibility that works well for many problems is to simply set the terminal cost approximation to zero. Alternatively, the terminal cost function approximation may be obtained by using some sophisticated offline training process that may involve an approximation architecture such as a neural network, or by using some heuristic calculation based on a simplified version of the problem. This form of truncated rollout may also be viewed as an intermediate approach between standard rollout where there is no truncation (and hence no cost function approximation), and approximation in value space without any rollout.

2.3.7 Rollout with an Expert - Model-Free Rollout

We will now consider a rollout algorithm for discrete deterministic optimization for the case where *we do not know the cost function and the constraints of the problem*. Instead we have access to a base heuristic, and also a human or software "expert" who can rank any two feasible solutions without assigning numerical values to them.

We consider the general discrete optimization problem of selecting a control sequence $u = (u_0, \ldots, u_{N-1})$ to minimize a function $G(u)$. For simplicity we assume that each component u_k is constrained to lie in a given constraint set U_k, but extensions to more general constraint sets are possible. We assume the following:

(a) A base heuristic with the following property is available: Given any $k < N-1$, and a partial solution $(u_0, \ldots, u_k)$, it generates, for every $\tilde{u}_{k+1} \in U_{k+1}$, a complete feasible solution by concatenating the given partial solution $(u_0, \ldots, u_k)$ with a sequence $(\tilde{u}_{k+1}, \ldots, \tilde{u}_{N-1})$. This complete feasible solution is denoted

$$S_k(u_0, \ldots, u_k, \tilde{u}_{k+1}) = (u_0, \ldots, u_k, \tilde{u}_{k+1}, \ldots, \tilde{u}_{N-1}).$$

The base heuristic is also used to start the algorithm from an artificial empty solution, by generating all components $\tilde{u}_0 \in U_0$ and a complete feasible solution $(\tilde{u}_0, \ldots, \tilde{u}_{N-1})$, starting from each $\tilde{u}_0 \in U_0$.

(b) An "expert" is available that can compare any two feasible solutions u and $\overline{u}$, in the sense that he/she can determine whether

$$G(u) > G(\overline{u}), \qquad \text{or} \qquad G(u) \leq G(\overline{u}).$$

It can be seen that deterministic rollout can be applied to this problem, even though the cost function G is unknown. The reason is that the

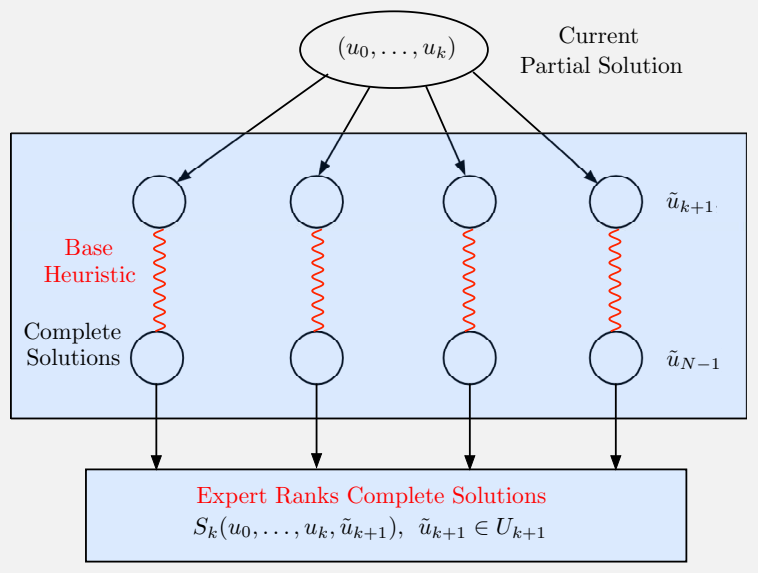

Figure 2.3.9 Schematic illustration of the rollout with an expert for minimizing $G(u)$ subject to

$$u \in U_0 \times \cdots \times U_{N-1}.$$

We assume that we do not know G and/or $U_0, \ldots, U_{N-1}$. Instead we have a base heuristic, which given a partial solution $(u_0, \ldots, u_k)$, outputs all next controls $\tilde{u}_{k+1} \in U_{k+1}$, and generates from each a complete solution

$$S_k(u_0, \ldots, u_k, \tilde{u}_{k+1}) = (u_0, \ldots, u_k, \tilde{u}_{k+1}, \ldots, \tilde{u}_{N-1}).$$

Also, we have a human or software "expert" that can rank any two complete solutions without assigning numerical values to them. The control that is selected from U_{k+1} by the rollout algorithm is the one whose corresponding complete solution is ranked best by the expert.

rollout algorithm uses the cost function only as a means of ranking complete solutions in terms of their cost. Hence, if the ranking of any two solutions can be revealed by the expert, this is all that is needed.† In fact, the constraint sets $U_0, \ldots, U_{N-1}$ need not be known either, as long as they can be generated by the base heuristic. Thus, the rollout algorithm can be described as follows (see Fig. 2.3.9):

We start with an artificial empty solution, and at the typical step, given the partial solution $(u_0, \ldots, u_k)$, $k < N-1$, we use the base heuristic

† Note that for this to be true, it is important that the problem is deterministic, and that the expert ranks solutions using some underlying (though unknown) cost function. In particular, the expert's rankings should have a transitivity property: if u is ranked better than u' and u' is ranked better than u'', then u is ranked better than u''.

to generate all possible one-step-extended solutions

$$(u_0, \ldots, u_k, \tilde{u}_{k+1}), \qquad \tilde{u}_{k+1} \in U_{k+1},$$

and the set of complete solutions

$$S_k(u_0, \ldots, u_k, \tilde{u}_{k+1}), \qquad \tilde{u}_{k+1} \in U_{k+1}.$$

We then use the expert to rank this set of complete solutions. Finally, we select the component u_{k+1} that is ranked best by the expert, extend the partial solution $(u_0, \ldots, u_k)$ by adding u_{k+1}, and repeat with the new partial solution $(u_0, \ldots, u_k, u_{k+1})$.

Except for the (mathematically inconsequential) use of an expert rather than a cost function, the preceding rollout algorithm can be viewed as a special case of the one given earlier. As a result several of the rollout variants that we have discussed so far (rollout with multiple heuristics, simplified rollout, and fortified rollout) can also be easily adapted.

Example 2.3.5 (Active Learning - Dataset Enrichment)

Let us consider a machine learning context whereby we have a dataset that we want to enrich with data selected from another dataset. The enrichment is to be done sequentially, one data point at a time, up to a given maximum size of N data points. We are interested in finding a sequential enrichment policy that optimizes some terminal cost function. This is the active learning context that we discussed briefly in Section 1.7.5.

We can approach the problem in terms of the discrete optimization framework of this section. The state is a partial dataset $(u_0, \ldots, u_k)$, consisting of the initial dataset, denoted u_0, and the k additional data points $u_1, \ldots, u_k$ that have been selected up to time k. The current state $(u_0, \ldots, u_k)$ is augmented with a data point u_{k+1} from the set of the remaining (unused) data points, and each possible new state is evaluated using a base heuristic and an expert, according to our framework of this section. An interesting variation here is to allow the option for an early termination of the dataset enrichment process; this can be done by allowing a dummy/empty data point selection at each time. An additional possibility for early termination is to introduce a cost for each new data point addition, so that the dataset enrichment process stops naturally, when there is little cost improvement.

Example 2.3.6 (Using a Large Language Model for Rollout)

The problem of minimizing $G(u)$ over a constraint set can be viewed as an N-gram optimization problem, discussed in Example 1.6.2, where the text window consists of a partial solution $(u_0, \ldots, u_k, u_{k+1})$ (preceded by $N-k-2$ "default" words to bring the total to N). We noted in that example that a GPT can be used as a policy that generates next words within the context of N-gram optimization. Thus the problem can be addressed within the model-free rollout framework of this section, whereby a GPT is used as a

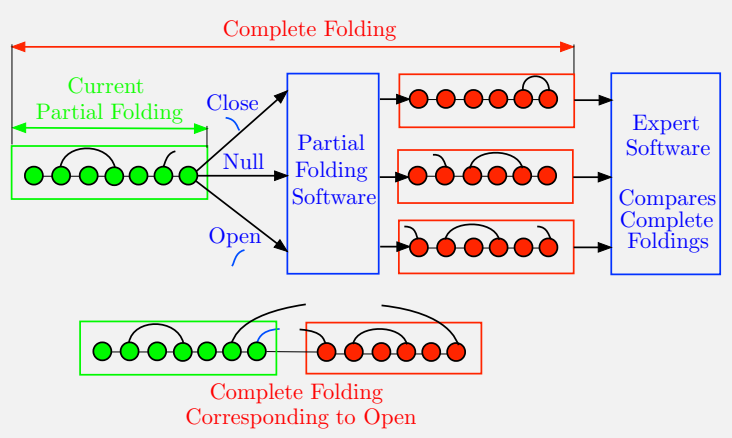

Figure 2.3.10 Schematic illustration of rollout for the RNA folding problem. The current state is the partial folding depicted on the left side. There are at most three choices for control at each state.

base heuristic for completion of partial solutions. The main issues with this approach are how to train a GPT for the problem at hand, and also in the absence of an explicit cost function $G(u)$, how to properly design the expert software for comparing complete solutions. Both of these issues are actively researched at present.

Example 2.3.7 (RNA Folding)

In a classical problem from computational biology, we are given a sequence of nucleotides, represented by circles in Fig. 2.3.10, and we want to "fold" the sequence in an "interesting" way (introduce pairings of nucleotides that result in an "interesting" structure). There are some constraints on which pairings are possible, but we will not go into the details of this (some types of constraints may require the use of the constrained rollout framework of Section 2.5). A common constraint is that the pairings should not "cross," i.e., given a pairing (i_1, i_2) there should be no pairing (i_3, i_4) where either $i_3 < i_1$ and $i_1 < i_4 < i_2$, or $i_1 < i_3 < i_2$ and $i_2 < i_4$. This type of problem has a long history of solution by DP, starting with the paper by Zuker and Stiegler [ZuS81]. There are several formulations, where the aim is to optimize some criterion, e.g., the number of pairings, or the "energy" of the folding. However, biologists do not agree on a suitable criterion, and have developed software to generate "reasonable" foldings, based on semi-heuristic reasoning. We will develop a rollout approach that makes use of such software without discussing their underlying principles.

We formulate the folding problem as a discrete optimization problem involving a pairing decision at each nucleotide in the sequence with at most three choices (open a pairing, close a pairing, do nothing); see Fig. 2.3.10. To apply rollout, we need a base heuristic, which given a partial folding, generates a complete folding (this is the *partial folding software* shown in Fig.

2.3.10). Two complete foldings can be compared by some other software, called the *expert software*. An interesting aspect of this problem is that there is no explicit cost function here (it is internal to the expert software). Thus by trying different partial folding and expert software, we may obtain multiple solutions, which may be used for further screening and/or experimental evaluation. For a recent implementation and variations, see Liu et al. [LPS22].

One more aspect of the problem that is worth noting is that there are at most three choices for control at each state, while the problem is deterministic. As a result, the problem is a good candidate for the use of multistep lookahead. In particular, with ℓ-step lookahead, the number of Q-factors to be computed at each state increases from 3 (or less) to 3^ℓ (or less).

Perpetual Rollout with an Expert

We have considered so far a rollout algorithm that makes a single pass through the variables $u_0, \ldots, u_{N-1}$ in a fixed order. However, it is clear that any order of the variables is possible, as long as the base heuristic can deal with that order. In an off-line setting, where rollout is used simply as a suboptimal method to minimize $G(u)$, this allows the possibility of repeating the rollout with multiple different orders. In particular, the final feasible solution obtained through each rollout pass can be used as the initial base heuristic solution for the next rollout pass. If the fortified version of rollout is used, then the algorithm produces a sequence of solutions with nonincreasing cost, and terminates when the set of selected variable orders is exhausted without any cost improvement.

Learning to Imitate the Expert

To implement model-free rollout, we need both a base heuristic and an expert. None of these may be readily available, particularly the expert, which may involve a hidden cost function that is implicitly used to rank complete solutions. Within this context, it is worth considering the case where an expert is not available but can be emulated by training with the use of data. In particular, suppose that we are given a set of control sequence pairs $(u^s, \overline{u}^s)$, $s = 1, \ldots, q$, with

$$G(u^s) > G(\overline{u}^s), \qquad s = 1, \ldots, q, \tag{2.19}$$

which we can use for training. Such a set may be obtained in a variety of ways, including querying the expert. We may then train a parametric approximation architecture such as a neural network to produce a function $\tilde{G}(u, r)$, where r is a parameter vector, and use this function in place of the unknown $G(u)$ to implement the preceding rollout algorithm.

A method, known as *comparison training*, has been suggested for this purpose, and has been used in a variety of game contexts, including backgammon and chess by Tesauro [Tes89b], [Tes01]. Briefly, given

the training set of pairs $(u^s, \overline{u}^s)$, $s = 1, \ldots, q$, which satisfy Eq. (2.19), we generate for each $(u^s, \overline{u}^s)$, two solution-cost pairs $(u^s, 1)$, $(\overline{u}^s, -1)$. A parametric architecture $\tilde{G}(\cdot, r)$, involving a parameter vector r, such as a neural network, is then trained by some form of regression with these data to produce an approximation $\tilde{G}(\cdot, \bar{r})$ to be used in place of $G(\cdot)$ in a rollout scheme. There is also a related methodology, known as *inverse reinforcement learning*, which aims to use an expert's history of choices to learn the expert's cost function; see the RL survey by Arora and Doshi [ArD21], and the influential early work in econometrics by Rust [Rus88].

Learning the Base Policy's Q-Factors

In another type of imitation approach, we view the base policy decisions as being selected by a process the mechanics of which are not observed except through its generated cost samples at the various stages. In particular, the stage costs starting from any given partial solution $(u_0, \ldots, u_k)$ are added to form samples of the base policy's Q-factors $Q_k(u_0, \ldots, u_k)$. In this way we can obtain Q-factor samples starting from many partial solutions $(u_0, \ldots, u_k)$. Moreover, a single complete solution $(u_0, \ldots, u_{N-1})$ generated by the base policy provides multiple Q-factor samples, one for each of the partial solutions $(u_0, \ldots, u_k)$.

We can then use the sample (partial solution, cost) pairs in conjunction with a training method (see Chapter 3) in order to construct parametric approximations $\tilde{Q}_k(u_0, \ldots, u_k, r_k)$, $k = 1, \ldots, N$, to the true Q-factors $Q_k(u_0, \ldots, u_k)$, where r_k is the parameter vector. Once the training has been completed and the Q-factors $\tilde{Q}_k(u_0, \ldots, u_k, r_k)$ have been obtained for all k, we can construct complete solutions step-by-step, by selecting the next component $\tilde{u}_{k+1}$, given the partial solution $(u_0, \ldots, u_k)$, through the minimization

$$\tilde{u}_{k+1} \in \arg \min_{u_{k+1} \in U_{k+1}} \tilde{Q}_{k+1}(\tilde{u}_0, \ldots, \tilde{u}_k, u_{k+1}, r_{k+1}).$$

Note that even though we are "learning" the base policy, our aim is not to imitate it, but rather to generate a rollout policy. The latter policy will make better decisions than the base policy, thanks to the cost improvement property of rollout. This points to an important issue of *exploration*: we must ensure that the training set of sample (partial solution, cost) pairs is broadly representative, in the sense that it is not unduly biased towards sample pairs that are generated by the base policy.

2.3.8 Using a World Model for Rollout

We now consider another optimal control problem context that bears some similarity with the model-free rollout methodology of the preceding section.

There is a true system and a cost per stage,

$$x_{k+1} = f_k(x_k, u_k), \qquad g_k(x_k, u_k), \tag{2.20}$$

but the controller does not have access to their mathematical description, so the system equation f_k and cost function g_k are unknown. Still, however, we assume that at time k, the controller can observe the current state x_k, and can predict all the possible next states x_{k+1} and the corresponding one-stage costs resulting from the possible control choices u_k. In other words, upon arrival to state x_k, the controller can calculate all the triplets of the form

$$\big(x_k, f_k(x_k, u_k), g_k(x_k, u_k)\big), \qquad u_k \in U_k(x_k), \tag{2.21}$$

corresponding to x_k [only these one-step triplets, and not full access to the true model (2.20)]. These local one-step triplet predictions may take a variety of forms in practical contexts. For example in robotics, the robot can typically predict accurately the immediate outcome of small control actions using its sensors, but its predictions about future environments may be much less precise. A similar situation occurs in many other contexts, such as autonomous driving, healthcare treatment planning, supply chain systems, financial trading and execution, etc.

The idea is that, by using data from past experience, or by pretraining a neural network or transformer, or by refinement with real-time sensor or other data, the controller can learn an approximate predictive model of system evolution and stage costs, possibly in a simplified or implicit representation:

$$x_{m+1} = \hat{f}_m(x_m, u_m), \quad \hat{g}_m(x_m, u_m), \qquad m = k+1, \ldots, N-1. \tag{2.22}$$

This type of learned model is an instance of what is called a *world model* in contemporary AI terminology (starting with the paper by Ha and Schmidhuber [HaS18]; see e.g., Hafner et al. [HLB19], and Moerland et al. [MBP23]), specialized here to a control-theoretic setting with explicit state and cost representations. It can be used as a computational surrogate for the true model for the purposes of approximation in value space or rollout.

In particular, at each state x_k, and given the possible next states upon arrival [cf. Eq. (2.21)], the world model (2.22) is used to construct system trajectories starting from each candidate next state. These trajectories are then used to compute the associated Q-factors under a given base policy, consistently with the approximation in value space framework of this chapter: the control that corresponds to the minimal Q-factor is applied at time k. The resulting next state is then observed and the process is repeated.

Note that there is no guarantee of cost improvement over the base policy, unless the exact model (2.20) and world model (2.22) act identically

at x_k. Nevertheless, in a sequential decision framework, state observations from the real system can compensate for world model prediction inaccuracies, consistent with the replanning character of approximation in value space and model predictive control (cf. Section 1.6.9).

In particular, our Newton step interpretation of approximation in value space, and the associated superlinear error bound guarantees apply; cf. our discussion of Section 1.5.3. As we have discussed earlier, good performance relies on an accurate model for the first step of the lookahead optimization, while the cost function approximation of the subsequent steps can be fairly inaccurate, subject to staying within the region of convergence of the Newton step.

We finally note that with appropriate modifications, a world model can still be used for approximate rollout in a stochastic problem setting, with system and cost per stage

$$x_{k+1} = f_k(x_k, u_k, w_k), \qquad g_k(x_k, u_k, w_k),$$

where f_k and g_k are unknown. What is necessary here is that at the current state x_k, the one-step triplets

$$\big(x_k, f_k(x_k, u_k, w_k), g_k(x_k, u_k, w_k)\big), \qquad u_k \in U_k(x_k),$$

are known to the controller, together with the probability distribution of w_k. A world model can then be used for approximate rollout by predicting trajectories starting from the next states. The world model may be simplified; for example it may be deterministic, based on the use of certainty equivalence (see our discussion of MPC in Section 1.6.9).

2.3.9 Local Search with Rollout for Discrete Optimization

We will now introduce a different rollout approach to the solution of the discrete optimization problem of Section 2.3.7:

$$\min_{u \in U} G(u),$$

where U is a given finite set of feasible solutions. In particular, we will adopt a perspective, which aligns with local search methods, such as those based on arc exchanges for scheduling and traveling salesman problems, tabu search, genetic algorithms, and others (see Ch. 10 of the author's network optimization book [Ber98], which contains an accessible account). These methods are iterative, and generate a sequence of solutions $\{u^k\} \subset U$, starting from some initial u^0. Note that the points u^k are feasible solutions (not partial solutions), and indeed u need not have the N-component structure $u = (u_0, \ldots, u_{N-1})$ of Section 2.3.7.

The general form of an iterative local search method is as follows: At the current iterate u it generates (in some unspecified way) a subset

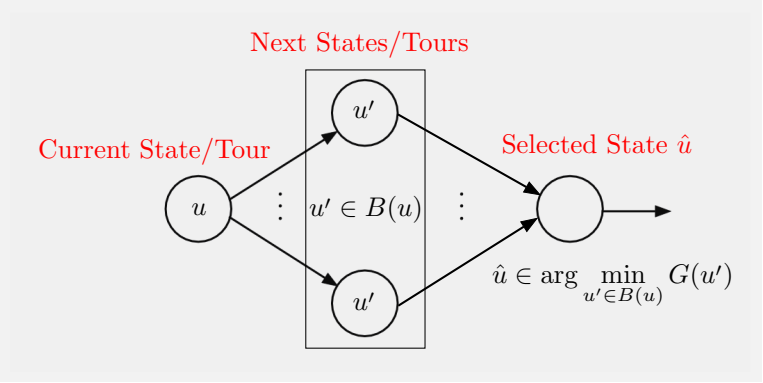

Figure 2.3.11 Schematic representation of a greedy base heuristic, such as k-OPT, for the traveling salesman problem. The current tour is u, the possible next tours $u' \in B(u)$, where $B(u)$ is the neighborhood of u, and the next tour selected by the base heuristic is

$$\hat{u} \in \arg \min_{u' \in B(u)} G(u').$$

$B(u) \subset U$, called the *neighborhood of* u, it evaluates the cost $G(u)$ for each point of $B(u)$, and it moves to the point $\hat{u} \in B(u)$ that has minimal cost. It then replaces u with $\hat{u}$, and continues up to when there is no further progress. We assume that each u is contained within its neighborhood, i.e., $u \in B(u)$ for all $u \in U$; this guarantees that there is no cost deterioration at any iteration, so the method terminates (some local search methods, like tabu search, allow a limited amount of cost deterioration, and introduce a suitable termination mechanism). Note the difference from the method of Section 2.3.7. Here, we are generating iteratively a sequence of complete solutions (not partial solutions).

An important point here is that a local search method of the type just described can be viewed as a dynamic system. The state is u, the control at u is the selection of a point $u' \in B(u)$, the set of next states is $B(u)$, and the system function generates the next state $\hat{u}$ according to the selection policy. Finally, once we choose a finite horizon for the corresponding optimal control problem, the cost function is $G(u_T)$, where u_T is the terminal state.

Example 2.3.8 (Traveling Salesman Problem)

For an example of a local search method, and the dynamic system and optimal control problem that it represents, consider the N-city traveling salesman problem that we discussed in Example 1.2.2. Here the states are the complete tours u (cycles that contain all of the N cities exactly once). In a popular approach, a neighborhood of a tour u is defined as the set $B(u)$ of all tours that can be obtained from u by exchanging k arcs that belong to u with another k arcs that do not belong to u (where k is some positive integer). This neighborhood defines a system equation where the decision/control variable

consists of the two sets of k arcs that are being exchanged. Such a transition from the current state/tour to the next state/tour is called a *k-interchange*. A choice, known as the *k-OPT heuristic*, which has been successful in practice, is a greedy heuristic, whereby the two sets of k arcs are chosen to optimize the cost of the new tour; see Fig. 2.3.11. The method stops when no improvement of the current tour is possible through a k-interchange.

The k-OPT heuristic or similar heuristics can also be used to define a rollout algorithm, which improves the cost of the final tour, based on our earlier analysis. On the other hand, this requires extra computation, which is potentially significant (an increase by a factor proportional to N^2). A truncated rollout approach, coupled with parallel computation, can reduce this extra computation drastically.

There are other possible methods to construct tour neigborhoods and base heuristics for the traveling salesman problem and its variants, including some that are defined with the assistance of trained neural networks or large language models.

Here is another example of local search in the context of prompting a pretrained large language model (LLM):

Example 2.3.9 (Context Adaptation as Local Search in LLM)

Let us discuss how our local search and rollout framework applies to a class of optimization problems arising in AI systems based on LLMs, originally addressed by Zhang et al. [ZHU25]. Here the object being optimized is not a direct solution to a combinatorial problem (such as a tour or a schedule), but rather a *context* consisting of system instructions (prompts) to the LLM. Examples of such instructions include reusable solution fragments or templates, or domain-specific constraints or failure modes, which guide the generation of solutions by the LLM.

In this setting, the state is a context, which plays a role analogous to a tour in the traveling salesman Example 2.3.8, except that it represents a mechanism that generates an output trajectory from the LLM. In particular, each context c induces a method for performing a task via the LLM. The heuristic, starting from c, produces a trajectory of reasoning steps, actions, and intermediate results, culminating in an output $O(c)$. The performance of this output can be evaluated through a cost function $G(\cdot)$, which may be defined via task success, constraint satisfaction, execution correctness, or other externally observable criteria. This process defines the discrete optimization problem

$$\min_{c \in C} G\big(O(c)\big),$$

where C is a known set of feasible contexts, which is structured according to the task at hand, and can be very large.

The neighborhood of a context c, $B(c)$, is a subset of the set of possible contexts C, and is defined by a set of local modifications of c, such as: adding a new rule, refining or specializing an existing rule, removing or disabling a rule, merging or deduplicating similar context elements.

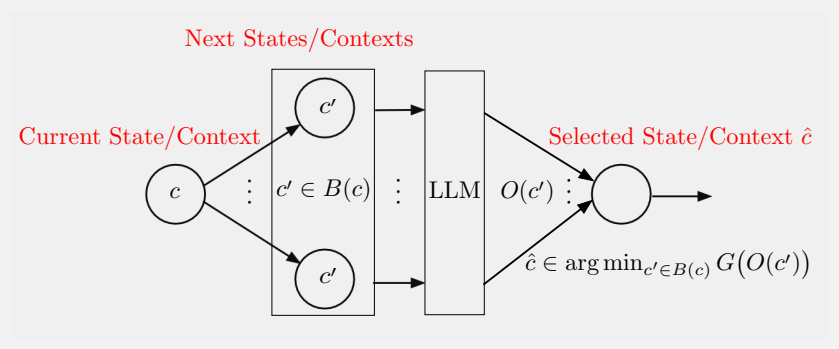

Figure 2.3.12 Schematic representation of a greedy base heuristic defined by an LLM for the context adaptation problem. The current state is c, the possible next states are denoted $c' \in B(c)$, and the next state selected by the base heuristic is

$$\hat{c} \in \arg\min_{c' \in B(c)} G\big(O(c')\big),$$

where $O(c')$ is the output of the LLM when prompted by c'.

These neighborhood operations are analogous to the arc exchanges described for the preceding traveling salesman example. Similar to successful local search methods, the neighborhood is restricted, and arbitrary global rewrites of the context are typically excluded, since they may destroy useful structure accumulated through prior iterations. In fact the neighborhood $B(c)$ may be constructed by using some kind of neural network.

For a given LLM, a base heuristic in this setting consists of a procedure for executing tasks and observing feedback; see Fig. 2.3.12. Starting from a given context/state c, the base heuristic generates task executions by applying the LLM to the possible next contexts.

From the viewpoint of local search and rollout methods, context adaptation for LLMs represents a form of meta-level discrete optimization, where the objects being optimized are contexts, that are evaluated using the LLM. The same principles that govern effective local search in classical combinatorial optimization - restricted neighborhoods, incremental improvement, memory of past failures, and selective use of rollout - apply to this LLM setting as well.

The preceding example illustrates a mechanism that can bring to bear neural network technology on local search methods. In particular, the elements of the neighborhood of the current iterate can be evaluated with a neural network, which may take into account the potential of the next iterate to lead to improved solutions at future iterations.

Rollout

Our discussion so far has focused on how a local search algorithm can be viewed in terms of an optimal control problem, involving a dynamic

system and a base policy. We can now take this view as a starting point for approximation in value space and rollout.

In particular, starting with the current point u, the base heuristic underlying the method can generate truncated trajectories of points, with some terminal cost function approximation. For each truncated trajectory, the terminal cost at the final point of the trajectory serves as its Q-factor. The next point selected by the rollout method is the point in $B(u)$ that corresponds to the minimal Q-factor. This type of algorithm brings to bear the benefits of a longer prediction horizon that underlies rollout and the model predictive control methodology (cf. Section 1.6.9).

2.3.10 Inference in n-Grams, Transformers, HMMs, and Markov Chains

In this section we consider a type of deterministic sequential decision problem involving n-grams and transformers (cf. Section 1.6, Example 1.6.2). We assume that the n-gram provides next word probabilities that can be used to generate word sequences, and we consider methods for computing N-step word sequences that are highly likely, based on these probabilities.† Computing the optimal (i.e., most likely) word sequence starting with a given initial state is an intractable problem for n-grams with large vocabularies, so we consider rollout algorithms that compute highly likely N-word sequences in time that is a low order polynomial in N and in the vocabulary size of the n-gram. This may be useful in several contexts: for example when the n-gram is a transformer that serves as a predictive model of some dynamic process. Then, a likely future word sequence can serve as a basis for present action selection, e.g., in model predictive control.

Our n-gram model generates a sequence $\{x_1, \ldots, x_N\}$ of text strings, starting from some initial string x_0 (here n and N are fixed positive integers). Each string x_k consists of a sequence of n words, chosen from a given list (the *vocabulary* of the n-gram). The kth string x_k is transformed into the next string x_{k+1} by adding a word at the front end of x_k and deleting the word at the back end of x_k; see Example 1.6.2.

Given a text string x_k, the n-gram provides probabilities $p(x_{k+1} \mid x_k)$ for the next text string x_{k+1}. The probabilities can be viewed as the transition probabilities of a stationary Markov chain, whose state space is the set of all n-word sequences x_k.‡ Bearing this context in mind, we also refer to x_k as the *state* (of the underlying Markov chain).

† This section is based on joint work with Yuchao Li; see the paper by Li and Bertsekas [LiB24], which also contains extensive computational experimentation.

‡ The stationarity assumption simplifies our notation, but is not essential to our methodology, as we will discuss later. The probabilities may also depend on external information, such as user-provided prompts.

The transition probabilities $p(x_{k+1} \mid x_k)$ provide guidance for generating state sequences with some specific purpose in mind. To this end, a transformer may use a (next word) *selection policy*, i.e., a (possibly time-dependent) function μ_k, which selects the text string that follows x_k as

$$x_{k+1} = \mu_k(x_k).$$

We are generally interested in selection policies that give preference to high-probability future words. In particular, we will focus on the computation of the most likely N-step state sequence generated by a general finite-state Markov chain. This is the sequence that maximizes $p(x_1, \ldots, x_N \mid x_0)$, where x_0 is a given initial condition. Equivalently, it is the sequence that maximizes

$$p(x_1 \mid x_0) \cdot p(x_2 \mid x_1) \cdots p(x_N \mid x_{N-1}).$$

We will see that this is a problem that can be addressed by DP, despite the multiplicative character of the above expression. Indeed DP applies to problems of multiplicative (rather than additive) cost functions, provided the multiplicative terms are positive. The reason is that maximization (or minimization) of the product of positive terms is equivalent to maximization (or minimization) of the sum of the corresponding logarithms.

State Estimation in Hidden Markov Models

The most likely N-step sequence problem also arises in inference problems involving Hidden Markov Models (HMM), a problem area with a vast range of applications. We will briefly describe this context, before focusing on the problem of generating a most likely sequence in a Markov chain.

In the HMM context, again we have a Markov chain with state at time k denoted by x_k, and transition probabilities, which are known and are denoted by

$$p(x_{k+1} \mid x_k).$$

However, the states x_k are not observable (i.e., they are "hidden"). Instead, when a transition from x_k to x_{k+1} occurs, we obtain some data about the transition (cf. Fig. 2.3.13). In particular, we observe the value of a random variable z_k generated with probability

$$p(z_k \mid x_k, x_{k+1}).$$

The problem is to estimate the N-step trajectory $(x_1, x_2, \ldots, x_N)$ given an observed data sequence $(z_0, z_1, \ldots, z_{N-1})$.

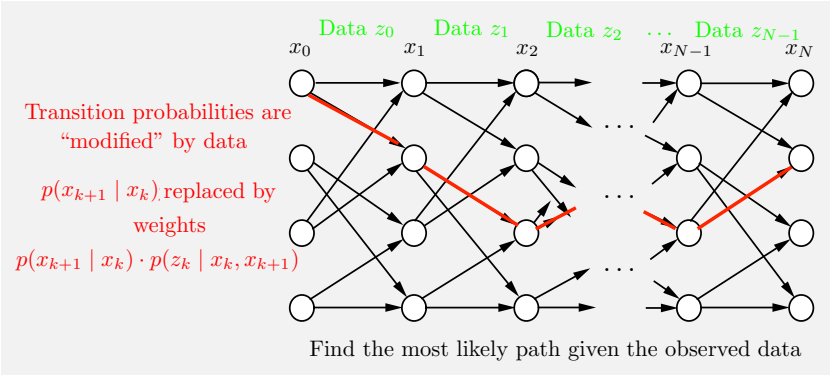

Figure 2.3.13 Illustration of the state trajectory generated by an HMM. Given a data sequence $(z_0, z_1, \ldots, z_{N-1})$, the most likely trajectory is the one that maximizes the product of weights

$$w(z_0, x_0, x_1) \cdot w(z_1, x_1, x_2) \cdots w(z_{N-1}, x_{N-1}, x_N),$$

where

$$w(z_k, x_k, x_{k+1}) = p(x_{k+1} \mid x_k) \cdot p(z_k \mid x_k, x_{k+1}).$$

Example 2.3.10 (Language Translation)

Suppose that we wish to translate an English phrase consisting of N words into French. The N words of the English phrase are the known data z_k, $k = 0, \ldots, N-1$, the hidden states x_k of the HMM are French words that may possibly correspond to these English words, and any sequence $(x_1, x_2, \ldots, x_N)$ provides a translation of the English phrase into French. The transition probabilities $p(x_{k+1} \mid x_k)$ and observation probabilities $p(z_k \mid x_k, x_{k+1})$ are assumed known (they can be derived from linguistic data). We wish to find the "most likely" French translation of the given English phrase, in the sense that it corresponds to the most likely path, the one that maximizes the product of "modified probabilities"/weights

$$p(x_{k+1} \mid x_k) \cdot p(z_k \mid x_k, x_{k+1}).$$

The product of the weights expresses the likelihood of the state sequence given the observed data sequence. This approach to language translation is known as *statistical machine translation*, and has been used for many years. Over time it has been replaced by deep neural network and transformer-based translation approaches, which can capture the interdependencies between multiple adjacent phrases.

The HMM inference problem has a long history. It has been addressed principally with the Viterbi algorithm, a shortest path algorithm originally developed for coding/decoding in an information theory context.

A DP-oriented discussion of the Viterbi algorithm, based on the original sources [Vit67], [For73], is given in Section 2.2.2 of the author's book [Ber17a]. Related methods such as Viterbi training, the Baum-Welch algorithm, and others, are used widely for the estimation of the most likely sequence, and also for the problem of estimating the transition probabilities $p(x_{k+1} \mid x_k)$ using data. These methods play an important role in several diverse fields, such as speech recognition [Rab89], [EpM02], computational linguistics [JuM23], [MaS99], coding and error correction [PrS01], [PrS08], bioinformatics [Edd96], [DEK98], computational finance [MaE07], and others; see the edited volumes by Bouguila, Fan, and Amayri [BFA22], Mamon and Elliott [MaE14], and Westhead and Vijayabaskar [WeV17]. Compared to these fields, the transformer/n-gram context tends to involve a much larger state space. Still, however, there are HMM problems where the state space is intractably large, and which can benefit from the rollout methodology to be developed in this section.

In what follows, we will not consider explicitly issues of inference in HMMs. However, our Markov chain/most likely sequence generation methods fully apply to this context, once we modify appropriately the transition probabilities to account for the observed data. In particular, the transition probabilities $p(x_{k+1} \mid x_k)$ should be replaced by the data-dependent weights

$$w(z_k, x_k, x_{k+1}) = p(x_{k+1} \mid x_k) \cdot p(z_k \mid x_k, x_{k+1}).$$

Then, given the data sequence, $(z_0, \ldots, z_{N-1})$, the problem is to find the most likely state sequence $(x_1, \ldots, x_N)$ that corresponds to this data, i.e., the one that maximizes

$$w(z_0, x_0, x_1) \cdot w(z_1, x_1, x_2) \cdots w(z_{N-1}, x_{N-1}, x_N);$$

cf. Fig. 2.3.13. The exact and approximate DP algorithms that we will discuss in this section fully apply to this problem.

Computing the Most Likely Sequence in a Markov Chain

We will next consider a finite-state stationary Markov chain and various policies for generating highly likely sequences according to the transition probabilities of the Markov chain. We will generally use the symbols x and y for states, and we will denote the chain's transition probabilities by $p(y \mid x)$. We assume that given a state x, the probabilities $p(y \mid x)$ are either known or can be generated on-line by means of software such as a transformer.

We assume stationarity of the Markov chain in part to alleviate an overburdened notation, and also because n-gram and transformer models are typically assumed to be stationary. However, *the rollout methodology*

and the manner in which we use it do not depend at all on stationarity of the transition probabilities, or infinite horizon properties of Markov chains, such as ergodic classes, transient states, etc. In fact, they also do not depend on the stationarity of the state space either. Only the Markov property is used in our discussion, i.e., the probability of the next state depends on the immediately preceding state, and not on earlier states.

A *selection policy* π is a sequence of functions $\{\mu_0, \ldots, \mu_{N-1}\}$, which given the current state x_k, determines the next state x_{k+1} as

$$x_{k+1} = \mu_k(x_k).$$

Note that for a given π, the state evolution is *deterministic*; so for a given π and x_0, the generated state sequence $\{x_1, \ldots, x_N\}$ is fully determined. Moreover the choice of the policy π is arbitrary, although we are primarily interested in π that give preference to high-probability next states.

Given a policy $\pi = \{\mu_0, \ldots, \mu_{N-1}\}$ and a starting state x at time k, the state at future times $m > k$ is denoted by $y_{m,k}(x, \pi)$:

$$y_{m,k}(x, \pi) = \text{ state at time } m > k \text{ starting at state } x \text{ and using } \pi.$$

The *state trajectory generated by a policy π, starting at state x at time k*, is the sequence

$$y_{k+1,k}(x, \pi), \ldots, y_{N,k}(x, \pi),$$

(cf. Fig. 2.3.14), and the probability of its occurrence in the given Markov chain is

$$P_k(x, \pi) = p\big(y_{k+1,k}(x, \pi) \mid x\big) \cdot \prod_{i=k+1}^{N-1} p\big(y_{i+1,k}(x, \pi) \mid y_{i,k}(x, \pi)\big), \quad (2.23)$$

according to the multiplication rule for conditional probabilities.

Optimal/Most Likely Selection Policy

The *most likely selection policy*, denoted by $\pi^* = \{\mu_0^*, \ldots, \mu_{N-1}^*\}$, maximizes over all policies π the probabilities $P_k(x, \pi)$ for every initial state x and time k. The corresponding probabilities of π^*, starting at state x at time k, are denoted by $P_k^*(x)$:

$$P_k^*(x) = P_k(x, \pi^*) = \max_{\pi} P_k(x, \pi).$$

This is similar to our finite-horizon DP context, except that we consider a multiplicative reward function, instead of an additive cost function [$P_k^*(x)$ can be viewed as an optimal reward-to-go from state x at time k, in place of

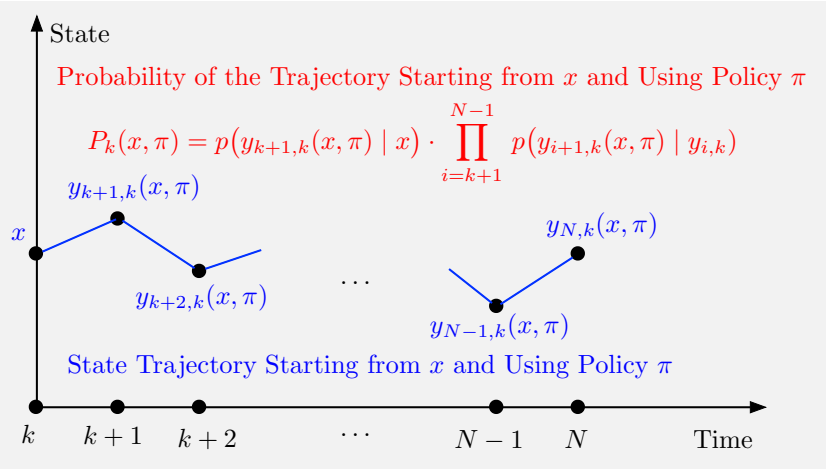

Figure 2.3.14 Illustration of the state trajectory generated by a policy π, starting at state x at time k. The probability of its occurrence, $P_k(x,\pi)$, is the product of the transition probabilities along the $N-k$ steps of the trajectory [cf. Eq. (2.23)].

the optimal cost-to-go $J_k^*(x)$ that we have considered so far in the additive DP case].†

The policy π^* and its probabilities $P_k^*(x)$ can be generated by the following DP-like algorithm, which operates in two stages:

(a) It first computes the probabilities $P_k^*(x)$ backwards, for all x, according to

$$P_k^*(x) = \max_y p(y \mid x) P_{k+1}^*(y), \qquad k = N-1, \ldots, 0, \tag{2.24}$$

starting with

$$P_N^*(x) \equiv 1.$$

(b) It then generates sequentially the selections $x_1^*, \ldots, x_N^*$ of π^* according to

$$x_{k+1}^* = \mu_k^*(x_k^*) \in \arg\max_y p(y \mid x_k^*) P_{k+1}^*(y), \tag{2.25}$$

starting with $x_0^* = x_0$.

This algorithm is equivalent to the usual DP algorithm for multistage additive costs, after we take logarithms of the multiplicative expressions defining the probabilities $P_k(x,\pi)$.

† Generally multiplicative reward problems can be converted to additive cost DP problems involving negative logarithms of the multiplicative reward factors (assuming they are positive).

Greedy Policy

At any given state x_k, the *greedy policy* produces the next state by maximization of the corresponding transition probability over all y:

$$\max_y p(y \mid x_k).$$

We assume that ties in the above maximization are broken according to some prespecified deterministic rule. For example if the states are labeled by distinct integers, one possibility is to specify the greedy selection at x_k as the state y with minimal label, among those that attain the maximum above. Note that the greedy policy is not only deterministic, but it is also stationary (its selections depend only on the current state and not on the time k). We will consequently use the notation $\overline{\pi} = \{\overline{\mu}, \ldots, \overline{\mu}\}$ for the greedy policy, where

$$\overline{\mu}(x_k) \in \arg\max_y p(y \mid x_k), \tag{2.26}$$

and $\overline{\mu}(x_k)$ is uniquely defined according to our deterministic convention for breaking ties in the maximization above. The corresponding probabilities $P_k(x_k, \overline{\pi})$ are given by the DP-like algorithm

$$P_k(x, \overline{\pi}) = p\big(\overline{\mu}(x) \mid x\big) P_{k+1}\big(\overline{\mu}(x), \overline{\pi}\big), \qquad k = N-1, \ldots, 0, \tag{2.27}$$

starting with

$$P_N(x, \overline{\pi}) \equiv 1.$$

Equivalently, we can compute $P_k(x, \overline{\pi})$ by using forward multiplication of the transition probabilities along the trajectory generated by the greedy policy, starting from x; cf. Eq. (2.23).

The limitation of the greedy policy is that it chooses the locally optimal next state without considering the impact of this choice on future state selections. The rollout approach, to be discussed next, mitigates this limitation with a mechanism for looking into the future, and balancing the desire for a high-probability next transition with the potential undesirability of low-probability future transitions.

Rollout Policy

At a given state x_k, the *rollout policy with one-step lookahead* produces the next state, denoted $\tilde{\mu}_k(x_k)$, by maximizing $p(y \mid x_k) P_{k+1}(y, \overline{\pi})$ over all y:

$$\tilde{\mu}_k(x_k) \in \arg\max_y p(y \mid x_k) P_{k+1}(y, \overline{\pi}). \tag{2.28}$$

Thus *it optimizes the selection of the first state y, assuming that the subsequent states will be chosen using the greedy policy.*

By comparing the maximization (2.28) with the one for the most likely selection policy [cf. Eq. (2.25)], we see that it chooses the next state similarly, except that $P^*_{k+1}(y)$ (which is hard to compute) is replaced by the (much more easily computable) probability $P_{k+1}(y, \overline{\pi})$. In particular, the latter probability is computed for every y by running the greedy policy forward starting from y and multiplying the corresponding transition probabilities along the generated state trajectory. This is a polynomial computation, which is roughly larger by a factor N over the greedy selection method. However, there are ways to reduce this computation, including the use of parallel processing and other possibilities, which we will discuss later.

The expression $p(y \mid x_k)P_{k+1}(y, \overline{\pi})$ that is maximized over y in Eq. (2.28) can be viewed as the *Q-factor of the pair* (x_k, y) *corresponding to the base policy* $\overline{\pi}$, and is denoted by $Q_{\overline{\pi},k}(x_k, y)$:

$$Q_{\overline{\pi},k}(x_k, y) = p(y \mid x_k)P_{k+1}(y, \overline{\pi}). \tag{2.29}$$

This is similar to the approximation in value space context, except that we consider a multiplicative reward function, whereby at state x_k we choose the action y that yields the maximal Q-factor.

Rollout Policy with ℓ-Step Lookahead

Another rollout possibility includes *rollout with ℓ-step lookahead* ($\ell > 1$), whereby given x_k we maximize over all sequences $\{y_1, y_2, \ldots, y_\ell\}$ up to ℓ steps ahead, the *ℓ-step Q-factor*

$$Q_{\overline{\pi},k,\ell}(x_k, y_1, \ldots, y_\ell) = p(y_1 \mid x_k)p(y_2 \mid y_1) \cdots p(y_\ell \mid y_{\ell-1})P_{k+\ell}(y_\ell, \overline{\pi}), \tag{2.30}$$

and if $\{\tilde{y}_1, \tilde{y}_2, \ldots, \tilde{y}_\ell\}$ is the maximizing sequence, we select $\tilde{y}_1$ at x_k, and discard the remaining states $\tilde{y}_2, \ldots, \tilde{y}_\ell$.† In practice the performance of ℓ-step lookahead rollout policies almost always improves with increasing ℓ. However, artificial examples have been constructed where this is not so; see the book [Ber19a], Section 2.1.1. Moreover, the computational overhead of ℓ-step lookahead increases with ℓ.

Simplified and Truncated Rollout

As we have already noted, one of the difficulties that arises in the application of rollout is the potentially very large number of the Q-factors that need to be calculated at each time step at the current state x [it is equal to the number of states y for which $p(y \mid x) > 0$]. In practice the computation of Q-factors can be restricted to a subset of most probable next

† If $\ell > N - k$, then ℓ must be reduced to $N - k$, to take into account end-of-horizon effects.

states, as per the transition probabilities $p(y \mid x)$ (this is similar to simplified rollout that we discussed in Section 2.3.5). For example, often many of the transition probabilities $p(y \mid x)$ are very close to 0, and can be safely ignored.

Another possibility to reduce computation is to truncate the trajectories generated from the next states y by the greedy policy, up to m steps (assuming that $k+m < N$, i.e., if we are more than m steps away from the end of the horizon). This is essentially the truncated rollout algorithm, discussed in Section 2.3.6, whereby we maximize over y the m-step Q-factor of the greedy policy $\overline{\pi}$:

$$p(y \mid x_k) P_{k+1,m}(y, \overline{\pi}), \tag{2.31}$$

where

$$P_{k+1,m}(y, \overline{\pi}) = p\big(y_{k+2,k+1}(y, \overline{\pi}) \mid y\big) \cdot \prod_{i=k+2}^{k+m} p\big(y_{i+1,k+1}(y, \overline{\pi}) \mid y_{i,k+1}(y, \overline{\pi})\big)$$

is the m-step product of probabilities along the path generated by the greedy policy $\overline{\pi}$ starting from y at time $k+1$ [cf. Eq. (2.23)]. By contrast, in rollout without truncation, we maximize over y

$$p(y \mid x_k) P_{k+1}(y, \overline{\pi}),$$

where

$$P_{k+1}(y, \overline{\pi}) = p\big(y_{k+2,k+1}(y, \overline{\pi}) \mid y\big) \cdot \prod_{i=k+2}^{N-1} p\big(y_{i+1,k+1}(y, \overline{\pi}) \mid y_{i,k+1}(y, \overline{\pi})\big)$$

is the $(N-k-1)$-step product of probabilities along the path generated by the greedy policy starting from y at time $k+1$; cf. Eqs. (2.23) and (2.28).

Multiple On-line Policy Iterations - Double Rollout

Still another possibility is to apply the rollout approach successively, in *multiple policy iterations*, by using the rollout policy obtained at each iteration as base policy for the next iteration. This corresponds to the fundamental DP algorithm of *policy iteration*.

Performing on-line just two policy iterations amounts to using the rollout algorithm as a base policy for another rollout algorithm. This has been called *double rollout*, and it has been discussed in Section 2.3.5 of the book [Ber20a] and Section 6.5 of the book [Ber22a]. It amounts to two successive Newton steps for solving Bellman's equation. Generally, one-step lookahead rollout requires $O(q \cdot N)$ applications of the base policy where q

is the number of Q-factors calculated at each time step.† Thus with each new policy iteration, there is an amplification factor $O(q \cdot N)$ of the computational requirements. Still, however, the multiple iteration approach may be viable, even on-line, when combined with some of the other time-saving computational devices described above (e.g., truncation and simplification to reduce q), in view of the relative simplicity of the calculations involved and their suitability for parallel computation. This is particularly so for double rollout. Policy iteration/double rollout is discussed by Yan et al. [YDR04] in the context of the game of solitaire, and by Silver and Barreto [SiB22] in the context of a broader class of search methods.

We next show that the rollout selection policy with one-step lookahead has a *performance improvement property*: it generates more likely state sequences than the greedy policy, starting from any state, and the improvement is often very substantial.

Performance Improvement Properties of Rollout Policies

We will show by induction a performance improvement property of the rollout algorithm with one-step lookahead, namely that for all states $x \in X$ and k, we have

$$P_k(x, \overline{\pi}) \leq P_k(x, \tilde{\pi}), \tag{2.32}$$

i.e., the probability of the sequence generated by the rollout policy is greater or equal to the probability of the sequence generated by the greedy policy; this is true for any starting state x at any time k. This is similar to our earlier rollout cost improvement results in Section 2.3.1.

Indeed, for $k = N$ this relation holds, since we have

$$P_N(x, \overline{\pi}) = P_N(x, \tilde{\pi}) \equiv 1.$$

Assuming that

$$P_{k+1}(x, \overline{\pi}) \leq P_{k+1}(x, \tilde{\pi}), \qquad \text{for all } x,$$

† For a more accurate estimate of the complexity of the greedy, rollout, and double rollout algorithms, note that the basic operation of the greedy operation is the maximization over the q numbers $p(y \mid x_k)$. Thus m steps of the greedy algorithm, as in an m-step Q-factor calculation, costs $q \cdot m$ comparisons. In m-step truncated rollout, we compare q greedy Q-factors so the number of comparisons per rollout time step is $q^2m + q$. Over N time steps the total is $(q^2m + q) \cdot N$ comparisons, while for the greedy algorithm starting from the initial state x_0, the corresponding number is $q \cdot N$. Thus there is an amplification factor of $qm + 1$ for the computation of simplified m-step truncated rollout over the greedy policy. Similarly it can be estimated that there is an extra amplification factor of no more than $qm + 1$ for using double rollout with (single) rollout as a base policy.

we will show that

$$P_k(x,\overline{\pi}) \leq P_k(x,\tilde{\pi}), \qquad \text{for all } x.$$

Indeed, we use the preceding relations to write

$$\begin{aligned} P_k(x,\tilde{\pi}) &= p\big(\tilde{\mu}_k(x) \mid x\big)P_{k+1}\big(\tilde{\mu}_k(x),\tilde{\pi}\big) \\ &\geq p\big(\tilde{\mu}_k(x) \mid x\big)P_{k+1}\big(\tilde{\mu}_k(x),\overline{\pi}\big) \\ &\geq p\big(\tilde{\mu}_k(x) \mid x\big)P_{k+1}\big(\overline{\mu}_k(x),\overline{\pi}\big) \\ &= P_k(x,\overline{\pi}), \end{aligned}$$

where

- The first equality holds from the definition of the probabilities corresponding to the rollout policy $\tilde{\pi}$.
- The first inequality holds by the induction hypothesis.
- The second inequality holds from the fact that the rollout choice $\tilde{\mu}_k(x)$ maximizes the Q-factor $p\big(\tilde{y} \mid x\big)P_{k+1}\big(y,\overline{\pi}\big)$ over y.
- The second equality holds from the definition of the probabilities corresponding to the greedy policy $\overline{\pi}$.

Thus the induction proof of the improvement property (2.32) is complete.

Clearly, the performance improvement property continues to hold in successive multiple iterations of the rollout policy, and in fact it can be shown that after a sufficiently large number of iterations it yields the most likely selection policy. This is a consequence of classical policy iteration convergence results, which establish the finite convergence of the policy iteration algorithm for finite-state Markovian decision problems, see e.g., [Ber12], [Ber17a], [Ber19a].

Performance improvement can also be established for the ℓ-step lookahead version of the rollout policy, using an induction proof that is similar to the one given above for the one-step lookahead case. Moreover, the books [Ber20a], [Ber22a] describe conditions under which simplified rollout maintains the performance improvement property. However, it is not necessarily true that the performance of the ℓ-step lookahead rollout policy improves as ℓ increases; see an example in the book [Ber19a], Section 2.1.1. Similarly, it is not necessarily true that the m-step truncated rollout policy performs better than the greedy policy.† On the other hand, known performance deterioration examples of this type are artificial and are apparently rare in practice.

† It performs better than an m-step version of the greedy policy, which generates a sequence of $m+1$ states, starting from the current state and using the greedy policy.

Computational Comparison of Greedy, Optimal, and Rollout Policies

We will now present some illustrative computational comparisons, using Markov chains that are small enough for the optimal/most likely selection policy to be computed exactly via the DP-like algorithm (2.24)-(2.25). Thus, the performance differences between the rollout, greedy, and optimal policies can be accurately assessed. The experiments are presented in more detail in the paper [LiB24], which also contains qualitatively similar computational results with much larger Markov chains involving n-grams, and a trained transformer neural network. We used Markov chains where there is a fixed number q of distinct states y such that $p(y \mid x) > 0$, with q being the same for all states x. These states were selected according to a uniform distribution, whereby all y with $p(y \mid x) > 0$ are equally likely. The probabilities $p(y \,|\, x)$ were also generated according to a uniform distribution.

Let us denote the state space by X and the number of states by $|X|$. We refer to the ratio $q/|X|$ (in percent) as the *branching factor* of the chain. We represent the probability of an entire sequence generated by a policy as the average of its constituent transition probabilities (i.e., a geometric mean over N as will be described below). In particular, given a sample set C of Markov chains, we compute the optimal occurrence probability of generated sequences, averaged over all chains, states, and transitions, and denoted by $(P_0^*)^{1/N}$, according to the average geometric mean formula

$$(P_0^*)^{1/N} = \frac{\sum_{c \in C} \sum_{x \in X} \left(P_{0,c}^*(x)\right)^{1/N}}{|C| \cdot |X|},$$

where $P_{0,c}^*(x)$ is the optimal occurrence probability with $x_0 = x$ and Markov chain c in the sample set. Similarly, we compute the occurrence probabilities of sequences generated by the greedy policy averaged over all chains, states, and transitions, and denoted by $(\overline{P}_0)^{1/N}$, according to

$$(\overline{P}_0)^{1/N} = \frac{\sum_{c \in C} \sum_{x \in X} \left(P_{0,c}(x, \overline{\pi})\right)^{1/N}}{|C| \cdot |X|}, \tag{2.33}$$

where $P_{0,c}(x, \overline{\pi})$ is the transition probability of the sequence generated by the greedy policy with $x_0 = x$ and Markov chain indexed by c. For the rollout algorithm (or variants thereof), we compute its averaged occurrence probability $(\tilde{P}_0)^{1/N}$ similar to Eq. (2.33) with $\tilde{\pi}$ in place of $\overline{\pi}$. Then the performance of the rollout algorithm can be measured by its *percentage recovery* of the optimality loss of the greedy policy, given by

$$\frac{(\tilde{P}_0)^{1/N} - (\overline{P}_0)^{1/N}}{(P_0^*)^{1/N} - (\overline{P}_0)^{1/N}} \times 100 \quad (\%). \tag{2.34}$$

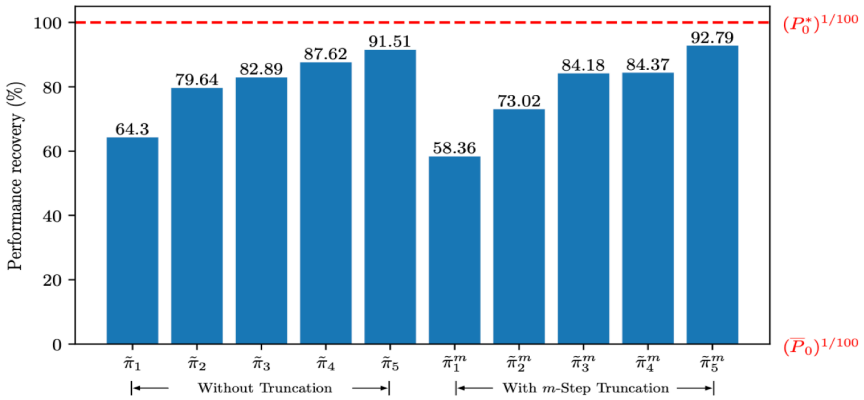

Figure 2.3.15 Percentage recovery of the optimality loss of the greedy policy through the use of rollout and its variants, applied to sequence selection problems with $N = 100$ for 50 randomly generated Markov chains with 100 states and 5% branching factor. Here $\tilde{\pi}_\ell$ is rollout with ℓ-step lookahead, and $\tilde{\pi}_\ell^m$ represents m-step truncated rollout for $m = 10$ with ℓ-step lookahead. It can be seen that on average, rollout and its variants provide a substantial improvement over the greedy policy, the improvement increases with the size of the lookahead, and truncated rollout methods perform comparably to their exact counterparts.

This performance measure describes accurately how the rollout performance compares with the greedy policy and how close it comes to optimality.

In the experiments presented here we have used small Markov chains with $|X| = 100$ states, branching factor $q = 5\%$, and sequence length $N = 100$. In summary, the percentage recovery has ranged roughly from 60% to 90% for one-step to five-step lookahead, untruncated and truncated rollout with $m = 10$ steps up to truncation; see Fig. 2.3.15. The performance improves as the length of the lookahead increases, but seems remarkably unaffected by the 90% truncation of the rollout horizon (the relative insensitivity of the performance of truncated rollout to the number of rollout steps m has been observed in other application contexts as well). The figure has been generated with a sample of 50 different Markov chains with $|X| = 100$ states, branching factor equal to 5%, and sequence length $N = 100$. The figure shows the results obtained by rollout with one-step and multi-step lookahead (ranging from 2 to 5 steps), and their m-step truncated counterparts with $m = 10$. Their percentage recovery, evaluated according to Eq. (2.34), is given in Fig. 2.3.15, where $\tilde{\pi}_\ell$ denotes rollout with ℓ-step lookahead, and $\tilde{\pi}_\ell^m$ denotes m-step truncated rollout with ℓ-step lookahead.

Figure 2.3.16 provides corresponding results using double rollout, which show a significant improvement over the case of single rollout. Moreover, truncating the rollout horizon by 90% has remarkably small effect on

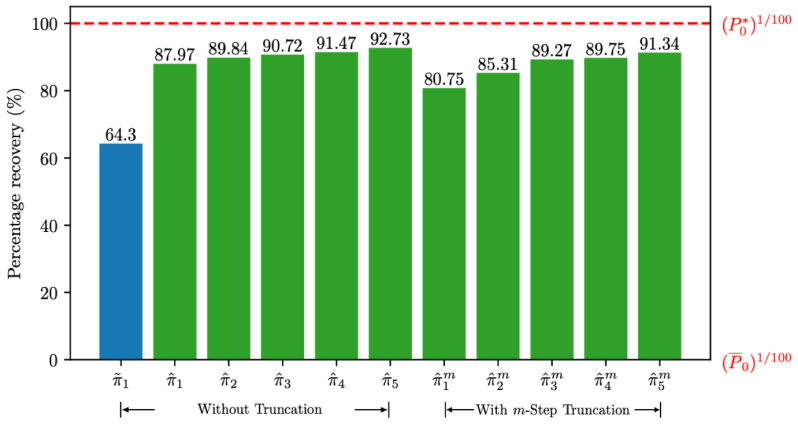

Figure 2.3.16 Percentage recovery of the optimality loss of the greedy policy through the use of double rollout and its variants, applied to sequence selection problems with $N = 100$ for 50 randomly generated Markov chains with 100 states and 5% branching factor. Here $\tilde{\pi}_\ell$ is rollout with ℓ-step lookahead, and $\tilde{\pi}_\ell^m$ represents m-step truncated rollout for $m = 10$ with ℓ-step lookahead.

the percentage recovery, similar to the case of a single rollout.

The preceding results are consistent with those of other computational studies using rollout and its variants. The sequences produced by rollout improve substantially over those generated by the base policy. Moreover, there is typically a relatively small degradation of performance when applying the truncated rollout compared with untruncated rollout. This is significant as truncated rollout greatly reduces the computation if m is substantially smaller than N, and also makes possible the use of double rollout.

2.4 APPROXIMATION IN VALUE SPACE WITH MULTISTEP LOOKAHEAD

We will now consider approximation in value space with multistep lookahead minimization, possibly also involving some form of rollout. Figure 2.4.1 describes the case of pure (nontruncated) form of rollout with two-step lookahead for deterministic problems. In particular, suppose that after k steps we have reached state x_k. We then consider the set of all possible two-step-ahead states x_{k+2}, we run the base heuristic starting from each of them, and compute the two-stage cost to get from x_k to x_{k+2}, plus the cost of the base heuristic from x_{k+2}. We select the state, say $\tilde{x}_{k+2}$, that is associated with minimum cost, compute the controls $\tilde{u}_k$ and $\tilde{u}_{k+1}$ that lead from x_k to $\tilde{x}_{k+2}$, choose $\tilde{u}_k$ as the next control and $x_{k+1} = f_k(x_k, \tilde{u}_k)$ as the next state, and discard $\tilde{u}_{k+1}$.

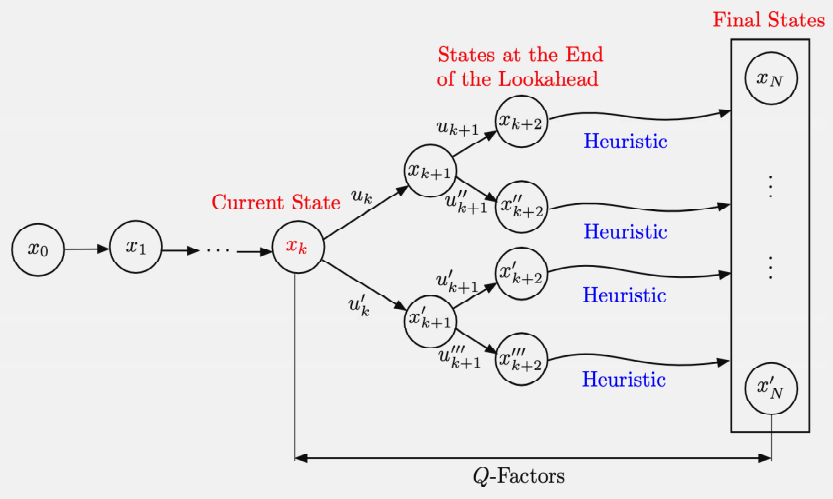

Figure 2.4.1 Illustration of multistep rollout with $\ell = 2$ for deterministic problems. We run the base heuristic from each leaf $x_{k+\ell}$ at the end of the lookahead graph. We then construct an optimal solution for the lookahead minimization problem, where the heuristic cost is used as terminal cost approximation. We thus obtain an optimal ℓ-step control sequence through the lookahead graph, use the first control in the sequence as the rollout control, discard the remaining controls, move to the next state, and repeat. Note that the multistep lookahead minimization may involve approximations aimed at simplifying the associated computations.

The extension of the algorithm to lookahead of more than two steps is straightforward: instead of the two-step-ahead states x_{k+2}, we run the base heuristic starting from all the possible ℓ-step ahead states $x_{k+\ell}$, etc. For cases where the ℓ-step lookahead minimization is very time consuming, we may consider variants involving approximations aimed at simplifying the associated computations.

An important variation is *truncated rollout with terminal cost approximation*. Here the rollout trajectories are obtained by running the base heuristic from the leaf nodes of the lookahead graph, and they are truncated after a given number of steps, while a terminal cost approximation is added to the heuristic cost to compensate for the resulting error; see Fig. 2.4.2. One possibility that works well for many problems, particularly when the combined lookahead for minimization and base heuristic simulation is long, is to simply set the terminal cost approximation to zero. Alternatively, the terminal cost function approximation can be obtained by problem approximation or by using some sophisticated off-line training process that may involve an approximation architecture such as a neural network. Generally, the terminal cost approximation is especially important if a large portion of the total cost is incurred upon termination (this

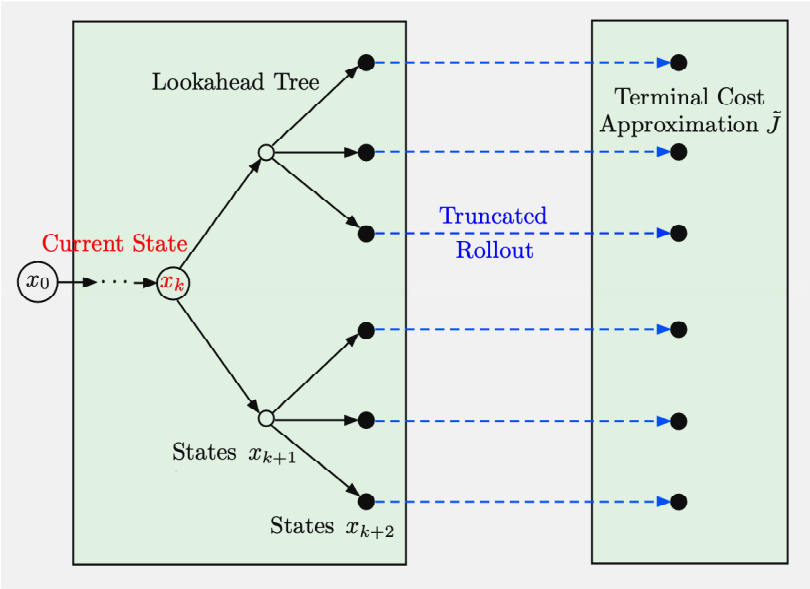

Figure 2.4.2 Illustration of truncated rollout with two-step lookahead and a terminal cost approximation $\tilde{J}$. The base heuristic is used for a limited number of steps and the terminal cost is added to compensate for the remaining steps.

is true for example in games).

Note that the preceding algorithmic scheme can be viewed as multistep approximation in value space, and *it can be interpreted as a Newton step*, with suitable starting point that is determined by the truncated rollout with the base heuristic, and the terminal cost approximation. This interpretation is possible once the discrete optimal control problem is reformulated to an equivalent infinite horizon SSP problem; cf. the discussion of Sections 1.6.2 and 2.1. Thus the algorithm inherits the fast convergence property of the Newton step, which we have discussed in the context of infinite horizon problems in Section 1.5; see also the book [Ber22a].

The architecture of Fig. 2.4.2 contains as a special case the general multistep approximation in value space scheme, where there is no rollout at all; i.e., the leaves of the multistep lookahead tree are evaluated with the function $\tilde{J}$. Figure 2.4.3 illustrates this special case, where for notational simplicity we have denoted the current state by x_0. The illustration involves an acyclic graph with a single root (the current state) and ℓ layers, with the nth layer consisting of the states x_n that are reachable from x_0 with a feasible sequence of n controls. In particular, there is an arc for every state x_1 of the 1st layer that can be reached from x_0 with a feasible control, and similarly an arc for every pair of states (x_n, x_{n+1}), of layers n and $n+1$, respectively, for which x_{n+1} can be reached from x_n with a feasible control. The cost of each of these arcs is the stage cost of the corresponding state-

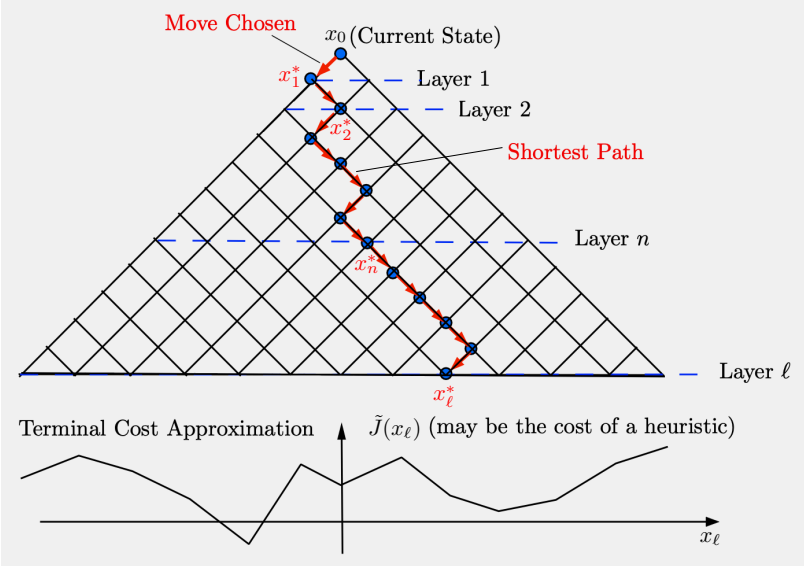

Figure 2.4.3 Illustration of the general ℓ-step approximation in value space scheme with a terminal cost approximation $\tilde{J}$ where x_0 denotes the current state. It involves an acyclic graph of ℓ layers, with layer n, $n = 1, \ldots, \ell$, consisting of all the states x_n that can be reached from x_0 with a sequence of n feasible controls. In ℓ-step approximation in value space, we obtain a trajectory

$$\{x_0, x_1^*, \ldots, x_\ell^*\}$$

that minimizes the shortest distance from x_0 to x_ℓ plus $\tilde{J}(x_\ell)$. We then use the control that corresponds to the first move $x_0 \to x_1^*$.

control pair, minimized over all possible controls that correspond to the same pair (x_n, x_{n+1}). Mathematically, the cost of the arc (x_n, x_{n+1}) is

$$\hat{g}_n(x_n, x_{n+1}) = \min_{\{u_n \in U_n(x_n) \mid x_{n+1} = f_n(x_n, u_n)\}} g_n(x_n, u_n). \tag{2.35}$$

For the states x_ℓ of the last layer there is also the terminal cost approximation $\tilde{J}(x_\ell)$, which may be obtained through off-line training or some other means. It can be thought of as the cost of an artificial arc connecting x_ℓ to an artificial termination state.

Once we have computed all the shortest distances $D(x_\ell)$ from x_0 to all states x_ℓ of the last layer ℓ, we obtain the ℓ-step lookahead control to be applied at the current state x_0, by minimizing over x_ℓ the sum

$$D(x_\ell) + \tilde{J}(x_\ell).$$

If x_ℓ^* is the state that attains the minimum, we generate the corresponding trajectory $(x_0, x_1^*, \ldots, x_\ell^*)$, and then use the control that corresponds to

the first move $x_0 \to x_1^*$; see Fig. 2.4.3. Note that the shortest path problems from x_0 to all states x_n of all the layers $n = 1, \ldots, \ell$ can be solved simultaneously by backward DP (start from layer ℓ and go back toward x_0).

Long Lookahead for Deterministic Problems

The architecture of Figs. 2.4.2 and 2.4.3 is similar to the one we discussed in Section 1.1 for AlphaZero and related programs. However, because it is adapted to deterministic problems, it is much simpler to implement and to use. In particular, the truncated rollout portion does not involve expensive Monte Carlo simulation, while the multistep lookahead minimization portion involves a deterministic shortest path problem, which is much easier to solve than its stochastic counterpart. These favorable characteristics can be exploited to facilitate implementations that involve very long lookahead.

Generally speaking, *longer lookahead is desirable because it typically results in improved performance*. We will adopt this as a working hypothesis. It is typically true in practice, although it cannot be established analytically in the absence of additional assumptions.† On the other hand, the on-line computational cost of multistep lookahead increases, often exponentially, with the length of lookahead. We conclude that *we should aim to use a lookahead that is as long as is allowed by the on-line computational budget* (the amount of time that is available for calculating a control to apply at the current state).

Long Lookahead by Rollout is Far More Economical than Long Lookahead Minimization

Our preceding discussion leads to the question of how to economize in computation in order to effectively increase the length of the multistep lookahead within a given on-line computational budget. One way to do this, which we have already discussed, is the use of truncated rollout that explores forward through a deterministic base policy at far less computational cost than lookahead minimization of equal length. As an example, let us consider the possibility of starting with a terminal cost function $\tilde{J}$, possibly generated by off-line training, and use as base policy for rollout the one-step lookahead policy $\tilde{\mu}$, defined by $\tilde{J}$ using the equation‡

$$\tilde{\mu}(x) \in \arg\min_{u \in U(x)} \Big[g(x,u) + \tilde{J}\big(f(x,u)\big) \Big]. \tag{2.36}$$

† Indeed, there are examples where as the size ℓ of the lookahead becomes longer, the performance of the multistep lookahead policy deteriorates (see [Ber17a], Section 6.1.2, or [Ber19a], Section 2.2.1). However, these examples are isolated and artificial. They are not representative of practical experience.

‡ For simplicity, we use stationary system notation, omitting the time subscripts of U, g, and f.

Let us assume that the principal computation in the minimization of Eq. (2.36) is the calculation of $\tilde{J}\big(f(x,u)\big)$, and compare two possibilities:

(a) Using *ℓ-step lookahead minimization with $\tilde{J}$ as the terminal cost approximation without any rollout*; cf. Fig. 2.4.3.

(b) Using *one-step lookahead minimization, with $(\ell-1)$-step truncated rollout and $\tilde{J}$ as the terminal cost approximation.*

Note that scheme (b) is the one used by the TD-Gammon program of Tesauro and Galperin [TeG96], out of necessity: multistep lookahead minimization is very expensive in backgammon, due to the rapid growth of the lookahead graph as ℓ increases (cf. the discussion of Section 1.1).

Suppose that the control set $U(x)$ has m elements for every x. Then the ℓ-step lookahead minimization *scheme (a) requires the calculation of as many as m^ℓ values of $\tilde{J}$*, because the number of leaves of the m-step lookahead graph are as many as m^ℓ.

The corresponding number of calculations of the value of $\tilde{J}$ for scheme (b) is clearly much smaller. To verify this, let us calculate this number as a function of m and ℓ. In particular, the first lookahead stage starting from the current state x_k requires m calculations corresponding to the m controls in $U(x_k)$, and yields corresponding states x_{k+1}, which are as many as m. For each of these states x_{k+1}, we must calculate a sequence of $\ell-1$ controls using the base policy (2.36) for stages $(k+1)$ to $(k+\ell)$. Each of these $\ell-1$ controls requires m calculations of the value of $\tilde{J}$. Thus, for the $\ell-1$ stages of truncated rollout, there are $m\cdot(\ell-1)$ calculations of the value of $\tilde{J}$ per state x_{k+1}, for a total of as many as $m^2\cdot(\ell-1)$ calculations. Adding the m calculations at state x_k, we conclude that *scheme (b) requires a total of as many as $m^2\cdot\ell$ calculations of the value of $\tilde{J}$.*

In conclusion, *both schemes (a) and (b) above look forward for ℓ stages, but their associated total computation grows exponentially and linearly with ℓ, respectively.* Thus, for a given computational budget, short lookahead minimization with long truncated rollout, can increase the total amount of lookahead and improve the performance of approximation in value space schemes. This is particularly so since based on the Newton step interpretations of approximation in value space of Section 1.5, truncated rollout with a reasonably good (e.g., stable) base policy often works about as well as long lookahead minimization. Extensive computational practice, starting with the rollout/TD-Gammon scheme of [TeG96], is consistent with this assessment.

In the following two sections, we will explore two alternative ways to speed up the lookahead minimization calculation, thereby allowing a larger number ℓ of computational stages for a given on-line computational budget. These are based on *iterative deepening* of the shortest path computation, and *pruning* of the lookahead minimization graph.

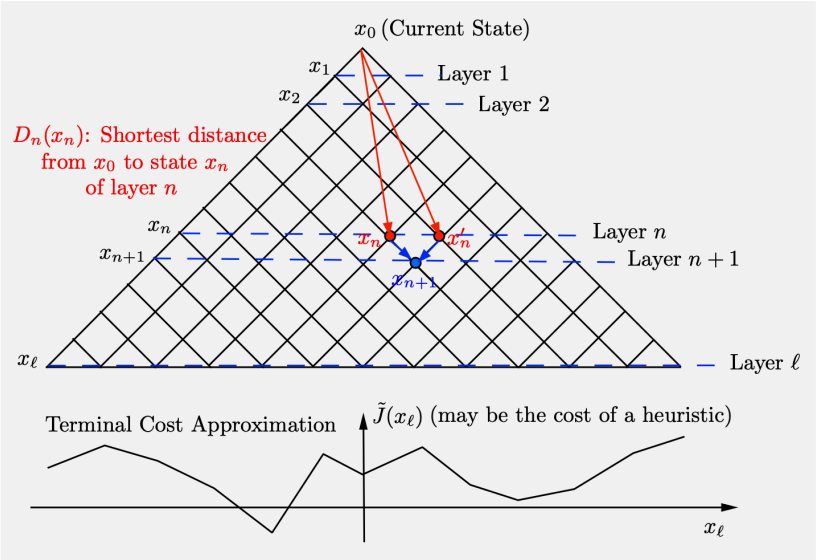

Figure 2.4.4 Illustration of the forward DP algorithm for computing the shortest distances from the current state x_0 to all the states x_n of the layers $n = 1, \ldots, \ell$. The shortest distance $D_{n+1}(x_{n+1})$ to a state x_{n+1} of layer $n+1$ is obtained by minimizing over all predecessor states x_n the sum

$$\hat{g}_n(x_n, x_{n+1}) + D_n(x_n).$$

2.4.1 Iterative Deepening Using Forward Dynamic Programming

As noted earlier, the shortest path problems from x_0 to x_ℓ in Fig. 2.4.3 can be solved simultaneously by the familiar backward DP that starts from layer ℓ and goes towards x_0. An important alternative for solving these problems is the *forward DP* algorithm. This is the same as the backwards DP algorithm with the *direction of the arcs reversed* (start from x_0 and go towards layer ℓ). In particular, the shortest distances $D_{n+1}(x_{n+1})$ to layer $n+1$ states are obtained from the shortest distances $D_n(x_n)$ to layer n states through the equation

$$D_{n+1}(x_{n+1}) = \min_{x_n} \Big[\hat{g}_n(x_n, x_{n+1}) + D_n(x_n)\Big],$$

which is also illustrated in Fig. 2.4.4. Here $\hat{g}_n(x_n, x_{n+1})$ is the cost (or length) of the arc (x_n, x_{n+1}); cf. Eq. (2.35).

In particular, the solution of the ℓ-step lookahead problem is obtained from the shortest path to the state x_ℓ^* of layer ℓ that minimizes $D_\ell(x_\ell) + \tilde{J}(x_\ell)$. The idea of iterative deepening is to *progressively solve the n-step lookahead problem first for $n = 1$, then for $n = 2$, and so on, until our*

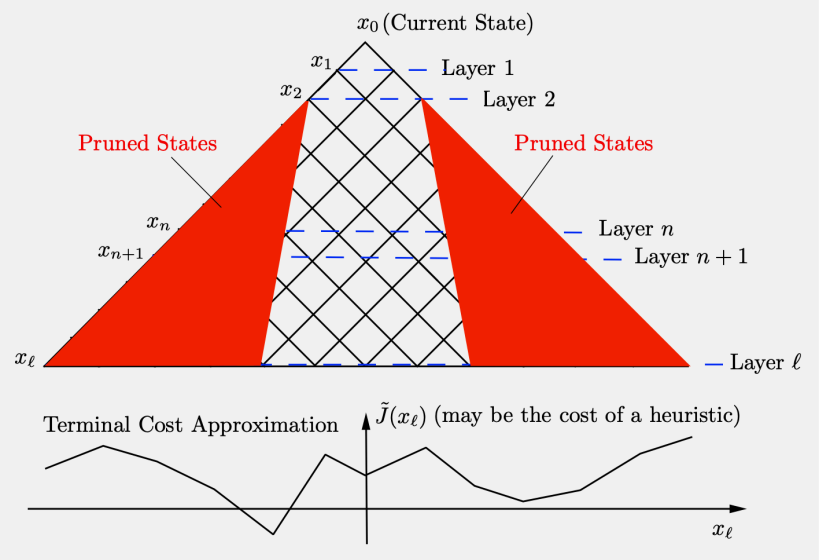

Figure 2.4.5 Illustration of iterative deepening with pruning within the context of forward DP.

on-line computational budget is exhausted. In addition to fitting perfectly the mechanism of the forward DP algorithm, this scheme has the character of an *"anytime" algorithm*; i.e., it returns the shortest distances to some depth $n \leq \ell$, which can in turn yield a solution of an n-step lookahead minimization after adding a suitable terminal cost function. In practice, this is an important advantage, well known from chess programming, which allows us to keep on aiming for longer lookahead minimization, within the limit imposed by our computational budget constraint.

Iterative Deepening Combined with Pruning

A principal difficulty in approximation in value space with ℓ-step lookahead stems from the rapid expansion of the lookahead graph as ℓ increases. One way to mitigate this difficulty is to "prune" the lookahead minimization graph, i.e., to delete some of its arcs in order to expedite the shortest path computations from the current state to the states of subsequent layers; see Fig. 2.4.5. One possibility is to combine pruning with iterative deepening by eliminating from the computation states $\hat{x}_n$ of layer n such that the n-step lookahead cost $D_n(\hat{x}_n) + \tilde{J}(\hat{x}_n)$ is "far from the minimum" over x_n. This in turn prunes automatically some of the states of the next layer $n+1$. The rationale is that such states are "unlikely" to be part of the shortest path that we aim to compute. Note that this type of pruning is progressive, i.e., we prune states in layer n before pruning states in layer $n+1$.

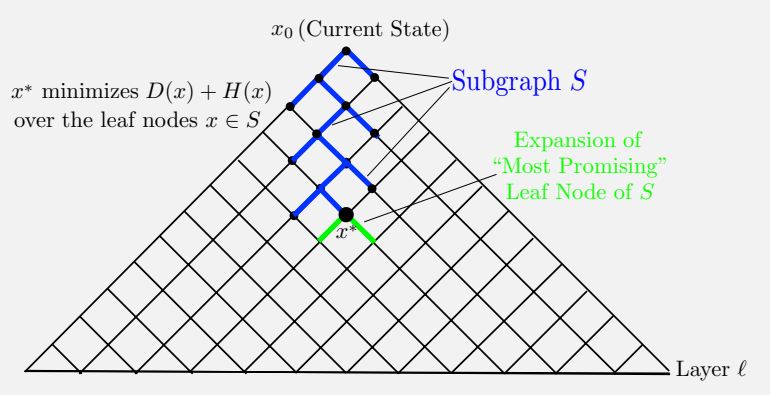

Figure 2.4.6 Illustration of the ℓ-step lookahead minimization problem and its suboptimal solution with the IMR algorithm. The algorithm maintains a connected acyclic subgraph S as shown. At each iteration it expands S by selecting a leaf node of S and by adding its neighbor nodes to S (if not already in S). The leaf node, denoted x^*, is the "most promising": the one that minimizes over all leaf nodes x of S the sum of the shortest distance $D(x)$ from x_0 to x and a "heuristic cost" $H(x)$.

2.4.2 Incremental Multistep Rollout

We will now consider a more flexible form of the rollout scheme, which we call *incremental multistep rollout* (IMR). It applies a base heuristic and a forward DP computation to a sequence of subgraphs of a multistep lookahead graph, with the size of the subgraphs expanding iteratively. In particular, in incremental rollout a connected subgraph of multiple paths is iteratively extended starting from the current state going towards the end of the lookahead horizon, instead of extending a single path as in rollout. This is similar to what is done in Monte Carlo Tree Search (MCTS, to be discussed later), which is also designed to solve approximately general multistep lookahead minimization problems (including stochastic ones), and involves iterative expansion of an acyclic lookahead graph to new nodes, as well as backtracking to previously encountered nodes. However, incremental rollout seems to be more appropriate than MCTS for deterministic problems, where there are no random variables in the problem's model and therefore Monte Carlo simulation does not make sense.

The IMR algorithm starts with and maintains a connected acyclic subgraph S of the given multistep lookahead graph G, which contains x_0. At each iteration it expands S by selecting a leaf node of S and by adding its neighbor nodes to S (if not already in S); see Fig. 2.4.6. This leaf node, denoted x^*, is the "most promising" one, in the sense that it minimizes (over all leaf nodes x of S) the sum

$$D(x) + H(x),$$

where

$D(x)$ is the shortest distance from x_0 to the leaf node x using only arcs that belong to S. This can be computed by forward DP. [A noteworthy possibility is to replace $D(x)$ with a conveniently computed approximation, which may be problem-specific. We will discuss such schemes later in this section.]

$H(x)$ is a "heuristic cost" corresponding to x. This is defined as the sum of three terms:

(a) The cost of the base heuristic starting from node x and ending at one of the states x_ℓ in the last layer ℓ.

(b) The terminal cost approximation $\tilde{J}(x_\ell)$, where x_ℓ is the state obtained via the base heuristic as in (a) above.

(c) An additional loss term $L(x)$ that depends on the layer to which x belongs. As an example, we will assume here that

$$L(x) = \delta \cdot (\text{the layer index of } x),$$

where δ is a positive parameter. Thus $L(x)$ adds a cost of δ for each extra arc to reach x from x_0, and penalizes nodes x that lie in more distant layers from the root x_0. It encourages the algorithm to "backtrack" and select nodes x^* that lie in layers closer to x_0. [Other ways to define $L(x)$ may also be considered.]

The algorithm starts with a connected acyclic subgraph S of the given multistep lookahead graph G, which contains x_0, and with a path P that starts at x_0, goes through S and ends at one of the terminal nodes of S (one possibility is to take S and P equal to just the root node x_0). It terminates when the end node of P belongs to layer ℓ (or earlier if a computational budget constraint is reached).

The role of the parameter δ is noteworthy and affects significantly the nature of the algorithm. When $\delta = 0$, the initial graph S consists of the single state x_0, and the base heuristic is sequentially improving, it can be seen that IMR performs exactly like the rollout algorithm for solving the ℓ-step lookahead minimization problem. On the other hand when δ is large enough, the algorithm operates like the forward DP algorithm. The reason is that a very large value of δ forces the algorithm to expand all nodes of a given layer before proceeding to the next layer.

Generally, as δ increases, the algorithm tends to backtrack more often, and to generate more paths through the graph, thereby visiting more nodes and increasing the number of applications of the base heuristic. Thus δ may be viewed as an *exploration parameter*; when δ is large the algorithm tends to explore more paths thereby improving the quality of the multistep lookahead minimization, at the expense of greater computational effort. In the absence of additional problem-specific information, favorable values

of δ should be obtained through experimentation. One may also consider alternative and more adaptive schemes; for example with a δ that depends on x_0, and is adjusted in the course of the computation. Finally, we note that if the base heuristic used in the calculation of $H(x)$ is sequentially consistent, a local optimality property of the incremental rollout trajectory can be shown (cf. Section 2.3.1).

Approximations in the Incremental Multistep Rollout Scheme

Let us now consider variants of the IMR scheme, which are aimed towards expediting its calculations, possibly at the expense of some degradation in its performance guarantees. To this end, it is useful to view the algorithm as maintaining a list L of the current leaf nodes of S. At each step a node is removed from L, and its neighbor nodes are added to S and to L, if not already there. This process continues until a path through S that starts at the root node x_0 and ends at some node of the last layer ℓ is constructed. In particular, each step of the IMR algorithm consists of:

(a) The selection of a node x^* of L for expansion.

(b) The addition to S and to L of each neighbor node x of the expanded node x^*, if x does not already belong to S and/or L, respectively. The values $D(x)$ and $H(x)$ are also computed at the time when x enters the list L.

(c) The removal of x^* from L.

Since evidently a node can enter the list L at most once, the algorithm will terminate with a path that starts at x_0 and ends at some node of layer ℓ. Moreover, this will happen regardless of which leaf node of the current subgraph S is chosen at each step for expansion.

It follows from the preceding argument that selecting the leaf node x^* that minimizes $D(x)+H(x)$ at each step is not essential to the algorithm's termination. This allows some flexibility in designing variations of the IMR algorithm, which aim at expediting the computation. In particular, we may consider selecting a leaf node x that has a "low" value of $D(x)+H(x)$ instead of selecting the one that has minimum value, while maintaining an appropriate cost improvement property. Two possibilities of this type are as follows:

(1) Replace $D(x)$, the shortest distance from x_0 to x through the subgraph S, with an upper bound $\overline{D}(x)$, which is the shortest distance from x_0 to x through the subgraph S *at the time that* x *becomes a leaf node of* S *and enters the list* L of leaf nodes. Note that as the set S grows, $D(x)$ may become smaller than $\overline{D}(x)$, as more nodes are added to S and additional paths from x_0 to x through S are created. The important point here is that in general $\overline{D}(x)$ is in many cases likely to be either equal or not too different from $D(x)$. Moreover,

$\overline{D}(x)$ is computed only once, at the time when x becomes a leaf node of S, thus saving in computational overhead.

(2) Organize L as a priority queue and instead of selecting a node x^* that minimizes $D(x)+H(x)$ or $\overline{D}(x)+H(x)$ over all nodes x of L, simply let x^* be the top node of the queue. This will save the overhead for minimizing over the nodes of L, which can be very large in number, depending on the problem at hand. We note here that it is possible to reduce this overhead by organizing L with a heap or bucket data structure, whereby the minimizing node is efficiently selected; such schemes are well known from implementations of label setting shortest path computations (i.e., Dijkstra's algorithm, see e.g., [Ber98], Ch. 2). However, simpler schemes to organize L are possible, which place nodes with small value of $D(x)+H(x)$ near the top of the queue. Such schemes have been proposed by the author for label correcting methods for shortest paths, in the context of approximations to Dijkstra's algorithm. We refer to the SLF (Small Label First), LLL (Last Label Last), and threshold shortest path algorithms, which are described in the network optimization book [Ber98] (Chapter 2) and the references given in that book. In practice, these algorithms work very well, and typically better than the heap or bucket schemes.

An important point is that when δ is small or is 0, the preceding variants of the IMR algorithm can be easily modified to coincide with the rollout algorithm, and inherit the corresponding cost improvement property. Moreover the cost improvement property can be restored by using the fortified rollout ideas of Section 2.3. In addition, the flexibility afforded by the modifications (1) and (2) above allow variations of the IMR scheme that are tailored to the problem at hand.

2.5 CONSTRAINED FORMS OF ROLLOUT ALGORITHMS

In this section we will discuss constrained deterministic DP problems, including challenging combinatorial optimization and integer programming problems. We introduce a rollout algorithm, which relies on a base heuristic and applies to problems with general trajectory constraints. Under suitable assumptions, we will show that if the base heuristic produces a feasible solution, the rollout algorithm has a cost improvement property: it produces a feasible solution, whose cost is no worse than the base heuristic's cost.

Before going into formal descriptions of the constrained DP problem formulation and the corresponding algorithms, it is worth revisiting the broad outline of the rollout algorithm for deterministic DP:

(a) It constructs a sequence $\{T_0, T_1, \ldots, T_N\}$ of complete system trajectories with monotonically nonincreasing cost (assuming a sequential improvement condition).

(b) The initial trajectory T_0 is the one generated by the base heuristic starting from x_0, and the final trajectory T_N is the one generated by the rollout algorithm.

(c) For each k, the trajectories $T_k, T_{k+1}, \ldots, T_N$ share the same initial portion $(x_0, \tilde{u}_0, \ldots, \tilde{u}_{k-1}, \tilde{x}_k)$.

(d) For each k, the base heuristic is used to generate a number of candidate trajectories, all of which share the initial portion with T_k, up to state $\tilde{x}_k$. These candidate trajectories correspond to the controls $u_k \in U_k(x_k)$. (In the case of fortified rollout, these trajectories include the current "tentative best" trajectory.)

(e) For each k, the next trajectory T_{k+1} is the candidate trajectory that is best in terms of total cost.

In our constrained DP formulation, to be described shortly, we introduce a trajectory constraint $T \in C$, where C is some subset of admissible trajectories. A consequence of this is that some of the candidate trajectories in (d) above, may be infeasible. Our modification to deal with this situation is simple: *we discard all the candidate trajectories that violate the constraint, and we choose T_{k+1} to be the best of the remaining candidate trajectories, the ones that are feasible.*

Of course, for this modification to be viable, we have to guarantee that at least one of the candidate trajectories will satisfy the constraint for every k. For this we will rely on a sequential improvement condition that we will introduce shortly. For the case where this condition does not hold, we will introduce a fortified version of the algorithm, which requires only that the base heuristic generates a feasible trajectory T_0 starting from the initial condition x_0. Thus *to apply reliably the constrained rollout algorithm, we only need to know a single feasible solution*, i.e., a trajectory T_0 that starts at x_0 and satisfies the constraint $T_0 \in C$.

Constrained Problem Formulation

We assume that the state x_k takes values in some (possibly infinite) set and the control u_k takes values in some finite set. The finiteness of the control space is only needed for implementation purposes of the rollout algorithms to be described shortly. The algorithm can be defined without the finiteness condition, and makes sense, provided the implementation issues associated with infinite control spaces can be dealt with. A sequence of the form

$$T = (x_0, u_0, x_1, u_1, \ldots, u_{N-1}, x_N), \tag{2.37}$$

where

$$x_{k+1} = f_k(x_k, u_k), \qquad k = 0, 1, \ldots, N-1, \tag{2.38}$$

is referred to as a *complete trajectory*. Our problem is stated succinctly as

$$\min_{T \in C} G(T), \tag{2.39}$$

where G is some cost function and C is the constraint set.

Note that G need not have the additive form

$$G(T) = g_N(x_N) + \sum_{k=0}^{N-1} g_k(x_k, u_k), \tag{2.40}$$

which we have assumed so far. Thus, except for the finiteness of the control space, which is needed for implementation of rollout, this is a very general optimization problem. In fact, later we will simplify the problem further by eliminating the state transition structure of Eq. (2.38).†

Trajectory constraints can arise in a number of ways. A relatively simple example is the standard problem formulation for deterministic DP: an additive cost of the form (2.40), where the controls satisfy the time-uncoupled constraints $u_k \in U_k(x_k)$ [so here C is the set of trajectories that are generated by the system equation with controls satisfying $u_k \in U_k(x_k)$]. In a more complicated constrained DP problem, there may be constraints that couple the controls of different stages such as

$$g_N^m(x_N) + \sum_{k=0}^{N-1} g_k^m(x_k, u_k) \leq b^m, \qquad m = 1, \ldots, M, \tag{2.41}$$

where g_k^m and b^m are given functions and scalars, respectively. Examples of this type include *multiobjective* or *Pareto* optimization problems, where there are multiple cost functions of interest, and all but one of the cost functions are treated through constraints (see e.g., [Ber17a], Ch. 2). Examples where difficult trajectory constraints arise also include situations where the control contains some discrete components, which once chosen must remain fixed for multiple time periods.

Here is a discrete optimization example involving the traveling salesman problem. A related classical example is the knapsack problem, described in most books on discrete optimization algorithms.

Example 2.5.1 (A Constrained Form of the Traveling Salesman Problem)

Let us consider a constrained version of the traveling salesman problem of Example 1.2.2. We want to find a minimum travel cost tour that additionally

† Actually, similar to our discussion on model-free rollout in Section 2.3.6, it is not essential that we know the explicit form of the cost function G and the constraint set C. For our constrained rollout algorithms, it is sufficient to have access to a human or software expert that can determine whether a given trajectory T is feasible, i.e., satisfies the constraint $T \in C$, and also to be able to compare any two feasible trajectories T_1 and T_2, based on some internal process that is unknown to us, without assigning numerical values to them.

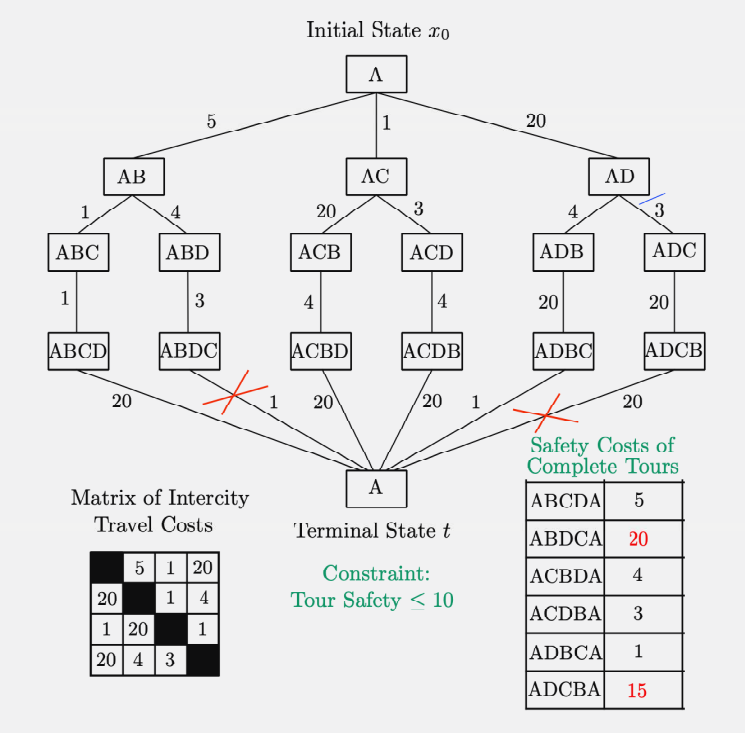

Figure 2.5.1 An example of a constrained traveling salesman problem; cf. Example 2.5.1. We want to find a minimum cost tour that has safety cost less or equal to 10. The safety costs of the six possible tours are given in the table on the right. The (unconstrained) minimum cost tour, ABDCA, does not satisfy the safety constraint. The optimal constrained tour is ABCDA.

satisfies a safety constraint that the "safety cost" of the tour should be less than a certain threshold; see Fig. 2.5.1. This constraint need not have the additive structure of Eq. (2.41). We are simply given a safety cost for each tour (see the table at the bottom right), which is calculated in a way that is of no further concern to us. In this example, for a tour to be admissible, its safety cost must be less than or equal to 10. Note that the (unconstrained) minimum cost tour, ABDCA, does not satisfy the safety constraint.

Using a Base Heuristic for Constrained Rollout

We will now describe formally the constrained rollout algorithm. We assume the availability of a base heuristic, which for any given partial trajectory

$$y_k = (x_0, u_0, x_1, \ldots, u_{k-1}, x_k),$$

can produce a (complementary) partial trajectory

$$R(y_k) = (x_k, u_k, x_{k+1}, u_{k+1}, \ldots, u_{N-1}, x_N),$$

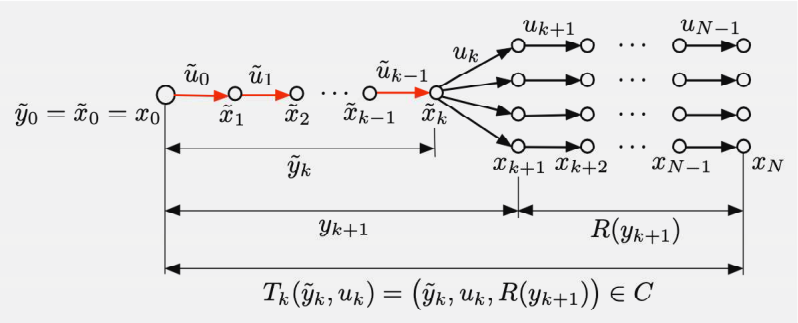

Figure 2.5.2 The trajectory generation mechanism of the rollout algorithm. At stage k, and given the current partial trajectory

$$\tilde{y}_k = (\tilde{x}_0, \tilde{u}_0, \tilde{x}_1, \ldots, \tilde{u}_{k-1}, \tilde{x}_k),$$

which starts at $\tilde{x}_0$ and ends at $\tilde{x}_k$, we consider all possible next states $x_{k+1} = f_k(\tilde{x}_k, u_k)$, run the base heuristic starting at

$$y_{k+1} = (\tilde{y}_k, u_k, x_{k+1}),$$

and form the complete trajectory $T_k(\tilde{y}_k, u_k)$. Then the rollout algorithm:

(a) Finds $\tilde{u}_k$, the control that minimizes the cost $G\big(T_k(\tilde{y}_k, u_k)\big)$ over all u_k for which the complete trajectory $T_k(\tilde{y}_k, u_k)$ is feasible.

(b) Extends $\tilde{y}_k$ by $\big(\tilde{u}_k, f_k(\tilde{x}_k, \tilde{u}_k)\big)$ to form $\tilde{y}_{k+1}$.

that starts at x_k and satisfies the system equation

$$x_{t+1} = f_t(x_t, u_t), \qquad t = k, \ldots, N-1.$$

Thus, given y_k and any control u_k, we can use the base heuristic to obtain a complete trajectory as follows:

(a) Generate the next state $x_{k+1} = f_k(x_k, u_k)$.

(b) Extend y_k to obtain the partial trajectory

$$y_{k+1} = \big(y_k, u_k, f_k(x_k, u_k)\big).$$

(c) Run the base heuristic from y_{k+1} to obtain the partial trajectory $R(y_{k+1})$.

(d) Join the two partial trajectories y_{k+1} and $R(y_{k+1})$ to obtain the complete trajectory $\big(y_k, u_k, R(y_{k+1})\big)$, which is denoted by $T_k(y_k, u_k)$:

$$T_k(y_k, u_k) = \big(y_k, u_k, R(y_{k+1})\big). \tag{2.42}$$

This process is illustrated in Fig. 2.5.2. Note that the partial trajectory $R(y_{k+1})$ produced by the base heuristic depends on the entire partial trajectory y_{k+1}, not just the state x_{k+1}.

A complete trajectory $T_k(y_k, u_k)$ of the form (2.42) is generally feasible for only the subset $\hat{U}_k(y_k)$ of controls u_k that maintain feasibility:

$$\hat{U}_k(y_k) = \big\{u_k \mid T_k(y_k, u_k) \in C\big\}. \tag{2.43}$$

Our rollout algorithm starts from a given initial state $\tilde{y}_0 = \tilde{x}_0$, and generates successive partial trajectories

$$(\tilde{y}_1, \ldots, \tilde{y}_N),$$

of the form

$$\tilde{y}_{k+1} = \big(\tilde{y}_k, \tilde{u}_k, f_k(\tilde{x}_k, \tilde{u}_k)\big), \qquad k = 0, \ldots, N-1, \tag{2.44}$$

where $\tilde{x}_k$ is the last state component of $\tilde{y}_k$, and $\tilde{u}_k$ is a control that minimizes the heuristic cost $G\big(T_k(\tilde{y}_k, u_k)\big)$ over all u_k for which $T_k(\tilde{y}_k, u_k)$ is feasible. Thus at stage k, the algorithm forms the set $U_k(\tilde{y}_k)$ [cf. Eq. (2.43)] and selects from $U_k(\tilde{y}_k)$ a control $\tilde{u}_k$ that minimizes the cost of the complete trajectory $T_k(\tilde{y}_k, u_k)$:

$$\tilde{u}_k \in \arg \min_{u_k \in U_k(\tilde{y}_k)} G\big(T_k(\tilde{y}_k, u_k)\big); \tag{2.45}$$

see Fig. 2.5.2. The objective is to produce a feasible final complete trajectory $\tilde{y}_N$, which has a cost $G(\tilde{y}_N)$ that is no larger than the cost of $R(\tilde{y}_0)$ produced by the base heuristic starting from $\tilde{y}_0$, i.e.,

$$G(\tilde{y}_N) \leq G\big(R(\tilde{y}_0)\big).$$

Note that $T_k(\tilde{y}_k, u_k)$ is not guaranteed to be feasible for any given u_k (i.e., may not belong to C), but we will assume that the constraint set $U_k(\tilde{y}_k)$ of problem (2.45) is nonempty, so that our rollout algorithm is well-defined. We will later modify our algorithm so that it is well-defined under the weaker assumption that just *the complete trajectory generated by the base heuristic starting from the initial state $\tilde{y}_0$ is feasible*, i.e., $R(\tilde{y}_0) \in C$.

Constrained Rollout Algorithm

The algorithm starts at stage 0 and sequentially proceeds to the last stage. At the typical stage k, it has constructed a partial trajectory

$$\tilde{y}_k = (\tilde{x}_0, \tilde{u}_0, \tilde{x}_1, \ldots, \tilde{u}_{k-1}, \tilde{x}_k) \tag{2.46}$$

that starts at the given initial state $\tilde{y}_0 = \tilde{x}_0$, and is such that

$$\tilde{x}_{t+1} = f_t(\tilde{x}_t, \tilde{u}_t), \qquad t = 0, 1, \ldots, k-1.$$

The algorithm then forms the set of controls

$$U_k(\tilde{y}_k) = \big\{u_k \mid T_k(\tilde{y}_k, u_k) \in C\big\}$$

that is consistent with feasibility [cf. Eq. (2.43)], and chooses a control $\tilde{u}_k \in U_k(\tilde{y}_k)$ according to the minimization

$$\tilde{u}_k \in \arg \min_{u_k \in U_k(\tilde{y}_k)} G\big(T_k(\tilde{y}_k, u_k)\big), \tag{2.47}$$

[cf. Eq. (2.45)], where

$$T_k(\tilde{y}_k, u_k) = \Big(\tilde{y}_k, u_k, R\big(\tilde{y}_k, u_k, f_k(\tilde{x}_k, u_k)\big)\Big);$$

[cf. Eq. (2.42)]. Finally, the algorithm sets

$$\tilde{x}_{k+1} = f_k(\tilde{x}_k, \tilde{u}_k), \qquad \tilde{y}_{k+1} = (\tilde{y}_k, \tilde{u}_k, \tilde{x}_{k+1}),$$

[cf. Eq. (2.44)], thus obtaining the partial trajectory $\tilde{y}_{k+1}$ to start the next stage.

It can be seen that our constrained rollout algorithm is not much more complicated or computationally demanding than its unconstrained version where the constraint $T \in C$ is not present (as long as checking feasibility of a complete trajectory T is not computationally demanding). Note, however, that our algorithm makes essential use of the deterministic character of the problem, and does not admit a straightforward extension to stochastic problems, since checking feasibility of a complete trajectory is typically difficult in the context of these problems.

The rollout algorithm just described is illustrated in Fig. 2.5.3 for our earlier traveling salesman Example 2.5.1. Here we want to find a minimum travel cost tour that additionally satisfies a safety constraint, namely that the "safety cost" of the tour should be less than a certain threshold. This cost may be given by a table as in Fig. 2.5.3, or it may be generated on line with an algorithm for any given tour. Note that the minimum cost tour, ABDCA, in this example does not satisfy the safety constraint. Moreover, the tour ABCDA obtained by the rollout algorithm has barely smaller cost than the tour ACDBA generated by the base heuristic starting from A. In fact if the travel cost D→A were larger, say 25, the tour produced by constrained rollout would be more costly than the one produced by the base heuristic starting from A. This points to the need for a constrained

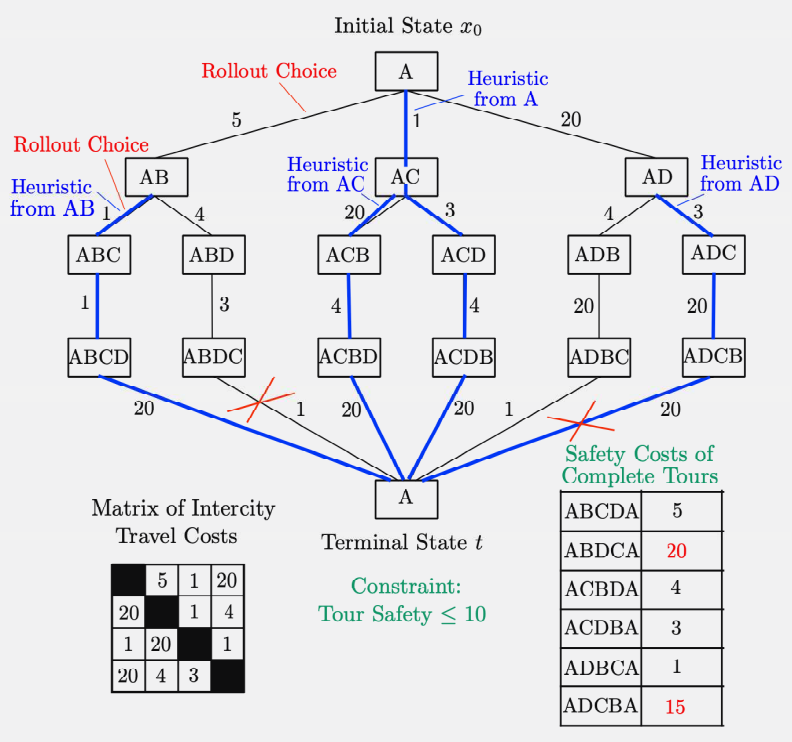

Figure 2.5.3 The constrained traveling salesman problem; cf. Example 2.5.1, and its rollout solution using the base heuristic shown, which completes a partial tour as follows:

At A it yields ACDBA.
At AB it yields ABCDA.
At AC it yields ACBDA.
At AD it yields ADCBA.

This base heuristic is not assumed to have any special structure. It is just capable of completing every partial tour without regard to any additional considerations. Thus for example the heuristic generates at A the complete tour ACDBA, and it switches to the tour ACBDA once the salesman moves to AC.

At city A, the rollout algorithm:

(a) Considers the partial tours AB, AC, and AD.

(b) Uses the base heuristic to obtain the corresponding complete tours ABCDA, ACBDA, and ADCBA.

(c) Discards ADCBA as being infeasible.

(d) Compares the other two tours, ABCDA and ACBDA, finds ABCDA to have smaller cost, and selects the partial tour AB.

(e) At AB, it considers the partial tours ABC and ABD.

(f) It uses the base heuristic to obtain the corresponding complete tours ABCDA and ABDCA, and discards ABDCA as being infeasible.

(g) It finally selects the complete tour ABCDA.

version of the notion of sequential improvement and for a fortified variant of the algorithm, which we discuss next.

Sequential Consistency, Sequential Improvement, and the Cost Improvement Property

We will now introduce sequential consistency and sequential improvement conditions guaranteeing that the control set $U_k(\tilde{y}_k)$ in the minimization (2.47) is nonempty, and that the costs of the complete trajectories

$$T_k(\tilde{y}_k, \tilde{u}_k)$$

are improving with each k in the sense that

$$G\big(T_{k+1}(\tilde{y}_{k+1}, \tilde{u}_{k+1})\big) \leq G\big(T_k(\tilde{y}_k, \tilde{u}_k)\big), \qquad k = 0, 1, \ldots, N-1,$$

while at the first step of the algorithm we have

$$G\big(T_0(\tilde{y}_0, \tilde{u}_0)\big) \leq G\big(R(\tilde{y}_0)\big).$$

It will then follow that the cost improvement property

$$G(\tilde{y}_N) \leq G\big(R(\tilde{y}_0)\big)$$

holds.

Definition 2.5.1: We say that the base heuristic is *sequentially consistent* if whenever it generates a partial trajectory

$$(x_k, u_k, x_{k+1}, u_{k+1}, \ldots, u_{N-1}, x_N),$$

starting from a partial trajectory y_k, it also generates the partial trajectory

$$(x_{k+1}, u_{k+1}, x_{k+2}, u_{k+2}, \ldots, u_{N-1}, x_N),$$

starting from the partial trajectory $y_{k+1} = \big(y_k, u_k, x_{k+1}\big)$.

As we have noted in the context of unconstrained rollout, greedy heuristics tend to be sequentially consistent. Also any policy [a sequence of feedback control functions $\mu_k(y_k)$, $k = 0, 1, \ldots, N-1$] for the DP problem of minimizing the terminal cost $G(y_N)$ subject to the system equation

$$y_{k+1} = \big(y_k, u_k, f_k(x_k, u_k)\big)$$

and the feasibility constraint $y_N \in C$ can be seen to be sequentially consistent. For an example where sequential consistency is violated, consider the base heuristic of the traveling salesman Example 2.5.1. From Fig. 2.5.3, it can be seen that the base heuristic at A generates ACDBA, but from AC it generates ACBDA, thus violating sequential consistency.

For a given partial trajectory y_k, let us denote by $y_k \cup R(y_k)$ the complete trajectory obtained by joining y_k with the partial trajectory generated by the base heuristic starting from y_k. Thus if

$$y_k = (x_0, u_0, \ldots, u_{k-1}, x_k)$$

and

$$R(y_k) = (x_k, u_k, \ldots, u_{N-1}, x_N),$$

we have

$$y_k \cup R(y_k) = (x_0, u_0, \ldots, u_{k-1}, x_k, u_k, \ldots, u_{N-1}, x_N).$$

Definition 2.5.2: We say that the base heuristic is *sequentially improving* if for every $k = 0, 1, \ldots, N-1$ and partial trajectory y_k for which $y_k \cup R(y_k) \in C$, the set $\hat{U}_k(y_k)$ is nonempty, and we have

$$G\big(y_k \cup R(y_k)\big) \geq \min_{u_k \in \hat{U}_k(y_k)} G\big(T_k(y_k, u_k)\big). \tag{2.48}$$

Note that for a base heuristic that is not sequentially consistent, the condition $y_k \cup R(y_k) \in C$ does not imply that the set $\hat{U}_k(y_k)$ is nonempty. The reason is that starting from the next state

$$y_{k+1} = \big(y_k, u_k, f_k(x_k, u_k)\big),$$

the base heuristic may generate a different trajectory than from y_k, even if it applies u_k at y_k. Thus we need to include nonemptiness of $\hat{U}_k(y_k)$ as a requirement in the preceding definition of sequential improvement (in the fortified version of the algorithm to be discussed shortly, this requirement will be removed).

On the other hand, if the base heuristic is sequentially consistent, it is also sequentially improving. The reason is that for a sequentially consistent heuristic, $y_k \cup R(y_k)$ is equal to one of the trajectories contained in the set

$$\big\{T_k(y_k, u_k) \mid u_k \in \hat{U}_k(y_k)\big\}.$$

Our main result is contained in the following proposition.

Proposition 2.5.1: (Cost Improvement for Constrained Rollout) Assume that the base heuristic is sequentially improving and generates a feasible complete trajectory starting from the initial state $\tilde{y}_0 = \tilde{x}_0$, i.e., $R(\tilde{y}_0) \in C$. Then for each k, the set $U_k(\tilde{y}_k)$ is nonempty, and we have

$$\begin{aligned} G\big(R(\tilde{y}_0)\big) &\geq G\big(T_0(\tilde{y}_0, \tilde{u}_0)\big) \\ &\geq G\big(T_1(\tilde{y}_1, \tilde{u}_1)\big) \\ &\geq \cdots \\ &\geq G\big(T_{N-1}(\tilde{y}_{N-1}, \tilde{u}_{N-1})\big) \\ &= G(\tilde{y}_N), \end{aligned}$$

where

$$T_k(\tilde{y}_k, \tilde{u}_k) = \big(\tilde{y}_k, \tilde{u}_k, R(\tilde{y}_{k+1})\big);$$

cf. Eq. (2.42). In particular, the final trajectory $\tilde{y}_N$ generated by the constrained rollout algorithm is feasible and has no larger cost than the trajectory $R(\tilde{y}_0)$ generated by the base heuristic starting from the initial state.

Proof: Consider $R(\tilde{y}_0)$, the complete trajectory generated by the base heuristic starting from $\tilde{y}_0$. Since $\tilde{y}_0 \cup R(\tilde{y}_0) = R(\tilde{y}_0) \in C$ by assumption, it follows from the sequential improvement definition, that the set $U_0(\tilde{y}_0)$ is nonempty and we have

$$G\big(R(\tilde{y}_0)\big) \geq G\big(T_0(\tilde{y}_0, \tilde{u}_0)\big),$$

[cf. Eq. (2.48)], while $T_0(\tilde{y}_0, \tilde{u}_0) \in C$.

The preceding argument can be repeated for the next stage, by replacing $\tilde{y}_0$ with $\tilde{y}_1$, and $R(\tilde{y}_0)$ with $T_0(\tilde{y}_0, \tilde{u}_0)$. Since $\tilde{y}_1 \cup R(\tilde{y}_1) = T_0(\tilde{y}_0, \tilde{u}_0) \in C$, from the sequential improvement definition, the set $U_1(\tilde{y}_1)$ is nonempty and we have

$$G\big(T_0(\tilde{y}_0, \tilde{u}_0)\big) = G\big(\tilde{y}_1 \cup R(\tilde{y}_1)\big) \geq G\big(T_1(\tilde{y}_1, \tilde{u}_1)\big),$$

[cf. Eq. (2.48)], while $T_1(\tilde{y}_1, \tilde{u}_1) \in C$. Similarly, the argument can be successively repeated for every k, to verify that $U_k(\tilde{y}_k)$ is nonempty and that $G\big(T_k(\tilde{y}_k, \tilde{u}_k)\big) \geq G\big(T_{k+1}(\tilde{y}_{k+1}, \tilde{u}_{k+1})\big)$ for all k. **Q.E.D.**

Proposition 2.5.1 establishes the fundamental cost improvement property for constrained rollout under the sequential improvement condition. On the other hand we may construct examples where the sequential improvement condition (2.48) is violated and the cost of the solution produced by rollout is larger than the cost of the solution produced by the

base heuristic starting from the initial state (cf. the unconstrained rollout Example 2.3.3).

In the case of the traveling salesman Example 2.5.1, it can be verified that the base heuristic specified in Fig. 2.5.3 is sequentially improving. However, if the travel cost D→A were larger, say 25, then it can be verified that the definition of sequential improvement would be violated at A, and the tour produced by constrained rollout would be more costly than the one produced by the base heuristic starting from A.

The Fortified Rollout Algorithm and Other Variations

We will now discuss some variations and extensions of the constrained rollout algorithm. Let us first consider the case where the sequential improvement assumption is not satisfied. Then it may happen that given the current partial trajectory $\tilde{y}_k$, the set of controls $U_k(\tilde{y}_k)$ that corresponds to feasible trajectories $T_k(\tilde{y}_k, u_k)$ [cf. Eq. (2.43)] is empty, in which case the rollout algorithm cannot extend the partial trajectory $\tilde{y}_k$ further. To bypass this difficulty, we introduce a *fortified constrained rollout algorithm*, patterned after the fortified algorithm given earlier. For validity of this algorithm, *we require that the base heuristic generates a feasible complete trajectory $R(\tilde{y}_0)$ starting from the initial state $\tilde{y}_0$.*

The fortified constrained rollout algorithm, in addition to the current partial trajectory

$$\tilde{y}_k = (\tilde{x}_0, \tilde{u}_0, \tilde{x}_1, \ldots, \tilde{u}_{k-1}, \tilde{x}_k),$$

maintains a complete trajectory $\hat{T}_k$, called *tentative best trajectory*, which is feasible (i.e., $\hat{T}_k \in C$) and agrees with $\tilde{y}_k$ up to state $\tilde{x}_k$, i.e., $\hat{T}_k$ has the form

$$\hat{T}_k = (\tilde{x}_0, \tilde{u}_0, \tilde{x}_1, \ldots, \tilde{u}_{k-1}, \tilde{x}_k, \overline{u}_k, \overline{x}_{k+1}, \ldots, \overline{u}_{N-1}, \overline{x}_N), \tag{2.49}$$

for some $\overline{u}_k, \overline{x}_{k+1}, \ldots, \overline{u}_{N-1}, \overline{x}_N$ such that

$$\overline{x}_{k+1} = f_k(\tilde{x}_k, \overline{u}_k), \qquad \overline{x}_{t+1} = f_t(\overline{x}_t, \overline{u}_t), \quad t = k+1, \ldots, N-1.$$

Initially, $\hat{T}_0$ is the complete trajectory $R(\tilde{y}_0)$, generated by the base heuristic starting from $\tilde{y}_0$, which is assumed to be feasible. At stage k, the algorithm forms the subset $\hat{U}_k(\tilde{y}_k)$ of controls $u_k \in U_k(\tilde{y}_k)$ such that the corresponding $T_k(\tilde{y}_k, u_k)$ is not only feasible, but also has cost that is no larger than the one of the current tentative best trajectory:

$$\hat{U}_k(\tilde{y}_k) = \Big\{ u_k \in U_k(\tilde{y}_k) \mid G\big(T_k(\tilde{y}_k, u_k)\big) \leq G(\hat{T}_k) \Big\}.$$

There are two cases to consider at state k:

(1) *The set* $\hat{U}_k(\tilde{y}_k)$ *is nonempty.* Then the algorithm forms the partial trajectory $\tilde{y}_{k+1} = (\tilde{y}_k, \tilde{u}_k, \tilde{x}_{k+1})$, where

$$\tilde{u}_k \in \arg \min_{u_k \in \hat{U}_k(\tilde{y}_k)} G\big(T_k(\tilde{y}_k, u_k)\big), \qquad \tilde{x}_{k+1} = f_k(\tilde{x}_k, \tilde{u}_k),$$

and sets $T_k(\tilde{y}_k, \tilde{u}_k)$ as the new tentative best trajectory, i.e.,

$$\hat{T}_{k+1} = T_k(\tilde{y}_k, \tilde{u}_k).$$

(2) *The set* $\hat{U}_k(\tilde{y}_k)$ *is empty.* Then, the algorithm forms the partial trajectory $\tilde{y}_{k+1} = \big(\tilde{y}_k, \tilde{u}_k, \tilde{x}_{k+1}\big)$, where

$$\tilde{u}_k = \overline{u}_k, \qquad \tilde{x}_{k+1} = \overline{x}_{k+1},$$

and $\overline{u}_k, \overline{x}_{k+1}$ are the control and state subsequent to $\tilde{x}_k$ in the current tentative best trajectory $\hat{T}_k$ [cf. Eq. (2.49)], and leaves $\hat{T}_k$ unchanged, i.e.,

$$\hat{T}_{k+1} = \hat{T}_k.$$

It can be seen that the fortified constrained rollout algorithm will follow the initial complete trajectory $\hat{T}_0$, the one generated by the base heuristic starting from $\tilde{y}_0$, up to a stage k where it will discover a new feasible complete trajectory with smaller cost to replace $\hat{T}_0$ as the tentative best trajectory. Similarly, the new tentative best trajectory $\hat{T}_k$ may be subsequently replaced by another feasible trajectory with smaller cost, etc.

Note that if the base heuristic is sequentially improving, the fortified rollout algorithm will generate the same complete trajectory as the (nonfortified) rollout algorithm given earlier, with the tentative best trajectory $\hat{T}_{k+1}$ being equal to the complete trajectory $T_k(\tilde{y}_k, \tilde{u}_k)$ for all k. The reason is that if the base heuristic is sequentially improving, the controls $\tilde{u}_k$ generated by the nonfortified algorithm belong to the set $\hat{U}_k(\tilde{y}_k)$ [by Prop. 2.5.1, case (1) above will hold].

However, it can be verified that even when the base heuristic is not sequentially improving, the fortified rollout algorithm will generate a complete trajectory that is feasible and has cost that is no worse than the cost of the complete trajectory generated by the base heuristic starting from $\tilde{y}_0$. This is because each tentative best trajectory has a cost that is no worse than the one of its predecessor, and the initial tentative best trajectory is just the trajectory generated by the base heuristic starting from the initial condition $\tilde{y}_0$.

Tree-Based Constrained Rollout Algorithms

It is possible to improve the performance of the rollout algorithm at the expense of maintaining more than one partial trajectory. In particular,

instead of the partial trajectory $\tilde{y}_k$ of Eq. (2.46), we can maintain a *tree* of partial trajectories that is rooted at $\tilde{y}_0$. These trajectories need not have equal length, i.e., they need not involve the same number of stages. At each step of the algorithm, we select a single partial trajectory from this tree, and execute the rollout algorithm's step as if this partial trajectory were the only one. Let this partial trajectory have k stages and denote it by $\tilde{y}_k$. Then we extend $\tilde{y}_k$ similar to our earlier rollout algorithm, with possibly multiple feasible trajectories. There is also a fortified version of this algorithm where a tentative best trajectory is maintained, which is the minimum cost complete trajectory generated thus far.

The aim of the tree-based algorithm is to obtain improved performance, essentially because it can go back and extend partial trajectories that were generated and temporarily abandoned at previous stages. The net result is a more flexible algorithm that is capable of examining more alternative trajectories. Note also that there is considerable freedom to select the number of partial trajectories maintained in the tree.

We finally mention a drawback of the tree-based algorithm: it is suitable for off-line computation, but it cannot be applied in an on-line context, where the rollout control selection is made after the current state becomes known as the system evolves in real-time .

2.5.1 Constrained Rollout for Discrete Optimization and Integer Programming

As noted in Section 2.1, general discrete optimization problems may be formulated as DP problems, which in turn can be addressed with rollout. The following is an example of a classical problem that involves both discrete and continuous variables. It can also be viewed as an instance of a 0-1 integer programming problem, and in fact this is the way it is usually addressed in the literature; see e.g., the book [DrH01]. The author's rollout book [Ber20a] contains additional examples.

Example 2.5.2 (Facility Location)

We are given a candidate set of N locations, and we want to place in some of these locations a "facility" that will serve the needs of a total of M "clients." Each client $i = 1, \ldots, M$ has a demand d_i for services that may be satisfied at a location $k = 0, \ldots, N-1$ at a cost a_{ik} per unit. If a facility is placed at location k, it has capacity to serve demand up to a known level c_k.

We introduce a 0-1 integer variable u_k to indicate with $u_k = 1$ that a facility is placed at location k at a cost b_k and with $u_k = 0$ that a facility is not placed at location k. Thus if y_{ik} denotes the amount of demand of client i to be served at facility k, the constraints are

$$\sum_{k=0}^{N-1} y_{ik} = d_i, \qquad i = 1, \ldots, M, \tag{2.50}$$

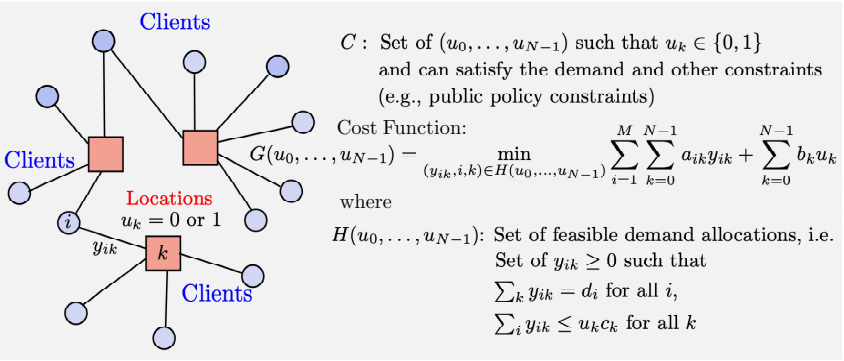

Figure 2.5.4 Schematic illustration of the facility location problem; cf. Example 2.5.2. Clients are matched to facilities, and the locations of the facilities are subject to optimization.

$$\sum_{i=1}^{M} y_{ik} \leq c_k u_k, \qquad k = 0, \ldots, N-1, \tag{2.51}$$

together with

$$y_{ik} \geq 0, \quad u_k \in \{0, 1\}, \qquad i = 1, \ldots, M, \ k = 0, \ldots, N-1. \tag{2.52}$$

We wish to minimize the cost

$$\sum_{i=1}^{M} \sum_{k=0}^{N-1} a_{ik} y_{ik} + \sum_{k=0}^{N-1} b_k u_k \tag{2.53}$$

subject to the preceding constraints; see Fig. 2.5.4. The essence of the problem is to place enough facilities at favorable locations to satisfy the clients' demand at minimum cost. This can be a very difficult mixed integer programming problem.

On the other hand, when all the variables u_k are fixed at some 0 or 1 values, the problem belongs to the class of *linear transportation* problems (see e.g., [Ber98]), and can be solved by fast polynomial algorithms. Thus the essential difficulty of the problem is how to select the integer variables u_k, $k = 0, \ldots, N-1$. This can be viewed as a discrete optimization problem of the type discussed in Section 1.6.3 (cf. Fig. 1.6.2). In terms of the notation of this figure, the control components are $u_0, \ldots, u_{N-1}$, where u_k can take the values 0 or 1.

To address the problem suboptimally by rollout, we must define a base heuristic at a "state" $(u_0, \ldots, u_k)$, where $u_j = 1$ or $u_j = 0$ specifies that a facility is or is not placed at location j, respectively. A suitable base heuristic at that state is to place a facility at all of the remaining locations (i.e., $u_j = 1$ for $j = k+1, \ldots, N-1$), and its cost is obtained by solving the corresponding linear transportation problem of minimizing the cost (2.53) subject to the constraints (2.50)-(2.52), with the variables u_j, $j = 0, \ldots, k$, fixed at the previously chosen values, and the variables u_j, $j = k+1, \ldots, N$, fixed at 1.

To illustrate, at the initial state where no placement decision has been made, we set $u_0 = 1$ (a facility is placed at location 0) or $u_0 = 0$ (a facility is not placed at location 0), we solve the two corresponding transportation problems, and we fix u_0, depending on which of the two resulting costs is smallest. Having fixed the status of location 0, we repeat with location 1: set the variable u_1 to 1 and to 0, solve the corresponding two transportation problems, and fix u_1, depending on which of the two resulting costs is smallest, etc.

It is easily seen that if the initial base heuristic choice (placing a facility at every candidate location) is feasible, i.e.,

$$\sum_{i=1}^{M} d_i \leq \sum_{k=0}^{N-1} c_k,$$

the rollout algorithm will yield a feasible solution with cost that is no larger than the cost corresponding to the initial application of the base heuristic. In fact it can be verified that the base heuristic here is sequentially consistent, so it is not necessary to use the fortified version of the algorithm. Regarding computational costs, the number of transportation problems to be solved is at first count $2N$, but it can be reduced to $N + 1$ by exploiting the fact that one of the two transportation problems at each stage after the first has been solved at an earlier stage.

It is worth noting, for readers that are familiar with the integer programming method of branch-and-bound, that the graph of Fig. 2.1.4 corresponds to the branch-and-bound tree for the problem, so the rollout algorithm amounts to a quick (and imperfect) method to traverse the branch-and-bound tree. This observation may be useful if we wish to use integer programming techniques to add improvements to the rollout algorithm.

We finally note that the rollout algorithm requires the solution of many linear transportation problems, which are defined by fairly similar data. It is thus important to use an algorithm that is capable of using effectively the final solution of one transportation problem as a starting point for the solution of the next. The auction algorithm for transportation problems (Bertsekas and Castañon [BeC89]) is particularly well-suited for this purpose.

Example 2.5.3 (Constrained Shortest Paths and Directed Spanning Trees)

Let us consider a spanning tree-type problem involving a directed graph with nodes $0, 1, \ldots, N$. At each node $k \in \{0, \ldots, N-1\}$ there is a set of outgoing arcs $u_k \in U_k$. Node N is special: it is viewed as a "root" node and has no outgoing arc. We are interested in collections of arcs involving a single outgoing arc per node,

$$u = (u_0, \ldots, u_{N-1})$$

with $u_k \in U_k$, $k = 0, \ldots, N-1$. We require that these arcs do not form a cycle, so that u specifies a directed spanning tree that is rooted at node

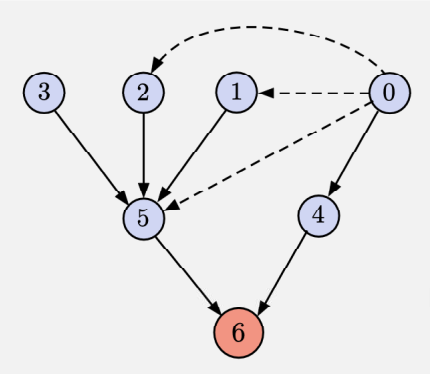

Figure 2.5.5 Schematic illustration of a constrained shortest path problem with root node $N = 6$. Given the current feasible spanning tree solution (indicated with solid line arcs), the rollout algorithm, considers a node k (in the figure $k = 0$) and the spanning tree arcs $\{u_i \mid i \neq k\}$ that are outgoing from the nodes $i \neq k$. It then considers the spanning trees that correspond to the outgoing arcs u_k from k that do not close a cycle with the set $\{u_i \mid i \neq k\}$ and are feasible [in the figure, these are the arcs indicated with broken lines, plus the arc (0,4)], and selects the arc that forms a spanning tree solution of minimum cost.

N. Note that for every node k, such a spanning tree specifies a unique path that starts at k, lies on the spanning tree, and ends at node N. We wish to find u that minimizes a given cost function $G(u)$ subject to certain additional constraints, which we do not specify further. The set of all constraints on u (including the constraint that the arcs form a directed spanning tree) is denoted abstractly as $u \in U$, so the problem comes within our constrained optimization framework of this section.

Note that this problem contains as a special case the classical shortest path problem, where we have a length for every arc and the objective is to find a tree of shortest paths to node N from all the nodes $0, \ldots, N-1$. Here U is just the constraint that the set of arcs $u = (u_0, \ldots, u_{N-1})$ form a directed spanning tree that is rooted at node N, and $G(u)$ is the sum of the lengths of all the paths specified by u, summed over all the start nodes $k = 0, \ldots, N-1$. Other shortest path-type problems, involving constraints, are included as special cases. For example, there may be a constraint that all the paths to N that are specified by the spanning tree corresponding to u contain a number of arcs that does not exceed a given upper bound.

Suppose that we have an initial solution/directed spanning tree

$$\bar{u} = (\bar{u}_0, \ldots, \bar{u}_{N-1}),$$

which is feasible (note here that finding such an initial solution may be a challenge). Let us apply the constrained rollout algorithm with a base heuristic that operates as follows: given a partial trajectory

$$y_k = (u_0, \ldots, u_{k-1}),$$

i.e., a sequence of k arcs, each outgoing from one of the nodes $0, \ldots, k-1$, it generates the complete trajectory/directed spanning tree

$$(u_0, \ldots, u_{k-1}, \bar{u}_k, \ldots, \bar{u}_{N-1}).$$

Thus the rollout algorithm, given a partial trajectory

$$\tilde{y}_k = (\tilde{u}_0, \ldots, \tilde{u}_{k-1}),$$

considers the set $\hat{U}_k(\tilde{y}_k)$ of all outgoing arcs u_k from node k, such that the complete trajectory

$$(\tilde{y}_k, u_k, \bar{u}_{k+1}, \ldots, \bar{u}_{N-1})$$

is feasible. It then selects the arc $u_k \in \hat{U}_k(\tilde{y}_k)$ that minimizes the cost

$$G(\tilde{y}_k, u_k, \bar{u}_{k+1}, \ldots, \bar{u}_{N-1});$$

see Fig. 2.5.5. It can be seen by induction, starting from $\bar{u}$, that the set of arcs $\hat{U}_k(\tilde{y}_k)$ is nonempty, and that the algorithm generates a sequence of feasible solutions/spanning trees, each with cost no worse than the preceding one.

Note that throughout the rollout process, a rooted spanning tree is maintained, and at each stage k, a single arc $\bar{u}_k$ that is outgoing from node k is replaced by the outgoing arc $\tilde{u}_k$. Thus two successive rooted spanning trees generated by the algorithm, differ by at most a single arc.

An interesting aspect of this rollout algorithm is that it can be applied multiple times with the final solution of one rollout application used to specify the base heuristic of the next rollout application. Moreover, a different order of nodes may be used in each rollout application. This can be viewed as a form of policy iteration, of the type that we have discussed. The algorithm will eventually terminate, in the sense that it can make no further progress. More irregular/heuristic orders of node selections are also possible; for example some nodes may be selected multiple times before others will be selected for the first time. However, there is no guarantee that the final solution thus obtained will be optimal.

2.6 SMALL STAGE COSTS AND LONG HORIZON - CONTINUOUS-TIME ROLLOUT

Let us consider the deterministic one-step approximation in value space scheme

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} \Big[g_k(x_k, u_k) + \tilde{J}_{k+1}\big(f_k(x_k, u_k)\big) \Big]. \tag{2.54}$$

In the context of rollout, $\tilde{J}_{k+1}\big(f_k(x_k, u_k)\big)$ is either the cost of the trajectory generated by the base heuristic starting from the next state $f_k(x_k, u_k)$, or some approximation that may involve truncation and terminal cost function approximation, as in the truncated rollout scheme of Section 2.3.6.

There is a special difficulty in this context, which is often encountered in practice. It arises when the cost per stage $g_k(x_k, u_k)$ is either 0 or is small relative to the cost-to-go approximation $\tilde{J}_{k+1}\big(f_k(x_k, u_k)\big)$. Then there is a potential pitfall to contend with: *the cost approximation errors that are inherent in the term* $\tilde{J}_{k+1}\big(f_k(x_k, u_k)\big)$ *may overwhelm the first stage cost term* $g_k(x_k, u_k)$, with unpredictable consequences for the quality of the one-step-lookahead policy $\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$.

The most straightforward way to address this issue is to use longer lookahead; this is typically what is done in the context of MPC (cf. Section 1.6.9). The difficulty here is that long lookahead minimization may require extensive on-line computation. In this case, creative application of the lookahead tree pruning and incremental rollout ideas, discussed in Section 2.4.2, may be helpful.

We will next discuss the difficulty with small stage costs for an alternative context, which arises from discretization of a continuous-time optimal control problem.

Continuous-Time Optimal Control and Approximation in Value Space

Consider a problem that involves a vector differential equation of the form

$$\dot{x}(t) = h\big(x(t), u(t), t\big), \qquad 0 \leq t \leq T, \tag{2.55}$$

where $x(t) \in \Re^n$ is the state vector at time t, $\dot{x}(t) \in \Re^n$ is the vector of first order time derivatives of the state at time t, $u(t) \in U \subset \Re^m$ is the control vector at time t, where U is the control constraint set, and T is a given terminal time. Starting from a given initial state $x(0)$, we want to find a feasible control trajectory $\big\{u(t) \mid t \in [0, T]\big\}$, which together with its corresponding state trajectory $\big\{x(t) \mid t \in [0, T]\big\}$, minimizes a cost function of the form

$$G\big(x(T)\big) + \int_0^T g\big(x(t), u(t), t\big)\, dt, \tag{2.56}$$

where g represents cost per unit time, and G is a terminal cost function. This is a classical problem with a long history.

Let us consider a simple conversion of the preceding continuous-time problem to a discrete-time problem, while treading lightly over some of the associated mathematical fine points. We introduce a small discretization increment $\delta > 0$, such that $T = \delta N$ where N is a large integer, and we replace the differential equation (2.55) by

$$x_{k+1} = x_k + \delta \cdot h_k(x_k, u_k), \qquad k = 0, \ldots, N-1.$$

Here the function h_k is given by

$$h_k(x_k, u_k) = h\big(x(k\delta), u(k\delta), k\delta\big),$$

where we view $\{x_k \mid k = 0, \ldots, N-1\}$ and $\{u_k \mid k = 0, \ldots, N-1\}$ as state and control trajectories, respectively, which approximate the corresponding continuous-time trajectories:

$$x_k \approx x(k\delta), \qquad u_k \approx u(k\delta).$$

We also replace the cost function (2.56) by

$$g_N(x_N) + \sum_{k=0}^{N-1} \delta \cdot g_k(x_k, u_k),$$

where

$$g_N(x_N) = G\big(x(N\delta)\big), \qquad g_k(x_k, u_k) = g\big(x(k\delta), u(k\delta), k\delta\big).$$

Thus the approximation in value space scheme with time discretization takes the form

$$\tilde{\mu}_k(x_k) \in \arg\min_{u_k \in U} \Big[\delta \cdot g_k(x_k, u_k) + \tilde{J}_{k+1}\big(x_k + \delta \cdot h_k(x_k, u_k)\big)\Big]; \tag{2.57}$$

where $\tilde{J}_{k+1}$ is the function that approximates the cost-to-go starting from a state at time $k+1$. We note here that the ratio of the terms $\delta \cdot g_k(x_k, u_k)$ and $\tilde{J}_{k+1}\big(x_k + \delta \cdot h_k(x_k, u_k)\big)$ is likely to tend to 0 as $\delta \to 0$, since $\tilde{J}_{k+1}\big(x_k + \delta \cdot h_k(x_k, u_k)\big)$ ordinarily stays roughly constant at a nonzero level as $\delta \to 0$. This suggests that the one-step lookahead minimization may be degraded substantially by discretization, and other errors, including rollout truncation and terminal cost approximation. Note that a similar sensitivity to errors may occur in other discrete-time models that involve frequent selection of decisions, with cost per stage that is very small relative to the cumulative cost over many stages and/or the terminal cost.

To deal with this difficulty, we subtract the constant $\tilde{J}_k(x_k)$ in the one-step-lookahead minimization (2.57), and write

$$\tilde{\mu}_k(x_k) \in \arg\min_{u_k \in U} \Big[\delta \cdot g_k(x_k, u_k) + \Big(\tilde{J}_{k+1}\big(x_k + \delta \cdot h_k(x_k, u_k)\big) - \tilde{J}_k(x_k)\Big)\Big]; \tag{2.58}$$

since $\tilde{J}_k(x_k)$ does not depend on u_k, the results of the minimization are not affected. Assuming $\tilde{J}_k$ is differentiable with respect to its argument, we can write

$$\tilde{J}_{k+1}\big(x_k + \delta \cdot h_k(x_k, u_k)\big) - \tilde{J}_k(x_k) \approx \delta \cdot \nabla_x \tilde{J}_k(x_k)' h_k(x_k, u_k),$$

where $\nabla_x \tilde{J}_k$ denotes the gradient of J_k (a column vector), and prime denotes transposition. By dividing with δ, and taking informally the limit as $\delta \to 0$, we can write the one-step lookahead minimization (2.58) as

$$\tilde{\mu}(t) \in \arg\min_{u(t) \in U} \Big[g\big(x(t), u(t), t\big) + \nabla_x \tilde{J}_t\big(x(t)\big)' h\big(x(t), u(t), t\big)\Big], \tag{2.59}$$

where $\tilde{J}_t(x)$ is the continuous-time cost function approximation and $\nabla_x \tilde{J}_t(x)$ is its gradient with respect to x. This is the correct analog of the approximation in value space scheme (2.54) for continuous-time problems.

Rollout for Continuous-Time Optimal Control

In view of the value approximation scheme of Eq. (2.59), it is natural to speculate that the continuous-time analog of rollout with a base policy of the form

$$\pi = \Big\{\mu_t\big(x(t)\big) \mid 0 \leq t \leq T\Big\}, \tag{2.60}$$

where $\mu_t\big(x(t)\big) \in U$ for all $x(t)$ and t, has the form

$$\tilde{\mu}_t\big(x(t)\big) \in \arg\min_{u(t)\in U}\Big[g\big(x(t),u(t),t\big) + \nabla_x J_{\pi,t}\big(x(t)\big)'h\big(x(t),u(t),t\big)\Big]. \tag{2.61}$$

Here $J_{\pi,t}\big(x(t)\big)$ is the cost of the base policy π starting from state $x(t)$ at time t, and satisfies the terminal condition

$$J_{\pi,T}\big(x(T)\big) = G\big(x(T)\big).$$

Computationally, the inner product in the right-hand side of the above minimization can be approximated using the finite difference formula

$$\nabla_x J_{\pi,t}\big(x(t)\big)'h\big(x(t),u(t),t\big) \approx \frac{J_{\pi,t}\Big(x(t)+\delta\cdot h\big(x(t),u(t),t\big)\Big) - J_{\pi,t}\big(x(t)\big)}{\delta},$$

which can be calculated by running the base policy π starting from $x(t)$ and from $x(t)+\delta\cdot h\big(x(t),u(t),t\big)$. (This finite differencing operation may involve tricky computational issues, but we will not get into this.)

An important question is how to select the base policy π. A choice that is often sensible and convenient is to choose π to be a "short-sighted" policy, which takes into account the "short term" cost from the current state (say for a very small horizon starting from the current time t), but ignores the remaining cost. An extreme case is the *myopic* policy, given by

$$\mu_t\big(x(t)\big) \in \arg\min_{u\in U} g\big(x(t),u(t),t\big).$$

This policy is the continuous-time analog of the greedy policy that we discussed in the context of discrete-time problems, and the traveling salesman Example 1.2.3 in particular.

The following example illustrates the rollout algorithm (2.61) with a problem that has a special property: the base policy cost $J_{\pi,t}\big(x(t)\big)$ is independent of $x(t)$ (it depends only on t), so that $\nabla_x J_{\pi,t}\big(x(t)\big) \equiv 0$. In this case, in view of Eq. (2.59), the rollout policy is myopic. It turns out that the optimal policy in this example is also myopic, so that the rollout policy is optimal, even though the base policy is very poor.

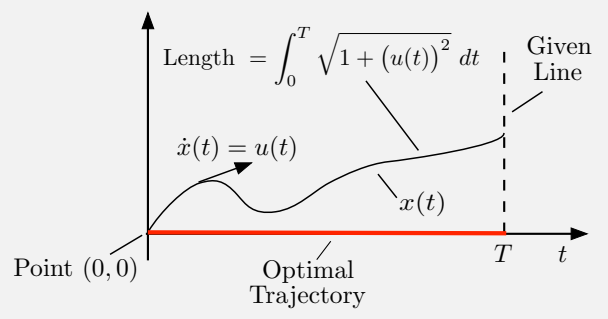

Figure 2.6.1 Problem of finding a curve of minimum length from a given point to a given line, and its formulation as a calculus of variations problem.

Example 2.6.1 (A Calculus of Variations Problem)

This is a simple example from the classical context of calculus of variations (see [Ber17a], Example 7.1.3). The problem is to find a minimum length curve that starts at a given point and ends at a given line. Without loss of generality, let $(0,0)$ be the given point, and let the given line be the vertical line that passes through $(T,0)$, as shown in Fig. 2.6.1.

Let $\big(t,x(t)\big)$ be the points of the curve, where $0 \le t \le T$. The portion of the curve joining the points $\big(t,x(t)\big)$ and $\big(t+dt,x(t+dt)\big)$ can be approximated, for small dt, by the hypotenuse of a right triangle with sides dt and $\dot{x}(t)dt$. Thus the length of this portion is

$$\sqrt{(dt)^2 + \big(\dot{x}(t)\big)^2(dt)^2},$$

which is equal to

$$\sqrt{1+\big(\dot{x}(t)\big)^2}\,dt.$$

The length of the entire curve is the integral over $[0,T]$ of this expression, so the problem is to

$$\text{minimize} \quad \int_0^T \sqrt{1+\big(\dot{x}(t)\big)^2}\,dt$$
$$\text{subject to} \quad x(0) = 0.$$

To reformulate the problem as a continuous-time optimal control problem, we introduce a control u and the system equation

$$\dot{x}(t) = u(t), \qquad x(0) = 0.$$

Our problem then takes the form

$$\text{minimize} \quad \int_0^T \sqrt{1+\big(u(t)\big)^2}\,dt.$$

This is a problem that fits our continuous-time optimal control framework, with

$$h\big(x(t),u(t),t\big)=u(t),\quad g\big(x(t),u(t),t\big)=\sqrt{1+\big(u(t)\big)^2},\quad G\big(x(T)\big)=0.$$

Consider now a base policy π whereby the control depends only on t and not on x. Such a policy has the form

$$\mu_t\big(x(t)\big)=\beta(t),\qquad \text{for all } x(t),$$

where $\beta(t)$ is some scalar function. For example, $\beta(t)$ may be constant, $\beta(t)\equiv\bar{\beta}$ for some scalar $\bar{\beta}$, which yields a straight line trajectory that starts at $(0,0)$ and makes an angle ϕ with the horizontal with $\tan(\phi)=\bar{\beta}$. The cost function of the base policy is

$$J_{\pi,t}\big(x(t)\big)=\int_t^T\sqrt{1+\beta(\tau)^2}\,d\tau,$$

which is independent of $x(t)$, so that $\nabla_x J_{\pi,t}\big(x(t)\big)\equiv 0$. Thus, from the minimization of Eq. (2.61), we have

$$\tilde{\mu}_t\big(x(t)\big)\in\arg\min_{u(t)\in\Re}\sqrt{1+\big(u(t)\big)^2},$$

and the rollout policy is

$$\tilde{\mu}_t\big(x(t)\big)\equiv 0.$$

This is the optimal policy: it corresponds to the horizontal straight line that starts at $(0,0)$ and ends at $(T,0)$.

Rollout with General Base Heuristics - Sequential Improvement

An extension of the rollout algorithm (2.61) is to use a more general base heuristic whose cost function $H_t\big(x(t)\big)$ can be evaluated by simulation. This rollout algorithm has the form

$$\tilde{\mu}(t)\in\arg\min_{u(t)\in U}\Big[g\big(x(t),u(t),t\big)+\nabla_x H_t\big(x(t)\big)'h\big(x(t),u(t),t\big)\Big].$$

Here the policy cost function $J_{\pi,t}$ is replaced by a more general differentiable function H_t, obtainable through a base heuristic, which may lack the sequential consistency property that is inherent in policies.

We will now show a cost improvement property of the rollout algorithm based on the natural condition

$$H_T\big(\tilde{x}(T)\big)=G\big(\tilde{x}(T)\big),\tag{2.62}$$

and the assumption

$$\min_{u(t)\in U}\Big[g\big(x(t),u(t),t\big)+\nabla_t H_t\big(x(t)\big)+\nabla_x H_t\big(x(t)\big)'h\big(x(t),u(t),t\big)\Big]\leq 0, \tag{2.63}$$

for all $\big(x(t),t\big)$, where $\nabla_x H_t$ denotes gradient with respect to x, and $\nabla_t H_t$ denotes gradient with respect to t. This assumption is the continuous-time analog of the sequential improvement condition of Definition 2.3.2 [cf. Eq. (2.13)]. Under this assumption, we will show that

$$J_{\tilde\pi,0}\big(x(0)\big)\leq H_0\big(x(0)\big), \tag{2.64}$$

i.e., the cost of the rollout policy starting from the initial state $x(0)$ is no worse than the base heuristic cost starting from the same initial state.

Indeed, let $\big\{\tilde x(t)\mid t\in[0,T]\big\}$ and $\big\{\tilde u(t)\mid t\in[0,T]\big\}$ be the state and control trajectories generated by the rollout policy starting from $x(0)$. Then the sequential improvement condition (2.63) yields

$$g\big(\tilde x(t),\tilde u(t),t\big)+\nabla_t H_t\big(\tilde x(t)\big)+\nabla_x H_t\big(\tilde x(t)\big)'h\big(\tilde x(t),\tilde u(t),t\big)\leq 0$$

for all t, and by integration over $[0,T]$, we obtain

$$\int_0^T g\big(\tilde x(t),\tilde u(t),t\big)\,dt+\int_0^T\Big(\nabla_t H_t\big(\tilde x(t)\big)+\nabla_x H_t\big(\tilde x(t)\big)'h\big(\tilde x(t),\tilde u(t),t\big)\Big)\,dt\leq 0. \tag{2.65}$$

The second integral above can be written as

$$\int_0^T\Big(\nabla_t H_t\big(\tilde x(t)\big)+\nabla_x H_t\big(\tilde x(t)\big)'h\big(\tilde x(t),\tilde u(t),t\big)\Big)\,dt = \int_0^T\Big(\nabla_t H_t\big(\tilde x(t)\big)+\nabla_x H_t\big(\tilde x(t)\big)'\frac{d\tilde x(t)}{dt}\Big)\,dt,$$

and its integrand is the total differential with respect to time: $\frac{d}{dt}\Big(H_t\big(\tilde x(t)\big)\Big)$. Thus we obtain from Eq. (2.65)

$$\begin{aligned}\int_0^T g\big(\tilde x(t),\tilde u(t),t\big)\,dt+\int_0^T\frac{d}{dt}\Big(H_t\big(\tilde x(t)\big)\Big)\,dt\\ =\int_0^T g\big(\tilde x(t),\tilde u(t),t\big)\,dt+H_T\big(\tilde x(T)\big)-H_0\big(\tilde x(0)\big)\leq 0.\end{aligned} \tag{2.66}$$

Since

$$H_T\big(\tilde x(T)\big)=G\big(\tilde x(T)\big),$$

[cf. Eq. (2.62)] and $\tilde x(0)=x(0)$, from Eq. (2.66) [which is a direct consequence of the sequential improvement condition (2.63)], it follows that

$$J_{\tilde\pi,0}\big(x(0)\big)=\int_0^T g\big(\tilde x(t),\tilde u(t),t\big)\,dt+G\big(\tilde x(T)\big)\leq H_0\big(x(0)\big),$$

thus proving the cost improvement property (2.64).

Note that the sequential improvement condition (2.63) is satisfied if H_t is the cost function $J_{\pi,t}$ corresponding to a base policy π. The reason is that for any policy $\pi = \{\mu_t(x(t)) \mid 0 \leq t \leq T\}$ [cf. Eq. (2.60)], the analog of the DP algorithm (under the requisite mathematical conditions) is

$$0 = g\big(x(t), \mu_t(x(t)), t\big) + \nabla_t J_{\pi,t}\big(x(t)\big) + \nabla_x J_{\pi,t}\big(x(t)\big)' h\big(x(t), \mu_t(x(t)), t\big). \tag{2.67}$$

In continuous-time optimal control theory, this is known as the *Hamilton-Jacobi-Bellman equation*. It is a partial differential equation, which may be viewed as the continuous-time analog of the DP algorithm for a single policy; there is also a Hamilton-Jacobi-Bellman equation for the optimal cost function $J_t^*\big(x(t)\big)$ (see optimal control textbook accounts, such as [Ber17a], Section 7.2, and the references cited there). As illustration, the reader may verify that the cost function of the base policy used in the calculus of variations problem of Example 2.6.1 satisfies this equation. It can be seen from the Hamilton-Jacobi-Bellman Eq. (2.67) that when $H_t = J_{\pi,t}$, the sequential improvement condition (2.63) and the cost improvement property (2.64) hold.

Approximating Cost Function Differences

The preceding analysis suggests that when dealing with a discrete-time problem with a long horizon N, a system equation $x_{k+1} = f_k(x_k, u_k)$, and a small cost per stage $g_k(x_k, u_k)$ relative to the optimal cost-to-go function $J_{k+1}^*\big(f_k(x_k, u_k)\big)$, it is worth considering an alternative implementation of the approximation in value space scheme. In particular, we should consider approximating the cost differences

$$D_k^*(x_k, u_k) = J_{k+1}^*\big(f_k(x_k, u_k)\big) - J_k^*(x_k)$$

instead of approximating the optimal cost-to-go functions $J_{k+1}^*\big(f_k(x_k, u_k)\big)$. The one-step-lookahead minimization (2.54) should then be replaced by

$$\tilde{\mu}_k(x_k) \in \arg\min_{u_k \in U_k(x_k)} \big[g_k(x_k, u_k) + \tilde{D}_k(x_k, u_k)\big],$$

where $\tilde{D}_k$ is the approximation to D_k^*.

Note also that while for continuous-time problems, the idea of approximating the gradient of the optimal cost function is essential and comes out naturally from the analysis, for discrete-time problems, approximating cost-to-go differences rather than cost functions is optional and should be considered in the context of a given problem, possibly in conjunction with increased lookahead. Methods along this line include advantage updating, cost shaping, biased aggregation, and the use of baselines, for which we refer to the books [BeT96], [Ber19a], and [Ber20a]. A special method to explicitly approximate cost function differences is *differential training*, which was proposed in the author's paper [Ber97b], and was also discussed in Section 4.3.4 of the book [Ber20a].

The Case of Zero Cost per Stage

The most extreme and challenging case of small stage costs arises when the cost per stage is zero for all states, while a nonzero cost may be incurred only at termination. This type of cost structure occurs, among others, in games such as chess and backgammon (another interesting context is solving the Rubik's cube [AMS19], [MAS19]). It also occurs in several other contexts, including constraint programming problems (Section 2.1), where there is not even a terminal cost, just constraints to be satisfied.

Under these circumstances, the idea of approximating cost-to-go differences that we have just discussed may not be effective, and applying approximation in value space may involve serious challenges. An advisable remedy is to resort to long lookahead, either through multistep lookahead minimization (possibly augmented by pruning or incremental rollout ideas; cf. Section 2.4), or through some form of truncated rollout.

It may also be important to introduce a terminal cost function approximation by using problem simplification (solving a simpler problem, in place of the original), or neural network training. Chess, Go, backgammon, and other games are prime examples of zero cost per stage problems, where the use of a terminal cost function approximation, obtained through a neural network training, has been a critical part of the solution methodology.

2.7 APPROXIMATION IN VALUE SPACE - STOCHASTIC PROBLEMS

We will now discuss approximation in value space for stochastic problems. We first focus on rollout algorithms for finite state, control, and disturbance spaces. We will restrict ourselves to the case where the base heuristic is a policy $\pi = \{\mu_0, \ldots, \mu_{N-1}\}$. The rollout policy applies at state x_k the control $\tilde{\mu}_k(x_k)$ given by the minimization

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} E\Big\{ g_k(x_k, u_k, w_k) + J_{k+1,\pi}\big(f_k(x_k, u_k, w_k)\big) \Big\}.$$

Equivalently, the rollout policy $\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$ is obtained by minimization over the Q-factors $Q_{k,\pi}(x_k, u_k)$ of the base policy:

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} Q_{k,\pi}(x_k, u_k),$$

where

$$Q_{k,\pi}(x_k, u_k) = E\Big\{ g_k(x_k, u_k, w_k) + J_{k+1,\pi}\big(f_k(x_k, u_k, w_k)\big) \Big\}.$$

We first establish that the cost improvement property that we showed for deterministic problems under the sequential consistency condition carries through for stochastic problems. In particular, let us denote by $J_{k,\pi}(x_k)$

the cost corresponding to starting the base policy at state x_k, and by $J_{k,\tilde{\pi}}(x_k)$ the cost corresponding to starting the rollout algorithm at state x_k. We claim that

$$J_{k,\tilde{\pi}}(x_k) \leq J_{k,\pi}(x_k), \qquad \text{for all } x_k \text{ and } k. \tag{2.68}$$

We prove this inequality by induction similar to the deterministic case [cf. Eq. (2.12)]. Clearly it holds for $k = N$, since

$$J_{N,\tilde{\pi}} = J_{N,\pi} = g_N.$$

Assuming that it holds for index $k+1$, we have for all x_k,

$$\begin{aligned} J_{k,\tilde{\pi}}(x_k) &= E\Big\{ g_k\big(x_k, \tilde{\mu}_k(x_k), w_k\big) + J_{k+1,\tilde{\pi}}\Big(f_k\big(x_k, \tilde{\mu}_k(x_k), w_k\big)\Big)\Big\} \\ &\leq E\Big\{ g_k\big(x_k, \tilde{\mu}_k(x_k), w_k\big) + J_{k+1,\pi}\Big(f_k\big(x_k, \tilde{\mu}_k(x_k), w_k\big)\Big)\Big\} \\ &= \min_{u_k \in U_k(x_k)} E\Big\{ g_k(x_k, u_k, w_k) + J_{k+1,\pi}\big(f_k(x_k, u_k, w_k)\big)\Big\} \\ &\leq E\Big\{ g_k\big(x_k, \mu_k(x_k), w_k\big) + J_{k+1,\pi}\Big(f_k\big(x_k, \mu_k(x_k), w_k\big)\Big)\Big\} \\ &= J_{k,\pi}(x_k), \end{aligned} \tag{2.69}$$

where:

(a) The first equality is the DP equation for the rollout policy $\tilde{\pi}$.

(b) The first inequality holds by the induction hypothesis.

(c) The second equality holds by the definition of the rollout algorithm.

(d) The final equality is the DP equation for the base policy π.

The induction proof of the cost improvement property is thus complete.

The preceding cost improvement argument assumes that the cost functions $J_{k+1,\pi}$ of the base policy are calculated exactly. In practice, truncated rollout with terminal cost function approximation and limited simulation may be used to approximate $J_{k+1,\pi}$. In this case the cost function of the rollout policy can still be viewed as the result of a Newton step in the context of an approximation in value space scheme. Moreover, the cost improvement property can still be proved under some conditions that we will not discuss in this book; see the books [Ber12], [Ber19a], and [Ber20a].

Some Rollout Examples

Similar to deterministic problems, it has been observed empirically that for stochastic problems the rollout policy not only does not degrade the

performance of the base policy, but also typically produces substantial cost improvement, thanks to its underlying Newton step; see also the case studies referenced at the end of the chapter. To emphasize this point, we provide here an example of a nontrivial optimal stopping problem where the rollout policy is actually optimal, despite the fact that the base policy is rather naive. Such behavior is of course special and nontypical, but highlights the nature of the cost improvement property of rollout.

Example 2.7.1 (Optimal Stopping and Rollout Optimality)

Optimal stopping problems are characterized by the availability, at each state, of a control that stops the evolution of the system. We will consider a problem with two control choices: at each stage we observe the current state of the system and decide whether to continue or to stop the process. We formulate this as an N-stage problem where stopping is mandatory at or before stage N.

Consider a stationary version of the problem (state and disturbance spaces, disturbance distribution, control constraint set, and cost per stage are the same for all times). At each state x_k and at time k, if we stop, the system moves to a termination state at a cost $C(x_k)$ and subsequently remains there at no cost. If we do not stop, the system moves to state $x_{k+1} = f(x_k, w_k)$ at cost $g(x_k, w_k)$. The terminal cost, assuming stopping has not occurred by the last stage, is $C(x_N)$. An example is a problem of optimal exercise of a financial option where x is the asset's price, $C(x) = x$, and $g(x, w) \equiv 0$.

The DP algorithm (for states other than the termination state) is given by

$$J_N^*(x_N) = C(x_N), \tag{2.70}$$

$$J_k^*(x_k) = \min\Big[C(x_k),\, E\Big\{g(x_k, w_k) + J_{k+1}^*\big(f(x_k, w_k)\big)\Big\}\Big], \tag{2.71}$$

and it is optimal to stop at time k for states x in the set

$$S_k = \Big\{x \;\Big|\; C(x) \leq E\Big\{g(x, w) + J_{k+1}^*\big(f(x, w)\big)\Big\}\Big\}.$$

Consider now the rather primitive base policy π, whereby *we stop at every state* x. Thus we have for all x_k and k,

$$J_{k,\pi}(x_k) = C(x_k).$$

The rollout policy is stationary and can be computed on-line relatively easily, since $J_{k,\pi}$ is available in closed form. In particular, the rollout policy is to stop at x_k if

$$C(x_k) \leq E\Big\{g(x_k, w_k) + C\big(f(x_k, w_k)\big)\Big\},$$

i.e., if x_k is in the set S_{N-1}, and otherwise to continue.

The rollout policy also has an intuitive interpretation: it stops at the states for which it is better to stop rather than continue for one more stage

and then stop. A policy of this type turns out to be optimal in several types of stopping applications. Let us provide a condition that guarantees its optimality.

We have from the DP Eqs. (2.70)-(2.71),

$$J^*_{N-1}(x) \leq J^*_N(x), \qquad \text{for all } x,$$

and using this fact in the DP equation (2.71), we obtain inductively

$$J^*_k(x) \leq J^*_{k+1}(x), \qquad \text{for all } x \text{ and } k.$$

Using this fact and the definition of S_k we see that

$$S_0 \subset \cdots \subset S_k \subset S_{k+1} \subset \cdots \subset S_{N-1}. \tag{2.72}$$

We will now consider a condition guaranteeing that all the stopping sets S_k are equal. Suppose that the set S_{N-1} is *absorbing* in the sense that if a state belongs to S_{N-1} and we decide to continue, the next state will also be in S_{N-1}:

$$f(x,w) \in S_{N-1}, \qquad \text{for all } x \in S_{N-1},\ w. \tag{2.73}$$

We will show that equality holds in Eq. (2.72) and for all k we have

$$S_k = S_{N-1} = \left\{ x \in S \;\middle|\; C(x) \leq E\Big\{ g(x,w) + C\big(f(x,w)\big) \Big\} \right\}.$$

Indeed, by the definition of S_{N-1}, we have

$$J^*_{N-1}(x) = C(x), \qquad \text{for all } x \in S_{N-1},$$

and using Eq. (2.73) we obtain for $x \in S_{N-1}$

$$E\Big\{ g(x,w) + J^*_{N-1}\big(f(x,w)\big) \Big\} = E\Big\{ g(x,w) + C\big(f(x,w)\big) \Big\} \geq C(x).$$

Therefore, stopping is optimal for all $x_{N-2} \in S_{N-1}$ or equivalently $S_{N-1} \subset S_{N-2}$. This together with Eq. (2.72) implies $S_{N-2} = S_{N-1}$. Proceeding similarly, we obtain $S_k = S_{N-1}$ for all k. Thus the optimal policy is to stop if and only if the state is within the set S_{N-1}, which is precisely the set of states where the rollout policy stops.

In conclusion, if condition (2.73) holds (the one-step stopping set S_{N-1} is absorbing), the rollout policy is optimal. Moreover, the preceding analysis [cf. Eq. (2.72)] can be used to show that even if the one-step stopping set S_{N-1} is not absorbing, the rollout policy stops and is optimal within the set of states $x \in \cap_k S_k$, and correctly continues within the set of states $x \notin S_{N-1}$. Contrary to the optimal policy, it also stops within the subset of states $x \in S_{N-1}$ that are not in $\cap_k S_k$. Thus, even in the absence of condition (2.73), the rollout policy is quite sensible even though the base policy is not.

We next discuss a special case of the preceding example. Again the one-step lookahead/rollout policy is optimal, despite the fact that the base policy is poor. Related examples can be found in Chapter 3 of the DP textbook [Ber17a].

Example 2.7.2 (The Rational Burglar)

A burglar may at any night k choose to retire with his accumulated earnings x_k or enter a house and bring home a random amount w_k. However, in the latter case he gets caught with probability p, and then he is forced to terminate his activities and forfeit all of his earnings thus far. The amounts w_k are independent, identically distributed with mean $\overline{w}$. The problem is to find a policy that maximizes the burglar's expected earnings over N nights.

We can formulate this problem as a stopping problem with two actions (retire or continue) and a state space consisting of the real line, the retirement state, and a special state corresponding to the burglar getting caught. The DP algorithm is given by

$$J_N^*(x_N) = x_N,$$

$$J_k^*(x_k) = \max\Big[x_k,\, (1-p)E\big\{J_{k+1}^*(x_k + w_k)\big\}\Big].$$

The one-step stopping set is

$$S_{N-1} = \big\{x \mid x \geq (1-p)(x+\overline{w})\big\} = \left\{x \;\middle|\; x \geq \frac{(1-p)\overline{w}}{p}\right\},$$

(more accurately this set together with the special state corresponding to the burglar's arrest). Since this set is absorbing in the sense of Eq. (2.73), we see that the one-step lookahead/rollout policy by which the burglar retires when his earnings reach or exceed $(1-p)\overline{w}/p$ is optimal. Note that the base policy of the burglar is the "timid" policy of always retiring, regardless of his accumulated earnings, which is far from optimal.

2.7.1 Simplified Rollout and Policy Iteration

The cost improvement property (2.68) also holds for the simplified version of the rollout algorithm (cf. Section 2.3.4) where the rollout policy is defined by

$$\tilde{\mu}_k(x_k) \in \arg\min_{u_k \in \overline{U}_k(x_k)} Q_{k,\pi}(x_k, u_k), \tag{2.74}$$

for a subset $\overline{U}_k(x_k) \subset U_k(x_k)$ that contains the base policy control $\mu_k(x_k)$. The proof is obtained by replacing the last inequality in the argument of Eq. (2.69),

$$\min_{u_k \in U_k(x_k)} Q_{k,\pi}(x_k, u_k) \leq Q_{k,\pi}\big(x_k, \mu_k(x_k)\big),$$

with the inequality

$$\min_{u_k \in \overline{U}_k(x_k)} Q_{k,\pi}(x_k, u_k) \leq Q_{k,\pi}\big(x_k, \mu_k(x_k)\big).$$

The simplified rollout algorithm (2.74) may be implemented in a number of ways, including control constraint discretization/approximation, a random search algorithm, or a one-agent-at-a-time minimization process, as in multiagent rollout.

The simplified rollout idea can also be used within the infinite horizon policy iteration (PI) context. In particular, instead of the minimization

$$\tilde{\mu}(x) \in \arg\min_{u \in U(x)} E_w\Big\{g(x,u,w) + \alpha J_\mu\big(f(x,u,w)\big)\Big\}, \qquad \text{for all } x, \quad (2.75)$$

in the policy improvement operation, it is sufficient for cost improvement to generate a new policy $\tilde{\mu}$ that satisfies for all x,

$$E_w\Big\{g\big(x,\tilde{\mu}(x),w\big)+\alpha J_\mu\big(f(x,\tilde{\mu}(x),w)\big)\Big\} \leq E_w\Big\{g(x,u,w)+\alpha J_\mu\big(f(x,u,w)\big)\Big\}.$$

This cost improvement property is the critical argument for proving convergence of the PI algorithm and its variations to the optimal cost function and policy; see the corresponding proofs in the books [Ber17a] and [Ber19a].

2.7.2 Certainty Equivalence Approximations

As in the case of deterministic DP problems, it is possible to use ℓ-step lookahead, with the aim to improve the performance of the policy obtained through approximation in value space. This, however, can be computationally expensive, because the lookahead graph expands fast as ℓ increases, due to the stochastic character of the problem. Using *certainty equivalence* (CE for short) is an important approximation approach for dealing with this difficulty, as it reduces the size of the ℓ-step lookahead graph. Moreover, CE mitigates the potentially excessive simulation because it reduces the stochastic variance of the Q-factors calculated at each stage.

In the pure but somewhat flawed version of this approach, when solving the ℓ-step lookahead minimization problem, we simply replace *all* of the uncertain quantities $w_k, w_{k+1}, \ldots, w_{k+\ell-1}, \ldots, w_{N-1}$ by some nominal value $\overline{w}$, thus making that problem fully deterministic. Unfortunately, this affects significantly the character of the approximation: when w_k is replaced by a deterministic quantity the Newton step interpretation of the underlying approximation in value space scheme is lost to a great extent.

Still, we may largely correct this difficulty, while retaining substantial simplification, by using CE *only after the first stage* of the ℓ-step lookahead. We can do this with a CE scheme whereby only the uncertain quantities $w_{k+1}, \ldots, w_{N-1}$ are replaced by a deterministic value $\overline{w}$, while w_k is treated as a stochastic quantity.†

† Variants of the CE approach, based on less drastic simplifications of the probability distributions of the uncertain quantities, are given in the books [Ber17a], Section 6.2.2 and [Ber19a], Section 2.3.2.

The CE approach, first proposed in the paper by Bertsekas and Castañon [BeC99], has an important property: *it maintains the Newton step character of the approximation in value space scheme*. In particular, the function $J_{\tilde{\mu}}$ of the ℓ-step lookahead policy $\tilde{\mu}$ obtained is generated by a Newton step, applied to the function obtained by the last $\ell-1$ minimization steps (modified by CE, and applied to the terminal cost function approximation); see the monograph [Ber20a] for a discussion. Thus the benefit of the fast convergence of Newton's method is restored. In fact based on insights derived from this Newton step interpretation, it appears that the performance penalty for the CE approximation is typically small. At the same time the ℓ-step lookahead minimization involves only one stochastic step, the first one, and hence potentially a much "thinner" lookahead graph, than the ℓ-step minimization that does not involve any CE-type approximations; see Fig. 2.7.1. Moreover, the ideas of tree pruning and iterative deepening, which we have discussed in Section 2.4 for deterministic multistep lookahead, come into play when the CE approximation is used.

Certainty Equivalence Approximations in Rollout

Certainty equivalence ideas can also be used in the context of rollout with a base policy. The cost function $J_\mu\big(f(x,u,w)\big)$ of the base policy μ in the rollout control calculation

$$\tilde{\mu}(x) \in \arg\min_{u\in U(x)} E_w\Big\{g(x,u,w) + \alpha J_\mu\big(f(x,u,w)\big)\Big\}, \quad \text{for all } x, \quad (2.76)$$

[cf. Eq. (2.75)], can be approximated by the cost function

$$\bar{J}_\mu\big(f(x,u,w)\big)$$

of μ for a deterministic problem where the stochastic disturbances w are replaced by "typical" deterministic quantities $\overline{w}$. If w is a continuous random variable, then it is natural to use $\overline{w} = E\{w\}$. If w is a discrete random variable, then one possibility is to use as $\overline{w}$ the most probable value of w under the state-control pair $\big(x,\mu(x)\big)$. The resulting scheme can be viewed as an approximate rollout algorithm in the context of approximation in value space, but with exact minimization at the first step of lookahead [cf. Eq. (2.76)]. For this reason it still implements a Newton step for solving the Bellman equation and inherits the attendant cost improvement properties.

Another possibility, which applies to problems of finite horizon N, is to calculate the rollout control at state x_k by computing the most likely future disturbance trajectory $(w_{k+1},\ldots,w_{N-1})$ under the base policy; cf. Section 2.3.7. Similar ideas also apply to truncated rollout schemes that incorporate CE approximations.

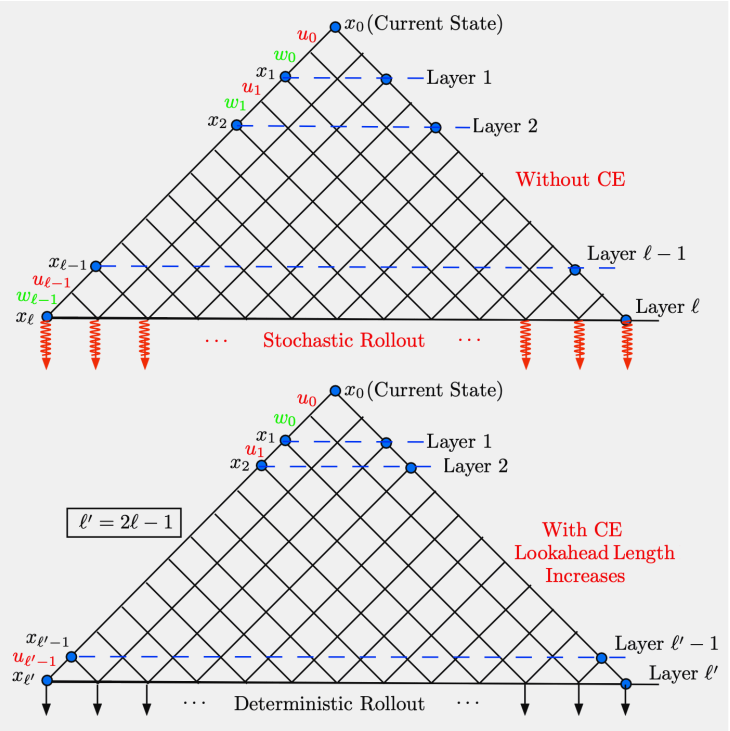

Figure 2.7.1 Illustration of the computational savings with the CE approximation, applied at the states after the first layer of states of the multistep lookahead tree. The figure on the top (or the bottom) illustrates the lookahead tree without (or with, respectively) CE. It can be seen that with CE, the lookahead tree grows much faster (the layers contain more states). In particular, the "height" of the ℓ-step lookahead graph without CE is the same as the "height" of a ℓ'-step lookahead graph with CE, where $\ell' = 2\ell - 1$. Moreover, with a number m of controls per state, and a number n of disturbances per state-control pair, the number of leaves of the ℓ-step lookahead tree is estimated as $O(mn\ell)$ without CE and $O\big(m(n+\ell)\big)$ with CE.

Example 2.7.3 (Markov Jump Parameter Problems)

Consider an N-stage horizon problem where the uncertainty in the system equation comes from a parameter θ_k that is perfectly observed, but evolves according to a Markov chain with transition probabilities

$$p_{ij}^k(x_k, u_k) = P(\theta_{k+1} = j \mid \theta_k = i, x_k, u_k), \qquad i, j = 1, \ldots, r,$$

where $1, \ldots, r$, are the possible values of the parameter. For example, in a maintenance-and-repair context, the system can be in one of two states, "good" and "bad", and the problem may be to schedule preventive maintenance of the system to maximize adherence to a certain production objective.

To capture the correlation between successive parameter values, we introduce an augmented state (x_k, θ_k), and policies $\mu_k(x_k, \theta_k)$ that depend on this augmented state. To this end, we must define a stochastic dynamic system that describes the evolution from the augmented state (x_k, θ_k) to the next augmented state (x_{k+1}, θ_{k+1}). We thus introduce a stochastic variable w_k that models the next value of the parameter, $\theta_{k+1} = w_k$, and changes according to the given Markov chain transition probabilities; see our discussion of state augmentation in Section 1.6.5. The augmented state evolves according to

$$x_{k+1} = f_k(x_k, u_k, \theta_k), \qquad \theta_{k+1} = w_k, \qquad k = 1, \ldots, N-1,$$

where w_k takes the values $i = 1, \ldots, r$ with probabilities $p^k_{i\theta_k}(x_k, u_k)$.

Suppose now that we want to apply rollout with a given base policy, which at state parameter pair (x_k, θ_k) applies control $\mu_k(x_k, \theta_k)$. To implement a CE approximation, we must calculate the base policy controls for the future time periods, $k+1, \ldots, N-1$, while assuming a deterministic trajectory of parameter values after θ_{k+1}. One possibility is to compute this trajectory as the most likely future trajectory $(w_{k+1}, \ldots, w_{N-1})$ under the base policy. This computation must be done for each of the possible pairs (x_{k+1}, θ_{k+1}), using the Markov chain probabilities that correspond to the state and control pairs that are generated by the base policy starting with (x_{k+1}, θ_{k+1}).

2.7.3 Simulation-Based Implementation of the Rollout Algorithm

A conceptually straightforward way to compute the rollout control at a given state x_k and time k is to consider each possible control $u_k \in U_k(x_k)$, and to generate a "large" number of simulated trajectories of the system starting from (x_k, u_k). Thus a simulated trajectory is obtained from

$$x_{i+1} = f_i\big(x_i, \mu_i(x_i), w_i\big), \qquad i = k+1, \ldots, N-1,$$

where $\{\mu_{k+1}, \ldots, \mu_{N-1}\}$ is the tail portion of the base policy, the starting state of the simulated trajectory is

$$x_{k+1} = f_k(x_k, u_k, w_k),$$

and the disturbance sequence $\{w_k, \ldots, w_{N-1}\}$ is obtained by random sampling. The costs of the trajectories corresponding to a pair (x_k, u_k) can be viewed as samples of the Q-factor

$$Q_{k,\pi}(x_k, u_k) = E\Big\{g_k(x_k, u_k, w_k) + J_{k+1,\pi}\big(f_k(x_k, u_k, w_k)\big)\Big\},$$

where $J_{k+1,\pi}$ is the cost-to-go function of the base policy, i.e., $J_{k+1,\pi}(x_{k+1})$ is the cost of using the base policy starting from x_{k+1}. For problems with a large number of stages, it is also common to truncate the rollout trajectories

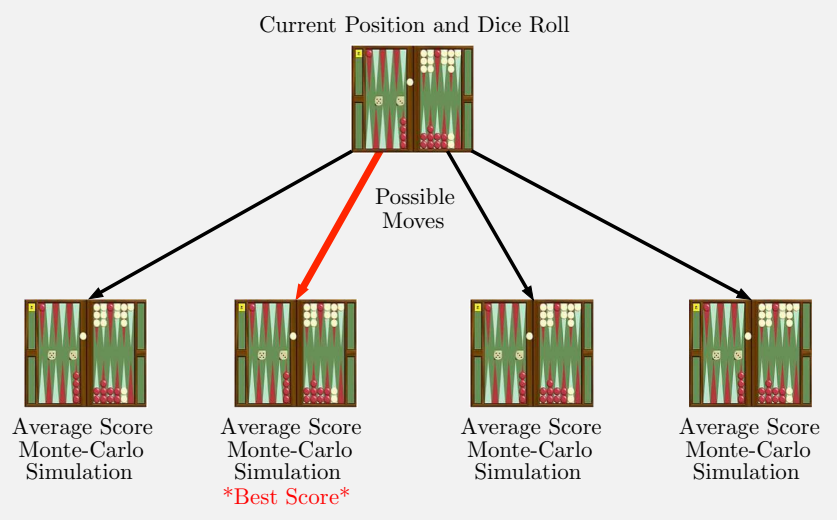

Figure 2.7.2 Illustration of rollout for backgammon. At a given position and roll of the dice, the set of all possible moves is generated, and the outcome of the game for each move is evaluated by "rolling out" (simulating to the end) many games using a suboptimal/heuristic backgammon player (the TD-Gammon player was used for this purpose in [TeG96]), and by Monte Carlo averaging the scores. The move that results in the best average score is selected for play.

and add a terminal cost function approximation as compensation for the resulting error.

By Monte Carlo averaging of the costs of the sample trajectories plus the terminal cost (if any), we obtain an approximation to the Q-factor $Q_{k,\pi}(x_k, u_k)$ for each $u_k \in U_k(x_k)$, denoted by $\tilde{Q}_{k,\pi}(x_k, u_k)$. We then compute the (approximate) rollout control $\tilde{\mu}_k(x_k)$ with the minimization

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} \tilde{Q}_{k,\pi}(x_k, u_k). \tag{2.77}$$

Example 2.7.4 (Backgammon)

The first impressive application of rollout was given for the ancient two-player game of backgammon, in the paper by Tesauro and Galperin [TeG96]; see Fig. 2.7.2. They implemented a rollout algorithm, which attained a level of play that was better than all computer backgammon programs, and eventually better than the best humans. Tesauro had proposed earlier the use of one-step and two-step lookahead with lookahead cost function approximation provided by a neural network, resulting in a backgammon program called TD-Gammon [Tes89a], [Tes89b], [Tes92], [Tes94], [Tes95], [Tes02]. TD-Gammon was trained with an approximate policy iteration method, and was used as the base policy (for each of the two players) to simulate game trajectories.

The rollout algorithm also involved truncation of long game trajectories, using a terminal cost function approximation based on TD-Gammon's position evaluation. Game trajectories are of course random, since they involve the use of dice at each player's turn. Thus the scores of many trajectories have to be generated and averaged with the Monte Carlo method to assess the probability of a win from a given position.

An important issue to consider here is that backgammon is a two-player game and not an optimal control problem that involves a single decision maker. While there is a DP theory for sequential zero-sum games, this theory has not been covered in this book. Thus how are we to interpret rollout algorithms in the context of two-player games, with both players using some base policy? The answer is to view the game as a (one-player) optimal control problem, where one of the two players passively uses the base policy exclusively (TD-Gammon in the present example). The other player takes the role of the optimizer, and actively tries to improve on his base policy (TD-Gammon) by using rollout. Thus "policy improvement" in the context of the present example means that when playing against a TD-Gammon opponent, the rollout player achieves a better score on the average than if he/she were to play with the TD-Gammon strategy. In particular, the theory does not guarantee that a rollout player that is trained using TD-Gammon for both players will do better than TD-Gammon would against a non-TD-Gammon opponent. While this is a plausible practical hypothesis, it is one that can only be tested empirically. In fact relevant counterexamples have been constructed for the game of Go using "adversarial" optimization techniques; see Wang et al. [WGB22], and also our discussion on minimax problems in Section 2.12.

Most of the currently existing computer backgammon programs descend from TD-Gammon. Rollout-based backgammon programs are the most powerful in terms of performance, consistent with the principle that a rollout algorithm performs better than its base heuristic. However, they are too time-consuming for real-time play (without parallel computing hardware), because of the extensive on-line simulation requirement at each move.† They have been used in a limited diagnostic way to assess the quality of neural network-based programs (many articles and empirical works on computer backgammon are posted on-line; see e.g., http://www.bkgm.com/articles/page07.html).

2.7.4 Variance Reduction in Rollout - Comparing Advantages

When using simulation, sampling is often organized to effect *variance reduction*. By this we mean that for a given problem, the collection and use of samples is structured so that the variance of the simulation error is made smaller, with the same amount of simulation effort. There are several methods of this type for which we refer to textbooks on simulation (see, e.g., Ross [Ros12], and Rubinstein and Kroese [RuK16]).

† The situation in backgammon is exacerbated by its high branching factor, i.e., for a given position, the number of possible successor positions is quite large, as compared for example with chess.

In this section we discuss a method to reduce the effects of the simulation error in the calculation of the Q-factors in the context of rollout. The key idea is that the selection of the rollout control depends on the values of the Q-factor differences

$$\tilde{Q}_{k,\pi}(x_k,u_k)-\tilde{Q}_{k,\pi}(x_k,\hat{u}_k)$$

for all pairs of controls $(u_k,\hat{u}_k)$. These values must be computed accurately, so that the controls u_k and $\hat{u}_k$ can be accurately compared. On the other hand, the simulation/approximation errors in the computation of the individual Q-factors $\tilde{Q}_{k,\pi}(x_k,u_k)$ may be magnified through the preceding differencing operation.

An approach to counteract this type of simulation error magnification is to approximate the Q-factor difference $\tilde{Q}_{k,\pi}(x_k,u_k)-\tilde{Q}_{k,\pi}(x_k,\hat{u}_k)$ by sampling the difference

$$C_k(x_k,u_k,\mathbf{w}_k)-C_k(x_k,\hat{u}_k,\mathbf{w}_k), \tag{2.78}$$

where $\mathbf{w}_k=(w_k,w_{k+1},\ldots,w_{N-1})$ is the *same disturbance sequence* for the two controls u_k and $\hat{u}_k$, and

$$C_k(x_k,u_k,\mathbf{w}_k)=g_N(x_N)+g_k(x_k,u_k,w_k)+\sum_{i=k+1}^{N-1}g_i\big(x_i,\mu_i(x_i),w_i\big),$$

with $\{\mu_{k+1},\ldots,\mu_{N-1}\}$ being the tail portion of the base policy. †

For a simple example that illustrates how this form of variance reduction works, suppose we want to calculate the difference q_1-q_2 of two numbers q_1 and q_2 by subtracting two simulation samples $s_1=q_1+w_1$ and $s_2=q_2+w_2$, where w_1 and w_2 are zero-mean random variables. Then s_1-s_2 is unbiased in the sense that its mean is equal to q_1-q_2. However, the variance of s_1-s_2 decreases as the correlation of w_1 and w_2 increases. It is maximized when w_1 and w_2 are uncorrelated, and it is minimized (it is equal to 0) when w_1 and w_2 are equal.

The preceding example suggests a simulation scheme that is based on the difference (2.78) and involves a common disturbance $\mathbf{w}_k$ for u_k and $\hat{u}_k$. In particular, it may be far more accurate than the one obtained by differencing samples of $C_k(x_k,u_k,\mathbf{w}_k)$ and $C_k(x_k,\hat{u}_k,\hat{\mathbf{w}}_k)$, which involve two different disturbances $\mathbf{w}_k$ and $\hat{\mathbf{w}}_k$. Indeed, by introducing the zero mean sample errors

$$D_k(x_k,u_k,\mathbf{w}_k)=C_k(x_k,u_k,\mathbf{w}_k)-\tilde{Q}_{k,\pi}(x_k,u_k),$$

† For this to be possible, we need to assume that the probability distribution of each disturbance w_i does not depend on x_i and u_i.

it can be seen that the variance of the error in estimating $\tilde{Q}_{k,\pi}(x_k, u_k) - \tilde{Q}_{k,\pi}(x_k, \hat{u}_k)$ with the former method will be no larger than with the latter method if and only if

$$E_{\mathbf{w}_k,\,\hat{\mathbf{w}}_k}\Big\{\big|D_k(x_k, u_k, \mathbf{w}_k) - D_k(x_k, \hat{u}_k, \hat{\mathbf{w}}_k)\big|^2\Big\} \geq E_{\mathbf{w_k}}\Big\{\big|D_k(x_k, u_k, \mathbf{w}_k) - D_k(x_k, \hat{u}_k, \mathbf{w}_k)\big|^2\Big\}.$$

By expanding the quadratic forms and using the fact $E\big\{D_k(x_k, u_k, \mathbf{w}_k)\big\} = 0$, we see that this condition is equivalent to

$$E\big\{D_k(x_k, u_k, \mathbf{w}_k)D_k(x_k, \hat{u}_k, \mathbf{w}_k)\big\} \geq 0; \tag{2.79}$$

i.e., the errors $D_k(x_k, u_k, \mathbf{w}_k)$ and $D_k(x_k, \hat{u}_k, \mathbf{w}_k)$ being nonnegatively correlated. A little thought should convince the reader that this property is likely to hold in many types of problems.

Roughly speaking, the relation (2.79) holds if changes in the value of u_k (at the first stage) have little effect on the value of the error $D_k(x_k, u_k, \mathbf{w}_k)$ relative to the effect induced by the randomness of $\mathbf{w}_k$. To see this, suppose that there exists a scalar $\gamma < 1$ such that, for all x_k, u_k, and $\hat{u}_k$, there holds

$$E\Big\{\big|D_k(x_k, u_k, \mathbf{w}_k) - D_k(x_k, \hat{u}_k, \mathbf{w}_k)\big|^2\Big\} \leq \gamma E\Big\{\big|D_k(x_k, u_k, \mathbf{w}_k)\big|^2\Big\}. \tag{2.80}$$

Then we have, by using the generic relation $ab \geq a^2 - |a| \cdot |b - a|$ for two scalars a and b,

$$\begin{aligned} &D_k(x_k, u_k, \mathbf{w}_k)D_k(x_k, \hat{u}_k, \mathbf{w}_k) \\ &\qquad \geq \big|D_k(x_k, u_k, \mathbf{w}_k)\big|^2 \\ &\qquad\quad - \big|D_k(x_k, u_k, \mathbf{w}_k)\big| \cdot \big|D_k(x_k, \hat{u}_k, \mathbf{w}_k) - D_k(x_k, u_k, \mathbf{w}_k)\big|, \end{aligned}$$

from which we obtain

$$\begin{aligned} &E\big\{D_k(x_k,u_k, \mathbf{w}_k)D_k(x_k, \hat{u}_k, \mathbf{w}_k)\big\} \\ &\qquad \geq E\Big\{\big|D_k(x_k, u_k, \mathbf{w}_k)\big|^2\Big\} \\ &\qquad\quad - E\Big\{\big|D_k(x_k, u_k, \mathbf{w}_k)\big| \cdot \big|D_k(x_k, \hat{u}_k, \mathbf{w}_k) - D_k(x_k, u_k, \mathbf{w}_k)\big|\Big\} \\ &\qquad \geq E\Big\{\big|D_k(x_k, u_k, \mathbf{w}_k)\big|^2\Big\} - \frac{1}{2}E\Big\{\big|D_k(x_k, u_k, \mathbf{w}_k)\big|^2\Big\} \\ &\qquad\quad - \frac{1}{2}E\Big\{\big|D_k(x_k, \hat{u}_k, \mathbf{w}_k) - D_k(x_k, u_k, \mathbf{w}_k)\big|^2\Big\} \\ &\qquad \geq \frac{1-\gamma}{2}E\Big\{\big|D_k(x_k, u_k, \mathbf{w}_k)\big|^2\Big\}, \end{aligned}$$

where for the second inequality we use the generic relation

$$-|a| \cdot |b| \geq -\frac{1}{2}(a^2 + b^2)$$

for two scalars a and b, and for the third inequality we use Eq. (2.80).

Thus, under the assumption (2.80), the condition (2.79) holds and guarantees that by averaging cost difference samples rather than differencing (independently obtained) averages of cost samples, the simulation error variance does not increase.

Let us finally note the potential benefit of using Q-factor differences in contexts other than rollout. In particular when approximating Q-factors $Q_{k,\pi}(x_k, u_k)$ using parametric architectures (Section 3.3 in Chapter 3), it may be important to approximate and compare instead the differences

$$A_{k,\pi}(x_k, u_k) = Q_{k,\pi}(x_k, u_k) - \min_{u_k \in U_k(x_k)} Q_{k,\pi}(x_k, u_k).$$

The function $A_{k,\pi}(x_k, u_k)$ is also known as the *advantage of the pair* (x_k, u_k), and can serve just as well as $Q_{k,\pi}(x_k, u_k)$ for the purpose of comparing controls, but may work better in the presence of approximation errors. The use of advantages will be discussed further in Chapter 3.

2.7.5 Monte Carlo Tree Search

In our earlier discussion of simulation-based rollout implementation, we implicitly assumed that once we reach state x_k, we generate the same large number of trajectories starting from each pair (x_k, u_k), with $u_k \in U(x_k)$, to the end of the horizon. The drawback of this is threefold:

(a) The trajectories may be too long because the horizon length N is large (or infinite, in an infinite horizon context).

(b) Some of the controls u_k may be clearly inferior to others, and may not be worth as much sampling effort.

(c) Some of the controls u_k that appear to be promising, may be worth exploring better through multistep lookahead.

This has motivated multistep lookahead variants, generally referred to as *Monte Carlo tree search* (MCTS for short), which aim to trade off computational economy with a hopefully small risk of degradation in performance. Such variants involve, among others, early discarding of controls that are deemed inferior based on the results of preliminary calculations, and simulation that is limited in scope (either because of a reduced number of simulation samples, or because of a shortened horizon of simulation, or both).

A simple remedy for (a) above is to use rollout trajectories of reasonably limited length, with some terminal cost approximation at the end (in

an extreme case, the rollout may be skipped altogether for some states, i.e., rollout trajectories have zero length). The terminal cost function may be very simple (such as zero) or may be obtained through some auxiliary calculation. In fact the base policy used for rollout may be used to construct the terminal cost function approximation, as noted for the rollout-based backgammon algorithm of Example 2.7.4. In particular, an approximation to the cost function of the base policy may be obtained by training some approximation architecture, such as a neural network (see Chapter 3), and may be used as a terminal cost function.

A simple but less straightforward remedy for (b) is to use some heuristic or statistical test to discard some of the controls u_k, as soon as this is suggested by the early results of simulation, or even before any simulation, based for example on the suggestions of a trained neural network. Similarly, to implement (c) one may use some heuristic to increase the length of lookahead selectively for some of the controls u_k. This is similar to the incremental multistep rollout scheme for deterministic problems that we discussed in Section 2.4.2; see Fig. 2.4.6.

The MCTS approach can be based on sophisticated procedures for implementing and combining the remedies just described. The general idea is to use the interim results of the computation and statistical tests to focus the simulation effort along the most promising directions. Thus to implement MCTS with multistep lookahead, one needs to maintain a lookahead tree, which is expanded as the relevant Q-factors are evaluated by simulation, and which balances *the competing desires of exploitation and exploration* (generate and evaluate controls that seem most promising in terms of performance versus assessing the potential of inadequately explored controls). Ideas that were developed in the context of multiarmed bandit problems have played an important role in the construction of this type of MCTS procedure (see the end-of-chapter references).

In the simple case of one-step lookahead, with Q-factors calculated by Monte Carlo simulation, MCTS fundamentally aims to find efficiently the minimum of the expected values of a finite number of random variables. This is illustrated in the following example.

Example 2.7.5 (Statistical Tests for Adaptive Sampling with One-Step Lookahead)

Let us consider a typical one-step lookahead selection strategy that is based on adaptive sampling. We are at a state x_k and we try to find a control $\tilde{u}_k$ that minimizes an approximate Q-factor

$$\tilde{Q}_k(x_k, u_k) = E\Big\{ g_k(x_k, u_k, w_k) + \tilde{J}_{k+1}\big(f_k(x_k, u_k, w_k)\big) \Big\}$$

over $u_k \in U_k(x_k)$, with $\tilde{Q}_k(x_k, u_k)$ computed by averaging samples of the expression within braces. We assume that $U_k(x_k)$ contains m elements, which

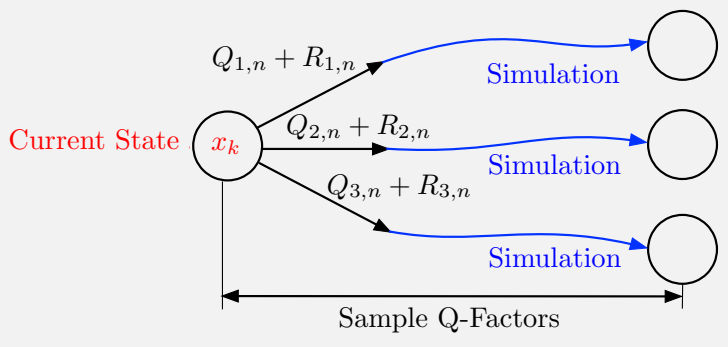

Figure 2.7.3 Illustration of one-step lookahead MCTS at a state x_k. The Q-factor sampled next corresponds to the control i with minimum sum of exploitation index (here taken to be the running average $Q_{i,n}$) and exploration index ($R_{i,n}$, possibly given by the UCB rule).

for simplicity are denoted $1, \ldots, m$. At the ℓth sampling period, knowing the outcomes of the preceding sampling periods, we select one of the m controls, say i_ℓ, and we draw a sample of $\tilde{Q}_k(x_k, i_\ell)$, whose value is denoted by S_{i_ℓ}. Thus after the nth sampling period we have an estimate $Q_{i,n}$ of the Q-factor of each control $i = 1, \ldots, m$ that has been sampled at least once, given by

$$Q_{i,n} = \frac{\sum_{\ell=1}^{n} \delta(i_\ell = i) S_{i_\ell}}{\sum_{\ell=1}^{n} \delta(i_\ell = i)},$$

where

$$\delta(i_\ell = i) = \begin{cases} 1 & \text{if } i_\ell = i, \\ 0 & \text{if } i_\ell \neq i. \end{cases}$$

Thus $Q_{i,n}$ is the *empirical mean* of the Q-factor of control i (total sample value divided by total number of samples), assuming that i has been sampled at least once.

After n samples have been collected, with each control sampled at least once, we may declare the control i that minimizes $Q_{i,n}$ as the "best" one, i.e., the one that truly minimizes the Q-factor $Q_k(x_k, i)$. However, there is a positive probability that there is an error: the selected control may not minimize the true Q-factor. In adaptive sampling, roughly speaking, we want to design the sample selection strategy and the criterion to stop the sampling, in a way that keeps the probability of error small (by allocating some sampling effort to all controls), and the number of samples limited (by not wasting samples on controls i that appear inferior based on their empirical mean $Q_{i,n}$).

Intuitively, a good sampling policy will balance at time n the desires of exploitation and exploration (i.e., sampling controls that seem most promising, in the sense that they have a small empirical mean $Q_{i,n}$, versus assessing the potential of inadequately explored controls, those i that have been sampled a small number of times). Thus it makes sense to sample next the control i that minimizes the sum

$$T_{i,n} + R_{i,n}$$

of two indices: an *exploitation index* $T_{i,n}$ and an *exploration index* $R_{i,n}$. Usually the exploitation index is chosen to be the empirical mean $Q_{i,n}$; see Fig. 2.7.3. The exploration index is based on a confidence interval formula and depends on the sample count

$$s_i = \sum_{\ell=1}^{n} \delta(i_\ell = i)$$

of control i. A frequently suggested choice is the UCB rule (upper confidence bound), which sets

$$R_{i,n} = -c\sqrt{\frac{\log n}{s_i}},$$

where c is a positive constant that is selected empirically (some analysis suggests values near $c = \sqrt{2}$, assuming that $Q_{i,n}$ is normalized to take values in the range $[-1, 0]$). The UCB rule, first proposed in the paper by Auer, Cesa-Bianchi, and Fischer [ACF02], has been extensively discussed in the literature both for one-step and for multistep lookahead [where it is called UCT (UCB applied to trees; see Kocsis and Szepesvari [KoS06])].†

Its justification is based on probabilistic analyses that relate to the multiarmed bandit problem, and is beyond our scope. Alternatives to the UCB formula have been suggested, and in fact in the AlphaZero program, the exploitation term has a different form than the one above, and depends on the depth of lookahead (see Silver et al. [SHS17]).

Sampling policies for MCTS with multistep lookahead are based on similar sampling ideas to the case of one-step lookahead. A simulated trajectory is run from a node i of the lookahead tree that minimizes the sum $T_{i,n} + R_{i,n}$ of an exploitation index and an exploration index. There are several schemes of this type, but the details are beyond our scope and are often problem-dependent (see the end-of-chapter references).

A major success has been the use of MCTS in two-player game contexts, such as the AlphaGo program (Silver et al. [SHM16]), which performs better than the best humans in the game of Go. This program integrates several of the techniques discussed in this book, including MCTS and rollout using a base policy that is trained off-line using a deep neural network. The AlphaZero program, which has performed spectacularly well against humans and other programs in the games of Go and chess (Silver et al. [SHS17]), bears some similarity with AlphaGo, and critically relies on MCTS, but does not use rollout in its on-line playing mode (it relies primarily on very long lookahead). Let us also note that based on early tests, MCTS is not effective in computer backgammon and it has not been adopted in commercial programs. More generally, the success of MCTS appears to be problem-dependent as well as implementation-dependent.

† The paper [ACF02] refers to the rule given here as UCB1 and credits its motivation to the paper by Agrawal [Agr95]. The book by Lattimore and Szepesvari [LaS20] provides an extensive discussion of the UCB rule and its generalizations.

2.7.6 Randomized Policy Improvement by Monte Carlo Tree Search

We have described rollout and MCTS as schemes for policy improvement: start with a base policy, and compute an improved policy based on the results of one-step lookahead or multistep lookahead followed by simulation with the base policy. We have implicitly assumed that both the base policy and the rollout policy are deterministic in the sense that they map each state x_k into a unique control $\tilde{\mu}_k(x_k)$ [cf. Eq. (2.77)]. In some (even nonstochastic) contexts, success has been achieved with *randomized policies*, which map a state x_k to a probability distribution over the set of controls $U_k(x_k)$, rather than mapping onto a single control. In particular, the AlphaGo and AlphaZero programs use MCTS to generate and use for training purposes randomized policies, which specify at each board position the probabilities with which the various moves are selected.

A randomized policy can be used as a base policy in a rollout context in exactly the same way as a deterministic policy: for a given state x_k, we just generate sample trajectories and associated sample Q-factors, using probabilistically selected controls, starting from each leaf-state of the lookahead tree that is rooted at x_k. We then average the corresponding Q-factor samples. The rollout/improved policy, as described here, is a deterministic policy, i.e., it applies at x_k the control $\tilde{\mu}_k(x_k)$ that is "best" according to the results of the rollout [cf. Eq. (2.77)]. Still, however, if we wish to generate an improved policy that is randomized, we can simply change the probabilities of different controls in the direction of the deterministic rollout policy. This can be done by increasing by some amount the probability of the "best" control $\tilde{\mu}_k(x_k)$ from its base policy level, while proportionally decreasing the probabilities of the other controls.

The use of MCTS provides a related method to "improve" a randomized policy. In the process of the adaptive simulation that is used in MCTS, we generate *frequency counts* of the different controls in $U_k(x_k)$, i.e., the proportion of rollout trajectories associated with each $u_k \in U_k(x_k)$. We can then obtain the rollout randomized policy by moving the probabilities of the base policy in the direction suggested by the frequency counts, i.e., increase the probability of high-count controls and reduce the probability of the others. This type of policy improvement is reminiscent of gradient-type methods, and has been successful in some contexts; see the end-of-chapter references for such policy improvement implementations in AlphaGo, AlphaZero, and other applications.

2.8 ROLLOUT FOR INFINITE-SPACES PROBLEMS - OPTIMIZATION HEURISTICS

We have considered so far finite control space applications of rollout, so there is a finite number of relevant Q-factors at each state x_k, which are

evaluated by simulation and are exhaustively compared. When the control constraint set is infinite, to implement this approach the constraint set must be replaced by a finite set, obtained by some form of discretization or random sampling, which can be inconvenient and ineffective. In this section we will discuss an alternative approach to deal with an infinite number of controls and Q-factors at x_k. The idea is to *use a base heuristic that involves a continuous optimization*, and to rely on a linear or nonlinear programming method to solve the corresponding lookahead optimization problem.

2.8.1 Rollout for Infinite-Spaces Deterministic Problems

To develop the basic idea of how to deal with infinite control spaces, we first consider deterministic problems, involving a system $x_{k+1} = f_k(x_k, u_k)$, and a cost per stage $g_k(x_k, u_k)$. The rollout minimization is

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} \tilde{Q}_k(x_k, u_k), \tag{2.81}$$

where $\tilde{Q}_k(x_k, u_k)$ is the approximate Q-factor

$$\tilde{Q}_k(x_k, u_k) = g_k(x_k, u_k) + H_{k+1}\big(f_k(x_k, u_k)\big), \tag{2.82}$$

with $H_{k+1}(x_{k+1})$ being the cost of the base heuristic starting from state x_{k+1} [cf. Eq. (2.10)]. Suppose that we have a differentiable closed-form expression for H_{k+1}, and the functions g_k and f_k are known and are differentiable with respect to u_k. Then the Q-factor $\tilde{Q}_k(x_k, u_k)$ of Eq. (2.82) is also differentiable with respect to u_k, and its minimization (2.81) may be addressed with one of the many gradient-based methods that are available for differentiable unconstrained and constrained optimization.

The preceding approach requires that the heuristic cost $H_{k+1}(x_{k+1})$ can be differentiated, so it should either be available in closed form, which is quite restrictive, or that it can be differentiated numerically, which may be inconvenient and/or unreliable. These difficulties can be circumvented by *using a base heuristic that is itself based on multistep optimization*. In particular, suppose that $H_{k+1}(x_{k+1})$ is the optimal cost of some $(\ell - 1)$-stage deterministic optimal control problem that is related to the original problem. Then the rollout algorithm (2.81)-(2.82) can be implemented by solving the *ℓ-stage deterministic optimal control problem, which seamlessly concatenates the first stage minimization over u_k [cf. Eq. (2.81)], with the $(\ell-1)$-stage minimization of the base heuristic*; see Fig. 2.8.1. This ℓ-stage problem may be solvable on-line by standard continuous spaces nonlinear

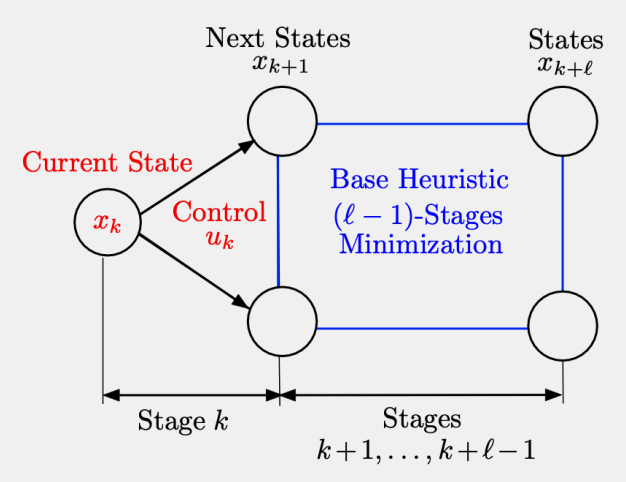

Figure 2.8.1 Schematic illustration of rollout for a deterministic problem with infinite control spaces. The base heuristic is to solve an $(\ell-1)$-stage deterministic optimal control problem, which together with the kth stage minimization over $u_k \in U_k(x_k)$, seamlessly forms an ℓ-stage continuous spaces optimal control/nonlinear programming problem that starts at state x_k.

programming or optimal control methods.† A major paradigm of methods of this type is model predictive control, which we have discussed in Chapter 1 (cf. Section 1.6.9). In the present section we will discuss a few other possibilities. The following is a simple example of an important class of inventory storage and supply chain management processes.

Example 2.8.1 (Supply Chain Management)

Let us consider a supply chain system, where a certain item is produced at a production center and fulfilled at a retail center. Stock of the item is shipped from the production center to the retail center, where it arrives with a delay of $\tau \geq 1$ time units, and is used to fulfill a known stream of demands d_k over an N-stage horizon; see Fig. 2.8.2. We denote:

x_k^1: The stock at hand at the production center at time k.

x_k^2: The stock at hand at the retail center at time k, used to fulfill demand (both positive and negative x_k^2 are allowed; a negative value indicates that there is backordered demand).

† Note, however, that for this to be possible, it is necessary to have a mathematical model of the system; a simulator is not sufficient. Another difficulty occurs when the control space is the union of a discrete set and a continuous set. Then it may be necessary to use some type of mixed integer programming technique to solve the ℓ-stage problem. Alternatively, it may be possible to handle the discrete part by brute force enumeration, followed by continuous optimization.

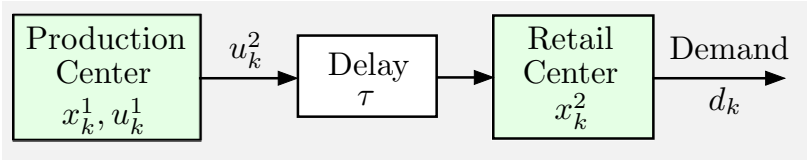

Figure 2.8.2. Illustration of a simple supply chain system for Example 2.8.1.

u_k^1: The amount produced at time k.

u_k^2: The amount shipped at time k (and arriving at the retail center τ time units later).

The state at time k is the stock available at the production and retail centers, x_k^1, x_k^2, plus the stock amounts that are in transit and have not yet arrived at the retail center $u_{k-\tau-1}^2, \ldots, u_{k-1}^2$. The control $u_k = (u_k^1, u_k^2)$ is chosen from some constraint set that may depend on the current state, and is subject to production capacity and transport availability constraints. The system equation is

$$x_{k+1}^1 = x_k^1 + u_k^1 - u_k^2, \qquad x_{k+1}^2 = x_k^2 + u_{k-\tau}^2 - d_k,$$

and involves the delayed control component $u_{k-\tau}^2$. Thus the exact DP algorithm involves state augmentation as introduced in Section 1.6.5, and may thus be much more complicated than in the case where there are no delays.†

The cost at time k consists of three components: a production cost that depends on x_k^1 and u_k^1, a transportation cost that depends on u_k^2, and a fulfillment cost that depends on x_k^2 [which includes positive costs for both excess inventory (i.e., $x_k^2 > d_k$) and for backordered demand (i.e., $x_k^2 < d_k$)]. The precise forms of these cost components are immaterial for the purposes of this example.

Here the control vector u_k is often continuous (or a mixture of discrete and continuous components), so it may be essential for the purposes of rollout to use the continuous optimization framework of this section. In particular, at the current stage k, we know the current state, which includes x_k^1, x_k^2, and the amounts of stock in transit together with their scheduled arrival times at the retail center. We then apply some heuristic optimization to determine the stream of future production and shipment levels over ℓ steps, and use the first component of this stream as the control applied by rollout. As an example we may use as base policy one that brings the retail inventory to some target value ℓ stages ahead, and possibly keep it at that value for a portion of the remaining periods. This is a nonlinear programming or mixed integer programming problem that may be solvable with available software far more efficiently than by a discretized form of DP.

† Despite the fact that with large delays, the size of the augmented state space can become very large (cf. Section 1.6.5), the implementation of rollout schemes is not affected much by this increase in size. For this reason, rollout can be very well suited for problems involving delayed effects of past states and controls.

A major benefit of rollout in the supply chain context is that it can readily incorporate on-line replanning. This is necessary when unexpected demand changes, production or transport equipment failures occur, or updated forecasts become available.

The following example deals with a common class of problems of resource allocation over time.

Example 2.8.2 (Multistage Linear and Mixed Integer Programming)

Let us consider a deterministic optimal control problem with linear system equation

$$x_{k+1} = A_k x_k + B_k u_k + d_k, \qquad k = 0, \ldots, N-1,$$

where A_k and B_k are known matrices of appropriate dimension, d_k is a known vector, and x_k and u_k are column vectors. The cost function is linear of the form

$$c_N{}' x_N + \sum_{k=0}^{N-1} (c_k{}' x_k + d_k{}' u_k),$$

where c_k and d_k are known column vectors of appropriate dimension, and a prime denotes transpose. The terminal state and state-control pairs (x_k, u_k) are constrained by

$$x_N \in T, \qquad (x_k, u_k) \in P_k, \qquad k = 0, \ldots, N-1,$$

where T and P_k, $k = 0, \ldots, N-1$, are given sets, which are specified by linear and possibly integer constraints.

As an example, consider a multi-item production system, where the state is $x_k = (x_k^1, \ldots, x_k^n)$ and x_k^i represents stock of item i available at the start of period k. The state evolves according to the system equation

$$x_{k+1}^i = \sum_{j=1}^{n} a_k^{ij} u_k^{ij} - d_k^i, \qquad i = 1, \ldots, n,$$

where u_k^{ij} is the amount of product i that is used during time k for the manufacture of product j, a_k^{ij} are known scalars that are related to the underlying production process, and d_k^i is a deterministic demand of product i that is fulfilled at time k. One constraint here is that

$$\sum_{j=1}^{n} u_k^{ij} \leq x_k^i, \qquad i = 1, \ldots, n,$$

and there are additional linear and integer constraints on (x_k, u_k), which are collected in a general constraint of the form $(x_k, u_k) \in P_k$ (e.g., nonnegativity, production capacity, storage constraints, etc). Note that the problem

may be further complicated by production delays, as in the preceding supply chain Example 2.8.1. Moreover, while in this section we focus on deterministic problems, we may envision a stochastic version of the problem where the demands d_k^i are random with given probability distributions, which are subject to revisions based on randomly received forecasts.

The problem may be solved using a linear or mixed integer programming algorithm, but this may be very time-consuming when N is large. Moreover, the problem will need to be resolved on-line if some of the problem data changes and replanning is necessary. A suboptimal alternative is to use truncated rollout with an ℓ-stage mixed integer optimization, and a polyhedral terminal cost function $\tilde{J}_{k+\ell}$ to provide a terminal cost optimization. A simple possibility is no terminal cost $[\tilde{J}_{k+\ell}(x_{k+\ell}) \equiv 0]$, and another possibility is a polyhedral lower bound approximation that can be based on relaxing the integer constraints after stage $k+\ell$, or some kind of training approach that uses data.

We will next discuss how rollout can accommodate stochastic disturbances by using deterministic optimization ideas based on certainty equivalence and the methodology of stochastic programming.

2.8.2 Rollout Based on Stochastic Programming

We have focused so far in this section on rollout that relies on deterministic continuous optimization. There is an important class of methods, known as *stochastic programming*, which can be used for stochastic optimal control, but bears a close connection to continuous spaces deterministic optimization. We will first describe this connection for two-stage problems, then discuss extensions to many-stages problems, and finally show how rollout can be brought to bear for their approximate solution.

Example 2.8.3 (Two-Stage Stochastic Programming)

Consider a stochastic problem of optimal decision making over two stages: In the first stage we will choose a finite-dimensional vector u_0 from a subset U_0 with cost $g_0(u_0)$. Then an uncertain event represented by a random variable w_0 will occur, whereby w_0 will take one of the values $w^1, \ldots, w^m$ with corresponding probabilities $p^1, \ldots, p^m$. Once w_0 occurs, we will know its value w^i, and we must then choose at the second stage a vector u_1^i from a subset $U_1(u_0, w^i)$ at a cost $g_1(u_1^i, w^i)$. The objective is to minimize the expected cost

$$g_0(u_0) + \sum_{i=1}^{m} p^i g_1(u_1^i, w^i),$$

subject to

$$u_0 \in U_0, \qquad u_1^i \in U_1(u_0, w^i), \quad i = 1, \ldots, m.$$

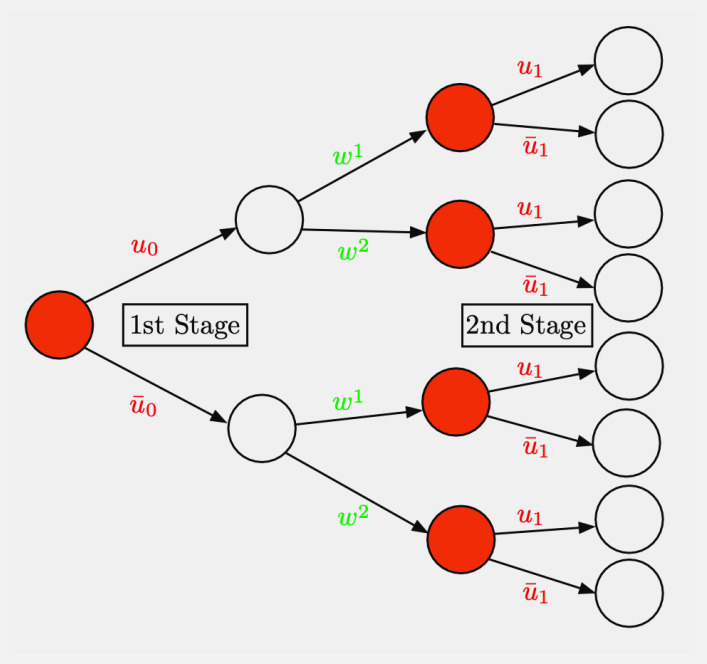

Figure 2.8.3. Illustration of the DP problem associated with two-stage stochastic programming; cf. Example 2.8.3. The figure depicts the case where each variable u_0, w_0, and u_1 can take only two values. A similar conversion to a DP problem is possible for a multistage stochastic programming problem, involving multiple choices of decisions, each followed by an uncertain event whose outcome is perfectly observed by the decision maker.

We can view this problem as a two-stage DP problem, where $x_1 = w_0$ is the system equation, the disturbance w_0 can take the values $w^1, \ldots, w^m$ with probabilities $p^1, \ldots, p^m$, the cost of the first stage is $g_0(u_0)$, the cost of the second stage is $g_1(x_1, u_1)$, and the terminal cost is 0. The intuitive meaning is that since at time 0 we don't know yet which of the m values w^i of w_0 will occur, we must calculate (in addition to u_0) a separate second stage decision u_1^i for each i, which will be used after we know that the value of w_0 is w^i.

However, if u_0 and u_1 take values in a continuous space such as the Euclidean spaces $\Re^{d_0}$ and $\Re^{d_1}$, respectively, we can also equivalently view the problem as a nonlinear programming problem of dimension $(d_0 + md_1)$ (the optimization variables are u_0 and u_1^i, $i = 1, \ldots, m$).

For a generalization of the preceding example, consider the stochastic DP problem of Section 1.3 for the case where there are only two stages, and the disturbances w_0 and w_1 can independently take one of the m values $w^1, \ldots, w^m$ with corresponding probabilities $p_0^1, \ldots, p_0^m$ and $p_1^1, \ldots, p_1^m$, respectively. The optimal cost function $J_0(x_0)$ is given by the two-stage

DP algorithm

$$J_0(x_0) = \min_{u_0 \in U_0(x_0)} \left[\sum_{i=1}^{m} p_0^i \Big\{ g_0(x_0, u_0, w^i) + \min_{u_1^i \in U_1(f_0(x_0,u_0,w^i))} \left[\sum_{j=1}^{m} p_1^j \Big\{ g_1\big(f_0(x_0, u_0, w^i), u_1^i, w^j\big) + g_2\big(f_1\big(f_0(x_0, u_0, w^i), u_1^i, w^j\big)\big) \Big\} \right] \Big\} \right].$$

By bringing the inner minimization outside the inner brackets, we see that this DP algorithm is equivalent to solving the nonlinear programming problem

$$\begin{aligned} \text{minimize } & \sum_{i=1}^{m} p_0^i \Big\{ g_0(x_0, u_0, w^i) + \sum_{j=1}^{m} p_1^j \Big\{ g_1\big(f_0(x_0, u_0, w^i), u_1^i, w^j\big) + g_2\big(f_1\big(f_0(x_0, u_0, w^i), u_1^i, w^j\big)\big) \Big\} \Big\} \\ \text{subject to } & u_0 \in U_0(x_0), \qquad u_1^i \in U_1\big(f_0(x_0, u_0, w^i)\big), \quad i = 1, \ldots, m. \end{aligned} \tag{2.83}$$

If the controls u_0 and u_1^i are elements of $\Re^d$, this problem involves $d(1 + m)$ scalar variables. An example is the multi-item production problem described in Example 2.8.2 in the case where the demands w_k^i and/or the production coefficients a_k^{ij} are stochastic.

We can also consider an N-stage stochastic optimal control problem. A similar reformulation as a nonlinear programming problem is possible. It converts the N-stage stochastic problem into a deterministic optimization problem of dimension that grows exponentially with the number of stages N. In particular, for an N-stage problem, the number of control variables expands by a factor m with each additional stage. The total number of variables is bounded by

$$d\big(1 + m + m^2 + \cdots + m^{N-1}\big),$$

where m is the maximum number of values that a disturbance can take at each stage and d is the dimension of the control vector.

2.8.3 Stochastic Rollout with Certainty Equivalence

The dimension of the preceding nonlinear programming formulation of the multistage stochastic optimal control problem with continuous control spaces can be very large. This motivates a variant of a rollout algorithm

that relies on a stochastic optimization for the current stage, and a deterministic optimization that relies on (assumed) certainty equivalence for the remaining stages, where the base policy is used. In this way, the dimension of the nonlinear programming problem to be solved by rollout is drastically reduced.

This rollout algorithm operates as follows: Given a state x_k and control $u_k \in U_k(x_k)$, we consider the next states x_{k+1}^i that correspond to the m possible values w_k^i, $i = 1, \ldots, m$, which occur with the known probabilities p_k^i, $i = 1, \ldots, m$. We then consider the approximate Q-factors

$$\tilde{Q}_k(x_k, u_k) = \sum_{i=1}^{m} p_k^i \big(g_k(x_k, u_k, w_k^i) + \tilde{H}_{k+1}(x_{k+1}^i) \big), \tag{2.84}$$

where $\tilde{H}_{k+1}(x_{k+1}^i)$ is the cost of a base policy, which starting at stage $k+1$ from

$$x_{k+1}^i = f_k(x_k, u_k, w_k^i),$$

optimizes the cost-to-go starting from x_{k+1}^i, while assuming that the future disturbances $w_{k+1}, \ldots, w_{N-1}$, will take some nominal (nonrandom) values $\overline{w}_{k+1}, \ldots, \overline{w}_{N-1}$. The rollout control $\tilde{\mu}_k(x_k)$ computed by this algorithm is

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} \tilde{Q}_k(x_k, u_k). \tag{2.85}$$

Note that this rollout algorithm does not have the cost improvement property, because it involves an approximation: the cost $\tilde{H}_{k+1}(x_{k+1}^i)$ used in Eq. (2.84) is an approximation to the cost of a policy. It is the cost of a policy applied to the certainty equivalent version of the original stochastic problem.

The key fact now is that the problem (2.85) can be viewed as a seamless $(N-k)$-stage deterministic optimization, which involves the control u_0, and for each value w_k^i of the disturbance w_k, the sequence of controls $(u_{k+1}^i, \ldots, u_{N-1}^i)$. If the controls are elements of $\Re^d$, this deterministic optimization involves a total of

$$d\big(1 + (N-k-1)m\big) \tag{2.86}$$

scalar variables. Currently available deterministic optimization software can deal with quite large numbers of variables, particularly in the context of linear programming, so by using rollout in combination with certainty equivalence, very large problems with continuous state and control variables may be addressed. We refer to the paper by Hu et al. [HWP22] for an application of this idea to problems of maintenance scheduling.

Another possibility is to use multistep lookahead that aims to represent better the stochastic character of the uncertainty. Here at state x_k we solve an $(N-k)$-stage optimal control problem, where the uncertainty

is fully taken into account in the first ℓ stages, similar to stochastic programming, and in the remaining $N-k-\ell$ stages, the uncertainty is dealt with certainty equivalence, by fixing the disturbances $w_{k+\ell},\ldots,w_{N-1}$ at some nominal values (we assume here for simplicity that $\ell < N-k$). If the controls are elements of $\Re^d$, and the number of values that the disturbances $w_0,\ldots,w_{N-1}$ can take is m, the total number of control variables of this problem is

$$d\big(1+m+\cdots+m^{\ell-1}+(N-k-\ell)m^{\ell}\big),$$

[this is the ℓ-step lookahead generalization of the formula (2.86)]. Once the optimal policy $\{\tilde{u}_k,\tilde{\mu}_{k+1},\tilde{\mu}_{k+2},\ldots\}$ for this problem is obtained, the first control component $\tilde{u}_k$ is applied at x_k and the remaining components $\{\tilde{\mu}_{k+1},\tilde{\mu}_{k+2},\ldots\}$ are discarded. Note also that this multistep lookahead approach may be combined with the ideas of multiagent rollout, which will be discussed in the next section.

2.9 MULTIAGENT ROLLOUT

We will now consider a special structure of the control space, whereby the control u_k consists of m components, $u_k=(u_k^1,\ldots,u_k^m)$, with a separable control constraint structure $u_k^\ell \in U_k^\ell(x_k)$, $\ell=1,\ldots,m$. The control constraint set is the Cartesian product

$$U_k(x_k)=U_k^1(x_k)\times\cdots\times U_k^m(x_k). \tag{2.87}$$

Conceptually, each component u_k^ℓ, $\ell=1,\ldots,m$, is chosen at stage k by a separate "agent" (a decision making entity), and for the sake of the following discussion, we assume that each set $U_k^\ell(x_k)$ is finite. We discussed this type of problem briefly in Section 1.6.7, and we will discuss it in this section in greater detail.

The one-step lookahead minimization

$$\tilde{u}_k \in \arg\min_{u_k\in U_k(x_k)} E\Big\{g_k(x_k,u_k,w_k)+J_{k+1,\pi}\big(f_k(x_k,u_k,w_k)\big)\Big\}, \tag{2.88}$$

where π is a base policy, involves as many as n^m Q-factors, where n is the maximum number of elements of the sets $U_k^\ell(x_k)$ [so that n^m is an upper bound to the number of controls in $U_k(x_k)$, in view of the Cartesian product structure (2.87)]. As a result, the standard rollout algorithm requires an exponential [order $O(n^m)$] number of base policy cost computations per stage, which can be overwhelming even for moderate values of m.

This motivates an alternative and more efficient rollout algorithm, called *multiagent rollout* also referred to as *agent-by-agent rollout*, that still achieves the cost improvement property

$$J_{k,\tilde{\pi}}(x_k)\le J_{k,\pi}(x_k),\qquad \forall\ x_k,k, \tag{2.89}$$

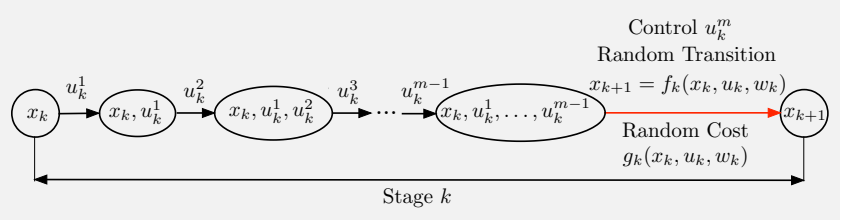

Figure 2.9.1 Equivalent formulation of the N-stage stochastic optimal control problem for the case where the control u_k consists of m components $u_k^1, u_k^2, \ldots, u_k^m$:

$$u_k = (u_k^1, \ldots, u_k^m) \in U_k^1(x_k) \times \cdots \times U_k^m(x_k);$$

cf. Section 1.6.7. The figure depicts the kth stage transitions. Starting from state x_k, we generate the intermediate states

$$(x_k, u_k^1), (x_k, u_k^1, u_k^2), \ldots, (x_k, u_k^1, \ldots, u_k^{m-1}),$$

using the respective controls $u_k^1, \ldots, u_k^{m-1}$. The final control u_k^m leads from $(x_k, u_k^1, \ldots, u_k^{m-1})$ to

$$x_{k+1} = f_k(x_k, u_k, w_k),$$

and a stage cost $g_k(x_k, u_k, w_k)$ is incurred. All of the preceding transitions, which involve the controls $u_k^1, \ldots, u_k^{m-1}$, incur zero cost.

where $J_{k,\tilde{\pi}}(x_k)$, $k = 0, \ldots, N$, is the cost-to-go of the rollout policy $\tilde{\pi}$ starting from state x_k. Indeed we will exploit the multiagent structure to construct an algorithm that maintains the cost improvement property at much smaller computational cost, namely requiring order $O(nm)$ base policy cost computations per stage.

A key idea here is that the computational requirements of the rollout one-step minimization (2.88) are proportional to the size of the control space and are independent of the size of the state space. We consequently reformulate the problem so that control space complexity is traded off with state space complexity, as discussed in Section 1.6.7. This is done by "unfolding" the control u_k into its m components $u_k^1, u_k^2, \ldots, u_k^m$. At the same time, between x_k and the next state $x_{k+1} = f_k(x_k, u_k, w_k)$, we introduce artificial intermediate "states" and corresponding transitions; see Fig. 2.9.1, given in Section 1.6.7 and repeated here for convenience.

It can be seen that this reformulated problem is equivalent to the original, since any control choice that is possible in one problem is also possible in the other problem, while the cost structure of the two problems is the same: each policy of the reformulated problem corresponds to a policy of the original problem, with the same cost function, and reversely.†

† A fine point here is that policies of the original problem involve functions

Consider now the standard rollout algorithm applied to the reformulated problem of Fig. 2.9.1, with a given base policy $\pi = \{\mu_0, \ldots, \mu_{N-1}\}$, which is also a policy of the original problem [so that $\mu_k = (\mu_k^1, \ldots, \mu_k^m)$, with each μ_k^ℓ, $\ell = 1, \ldots, m$, being a function of just x_k]. The algorithm involves a minimization over only one control component at the states x_k and at the intermediate states

$$(x_k, u_k^1), (x_k, u_k^1, u_k^2), \ldots, (x_k, u_k^1, \ldots, u_k^{m-1}).$$

In particular, *for each stage k, the algorithm requires a sequence of m minimizations, once over each of the agent controls $u_k^1, \ldots, u_k^m$, with the past controls determined by the rollout policy, and the future controls determined by the base policy.* Assuming a maximum of n elements in the constraint sets $U_k^\ell(x_k)$, the computation required at each stage k is of order $O(n)$ for each of the "states"

$$x_k, (x_k, u_k^1), \ldots, (x_k, u_k^1, \ldots, u_k^{m-1}),$$

for a total of order $O(nm)$ computation.

To elaborate, at $(x_k, u_k^1, \ldots, u_k^{\ell-1})$ with $\ell \leq m$, and for each of the controls $u_k^\ell \in U_k^\ell(x_k)$, we generate by simulation a number of system trajectories up to stage N, with all future controls determined by the base policy. We average the costs of these trajectories, thereby obtaining the Q-factors corresponding to $(x_k, u_k^1, \ldots, u_k^{\ell-1}, u_k^\ell)$, for all values $u_k^\ell \in U_k^\ell(x_k)$ (with the preceding controls $u_k^1, \ldots, u_k^{\ell-1}$ held at the values computed earlier, and the future controls $u_k^{\ell+1}, \ldots, u_k^m, u_{k+1}, \ldots, u_{N-1}$ determined by the base policy). We then select the control $u_k^\ell \in U_k^\ell(x_k)$ that corresponds to the minimal Q-factor.

Prerequisite assumptions for the preceding algorithm to work in an on-line multiagent setting are:

(a) All agents have access to the current state x_k as well as the base policy (including the control functions μ_n^ℓ, $\ell = 1, \ldots, m$, $n = 0, \ldots, N-1$ of all agents).

(b) There is an order in which agents compute and apply their local controls.

(c) The agents share their information, so agent ℓ knows the local controls $u_k^1, \ldots, u_k^{\ell-1}$ computed by the predecessor agents $1, \ldots, \ell-1$ in the given order.

---

of x_k, while policies of the reformulated problem involve functions of the choices of the preceding agents, as well as x_k. However, by successive substitution of the control functions of the preceding agents, we can view control functions of each agent as depending exclusively on x_k. It follows that the multi-transition structure of the reformulated problem cannot be exploited to reduce the cost function beyond what can be achieved with a single-transition structure.

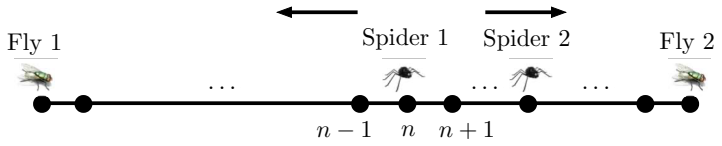

Figure 2.9.2 Illustration of the two-spiders and two-flies problem. The spiders move along integer points of a line. The two flies stay still at some integer locations. The character of the optimal policy is to move the two spiders towards two different flies.

Multiagent rollout with the given base policy starts with spider 1 at location n, and calculates the two Q-factors of moving to locations $n-1$ and $n+1$, assuming that the remaining moves of the two spiders will be made using the go-towards-the-nearest-fly base policy. The Q-factor of going to $n-1$ is smallest because it saves in unnecessary moves of spider 1 towards fly 2, so spider 1 will move towards fly 1. The trajectory generated by multiagent rollout is to move spiders 1 and 2 towards flies 1 and 2, respectively, then spider 2 first captures fly 2, and then spider 1 captures fly 1.

Note that the rollout policy obtained from the reformulated problem may be different from the rollout policy obtained from the original problem. However, the former rollout algorithm is far more efficient than the latter in terms of required computation, while still maintaining the cost improvement property (2.89).

Illustrative Examples

The following spiders-and-flies example illustrates how multiagent rollout may exhibit intelligence and agent coordination that is totally lacking from the base policy. This behavior has been supported by computational experiments and analysis with larger (two-dimensional) spiders-and-flies problems.

Example 2.9.1 (Spiders and Flies)

We have two spiders and two flies moving along integer locations on a straight line. For simplicity we assume that the flies' positions are fixed at some integer locations, although the problem is qualitatively similar when the flies move randomly. The spiders have the option of moving either left or right by one unit; see Fig. 2.9.2. The objective is to minimize the time to capture both flies. The problem has essentially a finite horizon since the spiders can force the capture of the flies within a known number of steps.

The salient feature of the optimal policy here is to move the two spiders towards different flies. The minimal time to capture is the maximum of the initial distances of the two spider-fly pairs of the optimal policy.

Let us apply multiagent rollout with the base policy that directs each spider to move one unit towards the closest fly position (a tie is broken by moving towards the right-side fly). The base policy is poor because it may unnecessarily move both spiders in the same direction, when in fact only one

is needed to capture the fly. This limitation is due to the lack of coordination between the spiders: each acts selfishly, ignoring the presence of the other. We will see that rollout restores a significant degree of coordination between the spiders through an optimization that takes into account the long-term consequences of the spider moves.

According to the multiagent rollout mechanism, the spiders choose their moves one-at-a-time, optimizing over the two Q-factors corresponding to the right and left moves, while assuming that future moves will be chosen according to the base policy. Let us consider a stage, where the two flies are alive, while both spiders are closest to fly 2, as in Fig. 2.9.2. Then the rollout algorithm will start with spider 1 and calculate two Q-factors corresponding to the right and left moves, while using the base heuristic to obtain the next move of spider 2, and the remaining moves of the two spiders. Depending on the values of the two Q-factors, spider 1 will move to the right or to the left, and it can be seen that it will choose to *move away from spider 2* even if doing so increases its distance to its closest fly *contrary to what the base heuristic will do*. Then spider 2 will act similarly and the process will continue. Intuitively, at the state of Fig. 2.9.2, spider 1 moves away from spider 2 and fly 2, because it recognizes that spider 2 will capture fly 2 earlier, so it might as well move towards the other fly.

Thus *the multiagent rollout algorithm induces implicit move coordination*, i.e., each spider moves in a way that takes into account future moves of the other spider. In fact it can be verified that the algorithm will produce an optimal sequence of moves starting from any initial spider positions. It can also be seen that ordinary rollout (both spiders move at once) will also produce an optimal move sequence.

The example illustrates how a poor base heuristic can produce an excellent rollout solution, something that can be observed frequently in many other problems. Intuitively, the key fact is that rollout is "farsighted" in the sense that it can benefit from control calculations that reach far into future stages.

A two-dimensional generalization of the example is also interesting. Here the flies are at two corners of a square in the plane. It can be shown that the two spiders, starting from the same position within the square, will separate under the rollout policy, with each moving towards a different spider, while under the base policy, they will move in unison along the shortest path to the closest surviving fly. Again this will happen for both standard and multiagent rollout.

Let us consider another example of a discrete optimization problem that can be solved efficiently with multiagent rollout.

Example 2.9.2 (Multi-Vehicle Routing)

Consider a multi-vehicle routing problem, whereby m vehicles move along the arcs of a given graph, aiming to perform tasks located at the nodes of the graph; see Fig. 2.9.3. When a vehicle reaches a task, it performs it, and can move on to perform another task. We wish to perform the tasks in a minimum number of individual vehicle moves.

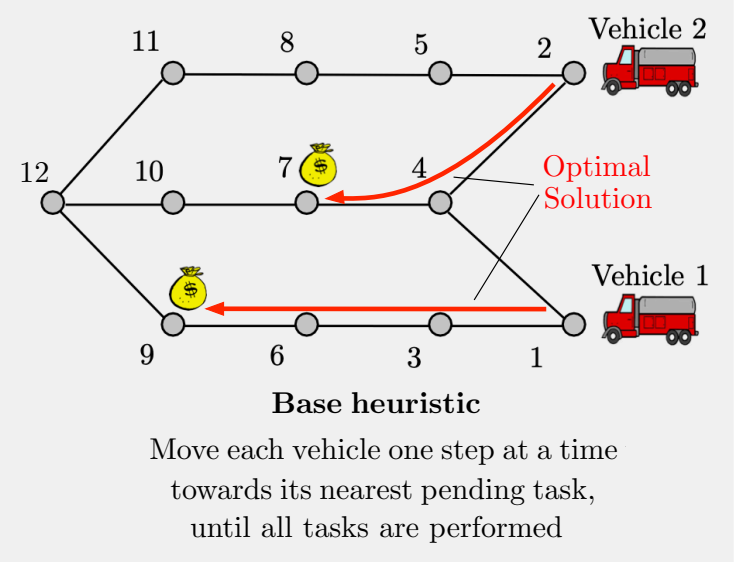

Figure 2.9.3 An instance of the vehicle routing problem of Example 2.9.2, and the multiagent rollout approach. The two vehicles aim to collectively perform the two tasks as fast as possible. Here, we should avoid sending both vehicles to node 4, towards the task at node 7; sending only vehicle 2 towards that task, while sending vehicle 1 towards the task at node 9 is clearly optimal. However, the base heuristic has "limited vision" and does not perceive this. By contrast the standard and the one-vehicle-at-a-time rollout algorithms look beyond the first move and avoid this inefficiency: they examine both moves of vehicle 1 to nodes 3 and 4, and use the base heuristic to explore the corresponding trajectories to the end of the horizon, and discover that vehicle 2 can reach quickly node 7, and that it is best to send vehicle 1 towards node 9.

In particular, the one-vehicle-at-a-time rollout algorithm will operate as follows: given the starting position pair $(1, 2)$ of the vehicles and the current pending tasks at nodes 7 and 9, we first compare the Q-factors of the two possible moves of vehicle 1 (to nodes 3 and 4), assuming that all the remaining moves will be selected by the base heuristic at the beginning of each stage. Thus vehicle 1 will choose to move to node 3. Then with knowledge of the move of vehicle 1 from 1 to 3, we select the move of vehicle 2 by comparing the Q-factors of its two possible moves (to nodes 4 and 5), taking also into account the fact that the remaining moves will be made according to the base heuristic. Thus vehicle 2 will choose to move to node 4.

We then continue at the next state [vehicle positions at (3,4) and pending tasks at nodes 7 and 9], select the base heuristic moves of vehicles 1 and 2 on the path to the closest pending tasks [(9 and 7), respectively], etc. Eventually the rollout finds the optimal solution (move vehicle 1 to node 9 in three moves and move vehicle 2 to node 7 in two moves), which has a total cost of 5. By contrast it can be seen that the base heuristic at the initial state will move both vehicles to node 4 (towards the closest pending task), and generate a trajectory that moves vehicle 1 along the path $1 \to 4 \to 7$ and vehicle 2 along the path $2 \to 4 \to 7 \to 10 \to 12 \to 9$, while incurring a total cost of 7.

For a large number of vehicles and a complicated graph, this is a nontrivial combinatorial problem. The problem can be formulated as a discrete deterministic optimization problem, and addressed by approximate DP methods. The state at a given stage is the m-tuple of current positions of the vehicles together with the list of pending tasks, but the number of these states can be enormous (it increases exponentially with the number of nodes and the number of vehicles). Moreover the number of joint move choices by the vehicles also increases exponentially with the number of vehicles.

We are thus motivated to use a multiagent rollout approach. We define a base heuristic as follows: at a given stage and state (vehicle positions and pending tasks), it finds the closest pending task (in terms of number of moves needed to reach it) for each of the vehicles and moves each vehicle one step towards the corresponding closest pending task (this is a legitimate base heuristic: it assigns to each state a vehicle move for every vehicle).†

In the multiagent rollout algorithm, at a given stage and state, we take each vehicle in the order $1, \ldots, n$, and we compare the Q-factors of the available moves to that vehicle while assuming that all the remaining moves will be made according to the base heuristic, and taking into account the moves that have been already made and the tasks that have already been performed; see the illustration of Fig. 2.9.3. In contrast to all-vehicles-at-once rollout, the one-vehicle-at-a-time rollout algorithm considers a polynomial (in m) number of moves and corresponding shortest path problems at each stage. In the example of Fig. 2.9.3, the one-vehicle-at-a-time rollout finds the optimal solution, while the base heuristic starting from the initial state does not.

The Cost Improvement Property

Generally, it is unclear how the two rollout policies (standard/all-agents-at-once and agent-by-agent) perform relative to each other in terms of attained cost.‡ On the other hand, both rollout policies perform no worse than the

† There is an alternative version of the base heuristic, which makes selections one-vehicle-at-a-time: at a given stage and state (vehicle positions and pending tasks), it finds the closest pending task (in terms of number of moves needed to reach it) for vehicle 1 and moves this vehicle one step towards this closest pending task. Then it finds the closest pending task for vehicle 2 (the pending status of the tasks, however, may have been affected by the move of vehicle 1) and moves this vehicle one step towards this closest pending task, and continues similarly for vehicles $3, \ldots, n$. There is a subtle difference between the two base heuristics: for example they may make different choices when vehicle 1 reaches a pending task in a single move, thereby changing the status of that task, and affecting the choice of the base heuristic for vehicle 2, etc.

‡ For an example where the standard rollout algorithm works better, consider a single-stage problem, where the objective is to minimize the first stage cost $g_0(u_0^1, \ldots, u_0^m)$. Let $\overline{u}_0 = (\overline{u}_0^1, \ldots, \overline{u}_0^m)$ be the control applied by the base policy, and assume that $\overline{u}_0$ is not optimal. Suppose that starting at $\overline{u}_0$, the cost cannot be improved by varying any single control component. Then the multiagent rollout algorithm stays at the suboptimal $\overline{u}_0$, while the standard rollout algorithm finds

base policy, since the performance of the base policy is identical for both the reformulated and the original problems. This cost improvement property can also be shown analytically as follows by induction, by modifying the standard rollout cost improvement proof; cf. Section 2.7.

Proposition 2.9.1: (Cost Improvement for Multiagent Rollout) The rollout policy $\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$ obtained by multiagent rollout satisfies

$$J_{k,\tilde{\pi}}(x_k) \leq J_{k,\pi}(x_k), \qquad \text{for all } x_k \text{ and } k, \tag{2.90}$$

where π is the base policy.

Proof: We will show the inequality (2.90) by induction, but for simplicity, we will give the proof for the case of just two agents, i.e., $m = 2$. Clearly the inequality holds for $k = N$, since $J_{N,\tilde{\pi}} = J_{N,\pi} = g_N$. Assuming that it holds for index $k+1$, we have for all x_k,

$$\begin{aligned}
J_{k,\tilde{\pi}}(x_k) &= E\Big\{g_k\big(x_k, \tilde{\mu}_k^1(x_k), \tilde{\mu}_k^2(x_k), w_k\big) \\
&\qquad + J_{k+1,\tilde{\pi}}\Big(f_k\big(x_k, \tilde{\mu}_k^1(x_k), \tilde{\mu}_k^2(x_k), w_k\big)\Big)\Big\} \\
&\leq E\Big\{g_k\big(x_k, \tilde{\mu}_k^1(x_k), \tilde{\mu}_k^2(x_k), w_k\big) \\
&\qquad + J_{k+1,\pi}\Big(f_k\big(x_k, \tilde{\mu}_k^1(x_k), \tilde{\mu}_k^2(x_k), w_k\big)\Big)\Big\} \\
&= \min_{u_k^2 \in U_k^2(x_k)} E\Big\{g_k(x_k, \tilde{\mu}_k^1(x_k), u_k^2, w_k) \\
&\qquad + J_{k+1,\pi}\Big(f_k\big(x_k, \tilde{\mu}_k^1(x_k), u_k^2, w_k\big)\Big)\Big\} \\
&\leq E\Big\{g_k\big(x_k, \tilde{\mu}_k^1(x_k), \mu_k^2(x_k), w_k\big) \\
&\qquad + J_{k+1,\pi}\Big(f_k\big(x_k, \tilde{\mu}_k^1(x_k), \mu_k^2(x_k), w_k\big)\Big)\Big\}
\end{aligned}$$

an optimal control. Thus, for one-stage problems, the standard rollout algorithm will perform no worse than the multiagent rollout algorithm.

The example just given is best seen within the framework of the classical coordinate descent method for minimizing a function of m components. This method can get stuck at a nonoptimal point in the absence of appropriate conditions on the cost function, such as differentiability and/or convexity. However, within our context of multistage rollout and possibly stochastic disturbances, it appears that the consequences of such a phenomenon may not be serious. In fact, one can construct multi-stage examples where multiagent rollout performs better than the standard rollout.

$$
\begin{aligned}
&= \min_{u_k^1 \in U_k^1(x_k)} E\Big\{ g_k(x_k, u_k^1, \mu_k^2(x_k), w_k) \\
&\qquad\qquad + J_{k+1,\pi}\Big(f_k\big(x_k, u_k^1, \mu_k^2(x_k), w_k\big)\Big)\Big\} \\
&\leq E\Big\{ g_k\big(x_k, \mu_k^1(x_k), \mu_k^2(x_k), w_k\big) \\
&\qquad\qquad + J_{k+1,\pi}\Big(f_k\big(x_k, \mu_k^1(x_k), \mu_k^2(x_k), w_k\big)\Big)\Big\} \\
&= J_{k,\pi}(x_k),
\end{aligned}
$$

where:

(a) The first equality is the DP equation for the rollout policy $\tilde{\pi}$.

(b) The first inequality holds by the induction hypothesis.

(c) The second equality holds by the definition of the rollout algorithm as it pertains to agent 2.

(d) The third equality holds by the definition of the rollout algorithm as it pertains to agent 1.

(e) The fourth equality is the DP equation for the base policy π.

The induction proof of the cost improvement property (2.90) is thus complete for the case $m = 2$. The proof for an arbitrary number of agents m is entirely similar. **Q.E.D.**

Optimizing the Agent Order in Agent-by-Agent Rollout - Multiagent Parallelization

In the multiagent rollout algorithm described so far, the agents optimize the control components sequentially in a fixed order. It is possible to improve performance by trying to optimize at each stage k the order of the agents.

An efficient way to do this is to first optimize over all single agent Q-factors, by solving the m minimization problems that correspond to each of the agents $\ell = 1, \ldots, m$ being first in the multiagent rollout order. If ℓ_1 is the agent that produces the minimal Q-factor, we fix ℓ_1 to be the first agent in the multiagent rollout order. Then we optimize over all single agent Q-factors, by solving the $m - 1$ minimization problems that correspond to each of the agents $\ell \neq \ell_1$ being second in the multiagent rollout order. Let ℓ_2 be the agent that produces the minimal Q-factor, fix ℓ_2 to be the second agent in the multiagent rollout order, and continue in this manner. In the end, after

$$m + (m-1) + \cdots + 1 = \frac{m(m+1)}{2} \tag{2.91}$$

minimizations, we obtain an agent order $\ell_1, \ldots, \ell_m$ that produces a potentially much reduced Q-factor value, as well as the corresponding rollout control component selections.

The method just described likely produces substantially better performance, and eliminates the need for guessing a good agent order, but it increases the number of Q-factor calculations needed per stage roughly by a factor $(m+1)/2$. Still this is much better than the all-agents-at-once approach, which requires an exponential number of Q-factor calculations. Moreover, the Q-factor minimizations of the above process can be parallelized, so with m parallel processors, we can perform the number of $m(m+1)/2$ minimizations derived above in just m batches of parallel minimizations, which require about the same time as in the case where the agents are selected for Q-factor minimization in a fixed order. We finally note that our earlier cost improvement proof goes through again by induction, when the order of agent selection is variable at each stage k.

Multiagent Rollout Variants

The agent-by-agent rollout algorithm admits several variants. We describe briefly a few of these variants.

(a) We may use multiagent truncated rollout and terminal cost function approximation, as described earlier. Of course, in this case the cost improvement property need not hold.

(b) When the control constraint sets $U_k^\ell(x_k)$ are infinite, multiagent rollout still applies, based on the tradeoff between control and state space complexity, cf. Fig. 2.9.1. In particular, when the sets $U_k^\ell(x_k)$ are intervals of the real line, each agent's lookahead minimization problem can be performed with the aid of one-dimensional search methods.

(c) When the problem is deterministic there are additional possible variants of the multiagent rollout algorithm. In particular, for deterministic problems, we may use a more general base policy, i.e., a heuristic that is not defined by an underlying policy; cf. Section 2.3.1. In this case, if the sequential improvement assumption for the modified problem of Fig. 2.9.1 is not satisfied, then the cost improvement property may not hold. However, cost improvement may be restored by introducing fortification, as discussed in Section 2.3.2.

(d) The multiagent rollout algorithm can be modified simply to apply to infinite horizon problems. In this context, we may also consider policy iteration methods, which can be viewed as repeated rollout. These methods may involve agent-by-agent policy improvement, and value and policy approximations of intermediately generated policies (see the RL book [Ber19a], Section 5.7.3; also the paper [MLB25]). The paper [Ber21a] provides an analysis of methods involving exact policy evaluation and agent-by-agent policy improvement, and shows convergence to an agent-by-agent optimal policy, a form of locally optimal policy.

(e) The multiagent rollout algorithm can be modified simply to apply to deterministic continuous-time optimal control problems; cf. Section 2.6. The idea is again to simplify the minimization over $u(t)$ in the case where $u(t)$ consists of multiple components $u^1(t), \ldots, u^m(t)$.

(f) We can implement within the agent-by-agent rollout context the use of Q-factor differences. The motivation is similar: deal with the approximation errors that are inherent in the estimated cost of the base policy, $\tilde{J}_{k+1,\pi}\big(f_k(x_k, u_k)\big)$, and may overwhelm the current stage cost term $g_k(x_k, u_k)$. This may seriously degrade the quality of the rollout policy; see also the discussion of advantage updating and differential training in Chapter 3.

Constrained Multiagent Rollout

Let us consider a special structure of the control space, where the control u_k consists of m components, $u_k = (u_k^1, \ldots, u_k^m)$, each belonging to a corresponding set $U_k^\ell(x_k)$, $\ell = 1, \ldots, m$. Thus the control space at stage k is the Cartesian product

$$U_k(x_k) = U_k^1(x_k) \times \cdots \times U_k^m(x_k).$$

We refer to this as the *multiagent case*, motivated by the special case where each component u_k^ℓ is chosen by a separate agent ℓ at stage k.

Similar to the unconstrained case, we can introduce a modified but equivalent problem, involving one-at-a-time agent control selection. In particular, at the generic state x_k, we break down the control u_k into the sequence of the m controls $u_k^1, u_k^2, \ldots, u_k^m$, and between x_k and the next state $x_{k+1} = f_k(x_k, u_k)$, we introduce artificial intermediate "states"

$$(x_k, u_k^1), (x_k, u_k^1, u_k^2), \ldots, (x_k, u_k^1, \ldots, u_k^{m-1}),$$

and corresponding transitions. The choice of the last control component u_k^m at "state" $(x_k, u_k^1, \ldots, u_k^{m-1})$ marks the transition at cost $g_k(x_k, u_k)$ to the next state $x_{k+1} = f_k(x_k, u_k)$ according to the system equation. It is evident that this reformulated problem is equivalent to the original, since any control choice that is possible in one problem is also possible in the other problem, with the same cost.

By working with the reformulated problem, we can consider a rollout algorithm requires a sequence of m minimizations per stage, one over each of the control components $u_k^1, \ldots, u_k^m$, with the past controls already determined by the rollout algorithm, and the future controls determined by running the base heuristic. Assuming a maximum of n elements in the control component spaces $U_k^\ell(x_k)$, $\ell = 1, \ldots, m$, the computation required for the m single control component minimizations is of order $O(nm)$ per stage. By contrast the standard rollout minimization (2.47) involves the computation of as many as n^m terms $G\big(T_k(\tilde{y}_k, u_k)\big)$ per stage.

2.9.1 Asynchronous and Autonomous Multiagent Rollout

In this section we consider multiagent rollout algorithms that are distributed and asynchronous in the sense that the agents may compute their rollout controls in parallel rather than in sequence, aiming at computational speedup. An example of such an algorithm is obtained when at a given stage, agent ℓ computes the rollout control $\tilde{u}_k^\ell$ before knowing the rollout controls of some of the agents $1, \ldots, \ell - 1$, and uses the controls $\mu_k^1(x_k), \ldots, \mu_k^{\ell-1}(x_k)$ of the base policy in their place.

This algorithm often works well and it may be worth trying on a given problem. However, it does not possess the cost improvement property, and may not work well at all for some problems and specially selected initial states. In fact we can construct a simple example with a single state, two agents, and two controls per agent, where the second agent does not take into account the control applied by the first agent, and as a result the rollout policy performs worse than the base policy for some initial states.

Example 2.9.3 (Cost Deterioration in the Absence of Adequate Agent Coordination)

Consider a problem with two agents ($m = 2$) and a single state. Thus the state does not change and the costs of different stages are decoupled (the problem is essentially static). Each of the two agents has two controls: $u_k^1 \in \{0, 1\}$ and $u_k^2 \in \{0, 1\}$. The cost per stage g_k is equal to 0 if $u_k^1 \neq u_k^2$, is equal to 1 if $u_k^1 = u_k^2 = 0$, and is equal to 2 if $u_k^1 = u_k^2 = 1$. Suppose that the base policy applies $u_k^1 = u_k^2 = 0$. Then it can be seen that when executing rollout, the first agent applies $u_k^1 = 1$, and in the absence of knowledge of this choice, the second agent also applies $u_k^2 = 1$ (thinking that the first agent will use the base policy control $u_k^1 = 0$). Thus the cost of the rollout policy is 2 per stage, while the cost of the base policy is 1 per stage. By contrast the rollout algorithm that takes into account the first agent's control when selecting the second agent's control applies $u_k^1 = 1$ and $u_k^2 = 0$, thus resulting in a rollout policy with the optimal cost of 0 per stage.

The difficulty here is inadequate coordination between the two agents. In particular, each agent uses rollout to compute the local control, thinking that the other will use the base policy control. If instead the two agents coordinated their control choices, they would have applied an optimal policy.

The simplicity of the preceding example raises serious questions as to whether the cost improvement property (2.90) can be easily maintained by a distributed rollout algorithm where the agents do not know the controls applied by the preceding agents in the given order of local control selection, and use instead the controls of the base policy. One may speculate that if the agents are naturally "weakly coupled" in the sense that their choice of control has little impact on the desirability of various controls of other

agents, then a more flexible inter-agent communication pattern may be sufficient for cost improvement.†

An important question is to clarify the extent to which agent coordination is essential. In what follows in this section, we will discuss a distributed asynchronous multiagent rollout scheme, which is based on the use of a signaling policy that provides estimates of coordinating information once the current state is known.

Autonomous Multiagent Rollout - Signaling Policies

An interesting possibility for autonomous control selection by the agents is to use a distributed rollout algorithm, which is augmented by a precomputed signaling policy that embodies agent coordination.‡ The idea is to assume that the agents do not communicate their computed rollout control components to the subsequent agents in the given order of local control selection. Instead, *once the agents know the state, they use precomputed (or easily computed) approximations to the control components of the preceding agents*, and compute their own control components in parallel and asynchronously. We call this algorithm *autonomous multiagent rollout*. While this type of algorithm involves a form of redundant computation, it allows for additional speedup through parallelization.

The algorithm at the kth stage uses a base policy $\mu_k = \{\mu_k^1, \ldots, \mu_k^{m-1}\}$, but also uses *a second policy* $\widehat{\mu}_k = \{\widehat{\mu}_k^1, \ldots, \widehat{\mu}_k^{m-1}\}$, called the *signaling policy*, which is computed off-line, is known to all the agents for on-line use, and is designed to play an agent coordination role. Intuitively, $\widehat{\mu}_k^\ell(x_k)$ *provides an intelligent "guess" about what agent ℓ will do at state x_k*. This is used in turn by all other agents $i \neq \ell$ to compute asynchronously their own rollout control components on-line.

More precisely, the autonomous multiagent rollout algorithm uses the base and signaling policies to generate a rollout policy $\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$

† In particular, one may divide the agents in "coupled" groups, and require coordination of control selection only within each group, while the computation of different groups may proceed in parallel. Note that the "coupled" group formations may change over time, depending on the current state. For example, in applications where the agents' locations are distributed within some geographical area, it may make sense to form agent groups on the basis of geographic proximity, i.e., one may require that agents that are geographically near each other (and hence are more coupled) coordinate their control selections, while agents that are geographically far apart (and hence are less coupled) forego any coordination.

‡ The general idea of coordination by sharing information about the agents' policies arises also in other multiagent algorithmic contexts, including some that involve forms of policy gradient methods and Q-learning; see the surveys of the relevant research cited earlier. The survey by Matignon, Laurent, and Le Fort-Piat [MLL12] focuses on coordination problems from an RL point of view.

as follows. At stage k and state x_k, $\tilde{\mu}_k(x_k) = \big(\tilde{\mu}_k^1(x_k), \ldots, \tilde{\mu}_k^m(x_k)\big)$, is obtained according to

$$\tilde{\mu}_k^1(x_k) \in \arg\min_{u_k^1 \in U_k^1(x_k)} E\Big\{ g_k\big(x_k, u_k^1, \mu_k^2(x_k), \ldots, \mu_k^m(x_k), w_k\big) + J_{k+1,\pi}\Big(f_k\big(x_k, u_k^1, \mu_k^2(x_k), \ldots, \mu_k^m(x_k), w_k\big)\Big)\Big\},$$

$$\tilde{\mu}_k^2(x_k) \in \arg\min_{u_k^2 \in U_k^2(x_k)} E\Big\{ g_k\big(x_k, \widehat{\mu}_k^1(x_k), u_k^2, \ldots, \mu_k^m(x_k), w_k\big) + J_{k+1,\pi}\Big(f_k\big(x_k, \widehat{\mu}_k^1(x_k), u_k^2, \ldots, \mu_k^m(x_k), w_k\big)\Big)\Big\},$$

$$\ldots \qquad \ldots \qquad \ldots$$

$$\tilde{\mu}_k^m(x_k) \in \arg\min_{u_k^m \in U_k^m(x_k)} E\Big\{ g_k\big(x_k, \widehat{\mu}_k^1(x_k), \ldots, \widehat{\mu}_k^{m-1}(x_k), u_k^m, w_k\big) + J_{k+1,\pi}\Big(f_k\big(x_k, \widehat{\mu}_k^1(x_k), \ldots, \widehat{\mu}_k^{m-1}(x_k), u_k^m, w_k\big)\Big)\Big\}. \tag{2.92}$$

Note that the preceding computation of the controls $\tilde{\mu}_k^1(x_k), \ldots, \tilde{\mu}_k^m(x_k)$ can be done asynchronously and in parallel, and without direct agent coordination, since the signaling policy values $\widehat{\mu}_k^1(x_k), \ldots, \widehat{\mu}_k^{m-1}(x_k)$ are precomputed and are known to all the agents.

The simplest choice is to *use as signaling policy $\widehat{\mu}$ the base policy μ*. However, this choice does not guarantee policy improvement as evidenced by Example 2.9.3. In fact performance deterioration with this choice is not uncommon, and can be observed in more complicated examples, including the following.

Example 2.9.4 (Spiders and Flies - Use of the Base Policy for Signaling)

Consider the problem of Example 2.9.1, which involves two spiders and two flies on a line, and the base policy μ that moves a spider towards the closest surviving fly (and in case where a spider starts at the midpoint between the two flies, moves the spider to the right). Assume that we use as signaling policy $\widehat{\mu}$ the base policy μ. It can then be verified that if the spiders start from different positions, the rollout policy will be optimal (will move the spiders in opposite directions). If, however, the spiders start *from the same position*, a completely symmetric situation is created, whereby the rollout controls move both flies in the direction of the fly *furthest away* from the spiders' position (or to the left in the case where the spiders start at the midpoint between the two flies). Thus, the spiders end up oscillating around the middle of the interval between the flies and never catch the flies!

The preceding example is representative of a broad class of counterexamples that involve multiple identical agents. If the agents start at

the same initial state, with a base policy that has identical components, and use the base policy for signaling, the agents will select identical controls under the corresponding multiagent rollout policy, ending up with a potentially serious cost deterioration.

This example also highlights an effect of the sequential choice of the control components $u_k^1, \ldots, u_k^m$, based on the reformulated problem of Fig. 2.9.1: it tends to break symmetries and "groupthink" that guides the agents towards selecting the same controls under identical conditions. Generally, any sensible multiagent policy must be able to deal in some way with this "groupthink" issue. One simple possibility is for each agent ℓ to randomize somehow the control choices of other agents $j \neq \ell$ when choosing its own control, particularly in "tightly coupled" cases where the choice of agent ℓ is "strongly" affected by the choices of the agents $j \neq \ell$.

An alternative idea is to choose the signaling policy $\widehat{\mu}_k$ to approximate the sequential multiagent rollout policy (the one computed with each agent knowing the controls applied by the preceding agents), or some other policy that is known to embody coordination between the agents. In particular, we may obtain $\widehat{\mu}_k$ as the multiagent rollout policy for a related but simpler problem, such as a certainty equivalent version of the original problem, whereby the stochastic system is replaced by a deterministic one.

Another interesting possibility is to compute $\widehat{\mu}_k = (\widehat{\mu}_k^1, \ldots, \widehat{\mu}_k^m)$ by off-line training of a neural network (or m networks, one per agent) with training samples generated through the sequential multiagent rollout policy. We intuitively expect that if the neural network provides a signaling policy that approximates well the sequential multiagent rollout policy, we would obtain better performance than the base policy. This expectation was confirmed in a case study involving a large-scale multi-robot repair application (see [BKB23]).

The advantage of autonomous multiagent rollout with neural network or other approximations is that it may lead to approximate policy improvement, while at the same time allowing asynchronous agent operation without coordination through communication of their rollout control values (but still assuming knowledge of the exact state by all agents).

2.10 ROLLOUT FOR BAYESIAN OPTIMIZATION AND SEQUENTIAL ESTIMATION

In this section, we discuss a wide class of problems that have been studied intensively in statistics and related fields since the 1940s. Roughly speaking, in these problems we use observations and sampling for the purpose of inference, but the number and the type of observations are not fixed in advance. Instead, the outcomes of the observations are sequentially evaluated on-line with a view toward stopping or modifying the observation process. This involves sequential decision making, thus bringing to bear exact and

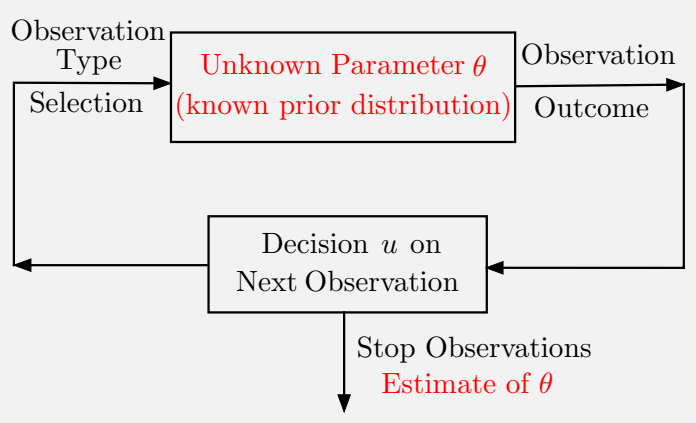

Figure 2.10.1 Illustration of sequential estimation of a parameter θ. At each time a decision is made to select one of several observation types relating to θ, each of different cost, or stop the observations and provide a final estimate of θ.

approximate DP. A central issue here is to estimate an m-dimensional random vector θ, using optimal sequential selection of observations, which are based on feedback from preceding observations; see Fig. 2.10.1. Here is a simple but historically important illustrative example, where θ represents a binary hypothesis.

Example 2.10.1 (Hypothesis Testing - Sequential Probability Ratio Test)

Consider a hypothesis testing problem whereby we can make observations, at a cost C each, relating to two hypotheses. Given a new observation, we can either accept one of the hypotheses or delay the decision for one more period, pay the cost C, and obtain a new observation. At issue is trading off the cost of observation with the higher probability of accepting the wrong hypothesis. As an example, in a quality control setting, the two hypotheses may be that a certain product meets or does not meet a certain level of quality, while the observations may consist of quantitative tests of the quality of the product.

Intuitively, one expects that once the conditional probability of one of the hypotheses, given the observations thus far, gets sufficiently close to 1, we should stop the observations. Indeed classical DP analyses bear this out; see e.g., the books by Chernoff [Che72], DeGroot [DeG70], Whittle [Whi82], and the references quoted therein. In particular, the simple version of the hypothesis testing problem just described admits a simple and elegant optimal solution, known as the *sequential probability ratio test*. On the other hand more complex versions of the problem, involving for example multiple hypotheses and/or multiple types of observations, are computationally intractable, thus necessitating the use of suboptimal approaches.

An important distinction in sequential estimation problems is whether the current choice of observation affects the cost and the availability of future observations. If this is so, the problem can often be viewed most fruitfully as a *combined estimation and control problem*, and is related to a type of *adaptive control* problem that we will discuss in the next section. As an example we will consider there sequential decoding, whereby we search for a hidden code word by using a sequence of queries, in the spirit of the Wordle puzzle and the family of Mastermind games [see, e.g., the Wikipedia page for "Mastermind (board game)"].

If the observation choices are "independent" and do not affect the cost or availability of future observations, the problem is substantially simplified. We will discuss problems of this type in the present section, starting with the cases of surrogate and Bayesian optimization.

Surrogate Optimization

Surrogate optimization refers to a collection of methods, which address suboptimally a broad range of minimization problems, beyond the realm of DP. The problem is to minimize approximately a function that is given as a "black box." By this we mean a function whose analytical expression is unknown, and whose values at any one point may be hard-to-compute, e.g., may requite costly simulation or experimentation. The idea is to replace such a cost function with a "surrogate" whose values are easier to compute.

Here we introduce a model of the cost function that is parametrized by a parameter θ; see Fig. 2.10.2. We observe sequentially the cost function at a few observation points, construct a model of the cost function (the surrogate) by estimating θ based on the results of the observations, and minimize the surrogate to obtain a suboptimal solution. The question is how to select observation points sequentially, using feedback from previous observations. This selection process often embodies an *exploration-exploitation tradeoff*: Observing at points likely to have near-optimal value vs observing at points in relatively unexplored areas of the search space.

Surrogate optimization at its core involves construction from data of functions of interest. Thus the ideas to be presented apply to other domains, e.g., the construction of probability density functions from data.

Bayesian Optimization

Bayesian optimization (BO) has been widely used for the approximate optimization of functions whose values at given points can only be obtained through time-consuming calculation, simulation, or experimentation. A classical application from geostatistical interpolation, pioneered by the statisticians Matheron and Krige, was to identify locations of high gold distribution in South Africa based on samples from a few boreholes (the name "kriging" is often used to refer to this type of application; see the review by Kleijnen [Kle09]). As another example, BO has been used to select the

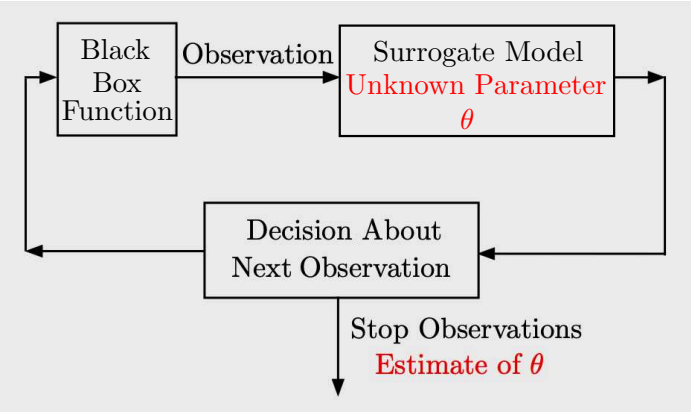

Figure 2.10.2 Illustration of the construction of a surrogate for a "black box" function f whose values are hard-to-compute. We replace f with a parametric model that involves a parameter θ to be estimated by using observations at some points. The points are selected sequentially, using the results of earlier observations. Eventually, the observation process is stopped (often when an observation/computation budget limit is reached), and the final estimate of θ is used to construct the surrogate to be minimized in place of f.

hyperparameters of machine-learning models, including the architectural parameters of the deep neural network of AlphaZero; see [SHS17].

In this section, we will focus on a relatively simple BO formulation that can be viewed as a special case of surrogate optimization. In particular, we will discuss the case where the surrogate function is parametrized by the collection of its values at the points where it is defined.† See the references cited later in this section. Formally, we want to minimize a real-valued function f, defined over a set of m points, which we denote by $1, \ldots, m$. These m points lie in some space, which we leave unspecified for the moment.‡ The values of the function are not readily available, but can be estimated with observations that may be imperfect. However, the observations are so costly that we can only hope to observe the function at a limited number of points. Once the function has been estimated with this type of observation process, we obtain a surrogate cost function, which may be minimized to obtain an approximately optimal solution.

† More complex forms of surrogates are obtained through linear combinations of some basis functions, with the parameter vector θ consisting of the weights of the basis functions.

‡ We restrict the domain of definition of f to be the finite set $\{1, \ldots, m\}$ in order to facilitate the implementation of the rollout algorithm to be discussed in what follows. However, in a more general and sometimes more convenient formulation, the domain of f can be an infinite set, such as a subset of a finite-dimensional Euclidean space.

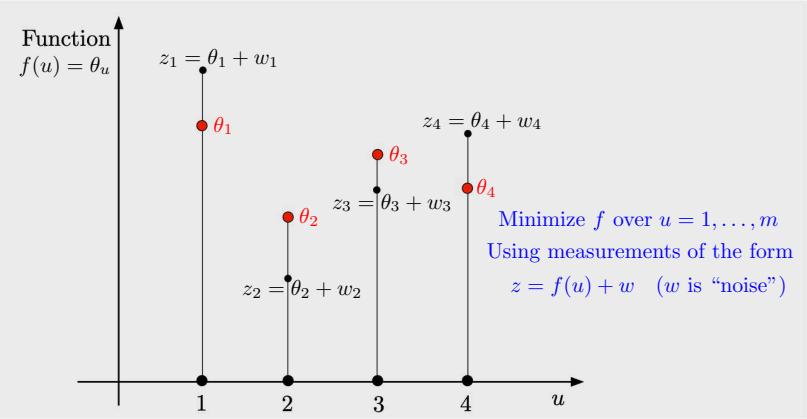

Figure 2.10.3 Illustration of a function f that we wish to estimate. The function is defined at the points $u = 1, 2, 3, 4$, and is represented by a vector $\theta = (\theta_1, \theta_2, \theta_3, \theta_4) \in \Re^4$, in the sense that $f(u) = \theta_u$ for all u. The prior distribution of θ is given, and is used to construct the posterior distribution of θ given noisy observations $z_u = \theta_u + w_u$ at some of the points u.

We denote the value of f at a point u by θ_u:

$$\theta_u = f(u), \qquad \text{for all } u = 1, \ldots, m.$$

Thus the m-dimensional vector $\theta = (\theta_1, \ldots, \theta_m)$ belongs to $\Re^m$ and represents the function f. We assume that we obtain sequentially noisy observations of values $f(u) = \theta_u$ at suitably selected points u. These values are used to estimate the vector θ (i.e., the function f), and to ultimately minimize (approximately) f over the m points $u = 1, \ldots, m$. The essence of the problem is to select points for observation based on an exploration-exploitation tradeoff (exploring the potential of relatively unexplored candidate solutions and improving the estimate of promising candidate solutions). The fundamental idea of the BO methodology is that the function value changes relatively slowly, so that observing the function value at some point provides information about the function values at neighboring points. Thus a limited number of strategically chosen observations can provide reasonable approximation to the true cost function over a large portion of the search space.

For a mathematical formulation of a BO framework, we assume that at each of N successive times $k = 1, \ldots, N$, we select a single point $u_k \in \{1, \ldots, m\}$, and observe the corresponding component θ_{u_k} of θ (i.e., the function value at u_k) with some noise w_{u_k}, i.e.,

$$z_{u_k} = \theta_{u_k} + w_{u_k}; \tag{2.93}$$

see Fig. 2.10.3. We view the observation points $u_1, \ldots, u_N$ as the optimization variables (or controls/actions in a DP/RL context), and consider policies for selecting u_k with knowledge of the preceding observations

$z_{u_1}, \ldots, z_{u_{k-1}}$ that have resulted from the selections $u_1, \ldots, u_{k-1}$. We assume that the noise random variables w_u, $u \in \{1, \ldots, m\}$ are independent and that their distributions are given. Moreover, we assume that θ has a given a priori distribution on the space of m-dimensional vectors $\Re^m$, which we denote by b_0. The posterior distribution of θ, given any subset of observations

$$\{z_{u_1}, \ldots, z_{u_k}\},$$

is denoted by b_k.

An important special case arises when b_0 and the distributions of w_u, $u \in \{1, \ldots, m\}$, are Gaussian. In this case b_0 is a multidimensional Gaussian distribution, defined by its mean (based on prior knowledge, or an equal value for all $u = 1, \ldots, m$ in the absence of such knowledge) and its covariance matrix [implying greater correlation for pairs (u, u') that are "close" to each other in some problem-specific sense, e.g., exponentially decreasing with the Euclidean distance between u and u']. A key consequence of this assumption is that the posterior distribution b_k is multidimensional Gaussian, and can be calculated in closed form by using well-known formulas.

More generally, b_k evolves according to an equation of the form

$$b_{k+1} = B_k(b_k, u_{k+1}, z_{u_{k+1}}), \qquad k = 0, \ldots, N-1. \tag{2.94}$$

Thus given the set of observations up to time k, and the next choice u_{k+1}, resulting in an observation value $z_{u_{k+1}}$, the function B_k gives the formula for updating b_k to b_{k+1}, and may be viewed as a recursive estimator of b_k. In the Gaussian case, the function B_k can be written in closed form, using standard formulas for Gaussian random vector estimation. In other cases where no closed form expression is possible, B_k can be implemented through simulation that computes (approximately) the new posterior b_{k+1} using samples generated from the current posterior b_k.

At the end of the sequential estimation process, after the complete observation set

$$\{z_{u_1}, \ldots, z_{u_N}\}$$

has been obtained, we have the posterior distribution b_N of θ, which we can use to compute a surrogate of f. As an example we may use as surrogate the posterior mean

$$\hat{\theta} = (\hat{\theta}_1, \ldots, \hat{\theta}_m),$$

and declare as minimizer of f over u the point u^* with minimum posterior mean:

$$u^* \in \arg\min\{\hat{\theta}_u \mid u = 1, \ldots, m\};$$

see Fig. 2.10.4.

There is a large literature relating to the surrogate and Bayesian optimization methodology and its applications, particularly for the Gaussian

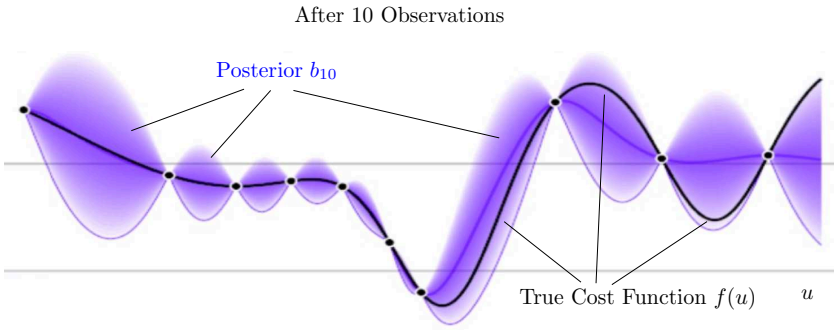

Figure 2.10.4 Illustration of the true cost function f, defined over an interval of the real line, and the posterior distribution b_{10} after noise-free measurements at 10 points. The shaded area represents the interval of the mean plus/minus the standard deviation of the posterior b_{10} at the points u. The mean of the finally obtained posterior, as a function of u, may be viewed as a surrogate cost function that can be minimized in place of f. Note that since the observations are assumed noise-free, the mean of the posterior is exact at the observation points.

case. We refer to the books by Rasmussen and Williams [RaW06], Powell and Ryzhov [PoR12], the highly cited papers by Saks et al. [SWM89], Jones, Schonlau, and Welch [JSW98], and Queipo et al. [QHS05], the reviews by Sasena [Sas02], Powell and Frazier [PoF08], Forrester and Keane [FoK09], Kleijnen [Kle09], Brochu, Cora, and De Freitas [BCD10], Ryzhov, Powell, and Frazier [RPF12], Ghavamzadeh, Mannor, Pineau, and Tamar [GMP15], Shahriari et al. [SSW15], and Frazier [Fra18], and the references quoted there. Our purpose here is to focus on the aspects of the subject that are most closely connected to exact and approximate DP.

A Dynamic Programming Formulation

The sequential estimation problem just described, viewed as a DP problem, involves a state at time k, which is the posterior (or belief state) b_k, and a control/action at time k, which is the point index u_{k+1} selected for observation. The transition equation according to which the state evolves is

$$b_{k+1} = B_k(b_k, u_{k+1}, z_{u_{k+1}}), \qquad k = 0, \ldots, N-1;$$

cf. Eq. (2.94). To complete the DP formulation, we need to introduce a cost structure. To this end, we assume that observing θ_u, as per Eq. (2.93), incurs a cost $c(u)$, and that there is a terminal cost $G(b_N)$ that depends of the final posterior distribution; as an example, the function G may involve the mean and covariance corresponding to b_N.

The corresponding DP algorithm is given by

$$J_k^*(b_k) = \min_{u_{k+1}\in\{1,\ldots,m\}} \Big[c(u_{k+1}) + E_{z_{u_{k+1}}} \Big\{ J_{k+1}^*\big(B_k(b_k, u_{k+1}, z_{u_{k+1}})\big) \mid b_k, u_{k+1} \Big\} \Big], \tag{2.95}$$

and proceeds backwards from the terminal condition

$$J_N^*(b_N) = G(b_N). \tag{2.96}$$

The expected value in the right side of the DP equation (2.95) is taken with respect to the conditional distribution of $z_{u_{k+1}}$, given b_k and the choice u_{k+1}. The observation cost $c(u)$ may be 0 or a constant for all u, but it can also have a more complicated dependence on u. The terminal cost $G(b_N)$ may be a suitable measure of surrogate "fidelity" that depends on the posterior mean and covariance of θ corresponding to b_N.

Generally, executing the DP algorithm (2.95) is practically infeasible, because the space of posterior distributions is infinite-dimensional. In the Gaussian case where the a priori distribution b_0 is Gaussian and the noise variables w_u are Gaussian, the posterior b_k is m-dimensional Gaussian, so it is characterized by its mean and covariance, and can be specified by a finite set of numbers. Despite this simplification, the DP algorithm (2.95) is prohibitively time-consuming even under Gaussian assumptions, except for simple special cases. We consequently resort to approximation in value space, whereby the function J_{k+1}^* in the right side of Eq. (2.95) is replaced by an approximation $\tilde{J}_{k+1}$.

Approximation in Value Space

The most popular BO methodology makes use of a myopic/greedy policy μ_{k+1}, which at each time k and given b_k, selects a point $\hat{u}_{k+1} = \mu_{k+1}(b_k)$ for the next observation, using some calculation involving an *acquisition function*. This function, denoted $A_k(b_k, u_{k+1})$, quantifies some form of "expected benefit" for an observation at u_{k+1}, given the current posterior b_k.† The myopic policy selects the next point at which to observe, $\hat{u}_{k+1}$,

† A common type of acquisition function is the *upper confidence bound*, which has the form

$$A_k(b_k, u) = T_k(b_k, u) + \beta R_k(b_k, u),$$

where $T_k(b_k, u)$ is the negative of the mean of $f(u)$ under the posterior distribution b_k, $R_k(b_k, u)$ is the standard deviation of $f(u)$ under the posterior distribution b_k, and β is a tunable positive scalar parameter. Thus $T_k(b_k, u)$ can be

by maximizing the acquisition function:

$$\hat{u}_{k+1} \in \arg \max_{u_{k+1} \in \{1,\ldots,m\}} A_k(b_k, u_{k+1}). \tag{2.97}$$

Several ways to define suitable acquisition functions have been proposed, and an important issue is to be able to calculate economically its values $A_k(b_k, u_{k+1})$ for the purposes of the maximization in Eq. (2.97). Another important issue of course is to be able to calculate the posterior b_k economically.

Approximation in value space is an alternative approach, which is based on the DP formulation of the preceding section. In particular, in this approach we approximate the DP algorithm (2.95) by replacing J^*_{k+1} with an approximation $\tilde{J}_{k+1}$ in the minimization of the right side. Thus we select the next observation at point $\tilde{u}_{k+1}$ according to

$$\tilde{u}_{k+1} \in \arg \min_{u_{k+1} \in \{1,\ldots,m\}} Q_k(b_k, u_{k+1}), \qquad k = 0, \ldots, N-1, \tag{2.98}$$

where $Q_k(b_k, u_{k+1})$ is the Q-factor corresponding to the pair (b_k, u_{k+1}), given by

$$Q_k(b_k, u_{k+1}) = c(u_{k+1}) + E_{z_{u_{k+1}}} \Big\{ \tilde{J}_{k+1}\big(B_k(b_k, u_{k+1}, z_{u_{k+1}})\big) \mid b_k, u_{k+1} \Big\}. \tag{2.99}$$

The expected value in the preceding equation is taken with respect to the conditional probability distribution of $z_{u_{k+1}}$ given (b_k, u_{k+1}), which can be computed using b_k and the given distribution of the noise $w_{u_{k+1}}$. Thus if b_k and $\tilde{J}_{k+1}$ are available, we may use Monte Carlo simulation to determine the Q-factors $Q_k(b_k, u_{k+1})$ for all $u_{k+1} \in \{1, \ldots, m\}$, and select as next point for observation the one that corresponds to the minimal Q-factor [cf. Eq. (2.98)].

Rollout Algorithms for Bayesian Optimization

A special case of approximation in value space is the rollout algorithm, whereby the function J^*_{k+1} in the right side of the DP Eq. (2.95) is replaced by the cost function of some base policy $\mu_{k+1}(b_k)$, $k = 0, \ldots, N-1$. Thus,

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viewed as an *exploitation index* (encoding our desire to search within parts of the space where f takes low value), while $R_k(b_k, u)$ can be viewed as an *exploration index* (encoding our desire to search within parts of the space that are relatively unexplored). There are several other popular acquisition functions, which directly or indirectly embody a tradeoff between exploitation and exploration. A popular example is the *expected improvement* acquisition function, which is equal to the expected value of the reduction of $f(u)$ relative to the minimal value of f obtained up to time k (under the posterior distribution b_k).

given a base policy the rollout algorithm uses the cost function of this policy as the function $\tilde{J}_{k+1}$ in the approximation in value space scheme (2.98)-(2.99). The values of $\tilde{J}_{k+1}$ needed for the Q-factor calculations in Eq. (2.99) can be computed or approximated by simulation. Greedy/myopic policies based on an acquisition function [cf. Eq. (2.97)] have been suggested as base policies in various rollout proposals.†

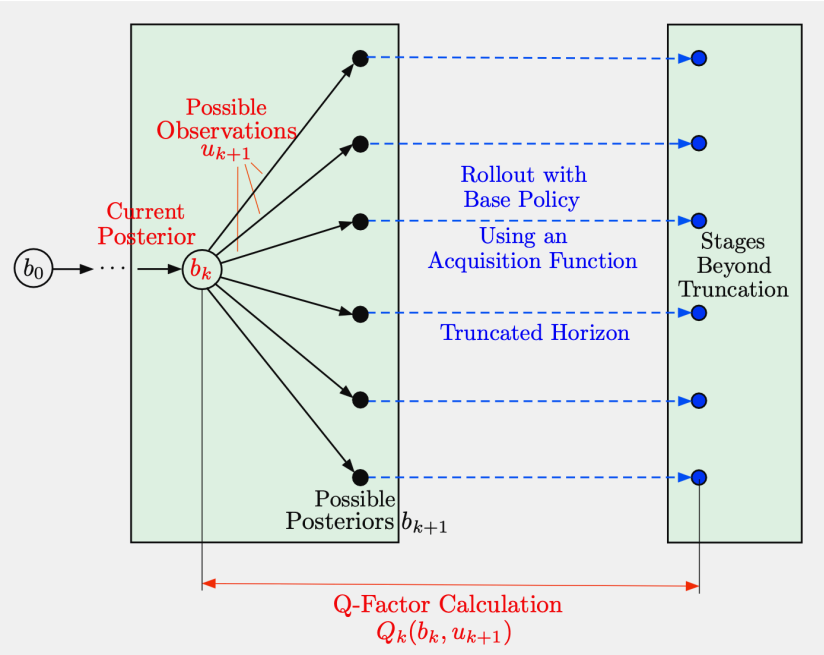

Figure 2.10.5 Illustration of rollout at the current posterior b_k. For each $u_{k+1} \in \{1, \ldots, m\}$, we compute the Q-factor $Q_k(b_k, u_{k+1})$ by using Monte Carlo simulation with samples from $w_{u_{k+1}}$ and a base heuristic that uses an acquisition function starting from each possible posterior b_{k+1}. The rollout may extend to the end of the horizon N, or it may be truncated after a few steps.

In particular, given b_k, the rollout algorithm computes for each $u_{k+1} \in$

† The rollout algorithm for BO was first proposed under Gaussian assumptions by Lam, Wilcox, and Wolpert [LWW16]. It was further discussed by Jiang et al. [JJB20], [JCG20], Lee et al. [LEC20], Lee [Lee20], Yue and Kontar [YuK20], Lee et al. [LEP21], Paulson, Sorouifar, and Chakrabarty [PSC22], where it is also referred to as "nonmyopic BO" or "nonmyopic sequential experimental design." For related work, see Gerlach, Hoffmann, and Charlish [GHC21]. These papers also discuss various approximations to the rollout approach, and generally report encouraging computational results. Section 3.5 of the author's book [Ber20a] focuses on rollout algorithms for surrogate and Bayesian optimization.

$\{1, \ldots, m\}$ a Q-factor value $Q_k(b_k, u_{k+1})$ by simulating the base policy for multiple time steps starting from all possible posteriors b_{k+1} that can be generated from (b_k, u_{k+1}), and by accumulating the corresponding cost [including a terminal cost such as $G(b_N)$]; see Fig. 2.10.5. It then selects the next point $\tilde{u}_{k+1}$ for observation by using the Q-factor minimization of Eq. (2.98).

Note that the equation

$$b_{k+1} = B_k(b_k, u_{k+1}, z_{u_{k+1}}), \qquad k = 0, \ldots, N-1,$$

which governs the evolution of the posterior distribution (or belief state), is stochastic because $z_{u_{k+1}}$ involves the stochastic noise $w_{u_{k+1}}$. Thus some Monte Carlo simulation is unavoidable in the calculation of the Q-factors $Q_k(b_k, u_{k+1})$. On the other hand, one may greatly reduce the Monte Carlo computational burden by employing a certainty equivalence approximation, which at stage k, treats only the noise $w_{u_{k+1}}$ as stochastic, and replaces the noise variables $w_{u_{k+2}}, w_{u_{k+3}}, \ldots$, after the first stage of the calculation, by deterministic quantities such as their means $\hat{w}_{u_{k+2}}, \hat{w}_{u_{k+3}}, \ldots$.

The simulation of the Q-factor values may also involve other approximations, some of which have been suggested in various proposals for rollout-based BO. For example, if the number of possible observations m is very large, we may compute and compare the Q-factors of only a subset. In particular, at a given time k, we may rank the observations by using an acquisition function, select a subset U_{k+1} of most promising observations, compute their Q-factors $Q_k(b_k, u_{k+1})$, $u_{k+1} \in U_{k+1}$, and select the observation whose Q-factor is minimal; this idea has been used in the case of the Wordle puzzle in the papers by Bhambri, Bhattacharjee, and Bertsekas [BBB22], [BBB23], which will be discussed in the next section.

Multiagent Rollout for Bayesian Optimization

In some BO applications there arises the possibility of simultaneously performing multiple observations before receiving feedback about the corresponding observation outcomes. This occurs, among others, in two important contexts:

(a) In parallel computation settings, where multiple processors are used to perform simultaneously expensive evaluations of the function f at multiple points u. These evaluations may involve some form of truncated simulation, so they yield evaluations of the form $z_u = \theta_u + w_u$, where w_u is the simulation noise.

(b) In distributed sensor systems, where a number of sensors provide in parallel relevant information about the random vector θ that we want to estimate; see e.g., the recent paper by Li, Krakow, and Gopalswamy [LKG21], which describes related multisensor estimation problems, based on the multiagent rollout methodology of Section 2.9.

Of course in such cases we may treat the entire set of simultaneous observations as a single observation within an enlarged Cartesian product space of observations, but there is a fundamental difficulty: the size of the observation space (and hence the number of Q-factors to be calculated by rollout at each time step) grows exponentially with the number of simultaneous observations. This in turn greatly increases the computational requirements of the rollout algorithm.

To address this difficulty, we may employ the methodology of multiagent rollout whereby the policy improvement is done one-agent-at-a-time in a given order, with (possibly partial) knowledge of the choices of the preceding agents in the order. As a result, the amount of computation for each policy improvement grows linearly with the number of agents, as opposed to exponentially for the standard all-agents-at-once method. At the same time the theoretical cost improvement property of the rollout algorithm can be shown to be preserved, while the empirical evidence suggests that great computational savings are achieved with hardly any performance degradation.

Generalization to Sequential Estimation of Random Vectors

Aside from BO, there are several other types of simple sequential estimation problems, which involve "independent sampling," i.e., problems where the choice of an observation type does not affect the quality, cost, or availability of observations of other types. A common class of problems that contains BO as a special case and admits a similar treatment, is to sequentially estimate an m-dimensional random vector $\theta = (\theta_1, \ldots, \theta_m)$ by using N linear observations of θ of the form

$$z_u = a_u' \theta + w_u, \qquad u \in \{1, \ldots, n\},$$

where n is some integer. Here w_u are independent random variables with given probability distributions, the m-dimensional vectors a_u are known, and $a_u'\theta$ denotes the inner product of a_u and θ. Similar to the case of BO, the problem simplifies if the given a priori distribution of θ is Gaussian, and the random variables w_u are independent and Gaussian. Then, the posterior distribution of θ, given any subset of observations, is Gaussian (thanks to the linearity of the observations), and can be calculated in closed form.

Observations are generated sequentially at times $1, \ldots, N$, one at a time and with knowledge of the outcomes of the preceding observations, by choosing an index $u_k \in \{1, \ldots, n\}$ at time k, at a cost $c(u_k)$. Thus u_k are the optimization variables, and affect both the quality of estimation of θ and the observation cost. The objective, roughly speaking, is to select N observations to estimate θ in a way that minimizes an appropriate cost function; for example, one that penalizes some form of estimation

error plus the cost of the observations. We can similarly formulate the corresponding optimization problem in terms of N-stage DP, and develop rollout algorithms for its approximate solution.

2.11 ADAPTIVE CONTROL BY ROLLOUT WITH A POMDP FORMULATION

In this section, we discuss various approaches for the approximate solution of Partially Observed Markovian Decision Problems (POMDP) with a special structure, which is well-suited for adaptive control, as well as other contexts that involve search for a hidden object.† It is well known that POMDP are among the most challenging DP problems, and nearly always require the use of approximations for (suboptimal) solution.

The application and implementation of rollout and approximate PI methods to general finite-state POMDP is described in the author's RL book [Ber19a] (Section 5.7.3). Here we will focus attention on a special class of POMDP where the state consists of two components:

(a) A perfectly observed component x_k that evolves over time according to a discrete-time equation.

(b) An unobserved component θ that stays constant and is estimated through the perfect observations of the component x_k.

We view θ as a parameter in the system equation that governs the evolution of x_k, hence the connection with adaptive control. Thus we have

$$x_{k+1} = f_k(x_k, \theta, u_k, w_k), \tag{2.100}$$

where u_k is the control at time k, selected from a set $U_k(x_k)$, and w_k is a random disturbance with given probability distribution that depends on (x_k, θ, u_k). We will assume that θ can take one of m known values $\theta^1, \ldots, \theta^m$:

$$\theta \in \{\theta^1, \ldots, \theta^m\},$$

see Fig. 2.11.1.

The a priori probability distribution of θ is given and is updated based on the observed values of the state components x_k and the applied controls u_k. In particular, we assume that the information vector

$$I_k = \{x_0, \ldots, x_k, u_0, \ldots, u_{k-1}\}$$

† In Section 1.6.8, we discussed the indirect adaptive control approach, which enforces a separation of the controller into a system identification algorithm and a policy reoptimization algorithm. The POMDP approach of this section (also summarized in Section 1.6.6), does not assume such an a priori separation, and is thus founded on a more principled algorithmic framework.

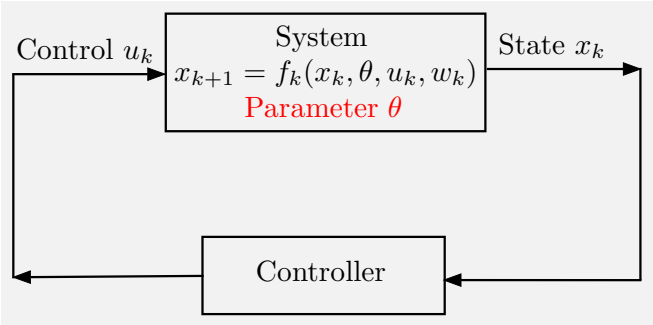

Figure 2.11.1 Illustration of an adaptive control scheme involving perfect state observation of a system with an unknown parameter θ. At each time a decision is made to select a control and (possibly) one of several observation types, each of different cost.

is available at time k, and is used to compute the conditional probabilities

$$b_{k,i} = P\{\theta = \theta^i \mid I_k\}, \qquad i = 1, \ldots, m.$$

These probabilities form a vector

$$b_k = (b_{k,1}, \ldots, b_{k,m}),$$

which together with the perfectly observed state x_k, forms the pair (x_k, b_k) that is commonly called the *belief state* of the POMDP at time k.

Note that according to the classical methodology of POMDP (see e.g., [Ber17a], Chapter 4), the belief component b_{k+1} is determined by the belief state (x_k, b_k), the control u_k, and the observation obtained at time $k+1$, i.e., x_{k+1}. Thus b_k can be updated according to an equation of the form

$$b_{k+1} = B_k(x_k, b_k, u_k, x_{k+1}),$$

where B_k is an appropriate function, which can be viewed as a recursive estimator of θ. There are several approaches to implement this estimator (perhaps with some approximation error), including the use of Bayes' rule and the simulation-based method of particle filtering.

The preceding mathematical model forms the basis for a classical adaptive control formulation, where each θ^i represents a set of system parameters, and the computation of the belief probabilities $b_{k,i}$ can be viewed as the outcome of a system identification algorithm. In this context, the problem becomes one of *dual control*, a classical type of combined identification and control problem, whose optimal solution is notoriously difficult.

Another interesting context arises in search problems, where θ specifies the locations of one or more objects of interest within a given space. Some puzzles, including the popular Wordle game, fall within this category, as we will discuss briefly later in this section.

The Exact DP Algorithm - Approximation in Value Space

We will now describe an exact DP algorithm that operates in the space of information vectors I_k. To describe this algorithm, let us denote by $J_k(I_k)$ the optimal cost starting at information vector I_k at time k. We can view I_k as a state of the POMDP, which evolves over time according to the equation

$$I_{k+1} = (I_k, x_{k+1}, u_k) = \big(I_k, f_k(x_k, \theta, u_k, w_k), u_k\big).$$

Viewing this as a system equation, whose right-hand side involves the state I_k, the control u_k, and the disturbance w_k, the DP algorithm takes the form

$$\begin{aligned} J_k^*(I_k) &= \min_{u_k \in U_k(x_k)} E_{\theta, w_k}\Big\{ g_k(x_k, \theta, u_k, w_k) + J_{k+1}^*(I_{k+1}) \mid I_k, u_k \Big\} \\ &= \min_{u_k \in U_k(x_k)} E_{\theta, w_k}\Big\{ g_k(x_k, \theta, u_k, w_k) + \\ &\qquad\qquad J_{k+1}^*\big(I_k, f_k(x_k, \theta, u_k, w_k), u_k\big) \mid I_k, u_k \Big\}, \end{aligned} \tag{2.101}$$

for $k = 0, \ldots, N-1$, with $J_N(I_N) = g_N(x_N)$; see e.g., the DP textbook [Ber17a], Section 4.1.

The algorithm (2.101) is typically very hard to implement, in part because of the dependence of J_{k+1}^* on the entire information vector I_{k+1}, which expands in size according to

$$I_{k+1} = (I_k, x_{k+1}, u_k).$$

To address this difficulty, we may use approximation in value space, based on replacing $J_{k+1}^*(I_{k+1})$ in the DP algorithm (2.101) with some function $\tilde{J}_{k+1}(I_{k+1})$ such that the expected value

$$E_{\theta, w_k}\Big\{ \tilde{J}_{k+1}(I_{k+1}) \mid I_k, u_k \Big\} \tag{2.102}$$

can be obtained with a tractable computation for any (I_k, u_k). A useful possibility arises when the cost function approximations

$$E_{w_k}\Big\{ \tilde{J}_{k+1}(I_{k+1}) \mid I_k, \theta^i, u_k \Big\}$$

can be obtained for each *fixed value of* θ^i with a tractable computation. In this case, we may compute the cost function approximation (2.102) by using the formula

$$E_{\theta, w_k}\Big\{ \tilde{J}_{k+1}(I_{k+1}) \mid I_k, u_k \Big\} = \sum_{i=1}^{m} b_{k,i} E_{w_k}\big\{ \tilde{J}_{k+1}(I_{k+1}) \mid I_k, \theta, u_k \big\}$$

which follows from the law of iterated expectations,

$$E_{\theta,w_k}\{\cdot \mid I_k, u_k\} = E_\theta\big\{E_{w_k}\{\cdot \mid I_k, \theta, u_k\} \mid I_k, u_k\big\}.$$

We will now discuss some choices of functions $\tilde{J}_{k+1}$ with a structure that facilitates the implementation of the corresponding approximation in value space scheme. One possibility is to use the optimal cost functions corresponding to the m parameters θ^i,

$$\hat{J}^i_{k+1}(x_{k+1}), \qquad i = 1, \ldots, m. \tag{2.103}$$

In particular, $\hat{J}^i_{k+1}(x_{k+1})$ is the optimal cost that would be obtained starting from state x_{k+1} under the assumption that $\theta = \theta^i$; this corresponds to a perfect state information problem. Then an approximation in value space scheme with one-step lookahead minimization is given by

$$\tilde{u}_k \in \arg\min_{u_k \in U_k(x_k)} \sum_{i=1}^m b_{k,i} E_{w_k}\Big\{g_k(x_k, \theta^i, u_k, w_k) + \hat{J}^i_{k+1}\big(f_k(x_k, \theta^i, u_k, w_k)\big) \mid x_k, \theta^i, u_k\Big\}. \tag{2.104}$$

In particular, instead of the optimal control, which minimizes the optimal Q-factor of (I_k, u_k) appearing in the right side of Eq. (2.101), we apply control $\tilde{u}_k$ that minimizes the expected value over θ of the optimal Q-factors that correspond to *fixed values of* θ.

In the case where the horizon is infinite, it is reasonable to expect that an improving estimate of the parameter θ can be obtained over time, and that with a suitable estimation scheme, it converges asymptotically to the correct value of θ, call it θ^*, i.e.,

$$\lim_{k\to\infty} b_{k,i} = \begin{cases} 1 & \text{if } \theta^i = \theta^*, \\ 0 & \text{if } \theta^i \neq \theta^*. \end{cases}$$

Then it can be seen that the generated one-step lookahead controls $\tilde{u}_k$ are asymptotically obtained from the Bellman equation that corresponds to the correct parameter θ^*, and are typically optimal in some asymptotic sense. Schemes of this type have been extensively discussed in the adaptive control literature since the 70s; see the end-of-chapter references and discussion.

Generally, the optimal costs $\hat{J}^i_{k+1}(x_{k+1})$ of Eq. (2.103), which correspond to the different parameter values θ^i, may be hard to compute, despite the fact that they correspond to perfect state information problems.† An alternative possibility is to use off-line trained approximations to $\hat{J}^i_{k+1}(x_{k+1})$ involving neural networks or other approximation architectures. Still another possibility, described next, is to use a rollout approach.

† In favorable special cases, such as linear quadratic problems, the optimal costs $\hat{J}^i_{k+1}(x_{k+1})$ may be easily calculated in closed form. Still, however, even in such cases the calculation of the belief probabilities $b_{k,i}$ may not be simple, and may require the use of a system identification algorithm.

Rollout and Cost Improvement

A simpler possibility for approximation in value space is to use the cost of a given policy π^i in place of the optimal cost $\hat{J}^i_{k+1}(x_{k+1})$ of Eq. (2.103) that corresponds to θ^i. In this case the one-step lookahead scheme (2.104) takes the form

$$\tilde{u}_k \in \arg\min_{u_k \in U_k(x_k)} \sum_{i=1}^{m} b_{k,i} E_{w_k} \Big\{ g_k(x_k, \theta^i, u_k, w_k) + \hat{J}^i_{k+1,\pi^i}\big(f_k(x_k, \theta^i, u_k, w_k)\big) \mid x_k, \theta^i, u_k \Big\}, \tag{2.105}$$

with $\pi^i = \{\mu^i_0, \ldots, \mu^i_{N-1}\}$, $i = 1, \ldots, m$, being known policies, *with components μ^i_k that depend only on x_k*. Here, the term

$$\hat{J}^i_{k+1,\pi^i}\big(f_k(x_k, \theta^i, u_k, w_k)\big)$$

in Eq. (2.105) is the cost of the base policy π^i, calculated starting from the next state

$$x_{k+1} = f_k(x_k, \theta^i, u_k, w_k),$$

under the assumption that θ will stay fixed at the value $\theta = \theta^i$ until the end of the horizon. Note that the cost function of π^i, conditioned on $\theta = \theta^i$, x_k, and u_k, which is needed in Eq. (2.105), can be calculated by Monte Carlo simulation. This is made possible by the fact that the components μ^i_k of π^i depend only on x_k [rather than I_k or the belief state (x_k, b_k)].

The preceding scheme has the character of a rollout algorithm, but strictly speaking, it does not qualify as a rollout algorithm because the policy components μ^i_k involve a dependence on i in addition to the dependence on x_k. On the other hand if we restrict all the policies π^i to be the same for all i, the corresponding functions μ_k depend only on x_k and not on i, thus defining a legitimate base policy. In this case the rollout scheme (2.105) amounts to replacing

$$E_{w_k}\big\{ J^*_{k+1}(I_{k+1}) \mid I_k, u_k \big\}$$

in the DP algorithm (2.101) with

$$E_{w_k}\big\{ J_{k+1,\pi}(I_{k+1}) \mid I_k, u_k \big\}.$$

Similar to Section 2.7, a cost improvement property can then be shown.

Within our rollout context, a policy π such that $\pi^i = \pi$ for all i should be a *robust* policy, in the sense that it should work adequately well for all parameter values θ^i. The method to obtain such a policy is likely problem-dependent. On the other hand robust policies have a long history in the context of adaptive control, and have been discussed widely (see e.g., the book by Jiang and Jiang [JiJ17], and the references quoted therein).

The Case of a Deterministic System

Let us now consider the case where the system (2.100) is deterministic of the form

$$x_{k+1} = f_k(x_k, \theta, u_k). \tag{2.106}$$

Then, while the problem still has a stochastic character due to the uncertainty about the value of θ, the DP algorithm (2.101) and its approximation in value space counterparts are greatly simplified because there is no expectation over w_k to contend with. Indeed, given a state x_k, a parameter θ^i, and a control u_k, the on-line computation of the control of the rollout-like algorithm (2.105), takes the form

$$\tilde{u}_k \in \arg \min_{u_k \in U_k(x_k)} \sum_{i=1}^{m} b_{k,i} \Big(g_k(x_k, \theta^i, u_k) + \hat{J}^i_{k+1,\pi^i}\big(f_k(x_k, \theta^i, u_k)\big) \Big). \tag{2.107}$$

The computation of $\hat{J}^i_{k+1,\pi^i}\big(f_k(x_k, \theta^i, u_k)\big)$ involves a deterministic propagation from the state x_{k+1} of Eq. (2.106) up to the end of the horizon, using the base policy π^i, while assuming that θ is fixed at the value θ^i.

In particular, the term

$$Q_k(x_k, u_k, \theta^i) = g_k(x_k, \theta^i, u_k) + \hat{J}^i_{k+1,\pi^i}\big(f_k(x_k, \theta^i, u_k)\big) \tag{2.108}$$

appearing on the right side of Eq. (2.107) is viewed as a Q-factor that must be computed for every pair (u_k, θ^i), $u_k \in U_k(x_k)$, $i = 1, \ldots, m$, using the base policy π^i. The expected value of this Q-factor,

$$\hat{Q}_k(x_k, u_k) = \sum_{i=1}^{m} b_{k,i} Q_k(x_k, u_k, \theta^i),$$

must then be calculated for every $u_k \in U_k(x_k)$, and the computation of the rollout control $\tilde{u}_k$ is obtained from the minimization

$$\tilde{u}_k \in \arg \min_{u_k \in U_k(x_k)} \hat{Q}_k(x_k, u_k); \tag{2.109}$$

cf. Eq. (2.107). This computation is illustrated in Fig. 2.11.2.

The case of a deterministic system is particularly interesting because we can typically expect that the true parameter θ^* is identified in a finite number of stages, since at each stage k, we are receiving a noiseless measurement relating to θ, namely the state x_k. Once this happens, the problem becomes one of perfect state information.

An illustration similar to the one of Fig. 2.11.2 applies to the rollout scheme (2.105) for the case of a stochastic system. In this case, a Q-factor

$$Q_k(x_k, u_k, \theta^i, w_k) = g_k(x_k, \theta^i, u_k, w_k) + \hat{J}^i_{k+1,\pi^i}\big(f_k(x_k, \theta^i, u_k, w_k)\big)$$

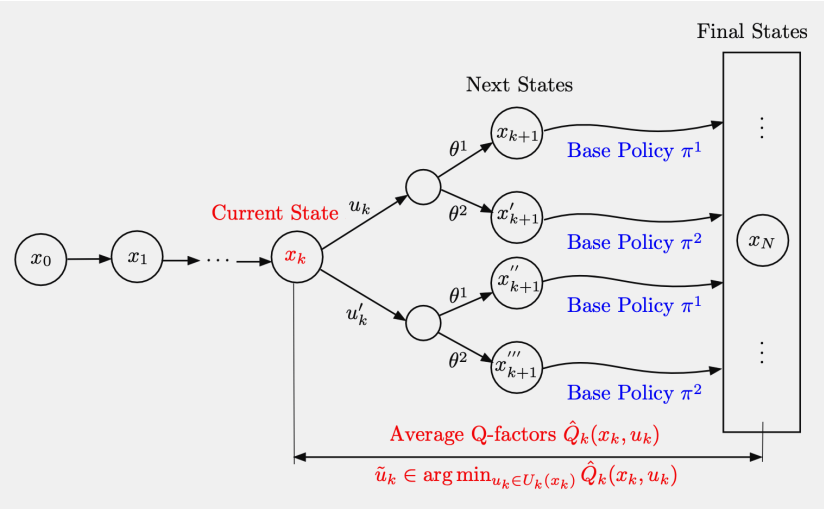

Figure 2.11.2 Schematic illustration of adaptive control by rollout for deterministic systems; cf. Eqs. (2.108) and (2.109). The Q-factors $Q_k(x_k, u_k, \theta^i)$ are averaged over θ^i, using the current belief distribution b_k, and the control applied is the one that minimizes over $u_k \in U_k(x_k)$ the averaged Q-factor

$$\hat{Q}_k(x_k, u_k) = \sum_{i=1}^{m} b_{k,i} Q_k(x_k, u_k, \theta^i).$$

must be calculated for every triplet (u_k, θ^i, w_k), using the base policy π^i. The rollout control $\tilde{u}_k$ is obtained by minimizing the expected value of this Q-factor [averaged using the distribution of (θ, w_k)]; cf. Eq. (2.105).

An interesting and intuitive example that demonstrates the deterministic system case is the popular Wordle puzzle.

Example 2.11.1 (The Wordle Puzzle)

In the classical form of this puzzle, we try to guess a mystery word θ^* out of a known finite collection of 5-letter words. This is done with sequential guesses each of which provides additional information on the correct word θ^*, by using certain given rules to shrink the current mystery list (the smallest list that contains θ^*, based on the currently available information). The objective is to minimize the number of guesses to find θ^* (using more than 6 guesses is considered to be a loss). This type of puzzle descends from the classical family of Mastermind puzzles that centers around decoding a secret sequence of objects (e.g., letters or colors) using partial observations.

The rules for shrinking the mystery list relate to the common letters between the word guesses and the mystery word θ^*, and they will not be described here (there is a large literature regarding the Wordle puzzle). More-

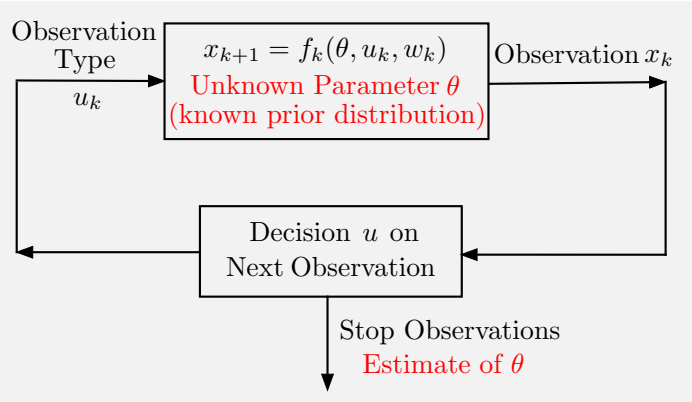

Figure 2.11.3 A view of sequential estimation as an adaptive control problem. The system function f_k does not depend on the current state x_k, so the system provides a decision-dependent noisy observation of θ.

over, θ^* is assumed to be chosen from the initial collection of 5-letter words according to a uniform distribution. Under this assumption, it can be shown that the belief distribution b_k at stage k continues to be uniform over the mystery list. As a result, we may use as state x_k the mystery list at stage k, which evolves deterministically according to an equation of the form (2.106), where u_k is the guess word at stage k. There are several base policies to use in the rollout-like algorithm (2.107), which are described in the papers by Bhambri, Bhattacharjee, and Bertsekas [BBB22], [BBB23], together with computational results, which show that the corresponding rollout algorithm (2.107) performs remarkably close to the optimal policy (first obtained with a very computationally intensive exact DP calculation by Selby in 2022).

The rollout approach also applies to several variations of the Wordle puzzle. Such variations may include for example a larger length $\ell > 5$ of mystery words, and/or a known nonuniform distribution over the initial collection of ℓ-letter words; see [BBB22].

The Case of Sequential Estimation - Alternative Base Policies

We finally note that the adaptive control framework of this section contains as a special case the sequential estimation framework of the preceding section. Here the problem formulation involves a dynamic system of the form

$$x_{k+1} = f_k(\theta, u_k, w_k),$$

where the state x_{k+1} is the observation at time $k+1$ and exhibits no explicit dependence on the preceding observation x_k, but depends on the stochastic disturbance w_k, and on the decision u_k; cf. Figs. 2.11.1 and 2.11.3. This decision may involve a cost and determines the type of next observation out of a collection of possible types.

While the rollout methodology of the present section applies to sequential estimation problems, other rollout algorithms may also be used, depending on the problem's detailed structure. In particular, the rollout algorithms for Bayesian optimization of the works noted in Section 2.10 involve base policies that depend on the current belief state b_k, rather than the current state x_k. Another example of rollout for adaptive control, which uses a base policy that depends on the current belief state is given in Section 6.7 of the book [Ber22a]. For work on related stochastic optimal control problems that involve observation costs and the rollout approach, see Antunes and Heemels [AnH14], and Khashooei, Antunes, and Heemels [KAH15].

2.12 MINIMAX CONTROL AND REINFORCEMENT LEARNING

The problem of optimal control of uncertain systems is usually treated within a stochastic framework, whereby all disturbances w_k are described by probability distributions, and the expected value of the cost is minimized. However, in many practical situations a stochastic description of the disturbances may not be available, but one may have information with less detailed structure, such as bounds on their magnitude. In other words, one may know a set within which the disturbances are known to lie, but may not know the corresponding probability distribution. Under these circumstances we can use a minimax approach, i.e., try to minimize a cost function assuming that the worst possible values of the disturbances will occur. In this approach, we assume an antagonistic opponent, called the *maximizer*, who chooses w_k with the aim to maximize the cost. The controller, who chooses the controls u_k with the aim to minimize the cost, will be referred to as the *minimizer*.†

In this section, we consider a variety of RL schemes for minimax control. We start with exact DP and the corresponding Bellman equation for finite and infinite horizon problems, in order to provide the foundation for approximate DP/RL approaches. The main difference from the stochastic control problems we have discussed earlier is that the disturbance choices w_k are made by maximization rather than by randomization. Accordingly, the expected value operation is replaced by maximization over w_k in the

† The minimax approach to decision and control has its origins in the 50s and 60s. It is also referred to by other names, depending on the underlying context, such as *robust control*, *robust optimization*, *control with a set membership description of the uncertainty*, and *games against nature*. In this book, we will be using the "minimax control" name. The minimax approach is also connected with two-player games, when in lack of information about the opponent, we adopt a worst case viewpoint during on-line play, as well as with contexts where we wish to guard against adversarial attacks.

DP algorithm and in Bellman's equation. We then turn to approximate DP and we discuss two distinct RL schemes.

The first RL scheme (Section 2.12.2) is similar to the ones we have considered so far in this chapter. We approximate the optimal cost function J^* (or cost functions $J_0^*, \ldots, J_{N-1}^*$, in the case of a finite horizon) by an approximation $\tilde{J}$ (or approximations $\tilde{J}_0, \ldots, \tilde{J}_{N-1}$, respectively), and we assume that the minimizer chooses the control u_k by one-step or multistep lookahead minimization, while the maximizer chooses the disturbances by exact cost maximization. Rollout is the special case where $\tilde{J}$ is equal to the cost function of some policy for the minimizer.

The second RL scheme (Section 2.12.3) is based on approximation in policy space, in addition to approximation in value space. We again introduce a cost function approximation $\tilde{J}$. However, instead of choice by maximization, we introduce a policy $\nu(x,u)$ for the maximizer, i.e., a rule by which the disturbance w is chosen at state-control pair (x,u). Naturally, the maximizer's policy ν is chosen to emulate approximately maximizing selections, and we will discuss a few possibilities along this line. However, once the maximizer's policy ν has been chosen, the minimizer's problem becomes a one-player optimization that can be dealt with by using the methods that we have discussed so far in this chapter. This brings to bear the full spectrum of approximation in value space techniques of the preceding sections, including problem approximation and various types of rollout. We discuss briefly an extension of this second RL scheme to distributionally robust control, a set of models that receives much attention at present. Finally, in Section 2.12.4, we illustrate the scheme of Section 2.12.3 within the context of computer chess and other two-person games for which suitable policies ν for the maximizer can be implemented through sophisticated computer engines.

2.12.1 Exact Dynamic Programming for Minimax Problems

Let us first consider a finite horizon case, and assume that the disturbances $w_0, w_1, \ldots, w_{N-1}$ do not have a probabilistic description, but rather are known to belong to corresponding given sets $W_k(x_k,u_k)$, $k=0,1,\ldots,N-1$, which may depend on the current state x_k and control u_k. The minimax control problem is to find a policy $\pi = \{\mu_0, \ldots, \mu_{N-1}\}$ with $\mu_k(x_k) \in U_k(x_k)$ for all x_k and k, which minimizes the cost function

$$J_\pi(x_0) = \max_{\substack{w_k \in W_k(x_k,\mu_k(x_k)) \\ k=0,1,\ldots,N-1}} \left[g_N(x_N) + \sum_{k=0}^{N-1} g_k\big(x_k, \mu_k(x_k), w_k\big) \right].$$

The DP algorithm for this problem takes the following form, which resembles the one corresponding to the stochastic DP problem (maximization is used in place of expectation):

$$J_N^*(x_N) = g_N(x_N), \tag{2.110}$$

$$J_k^*(x_k) = \min_{u_k \in U(x_k)} \max_{w_k \in W_k(x_k,u_k)} \Big[g_k(x_k,u_k,w_k) + J_{k+1}^*\big(f_k(x_k,u_k,w_k)\big) \Big]. \tag{2.111}$$

This algorithm can be explained by using a principle of optimality type of argument. In particular, we consider the tail subproblem whereby we are at state x_k at time k, and we wish to minimize the "cost-to-go"

$$\max_{\substack{w_t \in W_t(x_t,\mu_t(x_t)) \\ t=k,k+1,\ldots,N-1}} \left[g_N(x_N) + \sum_{t=k}^{N-1} g_t\big(x_t,\mu_t(x_t),w_t\big) \right].$$

We argue that if $\pi^* = \{\mu_0^*, \mu_1^*, \ldots, \mu_{N-1}^*\}$ is an optimal policy for the minimax problem, then the tail of the policy $\{\mu_k^*, \mu_{k+1}^*, \ldots, \mu_{N-1}^*\}$ is optimal for the tail subproblem. The optimal cost of this subproblem is $J_k^*(x_k)$, as given by the DP algorithm (2.110)-(2.111). The algorithm expresses the intuitive fact that when at state x_k at time k, then regardless of what happened in the past, we should choose u_k that minimizes the worst/maximum value over w_k of the sum of the current stage cost plus the optimal cost of the tail subproblem that starts from the next state. This argument requires a mathematical proof, which turns out to involve a few fine points. For a detailed mathematical derivation, we refer to the author's textbook [Ber17a], Section 1.6. However, the DP algorithm (2.110)-(2.111) is correct assuming finite state and control spaces, among other cases.

Minimax Control and Zero-Sum Game Theory

The theory of minimax control is intimately connected with the theory of dynamic zero-sum game problems, which essentially involve *two* minimax control problems:

(a) The *min-max problem*, where the minimizer chooses a policy first and the maximizer chooses a policy second with knowledge of the minimizer's policy. The DP algorithm for this problem has the form

$$J_N^*(x_N) = g_N(x_N),$$

$$J_k^*(x_k) = \min_{u_k \in U_k(x_k)} \max_{w_k \in W_k(x_k)} \Big[g_k(x_k,u_k,w_k) + J_{k+1}^*\big(f_k(x_k,u_k,w_k)\big) \Big].$$

(b) The *max-min problem*, where the maximizer chooses policy first and the minimizer chooses policy second with knowledge of the maximizer's policy. The DP algorithm for this problem has the form

$$\hat{J}_N(x_N) = g_N(x_N),$$

$$\hat{J}_k(x_k) = \max_{w_k \in W_k(x_k)} \min_{u_k \in U_k(x_k)} \Big[g_k(x_k,u_k,w_k) + \hat{J}_{k+1}\big(f_k(x_k,u_k,w_k)\big) \Big].$$

A basic and easily seen fact is that

$$\text{Max-Min optimal value} \leq \text{Min-Max optimal value}.$$

There is an extensive and time-honored theory for dynamic zero-sum games, which is focused on conditions that guarantee that

$$\text{Max-Min optimal value} = \text{Min-Max optimal value}. \tag{2.112}$$

However, this question is of limited interest in engineering contexts that involve worst-case design. Moreover, the validity of the minimax equality (2.112) is beyond the range of practical RL, and thus will not be discussed here. The main reason is that once approximations are introduced, the delicate assumptions that guarantee the minimax equality are disrupted.

Minimax Control Over an Infinite Horizon

The formulation of the infinite horizon version of the preceding minimax control problem follows similar lines to its stochastic counterpart (cf. Section 1.4). The system equation f, cost per stage g, control constraint sets U, and disturbance constraint sets W do not depend on the time k. For a discounted version of the problem, the cost function of a policy $\pi = \{\mu_0, \mu_1, \ldots\}$ has the form

$$J_\pi(x_0) = \lim_{N\to\infty} \max_{\substack{w_k \in W(x_k,u_k) \\ k=0,1,\ldots}} \left\{ \sum_{k=0}^{N-1} \alpha^k g\big(x_k, \mu_k(x_k), w_k\big) \right\}, \tag{2.113}$$

where $\alpha < 1$ is the discount factor. When the range of values of the stage cost g is bounded, the discount factor guarantees that the limit defining $J_\pi(x_0)$ exists and is finite. It can then be shown that the optimal cost function J^* also takes finite values, and is the unique solution of the Bellman equation

$$J^*(x) = \min_{u\in U(x)} \max_{w\in W(x,u)} \Big[g(x,u,w) + \alpha J^*\big(f(x,u,w)\big) \Big], \qquad \text{for all } x. \tag{2.114}$$

Moreover, a stationary policy μ^* policy is optimal if and only if $\mu^*(x)$ attains the minimum in Bellman's equation for all x. Straightforward analogs of the value and policy iteration algorithms are also valid under the same circumstances.† These results follow from general analyses of abstract DP models under conditions that guarantee that the Bellman operator is a contraction mapping; see the books [Ber12], [Ber22b].

† The policy iteration algorithm involves some subtleties and requires modifications, which have been the subject of quite a bit of research. We will not discuss this issue here; see the author's paper [Ber21b] and book [Ber22b] (Chapter 5).

Undiscounted problems, where $\alpha = 1$, usually involve a special cost-free termination state. They are generally more complicated than discounted problems, but the preceding results hold under the assumption that there is a bound on the number of stages needed to reach the termination state, regardless of the choices of the minimizer and the maximizer. This condition is fulfilled, for example, in many computer games, like chess, which will be discussed later in this section.

2.12.2 Minimax Approximation in Value Space and Rollout

The approximation ideas for stochastic optimal control that we have discussed in this chapter are also relevant within the minimax context. In particular, for finite horizon problems, approximation in value space with one-step lookahead applies at state x_k a control

$$\tilde{u}_k \in \arg \min_{u_k \in U_k(x_k)} \max_{w_k \in W_k(x_k,u_k)} \Big[g_k(x_k, u_k, w_k) + \tilde{J}_{k+1}\big(f_k(x_k, u_k, w_k)\big) \Big], \tag{2.115}$$

where $\tilde{J}_{k+1}(x_{k+1})$ is an approximation to the optimal cost-to-go $J^*_{k+1}(x_{k+1})$ from state x_{k+1}.

Rollout is obtained when this approximation is the tail cost of some base policy $\pi = \{\mu_0, \ldots, \mu_{N-1}\}$:

$$\tilde{J}_{k+1}(x_{k+1}) = J_{k+1,\pi}(x_{k+1}).$$

Given π, we can compute $J_{k+1,\pi}(x_{k+1})$ by solving a *deterministic maximization* DP problem with the disturbances $w_{k+1}, \ldots, w_{N-1}$ playing the role of "optimization variables/controls." For finite state, control, and disturbance spaces, this is a longest path problem defined on an acyclic graph, since the control variables $u_{k+1}, \ldots, u_{N-1}$ are determined by the base policy. It is then straightforward to implement rollout: at x_k we generate all next states of the form

$$x_{k+1} = f_k(x_k, u_k, w_k)$$

corresponding to all possible values of $u_k \in U_k(x_k)$ and $w_k \in W_k(x_k, u_k)$. We then run the maximization/longest path problem described above to compute $J_{k+1,\pi}(x_{k+1})$ from each of these possible next states x_{k+1}. Finally, we obtain the rollout control $\tilde{u}_k$ by solving the minimax problem in Eq. (2.115). Moreover, it is possible to use truncated rollout to approximate the tail cost of the base policy $J_{k+1,\pi}(x_{k+1})$.†

Note that like all rollout algorithms, the minimax rollout algorithm is well-suited for on-line replanning in problems where data may be changing or may be revealed during the process of control selection.

† For a more detailed discussion of this implementation, see the author's paper [Ber19b] (Section 5.4).

We mentioned earlier that deterministic problems allow a more general form of rollout, whereby we may use a base heuristic that need not be a legitimate policy, i.e., it need not be sequentially consistent. For cost improvement it is sufficient that the heuristic be sequentially improving. A similarly more general view of rollout is not easily constructed for stochastic problems, but is possible for minimax control.

In particular, suppose that at any state x_k there is a heuristic that generates a sequence of feasible controls and disturbances, and corresponding states,

$$\{u_k, w_k, x_{k+1}, u_{k+1}, w_{k+1}, x_{k+2}, \ldots, u_{N-1}, w_{N-1}, x_N\},$$

with corresponding cost

$$H_k(x_k) = g_k(x_k, u_k, w_k) + \cdots + g_{N-1}(x_{N-1}, u_{N-1}, w_{N-1}) + g_N(x_N).$$

Then the rollout algorithm applies at state x_k a control

$$\tilde{u}_k \in \arg \min_{u_k \in U_k(x_k)} \max_{w_k \in W_k(x_k, u_k)} \Big[g_k(x_k, u_k, w_k) + H_{k+1}\big(f_k(x_k, u_k, w_k)\big) \Big].$$

This does not preclude the possibility that the disturbances $w_k, \ldots, w_{N-1}$ are chosen by an antagonistic opponent, but allows more general choices of disturbances, obtained for example, by some form of approximate maximization. For example, when the disturbance involves multiple components, $w_k = (w_k^1, \ldots, w_k^m)$, corresponding to multiple opponent agents, *the heuristic may involve an agent-by-agent maximization strategy*.

The sequential improvement condition, similar to the deterministic case, is that for all x_k and k,

$$\min_{u_k \in U_k(x_k)} \max_{w_k \in W_k(x_k, w_k)} \Big[g_k(x_k, u_k, w_k) + H_{k+1}\big(f_k(x_k, u_k, w_k)\big) \Big] \leq H_k(x_k).$$

It guarantees cost improvement, i.e., that for all x_k and k, the rollout policy

$$\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$$

satisfies

$$J_{k,\tilde{\pi}}(x_k) \leq H_k(x_k).$$

Thus, generally speaking, minimax rollout is fairly similar to rollout for deterministic as well as stochastic DP problems. The main difference with deterministic (or stochastic) problems is that to compute the Q-factor of a control u_k, we need to solve a maximization problem, rather than carry out a deterministic (or Monte Carlo, respectively) simulation with the given base policy.

Example 2.12.1 (Pursuit-Evasion Problems)

Consider a pursuit-evasion problem with state $x_k = (x_k^1, x_k^2)$, where x_k^1 is the location of the minimizer/pursuer and x_k^2 is the location of the maximizer/evader, at stage k, in a (finite node) graph defined in two- or three-dimensional space. There is also a cost-free and absorbing termination state that consists of a subset of pairs (x^1, x^2) that includes all pairs with $x^1 = x^2$, and possibly some other pairs for which x^1 and x^2 are "close enough" for the pursuer to capture the evader.

The pursuer chooses one out of a finite number of actions $u_k \in U_k(x_k)$ at each stage k, when at state x_k, and if the state is x_k and the pursuer selects u_k, the evader may choose from a known set $X_{k+1}(x_k, u_k)$ of next states x_{k+1}, which depends on (x_k, u_k). The objective of the pursuer is to minimize a nonnegative terminal cost $g(x_N^1, x_N^2)$ at the end of N stages (or reach the termination state, which has cost 0 by assumption). A reasonable base policy for the pursuer can be precomputed by DP as follows: given the current (nontermination) state $x_k = (x_k^1, x_k^2)$, make a move along the path that starts from x_k^1 and minimizes the terminal cost after $N-k$ stages, under the assumption that the evader will stay motionless at his current location x_k^2. (In a variation of this policy, the DP computation is done under the assumption that the evader will follow some nominal sequence of moves.)

For the on-line computation of the rollout control, we need the maximal value of the terminal cost that the evader can achieve starting from every $x_{k+1} \in X_{k+1}(x_k, u_k)$, assuming that the pursuer will follow the base policy (which has already been computed). We denote this maximal value by $\tilde{J}_{k+1}(x_{k+1})$. The required values $\tilde{J}_{k+1}(x_{k+1})$ can be computed by an $(N-k)$-stage DP computation involving the optimal choices of the evader, while assuming the pursuer uses the (already computed) base policy. Then the rollout control for the pursuer is obtained from the minimization

$$\tilde{\mu}_k(x_k) \in \arg \min_{u_k \in U_k(x_k)} \max_{x_{k+1} \in X_{k+1}(x_k, u_k)} \tilde{J}_{k+1}(x_{k+1}).$$

Note that the preceding algorithm can be adapted for the imperfect information case where the pursuer knows x_k^2 imperfectly. This is possible by using a form of assumed certainty equivalence: the pursuer's base policy and the evader's maximization can be computed by using an estimate of the current location x_k^2 instead of the unknown true location.

In the preceding pursuit-evasion example, the choice of the base policy was facilitated by the special structure of the problem. Generally, however, finding a suitable base policy that can be conveniently implemented is an important problem-dependent issue.

Variants of Minimax Rollout

Several of the variants of rollout discussed earlier have analogs in the minimax context, e.g., truncation with terminal cost approximation, multistep

and selective step lookahead, and multiagent rollout. In particular, in the ℓ-step lookahead variant, we solve the ℓ-stage problem

$$\min_{u_k,\mu_{k+1},\ldots,\mu_{k+\ell-1}} \max_{\substack{w_k\in W_k(x_k,u_k)\\ w_t\in W_t(x_t,\mu_t(x_t))\\ t=k+1,\ldots,N-1}} \Big\{ g_k(x_k,u_k,w_k) + \sum_{t=k+1}^{k+\ell-1} g_t\big(x_t,\mu_t(x_t),w_t\big) + H_{k+\ell}(x_{k+\ell}) \Big\},$$

we find an optimal solution $\tilde{u}_k, \tilde{\mu}_{k+1}, \ldots, \tilde{\mu}_{k+\ell-1}$, and we apply the first component $\tilde{u}_k$ of that solution. As an example, this type of problem is solved at each move of some chess programs, where the terminal cost function is encoded through a position evaluator. In fact when multistep lookahead is used, special techniques such as *alpha-beta pruning* may be used to accelerate the computations by eliminating unnecessary portions of the lookahead graph. These techniques are well-known in the context of the two-person computer game methodology, and are used widely in games such as chess.

It is interesting to note that, contrary to the case of stochastic optimal control, there is an on-line *constrained form of rollout* for minimax control. Here there are some additional trajectory constraints of the form

$$(x_0, u_0, \ldots, u_{N-1}, x_N) \in C,$$

where C is an arbitrary set. The modification needed is similar to the one of Section 2.5: at partial trajectory

$$\tilde{y}_k = (\tilde{x}_0, \tilde{u}_0, \ldots, \tilde{u}_{k-1}, \tilde{x}_k),$$

generated by rollout, we use a heuristic with cost function H_{k+1} to compute the Q-factor

$$\tilde{Q}_k(\tilde{x}_k,u_k) = \max_{w_k,\ldots,w_{N-1}} \Big[g_k(\tilde{x}_k,u_k,w_k) + H_{k+1}\big(f_k(\tilde{x}_k,u_k,w_k),w_{k+1},\ldots,w_{N-1}\big) \Big]$$

for each u_k in the set $\tilde{U}_k(\tilde{y}_k)$ that guarantee feasibility [we can check feasibility here by running some algorithm that verifies whether the future disturbances $w_k,\ldots,w_{N-1}$ can be chosen to violate the constraint under the base policy, starting from $(\tilde{y}_k,u_k)$]. Once the set of "feasible controls" $\tilde{U}_k(\tilde{y}_k)$ is computed, we can obtain the rollout control by the Q-factor minimization:

$$\tilde{u}_k \in \arg\min_{u_k\in\tilde{U}_k(\tilde{y}_k)} \tilde{Q}_k(\tilde{x}_k,u_k).$$

We may also use fortified versions of the unconstrained and constrained rollout algorithms, which guarantee a feasible cost-improved rollout policy. This requires the assumption that the base heuristic at the initial state produces a trajectory that is feasible for all possible disturbance sequences.

Truncated and multiagent versions of the minimax rollout algorithm are possible. The following example describes the multiagent case.

Example 2.12.2 (Multiagent Minimax Rollout)

Let us consider a minimax problem where the minimizer's choice involves the collective decision of m agents, $u = (u^1, \ldots, u^m)$, with u^ℓ corresponding to agent ℓ, and constrained to lie within a finite set U^ℓ. Thus u must be chosen from within the set

$$U = U^1 \times \ldots \times U^m,$$

which is finite but grows exponentially in size with m. The maximizer's choice w is constrained to belong to a finite set W. We consider multiagent rollout for the minimizer, and for simplicity, we focus on a two-stage problem. However, there are straightforward extensions to a more general multistage framework.

In particular, we assume that the minimizer knowing an initial state x_0, chooses $u = (u^1, \ldots, u^m)$, with $u^\ell \in U^\ell$, $\ell = 1, \ldots, m$, and a state transition

$$x_1 = f_0(x_0, u)$$

occurs with cost $g_0(x_0, u)$. Then the maximizer, knowing x_1, chooses $w \in W$, and a terminal state

$$x_2 = f_1(x_1, w)$$

is generated with cost

$$g_1(x_1, w) + g_2(x_2).$$

The problem is to select $u \in U$, to minimize

$$g_0(x_0, u) + \max_{w \in W} \big[g(x_1, w) + g_2(x_2)\big].$$

The exact DP algorithm for this problem is given by

$$J_1^*(x_1) = \max_{w \in W} \Big[g_1(x_1, w) + g_2\big(f_1(x_1, w)\big)\Big],$$

$$J_0^*(x_0) = \min_{u \in U} \Big[g_0(x_0, u) + J_1^*\big(f_0(x_0, u)\big)\Big].$$

This DP algorithm is computationally intractable for large m. The reason is that the set of possible minimizer choices u grows exponentially with m, and for each of these choices the value of $J_1^*\big(f_0(x_0, u)\big)$ must be computed.

However, the problem can be solved approximately with multiagent rollout, using a base policy $\mu = (\mu^1, \ldots, \mu^m)$. Then the number of times

$J_1^*\big(f_0(x_0,u)\big)$ needs to be computed is dramatically reduced. This computation is done sequentially, one-agent-at-a-time, as follows:

$$\tilde{u}^1 \in \arg\min_{u^1\in U^1}\Big[g_0\big(x_0,u^1,\mu^2(x_0),\ldots,\mu^m(x_0)\big) + J_1^*\Big(f_0\big(x_0,u^1,\mu^2(x_0),\ldots,\mu^m(x_0)\big)\Big)\Big],$$

$$\tilde{u}^2 \in \arg\min_{u^2\in U^2}\Big[g_0\big(x,\tilde{u}^1,u^2,\mu^3(x_0),\ldots,\mu^m(x_0)\big) + J_1^*\Big(f_0\big(x_0,\tilde{u}^1,u^2,\mu^3(x_0),\ldots,\mu^m(x_0)\big)\Big)\Big],$$

$$\ldots \quad \ldots \quad \ldots \quad \ldots$$

$$\tilde{u}^m \in \arg\min_{u^m\in U^m}\Big[g_0\big(x_0,\tilde{u}^1,\tilde{u}^2,\ldots,\tilde{u}^{m-1},u^m\big) + J_1^*\Big(f_0\big(x_0,\tilde{u}^1,\tilde{u}^2,\ldots,\tilde{u}^{m-1},u^m\big)\Big)\Big].$$

In this algorithm, the number of times for which $J_1^*\big(f_0(x_0,u)\big)$ must be computed grows linearly with m.

When the number of stages is larger than two, a similar algorithm can be used. Essentially, the one-stage maximizer's cost function J_1^* must be replaced by the optimal cost function of a multistage maximization problem, where the minimizer is constrained to use the base policy (see also the paper [Ber19b], Section 5.4).

An interesting question is how do various algorithms work when approximations are used in the min-max and max-min problems? We can certainly improve the minimizer's policy *assuming a fixed policy for the maximizer*; this will be the basis for the scheme of the next section. However, it is unclear how to improve both the minimizer's and the maximizer's policies simultaneously. In practice, in *symmetric games*, like chess, a common policy is trained for both players. In particular, in the AlphaZero and TD-Gammon programs this strategy is computationally expedient and has worked well. However, there is no reliable theory to guide the simultaneous training of policies for both maximizer and minimizer, and it is quite plausible that unusual behavior may arise in exceptional cases.†

Another interesting fact is that even the *exact* policy iteration method (given in 1969 by Pollatschek and Avi-Itzhak [PoA69]) encounters serious convergence difficulties. The reason is that Newton's method applied to solution of the Bellman equation for minimax problems (which is equivalent to policy iteration) exhibits more complex and unreliable behavior than

† Indeed such exceptional cases have been reported for the AlphaGo program in late 2022, when humans defeated an AlphaGo look-alike, KataGo, "by using adversarial techniques that take advantage of KataGo's blind spots" (according to the reports); see Wang et al. [WGB22].

its one-player counterpart. Mathematically, this is because the Bellman operator T for infinite horizon problems, given by

$$(TJ)(x) = \min_{u\in U(x)} \max_{w\in W(x,u)} \Big[g(x,u,w) + \alpha J\big(f(x,u,w)\big)\Big], \qquad \text{for all } x,$$

is neither convex nor concave as a function of J. To see this, note that the function

$$\max_{w\in W(x,u)} \Big[g(x,u,w) + \alpha J\big(f(x,u,w)\big)\Big],$$

viewed as a function of J [for fixed (x,u)], is convex, and when minimized over $u \in U(x)$, it becomes neither convex nor concave. As a result there are special difficulties in connection with convergence of Newton's method and the natural form of policy iteration. The author's paper [Ber21b] and book [Ber22b] (Chapter 5) address these convergence issues with modified versions of the policy iteration method, and give many earlier references on policy iteration and other related methods for minimax problems.

2.12.3 Combined Approximation in Value and Policy Space for Minimax Control

We will now discuss ways to simplify the approximation in value space scheme of the preceding section. For an infinite horizon problem, this scheme applies at state x the control

$$\tilde{u} \in \arg\min_{u\in U(x)} \max_{w\in W(x,u)} \Big[g(x,u,w) + \alpha \tilde{J}\big(f(x,u,w)\big)\Big], \tag{2.116}$$

where $\tilde{J}$ is an approximation to the optimal cost function J^*. The difficulty here is that the maximization over $w \in W(x,u)$ must be performed for each $u \in U(x)$, and complicates the minimization over $u \in U(x)$.

In an alternative scheme that alleviates this difficulty, we may introduce an approximation of the maximizer's optimal policy in addition to the cost function approximation $\tilde{J}$. This is a policy $\nu(x,u)$ that depends on both x and u, and is used to approximate the minimax scheme of Eq. (2.116) with the minimization scheme

$$\tilde{u} \in \arg\min_{u\in U(x)} \Big[g\big(x,u,\nu(x,u)\big) + \alpha \tilde{J}\Big(f\big(x,u,\nu(x,u)\big)\Big)\Big]. \tag{2.117}$$

One possibility to obtain ν is off-line training, using samples from the maximization

$$\max_{w\in W(x,u)} \Big[g(x,u,w) + \alpha \tilde{J}\big(f(x,u,w)\big)\Big],$$

of Eq. (2.116). If the policy $\nu(x,u)$ is optimal for the maximizer, this scheme coincides with the minimax approximation in value space scheme

of the preceding section and inherits the corresponding Newton step-related superlinear cost improvement guarantees.† Otherwise, cost improvement is not guaranteed, but can be expected under favorable conditions where $\nu(x,u)$ is near-optimal.

Once $\nu(x,u)$ has been determined, the simplified scheme (2.117) is just a one-player approximation in value space for a problem defined by the system equation

$$x_{k+1} = f\big(x_k, u_k, \nu(x_k,u_k)\big), \tag{2.118}$$

and the cost per stage

$$g\big(x,u,\nu(x,u)\big).$$

It is important to note that this problem is deterministic, and is suitable for the special search techniques and simplifications that we have discussed in connection with deterministic problems, including the rollout schemes of Sections 2.3-2.6. Moreover the superlinear error bounds associated with Newton's method (cf. Section 1.5.3), apply to this problem.

Extension to Distributionally Robust Control Models

There is an important minimax-type model where the disturbance w_k is a probability distribution and the set $W_k(x_k,u_k)$ is a given set of probability distributions for each pair (x_k,u_k). This problem formulation is known as a *distributionally robust model*. It dates to the early days of statistics (see e.g., the classical book by Blackwell and Girshick [BlG54], and the more recent book by Chen and Paschalidis [ChP20]). Within the RL context of this section, it can be treated as a special case of the schemes that we have discussed. In particular, by allowing the policy $\nu(x,u)$ to take probability distribution values, the approximation in value space scheme (2.117) applies. Note, however, that in this case, the dynamic system (2.118) is stochastic, and as a result the implementation of the corresponding approximation in value space scheme is more complicated.

2.12.4 A Meta Algorithm for Computer Chess Based on Reinforcement Learning

In this section, we will apply the approach of approximating the maximizer's action with a suitably constructed policy to computer chess, based

† A subtle but important point here is that in single-player DP contexts [cf. Eq. (2.117)], the Bellman equation has concave components. This turns out to have a beneficial effect on the convergence properties of Newton's method. By contrast, in two-player games the components of Bellman's equation may be neither concave nor convex, and this complicates significantly the convergence properties of Newton's method; see the author's paper [Ber21b] and abstract DP book [Ber22b] (Chapter 5).

on the work by Gundawar, Li, and Bertsekas [GLB24].† A key fact is that for computer chess there are several programs (commonly called *engines*), which supply a move selection policy [this is the maximizer's policy approximation ν in Eq. (2.117)]. Engines can also be used to supply board position evaluations [this is the cost function approximation $\tilde{J}$ in Eq. (2.117)].‡

For a brief summary, our scheme is built around two components:

(a) The *position evaluator*. This is implemented with one of the many publicly available chess engines, such as the popular and freely available champion program Stockfish, which produces an evaluation of any given position (normalized in a way that is standard in computer chess).

(b) The *nominal opponent*. This is an approximation to the true opponent engine or human, whom we intend to play with. It outputs deterministically a move at each given position, against which we expect to play. In the absence of a known opponent, a reasonable choice is to use a competent chess engine as nominal opponent, such as for example the one used to provide position evaluations (e.g., Stockfish). In any case, it is important *not* to use a relative poor nominal opponent, which would lead us to underestimate the true opponent.

The nominal opponent and the position evaluator may be implemented with different chess engines. Moreover the nominal opponent may be changed from game to game to adapt to the real opponent at hand. An important fact is that stored knowledge of the opponent and evaluator engines, such as an opening book or an endgame database, are indirectly incorporated into the approximation in value space scheme.

To make the connection with our mathematical framework for minimax problems, we use the following notation:

- x_k is the chess position of our player at time k.
- u_k is the move choice of our player at time k in response to position x_k.
- w_k is the move choice of the nominal opponent at time k in response to position x_k followed by move u_k.

The resulting position at time $k+1$ is given by

$$x_{k+1} = f(x_k, u_k, w_k),$$

† The ideas of the present section apply more broadly to any two-person antagonistic game, which uses computer engines to supply cost function approximation and move selection policies.

‡ We view the approach of this section as a *meta algorithm*, which is a broad term that describes an algorithm that "provides a framework or strategy to develop, combine, or enhance other algorithms" (according to ChatGPT).

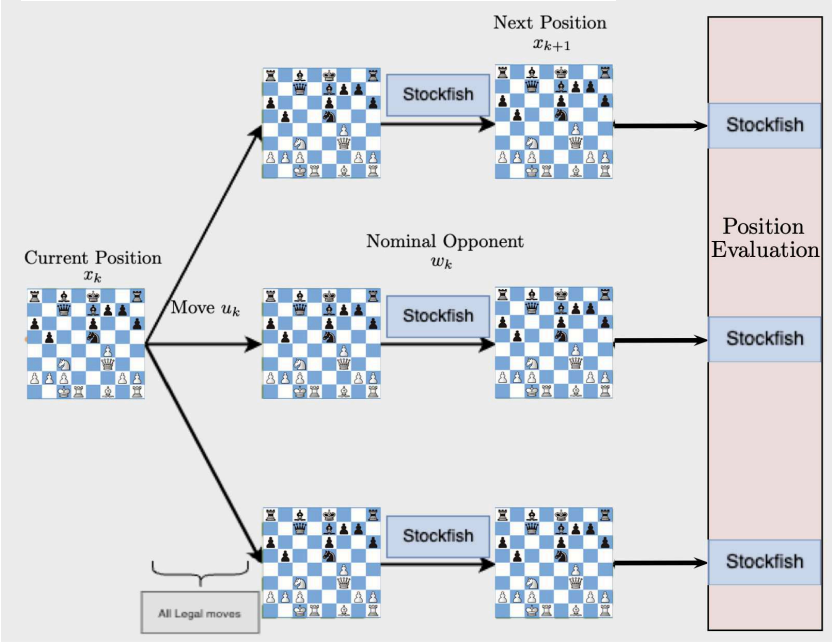

Figure 2.12.1 Illustration of the sequence of calculations of a one-step lookahead scheme for computer chess. Here we use the chess engine Stockfish for both opponent move generation and for position evaluation. At the current position x_k, we generate all legal moves u_k, and for each pair (x_k, u_k), the opponent engine generates a single best move w_k, resulting in the position

$$x_{k+1} = f(x_k, u_k, w_k).$$

We then use the position evaluation engine to evaluate each of the possible x_{k+1}.

where f is a known function. This corresponds to the dynamic system, where x_k is viewed as the state, u_k is viewed as the control, and w_k is viewed as a known or random disturbance.

At the current position x_k, our scheme operates as follows:

(1) We generate all legal moves u_k.

(2) For each pair (x_k, u_k), the nominal opponent generates a single best move w_k, resulting in the position

$$x_{k+1} = f(x_k, u_k, w_k).$$

(3) We evaluate each of the possible positions x_{k+1}.

Thus, there are a total of at most m position evaluations and m nominal opponent move generations, where m is an upper bound to the number of legal moves at any position.

Figure 2.12.1 describes our scheme for the case of one-step lookahead, with Stockfish used for nominal opponent moves and position evaluations. Note that the scheme is well suited for parallel computation. In particular, the moves of the nominal opponent can be computed in parallel, and the positions following the generation of the opponent's moves can also be evaluated in parallel. Thus, with sufficient parallel computing resources, our scheme requires roughly twice as much computation as the underlying engines to generate a move.

The paper [GLB24] presents test results using strong chess engines, such as Stockfish, as well as other weaker engines. These results show that, similar to rollout, our scheme improves the performance of the position evaluation engine on which it is based, and for relatively weak engines, dramatically so. This finding generally assumes a reasonable choice for opponent move selection, such as for example the engine that is used for position evaluation. In our tests, a scheme that uses Stockfish of various strength levels as the engine for both nominal opponent and position evaluator has beaten the engine itself by a significant margin, although the winning margin diminishes as the strength of the engine increases. This is to be expected since Stockfish plays near-perfect chess at its highest levels. Qualitatively, this is similar to the performance of approximation in value space and rollout methods, which emulate a Newton step and attain a superlinear convergence, but with the amount of cost improvement diminishing near J^*.

Multistep Lookahead and Other Extensions

It is possible to introduce two-step and multistep lookahead in the preceding algorithm: the lookahead stage of Fig. 2.12.1 is replicated over multiple stages; see Fig. 2.12.2. For an ℓ-step lookahead scheme, the computation needed per move is $m + m^2 + \cdots + m^\ell$ opponent move generations plus m^ℓ position evaluations. This computation time can be reduced if the lookahead tree is suitably pruned at the second stage. Moreover, it can be seen that given sufficient parallel computing resources, to generate a move, an ℓ-step lookahead scheme requires roughly $\ell + 1$ times as much computation as the underlying engines. For more details, extensions, and computational testing, see the paper [GLB24].

Let us also note the possibility of replacing the single nominal opponent move with multiple (presumably good) moves. This creates a reduced minimax tree, where the moves at each stage, after the first one, are pruned, except for a few top moves, selected using some form of position evaluation. The player's move at the current position is then found with a minimax search over the reduced tree that consists of the moves that have not been pruned. Note that it is important not to prune any moves of the first stage in order to preserve the Newton step character of the approximation in value space scheme. On the other hand, the (pruned) multistep minimax

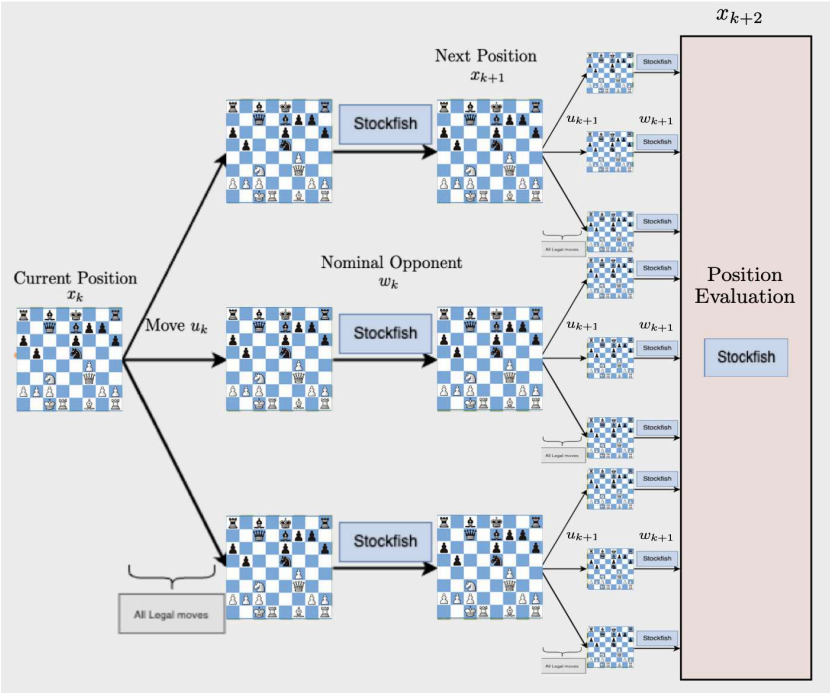

Figure 2.12.2 Illustration of the two-step lookahead version of the computer chess scheme of Fig. 2.12.1. Again we use Stockfish for both opponent move generation and for position evaluation.

search can be costly, and may require a lot of position evaluations and move generations, so for the scheme to be viable, the pruning should be aggressive and the length of the lookahead should be limited.

2.12.5 Combined Approximation in Value and Policy Space for Sequential Noncooperative Games

Noncooperative games (also called nonzero-sum games, or Nash games) represent a significant extension of the zero-sum games that we discussed earlier. Here there are a finite number of agents who choose distinct controls, interact with each other, and aim to optimize their private cost functions. The analysis of the agents' choices in the absence of complete or partial information regarding the choices of the other agents has fascinated engineers, mathematicians, and social scientists since the seminal (26 pages!) Ph.D. dissertation of J. Nash [Nas50].

In this section we will discuss a relatively simple special class of nonzero-sum games, which involves sequential choices of the controls of the agents, with complete communication of these controls to the other agents. Still their treatment by DP is very challenging in general. In fact,

contrary to the minimax problem, where there is a Bellman equation (albeit complicated by components that are neither convex nor concave; cf. the discussion at the end of Section 2.12.2), for the class of problems of this section there may not be a Bellman equation that can be used as a basis for approximation in value space.

We will now focus on an important example of sequential noncooperative game, a *leader-follower problem*, also known as a *Stackelberg game* in economics and as a *bilevel optimization* problem in mathematical programming. It involves a dominant decision maker, called the *leader*, and m other decision makers, called the *followers*. Once the followers observe the decision of the leader, they optimize their choices according to their private cost functions. We make no assumption on the state and control spaces, other than that the constraints of the followers are decoupled.

Let us first consider for simplicity a two-stage stochastic environment with perfect state information, and the special case where *both leader and followers act cooperatively* in the sense that they minimize a common cost function. Thus, we assume that the leader knowing an initial state x_0, makes a decision $u_0 \in U_0$, and a state transition

$$x_1 = f_0(x_0, u_0, w_0)$$

occurs with cost $g_0(x_0, u_0, w_0)$, where w_0 is a random disturbance. Then the followers, knowing x_1, choose decisions $u_1^\ell \in U_1^\ell$, $\ell = 1, \ldots, m$, and a terminal state

$$x_2 = f_1(x_1, u_1^1, \ldots, u_1^m, w_1)$$

is generated with cost

$$g_{1,1}(x_1, u_1^1, w_1) + \cdots + g_{1,m}(x_1, u_1^m, w_1) + g_2(x_2),$$

where w_1 is a random disturbance, and g_2 is the terminal cost function. The problem is to select u_0 and follower policies μ_1^ℓ, $\ell = 1, \ldots, m$, so as to minimize the total expected cost.

Clearly this is a two-stage multiagent DP problem, which can be very hard to solve for large m, because of the large size of the control space of the second stage and the coupling of the followers' actions through the terminal cost. This two-stage problem can be solved approximately with multiagent rollout; see Section 2.9. Moreover, the extension from a two-stage to a multistage framework is straightforward.

However, the multiagent formulation just described fails when the followers do not share with the leader the same cost function. In this case it seems very difficult to apply approximation in value space methods.† On the other hand, a convenient approach is to convert the problem to a single

† Approximation in policy space approaches are possible, and have been suggested in the literature, but have met with mixed success so far.

agent type of problem, involving just the leader and *nominal policies of the followers*, similar to the approach of Section 2.12.3, and the computer chess paradigm of Section 2.12.4. These policies, can be obtained using some form of policy training, and produce follower actions $u_1^\ell \in U_1^\ell$, $\ell = 1, \ldots, m$, that depend predictably on the current state x_1 and possibly the preceding action u_0 of the leader. The advantage of this approach is that the Newton step framework fully applies to the leader's problem, and the corresponding on-line implementation of the leader's policy is tractable, based on one-player approximation in value space schemes. Limited preliminary results using this approach have been encouraging, but more research along this line is needed.

2.13 NOTES, SOURCES, AND EXERCISES

In this chapter, we have first considered deterministic problems, then stochastic problems, and finally minimax problems. We have emphasized the effectiveness of approximation in value space, which derives from its connection to Newton's method. We have discussed several types of approximation in value space schemes, with an emphasis on rollout and its variations, which are very effective, reliable, and easily implementable.

Section 2.1: Our focus in this section has been on finite horizon problems, possibly involving a nonstationary system and cost per stage. However, the insights that can be obtained from the infinite horizon/stationary context fully apply. These include the interpretation of approximation in value space as a Newton step, and of rollout as a single step of the policy iteration method. The reason is that an N-step finite horizon/nonstationary problem can be converted to an infinite horizon/stationary problem with a termination state to which the system moves at the Nth stage; see Section 1.6.4.

Section 2.2: Approximation in value space has been considered in an ad hoc manner since the early days of DP, motivated by the curse of dimensionality. Moreover, the idea of ℓ-step lookahead minimization with horizon truncation beyond the ℓ steps has a long history and is often referred to as "rolling horizon" or "receding horizon" optimization. Approximation in value space was reframed in the late 80s and was coupled with model-free simulation methods that originated in AI.

Section 2.3: The main idea of rollout algorithms, obtaining an improved policy starting from some other suboptimal policy, has appeared in several DP contexts, including games; see e.g., Abramson [Abr90], and Tesauro and Galperin [TeG96]. The name "rollout" was coined in [TeG96] in the context of backgammon; see Example 2.7.4. The use of the name "rollout" has gradually expanded beyond its original context; for example samples

collected through trajectory simulation are referred to as "rollouts" by some authors.

In this book, we will adopt the original intended meaning: rollout is an algorithm that provides policy improvement starting from a base policy, which is evaluated with some form of Monte Carlo simulation, perhaps augmented by some other calculation that may include a terminal cost function approximation. The author's rollout book [Ber20a] provides a more extensive discussion of rollout algorithms and their applications.

The adaptation of rollout algorithms to discrete deterministic optimization problems, the notions of sequential consistency, sequential improvement, fortified rollout, and the use of multiple heuristics for parallel rollout were first given in the paper by Bertsekas, Tsitsiklis, and Wu [BTW97], and were also discussed in the neuro-dynamic programming book [BeT96]. Rollout algorithms for stochastic problems were further formalized in the papers by Bertsekas [Ber97b], and Bertsekas and Castañon [BeC99]. Extensions to constrained rollout were first given in the author's papers [Ber05a], [Ber05b]. A survey of rollout in discrete optimization was given by the author in [Ber13a].

Following the original works on rollout for discrete deterministic and stochastic optimization (Bertsekas, Tsitsiklis, and Wu [BTW97], Bertsekas and Castañon [BeC99], and the neuro-dynamic programming book [BeT96]), there has been a lot of research on rollout algorithms, which we list selectively in chronological order: Christodouleas [Chr97], Duin and Voss [DuV99], Secomandi [Sec00], [Sec01], [Sec03], Ferris and Voelker [FeV02], [FeV04], McGovern, Moss, and Barto [MMB02], Savagaonkar, Givan, and Chong [SGC02], Wu, Chong, and Givan [WCG02], [WCG03], Bertsimas and Popescu [BeP03], Guerriero and Mancini [GuM03], Tu and Pattipati [TuP03], Meloni, Pacciarelli, and Pranzo [MPP04], Yan et al. [YDR04], Nedich, Schneider, and Washburn [NSW05], Han, Lai, and Spivakovsky [HLS06], Lee et al. [LSG07], An et al. [ASP08], Berger et al. [BAP08], Besse and Chaib-draa [BeC08], Patek, Breton, and Kovatchev [PBK08], Sun et al. [SZL08], Tian, Bar-Shalom, and Pattipati [TBP08], Novoa and Storer [NoS09], Mishra et al. [MCT10], Malikopoulos [Mal10], Bertazzi et al. [BBG13], Sun et al. [SLJ13], Tesauro et al. [TGL13], Antunes and Heemels [AnH14], Beyme and Leung [BeL14], Goodson, Thomas, and Ohlmann [GTO15], [GTO17], Khashooei, Antunes, Heemels [KAH15], Li and Womer [LiW15], Mastin and Jaillet [MaJ15], Simroth, Holfeld, and Brunsch [SHB15], Huang, Jia, and Guan [HJG16], Lan, Guan, and Wu [LGW16], Lam, Willcox, and Wolpert [LWW16], Gommans et al. [GTA17], Lam and Willcox [LaW17], Ulmer [Ulm17], Bertazzi and Secomandi [BeS18], Sarkale et al. [SNC18], Ulmer et al. [UGM18], Zhang, Ohlmann, and Thomas [ZOT18], Arcari, Hewing, and Zeilinger [AHZ19], Chu, Xu, and Li [CXL19], Goodson, Bertazzi, and Levary [GBL19], Guerriero, Di Puglia, and Macrina [GDM19], Ho, Liu, and Zabinsky [HLZ19], Liu et al. [LLL19], Nozhati et al. [NSE19], Singh and Kumar [SiK19],

Yu et al. [YYM19], Yuanhong [Yua19], Andersen, Stidsen, and Reinhardt [ASR20], Issakkimuthu, Fern, and Tadepalli [IFT19], Lee et al. [LEC20], Li et al. [LZS20], Lee [Lee20], Montenegro et al. [MLM20], Meshram and Kaza [MeK20], Schope, Driessen, and Yarovoy [SDY21], Yan, Wang, and Xu [YWX20], Yue and Kontar [YuK20], Zhang, Kafouros, and Yu [ZKY20], Hoffman et al. [HCR21], Li, Krakow, and Gopalswamy [LKG21], Nozhati [Noz21], Rimélé et al. [RGG21], Tuncel et al. [TBP21], Xie, Li, and Xu [XLX21], Bertsekas [Ber22d], Paulson, Sonouifar, and Chakrabarty [PSC22], Bai et al. [BLJ23], Rusmevichientong et al. [RST23], Wu and Zeng [WuZ23], Gerlach and Piatkowski [GeP24], Samani, Hammar, and Stadler [SHS24], Samani et al. [SLD24], Yilmaz, Xiang, and Klein [YXK24], Zhang et al. [ZLZ24], Wang et al. [WTL25].

These references collectively include a large number of computational studies, discuss variants and problem-specific adaptations of rollout algorithms for a broad variety of practical problems, and consistently report favorable computational experience. The size of the cost improvement over the base policy is often impressive, evidently owing to the fast convergence rate of Newton's method that underlies rollout. Moreover these works illustrate some of the other important advantages of rollout: reliability, simplicity, suitability for on-line replanning, and the ability to interface with other RL techniques, such as neural network training, which can be used to provide suitable base policies and/or approximations to their cost functions.

The model-free rollout algorithm, in the form given here, was first discussed in the RL book [Ber19a]. It is related to the method of comparison training, proposed by Tesauro [Tes89a], [Tes89b], [Tes01], and discussed by several other authors (see [DNW16], [TCW19]). This is a general method for training an approximation architecture to choose between two alternatives, using a dataset of expert choices in place of an explicit cost function.

The material on most likely sequence generation for n-grams, HMMs, and Markov chains is recent, and was developed in the paper by Li and Bertsekas [LiB24]. As we have noted, our rollout-based most likely sequence generation algorithm can be useful to all contexts where the Viterbi algorithm is used. This includes algorithms for HMM parameter inference, which use the Viterbi algorithm as a subroutine.

Section 2.4: Our discussion of rollout, iterative deepening, and pruning in the context of multistep approximation in value space for deterministic problems contains some original ideas. In particular, the incremental multistep rollout algorithm and variations of Section 2.4.2 are presented here for the first time.

Note also that the multistep lookahead approximations described in Section 2.4 can be used more broadly within algorithms that employ forms of multistep lookahead search as subroutines. In particular, local search algorithms, such as tabu search, genetic algorithms, and others, which

are commonly used for discrete and combinatorial optimization, may be modified along the lines of Section 2.4 to incorporate RL and approximate DP ideas.

Section 2.5: Constrained forms of rollout were introduced in the author's papers [Ber05a] and [Ber05b]. The paper [Ber05a] also discusses rollout and approximation in value space for stochastic problems in the context of so-called *restricted structure policies*. The idea here is to simplify the problem by selectively restricting the information and/or the controls available to the controller, thereby obtaining a restricted but more tractable problem structure, which can be used conveniently in a one-step lookahead context. An example of such a structure is one where fewer observations are obtained, or one where the control constraint set is restricted to a single or a small number of given controls at each state.

Section 2.6: Rollout for continuous-time optimal control was first discussed in the author's rollout book [Ber20a]. A related discussion of policy iteration, including the motivation for approximating the gradient of the optimal cost-to-go $\nabla_x J_t$ rather than the optimal cost-to-go J_t, has been given in Section 6.11 of the neuro-dynamic programming book [BeT96]. This discussion also includes the use of value and policy networks for approximate policy evaluation and policy improvement for continuous-time optimal control. The underlying ideas have long historical roots, which are recounted in detail in the book [BeT96].

Section 2.7: The idea of the certainty equivalence approximation in the context of rollout for stochastic systems (Section 2.7.2) was proposed in the paper by Bertsekas and Castañon [BeC99], together with extensive empirical justification. However, the associated theoretical insight into this idea was established more recently, through the interpretation of approximation in value space as a Newton step, which suggests that the lookahead minimization after the first step can be approximated with small degradation of performance. This point is emphasized in the author's book [Ber22a], and the papers [Ber22c], [Ber24].

Markov jump problems (Example 2.7.3) have an interesting theory and several diverse applications (see e.g., Sworder [Swo69], Wonham [Won70], Chizeck, Willsky, and Castañon [CWC86], Abou-Kandil, Freiling, and Jank [AFJ95], Costa and Do Val [CDV02], and Li and Bertsekas [LiB25a]. For related monographs, see Mariton [Mar90], Costa, Fragoso, and Marques [CFM05]). Markov jump problems seem to be particularly well suited for the application of approximation in value space, rollout schemes, and the use of certainty equivalence.

The idea of variance reduction in the context of rollout (Section 2.7.4) was proposed by the author in the paper [Ber97b]. See also the DP textbook [Ber17a], Section 6.5.2.

The paper by Chang, Hu, Fu, and Marcus [CHF05], and the 2007

first edition of their monograph proposed and analyzed adaptive sampling in connection with DP, as well as early forms of Monte Carlo tree search, including statistical tests to control the sampling process (a second edition, [CHF13], appeared in 2013). The name "Monte Carlo tree search" has become popular, and in its current use, it encompasses a variety of methods that involve adaptive sampling, rollout, and extensions to sequential games. We refer to the papers by Coulom [Cou06], and Chang et al. [CHF13], the discussion by Fu [Fu17], and the survey by Browne et al. [BPW12].

Statistical tests for adaptive sampling has been inspired by works on multiarmed bandit problems; see Lai and Robbins [LaR85], Agrawal [Agr95], Burnetas and Katehakis [BuK97], Meuleau and Bourgine [MeB99], Auer, Cesa-Bianchi, and Fischer [ACF02], Kocsis and Szepesvari [KoS06], Dimitrakakis and Lagoudakis [DiL08], Audibert, Munos, and Szepesvari [AMS09], and Munos [Mun14]. The book by Lattimore and Szepesvari [LaS20] focuses on multiarmed bandit methods, and provides an extensive account of the UCB rule. For recent work on the theoretical properties of the UCB and UCT rules, see Shah, Xie, and Xu [SXX22], and Chang [Cha24].

Adaptive sampling and MCTS may be viewed within the context of a broader class of on-line lookahead minimization techniques, sometimes called *on-line search* methods. These techniques are based on a variety of ideas, such as random search and intelligent pruning of the lookahead tree. One may naturally combine them with approximation in value space and (possibly) rollout, although it is not necessary to do so (the multistep minimization horizon may extend to the terminal time N). For representative works, some of which apply to continuous spaces problems, including POMDP, see Peret and Garcia [PeG04], and Silver and Veness [SiV10]. The multistep lookahead approximation ideas of Section 2.4 may also be viewed within the context of on-line search methods.

Another rollout idea for stochastic problems, which we have not discussed in this book, is the *open-loop feedback controller* (OLFC), a suboptimal control scheme that dates to the 60s; see Dreyfus [Dre65]. The OLFC applies to POMDP as well, and uses an open-loop optimization over the future evolution of the system. In particular, it uses the current information vector I_k to determine the belief state b_k. It then solves the open-loop problem of minimizing

$$E\left\{g_N(x_N) + \sum_{i=k}^{N-1} g_i(x_i, u_i, w_i) \;\Big|\; I_k\right\}$$

subject to the constraints

$$x_{i+1} = f_i(x_i, u_i, w_i), \qquad u_i \in U_i, \qquad i = k, k+1, \ldots, N-1,$$

and applies the first control $\overline{u}_k$ in the optimal open-loop control sequence $\{\overline{u}_k, \overline{u}_{k+1}, \ldots, \overline{u}_{N-1}\}$. It is easily seen that the OLFC is a rollout algorithm

that uses as base policy the optimal open-loop policy for the problem (the one that ignores any state or observation feedback).

For a detailed discussion of the OLFC, we refer to the author's survey paper [Ber05a] (Section 4) and DP textbook [Ber17a] (Section 6.4.4). The survey [Ber05a] discusses also a generalization of the OLFC, called *partial open-loop-feedback-control*, which calculates the control input on the basis that *some* (but not necessarily all) of the observations will in fact be taken in the future, and the remaining observations will not be taken. This method often allows one to deal with those observations that are troublesome and complicate the solution, while taking into account the future availability of other observations that can be reasonably dealt with. A computational case study for hydrothermal power system scheduling is given by Martinez and Soares [MaS02]. A variant of the OLFC, which also applies to minimax control problems, is given in the author's paper [Ber72b], together with a proof of a cost improvement property over the optimal open-loop policy.

Section 2.8: The role of stochastic programming in providing a link between stochastic DP and continuous spaces deterministic optimization is well known; see the texts by Birge and Louveaux [BiL97], Kall and Wallace [KaW94], and Prekopa [Pre95], and the survey by Ruszczynski and Shapiro [RuS03]. Stochastic programming has been applied widely, and there is much to be gained from its combination with RL. The material of this section comes from the author's rollout book [Ber20a], Section 2.5.2. For a computational study that has tested the ideas of this section on a problem of maintenance scheduling, see Hu et al. [HWP22]. For a related application, see Gioia, Fadda, and Brandimarte [GFB24].

Section 2.9: The multiagent rollout algorithm was proposed in the author's papers [Ber19c], [Ber20b]. The paper [Ber21a] provides an extensive overview of this research. For followup work, see the sources given for Section 1.6 of Chapter 1.

Section 2.10: The material on rollout for Bayesian optimization and sequential estimation comes from a recent paper by the author [Ber22d]. This work is also the basis for the adaptive control material of Section 2.11, and has been included in the book [Ber22a]. The paper by Bhambri, Bhattacharjee, and Bertsekas [BBB22] discusses this material for the case of a deterministic system, applies rollout to sequential decoding in the context of the challenging Wordle puzzle, and provides an implementation using some popular base heuristics, with performance that is very close to optimal. For related work see Loxley and Cheung [LoC23].

Section 2.11: The POMDP framework for adaptive control dates to the 60s, and has stimulated substantial theoretical investigations; see Mandl [Man74], Borkar and Varaiya [BoV79], and the survey by Kumar [Kum85]. Some of the pitfalls of performing parameter estimation while simultane-

ously applying adaptive control have been described by Borkar and Varaiya [BoV79], and by Kumar [Kum83]; see [Ber17a], Section 6.8 for a related discussion.

The papers just mentioned have proposed on-line estimation of the unknown parameter θ and the use at each time period of a policy that is optimal for the current estimate. The papers provide nontrivial analyses that assert asymptotic optimality of the resulting adaptive control schemes under appropriate conditions. If parameter estimation schemes are similarly used in conjunction with rollout, as suggested in Section 2.11 [cf. Eq. (2.105)], one may conjecture that an asymptotic cost improvement property can be proved for the rollout policy, again under appropriate conditions.

Section 2.12: The treatment of sequential minimax problems by DP has a long history. For some early influential works, see Blackwell and Girshick [BlG54], Shapley [Sha53], and Witsenhausen [Wit66]. In minimax control problems, the maximizer is assumed to make choices with perfect knowledge of the minimizer's policy. If the roles of maximizer and minimizer are reversed, i.e., the maximizer has a policy (a sequence of functions of the current state) and the minimizer makes choices with perfect knowledge of that policy, the minimizer gains an advantage, the problem may genuinely change, and the optimal value may be reduced. Thus "min-max" and "max-min" are generally two different problems. In classical two-person zero-sum game theory, however, the main focus is on situations where the min-max and max-min are equal. By contrast, in engineering worst case design, the min-max and max-min values are typically unequal.

There is substantial literature on sequential zero-sum games in the context of DP, often called *Markov games*. The classical paper by Shapley [Sha53] addresses discounted infinite horizon games. A PI algorithm for finite-state Markov games was proposed by Pollatschek and Avi-Itzhak [PoA69], and was interpreted as a Newton method for solving the associated Bellman equation. They have also shown that the algorithm may not converge to the optimal cost function. Computational studies have verified that the Pollatschek and Avi-Itzhak algorithm converges much faster than its competitors, *when it converges* (see Breton et al. [BFH86], and also Filar and Tolwinski [FiT91], who proposed a modification of the algorithm). Related methods have been discussed for Markov games by van der Wal [Van78] and Tolwinski [Tol89]. Raghavan and Filar [RaF91], and Filar and Vrieze [FiV96] provide extensive surveys of the research up to 1996.

The author's paper [Ber21b] has explained the reason behind the unreliable behavior of the Pollatschek and Avi-Itzhak algorithm. This explanation relies on the Newton step interpretation of PI given in Chapter 1: in the case of Markov games, the Bellman operator does not have the concavity property that is typical of one-player games. The paper [Ber21b] has also provided a modified algorithm with solid convergence properties, which applies to very general types of sequential zero-sum games and minimax

control. The algorithm is based on the idea of constructing a special type of uniform contraction mapping, whose fixed point is the solution of the underlying Bellman equation; this idea was first suggested for discounted and stochastic shortest path problems by Bertsekas and Yu [BeY12], [YuB13]. Because the uniform contraction property is with respect to a sup-norm, the corresponding PI algorithm is convergent even under a totally asynchronous implementation. The algorithm, its variations, and the analysis of the paper [Ber21b] were incorporated as Chapter 5 in the 3rd edition of the author's abstract DP book [Ber22b].

The paper by Yu [Yu14] provides an analysis of stochastic shortest path games, where the termination state may not be reachable under some policies, following earlier work by Patek and Bertsekas [PaB99]. The paper [Yu14] also includes a rigorous analysis of the Q-learning algorithm for stochastic shortest path games (without any cost function approximation). The papers by Perolat et al. [PSP15], [PPG16], and the survey by Zhang, Yang, and Basar [ZYB21] discuss alternative RL methods for games.

The author's paper [Ber19b] develops VI, PI, and Dijkstra-like finitely terminating algorithms for exact solution of "robust shortest path planning" problems, which involve finding a shortest path assuming adversarial uncertainty in the state transitions. The paper also discusses related rollout algorithms for approximate solution.

Combining approximation in value and policy space for minimax control (Section 2.12.3) and noncooperative games (Section 2.12.5) is formally presented here for the first time, but it is a natural idea, which is likely to have occurred to other researchers. The computer chess methodology of Section 2.12.4 is based on this idea, and was introduced in the paper by Gundawar, Li, and Bertsekas [GLB24], as an application of model predictive control to computer chess.

E X E R C I S E S

2.1 (A Traveling Salesman Rollout Example with a Sequentially Improving Heuristic)

Consider the traveling salesman problem of Example 1.2.3 and Fig. 1.2.11, and the rollout algorithm starting from city A.

(a) Assume that the base heuristic is chosen to be the farthest neighbor heuristic, which completes a partial tour by successively moving to the farthest neighbor city not visited thus far. Show that this base heuristic is sequentially consistent. What are the tours produced by this base heuristic and the corresponding rollout algorithm? *Answer*: The base heuristic will produce the tour A→AD→ADB→ADBC→A with cost 45. The rollout algorithm will produce the tour A→AB→ABD→ABDC→A with cost 13.

(b) Assume that the base heuristic at city A is the nearest neighbor heuristic, while at the partial tours AB, AC, and AD it is the farthest neighbor heuristic. Show that this base heuristic is sequentially improving but not sequentially consistent. Compute the final tour generated by rollout.

Solution of part (b): Clearly the base heuristic is not sequentially consistent, since from A it generates

$$\text{A}\rightarrow\text{AC}\rightarrow\text{ACD}\rightarrow\text{ACDB}\rightarrow\text{A},$$

but from AC it generates

$$\text{AC}\rightarrow\text{ACB}\rightarrow\text{ACBD}\rightarrow\text{A}.$$

However, it is seen that the sequential improvement criterion (2.13) holds at each of the states A, AB, AC, and AD (and also trivially for the remaining states).

The base heuristic at A is the nearest neighbor heuristic so it generates

$$\text{A}\rightarrow\text{AC}\rightarrow\text{ACD}\rightarrow\text{ACDB}\rightarrow\text{A with cost 28.}$$

The rollout algorithm at state A looks at the three successor states AB, AC, AD, and runs the farthest neighbor heuristic from each, and generates:

$$\text{A}\rightarrow\text{AB}\rightarrow\text{ABD}\rightarrow\text{ABDC}\rightarrow\text{A with cost 13,}$$

$$\text{A}\rightarrow\text{AC}\rightarrow\text{ACB}\rightarrow\text{ACBD}\rightarrow\text{A with cost 45,}$$

$$\text{A}\rightarrow\text{AD}\rightarrow\text{ADB}\rightarrow\text{ADBC}\rightarrow\text{A with cost 45,}$$

so the rollout algorithm will move from A to AB.

Then the rollout algorithm looks at the two successor states ABC, ABD, and runs the base heuristic (whatever that may be; it does not matter) from each. The paths generated are:

$$\text{AB}\rightarrow\text{ABC}\rightarrow\text{ABCD}\rightarrow\text{A with cost 26,}$$

AB→ABD→ABDC→A with cost 13,

so the rollout algorithm will move from AB to ABD.

Thus the final tour generated by the rollout algorithm is

A→AB→ABD→ABDC→A, with cost 13.

2.2 (A Generic Example of a Base Heuristic that is not Sequentially Improving)

Consider a rollout algorithm for a deterministic problem with a base heuristic that produces an optimal control sequence at the initial state x_0, and uses the (optimal) first control u_0 of this sequence to move to the (optimal) next state x_1. Suppose that the base heuristic produces a strictly suboptimal sequence from every successor state $x_2 = f_1(x_1, u_1)$, $u_1 \in U_1(x_1)$, so that the rollout yields a control u_1 that is strictly suboptimal. Show that the trajectory produced by the rollout algorithm starting from the initial state x_0 is strictly inferior to the one produced by the base heuristic starting from x_0, while the sequential improvement condition does not hold.

2.3 (Computational Exercise - Parking with Problem Approximation and Rollout)

In this computational exercise we consider a more complex, imperfect state information version of the one-directional parking problem of Example 1.6.1. Recall that in this problem a driver is looking for a free parking space in an area consisting of N spaces arranged in a line, with a garage at the end of the line (space N). The driver starts at space 0 and traverses the parking spaces sequentially, i.e., from each space he/she goes to the next space, up to when he/she decides to park in space k at cost $c(k)$, if space k is free. Upon reaching the garage, parking is mandatory at cost C.

In Example 1.6.1, we assumed that the driver knows the probabilities $p(k+1), \ldots, p(N-1)$ of the parking spaces $(k+1), \ldots, (N-1)$, respectively, being free. Under this assumption, the state at stage k is either the termination state t (if already parked), or it is F (location k free), or it is $\overline{F}$ (location k taken), and the DP algorithm has the form

$$J_k^*(F) = \begin{cases} \min\Big[c(k),\, p(k+1)J_{k+1}^*(F) + \big(1-p(k+1)\big)J_{k+1}^*(\overline{F})\Big] & \text{if } k < N-1, \\ \min\big[c(N-1),\, C\big] & \text{if } k = N-1, \end{cases} \tag{2.119}$$

$$J_k^*(\overline{F}) = \begin{cases} p(k+1)J_{k+1}^*(F) + \big(1-p(k+1)\big)J_{k+1}^*(\overline{F}) & \text{if } k < N-1, \\ C & \text{if } k = N-1, \end{cases} \tag{2.120}$$

for the states other than the termination state t, while for t we have $J_k^*(t) = 0$ for all k.

We will now consider the more complex variant of the problem where the probabilities $p(0), \ldots, p(N-1)$ do not change over time, but are unknown to

the driver, so that he/she cannot use the exact DP algorithm (2.119)-(2.120). Instead, the driver considers a one-step lookahead approximation in value space scheme, which uses empirical estimates of these probabilities that are based on the ratio $\frac{f_k}{k+1}$, where f_k is the number of free spaces seen up to space k, after the free/taken status of spaces $0,\ldots,k$ has been observed. In particular, these empirical estimates are given by

$$b_k(m,f_k)=\gamma\bar{p}(m)+(1-\gamma)\frac{f_k}{k+1},\qquad m=k+1,\ldots,N-1, \tag{2.121}$$

where f_k is the number of free spaces seen up to space k, and γ and $\bar{p}(m)$ are fixed numbers between 0 and 1. Of course the values f_k observed by the driver evolve according to the true (and unknown) probabilities $p(0),\ldots,p(N-1)$ according to

$$f_{k+1}=\begin{cases} f_k+1 & \text{with probability } p(k+1),\\ f_k & \text{with probability } 1-p(k+1).\end{cases} \tag{2.122}$$

For the solution of this exercise you may assume any reasonable values you wish for N, $p(m)$, $\bar{p}(m)$, and γ. Recommended values are $N\geq 100$, and probabilities $p(m)$ and $\bar{p}(m)$ that are nonincreasing with m.

The decision made by the approximation in value space scheme is to park at space k if and only if it is free and in addition

$$c(k)\leq b_k(k+1,f_k)\tilde{J}_{k+1}(F)+\big(1-b_k(k+1,f_k)\big)\tilde{J}_{k+1}(\overline{F}), \tag{2.123}$$

where $\tilde{J}_{k+1}(F)$ and $\tilde{J}_{k+1}(\overline{F})$ are the cost-to-go approximations from stage $k+1$. Consider the following two different methods to compute $\tilde{J}_{k+1}(F)$ and $\tilde{J}_{k+1}(\overline{F})$ for use in Eq. (2.123):

(1) Here the approximate cost function values $\tilde{J}_{k+1}(F)$ and $\tilde{J}_{k+1}(\overline{F})$ are obtained by using problem approximation, whereby at time k it is assumed that the probabilities of free/taken status at the future spaces $m=k+1,\ldots,N-1$ are $b_k(m,f_k)$, $m=k+1,\ldots,N-1$, as given by Eq. (2.121).

More specifically, $\tilde{J}_{k+1}(F)$ and $\tilde{J}_{k+1}(\overline{F})$ are obtained by solving optimally the problem whereby we use the probabilities $b_k(m,f_k)$ of Eq. (2.121) in place of the unknown $p(m)$ in the DP algorithm (2.119)-(2.120):

$$\tilde{J}_{k+1}(F)=\hat{J}_{k+1}(F),\qquad \tilde{J}_{k+1}(\overline{F})=\hat{J}_{k+1}(\overline{F}),$$

where $\hat{J}_{k+1}(F)$ and $\hat{J}_{k+1}(\overline{F})$ are given at the last step of the DP algorithm

$$\hat{J}_{N-1}(F)=\min\big[c(N-1),\,C\big],\qquad \hat{J}_{N-1}(\overline{F})=C,$$

$$\hat{J}_m(F)=\min\big[c(m),\,b_k(m+1,f_k)\hat{J}_{m+1}(F)+\big(1-b_k(m+1,f_k)\big)\hat{J}_{m+1}(\overline{F})\big],$$
$$\text{if } k<m<N-1,$$

$$\hat{J}_m(\overline{F})=b_k(m+1,f_k)\hat{J}_{m+1}(F)+\big(1-b_k(m+1,f_k)\big)\hat{J}_{m+1}(\overline{F}),$$
$$\text{if } k<m<N-1.$$

(2) Here for each k, the approximate cost function values $\tilde{J}_{k+1}(F)$ and $\tilde{J}_{k+1}(\overline{F})$ are obtained by using rollout with a greedy base heuristic (park as soon as possible), and Monte Carlo simulation. In particular, according to this greedy heuristic, we have $\tilde{J}_{k+1}(F) = c(k+1)$. To compute $\tilde{J}_{k+1}(\overline{F})$ we generate many random trajectories by running the greedy heuristic forward from space $k+1$ assuming the probabilities $b_k(m+1, f_k)$ of Eq. (2.121) in place of the unknown $p(m+1)$, $m = k+1, \ldots, N-1$, and we average the cost results obtained.

(a) Use Monte Carlo simulation to compute the expected cost from spaces $0, \ldots, N-1$, when using each of the two schemes (1) and (2).

(b) Compare the performance of the schemes of part (a) with the following:

(i) The optimal expected costs $J_k^*(F)$ and $J_k^*(\overline{F})$ from $k = 0, \ldots, N-1$, using the DP algorithm (2.119)-(2.120), and the probabilities $p(m)$, $m = 0, \ldots, N-1$, that you used for the random generation of the numbers of free spaces f_k [cf. Eq. (2.122)].

(ii) The expected costs $\hat{J}_k(F)$ and $\hat{J}_k(\overline{F})$ from $k = 0, \ldots, N-1$ that are attained by using the greedy base heuristic. Argue that these are given by

$$\hat{J}_k(F) = c(k), \qquad k = 0, \ldots, N-1,$$

$$\hat{J}_k(\overline{F}) = p(k+1)c(k+1) + \big(1 - p(k+1)\big)\hat{J}_{k+1}(\overline{F}), \qquad k = 0, \ldots, N-2,$$

$$\hat{J}_{N-1}(\overline{F}) = C.$$

(c) Argue that scheme (1) becomes superior to scheme (2) in terms of cost attained as $\gamma \approx 1$ and $\bar{p}(m) \approx p(m)$. Are your computational results in rough agreement with this assertion?

(d) Argue that as $\gamma \approx 0$ and $N >> 1$, scheme (1) becomes superior to scheme (2) in terms of cost attained from parking spaces $k >> 1$.

(e) What happens if the probabilities $p(m)$ do not change much with m?

2.4 (Breakthrough Problem with a Random Base Heuristic)

Consider the breakthrough problem of Example 2.3.2 with the difference that instead of the greedy heuristic, we use the *random* heuristic, which at a given node selects one of the two outgoing arcs with equal probability. Denote by

$$D_k = p^k$$

the probability of success of the random heuristic in a graph of k stages, and by R_k the probability of success of the corresponding rollout algorithm. Show that for all k

$$R_k = p(2-p)R_{k-1} + p^2 D_{k-1}(1 - R_{k-1}).$$

and that

$$\frac{R_k}{D_k} = (2-p)\frac{R_{k-1}}{D_{k-1}} + p(1 - R_{k-1}).$$

Conclude that R_k/D_k increases exponentially with k.

2.5 (Breakthrough Problem with Truncated Rollout)

Consider the breakthrough problem of Example 2.3.2 and consider a truncated rollout algorithm that uses a greedy base heuristic with ℓ-step lookahead. This is the same algorithm as the one described in Example 2.3.2, except that if both outgoing arcs of the current node at stage k are free, the rollout algorithm considers the two end nodes of these arcs, and from each of them it runs the greedy algorithm for $\min\{l, N-k-1\}$ steps. Consider a Markov chain with $l+1$ states, where states $i = 0, \ldots, l-1$ correspond to the path generated by the greedy algorithm being blocked after i arcs. State ℓ corresponds to the path generated by the greedy algorithm being unblocked after ℓ arcs.

(a) Derive the transition probabilities for this Markov chain so that it models the operation of the rollout algorithm.

(b) Use computer simulation to generate the probability of a breakthrough, and to demonstrate that for large values of N, the optimal value of ℓ is roughly constant and much smaller than N (this can also be justified analytically, by using properties of Markov chains).

2.6 (Incremental Truncated Rollout Algorithm for Constraint Programming)

Consider a discrete N-stage optimization problem, involving a tree with a root node s that plays the role of an artificial initial state, and N layers of states $x_1 = (u_0), x_2 = (u_0, u_1), \ldots, x_N = (u_0, \ldots, u_{N-1})$, as shown in Fig. 2.1.4. We allow deadend nodes in the graph of this problem, i.e., states that have no successor states, and thus cannot be part of any feasible solution. We also assume that all stage costs as well as the terminal cost are 0. The problem is to find a feasible solution, i.e., a sequence of N transitions through the graph that starts at the initial state s and ends at some node of the last layer of states x_N.

(a) Argue that this is a discrete spaces formulation of a constraint programming problem, such as the one described in Section 2.1.

(b) Describe in detail an incremental rollout algorithm with ℓ-step lookahead minimization and m-step rollout truncation, which is similar to the IMR algorithm of Section 2.4.2 and operates as follows: The algorithm maintains a connected subtree S that contains the initial state s. The base policy at a state x_k either generates a feasible sequence of $\overline{m}$ arcs starting at x_k, where $\overline{m}$ is an integer that satisfies $1 \leq \overline{m} \leq \min\{m, N-k\}$, or determines that such a sequence does not exist. In the former case the node x_k is expanded by adding all of its neighbor nodes to S. In the latter case, the node x_k is deleted from S. The algorithm terminates once a state x_N of the last layer is added to S.

(c) Argue that since the algorithm cannot keep deleting nodes indefinitely, one of two things will eventually happen:

  (1) The graph S will be reduced to just the root node s, proving that there is no feasible solution.

  (2) The algorithm will terminate with a feasible solution.

(d) Suppose that the algorithm is operated so that the selected node x_k at each iteration is a leaf node of S, which is at maximum arc distance from s (the number of arcs of the path connecting s and x_k is maximized). Show that the subtree S always consists of just a path of nodes, together with all the neighbor nodes of the nodes of the path. Conclude that in this case, the algorithm can be implemented so that it requires $O(Nd)$ memory storage, where d is the maximum node degree. How does this algorithm compare with a depth-first search algorithm for finding a feasible solution?

(e) Describe an adaptation of the algorithm of part (d) for the case where most of the arcs have cost 0 but there are some arcs with positive cost.

2.7 (Purchasing Over Time with Multiagent Rollout)

Consider a market that makes available for purchase m products over N time periods, and a buyer that may or may not buy any one of these products subject to cash availability. For each product $i = 1, \ldots, m$ and time period $k = 0, \ldots, N-1$, we denote:

a_k^i: The asking price of product i at time k (the case where product i is unavailable for purchase is modeled by setting a_k^i to ∞).

v_k^i: The value to the buyer of product i at time k.

u_k^i: The decision to buy ($u_k^i = 1$) or not to buy ($u_k^i = 0$) product i at time k.

The conditional distributions $P(a_{k+1}^i \mid a_k^i, u_k^i = 1)$ and $P(a_{k+1}^i \mid a_k^i, u_k^i = 0)$ are given. (Thus when $u_k^i = 1$, product i will be made available at the next time period at a possibly different price a_{k+1}^i; however, it may also be unavailable, i.e., $a_{k+1}^i = \infty$.)

The amount of cash available to the buyer at time k is denoted by c_k, and evolves according to

$$c_{k+1} = c_k - \sum_{i=1}^{m} u_k^i a_k^i.$$

The initially available cash c_0 is a given positive number. Moreover, we have the constraint

$$\sum_{i=1}^{m} u_k^i a_k^i \leq c_k,$$

i.e., the buyer may not borrow to buy products. The buyer aims to maximize the total value obtained over the N time periods, plus the remaining cash at time N:

$$c_N + \sum_{k=0}^{N-1} \sum_{i=1}^{m} u_k^i a_k^i.$$

(a) Formulate the problem as a finite horizon DP problem by identifying the state, control, and disturbance spaces, the system equation, the cost function, and the probability distribution of the disturbance. Write the corresponding exact DP algorithm.

(b) Introduce a suitable base policy and formulate a corresponding multiagent rollout algorithm for addressing the problem.

2.8 (Treasure Hunting Using Adaptive Control and Rollout)

Consider a problem of sequentially searching for a treasure of known value v among n given locations. At each time period we may either select a location i to search at cost $c_i > 0$, or we may stop searching. Moreover, if the search location i is different from the location j where we currently are, we incur an additional switching cost $s_{ij} \geq 0$. If the treasure is at location i, a search at that location will find it with known probability $\beta_i < 1$. Our initial location is given, and the a priori probabilities p_i, $i = 1, \ldots, n$, that the treasure is at location i are also given. We assume that $\sum_{i=1}^{n} p_i < 0$, so there is positive probability that there is no treasure at any one of the n locations.

(a) Formulate the problem as a special case of the adaptive control problem of Section 2.11, with the parameter θ taking one of $(n + 1)$ values, $\theta^0, \theta^1, \ldots, \theta^n$, where θ^0 corresponds to the case where there is no treasure at any location, and θ^i, $i = 1, \ldots, n$, corresponds to the case where the treasure is at location i. Use as state the current location together with the current probability distribution of θ, and use as control the choice between stopping the search or continuing the search at one of the n locations.

(b) Consider the special case where there is only one location. Show that the optimal policy is to continue searching up to the point where the conditional expected benefit of the search falls below a certain threshold, and that the optimal cost can be computed very simply. (The proof is given in Example 4.3.1 of the DP textbook [Ber17a].)

(c) Formulate the rollout algorithm of Section 2.11 with two different base policies:

  (1) A policy that is optimal among the policies that never switch to another location (they continue to search the same location up to stopping).

  (2) A policy that is optimal among the policies that may stay at the current location or may switch to the location that is most likely to contain the treasure according to the current probability distribution of θ.

(d) Implement the preceding two rollout algorithms using reasonable problem data of your choice.

3

Learning Values and Policies

Contents

3.1. Parametric Approximation Architectures . . . . . p. 351
3.1.1. Cost Function Approximation . . . . . p. 352
3.1.2. Feature-Based Architectures . . . . . p. 353
3.1.3. Training of Linear and Nonlinear Architectures . p. 364
3.2. Neural Networks . . . . . p. 372
3.2.1. Training of Neural Networks . . . . . p. 377
3.2.2. Multilayer and Deep Neural Networks . . . . . p. 378
3.3. Learning Cost Functions in Approximate DP . . . . . p. 380
3.3.1. Fitted Value Iteration . . . . . p. 380
3.3.2. Q-Factor Parametric Approximation - Model-Free Implementation . . . . . p. 382
3.3.3. Parametric Approximation in Infinite Horizon Problems - Approximate Policy Iteration . . . . p. 384
3.3.4. Optimistic Policy Iteration with Parametric Q-Factor Approximation - SARSA and DQN . . . . . p. 388
3.3.5. Approximate Policy Iteration for Infinite Horizon POMDP . . . . . p. 390
3.3.6. Advantage Updating - Approximating Q-Factor Differences . . . . . p. 395
3.3.7. Differential Training of Cost Differences for Rollout p. 397
3.4. Learning a Policy in Approximate DP . . . . . p. 399
3.4.1. The Use of Classifiers for Approximation in Policy Space . . . . . p. 400
3.4.2. Policy Iteration with Value and Policy Networks p. 404
3.4.3. Why Use On-Line Play and Not Just Train a Policy Network to Emulate the Lookahead Minimization? p. 406

3.5. Policy Gradient and Related Methods . . . . . p. 408
3.5.1. Gradient Methods for Parametric Cost Optimization . . . . . p. 409
3.5.2. Policy Gradient-Like Methods . . . . . p. 412
3.5.3. Scaling and Proximal Policy Optimization Methods . . . . . p. 417
3.5.4. Random Direction Methods . . . . . p. 424
3.5.5. Random Search and Cross-Entropy Methods . . p. 426
3.5.6. Refining and Retraining Parametric Policies . . p. 428
3.6. Aggregation . . . . . p. 429
3.6.1. Aggregation with Representative States . . . . p. 430
3.6.2. Continuous Control Space Discretization . . . . p. 437
3.6.3. Continuous State Space - POMDP Discretization p. 438
3.6.4. General Aggregation . . . . . p. 440
3.6.5. Types of Aggregation and Error Bounds . . . . p. 443
3.6.6. Aggregation Using Features . . . . . p. 445
3.6.7. Biased Aggregation . . . . . p. 446
3.6.8. Asynchronous Distributed Aggregation . . . . . p. 449
3.7. Notes, Sources, and Exercises . . . . . p. 451

In this chapter, we will discuss the methods and objectives of off-line training through the use of parametric approximation architectures such as neural networks. We begin with a general discussion of parametric architectures and their training in Section 3.1. We then consider the training of neural networks in Section 3.2, and their use in the context of finite horizon approximate DP in Section 3.3. In Sections 3.4 and 3.5, we discuss the training of policies. Finally, in Section 3.6, we discuss aggregation methods.

3.1 PARAMETRIC APPROXIMATION ARCHITECTURES

For the success of approximation in value space, it is important to select a class of lookahead function approximations $\tilde{J}_k$ that is suitable for the problem at hand. In the preceding two chapters we discussed several methods for choosing $\tilde{J}_k$, based mostly on some form of rollout. We will now discuss how $\tilde{J}_k$ can be obtained by off-line training from a parametric class of functions, possibly involving a neural network, with the parameters "optimized" using some algorithm.

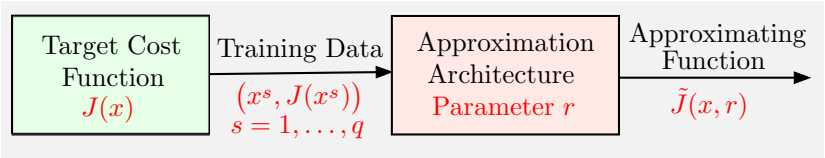

Figure 3.1.1 The general structure for parametric cost approximation. We approximate the target cost function $J(x)$ with a member from a parametric class $\tilde{J}(x, r)$ that depends on a parameter r. We use training data $\big(x^s, J(x^s)\big)$, $s = 1, \ldots, q$, and a form of optimization that aims to find a parameter $\hat{r}$ that "minimizes" the size of the errors $J(x^s) - \tilde{J}(x^s, \hat{r})$, $s = 1, \ldots, q$.

As we have noted in Chapter 1, the most popular structure for parametric cost function approximation involves a target function $J(x)$ that we want to approximate with a member of a parametric class of functions $\tilde{J}(x, r)$ that depends on a parameter r (see Fig. 3.1.1). In particular, we collect training data $\big(x^s, J(x^s)\big)$, $s = 1, \ldots, q$, which we use to determine a parameter $\hat{r}$ that leads to a good "fit" between the data $J(x^s)$ and the predictions $\tilde{J}(x^s, \hat{r})$ of the parametrized function. This is usually done through an optimization approach, aiming to minimize the size of the errors $J(x^s) - \tilde{J}(x^s, \hat{r})$, $s = 1, \ldots, q$.

Approximation of a target policy μ with a policy from a parametric class $\tilde{\mu}(x, r)$ is largely similar. Here, the training data may be obtained, for example, from rollout control calculations, thus enabling the construction of both value and policy networks that can be combined for use in a perpetual rollout scheme. An important difference, however, is that the

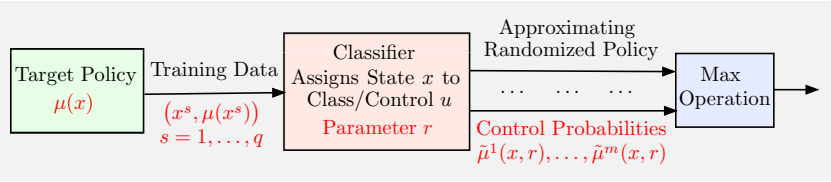

Figure 3.1.2 A general structure for parametric policy approximation for the case where the control space is finite, $U = \{u^1, \ldots, u^m\}$, and its relation to a classification scheme. It produces a randomized policy of the form (3.1), which is converted to a nonrandomized policy through the maximization operation (3.2).

approximate cost values $\tilde{J}(x, r)$ are real numbers, whereas the approximate policy values $\tilde{\mu}(x, r)$ are elements of a control space U. Thus if U consists of m-dimensional vectors, $\tilde{\mu}(x, r)$ consists of m numerical components. In this case the parametric approximation problems for cost functions and for policies are fairly similar, and both involve continuous space approximations.

On the other hand, the case where the control space is finite $U = \{u^1, \ldots, u^m\}$ is quite different. In this case, for any x, $\tilde{\mu}(x, r)$ consists of one of the m possible controls $u^1, \ldots, u^m$. This connects policy space approximation with traditional classification schemes, whereby objects x are classified as belonging to one of the categories $u^1, \ldots, u^m$. In particular, $\tilde{\mu}(x, r)$ *provides the category of* x*, and* $\tilde{\mu}(\cdot, r)$ *can be viewed as a classifier*. Some of the most prominent classification schemes actually produce randomized outcomes, i.e., x is associated with a probability distribution

$$\{\tilde{\mu}^1(x, r), \ldots, \tilde{\mu}^m(x, r)\}, \tag{3.1}$$

which is a randomized policy in our policy approximation context; see Fig. 3.1.2. This is often done for algorithmic convenience, since many optimization methods, including least squares regression, require that the optimization variables are continuous. Then, the randomized policy (3.1) can be converted to a nonrandomized policy using a maximization operation: associate x with the control of maximum probability (cf. Fig. 3.1.2),

$$\tilde{\mu}(x, r) \in \arg \max_{i=1,\ldots,m} \tilde{\mu}^i(x, r). \tag{3.2}$$

We will discuss the use of classification methods for approximation in policy space in Section 3.4, following our discussion of parametric approximation in value space.

3.1.1 Cost Function Approximation

For the remainder of this section, as well as Sections 3.2 and 3.3, we will focus on approximation in value space schemes, where the approximate cost

functions are selected from a parametric class of functions $\tilde{J}_k(x_k, r_k)$ that for each k, depend on the current state x_k and a vector $r_k = (r_{1,k}, \ldots, r_{m_k,k})$ of m_k "tunable" scalar parameters. By adjusting these parameters, the "shape" of $\tilde{J}_k$ can be adjusted to closely approximate a target function, typically the true optimal cost-to-go function J_k^* or the cost-to-go function $J_{k,\pi}$ of a given policy π. The class of functions $\tilde{J}_k(x_k, r_k)$ is called an *approximation architecture*, and the process of choosing the parameter vectors r_k is known as *training* or *tuning* the architecture. We will focus initially on approximation of cost functions, hence the use of the $\tilde{J}_k$ notation. In Section 3.4 we will consider the other major use of parametric approximation architectures, of the form $\tilde{\mu}_k(x_k, r_k)$, where the target function is a control function μ_k that is part of some policy.

The simplest training approach for parametric architectures is to do some form of semi-exhaustive or semi-random search in the space of parameter vectors and adopt the parameters that result in best performance of the associated one-step lookahead controller (according to some criterion). Such methods are mainly practical when the number of parameters is relatively small.

Random search and Bayesian optimization methods have also been used to tune *hyperparameters* of an approximation architecture. Examples include determining the number of layers in a neural network or selecting the number of clusters when partitioning discrete spaces. For further details on these methods, we refer to the research literature.

More systematic training methods are based on numerical optimization, such as a least squares fit that seeks to match the architecture's cost approximation to a "training set," i.e., a large number of state and cost value pairs obtained through some form of sampling. In Sections 3.1-3.3 we will focus primarily on this approach.

3.1.2 Feature-Based Architectures

There is a large variety of approximation architectures, including those based on polynomials, wavelets, radial basis functions, discretization/interpolation schemes, neural networks, and others. A particularly interesting type of cost approximation involves *feature extraction*, a process that maps the state x_k into some vector $\phi_k(x_k)$, called the *feature vector* associated with x_k at time k. The vector $\phi_k(x_k)$ consists of scalar components

$$\phi_{1,k}(x_k), \ldots, \phi_{m_k,k}(x_k),$$

called *features*. A feature-based cost approximation has the form

$$\tilde{J}_k(x_k, r_k) = \hat{J}_k\big(\phi_k(x_k), r_k\big),$$

where r_k is a parameter vector and $\hat{J}_k$ is some function. Thus, the cost approximation depends on the state x_k through its feature vector $\phi_k(x_k)$.

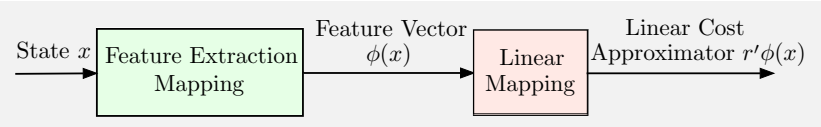

Figure 3.1.3 The structure of a linear feature-based architecture. At time k, we use a feature extraction mapping to generate an input $\phi_k(x_k)$ to a linear mapping defined by a parameter vector r_k.

Note that we are allowing for different features $\phi_k(x_k)$ and different parameter vectors r_k for each stage k. This is necessary for nonstationary problems (e.g., if the state space changes over time), and also to capture the effect of proximity to the end of the horizon. In contrast, for stationary problems with long or infinite horizons where the state space remains constant over k, it is common to use the same features and parameter vectors across all stages. The following discussion can be easily extended to accommodate infinite horizon methods, which we will explore in detail later.

Features are often designed manually, leveraging human intelligence, insights, and experience to capture the most significant characteristics of the current state. There are also systematic ways to construct features, including the use of data and neural networks, which we will discuss shortly. In this section, we provide a brief and selective presentation of architectures.

The rationale behind using features is that optimal cost-to-go functions J_k^* can be complex, nonlinear mappings. By decomposing their complexity into smaller, more manageable components, it becomes feasible to approximate these functions effectively. In particular, if the features encode much of the nonlinearity of J_k^*, we may be able to use a relatively simple architecture $\hat{J}_k$ to approximate J_k^*. For example, with a well-chosen feature vector $\phi_k(x_k)$, a good approximation to the cost-to-go is often provided by *linearly* weighting the features, i.e.,

$$\tilde{J}_k(x_k, r_k) = \hat{J}_k\big(\phi_k(x_k), r_k\big) = \sum_{\ell=1}^{m_k} r_{\ell,k}\phi_{\ell,k}(x_k) = r_k'\phi_k(x_k), \tag{3.3}$$

where $r_{\ell,k}$ and $\phi_{\ell,k}(x_k)$ are the ℓth components of r_k and $\phi_k(x_k)$, respectively, and $r_k'\phi_k(x_k)$ denotes the inner product of r_k and $\phi_k(x_k)$, viewed as column vectors of $\Re^{m_k}$ (a prime denotes transposition, so r_k' is a row vector); see Fig. 3.1.3.

This is called a *linear feature-based architecture*, and the scalar parameters $r_{\ell,k}$ are also called *weights*. Among other advantages, these architectures admit simpler training algorithms than their nonlinear counterparts; see the neuro-dynamic programming (NDP) book [BeT96]. Mathematically, the approximating function $\tilde{J}_k(x_k, r_k)$ can be viewed as a member of the subspace spanned by the features $\phi_{\ell,k}(x_k)$, $\ell = 1, \ldots, m_k$, which

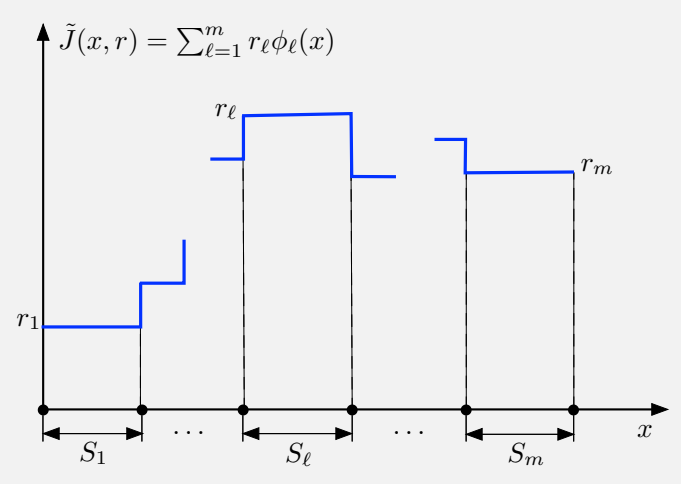

Figure 3.1.4 Illustration of a piecewise constant architecture. The state space is partitioned into subsets $S_1, \ldots, S_m$, with each subset S_ℓ defining the feature

$$\phi_\ell(x) = \begin{cases} 1 & \text{if } x \in S_\ell, \\ 0 & \text{if } x \notin S_\ell, \end{cases} \qquad \ell = 1, \ldots, m,$$

with its own weight r_ℓ.

for this reason are also referred to as *basis functions*. We provide a few examples, where for simplicity we drop the index k.

Example 3.1.1 (Piecewise Constant Approximation)

Suppose that the state space is partitioned into subsets $S_1, \ldots, S_m$, so that every state belongs to one and only one subset. Let the ℓth feature be defined by membership to the set S_ℓ, i.e.,

$$\phi_\ell(x) = \begin{cases} 1 & \text{if } x \in S_\ell, \\ 0 & \text{if } x \notin S_\ell, \end{cases} \qquad \ell = 1, \ldots, m.$$

Consider the architecture

$$\tilde{J}(x,r) = \sum_{\ell=1}^{m} r_\ell \phi_\ell(x),$$

where r is the vector consisting of the m scalar parameters $r_1, \ldots, r_m$. It can be seen that $\tilde{J}(x,r)$ is the piecewise constant function that has value r_ℓ for all states within the set S_ℓ; see Fig. 3.1.4.

The piecewise constant approximation is an example of a linear feature-based architecture that involves exclusively *local features*. These are

features that take a nonzero value only for a relatively small subset of states. Thus a change of a single weight causes a change of the value of $\tilde{J}(x,r)$ for relatively few states x. At the opposite end we have linear feature-based architectures that involve *global features*. These are features that take nonzero values for a large number of states. The following is a common example.

Example 3.1.2 (Polynomial Approximation)

An important case of linear architecture is one that uses polynomial basis functions. Suppose that the state consists of n components $x^1,\ldots,x^n$, each taking values within some range of integers. For example, in a queueing system, x^i may represent the number of customers in the ith queue, where $i=1,\ldots,n$. Suppose that we want to use an approximating function that is quadratic in the components x^i. Then we can define a total of $1+n+n^2$ basis functions that depend on the state $x=(x^1,\ldots,x^n)$ via

$$\phi_0(x)=1, \qquad \phi_i(x)=x^i, \qquad \phi_{ij}(x)=x^ix^j, \quad i,j=1,\ldots,n.$$

A linear approximation architecture that uses these functions is given by

$$\tilde{J}(x,r)=r_0+\sum_{i=1}^{n}r_ix^i+\sum_{i=1}^{n}\sum_{j=1}^{n}r_{ij}x^ix^j,$$

where the parameter vector r has components r_0, r_i, and r_{ij}, with $i,j=1,\ldots,n$. Indeed, any kind of approximating function that is polynomial in the components $x^1,\ldots,x^n$ can be constructed similarly.

A more general polynomial approximation may be based on some other known features of the state. For example, we may start with a feature vector

$$\phi(x)=\big(\phi_1(x),\ldots,\phi_m(x)\big)',$$

and transform it with a quadratic polynomial mapping. In this way we obtain approximating functions of the form

$$\tilde{J}(x,r)=r_0+\sum_{i=1}^{m}r_i\phi_i(x)+\sum_{i=1}^{m}\sum_{j=1}^{m}r_{ij}\phi_i(x)\phi_j(x),$$

where the parameter r has components r_0, r_i, and r_{ij}, with $i,j=1,\ldots,m$. This can also be viewed as a linear architecture that uses the basis functions

$$w_0(x)=1, \qquad w_i(x)=\phi_i(x), \qquad w_{ij}(x)=\phi_i(x)\phi_j(x), \quad i,j=1,\ldots,m.$$

The preceding example architectures are generic in the sense that they can be applied to many different types of problems. Other architectures rely on problem-specific insight to construct features, which are then combined into a relatively simple architecture. We present two examples involving games.

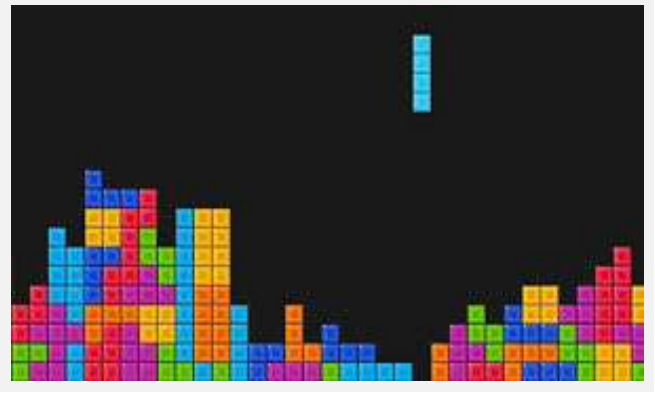

Figure 3.1.5 The board of the tetris game. The squares fill up as blocks of different shapes fall from the top of the grid and are added to the top of the wall. The shapes are generated according to some stochastic process. As a given block falls, the player can move horizontally and rotate the block in all possible ways, subject to the constraints imposed by the sides of the grid and the top of the wall. When a row of full squares is created, this row is removed, the bricks lying above this row move one row downward, and the player scores a point. The player's objective is to maximize the score attained (total number of rows removed) within N steps or up to termination of the game, whichever occurs first.

Example 3.1.3 (Tetris)

Let us consider the game of tetris, which we formulated in Example 1.6.2 as a stochastic shortest path problem with the termination state being the end of the game (see Fig. 3.1.5). The state is the pair of the board position x and the shape of the current falling block y. We viewed as control, the horizontal positioning and rotation applied to the falling block. The optimal cost-to-go function is a vector of huge dimension (there are 2^{200} board positions in a "standard" tetris board of width 10 and height 20). However, it has been successfully approximated in practice by low-dimensional linear architectures.

In particular, the following features have been proposed in the paper by Bertsekas and Ioffe [BeI96]: the heights of the columns, the height differentials of adjacent columns, the wall height (the maximum column height), the number of holes of the board, and the constant 1 (the unit is often included as a feature in cost approximation architectures, as it allows for a constant shift in the approximating function). These features are readily recognized by tetris players as capturing important aspects of the board position.† There

† The use of feature-based approximate DP methods for the game of tetris was first suggested in the paper by Tsitsiklis and Van Roy [TsV96], which introduced just two features (in addition to the constant 1): the wall height and the number of holes of the board. Most studies have used the set of features of [BeI96] described here, but other sets of features have also been used; see [ThS09] and the discussion in [GGS13].

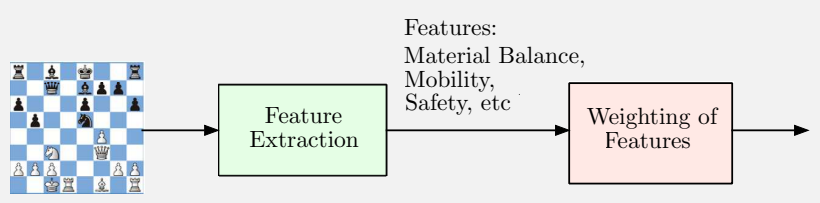

Figure 3.1.6 A feature-based architecture for computer chess.

are a total of 22 features for a "standard" board with 10 columns. Of course the $2^{200} \times 22$ matrix of feature values cannot be stored in a computer, but for any board position, the corresponding row of features can be easily generated, and this is sufficient for implementation of the associated approximate DP algorithms. For recent works involving approximate DP methods and the preceding 22 features, see [Sch13], [GGS13], and [SGG15], which reference several other related papers.

In the works mentioned above the shapes of the falling blocks are stochastically independent. In a more challenging version of the problem, which has not been considered in the literature thus far, successive shapes are correlated. Then the state of the problem would become more complex, since past shapes would be useful in predicting future shapes. As a result, we may need to introduce state estimation and additional features in order to properly deal with the effects of correlations.

Example 3.1.4 (Computer Chess)

Computer chess programs that involve feature-based architectures have existed for many years, and are still used widely (they have been upstaged in the mid-2010s by alternative types of chess programs, which use neural network techniques that will be discussed later). These programs are based on approximate DP for minimax problems, a feature-based parametric architecture, and multistep lookahead.

The fundamental principles on which all computer chess programs (as well as most two-person game programs) are based were laid out by Shannon [Sha50], before Bellman started his work on DP. Shannon proposed multistep lookahead and evaluation of the end positions by means of a "scoring function" (in our terminology this plays the role of a cost function approximation). This function may involve for example the calculation of a numerical value for each of a set of major features of a position that chess players easily recognize (such as material balance, mobility, pawn structure, and other positional factors), together with a method to combine these numerical values into a single score. Shannon then went on to describe various strategies of exhaustive and selective search over a multistep lookahead tree of moves.

We may view the scoring function as a feature-based architecture for evaluating a chess position/state (cf. Fig. 3.1.6). In most computer chess programs, the features are weighted linearly, i.e., the architecture $\tilde{J}(x, r)$ that is used for multistep lookahead is linear [cf. Eq. (3.3)]. In many cases, the weights have been determined manually, by trial and error based on experi-

ence. However, in some programs, the weights have been determined with supervised learning techniques that use examples of grandmaster play, i.e., by adjustment to bring the play of the program as close as possible to the play of chess grandmasters. This is a technique that applies more broadly in AI; see Tesauro [Tes89b], [Tes01].

In a recent computer chess breakthrough, the entire idea of extracting features of a position through human expertise was abandoned in favor of feature discovery through self-play and the use of neural networks. The first program of this type to attain supremacy over humans, as well as over the best computer programs that use human expertise-based features, was AlphaZero (Silver et al. [SHS17]). This program, described in Section 1.1, is based on DP principles of approximate policy iteration and multistep lookahead based on Monte Carlo tree search.

Our next example relates to a methodology for feature construction, where the number of features may increase as more data is collected. For a simple example, consider the piecewise constant approximation of Example 3.1.1, where more pieces are progressively added based on new data, possibly using some form of exploration-exploitation tradeoff.

Example 3.1.5 (Feature Extraction from Data)

We have viewed so far feature vectors $\phi(x)$ as functions of x, obtained through some unspecified process that is based on prior knowledge about the cost function being approximated. On the other hand, features may also be extracted from data. For example, suppose that with some preliminary calculation using data, we have identified some suitable states $x(\ell)$, $\ell = 1, \ldots, m$, that can serve as "anchors" for the construction of Gaussian basis functions of the form

$$\phi_\ell(x) = e^{-\frac{\|x - x(\ell)\|^2}{2\sigma^2}}, \qquad \ell = 1, \ldots, m, \tag{3.4}$$

where σ is a scalar "variance" parameter, and $\|\cdot\|$ denotes the standard Euclidean norm. This type of function is known as a *radial basis function*. It is concentrated around the state $x(\ell)$, and it is weighted with a scalar weight r_ℓ to form a parametric linear feature-based architecture, which can be trained using additional data. Several other types of data-dependent basis functions, such as support vector machines, are used in machine learning, where they are often referred to as *kernels*.

While it is possible to use a preliminary calculation to obtain the anchors $x(\ell)$ in Eq. (3.4), and then use additional data for training, one may also consider enrichment of the set of basis functions simultaneously with training. In this case the number of the basis functions increases as the training data is collected. A motivation here is that the quality of the approximation may increase with additional basis functions. This idea underlies a field of machine learning, known as *kernel methods*, or sometimes *nonparametric methods*.

A further discussion is outside our scope. We refer to the literature; see e.g., books such as Cristianini and Shawe-Taylor [ChS00], [ShC04], Scholkopf and Smola [ScS02], Bishop [Bis06], Kung [Kun14], surveys such as Hofmann,

Scholkopf, and Smola [HSS08], Pillonetto et al. [PDC14], RL-related discussions such as Dietterich and Wang [DiW02], Ormoneit and Sen [OrS02], Engel, Mannor, and Meir [EMM05], Jung and Polani [JuP07], Reisinger, Stone, and Miikkulainen [RSM08], Busoniu et al. [BBD10a], Bethke [Bet10], and recent developments such as Tu et al. [TRV16], Rudi, Carratino, and Rosasco [RCR17], Belkin, Ma, and Mandal [BMM18]. In what follows, for the sake of simplicity, we will focus on parametric architectures with a fixed and given feature vector, since the choice of approximation architecture is somewhat peripheral to our main focus.

The next example considers a feature extraction strategy that is particularly relevant to problems of partial state information.

Example 3.1.6 (Feature Extraction from Sufficient Statistics)

The concept of a sufficient statistic, which originated in inference methodologies, plays an important role in DP. As discussed in Section 1.6, it refers to quantities that summarize all the essential content of the state x_k for optimal control selection at time k.

In particular, consider a partial information context where at time k we have accumulated the *information vector* (also called the *past history*)

$$I_k = (z_0, \ldots, z_k, u_0, \ldots, u_{k-1}),$$

which consists of the past controls $u_0, \ldots, u_{k-1}$ and the state-related measurements $z_0, \ldots, z_k$ obtained at the times $0, \ldots, k$. The control u_k is allowed to depend only on I_k, and the optimal policy is a sequence of the form $\big\{\mu_0^*(I_0), \ldots, \mu_{N-1}^*(I_{N-1})\big\}$. We say that a function $S_k(I_k)$ is a *sufficient statistic at time k* if the control function μ_k^* depends on I_k only through $S_k(I_k)$, i.e., for some function $\hat{\mu}_k$, we have

$$\mu_k^*(I_k) = \hat{\mu}_k\big(S_k(I_k)\big),$$

where μ_k^* is optimal.

There are several examples of sufficient statistics, and they are typically problem-dependent. A trivial possibility is to view I_k itself as a sufficient statistic, and a more sophisticated possibility is to view the *belief state* b_k as a sufficient statistic (this is the conditional probability distribution of x_k given I_k; cf. Section 1.6.6). For a proof that b_k is indeed a sufficient statistic and for a more detailed discussion of other possible sufficient statistics, see [Ber17a], Chapter 4. For a mathematically more advanced discussion, see [BeS78], Chapter 10.

Since a sufficient statistic contains all the relevant information for optimal control purposes, it is natural to introduce features of a given sufficient statistic and to train a corresponding approximation architecture accordingly. Potentially effective features could include some special characteristic of I_k (such as whether some alarm-like "special" event has been observed), or a partial history (such as the last m measurements and controls in I_k, or more sophisticated versions based on the concept of a *finite-state controller* proposed

by White [Whi91], and White and Scherer [WhS94], and further discussed by Hansen [Han98], Kaelbling, Littman, and Cassandra [KLC98], Meuleau et al. [MPK99], Poupart and Boutilier [PoB04], Yu and Bertsekas [YuB08], Saldi, Yuksel, and Linder [SYL17]). In the case where the belief state b_k is used as a sufficient statistic, examples of good features may be a point estimate based on b_k, the variance of this estimate, and other quantities that can be simply extracted from b_k.

The paper by Bhattacharya et al. [BBW20] considers another type of feature vector that is related to the belief state. This is a sufficient statistic, denoted by y_k, which subsumes the belief state b_k, in the sense that b_k can be computed exactly knowing y_k. One possibility is for y_k to be the union of b_k and some identifiable characteristics of the belief state, or some compact representation of the measurement history up to the current time (such as a number of most recent measurements, or the state of a finite-state controller). Even though the information content of y_k is no different from the information content of b_k for the purposes of exact optimization, a sufficient statistic y_k that is specially designed for the problem at hand may lead to improved performance in the presence of cost and policy approximations.

We finally note a related idea, which is to supplement a sufficient statistic with features of other sufficient statistics, and thus obtain an enlarged/richer sufficient statistic. In problem-specific contexts, and in the presence of approximations, this may yield improved results.

Example 3.1.7 (Feature-Based Dimensionality Reduction by Aggregation)

The use of a feature vector $\phi(x)$ to represent the state x in an approximation architecture of the form $\tilde{J}\big(\phi(x), r\big)$ implicitly involves *state aggregation*, i.e., the grouping of states into subsets. We will discuss aggregation in some detail in Section 3.6. Here we will give a summary of a special type of aggregation architecture.

In particular, let us assume that the feature vector can take only a finite number of values, and define for each possible value v, the subset of states S_v whose feature vector is equal to v:

$$S_v = \big\{x \mid \phi(x) = v\big\}.$$

We refer to the sets S_v as the *aggregate states* induced by the feature vector. These sets form a partition of the state space. An approximate cost-to-go function of the form $\tilde{J}\big(\phi(x), r\big)$ is piecewise constant with respect to this partition; that is, it assigns the same cost-to-go value $\tilde{J}(v, r)$ to all states in the set S_v.

An often useful approach to deal with problem complexity in DP is to introduce an "aggregate" DP problem, whose states are some suitably defined feature vectors $\phi(x)$ of the original problem. The precise form of the aggregate problem may depend on intuition and/or heuristic reasoning, based on our understanding of the original problem. Suppose now that the aggregate problem is simple enough to be solved exactly by DP, and let $\hat{J}(v)$

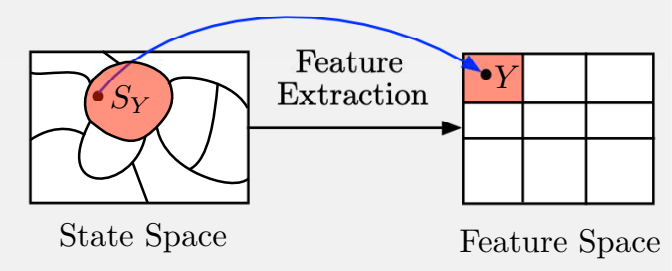

Figure 3.1.7 Feature-based state partitioning using a partition of the space of features. Each set Y of the feature space partition induces a set S_Y of the state space partition that consists of states with "similar" features, i.e., states that map into the same subset of the feature-space partition.

be its optimal cost-to-go when the initial value of the feature vector is v. Then $\hat{J}\big(\phi(x)\big)$ provides an approximation architecture for the original problem, i.e., the architecture that assigns to state x the (exactly) optimal cost-to-go $\hat{J}\big(\phi(x)\big)$ of the feature vector $\phi(x)$ in the aggregate problem. There is considerable freedom on how one formulates and solves aggregate problems. We refer to the DP textbooks [Ber12], [Ber17a], and the RL textbook [Ber19a], Chapter 6, for a detailed treatment; see also the discussion of Section 3.6.

The next example relates to an architecture that is particularly useful when parallel computation is available.

Example 3.1.8 (Feature-Based State Space Partitioning)

A simple method to construct complex and sophisticated approximation architectures, is to partition the state space into several subsets and construct a separate approximation for each subset. For example, by using a separate linear or quadratic polynomial approximation for each subset of the partition, we can construct piecewise linear or piecewise quadratic approximations over the entire state space. Similarly, we may use a separate neural network architecture for each set of the partition. An important issue here is the choice of the method for partitioning the state space. Regular partitions (e.g., grid partitions) may be used, but they often lead to a large number of subsets and very time-consuming computations.

Generally speaking, each subset of the partition should contain "similar" states so that the variation of the optimal cost-to-go over the states of the subset is relatively smooth and can be approximated with smooth functions. An interesting possibility is to use features as the basis for partition. In particular, one may use a more or less regular partition of the space of features, which induces a possibly irregular meaningful partition of the original state space. In this way, each subset of the irregular partition contains states with "similar features;" see Fig. 3.1.7.

As an illustration consider the game of chess. The state here consists of the board position, but the nature of the position evolves through distinct phases: opening, middlegame, and endgame. Each of these phases

may be affected differently by special features of the position. For example there are several different types of endgames (rook endgames, king-and-pawn endgames, minor-piece endgames, etc), which are characterized by identifiable features and call for different playing strategies. It would thus make sense to partition the set of chess positions according to their features, and use a separate strategy on each set of the partition. Indeed this is done to some extent in a number of chess programs.

A potential challenge with partitioned architectures is the discontinuity that can occur at the boundaries between subsets. To address this, a variant known as *soft partitioning*, is sometimes used, whereby the subsets of the partition are allowed to overlap and the discontinuity is smoothed out over their intersection. Once function approximations are computed for each subset, the approximate cost-to-go in the overlapping regions is represented as a smoothly varying linear combination of the corresponding subset approximations.

Partitioning and local approximations can also be used to enhance the quality of approximation in parts of the space where the target function has some special character. The next example addresses one such possibility.

Example 3.1.9 (Local-Global Architectures)

Suppose that the state space is partitioned into subsets $S_1, \ldots, S_M$ and consider approximations of the form

$$\tilde{J}(x,r) = \hat{J}(x,\hat{r}) + \sum_{m=1}^{M} \sum_{k=1}^{K_m} r_m(k)\phi_{k,m}(x), \tag{3.5}$$

where each $\phi_{k,m}(x)$ is a basis function which is local, in the sense that it contributes to the approximation only on the set S_m; that is, it takes the value 0 for $x \notin S_m$. Here $\hat{J}(x,\hat{r})$ is an architecture of the type discussed earlier. The parameter vector r consists of $\hat{r}$ and the coefficients $r_m(k)$ of the basis functions. Thus the portion $\hat{J}(x,\hat{r})$ of the architecture is used to capture "global" aspects of the target function, while each portion

$$\sum_{k=1}^{K_m} r_m(k)\phi_{k,m}(x)$$

is used to capture aspects of the target function that are "local" to the subset S_m.

Note that the training of local-global approximation architectures may potentially be enhanced by using specialized algorithms. The NDP book [BeT96] (Section 3.1.3) discusses computational methods that are tailored to the local-global structure.

Architectures with Automatic Feature Construction

Unfortunately, identifying an adequate set of features can be challenging, so it is important to have methods that construct features automatically, to

supplement any existing ones. Indeed, some architectures are specifically designed to function without pre-existing knowledge of suitable features. The kernel methods of Example 3.1.5 are one such approach. Another very popular possibility is *neural networks*, which we will describe in Section 3.2. Some of these architectures involve training that constructs simultaneously both the feature vectors $\phi(x)$ and the parameter vectors r that weigh them.

Importantly, architectures that generate features automatically can still incorporate additional features based on prior knowledge or problem-specific insights. For example, these architectures can take as input not only the state x, but also supplementary hand-crafted features deemed relevant for the problem. Another possibility is to combine automatically constructed features with other a priori known good features into a (mixed) linear architecture that involves both types of features. The weights of the latter linear architecture may be obtained with a separate second stage training process, following the first stage training process that automatically constructs suitable features using a nonlinear architecture such as a neural network. This approach can enhance the overall effectiveness of the approximation by merging the strengths of automatic feature construction with domain-specific expertise.

3.1.3 Training of Linear and Nonlinear Architectures

In this section, we discuss briefly the training process of choosing the parameter vector r of a parametric architecture $\tilde{J}(x, r)$, focusing primarily on incremental gradient methods. The most common type of training is based on a least squares optimization, also known as *least squares regression*. Here a set of state-cost training pairs

$$(x^s, \beta^s), \qquad s = 1, \ldots, q,$$

called the *training set*, is collected and r is determined by solving the problem

$$\min_r \sum_{s=1}^{q} \big(\tilde{J}(x^s, r) - \beta^s\big)^2. \tag{3.6}$$

Thus r is chosen to minimize the sum of squared errors between the sample costs β^s and the architecture-predicted costs $\tilde{J}(x^s, r)$. Here there is some target cost function J that we aim to approximate with $\tilde{J}(\cdot, r)$, and the sample cost β^s is the value $J(x^s)$ plus perhaps some error or "noise."

The cost function of the training problem (3.6) is generally nonconvex, and can be quite complicated. This may pose challenges, since there may exist multiple local minima. However, for a linear architecture the cost function is convex quadratic, and the training problem admits a closed-form solution. In particular, for the linear architecture

$$\tilde{J}(x, r) = r'\phi(x),$$

the problem becomes

$$\min_r \sum_{s=1}^{q} \big(r'\phi(x^s) - \beta^s\big)^2.$$

By setting the gradient of the quadratic objective to 0, we obtain

$$\sum_{s=1}^{q} \phi(x^s)\big(r'\phi(x^s) - \beta^s\big) = 0,$$

or

$$\sum_{s=1}^{q} \phi(x^s)\phi(x^s)'r = \sum_{s=1}^{q} \phi(x^s)\beta^s.$$

Thus by matrix inversion we obtain the minimizing parameter vector

$$\hat{r} = \left(\sum_{s=1}^{q} \phi(x^s)\phi(x^s)'\right)^{-1} \sum_{s=1}^{q} \phi(x^s)\beta^s. \tag{3.7}$$

If the inverse above does not exist, an additional quadratic in r, called a *regularization* function, is added to the least squares objective to deal with this, and also to help with other issues to be discussed later. A singular value decomposition approach may also be used to deal with the matrix inversion issue; see the NDP book [BeT96], Section 3.2.2.

Thus a linear architecture has the important advantage that the training problem can be solved exactly and conveniently with the formula (3.7) (of course it may be solved by any other algorithm that is suitable for linear least squares problems, including iterative algorithms). By contrast, if we use a nonlinear architecture, such as a neural network, the associated least squares problem is nonquadratic and also nonconvex, so it is hard to solve in principle. Despite this fact, through a combination of sophisticated implementation of special gradient algorithms, called *incremental*, and powerful computational resources, neural network methods have been successful in practice.

Incremental Gradient Methods

We will now discuss briefly special methods for solution of the nonlinear least squares training problem (3.6), assuming a parametric architecture that is differentiable in the parameter vector. This methodology can be properly viewed as a subject in nonlinear programming and iterative algorithms, and as such it can be studied independently of the approximate DP methods of this book. Thus the reader who has already some exposure to the subject may skip to the next section. The author's nonlinear programming textbook [Ber16] and the RL book [Ber19a] provide more detailed presentations.

We view the training problem (3.6) as a special case of the minimization of a sum of component functions

$$f(y) = \sum_{i=1}^{m} f_i(y), \tag{3.8}$$

where each f_i is a differentiable scalar function of the n-dimensional column vector y (this is the parameter vector). Thus we use the more common symbols y and m in place of r and q, respectively, and we replace the squared error terms

$$\big(\tilde{J}(x^s, r) - \beta^s\big)^2$$

in the training problem (3.6) with the generic terms $f_i(y)$.

The (ordinary) gradient method for problem (3.8) generates a sequence $\{y^k\}$ of iterates, starting from some initial guess y^0 for the minimum of the cost function f. It has the form†

$$y^{k+1} = y^k - \gamma^k \nabla f(y^k) = y^k - \gamma^k \sum_{i=1}^{m} \nabla f_i(y^k), \tag{3.9}$$

where γ^k is a positive stepsize parameter. The incremental gradient method is similar to the ordinary gradient method, but uses the gradient of a single component of f at each iteration. It has the general form

$$y^{k+1} = y^k - \gamma^k \nabla f_{i_k}(y^k), \tag{3.10}$$

where i_k is some index from the set $\{1, \ldots, m\}$, chosen by some deterministic or randomized rule. Thus a single component function f_{i_k} is used at iteration k, with great economies in gradient calculation cost over the ordinary gradient method (3.9), particularly when m is large. This is of course a radical simplification, which involves a large approximation error, yet it performs surprisingly well! The idea is to attain faster convergence when far from the solution as we will explain shortly; see the author's books [BeT96], [Ber16], and [Ber19a] for a more detailed discussion.

The method for selecting the index i_k of the component to be iterated on at iteration k is important for the performance of the method. We describe three common rules, the last two of which involve randomization:‡

(1) A *cyclic order*, the simplest rule, whereby the indices are taken up in the fixed deterministic order $1, \ldots, m$, so that i_k is equal to (k modulo

† We use standard calculus notation for gradients; see, e.g., [Ber16], Appendix A. In particular, $\nabla f(y)$ denotes the n-dimensional column vector whose components are the first partial derivatives $\partial f(y)/\partial y_i$ of f with respect to the components $y_1, \ldots, y_n$ of the column vector y.

‡ With these stepsize rules, the incremental gradient method is often called *stochastic gradient* or *stochastic gradient descent* method.

m) plus 1. A contiguous block of iterations involving the components $f_1, \ldots, f_m$ in this order and exactly once is called a *cycle*.

(2) A *uniform random order*, whereby the index i_k is chosen randomly by sampling over all indices with a uniform distribution, independently of the past history of the algorithm. This rule may perform better than the cyclic rule in some circumstances.

(3) A *cyclic order with random reshuffling*, whereby the indices are taken up one by one within each cycle, but their order after each cycle is reshuffled randomly (and independently of the past). This rule is used widely in practice, particularly when the number of components m is modest, for reasons to be discussed later.

Note that in the cyclic cases, it is essential to include all components in a cycle; otherwise some components will be sampled more often than others, leading to a bias in the convergence process. Similarly, it is necessary to sample according to the uniform distribution in the random order case.

Focusing for the moment on the cyclic rule (with or without reshuffling), we note that the motivation for the incremental gradient method is faster convergence: we hope that far from the solution, a single cycle of the method will be as effective as several (as many as m) iterations of the ordinary gradient method (think of the case where the components f_i are similar in structure). Near a solution, however, the incremental method may not be as effective.

In particular, we note that there are two complementary performance issues to consider in comparing incremental and nonincremental methods:

(a) *Progress when far from convergence.* Here the incremental method can be much faster. For an extreme case take m large and all components f_i identical to each other. Then an incremental iteration requires m times less computation than a classical gradient iteration, but gives exactly the same result, when the stepsize is scaled to be m times larger. While somewhat extreme, this example reflects the essential mechanism by which incremental methods can be much superior: far from the minimum a single component gradient will point to "more or less" in the right direction, at least most of the time; see the following example.

(b) *Progress when close to convergence.* Here the incremental method can be inferior. In particular, the ordinary gradient method (3.9) is convergent with a constant stepsize under reasonable assumptions, see e.g., [Ber16]. However, the incremental method requires a diminishing stepsize, and its ultimate rate of convergence can be much slower.

This behavior is illustrated in the following example, first given in the 1995 first edition of the author's nonlinear programming book [Ber16], and the NDP book [BeT96].

Example 3.1.10

Assume that y is a scalar, and that the problem is

$$\begin{aligned} &\text{minimize} \quad f(y) = \tfrac{1}{2}\sum_{i=1}^{m}(c_i y - b_i)^2 \\ &\text{subject to} \quad y \in \Re, \end{aligned}$$

where c_i and b_i are given scalars with $c_i \neq 0$ for all i. The minimum of each of the components $f_i(y) = \frac{1}{2}(c_i y - b_i)^2$ is

$$y_i^* = \frac{b_i}{c_i},$$

while the minimum of the least squares cost function f is

$$y^* = \frac{\sum_{i=1}^{m} c_i b_i}{\sum_{i=1}^{m} c_i^2}.$$

It can be seen that y^* lies within the range of the component minima

$$R = \left[\min_i y_i^*,\, \max_i y_i^*\right],$$

and that for all y *outside* the range R, the gradient

$$\nabla f_i(y) = c_i(c_i y - b_i)$$

has the same sign as $\nabla f(y)$ (see Fig. 3.1.8). As a result, when outside the region R, the incremental gradient method

$$y^{k+1} = y^k - \gamma^k c_{i_k}(c_{i_k} y^k - b_{i_k})$$

approaches y^* at each step, provided the stepsize γ^k is small enough. In fact it can be verified that it is sufficient that

$$\gamma^k \leq \min_i \frac{1}{c_i^2}.$$

However, for y *inside* the region R, the ith step of a cycle of the incremental gradient method need not make progress. It will approach y^* (for small enough stepsize γ^k) only if the current point y^k does not lie in the interval connecting y_i^* and y^*. This induces an oscillatory behavior within the region R, and as a result, the incremental gradient method will typically not converge to y^* unless $\gamma^k \to 0$. By contrast, the ordinary gradient method, which takes the form

$$y^{k+1} = y^k - \gamma \sum_{i=1}^{m} c_i(c_i y^k - b_i),$$

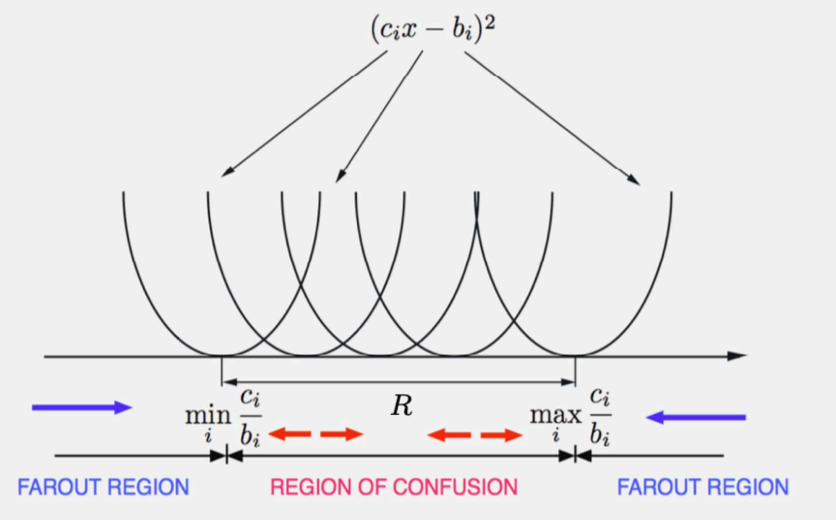

Figure 3.1.8. Illustrating the advantage of incrementalism when far from the optimal solution. The region of component minima

$$R = \left[\min_i y_i^*,\ \max_i y_i^*\right],$$

is labeled as the "region of confusion." It is the region where the method does not have a clear direction towards the optimum. The ith step in an incremental gradient cycle is a gradient step for minimizing $(c_i y - b_i)^2$, so if y lies outside the region of component minima $R = \left[\min_i y_i^*,\ \max_i y_i^*\right]$, (labeled as the "farout region") and the stepsize is small enough, progress towards the solution y^* is made.

can be verified to converge to y^* for any constant stepsize γ with

$$0 < \gamma \leq \frac{1}{\sum_{i=1}^m c_i^2}.$$

However, for y outside the region R, a full iteration of the ordinary gradient method need not make more progress towards the solution than a single step of the incremental gradient method. In other words, with comparably intelligent stepsize choices, *far from the solution (outside R), a single pass through the entire set of cost components by incremental gradient is roughly as effective as m passes by ordinary gradient.*

The preceding example assumes that each component function f_i has a minimum, so that the range of component minima is defined. In cases where the components f_i have no minima, a similar phenomenon may occur, as illustrated by the following example (the idea here is that we may combine several components into a single component that has a minimum).

Example 3.1.11:

Consider the case where f is the sum of increasing and decreasing convex exponentials, i.e.,

$$f_i(y) = a_i e^{b_i y}, \qquad y \in \Re,$$

where a_i and b_i are scalars with $a_i > 0$ and $b_i \neq 0$. Let

$$I^+ = \{i \mid b_i > 0\}, \qquad I^- = \{i \mid b_i < 0\},$$

and assume that I^+ and I^- have roughly equal numbers of components. Let also y^* be the minimum of $\sum_{i=1}^m f_i$.

Consider the incremental gradient method that given the current point, call it y^k, chooses some component f_{i_k} and iterates according to the incremental gradient iteration

$$y^{k+1} = y^k - \alpha^k \nabla f_{i_k}(y^k).$$

Then it can be seen that if $y^k >> y^*$, y^{k+1} will be substantially closer to y^* if $i \in I^+$, and negligibly further away than y^* if $i \in I^-$. The net effect, averaged over many incremental iterations, is that if $y^k >> y^*$, an incremental gradient iteration makes roughly one half the progress of a full gradient iteration, with m times less overhead for calculating gradients. The same is true if $y^k << y^*$. On the other hand as y^k gets closer to y^* the advantage of incrementalism is reduced, similar to the preceding example. In fact in order for the incremental method to converge, a diminishing stepsize is necessary, which will ultimately make the convergence slower than that of the nonincremental gradient method with a constant stepsize.

The discussion of the preceding examples relies on y being one-dimensional, but in many multidimensional problems the same qualitative behavior can be observed. In particular, a pass through the ith component f_i by the incremental gradient method can make progress towards the solution in the region where the component gradient $\nabla f_{i_k}(y^k)$ makes an angle less than 90 degrees with the cost function gradient $\nabla f(y^k)$. If the components f_i are not "too dissimilar," this is likely to happen in a region of points that are far from the optimal solution set. This behavior has been validated in numerous practical contexts, including the training of neural networks (see the following section), where incremental gradient methods, often referred to as *backpropagation methods*, have been widely employed.

Stepsize Choice

The choice of the stepsize γ^k (often referred to as the *learning rate*) plays an important role in the performance of incremental gradient methods. In practice, it is common to use a constant stepsize for a (possibly prespecified) number of iterations, then decrease the stepsize by a certain factor, and

repeat, up to the point where the stepsize reaches a prespecified floor value. An alternative possibility is to use a diminishing stepsize rule of the form

$$\gamma^k = \min\left\{\gamma,\ \frac{\beta_1}{k+\beta_2}\right\},$$

where γ, β_1, and β_2 are some positive scalars. There are also variants of the method that use a constant stepsize throughout, and can be shown to converge to a stationary point of f under reasonable assumptions. In one type of such method the degree of incrementalism gradually diminishes as the method progresses (see the author's paper [Ber97a]). Another incremental approach with similar aims is the aggregated gradient method, which is discussed in the author's textbooks [Ber15a], [Ber16], [Ber19a].

Regardless of whether a constant or a diminishing stepsize is ultimately used, the incremental method must use a much larger stepsize than the corresponding nonincremental gradient method (as much as m times larger, so that the size of the incremental gradient step is comparable to the size of the nonincremental gradient step).

One possibility is to use an adaptive stepsize rule, whereby, roughly speaking, the stepsize is reduced (or increased) when the progress of the method indicates that the algorithm is (or is not) oscillating. Formal ways to implement such stepsize rules with sound convergence properties have been proposed by Tseng [Tse98], and Moriyama, Yamashita, and Fukushima [MYF03].

Diagonal Scaling

The difficulty with stepsize selection may also be addressed with *diagonal scaling*, i.e., using a stepsize γ_j^k that is different for each of the components y_j of y. Second derivatives can be very useful for this purpose. A time-honored method in generic nonlinear programming problems of unconstrained minimization of a function f, is to use diagonal scaling with stepsizes

$$\gamma_j^k = \gamma\left(\frac{\partial^2 f(y^k)}{\partial^2 y_j}\right)^{-1}, \qquad j = 1, \ldots, n,$$

where γ is a constant that is nearly equal to 1 (the second derivatives may also be approximated by gradient difference approximations). However, in least squares training problems [cf. Eq. (3.6)], this type of scaling is inconvenient because of the additive form of f as a sum of a large number of component functions:

$$f(y) = \sum_{i=1}^{m} f_i(y),$$

cf. Eq. (3.8).

Thus incremental versions of diagonal scaling have been suggested in the neural network literature, and have been incorporated in publicly and commercially available software. Two influential papers are Duchi, Hazan, and Singer [DHS11], and Kingma and Ba [KiB14]. The latter paper has introduced the ADAM (Adaptive Moment Estimation) method, which is widely used at present for training neural networks. It adapts the stepsize for each component of the parameter vector r, based on estimates of the first and second moments of the gradient, and incorporates several other ideas that have been proposed in the incremental methods literature over the years.

There are also alternative approaches for scaling the direction of incremental methods to steer it closer to the Newton direction. The RL book [Ber19a] (Section 3.1.3) describes one such method that involves second derivatives, whereby the idea here is to write Newton's method in a format that is well suited to the additive character of the cost function f, and involves low-order matrix inversion. One can then implement diagonal scaling by setting to zero the off-diagonal terms of the inverted matrices, so that the algorithm involves no matrix inversion. There is also another related algorithm, which is based on the Gauss-Newton method and the extended Kalman filter; see the author's paper [Ber96], and the books [BeT96] and [Ber16]. Finally, we note an important scaling approach for stochastic parametric optimization problems, which will be discussed in Section 3.5.2.

3.2 NEURAL NETWORKS

There are several different types of neural networks that can be used for a variety of tasks, such as pattern recognition, classification, image and speech recognition, natural language processing, and others. In this section, we focus on our finite horizon DP context, and the role that neural networks can play in approximating the optimal cost-to-go functions J_k^*. As an example within this context, we may first use a neural network to construct an approximation to J_{N-1}^*. Then we may use this approximation to approximate J_{N-2}^*, and continue this process backwards in time, to obtain approximations to all the optimal cost-to-go functions J_k^*, $k = 1, \ldots, N-1$, as we will discuss in more detail in Section 3.3.

Throughout this section, we will focus on the type of neural network known as a *multilayer perceptron*, which is among the most widely used at present in the RL applications discussed in this book. Naturally, there are variations that are adapted to the problem at hand. For example AlphaZero uses a specialized neural network that takes advantage of the board-like structure of chess and Go to facilitate and expedite the associated computations.

To describe the use of neural networks in finite horizon DP, let us consider the typical stage k, and for convenience drop the index k; the

subsequent discussion applies to each value of k separately. We consider parametric architectures $\tilde{J}(x, v, r)$ of the form

$$\tilde{J}(x, v, r) = r'\phi(x, v) \tag{3.11}$$

that depend on two parameter vectors v and r. Our objective is to select v and r so that $\tilde{J}(x, v, r)$ approximates some target cost function that can be sampled (possibly with some error). The process is to collect a training set that consists of a large number of state-cost pairs (x^s, β^s), $s = 1, \ldots, q$, and to find a function $\tilde{J}(x, v, r)$ of the form (3.11) that matches the training set in a least squares sense, i.e., (v, r) minimizes

$$\sum_{s=1}^{q} \big(\tilde{J}(x^s, v, r) - \beta^s\big)^2.$$

We postpone for later the question of how the training pairs (x^s, β^s) are generated.† Notice the different roles of the two parameter vectors here: *v parametrizes $\phi(x, v)$*, which in some interpretation may be viewed as a feature vector, and *r is a vector of linear weighting parameters for the components of $\phi(x, v)$*.

Single Layer Perceptron

A neural network architecture provides a parametric class of functions $\tilde{J}(x, v, r)$ of the form (3.11) that can be used in the optimization framework just described. The simplest type of neural network is the *single layer perceptron*; see Fig. 3.2.1. Here the state x is encoded as a vector of numerical values $y(x)$ with components $y_1(x), \ldots, y_n(x)$, which is then transformed linearly as

$$Ay(x) + b,$$

where A is an $m \times n$ matrix and b is a vector in $\Re^m$.‡ This transformation is called the *linear layer* of the neural network. We view the components of A and b as parameters to be determined, and we group them together into the parameter vector $v = (A, b)$.

† The least squares training problem used here is based on *nonlinear regression*. This is a classical method for approximating the expected value of a function with a parametric architecture, and involves a least squares fit of the architecture to simulation-generated samples of the expected value. We refer to machine learning and statistics textbooks for more discussion.

‡ The method of encoding x into the numerical vector $y(x)$ is generally problem-dependent, but it can be critical for the success of the training process. We should note also that some of the components of $y(x)$ could be known interesting features of x that can be designed based on problem-specific knowledge.

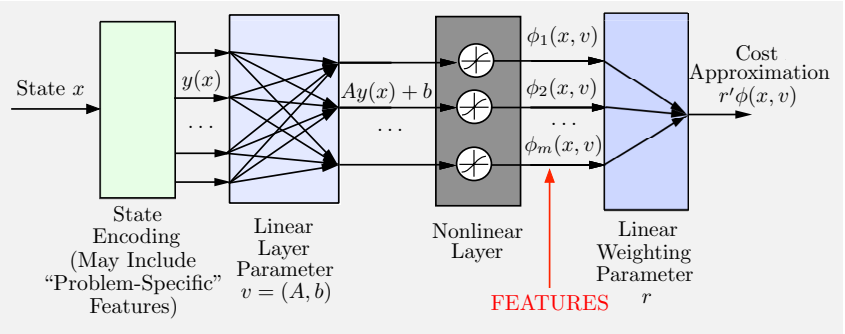

Figure 3.2.1 Schematic illustration of a single layer perceptron, a neural network consisting of a linear layer and a nonlinear layer. It provides a way to compute features of the state, which can be used for cost function approximation. The state x is encoded as a vector of numerical values $y(x)$, which is then transformed linearly as $Ay(x)+b$ in the linear layer. The m scalar output components of the linear layer, become the inputs to nonlinear one-dimensional functions $\sigma : \Re \mapsto \Re$, thus producing the m scalars

$$\phi_\ell(x,v) = \sigma\big((Ay(x)+b)_\ell\big),$$

which can be viewed as features that are in turn linearly weighted with parameters r_ℓ.

Each of the m scalar output components of the linear layer,

$$\big(Ay(x)+b\big)_\ell, \qquad \ell = 1,\ldots,m,$$

becomes the input to a nonlinear differentiable and monotonically increasing function σ that maps scalars to scalars. A simple and popular possibility is the *rectified linear unit* (ReLU for short), which is simply the function $\max\{0,\xi\}$, approximated by a differentiable function σ by some form of smoothing operation; for example $\sigma(\xi) = \ln(1+e^\xi)$, which is illustrated in Fig. 3.2.2. Other functions, used since the early days of neural networks, have the property

$$-\infty < \lim_{\xi\to-\infty} \sigma(\xi) < \lim_{\xi\to\infty} \sigma(\xi) < \infty;$$

see Fig. 3.2.3. Such functions are called *sigmoids*, and some common choices are the *hyperbolic tangent* function

$$\sigma(\xi) = \tanh(\xi) = \frac{e^\xi - e^{-\xi}}{e^\xi + e^{-\xi}},$$

and the *logistic* function

$$\sigma(\xi) = \frac{1}{1+e^{-\xi}}.$$

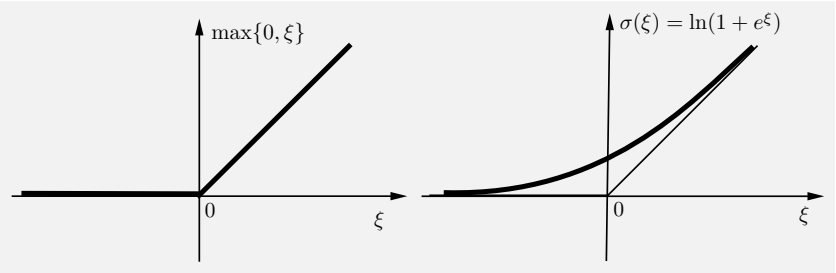

Figure 3.2.2 The rectified linear unit $\sigma(\xi) = \ln(1 + e^{\xi})$. It is the function $\max\{0, \xi\}$ with its corner "smoothed out." Its derivative is $\sigma'(\xi) = e^{\xi}/(1 + e^{\xi})$, and approaches 0 and 1 as $\xi \to -\infty$ and $\xi \to \infty$, respectively.

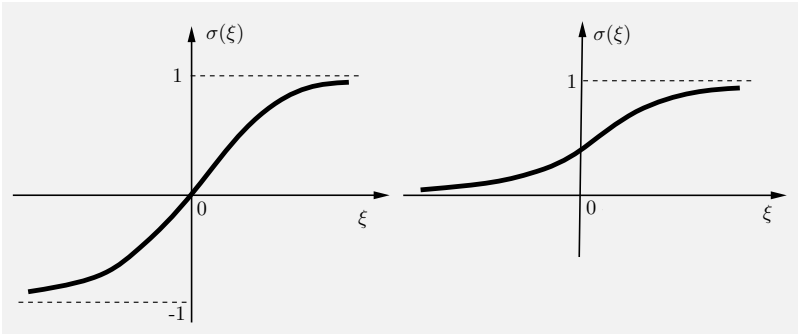

Figure 3.2.3 Some examples of sigmoid functions. The hyperbolic tangent function is on the left, while the logistic function is on the right.

In what follows, we will ignore the character of the function σ (except for differentiability), and simply refer to it as a "nonlinear unit" and to the corresponding layer as a "nonlinear layer."

At the outputs of the nonlinear units, we obtain the scalars

$$\phi_\ell(x, v) = \sigma\big((Ay(x) + b)_\ell\big), \qquad \ell = 1, \ldots, m.$$

One possible interpretation is to view $\phi_\ell(x, v)$ as features of x, which are linearly combined using weights r_ℓ, $\ell = 1, \ldots, m$, to produce the final output

$$\tilde{J}(x, v, r) = \sum_{\ell=1}^{m} r_\ell \phi_\ell(x, v) = \sum_{\ell=1}^{m} r_\ell \sigma\big((Ay(x) + b)_\ell\big).$$

Note that each value $\phi_\ell(x, v)$ depends on just the ℓth row of A and the ℓth component of b, not on the entire vector v. In some cases this motivates placing some constraints on individual components of A and b to achieve special problem-dependent "handcrafted" effects.

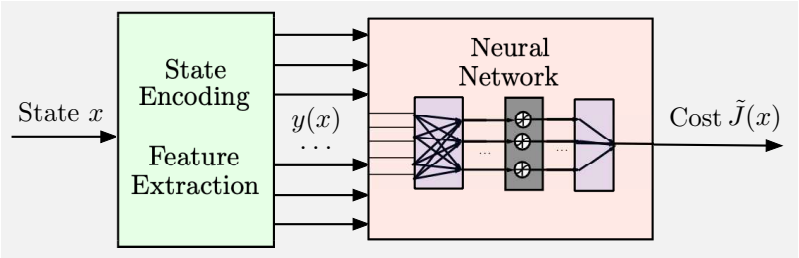

Figure 3.2.4 A nonlinear architecture with a view of the state encoding process as a feature extraction mapping preceding the neural network. The state encoder may also contain tunable parameters.

State Encoding and Direct Feature Extraction

The state-encoding operation that transforms x into the neural network input $y(x)$ can be instrumental in the success of the approximation scheme. Examples of state encodings are components of the state x, numerical representations of qualitative characteristics of x, and more generally features of x, i.e., functions of x that aim to capture "important nonlinearities" of the optimal cost-to-go function. With the latter view of state encoding, we may consider the approximation process as consisting of a feature extraction mapping, followed by a neural network with input the extracted features of x, and output the cost-to-go approximation; see Fig. 3.2.4.

The idea here is that with a well-designed feature extraction mapping, the neural network need not be very complicated and may be trained more easily. This intuition is borne out by simple examples and practical experience. However, as is often the case with neural networks, it is hard to support it with a quantitative analysis.

Universal Approximation Property of Neural Networks

An important question is how well we can approximate the target function J_k^* with a neural network architecture, assuming we can choose the number of the nonlinear units m to be as large as we want. The answer to this question is quite favorable and is provided by the so-called *universal approximation theorem*.

Roughly, the theorem says that assuming that x is an element of a Euclidean space X and $y(x) \equiv x$, a neural network of the form described can approximate arbitrarily closely (in an appropriate mathematical sense), over a compact subset $S \subset X$, any piecewise continuous function $J : S \mapsto \Re$, provided the number m of nonlinear units is sufficiently large. For proofs of the theorem, we refer to Cybenko [Cyb89], Funahashi [Fun89], Hornik, Stinchcombe, and White [HSW89], and Leshno et al. [LLP93]. For additional sources and intuitive explanations we refer to Bishop ([Bis95], pp. 129-130), Jones [Jon90], and the RL textbook [Ber19a], Section 3.2.1.

While the universal approximation theorem provides some reassurance about the adequacy of the neural network structure, it does not predict how many nonlinear units we may need for "good" performance in a given problem. Unfortunately, this is a difficult question to even pose precisely, let alone answer adequately. In practice, one is often reduced to trying increasingly larger values of m until one is convinced that satisfactory performance has been obtained for the task at hand. One may improve on trial-and-error schemes with more systematic hyperparameter search methods, such as Bayesian optimization, and in fact this idea has been used to tune the parameters of the deep network used by AlphaZero.

Experience has shown that in many cases the number of required nonlinear units and the corresponding dimension of the parameter space can be very large, adding significantly to the difficulty of solving the training problem. This has given rise to many suggestions for modifications of the perceptron structure. An important possibility is to concatenate multiple single layer perceptrons so that the output of the nonlinear layer of one perceptron becomes the input to the linear layer of the next, giving rise to deep neural networks, which we will discuss later.

3.2.1 Training of Neural Networks

Given a set of state-cost training pairs (x^s, β^s), $s = 1, \ldots, q$, the parameters of the neural network A, b, and r are obtained by solving the problem

$$\min_{A,b,r} \sum_{s=1}^{q} \left(\sum_{\ell=1}^{m} r_\ell \sigma\left(\big(Ay(x^s) + b\big)_\ell \right) - \beta^s \right)^2. \tag{3.12}$$

Note that the cost function of this problem is generally nonconvex, so there may exist multiple local minima.

In practice it is common to augment the cost function of this problem with a *regularization* function, such as a quadratic in the parameters A, b, and r. This is customary in least squares problems in order to make the problem easier to solve algorithmically. However, in the context of neural network training, regularization is primarily important for a different reason: it helps to avoid *overfitting*, which occurs when the number of parameters of the neural network is relatively large (comparable to the size of the training set). In this case a neural network model matches the training data very well but may not do as well on new data. This is a well-known difficulty, which has been researched extensively in machine learning. It is also the subject of much current research, particularly in the context of deep neural networks, as we will discuss later.

An important issue is to select a method to solve the training problem (3.12). While we can use any unconstrained optimization method that is based on gradients, in practice it is important to take into account the cost function structure of problem (3.12). The salient characteristic of this cost

function is that it is the sum of a potentially very large number q of component functions. This makes the computation of the cost function value of the training problem and/or its gradient very costly. For this reason the incremental methods of Section 3.1.3 are universally used for training.† Experience has shown that these methods can be vastly superior to their nonincremental counterparts in the context of neural network training.

The implementation of the training process has benefited from experience that has been accumulated over time, and has provided guidelines for scaling, regularization, initial parameter selection, and other practical issues; we refer to books on neural networks such as Bishop [Bis95], Bishop and Bishop [BiB24], Goodfellow, Bengio, and Courville [GBC16], and Haykin [Hay08], as well as to the overview paper on deep neural network training by Sun [Sun19], and the references quoted therein. Still, incremental methods can be quite slow, and training may be a time-consuming process. Fortunately, the training is ordinarily done off-line, possibly using parallel computation, in which case computation time may not be a serious issue. Moreover, in practice the neural network training problem typically need not be solved with great accuracy. This is also supported by the Newton step view of approximation in value space, which suggests that great accuracy in the off-line training of the terminal cost approximation is not critically important for good performance of the on-line play algorithm.

3.2.2 Multilayer and Deep Neural Networks

An important generalization of the single layer perceptron architecture involves a concatenation of multiple layers of linear and nonlinear functions; see Fig. 3.2.5. In particular the outputs of each nonlinear layer become the inputs of the next linear layer. In some cases it may make sense to add as additional inputs some of the components of the state x or the state encoding $y(x)$.

In the early days of neural networks practitioners tended to use few nonlinear layers (say one to three). For example, Tesauro's backgammon program and its descendants have performed well with one or two nonlinear layers [PaR12]. However, more recently a lot of success in certain problem domains (including image and speech processing, large language models, as well as approximate DP) has been achieved with *deep neural networks*, which involve a considerably larger number of layers.

† The incremental methods are valid for an arbitrary order of component selection within the cycle, but it is common to randomize the order at the beginning of each cycle. Also, in a variation of the basic method, we may operate on a batch of several components at each iteration, called a *minibatch*, rather than a single component. This has an averaging effect, which reduces the tendency of the method to oscillate and allows for the use of a larger stepsize; see the end-of-chapter references.

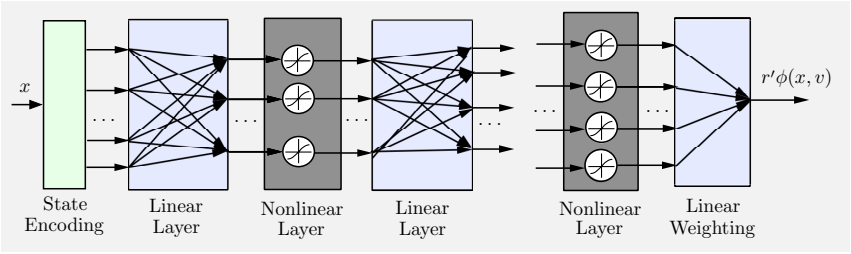

Figure 3.2.5 A deep neural network, with multiple layers. Each nonlinear layer constructs the inputs of the next linear layer.

There are a few questions to consider here. The first has to do with the reason for having multiple nonlinear layers, when a single one is sufficient to guarantee the universal approximation property. Here are some qualitative (and somewhat speculative) explanations:

(a) We may consider the possibility of using linear layers with a particular sparsity pattern, or other structure that embodies special linear operations such as convolution, which may be well-matched to the training problem at hand. Moreover, when such structures are used, the training problem often becomes easier, because the number of parameters in the linear layers is drastically decreased.

(b) If we view the outputs of each nonlinear layer as features, we see that the multilayer network produces a hierarchy of features, where each set of features is a function of the preceding set of features [except for the first set of features, which is a function of the encoding $y(x)$ of the state x]. In the context of specific applications, this hierarchical structure can be exploited to specialize the role of some of the layers and to enhance some characteristics of the state.

(c) Overparametrization (more weights than data, as in a deep neural network) helps to mitigate the detrimental effects of overfitting, and the attendant need for regularization. The explanation for this fascinating phenomenon (observed as early as the late 90s) is the subject of much recent research; see [ZBH16], [BMM18], [BRT18], [SJL18], [ADH19], [BLL19], [HMR19], [MVS19], [SuY19], [Sun19], [DMB21], [HaR21], [VLK21], [ZBH21] for representative works.

We finally note that the use of deep neural networks has been an important factor in the success of the AlphaGo and AlphaZero programs, as well as for large language models. New developments in hardware, software, architectural structures, training methodology, and integration with other types of neural networks such as transformers, are likely to improve the already enormous power of deep neural networks, and to allow the use of ever larger datasets, which are becoming increasingly available.

3.3 LEARNING COST FUNCTIONS IN APPROXIMATE DP

In the context of approximate DP/RL, architectures are mainly used to approximate either cost functions or policies. When a neural network is involved, the terms *value network* and *policy network* are commonly used, respectively.† In this section we will illustrate the use of value networks in finite horizon DP, while in the next section we will discuss the use of policy networks. We will also illustrate in Section 3.3.3 the combined use of policy and value networks within the context of an approximate policy iteration, whereby the policies and their cost functions are approximated by a policy and a value network, respectively, to generate a sequence of (approximately) improved policies. Finally, in Sections 3.3.6 and 3.3.7, we will describe how approximating Q-factor differences or cost differences (rather than Q-factors or costs) can be beneficial within our context of approximation in value space.

3.3.1 Fitted Value Iteration

Let us describe a popular approach for training an approximation architecture $\tilde{J}_k(x_k, r_k)$ for a finite horizon DP problem. The parameter vectors r_k are determined sequentially, starting from the end of the horizon, and proceeding backwards as in the DP algorithm: first r_{N-1} then r_{N-2}, and so on. The algorithm samples the state space for each stage k, and generates a large number of states x_k^s, $s = 1, \ldots, q$. It then determines sequentially the parameter vectors r_k to obtain a good "least squares fit" to the DP algorithm. The method can also be used in the infinite horizon case, in essentially identical form, and it is commonly called *fitted value iteration*.

In particular, each r_k is determined by generating a large number of sample states and solving a least squares problem that aims to minimize the error in satisfying the DP equation for these states at time k. At the typical stage k, having obtained r_{k+1}, we determine r_k from the least squares problem

$$r_k \in \arg\min_r \sum_{s=1}^{q} \left(\tilde{J}_k(x_k^s, r) - \min_{u \in U_k(x_k^s)} E\Big\{ g_k(x_k^s, u, w_k) + \tilde{J}_{k+1}\big(f_k(x_k^s, u, w_k), r_{k+1}\big) \Big\} \right)^2$$

where x_k^s, $s = 1, \ldots, q$, are the sample states that have been generated for the kth stage. Since r_{k+1} is assumed to be already known, the complicated

† The alternative terms *critic network* and *actor network* are also used often. In this book, we will use the terms "value network" and "policy network."

minimization term in the right side of this equation is the known scalar

$$\beta_k^s = \min_{u \in U_k(x_k^s)} E\Big\{ g_k(x_k^s, u, w_k) + \tilde{J}_{k+1}\big(f_k(x_k^s, u, w_k), r_{k+1}\big) \Big\}, \qquad (3.13)$$

so that r_k is obtained as

$$r_k \in \arg\min_r \sum_{s=1}^{q} \big(\tilde{J}_k(x_k^s, r) - \beta_k^s\big)^2. \qquad (3.14)$$

The algorithm starts at stage $N-1$ with the minimization

$$r_{N-1} \in \arg\min_r \sum_{s=1}^{q} \left(\tilde{J}_{N-1}(x_{N-1}^s, r) \right.$$
$$\left. - \min_{u \in U_{N-1}(x_{N-1}^s)} E\Big\{ g_{N-1}(x_{N-1}^s, u, w_{N-1}) + g_N\big(f_{N-1}(x_{N-1}^s, u, w_{N-1})\big) \Big\} \right)^2$$

and ends with the calculation of r_0 at $k = 0$.

In the case of a linear architecture, where the approximate cost-to-go functions are

$$\tilde{J}_k(x_k, r_k) = r_k' \phi_k(x_k), \qquad k = 0, \ldots, N-1,$$

the least squares problem (3.14) greatly simplifies, and admits the closed form solution

$$r_k = \left(\sum_{s=1}^{q} \phi_k(x_k^s) \phi_k(x_k^s)' \right)^{-1} \sum_{s=1}^{q} \beta_k^s \phi_k(x_k^s);$$

cf. Eq. (3.7). For a nonlinear architecture such as a neural network, incremental gradient algorithms may be used.

An important implementation issue is how to select the sample states x_k^s, $s = 1, \ldots, q$, $k = 0, \ldots, N-1$. In practice, they are typically obtained by some form of Monte Carlo simulation, but the distribution by which they are generated is important for the success of the method. In particular, it is important that the sample states are "representative" in the sense that they are visited often under a nearly optimal policy. More precisely, the frequencies with which various states appear in the sample should be roughly proportional to the probabilities of their occurrence under an optimal policy.

Aside from the issue of selection of the sampling distribution that we have just described, a difficulty with fitted value iteration arises when the horizon N is very long, since then the total number of parameters over the N stages may become excessive. In this case, however, the problem is often

stationary, in the sense that the system and cost per stage do not change as time progresses. Then it may be possible to treat the problem as one with an infinite horizon and bring to bear additional methods for training approximation architectures; see the relevant discussions in Chapter 5 of the book [Ber19a].

We finally note an important difficulty with the training method of this section: the calculation of each sample β_k^s of Eq. (3.13) requires a minimization of an expected value, which can be very time consuming. In the next section, we describe an alternative type of fitted value iteration, which uses Q-factors, and involves a simpler minimization, in which the order of the minimization and expectation operations in Eq. (3.13) is reversed.

3.3.2 Q-Factor Parametric Approximation - Model-Free Implementation

We will now consider an alternative form of approximation in value space and fitted value iteration, which involves approximation of the optimal Q-factors of state-control pairs (x_k, u_k) at time k, with no intermediate approximation of cost-to-go functions. An important characteristic of this algorithm is that it *allows for a model-free computation* (i.e., the use of a computer model in place of a mathematical model).

We recall that the optimal Q-factors are defined by

$$Q_k^*(x_k,u_k) = E\Big\{g_k(x_k,u_k,w_k)+J_{k+1}^*\big(f_k(x_k,u_k,w_k)\big)\Big\}, \quad k=0,\ldots,N-1, \tag{3.15}$$

where J_{k+1}^* is the optimal cost-to-go function for stage $k+1$. Thus $Q_k^*(x_k,u_k)$ is the cost attained by using u_k at state x_k, and subsequently using an optimal policy.

As noted in Section 1.3, the DP algorithm can be written as

$$J_k^*(x_k) = \min_{u\in U_k(x_k)} Q_k^*(x_k,u_k),$$

and by using this equation, we can write Eq. (3.15) in the following equivalent form that relates Q_k^* with Q_{k+1}^*:

$$Q_k^*(x_k,u_k) = E\Big\{g_k(x_k,u_k,w_k) + \min_{u\in U_{k+1}(f_k(x_k,u_k,w_k))} Q_{k+1}^*\big(f_k(x_k,u_k,w_k),u\big)\Big\}. \tag{3.16}$$

This suggests that in place of the Q-factors $Q_k^*(x_k,u_k)$, we may use Q-factor approximations as the basis for suboptimal control.

We can obtain such approximations by using methods that are similar to the ones we have considered so far. Parametric Q-factor approximations

$\tilde{Q}_k(x_k, u_k, r_k)$ may involve a neural network, or a feature-based linear architecture. The feature vector may depend on just the state, or on both the state and the control. In the former case, the architecture has the form

$$\tilde{Q}_k(x_k, u_k, r_k) = r_k(u_k)'\phi_k(x_k), \tag{3.17}$$

where $r_k(u_k)$ is a separate weight vector for each control u_k. In the latter case, the architecture has the form

$$\tilde{Q}_k(x_k, u_k, r_k) = r_k'\phi_k(x_k, u_k), \tag{3.18}$$

where r_k is a weight vector that is independent of u_k. The architecture (3.17) is suitable for problems with a relatively small number of control options at each stage. In what follows, we will focus on the architecture (3.18), but the discussion, with few modifications, also applies to the architecture (3.17) and to nonlinear architectures as well.

We may adapt the fitted value iteration approach of the preceding section to compute sequentially the parameter vectors r_k in Q-factor parametric approximations, starting from $k = N - 1$. This algorithm is based on Eq. (3.16), with r_k obtained by solving least squares problems similar to the ones of the cost function approximation case [cf. Eq. (3.14)]. As an example, the parameters r_k of the architecture (3.18) are computed sequentially by collecting sample state-control pairs (x_k^s, u_k^s), $s = 1, \ldots, q$, and solving the linear least squares problems

$$r_k \in \arg\min_r \sum_{s=1}^{q} \big(r'\phi_k(x_k^s, u_k^s) - \beta_k^s\big)^2, \tag{3.19}$$

where

$$\beta_k^s = E\left\{ g_k(x_k^s, u_k^s, w_k) + \min_{u \in U_{k+1}(f_k(x_k^s, u_k^s, w_k))} r_{k+1}'\phi_{k+1}\big(f_k(x_k^s, u_k^s, w_k), u\big) \right\}. \tag{3.20}$$

Thus, having obtained r_{k+1}, we obtain r_k through a least squares fit that aims to minimize the sum of the squared errors in satisfying Eq. (3.16). Note that the solution of the least squares problem (3.19) can be obtained in closed form as

$$r_k = \left(\sum_{s=1}^{q} \phi_k(x_k^s, u_k^s)\phi_k(x_k^s, u_k^s)' \right)^{-1} \sum_{s=1}^{q} \beta_k^s \phi_k(x_k^s, u_k^s);$$

[cf. Eq. (3.7)]. Once r_k has been computed, the one-step lookahead control $\tilde{\mu}_k(x_k)$ is obtained on-line as

$$\tilde{\mu}_k(x_k) \in \arg\min_{u \in U_k(x_k)} \tilde{Q}_k(x_k, u, r_k), \tag{3.21}$$

without the need to calculate any expected value. This latter property is a primary incentive for using Q-factors in approximate DP, particularly when there are tight constraints on the amount of on-line computation that is possible in the given practical setting.

The samples β_k^s of Eq. (3.20) involve the exact computation of an expected value. In an alternative implementation, we may replace β_k^s with an average of just a few samples (even a single sample) of the random variable

$$g_k(x_k^s, u_k^s, w_k) + \min_{u \in U_{k+1}(f_k(x_k^s, u_k^s, w_k))} r'_{k+1}\phi_{k+1}\big(f_k(x_k^s, u_k^s, w_k), u\big), \quad (3.22)$$

collected according to the probability distribution of w_k. This distribution may either be known explicitly, or in a model-free situation, through a computer simulator. In particular, *to implement this scheme, we only need a simulator that for any pair (x_k, u_k) generates samples of the next stage cost and state*

$$g_k(x_k, u_k, w_k), \qquad f_k(x_k, u_k, w_k),$$

according to the distribution of w_k.

Note that the samples of the random variable (3.22) do not require the computation of an expected value like the samples (3.13) in the cost approximation method of the preceding section. Moreover the samples of (3.22) involve a simpler minimization than the samples of (3.13). This is an important advantage of working with Q-factors rather than state costs.

Having obtained the weight vectors $r_0, \ldots, r_{N-1}$, and hence the one-step lookahead policy $\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$ through Eq. (3.21), a further possibility is to approximate this policy with a parametric architecture. This is *approximation in policy space built on top of approximation in value space*. The idea here is to simplify even further the on-line computation of the suboptimal controls by avoiding the minimization of Eq. (3.21); see the discussion in Section 1.3.3.

3.3.3 Parametric Approximation in Infinite Horizon Problems - Approximate Policy Iteration

In this section we will briefly discuss parametric approximation methods for infinite horizon problems, based on policy iteration (PI). We will focus on the finite-state version of the α-discounted problem of Section 1.4.1, using notation that is more convenient for such problems. In particular, states and successor states will be denoted by i and j, respectively. Moreover the system equation will be represented by control-dependent transition probabilities $p_{ij}(u)$ (the probability that the system will move to state j, given that it starts at state i and control u is applied). For a state-control pair (i, u), the average cost per stage is denoted by $g(i, u, j)$.

We recall that the PI algorithm in its exact form produces a sequence of stationary policies whose cost functions are progressively improving and

converge in a finite number of iterations to the optimal. The convergence proof relies on the fundamental cost improvement property of PI, and depends on the finiteness of the state and control spaces. This proof, together with other PI-related convergence proofs, can be found in the author's textbooks [Ber17a] or [Ber19a].

Let us state the exact form of the PI algorithm in terms of Q-factors, and in a way that facilitates approximations and simulation-based implementations. Given any policy μ, it generates the next policy $\tilde{\mu}$ with a two-step process as follows (cf. Section 1.4.1):

(a) *Policy evaluation*: We compute the cost function J_μ of μ and its associated Q-factors, which are given by

$$Q_\mu(i,u) = \sum_{j=1}^{n} p_{ij}(u)\big(g(i,u,j) + \alpha J_\mu(j)\big), \qquad i = 1,\ldots,n,\ u \in U(i).$$

Thus $Q_\mu(i,u)$ is the cost of starting at i, using u at the first stage, and then using μ for the remaining stages.

(b) *Policy improvement*: We compute the new policy $\tilde{\mu}$ according to

$$\tilde{\mu}(i) \in \arg\min_{u \in U(i)} Q_\mu(i,u), \qquad i = 1,\ldots,n.$$

Let us now describe one approach to approximating the two steps of the PI process:

(a) *Approximate policy evaluation*: Here we introduce a parametric architecture $\tilde{Q}_\mu(i,u,r)$ for the Q-factors of μ. We determine the value of the parameter vector r by generating (using a simulator of the system) a large number of training triplets (i^s,u^s,β^s), $s = 1,\ldots,q$, and by using a least squares fit:

$$\overline{r} \in \arg\min_{r} \sum_{s=1}^{q} \big(\tilde{Q}_\mu(i^s,u^s,r) - \beta^s\big)^2. \tag{3.23}$$

In particular, for a given pair (i^s,u^s), the scalar β^s is generated by starting at i^s, using u^s at the first stage, and simulating a trajectory of states and controls using μ for some number k of subsequent stages. Thus, β^s is a sample of $Q_\mu^k(i^s,u^s)$, the k-stage Q-factor of μ, which in the limit as $k \to \infty$ yields the infinite horizon Q-factor of μ. The number of stages k may be either large, or fairly small. However, in the latter case some terminal cost function approximation should be added at the end of the k-stage trajectory, to compensate for the difference $\big|Q_\mu(i,u) - Q_\mu^k(i,u)\big|$, which decreases in proportion to α^k, and may be large when k is small. Such a function may be obtained with additional training or from a previous iteration.

(b) *Approximate policy improvement*: Here we compute the new policy $\tilde{\mu}$ according to

$$\tilde{\mu}(i) \in \arg\min_{u \in U(i)} \tilde{Q}_\mu(i, u, \overline{r}), \qquad i = 1, \ldots, n, \tag{3.24}$$

where $\overline{r}$ is the parameter vector obtained from the policy evaluation formula (3.23).

An important alternative for approximate policy improvement, is to compute a set of pairs $\big(i^s, \tilde{\mu}(i^s)\big)$, $s = 1, \ldots, q$, using Eq. (3.24), and fit these pairs with a policy approximation architecture (see the next section on approximation in policy space). The overall scheme then becomes a form of PI that is based on approximation in both value and policy spaces.

At the end of the last policy evaluation step of PI, we have obtained a final Q-factor approximation $\tilde{Q}(i, u, \tilde{r})$. Then, in on-line play mode, we may apply the policy

$$\tilde{\mu}(i) \in \arg\min_{u \in U(i)} \tilde{Q}(i, u, \tilde{r}),$$

i.e., use the (would be) next policy iterate. Alternatively, we may apply the one-step lookahead policy

$$\tilde{\mu}(i) \in \arg\min_{u \in U(i)} \left[\sum_{j=1}^{n} p_{ij}(u) \left(g(i, u, j) + \alpha \min_{u' \in U(j)} \tilde{Q}(j, u', \tilde{r}) \right) \right], \tag{3.25}$$

or its multistep lookahead version. The latter alternative implements a Newton step and will likely result in substantially better performance. However, it is more time consuming, particularly if it is implemented by using a computer model and model-free simulation. Still another possibility, which also implements a Newton step, is to replace the function

$$\min_{u' \in U(j)} \tilde{Q}(j, u', \tilde{r})$$

in the preceding Eq. (3.25) with an off-line trained approximation.

Challenges Relating to Approximate Policy Iteration

Approximate PI in its various forms has been the subject of extensive theoretical and applied research. A comprehensive discussion is beyond our scope, and we refer to the literature, for detailed accounts, including the DP textbook [Ber12] or the RL textbook [Ber19a]. Let us provide a few comments relating to the challenges of approximate PI implementation.

(a) *Architectural issues*: The architecture $\tilde{Q}_\mu(i, u, r)$ may involve the use of features, and it could be linear, or it could be nonlinear such as a

neural network. A major advantage of a linear feature-based architecture is that the policy evaluation (3.23) is a linear least squares problem, which admits a closed-form solution. Moreover, when linear architectures are used, there is a broader variety of approximate policy evaluation methods with solid theoretical performance guarantees, such as TD(λ), LSTD(λ), and LSPE(λ), which are not described in this book, but are discussed extensively in the literature. Generally, identifying an architecture that fits the problem well and training it effectively can be a challenging and time-intensive task.

(b) *Exploration issues*: Generating an appropriate set of training triplets (i^s, u^s, β^s) at the policy evaluation step poses considerable challenges. A major difficulty has to do with *inadequate exploration*, the Achilles' heel of approximate PI. In particular, straightforward ways to evaluate a policy μ, typically rely on Q-factor samples of μ starting from states frequently visited by μ. Unfortunately, this may bias the simulation by underrepresenting states that are unlikely to occur under μ. As a result, the Q-factor estimates of these underrepresented states may be highly inaccurate, potentially causing serious errors in the calculation of the improved control policy $\tilde{\mu}$ via the policy improvement Eq. (3.24).

One way to address this issue is to use a large number of initial states to form a rich and representative subset of the state space. To keep simulation costs manageable, it may be necessary to use relatively short trajectories. However, when using short trajectories it may be important to introduce a terminal cost function approximation in the policy evaluation step in order to make the cost sample β^s more accurate. Other approaches to enhance exploration include the use of a so-called *off-policy*, i.e., a policy μ' other than the currently evaluated policy μ, which tends to visit states that are unlikely to be visited using μ. For further discussion, see Section 6.4 of the DP textbook [Ber12].

(c) *Oscillation issues*: Contrary to exact PI, which is guaranteed to yield an optimal policy, approximate PI produces a sequence of policies, which are only guaranteed to lie asymptotically within a certain error bound from the optimal; see the books [BeT96], Section 6.2.2, [Ber12], Section 2.5, and [Ber19a], Section 5.3.5. Moreover, the generated policies may oscillate. By this we mean that after a few iterations, policies tend to repeat in cycles.

The associated parameter vectors r may also tend to oscillate, although it is possible that there is convergence in parameter space and oscillation in policy space. This phenomenon, known as *chattering*, is explained in the author's survey papers [Ber10c], [Ber11b], and book [Ber12] (Section 6.4.3), and can be quite problematic, because

there is no guarantee that the policies involved in the oscillation are "good" policies, and it is often difficult to assess their performance relative to the optimal. We note, however, that oscillations can be avoided and approximate PI can be shown to converge under special conditions, which arise in particular when an aggregation approach is used; see the approximate PI survey [Ber11b].

We refer to the literature for further discussion of the preceding issues, as well as a variety of other approximate PI methods.

3.3.4 Optimistic Policy Iteration with Parametric Q-Factor Approximation - SARSA and DQN

There are also "optimistic" approximate PI methods with Q-factor approximation, and/or a few samples in between policy updates. We view these primarily as off-line training methods, but because of the limited number of samples between policy updates, they have the potential of on-line implementation. In this case, however, a number of difficulties must be overcome, as we will explain later in this section.

As an example, let us consider an extreme version of Q-factor parametric approximation that uses *a single sample* between policy updates. At the start of iteration k, we have the current parameter vector r^k, we are at some state i^k, and we have chosen a control u^k. Then:

(1) We simulate the next transition (i^k, i^{k+1}) using the transition probabilities $p_{i^k j}(u^k)$.

(2) We generate the control u^{k+1} with the minimization

$$u^{k+1} \in \arg \min_{u \in U(i^{k+1})} \tilde{Q}(i^{k+1}, u, r^k). \tag{3.26}$$

[In some schemes, to enhance exploration, u^{k+1} is chosen with a small probability to be a random element of $U(i^{k+1})$ or one that is "ϵ-greedy," i.e., attains within some ϵ the minimum above. This is commonly referred to as the use of an *off-policy*.]

(3) We update the parameter vector via

$$\begin{aligned} r^{k+1} = r^k - \gamma^k \nabla \tilde{Q}(i^k, u^k, r^k) \\ \cdot \big(\tilde{Q}(i^k, u^k, r^k) - g(i^k, u^k, i^{k+1}) - \alpha \tilde{Q}(i^{k+1}, u^{k+1}, r^k)\big), \end{aligned} \tag{3.27}$$

where γ^k is a positive stepsize, and $\nabla(\cdot)$ denotes gradient with respect to r evaluated at the current parameter vector r^k. In the special case where $\tilde{Q}$ is a linear feature-based architecture, $\tilde{Q}(i, u, r) = \phi(i, u)'r$, the gradient $\nabla \tilde{Q}(i^k, u^k, r^k)$ is just the feature vector $\phi(i^k, u^k)$, and iteration (3.27) becomes

$$r^{k+1} = r^k - \gamma^k \phi(i^k, u^k)\big(\phi(i^k, u^k)'r^k - g(i^k, u^k, i^{k+1}) - \alpha \phi(i^{k+1}, u^{k+1})'r^k\big).$$

Thus r^k *is changed in an incremental gradient direction*: the one opposite to the gradient (with respect to r) of the incremental error

$$\big(\phi(i^k,u^k)'r - g(i^k,u^k,i^{k+1}) - \alpha\phi(i^{k+1},u^{k+1})'r^k\big)^2,$$

evaluated at the current iterate r^k.

The process is now repeated with r^{k+1}, i^{k+1}, and u^{k+1} replacing r^k, i^k, and u^k, respectively.

Extreme optimistic schemes of the type just described have received a lot of attention, in part because *they admit a model-free implementation* [i.e., the use of a simulator, which provides for each pair (i^k,u^k), the next state i^{k+1} and corresponding cost $g(i^k,u^k,i^{k+1})$ that are needed in Eq. (3.27)]. They are often referred to as SARSA (State-Action-Reward-State-Action); see e.g., the books [BeT96], [BBD10a], [SuB18]. When Q-factor approximation is used, their behavior is very complex, their theoretical convergence properties are unclear, and there are no associated performance bounds in the literature. In practice, SARSA is often implemented in a less extreme/optimistic form, whereby several (perhaps many) state-control-transition cost-next state samples are batched together and suitably averaged before updating the vector r^k.

Other variants of the method attempt to reduce sampling effort by storing the generated samples in a buffer for reuse in subsequent iterations through some randomized process (cf. our earlier discussion of exploration). This is also sometimes called *experience replay*, an idea that has been used since the early days of RL, both to save sampling effort and to enhance exploration. The DQN (Deep Q Network) scheme, championed by DeepMind (see Mnih et al. [MKS15]), is based on this idea (the term "Deep" is a reference to DeepMind's affinity for deep neural networks, but the idea of experience replay does not depend on the use of a deep neural network architecture).

Another interesting idea from DeepMind [MRM16] is to introduce asynchronous parallel computation into the algorithm, based on the theory of distributed asynchronous methods in DP, gradient optimization, and RL, by Bertsekas and Tsitsiklis [Ber82a], [TBA86], [BeT89], [Tsi94], [BeT00], [Ber19a].

Q-Learning Algorithms and On-Line Play

Algorithms that approximate Q-factors, including SARSA and DQN, are fundamentally off-line training algorithms. This is because their training process is long and requires the collection of many samples before reaching a stage that resembles parameter convergence. It can thus be unreliable to use the interim approximate Q-factors for on-line decision making, particularly in an adaptive context that involves changing system parameters, thereby requiring on-line replanning.

On the other hand, compared to the approximate PI method of Section 3.3.3, SARSA and DQN are far better suited for on-line implementation, because the control generation process of Eq. (3.26) can also be used to select controls on-line, thereby facilitating the combination of training and on-line control selection. To this end, it is important, among others, to make sure that the parameters r_k stay at "safe" levels during the on-line control process, which can be a challenge. Still, even if this difficulty can be overcome in the context of a given problem, there are a number of other difficulties that SARSA and DQN can encounter during on-line play.

(a) *On-line exploration issues*: There is a need to occasionally select controls using an off-policy in order to enhance exploration, and finding an adequate off-policy in a given practical context can be a challenge. Moreover, the off-policy controls may improve exploration, but may be of poor quality, and in some contexts, may induce instability.

(b) *Robustness and replanning issues*: In an adaptive control context where the problem parameters are changing, the algorithm may be too slow to adapt to the changes.

(c) *Performance degradation issues*: Similar to our earlier discussion [cf. the comparison of Eqs. (3.24) and (3.25)], the minimization of Eq. (3.26) does not implement a Newton step, thereby resulting in performance loss. The alternative implementation

$$u^{k+1} \in \arg\min_{u \in U(i^{k+1})} \left[\sum_{j=1}^{n} p_{i^{k+1}j}(u) \Big(g(i^{k+1}, u, j) + \alpha \min_{u' \in U(j)} \tilde{Q}(j, u', r^k) \Big) \right],$$

which is patterned after Eq. (3.26), is better in this regard, but is computationally more costly, and thus less suitable for on-line implementation.

Generally speaking, the combination of off-line training and on-line play with the use of SARSA and DQN poses serious challenges. Nevertheless, encouraging results have been achieved in specific contexts, often with "manual tuning," i.e., tuning tailored to the problem at hand. Moreover, the popularity of these methods has been bolstered by the availability of open-source software that allows model-free implementations.

3.3.5 Approximate Policy Iteration for Infinite Horizon POMDP

In this section, we consider partial observation Markovian decision problems (POMDP) with a finite number of states and controls, and discounted additive cost over an infinite horizon. As discussed in Section 1.6.6, the optimal solution is typically intractable, so approximate DP/RL approaches

must be used. In this section we focus on PI methods that are based on rollout, and approximations in policy and value space. They update a policy by using truncated rollout with that policy and a terminal cost function approximation. We focus on cost function approximation schemes, but Q-factor approximation is also possible.

Due to its simulation-based rollout character, the methodology of this section relies critically on the finiteness of the control space. It can be extended to POMDP with infinite state space but finite control space, although we will not consider this possibility in this section.

We assume that there are n states denoted by $i \in \{1, \ldots, n\}$ and a finite set of controls U at each state. We denote by $p_{ij}(u)$ and $g(i, u, j)$ the transition probabilities and corresponding transition costs, from i to j under $u \in U$. The cost is accumulated over an infinite horizon and is discounted by $\alpha \in (0, 1)$. At each new generated state j, an observation z from a finite set Z is obtained with known probability $p(z \mid j, u)$ that depends on j and the control u that was applied prior to the generation of j. The objective is to select each control optimally as a function of the prior history of observations and controls.

A classical approach to this problem is to convert it to a perfect state information problem whose state is the current belief $b = (b_1, \ldots, b_n)$, where b_i is the conditional distribution of the state i given the prior history. As noted in Section 1.6.6, b is a sufficient statistic, which can serve as a substitute for the set of available observations, in the sense that optimal control can be achieved with knowledge of just b.

In this section, we consider a generalized form of sufficient statistic, which we call the *feature state* and we denote by y. *We require that the feature state y subsumes the belief state b.* By this we mean that b can be computed exactly knowing y. One possibility is for y to be the union of b and some distinguishable characteristics of the belief state, or some compact representation of the measurement history up to the current time (such as a number of most recent measurements, or the state of a finite-state controller).

We further assume that *y can be sequentially generated using a known feature estimator $F(y, u, z)$.* By this we mean that given that the current feature state is y, control u is applied, and observation z is obtained, the next feature can be exactly predicted as $F(y, u, z)$.

Clearly, since b is a sufficient statistic, the same is true for y. Thus the optimal costs achievable by the policies that depend on y and on b are the same. However, specific suboptimal schemes may become more effective with the use of the feature state y instead of just the belief state b.

The optimal cost $J^*(y)$, as a function of the sufficient statistic/feature state y, is the unique solution of the corresponding Bellman equation

$$J^*(y) = \min_{u \in U} \left[\hat{g}(y, u) + \alpha \sum_{z \in Z} \hat{p}(z \mid b_y, u) J^*\big(F(y, u, z)\big) \right].$$

Here we use the following notation:

b_y is the belief state that corresponds to feature state y, with components denoted by $b_{y,i}$, $i = 1, \ldots, n$.

$\hat{g}(y, u)$ is the expected cost per stage

$$\hat{g}(y, u) = \sum_{i=1}^{n} b_{y,i} \sum_{j=1}^{n} p_{ij}(u) g(i, u, j).$$

$\hat{p}(z \mid b_y, u)$ is the conditional probability that the next observation will be z given the current belief state b_y and control u

F is the feature state estimator. In particular, $F(y, u, z)$ is the next feature vector, when the current feature state is y, control u is applied, and observation z is obtained.

The feature space reformulation of the problem can serve as the basis for approximation in value space, whereby J^* is replaced in Bellman's equation by some function $\tilde{J}$ after one-step or multistep lookahead. For example a one-step lookahead scheme yields the suboptimal policy $\tilde{\mu}$ given by

$$\tilde{\mu}(y) \in \arg\min_{u \in U} \left[\hat{g}(y, u) + \alpha \sum_{z \in Z} \hat{p}(z \mid b_y, u) \tilde{J}\big(F(y, u, z)\big) \right].$$

In ℓ-step lookahead schemes, $\tilde{J}$ is used as terminal cost function in an ℓ-step horizon version of the original infinite horizon problem. In the standard form of a rollout algorithm, $\tilde{J}$ is the cost function of some base policy. We will next discuss a rollout scheme with ℓ-step lookahead, which involves rollout truncation and terminal cost approximation.

Truncated Rollout with Terminal Cost Function Approximation

In the pure form of the rollout algorithm, the cost function approximation $\tilde{J}$ is the cost function J_μ of a known base policy μ, and its value $\tilde{J}(y) = J_\mu(y)$ at any y is obtained by first extracting b from y, and then running a simulator starting from b, and using the system model, the feature generator, and μ. In the truncated form of rollout, $\tilde{J}(y)$ is obtained by running the simulator of μ for a given number of steps m, and then adding a terminal cost approximation $\hat{J}(\bar{y})$ for each terminal feature state $\bar{y}$ that is obtained at the end of the m steps of the simulation with μ (see Fig. 3.3.1).

Thus the rollout policy is defined by the base policy μ, the terminal cost function approximation $\hat{J}$, the number of steps m after which the simulated trajectory with μ is truncated, and the lookahead size ℓ. The choices of m and ℓ are typically made by trial and error, based on computational tractability among other considerations, while $\hat{J}$ may be chosen

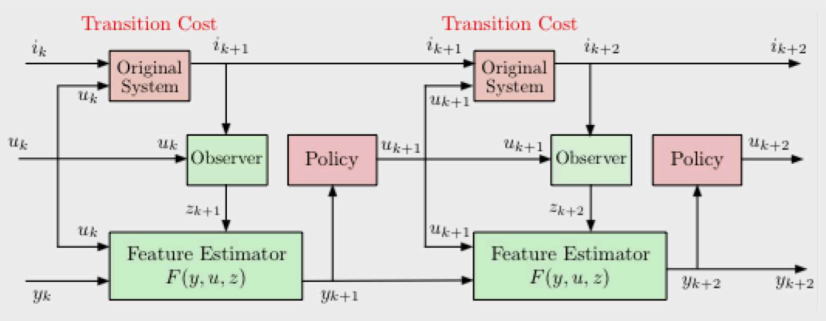

Figure 3.3.1 Composite system simulator for POMDP for a given policy. The starting state i_k at stage k of a trajectory is generated randomly using the belief state b_k, which is in turn computed from the feature state y_k.

on the basis of problem-dependent insight or through the use of some off-line approximation method. In some variants of the method, the multistep lookahead may be implemented approximately using a Monte Carlo tree search or adaptive sampling scheme.

Using m-step rollout between the ℓ-step lookahead and the terminal cost approximation gives the method the character of a single PI. We will use repeated truncated rollout as the basis for constructing a PI algorithm, which we will discuss next.

Supervised Learning of Rollout Policies and Cost Functions - Approximate Policy Iteration

The rollout algorithm uses multistep lookahead and on-line simulation of the base policy to generate the rollout control at any feature state of interest. To avoid the cost of on-line simulation, we can approximate the rollout policy off-line by using some approximation architecture, potentially involving a neural network. This is policy approximation built on top of the rollout scheme.

To this end, we may introduce a parametric family/architecture of policies of the form $\hat{\mu}(y, r)$, where r is a parameter vector. We then construct a training set that consists of a large number of sample feature state-control pairs (y^s, u^s), $s = 1, \ldots, q$, such that for each s, u^s is the rollout control at feature state y^s. We use this data set to obtain a parameter $\bar{r}$ by solving a corresponding classification problem, which associates each feature state y with a control $\hat{\mu}(y, \bar{r})$. The parameter $\bar{r}$ defines a classifier, which given a feature state y, classifies y as requiring control $\hat{\mu}(y, \bar{r})$ (see Section 3.4).

We can also apply the rollout policy approximation to the context of PI. The idea is to view rollout as a single policy improvement, and to view the PI algorithm as a *perpetual rollout process*, which performs multiple

policy improvements, using at each iteration the current policy as the base policy, and the next policy as the corresponding rollout policy.

In particular, we consider a PI algorithm where at the typical iteration we have a policy μ, which we use as the base policy to generate many feature state-control sample pairs (y^s, u^s), $s = 1, \ldots, q$, where u^s is the rollout control corresponding to feature state y^s. We then obtain an "improved" policy $\hat{\mu}(y, \overline{r})$ with an approximation architecture and a classification algorithm, as described above. The "improved" policy is then used as a base policy to generate samples of the corresponding rollout policy, which is approximated in policy space, etc.

To use truncated rollout in this PI scheme, a terminal cost approximation is required, which can take a variety of forms. Using zero is a simple possibility, which may work well if either the size ℓ of multistep lookahead or the length m of the rollout is relatively large. Another possibility is to use as terminal cost in the truncated rollout an approximation of the cost function of some base policy, which may be obtained with a neural network-based approximation architecture.

In particular, at any policy iteration with a given base policy, once the rollout data is collected, one or two neural networks are constructed: A *policy network* that approximates the rollout policy, and (in the case of rollout with truncation) a *value network* that constructs a cost function approximation for that rollout policy. Thus, we may consider two types of methods:

(a) *Approximate rollout and PI with truncation*, where each generated policy as well as its cost function are approximated by a policy and a value network, respectively. The cost function approximation of the current policy is used to truncate the rollout trajectories that are used to train the next policy.

(b) *Approximate rollout and PI without truncation*, where each generated policy is approximated using a policy network, but the rollout trajectories are continued up to a large maximum number of stages (enough to make the cost of the remaining stages insignificant due to discounting) or upon reaching a termination state. The advantage of this scheme is that only a policy network is needed; a value network is unnecessary since there is no rollout truncation with cost function approximation at the end.

Note that as in all approximate PI schemes, the sampling of feature states used for training is subject to exploration concerns. In particular, for each policy approximation, it is important to include in the sample set $\{y^s \mid s = 1, \ldots, q\}$, a subset of feature states that are "favored" by the rollout trajectories; e.g., start from some initial subset of feature states y^s and selectively add to this subset feature states that are encountered along the rollout trajectories. This is a challenging issue, which must be

approached with care.

An extensive case study of the methodology of this section was given in the paper by Bhattacharya et al. [BBW20] for the case of a pipeline repair problem. The implementation used there also includes the use of a partitioned state space architecture and an asynchronous distributed algorithm for off-line training; see Section 3.4.2. See also the paper by Bhattacharya et al. [BKB23], which deals with large multiagent POMDP problems, involving multiple robots operating on a network.

3.3.6 Advantage Updating - Approximating Q-Factor Differences

Let us now explore an important alternative to computing Q-factor approximations. It is motivated by the potential benefit of approximating Q-factor differences rather than Q-factors. In this method, called *advantage updating*, instead of computing and comparing $Q_k^*(x_k, u_k)$ for all $u_k \in U_k(x_k)$, we compute

$$A_k(x_k, u_k) = Q_k^*(x_k, u_k) - \min_{u \in U_k(x_k)} Q_k^*(x_k, u).$$

The function $A_k(x_k, u_k)$ can serve to compare controls, i.e., at state x_k select

$$\tilde{\mu}_k(x_k) \in \arg \min_{u \in U_k(x_k)} A_k(x_k, u),$$

and this can also be done when $A_k(x_k, u_k)$ is approximated with a value network.

Note that in the absence of approximations, selecting controls by advantage updating is equivalent to selecting controls by comparing their Q-factors. By contrast, when approximation is involved, comparing advantages instead of Q-factors can be important, because the former may have a much smaller range of values than the latter. In particular, Q_k^* may embody sizable quantities that depend on x_k but are independent of u_k, and which may interfere with algorithms such as the fitted value iteration (3.19)-(3.20). Thus, when training an architecture to approximate Q_k^*, the training algorithm may naturally try to capture the large scale behavior of Q_k^*, which may be irrelevant because it may not be reflected in the Q-factor differences A_k. However, with advantage updating, we may instead focus the training process on finer scale variations of Q_k^*, which may be all that matters. Here is an example (first given in the book [BeT96]) of what can happen when trained approximations of Q-factors are used.

Example 3.3.1

Consider the deterministic scalar linear system

$$x_{k+1} = x_k + \delta u_k,$$

and the quadratic cost per stage

$$g(x_k, u_k) = \delta(x_k^2 + u_k^2),$$

where δ is a very small positive constant [think of a δ-discretization of a continuous-time problem involving the differential equation $dx(t)/dt = u(t)$]. Let us focus on the stationary policy π, which applies at state x the control

$$\mu(x) = -2x,$$

and view it as the base policy of a rollout algorithm. The Q-factors of π over an infinite number of stages can be calculated to be

$$Q_\pi(x, u) = \frac{5x^2}{4} + \delta\left(\frac{9x^2}{4} + u^2 + \frac{5}{2}xu\right) + O(\delta^2).$$

(We omit the details of this calculation, which is based on the classical analysis of linear-quadratic optimal control problems; see e.g., Section 1.5, or [Ber17a], Section 3.1.) Thus the important part of $Q_\pi(x, u)$ for the purpose of rollout policy computation is

$$\delta\left(u^2 + \frac{5}{2}xu\right). \tag{3.28}$$

However, when a value network is trained to approximate $Q_\pi(x, u)$, the approximation will be dominated by $\frac{5x^2}{4}$, and the important part (3.28) will be "lost" when δ is very small. By contrast, the advantage function can be calculated to be

$$\begin{aligned} A_\mu(x, u) &= Q_\pi(x, u) - \min_v Q_\pi(x, v) + O(\delta^2) \\ &= \delta\left(u^2 + \frac{5}{2}xu - \min_v \left(v^2 + \frac{5}{2}xv\right)\right) + O(\delta^2) \\ &= \delta\left(u^2 + \frac{5}{2}xu + \frac{5^2}{4}x^2\right) + O(\delta^2), \end{aligned}$$

and when approximated with a value network, the approximation will be essentially unaffected by δ.

The Use of a Baseline

The idea of advantage updating is also related to the useful technique of subtracting a suitable constant (often called a *baseline*) from a quantity that is estimated; see Fig. 3.3.2 (in the case of advantage updating, the baselines depend on x_k, but the same general idea applies). This idea can also be used in the context of the fitted value iteration method given earlier, as well as in conjunction with other simulation-based methods in RL.

Example 3.3.1 also points to the connection between the ideas underlying advantage updating and the rollout methods for small stage costs relative to the cost function approximation, which we discussed in Section 2.6. In both cases it is necessary to avoid including terms of disproportionate size in the target function that is being approximated. The remedy in both cases is to subtract from the target function a suitable state-dependent baseline.

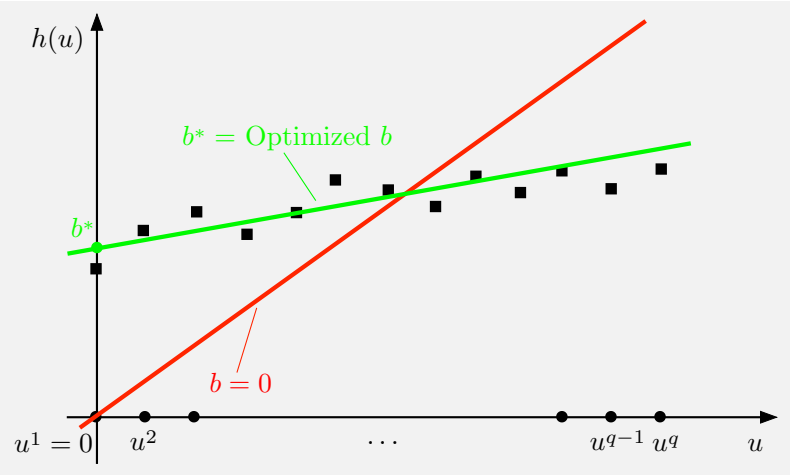

Figure 3.3.2 Illustration of the idea of subtracting a baseline constant from a cost or Q-factor approximation. Here we have samples $h(u^1), \ldots, h(u^q)$ of a scalar function $h(u)$ at sample points $u^1, \ldots, u^q$, and we want to approximate $h(u)$ with a linear function $\tilde{h}(u, r) = ru$, where r is a scalar tunable weight. We subtract a baseline constant b from the samples, and we solve the problem

$$\bar{r} \in \arg\min_r \sum_{s=1}^{q} \Big(\big(h(u^s) - b \big) - ru^s \Big)^2.$$

By properly adjusting b, we can improve the quality of the approximation, which after subtracting b from all the sample values, takes the form $\tilde{h}(u, b, r) = b + ru$. Conceptually, b serves as an additional weight (multiplying the basis function 1), which enriches the approximation architecture.

3.3.7 Differential Training of Cost Differences for Rollout

Let us now consider ways to approximate Q-factor differences (cf. our advantage updating discussion of the preceding section) by approximating cost function differences first. We recall here that given a base policy $\pi = \{\mu_0, \ldots, \mu_{N-1}\}$, the off-line computation of an approximate rollout policy $\tilde{\pi} = \{\tilde{\mu}_0, \ldots, \tilde{\mu}_{N-1}\}$ consists of two steps:

(1) In a preliminary phase, we compute approximations $\tilde{J}_k$ to the cost functions $J_{k,\pi}$ of the base policy π, possibly using simulation and a least squares fit from a parametrized class of functions.

(2) Given $\tilde{J}_k$ and a state x_k at time k, we compute the approximate Q-factor

$$\tilde{Q}_k(x_k, u) = E\Big\{ g_k(x_k, u, w_k) + \tilde{J}_{k+1}\big(f_k(x_k, u, w_k) \big) \Big\}$$

for all $u \in U_k(x_k)$, and we obtain the (approximate) rollout control $\tilde{\mu}_k(x_k)$ from the minimization

$$\tilde{\mu}_k(x_k) \in \arg \min_{u \in U_k(x_k)} \tilde{Q}_k(x_k, u).$$

Unfortunately, this method also suffers from the error magnification inherent in the Q-factor differencing operation. This motivates an alternative approach, called *differential training*, which is based on cost-to-go difference approximations. To this end, we note that to compute the rollout control $\tilde{\mu}_k(x_k)$, it is sufficient to have the cost-to-go differences

$$\tilde{J}_{k+1}\big(f_k(x_k, u, w_k)\big) - \tilde{J}_{k+1}\big(f_k(x_k, \mu_k(x_k), w_k)\big), \tag{3.29}$$

where $\mu_k(x_k)$ is the control applied by the base policy at x_k.

We thus consider a function approximation approach, whereby given any two states x_{k+1} and $\hat{x}_{k+1}$, we obtain an approximation $\tilde{G}_{k+1}(x_{k+1}, \hat{x}_{k+1})$ of the cost difference (3.29). We then compute the rollout control by

$$\begin{aligned}\tilde{\mu}_k(x_k) \in \arg \min_{u \in U_k(x_k)} E\Big\{ & g_k(x_k, u, w_k) - g_k(x_k, \mu_k(x_k), w_k) \\ & + \tilde{G}_{k+1}\big(f_k(x_k, u, w_k), f_k(x_k, \mu_k(x_k), w_k)\big)\Big\},\end{aligned} \tag{3.30}$$

where $\mu_k(x_k)$ is the control applied by the base policy at x_k. Note that the minimization (3.30) aims to simply subtract the approximate Q-factor of the base policy control $\mu_k(x_k)$ from the approximate Q-factor of every other control $u \in U_k(x_k)$.

An important point here is that the training of an approximation architecture to obtain $\tilde{G}_{k+1}$ can be done using any of the standard training methods, and a "differential" system, whose "states" are pairs $(x_k, \hat{x}_k)$ and will be described shortly. To see this, let us denote for all k and pair of states $(x_k, \hat{x}_k)$

$$G_k(x_k, \hat{x}_k) = J_{k,\pi}(x_k) - J_{k,\pi}(\hat{x}_k)$$

the cost function differences corresponding to the base policy π. We consider the DP equations corresponding to π, and to x_k and $\hat{x}_k$:

$$J_{k,\pi}(x_k) = E\Big\{ g_k\big(x_k, \mu_k(x), w_k\big) + J_{k+1,\pi}\big(f_k(x_k, \mu_k(x), w_k)\big)\Big\},$$

$$J_{k,\pi}(\hat{x}_k) = E\Big\{ g_k\big(\hat{x}_k, \mu_k(\hat{x}_k), w_k\big) + J_{k+1,\pi}\big(f_k(\hat{x}_k, \mu_k(\hat{x}_k), w_k)\big)\Big\},$$

and we subtract these equations to obtain

$$\begin{aligned}G_k(x_k, \hat{x}_k) = E\Big\{ & g_k\big(x_k, \mu_k(x_k), w_k\big) - g\big(\hat{x}_k, \mu_k(\hat{x}_k), w_k\big) \\ & + G_{k+1}\big(f_k(x_k, \mu_k(x_k), w_k), f_k(\hat{x}_k, \mu_k(\hat{x}_k), w_k)\big)\Big\},\end{aligned}$$

for all $(x_k, \hat{x}_k)$ and k. Therefore, G_k can be viewed as the cost-to-go function for a problem involving a fixed policy (the base policy), the state $(x_k, \hat{x}_k)$, the cost per stage

$$g_k\big(x_k, \mu_k(x_k), w_k\big) - g_k\big(\hat{x}_k, \mu_k(\hat{x}_k), w_k\big), \tag{3.31}$$

and the system equation

$$(x_{k+1}, \hat{x}_{k+1}) = \Big(f_k\big(x_k, \mu_k(x_k), w_k\big), f_k\big(\hat{x}_k, \mu_k(\hat{x}_k), w_k\big)\Big). \tag{3.32}$$

Thus, it can be seen that *any of the standard methods that can be used to train architectures that approximate $J_{k,\pi}$, can also be used for training architectures that approximate G_k*. For example, one may use simulation-based methods that generate pairs of trajectories starting at the pair of initial states $(x_k, \hat{x}_k)$, and generated according to Eq. (3.32) by using the base policy π. Note that a single random sequence $\{w_0, \ldots, w_{N-1}\}$ may be used to simultaneously generate samples of $G_k(x_k, \hat{x}_k)$ for several triples $(x_k, \hat{x}_k, k)$, and in fact this may have a substantial beneficial effect.

A special case of interest arises when a linear, feature-based architecture is used for the approximator $\tilde{G}_k$. In particular, let ϕ_k be a feature extraction mapping that associates a feature vector $\phi_k(x_k)$ with state x_k and time k, and let $\tilde{G}_k$ be of the form

$$\tilde{G}_k(x_k, \hat{x}_k) = r_k'\big(\phi_k(x_k) - \phi_k(\hat{x}_k)\big),$$

where r_k is a tunable weight vector of the same dimension as $\phi_k(x_k)$ and prime denotes transposition. The rollout policy is generated by

$$\tilde{\mu}_k(x_k) \in \arg\min_{u \in U_k(x_k)} E\Big\{g_k(x_k, u, w_k) + r_{k+1}'\phi_{k+1}\big(f_k(x_k, u, w_k)\big)\Big\},$$

which corresponds to using $r_{k+1}'\phi_{k+1}(x_{k+1})$ (plus an unknown inconsequential constant) as an approximation to $J_{k+1,\pi}(x_{k+1})$. Thus, in this approach, *we essentially use a linear feature-based architecture to approximate the cost functions $J_{k,\pi}$ of the base policy, but we train this architecture using the differential system (3.32) and the differential cost per stage of Eq. (3.31).* This is done by selecting pairs of initial states, running in parallel the corresponding trajectories using the base policy, and subtracting the resulting trajectory costs from each other.

3.4 LEARNING A POLICY IN APPROXIMATE DP

We have focused so far on approximation in value space using parametric architectures. In this section we will discuss how the cost function approximation methods discussed earlier in this chapter can be adapted for

the purpose of approximation in policy space, whereby we approximate a given policy by using optimization over a parametric family of some form. Throughout this section we focus on a fixed policy and we focus on the off-line training of that policy.

In particular, suppose that for a given stage k, we have access to a dataset of sample state-control pairs (x_k^s, u_k^s), $s = 1, \ldots, q$, obtained through some unspecified control process, such as rollout or problem approximation. We may then wish to "learn" this process by training the parameter vector r_k of a parametric family of policies $\tilde{\mu}_k(x_k, r_k)$, using least squares minimization/regression:

$$\overline{r}_k \in \arg\min_{r_k} \sum_{s=1}^{q} \left\| u_k^s - \tilde{\mu}_k(x_k^s, r_k) \right\|^2; \tag{3.33}$$

cf. our discussion of approximation in policy space in Section 1.3.3.

3.4.1 The Use of Classifiers for Approximation in Policy Space

As we have noted in Section 3.1, in the case of a continuous control space, training of a parametric architecture for policy approximation is similar to training for a cost approximation. In the case where the control space is finite, however, it is useful to make the connection of approximation in policy space with *classification*; cf. Fig. 3.1.2 and the discussion of Section 3.1.

Classification is an important subject in machine learning. The objective is to construct an algorithm, called a *classifier*, which assigns a given "object" to one of a finite number of "categories" based on its "characteristics." Here we use the term "object" generically. In some cases, the classification may relate to persons or situations. In other cases, an object may represent a hypothesis, and the problem is to decide which of the hypotheses is true, based on some data. In the context of approximation in policy space, *objects correspond to states, and categories correspond to controls to be applied at the different states*. Thus in this case, we view each sample $\left(x_k^s, u_k^s\right)$ as an object-category pair.

Generally, in classification we assume that we have a population of objects, each belonging to one of m categories $c = 1, \ldots, m$. We want to be able to assign a category to any object that is presented to us. Mathematically, we represent an object with a vector x (e.g., some raw description or a vector of features of the object), and we aim to construct a rule that assigns to every possible object x a unique category c.

To illustrate a popular classification method, let us assume that if we draw an object x at random from this population, the conditional probability of the object being of category c is $p(c \,|\, x)$. If we know the probabilities $p(c \,|\, x)$, we can use a classical statistical approach, whereby we assign x to

the category $c^*(x)$ that has maximal posterior probability, i.e.,

$$c^*(x) \in \arg \max_{c=1,\ldots,m} p(c \,|\, x). \tag{3.34}$$

This is called the Maximum a Posteriori rule (or MAP rule for short; see for example the book [BeT08], Section 8.2, for a discussion).

When the probabilities $p(c \,|\, x)$ are unknown, we may try to estimate them using a least squares optimization, based on the following property, whose proof is outlined in Exercise 3.1.

Proposition 3.4.1: (Least Squares Property of Conditional Probabilities) Let $\xi(x)$ be any prior distribution of x, so that the joint distribution of (c, x) is

$$\zeta(c, x) = \xi(x)p(c \,|\, x).$$

For a pair of classes (c, c'), define $z(c, c')$ by

$$z(c, c') = \begin{cases} 1 & \text{if } c = c', \\ 0 & \text{otherwise,} \end{cases}$$

and for a fixed class c and any function h of (c, x), consider

$$E\Big\{ \big(z(c, c') - h(c, x)\big)^2 \Big\},$$

the expected value with respect to the distribution $\zeta(c', x)$ of the random variable $\big(z(c, c') - h(c, x)\big)^2$. Then $p(c \,|\, x)$ minimizes this expected value over all functions $h(c, x)$, i.e., for all functions h and all classes c, we have

$$E\Big\{ \big(z(c, c') - p(c \,|\, x)\big)^2 \Big\} \leq E\Big\{ \big(z(c, c') - h(c, x)\big)^2 \Big\}. \tag{3.35}$$

The proposition states that $p(c \,|\, x)$ is the function of (c, x) that minimizes

$$E\Big\{ \big(z(c, c') - h(c, x)\big)^2 \Big\} \tag{3.36}$$

over all functions h of (c, x), *for any prior distribution of x and class c.* This suggests that we can obtain approximations to the probabilities $p(c \,|\, x)$, $c = 1, \ldots, m$, by minimizing an empirical/simulation-based approximation of the expected value (3.36).

More specifically, let us assume that we have a training set consisting of q object-category pairs (x^s, c^s), $s = 1, \ldots, q$, and corresponding vectors

$$z^s(c) = \begin{cases} 1 & \text{if } c^s = c, \\ 0 & \text{otherwise,} \end{cases} \qquad c = 1, \ldots, m,$$

and let us adopt a parametric approach. In particular, for each category $c = 1, \ldots, m$, we approximate the probability $p(c \,|\, x)$ with a function $\tilde{h}(c, x, r)$ that is parametrized by a vector r, and optimize over r the empirical approximation to the expected squared error of Eq. (3.36). Thus we can obtain r by the least squares regression:

$$\bar{r} \in \arg\min_r \sum_{s=1}^{q} \sum_{c=1}^{m} \big(z^s(c) - \tilde{h}(c, x^s, r)\big)^2, \tag{3.37}$$

perhaps with some quadratic regularization added. The functions $\tilde{h}(c, x, r)$ may be provided for example by a feature-based architecture or a neural network.

Note that each training pair (x^s, c^s) is used to generate m examples for use in the regression problem (3.37): $m - 1$ "negative" examples of the form $(x^s, 0)$, corresponding to the $m - 1$ categories $c \neq c^s$, and one "positive" example of the form $(x^s, 1)$, corresponding to $c = c^s$. Note also that the incremental gradient method can be applied to the solution of this problem.

The regression problem (3.37) approximates the minimization of the expected value (3.36), so we conclude that its solution $\tilde{h}(c, x, \bar{r})$, $c = 1, \ldots, m$, approximates the probabilities $p(c \,|\, x)$. Once this solution is obtained, we may use it to classify a new object x according to the rule

$$\text{Estimated Object Category} = \tilde{c}(x, \bar{r}) \in \arg\max_{c=1,\ldots,m} \tilde{h}(c, x, \bar{r}), \tag{3.38}$$

which approximates the MAP rule (3.34); cf. Fig. 3.4.1.

Returning to approximation in policy space, for a given training set (x_k^s, u_k^s), $s = 1, \ldots, q$, the classifier just described provides (approximations to) the "probabilities" of using the controls $u_k \in U_k(x_k)$ at the states x_k, so it yields a "randomized" policy $\tilde{h}(u, x_k, r_k)$ for stage k [once the values $\tilde{h}(u, x_k, r_k)$ are normalized so that, for any given x_k, they add to 1]; cf. Fig. 3.4.2. In practice, this policy is usually approximated by the deterministic policy $\tilde{\mu}_k(x_k, r_k)$ that uses at state x_k the control of maximal probability at that state; cf. Eq. (3.38).

For the simpler case of a classification problem with just two categories, say A and B, a similar formulation is to hypothesize a relation of the following form between object x and its category:

$$\text{Object Category} = \begin{cases} A & \text{if } \tilde{h}(x, r) = 1, \\ B & \text{if } \tilde{h}(x, r) = -1, \end{cases}$$

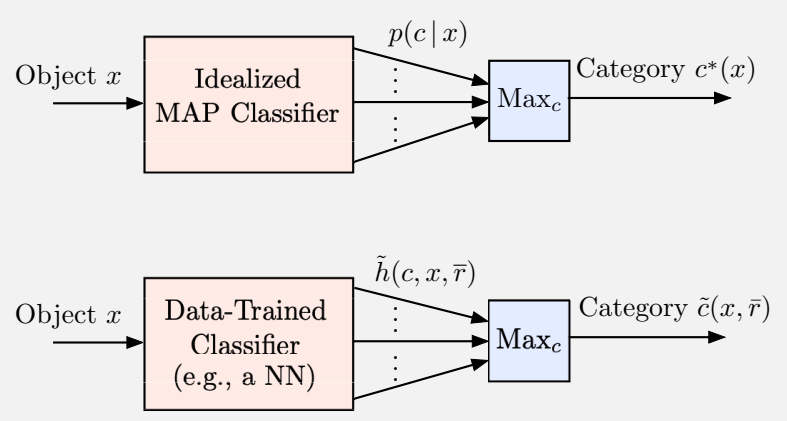

Figure 3.4.1 Illustration of the MAP classifier $c^*(x)$ for the case where the probabilities $p(c \mid x)$ are known [cf. Eq. (3.34)], and its data-trained version $\tilde{c}(x, \bar{r})$ [cf. Eq. (3.38)]. The classifier may be obtained by using the data set (x_k^s, u_k^s), $s = 1, \ldots, q$, and an approximation architecture such as a feature-based architecture or a neural network.

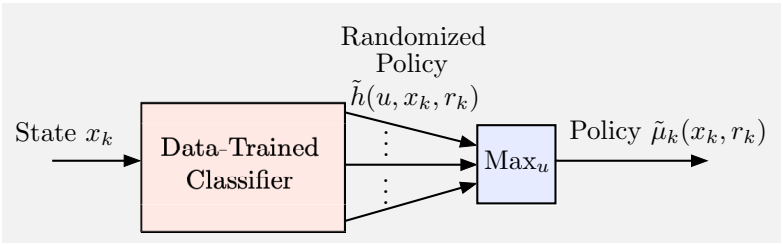

Figure 3.4.2 Illustration of classification-based approximation in policy space. The classifier, defined by the parameter r_k, is constructed by using the training set (x_k^s, u_k^s), $s = 1, \ldots, q$. It yields a randomized policy that consists of the probability $\tilde{h}(u, x_k, r_k)$ of using control $u \in U_k(x_k)$ at state x_k. This policy is approximated by the deterministic policy $\tilde{\mu}_k(x_k, r_k)$ that uses at state x_k the control that maximizes over $u \in U_k(x_k)$ the probability $\tilde{h}(u, x_k, r_k)$ [cf. Eq. (3.38)].

where $\tilde{h}$ is a given function and r is the unknown parameter vector. Given a set of q object-category pairs $(x^1, z^1), \ldots, (x^q, z^q)$ where

$$z^s = \begin{cases} 1 & \text{if } x \text{ is of category } A, \\ -1 & \text{if } x \text{ is of category } B, \end{cases}$$

we obtain r by the least squares regression:

$$\bar{r} \in \arg\min_r \sum_{s=1}^{q} \big(z^s - \tilde{h}(x^s, r)\big)^2.$$

The optimal parameter vector $\bar{r}$ is used to classify a new object with data vector x according to the rule

$$\text{Estimated Object Category} = \begin{cases} A & \text{if } \tilde{h}(x, \bar{r}) > 0, \\ B & \text{if } \tilde{h}(x, \bar{r}) < 0. \end{cases}$$

In the context of DP and approximation in policy space, this classifier may be used, among others, in stopping problems where there are just two controls available at each state: stopping (i.e., moving to a termination state) and continuing (i.e., moving to some nontermination state).

There are several variations of the preceding classification schemes, for which we refer to the specialized literature. Moreover, there are several commercially and publicly available software packages for solving the associated regression problems and their variants. They can be brought to bear on the problem of parametric approximation in policy space using any training set of state-control pairs, regardless of how it was obtained.

3.4.2 Policy Iteration with Value and Policy Networks

We noted earlier that contrary to rollout, approximate policy iteration (PI) is fundamentally an off-line training algorithm, because for a large scale problem, it is necessary to represent the cost functions or Q-factors of the successively generated policies with an approximation architecture. Thus, in a typical implementation, approximate PI involves the successive use of value networks to represent the cost functions of the generated policies, and one-step or multistep lookahead minimization to implement policy improvement.

On the other hand, it is also possible to use policy networks to approximate the results of policy improvement. In particular, we can start with a base policy and a terminal cost approximation, and generate state-control samples of the corresponding truncated rollout policy. These samples can be used with an approximation in policy space scheme to train a policy network that approximates the truncated rollout policy.

Then the cost function of the policy network can be approximated with a value network using the cost approximation methodology that we have discussed in this chapter. This value network can be used in turn as a terminal cost approximation in a truncated rollout algorithm where the previously obtained policy network can be used as a base policy. A new policy network can then be trained using samples of this rollout policy, etc. Thus a perpetual rollout scheme is obtained, which involves a sequence of value and policy networks.

One may also consider approximate PI algorithms that do not use a value network at all. Indeed the value network is only used to provide the approximate cost function values of the current policy, which are needed to calculate samples of the improved policy and train the corresponding

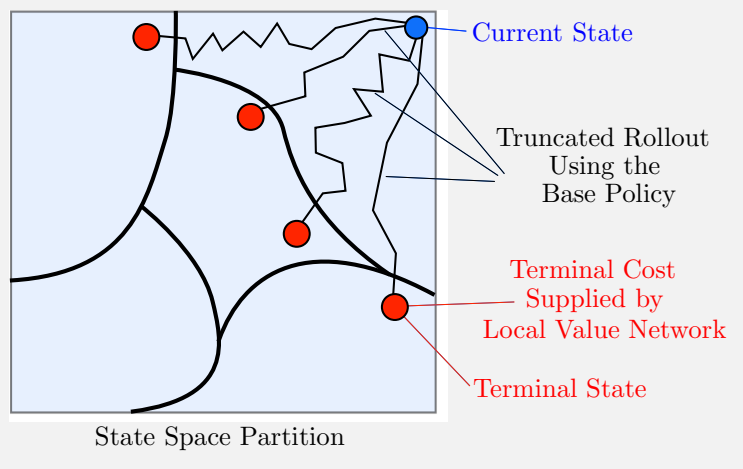

Figure 3.4.3 Illustration of a truncated rollout scheme with a partitioned architecture. A local value network is used for terminal cost function approximation for each subset of the partition.

policy network. On the other hand the samples of the improved policy can also be computed by rollout, using simulation-generated cost function values of the current policy. If the rollout can be suitably implemented with simulation, the training of a value network may be unnecessary.

Multiprocessor Parallelization

We have noted earlier that parallelization and distributed computation can be used in several different ways in rollout and PI schemes, including Q-factor, Monte Carlo, and multiagent parallelization. It is also possible to consider the use of multiple neural networks in the implementation of rollout or approximate PI. For example, when feature-based partitioning of the state space is used (cf. Example 3.1.8), we may consider a multiprocessor parallelization scheme, which involves multiple local value and/or policy networks, which operate locally within a subset of the state space partition; see Figs. 3.4.3 and 3.4.4.

Let us finally note that multiprocessor parallelization leads to the idea of an approximation architecture that involves a graph. Each node of the graph consists of a neural network and each arc connecting a pair of nodes corresponds to data transfer between the corresponding neural networks. The question of how to train such an architecture is quite complex and one may think of several alternative possibilities. For example the training may be collaborative with the exchange of training results and/or training data communicated periodically or asynchronously; see the book [Ber20a], Section 5.8.

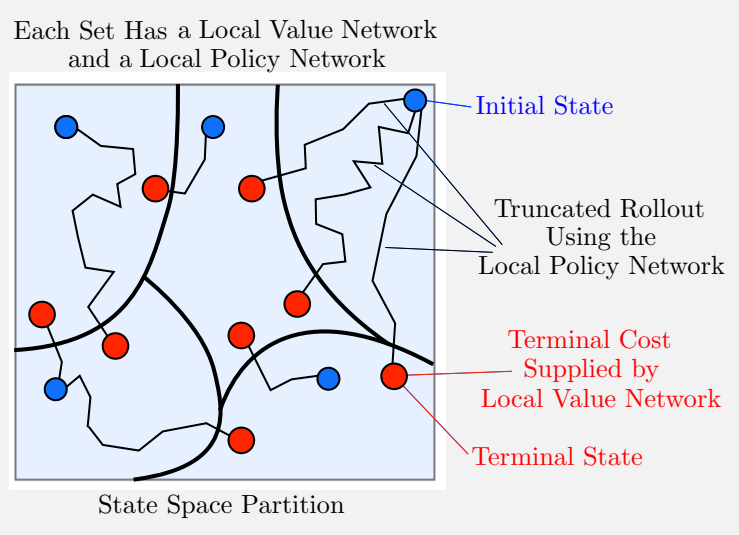

Figure 3.4.4 Illustration of a perpetual truncated rollout scheme with a partitioned architecture. A local value network and a local policy network are used for each subset of the partition. The policy network is used as the base policy and the value network is used to provide a terminal cost function approximation.

State-control training pairs for the corresponding rollout policy are obtained by starting at an initial state within some subset of the partition, generating rollout trajectories using the local policy network, which are truncated once the state enters a different subset of the partition, with the corresponding terminal cost function approximation supplied by the value network of that subset.

When a separate processor is used for each subset of the partition, the corresponding value networks are communicated between processors. This can be done asynchronously, with each processor sharing its value network as it becomes available. In a variation of this scheme, the local policy networks may also be shared selectively among processors for selective use in the truncated rollout process.

3.4.3 Why Use On-Line Play and Not Just Train a Policy Network to Emulate the Lookahead Minimization?

This is a sensible and common question, which stems from the mindset that neural networks have extraordinary function approximation properties. In other words, why go through the arduous and time-consuming process of on-line lookahead minimization, if we can do the same thing off-line and represent the lookahead policy with a trained policy network? In particular, we can select the policy from a suitably restricted class of policies, such as a parametric class of the form $\mu(x, r)$, where r is a parameter vector. We may then estimate r using some type of off-line training. Then the on-line computation of controls $\mu(x, r)$ can be much faster compared with on-line lookahead minimization.

On the negative side, because parametrized approximations often in-

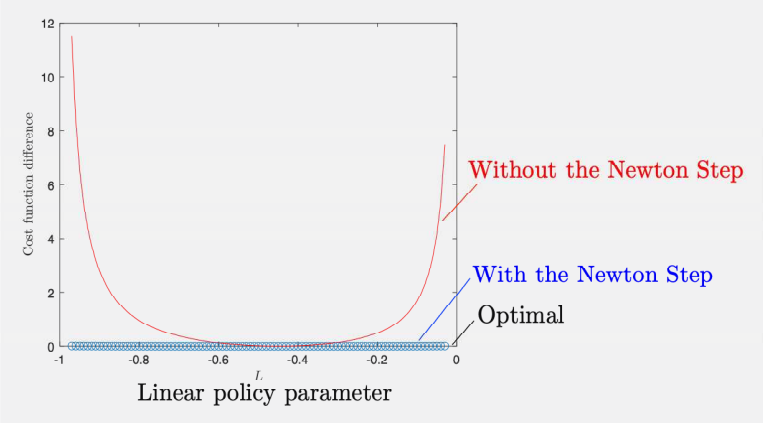

Figure 3.4.5 Illustration of the performance enhancement obtained when the Newton step/rollout is used in conjunction with an off-line trained (suboptimal) base policy for the linear quadratic problem; cf. Example 3.4.1. The figure shows the quadratic cost coefficient differences $K_L - K^*$ and $K_{\tilde{L}} - K^*$ as a function of L, where K_L is the quadratic cost coefficient of μ (without one-step lookahead/Newton step) and K_L is the quadratic cost coefficient of $\tilde{\mu}$ (with one-step lookahead/Newton step). The optimal performance is obtained for $L^* \approx -0.4$.

volve substantial calculations, they are not well suited for on-line replanning. In other words, *when the problem parameters change, as in adaptive control, there may not be enough time to retrain the parametric policy to deal with the changes*. From our point of view in this book, there is also another important reason why approximation in value space is needed on top of approximation in policy space: *the off-line trained policy may not perform nearly as well as the corresponding one-step or multistep lookahead/rollout policy*, because it lacks the extra power of the associated exact Newton step (cf. our discussion of AlphaZero and TD-Gammon in Section 1.1, and linear quadratic problems in Section 1.5).

Example 3.4.1 (A Linear-Quadratic Example)

Let us illustrate the benefit of the Newton step associated with one-step lookahead in an adaptive control setting. We consider a one-dimensional linear-quadratic problem, and we compare the performance of a linear policy with its corresponding one-step lookahead/rollout policy; cf. Fig. 3.4.5. In this example the system equation has the form

$$x_{k+1} = x_k + 2u_k,$$

and the quadratic cost function parameters are $q = 1$, $r = 0.5$. The optimal policy can be calculated to be

$$\mu^*(x) = L^* x,$$

with $L^* \approx -0.4$, and the optimal cost function is

$$J^*(x) = K^* x^2,$$

where $K^* \approx 1.1$. We want to explore what happens when we use a policy of the form

$$\mu_L(x) = Lx,$$

where $L \neq L^*$ (e.g., a policy that is optimal for another system equation or cost function parameters). The cost function of μ_L has the form

$$J_\mu(x) = K_L x^2,$$

where K_L is obtained by using the formulas given in Section 1.5. When J_μ is used as cost function approximation in a one-step lookahead/Newton step scheme, the cost function obtained is $K_{\tilde{L}} x^2$, where $\tilde{\mu}(x) = \tilde{L}x$ is the corresponding one-step lookahead policy. As the figure shows, when the change $L - L^*$ is substantial, the performance degradation $K_{\tilde{L}} - K^*$ is negligible, while the performance degradation $K_L - K^*$ is substantial.

3.5 POLICY GRADIENT AND RELATED METHODS

In this section we focus on infinite horizon problems and we discuss an alternative training approach for approximation in policy space, which is based on controller parameter optimization: we parametrize the policies by a vector r, and we optimize the corresponding expected cost over r. Thus, in contrast with the preceding section, we directly aim to approximate an optimal policy, rather than approximate a fixed policy.

For the most part in this section, we will assume no constraints on the parameter r, although we will occasionally comment regarding the presence of constraints. In particular, we determine r through the minimization

$$\min_r E\big\{J_{\tilde{\mu}(r)}(i_0)\big\}, \tag{3.39}$$

where $J_{\tilde{\mu}(r)}(i_0)$ is the cost of the policy $\tilde{\mu}(r)$ starting from the initial state i_0, and the expected value above is taken with respect to a suitable probability distribution of the initial state i_0 (cf. Fig. 3.5.1). This is to be contrasted with the classification approach of the preceding section, whereby we aim to learn a fixed policy, possibly as part of a policy iteration process. Here, we instead aim directly for an (approximately) optimal policy.

Note that in the case where the initial state i_0 is known and fixed, the method involves just minimization of $J_{\tilde{\mu}(r)}(i_0)$ over r. This simplifies the minimization a great deal, particularly when the problem is deterministic.

Similar to the preceding section, *we focus on off-line training of policies*. The on-line training of a policy involves a controller that interacts with the environment, applies controls, observes in real-time the effects

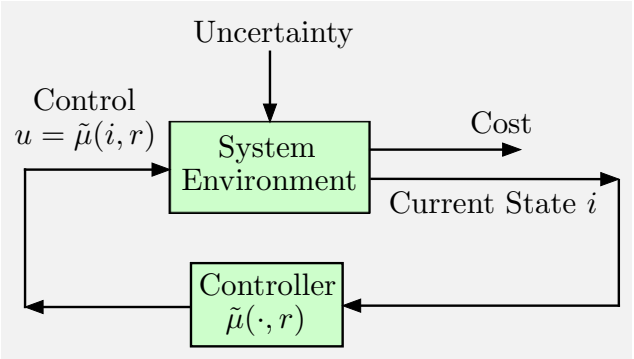

Figure 3.5.1 Illustration of the policy optimization framework for an infinite horizon problem. Policies are parametrized with a parameter vector r and denoted by $\tilde{\mu}(r)$, with components $\tilde{\mu}(i, r)$, $i = 1, \ldots, n$. Each parameter value r determines a policy $\tilde{\mu}(r)$, and a cost $J_{\tilde{\mu}(r)}(i_0)$ for each initial state i_0, as indicated in the figure. The optimization determines r through the minimization

$$\min_r E\Big\{J_{\tilde{\mu}(r)}(i_0)\Big\},$$

where the expected value above is taken with respect to a suitable probability distribution of i_0.

of these controls, and updates the policy parameters based on the observations. This challenging context is of considerable interest in practice, particularly in robotics, but is beyond our scope in this book.

Before delving into the details, it is worth reminding the reader that using an off-line trained policy without combining it with approximation in value space has the fundamental shortcoming, which we noted frequently in this book: *the off-line trained policy will not perform nearly as well as a scheme that uses this policy in conjunction with lookahead minimization*, because it lacks the extra power of the associated exact Newton step.

3.5.1 Gradient Methods for Parametric Cost Optimization

Let us first consider methods that perform the minimization (3.39) by using a gradient method, and for simplicity let us assume that the initial condition i_0 is known [otherwise an expected value over i_0 must be introduced in the cost function; cf. Eq. (3.39)]. Thus the aim is to minimize $J_{\tilde{\mu}(r)}(i_0)$ over r by using the gradient method

$$r^{k+1} = r^k - \gamma^k \nabla J_{\tilde{\mu}(r^k)}(i_0), \qquad k = 0, 1, \ldots, \tag{3.40}$$

assuming that $J_{\tilde{\mu}(r)}(i_0)$ is differentiable with respect to r. Here γ^k is a positive stepsize parameter, and $\nabla(\cdot)$ denotes gradient with respect to r evaluated at the current iterate r^k.

An important concern in this method is that the gradients $\nabla J_{\tilde{\mu}(r^k)}(i_0)$ may not be explicitly available. In this case, the gradients can be approximated by finite differences of cost function values $J_{\tilde{\mu}(r^k)}(i_0)$. Unfortunately, when the problem is stochastic, the cost function values may be computable only through Monte Carlo simulation. This may introduce a large amount of noise, so it is likely that we will need to average many samples in order to obtain sufficiently accurate gradients, thereby making the method inefficient. On the other hand, when the problem is deterministic, this difficulty does not arise, and the use of the gradient method (3.40) or other methods that do not rely on the use of gradients (such as coordinate descent) is facilitated.

In what follows in this section we will focus on alternative and typically more efficient gradient-like methods for stochastic problems, which are based on sampling, rather than the exact calculation of the gradient, as in Eq. (3.40). Before doing so, we address briefly the issue of nondifferentiabilities and constraints in gradient-based optimization.

Constraints and Nondifferentiabilities - Soft-Max Approximation

The gradient method (3.40) assumes that the cost function $J_{\tilde{\mu}(r)}$ is differentiable with respect to r. Frequently, however, this is not so, and to deal with nondifferentiabilities, a common practice has been to use the so-called *soft-max approximation* to smooth the nondifferentiabilities. As an example of this approach, nondifferentiable terms of the form

$$f(r) = \max\bigl\{f_1(r), \ldots, f_m(r)\bigr\};$$

can be approximated by

$$\frac{1}{c} \ln \left\{ \sum_{i=1}^{m} \lambda_i e^{c f_i(r)} \right\},$$

where c is a positive parameter, and λ_i are scalars satisfying

$$\sum_{i=1}^{m} \lambda_i = 1, \qquad \lambda_i > 0,\ i = 1, \ldots, m.$$

The parameter c controls the quality of the approximation (as c increases the approximation becomes better, but also the smoothed problem becomes more ill-conditioned and harder to solve by gradient methods). The parameters λ_i can be interpreted as Lagrange multipliers, which when chosen properly, improve the quality of the approximation of the smoothed func-

tion in the neighborhood of an optimal solution; see the book [Ber82a] for a detailed discussion.†

Another difficulty associated with the gradient method (3.40) is that there may be constraints on the parameter vector r. Gradient methods can be modified to deal with constraints, particularly in simple cases, such as nonnegativity or simplex constraints; see the discussion of Section 3.5.3 on gradient projection and proximal algorithms. However, constraints can also be dealt with by reparametrization of the policies in a way that eliminates the constraints. An important special case arises when the policies of a problem with finite state and control spaces are replaced by randomized policies, which are then reparametrized to eliminate the associated probability distribution constraints by using the soft-max operation.

In particular, suppose that the set of policies is enlarged to include all randomized policies. Then, for a problem with states $i = 1, \ldots, n$, controls $u = 1, \ldots, m$, and constraints $u \in U(i)$, an ordinary policy μ, which applies control $\mu(i) \in U(i)$, $i = 1, \ldots, n$, is replaced by a randomized policy that applies at state i the control u_i with probability $r_u(i)$. This randomized policy consists of a probability distribution $\big\{r_u(i) \mid u \in U(i)\big\}$ for each state i, where the probabilities/parameters $r_u(i)$ must satisfy

$$\sum_{u \in U(i)} r_u(i) = 1, \qquad r_u(i) \geq 0, \quad i = 1, \ldots, n. \tag{3.41}$$

The policy optimization aims to find an optimal probability distribution for each state i.

The important point here is that the set of randomized policies is continuous and thus far better suited for the application of the gradient optimization (3.40). On the other hand the parameters $r_u(i)$ must satisfy the simplex constraints (3.41), thus requiring a gradient method that can deal with constraints. The corresponding policy optimization problem can be solved by a gradient-like method that can deal with the probability constraints (3.41). Such methods will be briefly discussed in the next section. However, it is also possible to reparametrize the space of probability distributions with a transformation that approximates probabilities

† The soft-max approximation is discussed within the context of a broad class of smoothing schemes in the author's book [Ber82a], Chapters 3 and 5. It was first proposed in the author's papers [Ber75], [Ber77]; see also the paper by Poljak [Pol79], and the paper by Nesterov [Nes05], who used the smoothing approach for the complexity analysis of gradient methods. The soft-max approximation also bears a close connection to the exponential method of multipliers, first proposed by Kort and Bertsekas [KoB72]; see the books [Ber82a], Chapter 5, and [Ber16], Chapter 6, which also discuss a duality connection with the entropy minimization algorithm, a proximal minimization algorithm that is based on an exponential penalty function.

as follows:

$$r_u(i) \approx \frac{e^{\theta_u(i)\phi(i)}}{\sum_{v \in U(i)} e^{\theta_v(i)\phi(i)}}, \qquad i = 1, \ldots, n, \tag{3.42}$$

where $\theta_u(i)$ are the transformed parameters and $\phi(i)$ are some scalars that depend on i. In this way the policy optimization problem becomes unconstrained. We note, however, that there are some serious difficulties associated with this reparametrization. In particular, policy controls $r_u(i)$ that are applied with probability 1, are mapped to parameters $\theta_u(i)$ that are equal to ∞ in the reparametrized problem; see Exercise 3.2(b). Still the reparametrization $r_u(i) \mapsto \theta_u(i)$ of Eq. (3.42) is used widely in practice.

3.5.2 Policy Gradient-Like Methods

To get a sense of the general principle underlying the incremental gradient approach that uses randomization and sampling, let us digress from the DP context of this chapter, and consider the generic optimization problem

$$\min_{z \in Z} F(z), \tag{3.43}$$

where Z is some constraint set, and F is some real-valued function. Note that F may be nondifferentiable, and indeed the constraint set Z may be discrete (e.g., Z may be the finite set of stationary policies in a finite-spaces Markovian decision problem; see Exercise 3.2). Hence gradient methods would seem like an unlikely candidate for solution. However, we will transform the problem to one that is amenable to the use of the gradient methodology through an unusual procedure that we describe next.

We will convert the problem (3.43) to the *stochastic* problem

$$\min_{p \in \mathcal{P}_Z} E_p\big\{F(z)\big\}, \tag{3.44}$$

where z is viewed as a random variable, $\mathcal{P}_Z$ is the set of probability distributions over Z, p denotes the generic distribution in $\mathcal{P}_Z$, and $E_p\{\cdot\}$ denotes expected value with respect to p. Of course this enlarges the search space from Z to $\mathcal{P}_Z$, but it enhances the use of randomization schemes, simulation-based methods, even if the original problem is deterministic. Moreover, the cost function of the stochastic problem (3.44) may have some nice differentiability properties that are lacking in the original deterministic version (3.43). This is true in general, although under special assumptions on F, such as convexity, these differentiability properties are enhanced (see the paper [Ber73]). The reader may be able to appreciate the striking differences between the deterministic problem (3.43) and the stochastic problem (3.44) by working out simple examples; see Exercise 3.2.

At this point it is not clear how the stochastic optimization problem (3.44) relates to our stochastic DP context of this chapter. We will return to

this question later, but for the purpose of orientation, we note that to obtain a problem of the form (3.44), we must enlarge the set of policies to include *randomized policies*, mapping a state i into a probability distribution over the set of controls $U(i)$.

Suppose now that we restrict attention to a subset $\tilde{\mathcal{P}}_Z \subset \mathcal{P}_Z$ of probability distributions $p(z;r)$ that are parametrized by some continuous parameter r, e.g., a vector in some Euclidean space. In other words, we approximate the stochastic optimization problem (3.44) with the restricted problem

$$\min_r E_{p(z;r)}\big\{F(z)\big\}. \tag{3.45}$$

Then we may use a gradient method for solving this problem, such as

$$r^{k+1} = r^k - \gamma^k \nabla\Big(E_{p(z;r^k)}\big\{F(z)\big\}\Big), \qquad k = 0, 1, \ldots, \tag{3.46}$$

where $\nabla(\cdot)$ denotes gradient with respect to r of the function in parentheses, evaluated at the current iterate r^k.†

Likelihood-Ratio Policy Gradient Methods

We will first consider an incremental version of the gradient method (3.46). This method requires that $p(z;r)$ is differentiable with respect to r. It relies on a convenient gradient formula, sometimes referred to as the *log-likelihood trick*, which involves the natural logarithm of the sampling distribution.

This formula is obtained by the following calculation, which is based on interchanging gradient and expected value, and using the gradient formula $\nabla(\log p) = \nabla p/p$. We have

$$\begin{aligned}
\nabla\Big(E_{p(z;r)}\big\{F(z)\big\}\Big) &= \nabla\left(\sum_{z\in Z} p(z;r)F(z)\right) \\
&= \sum_{z\in Z} \nabla p(z;r)F(z) \\
&= \sum_{z\in Z} p(z;r)\frac{\nabla p(z;r)}{p(z;r)}F(z) \\
&= \sum_{z\in Z} p(z;r)\nabla\Big(\log\big(p(z;r)\big)\Big)F(z),
\end{aligned}$$

† The parametrization $p(z;r)$ must be differentiable with respect to r in order for the gradient to exist. One possibility, noted earlier, is to use a soft-max parametrization such as $p(z;r) = \frac{e^{r'\phi(z)}}{\sum_{y\in Z} e^{r'\phi(y)}}$ for all $z \in Z$, where $\phi(z)$ denotes a vector that suitably depends on z. This may involve some pitfalls, which are illustrated in Exercise 3.2(b).

and finally

$$\nabla\Big(E_{p(z;r)}\big\{F(z)\big\}\Big) = E_{p(z;r)}\Big\{\nabla\Big(\log\big(p(z;r)\big)\Big)F(z)\Big\}, \tag{3.47}$$

where for any given z, $\nabla\Big(\log\big(p(z;r)\big)\Big)$ is the gradient with respect to r of the function $\log\big(p(z;\cdot)\big)$, evaluated at r (the gradient is assumed to exist).

The preceding formula suggests an incremental implementation of the gradient iteration (3.46) that approximates the expected value in the right side in Eq. (3.47) with a single sample (cf. Section 3.1.3). The typical iteration of this method is as follows.

Sample-Based Gradient Method for Parametric Approximation of $\min_{z\in Z} F(z)$

Let r^k be the current parameter vector.

(a) Obtain a sample z^k according to the distribution $p(z;r^k)$.

(b) Compute the sampled gradient

$$\nabla\Big(\log\big(p(z^k;r^k)\big)\Big).$$

(c) Iterate according to

$$r^{k+1} = r^k - \gamma^k\nabla\Big(\log\big(p(z^k;r^k)\big)\Big)F(z^k). \tag{3.48}$$

The advantage of the preceding sample-based method is its simplicity and generality. It allows the use of parametric approximation for any minimization problem (well beyond DP), as long as the logarithm of the sampling distribution $p(z;r)$ can be conveniently differentiated with respect to r, and samples of z can be obtained using the distribution $p(z;r)$.

Note that in iteration (3.48) r is adjusted along a random direction. This direction does not involve at all the gradient of F (even if it exists), only the gradient of the logarithm of the sampling distribution! As a result the iteration has a *model-free character*: we don't need to know the mathematical form of the function F as long as we have a simulator that produces the cost function value $F(z)$ for any given z. This is also a major advantage offered by many random search methods.

An important issue is the efficient computation of the sampled gradient $\nabla\big(\log\big(p(z^k;r^k)\big)\big)$. In the context of DP, including the SSP and discounted problems that we have been dealing with, there are some specialized procedures and corresponding parametrizations to approximate this gradient conveniently. The following is an example.

Example 3.5.1 (Policy Gradient Method for Discounted DP)

Consider the α-discounted problem and denote by z the infinite horizon state-control trajectory:

$$z = \{i_0, u_0, i_1, u_1, \ldots\}.$$

We consider a parametrization of randomized policies with parameter r, so the control at state i is generated according to a distribution $p(u \mid i; r)$ over $U(i)$. Then for a given r, the state-control trajectory z is a random vector with probability distribution denoted $p(z; r)$. The cost corresponding to the trajectory z is

$$F(z) = \sum_{m=0}^{\infty} \alpha^m g(i_m, u_m, i_{m+1}), \tag{3.49}$$

and the problem is to minimize over r

$$E_{p(z;r)}\big\{F(z)\big\}.$$

To apply the sample-based gradient method (3.48), given the current iterate r^k, we must generate the sample state-control trajectory

$$z^k = \{i_0^k, u_0^k, i_1^k, u_1^k, \ldots\},$$

according to the distribution $p(z; r^k)$, we compute the corresponding cost $F(z^k)$, and also calculate the sampled gradient

$$\nabla\Big(\log\big(p(z^k; r^k)\big)\Big). \tag{3.50}$$

Let us assume that the logarithm of the randomized policy distribution $p(u \mid i; r)$ is differentiable with respect to r (a soft-max policy parametrization is often recommended for this purpose). Then the logarithm that is differentiated in Eq. (3.50) can be written as

$$\begin{aligned}\log\big(p(z^k; r^k)\big) &= \log \prod_{m=0}^{\infty} p_{i_m^k i_{m+1}^k}(u_m^k) p(u_m^k \mid i_m^k; r^k) \\ &= \sum_{m=0}^{\infty} \log\big(p_{i_m^k i_{m+1}^k}(u_m^k)\big) + \sum_{m=0}^{\infty} \log\big(p(u_m^k \mid i_m^k; r^k)\big),\end{aligned}$$

and its gradient (3.50), which is needed in the iteration (3.48), is given by

$$\nabla\Big(\log\big(p(z^k; r^k)\big)\Big) = \sum_{m=0}^{\infty} \nabla\Big(\log\big(p(u_m^k \mid i_m^k; r^k)\big)\Big). \tag{3.51}$$

This gradient involves the current randomized policy, but does not involve the system's transition probabilities and the costs per stage.

The policy gradient method (3.48) can now be implemented with a finite horizon approximation whereby r^k is changed after a finite number N

of time steps [so the infinite cost and gradient sums (3.49) and (3.51) are replaced by finite sums]. The method takes the form

$$r^{k+1} = r^k - \gamma^k \sum_{m=0}^{N-1} \nabla \log \big(p(u_m^k \mid i_m^k; r^k)\big) F_N(z_N^k),$$

where $z_N^k = (i_0^k, u_0^k, \ldots, i_{N-1}^k, u_{N-1}^k)$ is the generated N-step trajectory, $F_N(z_N^k)$ is the corresponding cost, and γ^k is the stepsize. The initial state i_0^k of the trajectory is chosen randomly, with due regard to exploration issues.

Policy gradient methods for other types of DP problems can be similarly developed, as well as variations involving a combination of policy and cost function parametrizations [e.g., replacing $F(z)$ of Eq. (3.49) by a parametrized estimate that has smaller variance, and possibly subtracting a suitable baseline, cf. the following discussion]. This leads to a class of *actor-critic* methods that differ from the PI-type methods that we discussed earlier in this chapter; we refer to the end-of-chapter literature for a variety of specific schemes. The policy gradient methods that we discussed in this section are sometimes called *actor-only* methods, as they involve only policy parametrizations.

Implementation Issues

There are several additional issues to consider in the implementation of the sample-based gradient method (3.48). The first of these is that the problem solved is a randomized version of the original. If the method produces a parameter $\overline{r}$ in the limit and the distribution $p(z;\overline{r})$ is not atomic (i.e., it is not concentrated at a single point), then a solution $\overline{z} \in Z$ must be extracted from $p(z;\overline{r})$. In the SSP and discounted problems that we have been dealing with, the subset $\tilde{\mathcal{P}}_Z$ of parametric distributions typically contains the atomic distributions, while it can be shown that minimization over the set of all distributions $\mathcal{P}_Z$ produces the same optimal value as minimization over Z (the use of randomized policies does not improve the optimal cost of the problem), so this difficulty does not arise.

Another issue is how to improve sampling efficiency. To this end, let us note a simple generalization of the gradient method (3.48), which can often improve its performance. It is based on the gradient formula

$$\nabla\Big(E_{p(z;r)}\big\{F(z)\big\}\Big) = E_{p(z;r)}\Big\{\nabla\Big(\log\big(p(z;r)\big)\Big)\big(F(z)-b\big)\Big\}, \qquad (3.52)$$

where b is any scalar. This formula generalizes Eq. (3.47), where $b = 0$, and holds in view of the following calculation, which shows that the term multiplying b in Eq. (3.52) is equal to 0:

$$E_{p(z;r)}\Big\{\nabla\Big(\log\big(p(z;r)\big)\Big)\Big\} = E_{p(z;r)}\left\{\frac{\nabla p(z;r)}{p(z;r)}\right\}$$

$$= \sum_{z \in Z} p(z;r) \frac{\nabla p(z;r)}{p(z;r)}$$

$$= \sum_{z \in Z} \nabla p(z;r) = \nabla \left(\sum_{z \in Z} p(z;r) \right) = 0,$$

where the last equality holds because $\sum_{z \in Z} p(z;r)$ is identically equal to 1 and hence does not depend on r.

Based on the gradient formula (3.52), we can modify the sampled gradient iteration (3.48) to read as follows:

$$r^{k+1} = r^k - \gamma^k \nabla \Big(\log \big(p(z^k; r^k) \big) \Big) \big(F(z^k) - b \big), \tag{3.53}$$

where b is some fixed scalar, called the *baseline*. Whereas the choice of b does not affect the gradient $\nabla \Big(E_{p(z;r)} \big\{ F(z) \big\} \Big)$ [cf. Eq. (3.52)], it affects the incremental gradient

$$\nabla \Big(\log \big(p(z^k; r^k) \big) \Big) \big(F(z^k) - b \big),$$

which is used in the iteration (3.53). Thus, by optimizing the baseline b, empirically or through a calculation (see e.g., [DNP11]), we can improve the performance of the algorithm. Moreover, in the context of discounted and SSP problems, state-dependent baseline functions have been used. Ideas of cost shaping are useful within this context; we refer to the RL textbook [Ber19a] and specialized literature for further discussion.

3.5.3 Scaling and Proximal Policy Optimization Methods

Gradient-like methods are notorious for their slow convergence, which is exacerbated by the detrimental effects of noise (when applied in a stochastic environment). A principal technique to deal with this difficulty is to scale the gradient by multiplication with a positive definite matrix.

In particular, scaled gradient methods for solving general nonlinear unconstrained optimization problems of the form

$$\min_{r \in \Re^m} f(r),$$

where f is a twice continuously differentiable objective function, have the form

$$r^{k+1} = r^k - (D^k)^{-1} \nabla f(r^k), \tag{3.54}$$

where D^k is a positive definite symmetric $m \times m$ matrix (see nonlinear programming books, such as [Ber16]). Two extreme cases are:

(a) Select D^k to be the Hessian matrix of f evaluated at x^k, in which case we obtain *Newton's method*. This method converges superlinearly when started sufficiently close to a strong local minimum (one where the Hessian is positive definite). An approximation to Newton's method for least squares problems where $f(x) = \sum_{i=1}^{N} \|g_i(x)\|^2$, is the *Gauss-Newton method*, which neglects the second derivatives of g_i (see e.g., [Ber16] for discussions of the Gauss-Newton method).

(b) Select D^k to be a diagonal approximation to the Hessian matrix of f evaluated at x^k. This method converges to a strong local minimum under appropriate conditions on D^k (it should not be too "small"). Its convergence rate is linear, and typically much slower than the superlinear convergence rate of Newton's method. On the other hand, a gradient method with diagonal scaling is much simpler than Newton's method, and more readily admits an incremental implementation of the type that we have discussed. Moreover, incremental versions of diagonally scaled gradient methods have worked well for the training of neural networks and other approximation architectures, particularly when supplemented with effective heuristics; cf. the discussion of Section 3.1.3 in connection with the training of neural networks and the ADAM algorithm.

The scaled gradient iteration (3.54) can also be written in an alternative and equivalent form in terms of a quadratic optimization:

$$r^{k+1} \in \arg\min_{r \in \Re^m} \left\{ \nabla f(r^k)'(r - r^k) + \frac{1}{2}(r - r^k)' D^k (r - r^k) \right\}. \qquad (3.55)$$

This is easily verified by setting to 0 the derivative of the quadratic cost above, thereby obtaining

$$0 = \nabla f(r^k) + D^k(r^{k+1} - r^k),$$

which is equivalent to Eq. (3.55).

The form (3.55) of the scaled gradient iteration is in turn related to two other important optimization methods (see nonlinear programming books such as [Ber16]). The first of these is the *gradient projection method*, which applies to constrained optimization problems of the form

$$\min_{r \in C} f(r), \qquad (3.56)$$

where C is some closed convex set. It takes the form

$$r^{k+1} \in \arg\min_{r \in C} \left\{ \nabla f(r^k)'(r - r^k) + \frac{1}{2}(r - r^k)' D^k (r - r^k) \right\}. \qquad (3.57)$$

This iteration can be equivalently written as

$$r^{k+1} = P_{D^k}\big(r^k - \nabla f(r^k)\big), \qquad (3.58)$$

where $P_{D^k}(\cdot)$ denotes projection onto C with respect to a norm that is defined by the scaling matrix D^k (hence the name "gradient projection"). We can see this by writing

$$P_{D^k}\big(r^k - \nabla f(r^k)\big) = \arg\min_{r\in C}\Big\{\big(r - r^k + \nabla f(r^k)\big)' D^k \big(r - r^k + \nabla f(r^k)\big)\Big\},$$

and expanding the quadratic form on the right-hand side, to verify that the iterations of Eqs. (3.57) and (3.58) are equivalent. When the problem is unconstrained, i.e., $C = \Re^m$, the gradient projection iteration (3.57) is equivalent to the scaled gradient iterations (3.54) and (3.55).

Let us also mention the *proximal minimization algorithm*, which applies to the constrained optimization problem (3.56), and has the form

$$r^{k+1} \in \arg\min_{r\in C}\left\{f(r) + \frac{1}{2}(r - r^k)' D^k (r - r^k)\right\}. \tag{3.59}$$

It can be seen that the gradient projection method (3.57) and the proximal algorithm (3.59) have strong similarities, and that they coincide when f is linear. Note also that the two algorithms can be suitably combined in ways that fit the structure of specific problems, and admit incremental and randomized/simulation-based implementations (see the author's survey paper [Ber10d]).

A second way to describe the gradient iteration (3.55) involves the "trust region" optimization

$$r^{k+1} \in \arg\min_{(r-r^k)' D^k (r-r^k)\le \beta^k} \nabla f(r^k)'(r - r^k), \tag{3.60}$$

where β^k is a positive scalar. It can be shown that the methods (3.55) and (3.60) are equivalent for an appropriate value of β^k (see [Ber16]). The role of β^k is to regulate the size of the constraint and ensure that the increment $r^{k+1} - r^k$ is appropriately small. This guards against oscillations of the iterates r^k, which may lead to divergence.

Natural Gradient Methods

While the scaling techniques discussed above have worked well for the general nonlinear optimization problem $\min_{r\in\Re^m} f(r)$, alternative scaling techniques have been suggested for stochastic gradient methods that involve optimization over parametrized probability distributions. In place of an inverse Hessian matrix, these methods use the Fisher Information Matrix (FIM), which accounts for the curvature of the cost function induced by the parametrization.

For the case of the parametric optimization problem

$$\min_r E_{p(z;r)}\big\{F(z)\big\} \tag{3.61}$$

that we have discussed earlier [cf. Eq. (3.45)], the FIM is given by

$$M(r) = E_{p(z;r)}\Big\{\nabla\big(\log\big(p(z;r)\big)\big)\nabla\big(\log\big(p(z;r)\big)\big)'\Big\} \tag{3.62}$$

where the gradient $\nabla\big(\log\big(p(z;r)\big)$ (a column vector) is computed with the formulas given earlier (cf. Example 3.5.1, for the case of discounted DP problems). The scaled version of the corresponding gradient method of Eq. (3.46) is given by

$$r^{k+1} = r^k - \gamma^k\big(M(r^k)\big)^{-1}\nabla\Big(E_{p(z;r^k)}\big\{F(z)\big\}\Big), \tag{3.63}$$

with the gradient above given by the earlier derived formula (3.47):

$$\nabla\Big(E_{p(z;r)}\big\{F(z)\big\}\Big) = E_{p(z;r)}\Big\{\nabla\Big(\log\big(p(z;r)\big)\Big)F(z)\Big\}.$$

The scaling induced by the FIM (3.62) in the scaled gradient method (3.63) is conceptually related to the Newton and the Gauss-Newton scaling noted earlier in connection to the scaled gradient method (3.54); see the relevant literature for more details, including the scholarly and comprehensive book by Amari [Ama16]. The scaled gradient

$$\big(M(r^k)\big)^{-1}E_{p(z;r^k)}\Big\{\nabla\Big(\log\big(p(z;r^k)\big)\Big)F(z)\Big\},$$

is also known as the *natural gradient* for the problem (3.61). Incremental/sampled versions of the scaled gradient method (3.63) are possible along the lines discussed earlier; cf. Eq. (3.48). In particular, for a discounted DP problem of Example 3.5.1, we can use the FIM formula (3.62) (or an approximation thereof), where

$$\nabla\Big(\log\big(p(z^k;r^k)\big)\Big) = \sum_{m=0}^{\infty}\nabla\Big(\log\big(p(u_m^k \mid i_m^k;r^k)\big)\Big);$$

cf. Eq. (3.51), and z^k is the kth sample state-control trajectory

$$z^k = \{i_0^k, u_0^k, i_1^k, u_1^k, \ldots\}.$$

The motivation for using the FIM in parametric optimization problems of the form (3.61), is that it defines an alternative norm in the space of probability distributions, which captures the sensitivity of the distribution $p(\cdot;r)$ with respect to the parameter r more effectively than the norm defined by the inverse Hessian approximations D^k discussed earlier. In this regard, it should be noted that the FIM is the Hessian matrix of the Kullback-Leibler (KL) divergence, an important metric for measuring

distance between two probability distributions. We refer to the papers by Amari [Ama98] and Kakade [Kak02] for further discussion. For more recent works that provide many additional references, see Deisenroth, Neumann, and Peters [DNP11], Grondman et al. [GBL12], Fazel et al. [FGK18], Cen et al. [CCC21], Bhandari and Russo [BhR24], and Muller and Montufar [MuM24].

Finally let us note that the trust region analog of the scaled iteration (3.63) takes the form

$$r^{k+1} \in \arg \min_{(r-r^k)' M(r^k)(r-r^k) \leq \beta^k} \nabla \Big(E_{p(z;r^k)} \big\{ F(z) \big\} \Big)' (r - r^k), \qquad (3.64)$$

where β^k is a positive scalar, in analogy with Eq. (3.60). Implementations of the policy gradient method with FIM scaling and with forms of trust region are provided by the proximal policy optimization (PPO) method, suggested by Shulman et al. [SWD17], which has been used widely, including for the training of large language models.

Constrained Natural Gradient Methods

Natural gradient methods can be extended to constrained optimization problems of the form

$$\min_{r \in C} E_{p(z;r)} \big\{ F(z) \big\}, \qquad (3.65)$$

where $C \subset \Re^m$ is a closed convex constraint set, in analogy with problem (3.56). Important special cases of constrained problems arise often in practice, such as when C consists of nonnegativity constraints:

$$C = \{ r \mid r \geq 0 \}, \qquad (3.66)$$

or has some other simple form, such as upper and lower bounds on the components of r, a simplex constraint, or a Cartesian product of simplices. In what follows, we will provide efficient variants of natural gradient methods for simple constraint sets, focusing primarily on the orthant constraint (3.66).

To simplify notation, let us denote by $r_1, \ldots, r_m$ the m components of r, and let us use an abbreviated notation for the gradient and partial derivatives of the cost function:

$$g(r) = \nabla \Big(E_{p(z;r)} \big\{ F(z) \big\} \Big), \qquad g_i(r) = \frac{\partial \Big(E_{p(z;r)} \big\{ F(z) \big\} \Big)}{\partial r_i}.$$

With this notation, the unconstrained natural gradient method is given by

$$r^{k+1} = r^k - \gamma^k \big(M(r^k) \big)^{-1} g(r^k), \qquad (3.67)$$

and its projected analog [cf. Eq. (3.57)] is given by

$$r^{k+1} \in \arg\min_{r \in C} \left\{ g(r^k)'(r - r^k) + \frac{1}{2\gamma^k}(r - r^k)'M(r^k)(r - r^k) \right\}. \quad (3.68)$$

This is a valid method with strong convergence guarantees, which follow from well-established analysis of gradient projection methods for general constrained nonlinear programming problems (see e.g., [Ber16]). Unfortunately, however, the practical implementation of this method runs into some serious difficulties. Chief among these is that the quadratic minimization in Eq. (3.68) may be much harder than the matrix inversion of Eq. (3.67). This is true even if C is the simple orthant constraint (3.66).

To address this issue we may consider a method that involves a simpler projection. For the case of the orthant constraint (3.66), one possibility that comes to mind is

$$r^{k+1} = \left[r^k - \gamma^k \tilde{M}(r^k)^{-1} g(r^k) \right]^+, \quad (3.69)$$

where $[\cdot]^+$ is the "clipping" operation, whereby the ith component $[r]_i^+$ of $[r]^+$ is obtained from the ith component r_i of r according to

$$[r]_i^+ = \max\{0, r_i\}, \qquad i = 1, \ldots, m,$$

and $\tilde{M}(r^k)$ is either the FIM $M(r^k)$ or a suitable approximation thereof.

Unfortunately when $\tilde{M}(r^k)$ is taken to be equal to the FIM $M(r^k)$, the algorithm (3.69) does not work. The reason is illustrated in Fig. 3.5.2, and has to do with the fact that when r^k lies at the boundary of the orthant, the iteration may not lead to cost reduction *regardless of how small the stepsize* γ^k *is*. In particular, it is possible that the unprojected iterate $r^k - \gamma^k \tilde{M}(r^k)^{-1} g(r^k)$ improves the cost function for g^k sufficiently small, but after projection onto the orthant it does not [see Fig. 3.5.2(a)]. In fact it is possible that r^k is optimal, but the iteration (3.69) moves r^k away from the optimal [see Fig. 3.5.2(b)].

It turns out, however, that there is a class of approximate FIM matrices $\tilde{M}(r^k)$ for which cost reduction can be guaranteed. This class is sufficiently wide to allow fast convergence when $\tilde{M}(r^k)$ properly embodies the essential content of the true FIM $M(r^k)$.

We say that a symmetric $m \times m$ matrix D with elements d_{ij} is *diagonal with respect to a subset of indices* $I \subset \{1, \ldots, n\}$, if

$$d_{ij} = 0, \qquad \forall\ i \in I,\ j = 1, \ldots, m,\ j \neq i.$$

Let us denote for all $r \geq 0$

$$I^+(r) = \big\{ i \in \{1, \ldots, m\} \ \big|\ r_i = 0,\ g_i(r) > 0 \big\}; \quad (3.70)$$

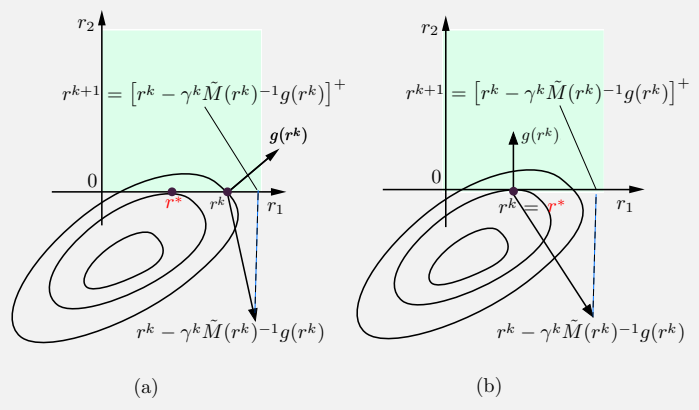

Figure 3.5.2. Two-dimensional examples where $I^+(r_k) = \{2\}$ [we have $r_1^k > 0$, while $r_2^k = 0$ and $g_2(r^k) > 0$]. The iteration is $r^{k+1} = \big[r^k - \gamma^k \tilde{M}(r^k)^{-1} g(r^k)\big]^+$. (a) A case where the iteration fails to make progress when $\tilde{M}(r^k)$ is not properly diagonalized. (b) A case where the iteration fails to stop at an optimal solution r^* when $\tilde{M}(r^k)$ is not properly diagonalized.

see Fig. 3.5.2 for an example. The following is a key proposition, due to the author's works [Ber82b], [Ber82c], which shows how to select an approximation to the FIM in order to guarantee the validity of the projected iteration (3.69).

Proposition 3.5.1: Let $r \geq 0$ and let $\tilde{M}(r)$ be a positive definite symmetric matrix that is diagonal with respect to $I^+(r)$. Denote by $r(\gamma)$ the iterate of the projected natural gradient iteration (3.69), starting from r, as a function of the stepsize γ:

$$r(\gamma) = \big[r - \gamma \tilde{M}(r)^{-1} g(r)\big]^+, \qquad \gamma \geq 0.$$

Consider a vector r that is nonstationary, in the sense that it violates the first order optimality condition

$$g_i(r) \geq 0 \quad \text{for all } i \text{ with } r_i = 0, \qquad g_i(r) = 0 \quad \text{for all } i \text{ with } r_i > 0.$$

Then there exists a scalar $\overline{\gamma} > 0$ such that the following cost improvement property holds:

$$E_{p(z;r(\gamma))}\big\{F(z)\big\} < E_{p(z;r)}\big\{F(z)\big\}, \qquad \text{for all } \gamma \text{ with } 0 < \gamma \leq \overline{\gamma}.$$

Based on Prop. 3.5.1 we conclude that to guarantee cost improvement, the matrix $\tilde{M}(r^k)$ in the iteration (3.69) should be chosen to be diagonal with respect to a subset of indices that contains

$$I^+(r^k) = \big\{ i \bigm| r_i^k = 0,\ g_i(r^k) > 0 \big\},$$

and the stepsize γ^k should be chosen sufficiently small to guarantee cost improvement. Actually, it turns out that to ensure convergence one should implement the iteration more carefully. The reason is that the set $I^+(r^k)$ exhibits an undesirable discontinuity at the boundary of the constraint set, whereby given a sequence $\{r^k\}$ of interior points that converges to a boundary point $\overline{r}$ the set $I^+(r^k)$ may be strictly smaller than the set $I^+(\overline{r})$. This causes difficulties in proving convergence of the algorithm and may have an adverse effect on its rate of convergence. To bypass these difficulties one may add to the set $I^+(r^k)$ the indices of those variables r_i^k that satisfy $g_i(r^k) > 0$ and are "near" zero (i.e., $0 \leq r_i^k \leq \epsilon$, where ϵ is a small fixed scalar).

The theoretical properties of the projection method (3.69), with the preceding modification and a properly defined stepsize rule are solid. Moreover, extensive computational practice has established its superior performance over alternative methods of the gradient type. We refer to the author's constrained optimization book [Ber82b] (Section 1.5) and the paper [Ber82c], where the method (3.69) was first proposed under the name "two-metric projection method;" see also Gafni and Bertsekas [GaB84]. These references, together with the book [Ber16] (Section 3.4), also discuss extensions to broader classes of problems involving linear constraints, such as upper and lower bounds on the parameters r_i, and simplex constraints. For more recent works that are focused on machine learning applications, see Schmidt, Kim, and Sra [SKS12], Xie and Wright [XiW21], [XiW24], and Wu and Xie [WuX24].

An important property of the projected natural gradient method (3.69) is that it can preserve the beneficial scaling properties of the Fisher information matrix. The reason is that the method tends to quickly identify the subspace of active constraints (the ones for which $r_i = 0$ at an optimal solution) and then reduces to the unconstrained natural gradient method on that subspace. For the same reason, the method is also well-suited for an incremental implementation.

3.5.4 Random Direction Methods

In this section, we will consider an alternative class of incremental versions of the policy gradient method (3.46), repeated here for convenience:

$$r^{k+1} = r^k - \gamma^k \nabla\Big(E_{p(z;r^k)}\big\{ F(z) \big\} \Big), \qquad k = 0, 1, \ldots. \tag{3.71}$$

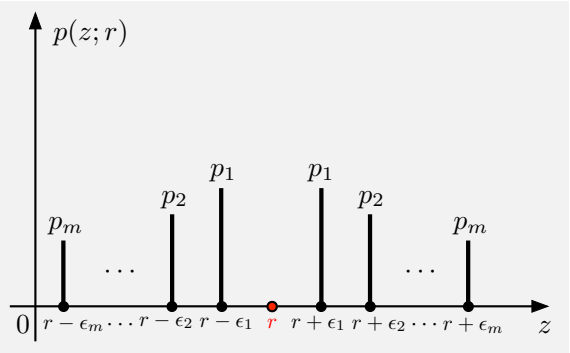

Figure 3.5.3 The distribution $p(z;r)$ used in the gradient iteration (3.72).

These methods are based on the use of a random search direction and only *two sample function values per iteration*. They are generally faster than methods that use a finite difference approximation of the entire cost function gradient; see the book by Spall [Spa03] for a detailed discussion, and the paper by Nesterov and Spokoiny [NeS17] for a more theoretical view. Moreover, these methods do not require the derivative of the sampling distribution or its logarithm. On the other hand, these methods have not been applied widely to the training of policies in the RL field.

For simplicity we first consider the case where z and r are scalars, and later discuss the multidimensional case. In particular, we assume that $p(z;r)$ is symmetric and is concentrated with probabilities p_i at the points $r+\epsilon_i$ and $r-\epsilon_i$, where $\epsilon_1,\ldots,\epsilon_m$ are some small positive scalars. Thus we have

$$E_{p(z;r)}\big\{F(z)\big\} = \sum_{i=1}^{m} p_i\big(F(r+\epsilon_i)+F(r-\epsilon_i)\big),$$

and

$$\nabla\Big(E_{p(z;r)}\big\{F(z)\big\}\Big) = \sum_{i=1}^{m} p_i\big(\nabla F(r+\epsilon_i)+\nabla F(r-\epsilon_i)\big).$$

The gradient iteration (3.71) becomes

$$r^{k+1} = r^k - \gamma^k \sum_{i=1}^{m} p_i\big(\nabla F(r^k+\epsilon_i)+\nabla F(r^k-\epsilon_i)\big), \qquad k=0,1,\ldots, \quad (3.72)$$

Let us now consider approximation of the gradient by finite differences:

$$\nabla F(r+\epsilon_i) \approx \frac{F(r+\epsilon_i)-F(r)}{\epsilon_i}, \quad \nabla F(r-\epsilon_i) \approx \frac{F(r)-F(r-\epsilon_i)}{\epsilon_i}.$$

We approximate the gradient iteration (3.72) by

$$r^{k+1} = r^k - \gamma^k \sum_{i=1}^{m} p_i \frac{F(r^k+\epsilon_i)-F(r^k-\epsilon_i)}{\epsilon_i}, \qquad k=0,1,\ldots. \quad (3.73)$$

One possible sample-based/incremental version of this iteration is

$$r^{k+1} = r^k - \gamma^k \frac{F(r^k + \epsilon_{i^k}) - F(r^k - \epsilon_{i^k})}{\epsilon_{i^k}}, \tag{3.74}$$

where i^k is an index generated with probabilities that are proportional to p_{i^k}. This algorithm uses one out of the m terms of the gradient in Eq. (3.73).

The extension to the case, where z and r are multidimensional, is straightforward. Here $p(z;r)$ is a probability distribution, whereby z takes values of the form $r + \epsilon d$, where d is a random vector that lies on the surface of the unit sphere, and ϵ (independently of d) takes scalar values according to a distribution that is symmetric around 0. The idea is that at r^k, we first choose randomly a direction d^k on the surface of the unit sphere, and then change r^k along d^k or along $-d^k$, depending on the sign of the corresponding directional derivative. For a finite difference approximation of this iteration, we sample z^k along the line $\{r^k + \epsilon d^k \mid \epsilon \in \Re\}$, and similar to the iteration (3.74), we set

$$r^{k+1} = r^k - \gamma^k \frac{F(r^k + \epsilon^k d^k) - F(r^k - \epsilon^k d^k)}{\epsilon^k} d^k, \tag{3.75}$$

where ϵ^k is the sampled value of ϵ.

Let us also discuss the case where $p(z;r)$ is a discrete but *nonsymmetric* distribution, i.e., z takes values of the form $r + \epsilon d$, where d is a random vector that lies on the surface of the unit sphere, and ϵ is a zero mean scalar. Then the analog of iteration (3.75) is

$$r^{k+1} = r^k - \gamma^k \frac{F(r^k + \epsilon^k d^k) - F(r^k)}{\epsilon^k} d^k, \tag{3.76}$$

where ϵ^k is the sampled value of ϵ. Thus in this case, we still require two function values per iteration. Generally, for a symmetric sampling distribution, iteration (3.75) tends to be more accurate than iteration (3.76), and is often preferred.

Algorithms of the form (3.75) and (3.76) are known as *random direction methods*. They use only two cost function values per iteration, and a direction d^k that need not be related to the gradient of F in any way. There is some freedom in selecting d^k, which could potentially be exploited in specific schemes. Selection of the stepsize γ^k and the sampling distribution for ϵ can be tricky, particularly when the values of F are noisy.

3.5.5 Random Search and Cross-Entropy Methods

The main drawback of the policy gradient methods that we have considered so far is the risk of unreliability due to the stochastic uncertainty corrupting the calculation of the gradients, the slow convergence that is typical of

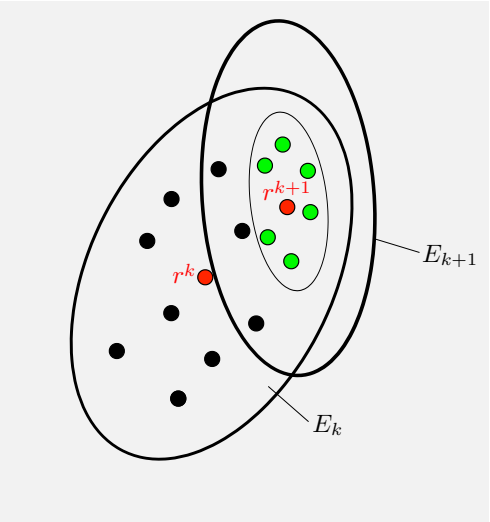

Figure 3.5.4 Schematic illustration of the cross-entropy method. At the current iterate r^k, we construct an ellipsoid E_k centered at r^k. We generate a number of random samples within E_k, and we "accept" a subset of the samples that have "low" cost. We then choose r^{k+1} to be the sample mean of the accepted samples, and construct a sample "covariance" matrix of the accepted samples. We then form the new ellipsoid E_{k+1} using this matrix and a suitably enlarged radius, and continue. Notice the resemblance with a policy gradient method: we move from r^k to r^{k+1} in a direction of cost improvement.

gradient methods in many settings, and the presence of local minima. For this reason, alternative methods based on random search are potentially more reliable alternatives. Viewed from a high level, random search methods are similar to policy gradient methods in that they aim at iterative cost improvement through sampling. However, they need not involve randomized policies, they are not subject to cost differentiability restrictions, and they offer some global convergence guarantees, so in principle they are not affected much by local minima.

Let us consider a parametric policy optimization approach based on solving the problem

$$\min_r E\big\{J_{\tilde{\mu}(r)}(i_0)\big\},$$

cf. Eq. (3.39). Random search methods for this problem explore the space of the parameter vector r in some randomized but intelligent fashion. There are several types of such methods for general optimization, and some of them have been suggested for approximate DP. We will briefly describe the *cross-entropy method*, which has gained considerable attention.

The method, when adapted to the approximate DP context, bears resemblance to policy gradient methods, in that it generates a parameter sequence $\{r^k\}$ by changing r^k to r^{k+1} along a direction of "improvement."

This direction is obtained by using the policy $\tilde{\mu}(r^k)$ to generate randomly cost samples corresponding to a set of sample parameter values that are concentrated around r^k. The current set of sample parameters are then screened: some are accepted and the rest are rejected, based on a cost improvement criterion. Then r^{k+1} is determined as a "central point" or as the "sample mean" in the set of accepted sample parameters, some more samples are generated randomly around r^{k+1}, and the process is repeated; see Fig. 3.5.4. Thus successive iterates r^k are "central points" of successively better groups of samples, so in some broad sense, the random sample generation process is guided by cost improvement. This is a general idea that is shared with other popular classes of random search methods.

The cross-entropy method is very simple to implement, does not suffer from the fragility of gradient-based optimization, does not involve randomized policies, and relies on some supportive theory. Importantly, the method does not require the calculation of gradients, and it does not require differentiability of the cost function. Moreover, it does not need a model to compute the required costs of different policies; a simulator is sufficient.

Like all random search methods, the convergence rate guarantees of the cross-entropy method are limited, and its success depends on domain-specific insights and the skilled use of heuristics. However, the method relies on solid ideas and has gained a favorable reputation. In particular, it was used with impressive success in the context of the game of tetris; see Szita and Lorincz [SzL06], and Thiery and Scherrer [ThS09]. There have also been reports of domain-specific successes with related random search methods; see Salimans et al. [SHC17]. We refer to the end-of-chapter literature for details and examples of implementation.

3.5.6 Refining and Retraining Parametric Policies

Suppose that we already have a pre-trained parametric policy, such as one implemented through a neural network, and we want to refine it. A possible motivation for this may be to correct some deficiencies of the policy, such as a bias towards undesirable behaviors. For example the policy may produce poor or unsafe responses at some sensitive states. We note that refining a pre-trained parametric policy has become the subject of intensive interest following the availability of highly capable pre-trained transformer-based software such as ChatGPT and DeepSeek.

A common way to retrain a policy is to modify the cost function of the problem in a way that penalizes the undesirable behaviors. To this end we need a special purpose/domain-specific dataset consisting of input-output pairs that are generated using human or machine feedback. The dataset may be supplemented with specific instructions on how to choose controls in special or critical situations.

Once the dataset has been chosen, we need a method that uses this dataset to retrain the policy. One possibility is to use a policy approximation method in combination with a parametric *comparison training* scheme, such as the one described in Section 2.3.6 in the context of learning how to imitate an expert. Another possibility is *adversarial training*, which involves exposing the policy to adversarial examples (involving states designed to trigger undesirable behaviors) and fine-tuning it to handle them correctly, using a policy gradient or random search method.

Another interesting retraining context, known as *knowledge distillation*, arises in situations where a policy must be simplified to be deployed in an environment where limited computational resources are available. This involves training a smaller, more computationally efficient policy (the "student") to mimic the behavior of a larger, more complex policy (the "teacher"). One possibility is to use the teacher as an expert that generates a dataset for training the student. We refer to Hinton, Vinyals, and Dean [HVD15], and the survey by Xu et al. [XLT24] for further discussion.

Mixture-of-Experts Techniques

A situation that often arises in practice is that the system may undergo significant structural changes following transition to some special states. For example significant parts of the system may be disabled or repurposed at random or scheduled times, in which case radical changes in the control policy may be needed. This situation comes under the general subject of adaptive control, but in cases where the structural changes are significant, it may be best handled by specialized *mixture-of-experts* techniques, which we will now explain briefly.

The main idea in a mixture-of-experts scheme is to have multiple policies (or experts) available, and to switch from one policy to another whenever the need arises. For example, when a structural change in the system occurs or when the environment goes through a significant change, simply adopt the policy that is best suited for the change that has occurred. The system change must be detected on line with a suitable algorithm, and then a gating mechanism must be used, which dynamically enables the most relevant policy. The different policies may be trained off line, possibly by refining some pre-trained policy.

As an example, we note the technique of multiple model control design that was mentioned in Section 1.6.8 in the context of adaptive control. Another more recent example is the mixture-of-experts scheme that was implemented with success in DeepSeek; see Dai et al. [DDZ24].

3.6 AGGREGATION

In this section we consider approximation in value space using a problem approximation approach that is based on aggregation. More specifically,

we construct a simpler and more tractable "aggregate" problem by creating special subsets of states, which we view as "aggregate states." We then solve the aggregate problem exactly by DP. This is the off-line training part of the aggregation approach, and it may be carried out with a variety of DP methods, including simulation-based value and policy iteration; we refer to the RL book [Ber19a] for a detailed account. Finally, we use the optimal cost-to-go function of the aggregate problem (or an approximation thereof) to construct a terminal cost approximation in a one-step or multistep lookahead approximation scheme for the original problem. Additionally, we may also use the optimal policy of the aggregate problem as a base policy for a truncated rollout scheme.

In addition to problem approximation, aggregation is related to feature-based parametric approximation. More specifically, it often produces a piecewise constant cost function approximation, which may be viewed as a linear feature-based parametrization, where the features are 0-1 membership functions; see Example 3.1.1. Aggregation can also be combined with other approximation schemes, which can produce a cost function approximation, possibly through the use of a neural network; see our subsequent discussion of biased aggregation.

Aggregation can be applied to both finite horizon and infinite horizon problems. In our discussion, we will focus primarily on the discounted infinite horizon problem with a finite number of states, although the ideas apply more broadly. In particular, we will focus on the standard discounted infinite horizon problem with the n states $1, \ldots, n$. States and successor states will be denoted by i and j. State transitions (i, j) under control u occur at discrete times according to transition probabilities $p_{ij}(u)$, and generate a cost $\alpha^k g(i, u, j)$ at time k, where $\alpha \in (0, 1)$ is the discount factor.

We consider deterministic stationary policies μ such that for each i, $\mu(i)$ is a control that belongs to a finite constraint set $U(i)$. We denote by $J_\mu(i)$ the total discounted expected cost of μ over an infinite number of stages starting from state i, by $J^*(i)$ the minimal value of $J_\mu(i)$ over all μ, and by J_μ and J^* the n-dimensional vectors that have components $J_\mu(i)$ and $J^*(i)$, $i = 1, \ldots, n$, respectively.

We will introduce aggregation in a simple intuitive form in Section 3.6.1, and generalize later to a more sophisticated form of feature-based aggregation, which we also discussed briefly in Example 3.1.7. Our coverage of aggregation in this section is somewhat abbreviated, and we refer to the books [Ber12], Section 6.5, and [Ber19a], Chapter 6, for a more detailed presentation.

3.6.1 Aggregation with Representative States

We will first focus on a relatively simple form of aggregation, which involves a special subset of states, called *representative*. Our approach is to view these states as the states of a smaller optimal control problem, the aggre-

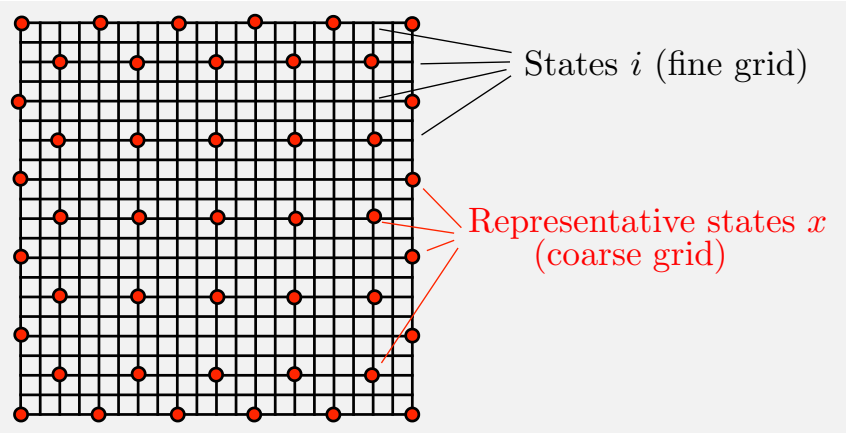

Figure 3.6.1 Illustration of aggregation with representative states; cf. Example 3.6.1. A relatively small number of states are viewed as representative. We define transition probabilities between pairs of aggregate states and we also define the associated expected transition costs. These specify a smaller DP problem, called the aggregate problem, which is solved exactly. The optimal cost function J^* of the original problem is approximated by interpolation from the optimal costs of the representative states r_y^* in the aggregate problem:

$$\tilde{J}(j) = \sum_{y \in \mathcal{A}} \phi_{jy} r_y^*, \qquad j = 1, \ldots, n,$$

and is used in a one-step or multistep lookahead scheme.

gate problem, which we will formulate and solve exactly in place of the original. We will then use the optimal aggregate costs of the representative states to approximate the optimal costs of the original problem states by interpolation. Let us describe a classical example.

Example 3.6.1 (Coarse Grid Approximation)

Consider a discounted problem where the state space is a grid of points $i = 1, \ldots, n$ on the plane. We introduce a coarser grid that consists of a subset of the states/points, which we call representative and denote by x; see Fig. 3.6.1. We now wish to formulate a lower-dimensional DP problem just on the coarse grid of states. The difficulty here is that there may be positive transition probabilities $p_{xj}(u)$ from some representative states x to some non-representative states j. To deal with this difficulty, we introduce artificial transition probabilities ϕ_{jy} from non-representative states j to representative states y, which we call *aggregation probabilities*. In particular, a transition from representative state x to a nonrepresentative state j, is followed by a transition from j to some other representative state y with probability ϕ_{jy}; see Fig. 3.6.2.

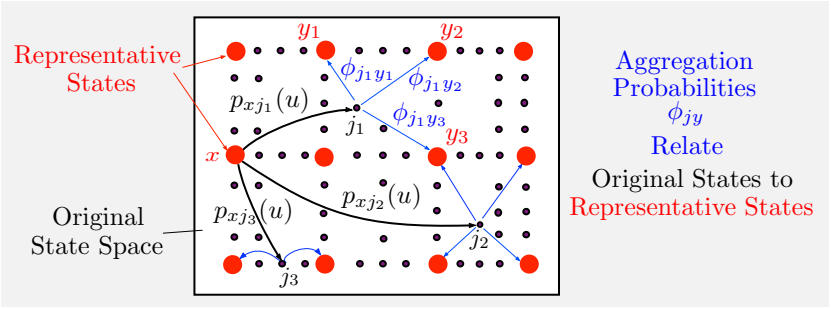

Figure 3.6.2 Illustration of the use of aggregation probabilities ϕ_{jy} from nonrepresentative states j to representative states y in Example 3.6.1. A transition from a state x to a nonrepresentative state j is followed by a transition to aggregate state y with probability ϕ_{jy}. In this figure, from representative state x, there are three possible transitions, to states j_1, j_2, and j_3, according to $p_{xj_1}(u), p_{xj_2}(u), p_{xj_3}(u)$, and each of these states is associated with a convex combination of representative states using the aggregation probabilities. For example, the state j_1 is associated with the aggregation probabilities $\phi_{j_1y_1}, \phi_{j_1y_2}, \phi_{j_1y_3}$, and the cost of j_1 is approximated by the corresponding convex combination of the costs of y_1, y_2, y_3; [cf. Eq. (3.78)].

This process involves approximation but constructs a transition mechanism for an *aggregate problem* whose states are just the representative ones. The transition probabilities between representative states x, y under control $u \in U(x)$ and the corresponding expected transition costs can be computed as

$$\hat{p}_{xy}(u) = \sum_{j=1}^{n} p_{xj}(u)\phi_{jy}, \qquad \hat{g}(x,u) = \sum_{j=1}^{n} p_{xj}(u)g(x,u,j). \tag{3.77}$$

We can solve the aggregate problem by any suitable exact DP method. Let $\mathcal{A}$ denote the set of representative states and let r_x^* denote the corresponding optimal cost of representative state x. We can then approximate the optimal cost function of the original problem with the interpolation

$$\tilde{J}(j) = \sum_{y \in \mathcal{A}} \phi_{jy} r_y^*, \qquad j = 1, \ldots, n. \tag{3.78}$$

This function may in turn be used in a one-step or multistep lookahead scheme for approximation in value space of the original problem.

Note that there is a lot of freedom in selecting the aggregation probabilities ϕ_{jy}. Intuitively, ϕ_{jy} should express a measure of proximity between j and y, e.g., ϕ_{jy} should be relatively large when y is geometrically close to j. For example, we could set $\phi_{jy_j} = 1$ for the representative state y_j that is "closest" to j, and $\phi_{jy_j} = 0$ for all other representative states $y \neq y_j$. In this case, Eq. (3.78) yields a piecewise constant cost function approximation $\tilde{J}$ (the constant values are the scalars r_y^* of the representative states y).

We will now formalize our framework for aggregation with representative states by generalizing the preceding example; see Fig. 3.6.3. We first consider the n-state version of the α-discounted problem of Section 1.4.1. We refer to this problem as the "original problem," to distinguish from the "aggregate problem," which we define next.

Aggregation Framework with Representative States

We introduce a finite subset $\mathcal{A}$ of the original system states, which we call *representative states*, and we denote them by symbols such as x and y. We construct an *aggregate problem*, with state space $\mathcal{A}$, and transition probabilities and transition costs defined as follows:

(a) We relate the original system states j to representative states $y \in \mathcal{A}$ with aggregation probabilities ϕ_{jy}; these are scalar "weights" satisfying for all j and $y \in \mathcal{A}$,

$$\sum_{y \in \mathcal{A}} \phi_{jy} = 1, \quad \phi_{jy} \geq 0, \ \phi_{yy} = 1.$$

(b) We define the transition probabilities between representative states x and y under control $u \in U(x)$ by

$$\hat{p}_{xy}(u) = \sum_{j=1}^{n} p_{xj}(u)\phi_{jy}. \tag{3.79}$$

(c) We define the expected transition costs at representative states x under control $u \in U(x)$ by

$$\hat{g}(x,u) = \sum_{j=1}^{n} p_{xj}(u)g(x,u,j). \tag{3.80}$$

The optimal costs of the representative states $y \in \mathcal{A}$ in the aggregate problem are denoted by r_y^*, and they define approximate costs for the original problem through the interpolation formula

$$\tilde{J}(j) = \sum_{y \in \mathcal{A}} \phi_{jy} r_y^*, \qquad j = 1, \ldots, n. \tag{3.81}$$

Aside from the selection of representative states, an important consideration is the choice of the aggregation probabilities. These probabilities

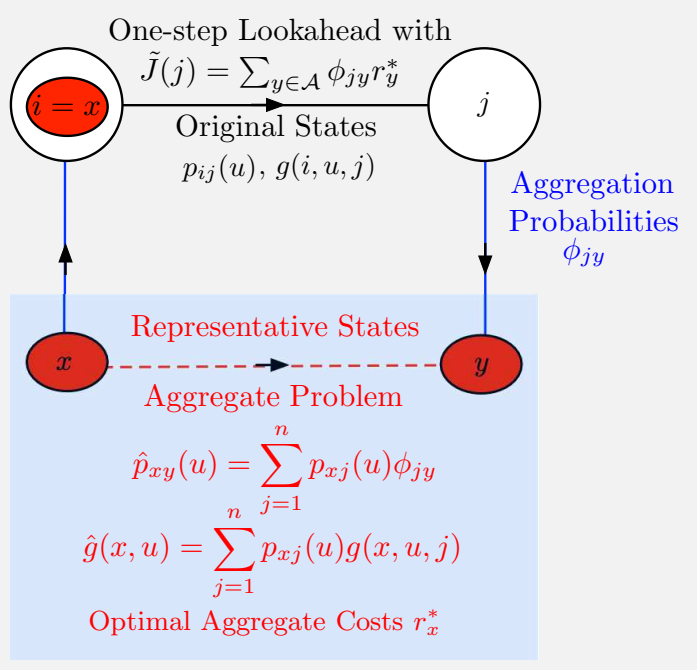

Figure 3.6.3 Illustration of the aggregate problem in the representative states framework. The transition probabilities $\hat{p}_{xy}(u)$ and transition costs $\hat{g}(x,u)$ are shown in the bottom part of the figure. Once the aggregate problem is solved (exactly) for its optimal costs r_y^*, we define approximate costs

$$\tilde{J}(j) = \sum_{y \in \mathcal{A}} \phi_{jy} r_y^*, \qquad j = 1, \ldots, n,$$

which are used for one-step lookahead approximation of the original problem.

express "similarity" or "proximity" of original to representative states (as in the case of the coarse grid Example 3.6.1), but in principle they can be arbitrary (as long as they are nonnegative and sum to 1 over y). Intuitively, ϕ_{jy} may be interpreted as some measure of "strength of relation" of j to y. The vectors $\{\phi_{jy} \mid j = 1, \ldots, n\}$ may also be viewed as basis functions for a linear cost function approximation via Eq. (3.81).

Hard Aggregation and Error Bound

A special case of interest, called *hard aggregation*, is when for every state j, we have $\phi_{jy} = 0$ for all representative states y, except a single one, denoted y_j, for which we have $\phi_{jy_j} = 1$ (we also require $\phi_{yy} = 1$ for all representative

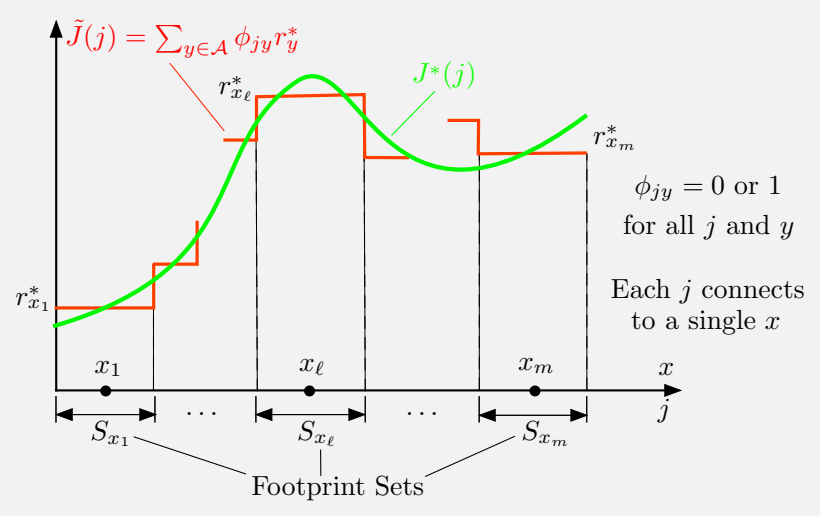

Figure 3.6.4 Illustration of the piecewise constant cost approximation

$$\tilde{J}(j) = \sum_{y \in \mathcal{A}} \phi_{jy} r_y^*, \qquad j = 1, \ldots, n,$$

in the hard aggregation case where we have $\phi_{jy} = 0$ for all representative states y, except a single one. Here $\tilde{J}$ is constant and equal to r_y^* for all j in the footprint set

$$S_y = \{j \mid \phi_{jy} = 1\}, \qquad y \in \mathcal{A}.$$

states y). In this case, the one-step lookahead approximation

$$\tilde{J}(j) = \sum_{y \in \mathcal{A}} \phi_{jy} r_y^*, \qquad j = 1, \ldots, n,$$

is *piecewise constant*; it is constant and equal to r_y^* for all j in the set

$$S_y = \{j \mid \phi_{jy} = 1\}, \qquad y \in \mathcal{A},$$

called the *footprint* of representative state y; see Fig. 3.6.4. Moreover the footprints of all the representative states are disjoint and form a partition of the state space, i.e.,

$$\cup_{x \in \mathcal{A}} S_x = \{1, \ldots, n\}.$$

The footprint sets can be used to define a bound for the error $(J^* - \tilde{J})$. In particular, it can be shown that

$$\left|J^*(j) - \tilde{J}(j)\right| \leq \frac{\epsilon}{1 - \alpha}, \qquad j = 1, \ldots, n,$$

where

$$\epsilon = \max_{y \in \mathcal{A}} \max_{i,j \in S_y} \left| J^*(i) - J^*(j) \right|$$

is the *maximum variation of J^* within the footprint sets S_y*. This error bound result can be extended to the more general aggregation framework that will be given in the next section. Note the primary intuition derived from the bound: *the error due to hard aggregation is small if J^* varies little within each set S_y*.

For a special hard aggregation case of interest, consider the geometrical context of Example 3.6.1. There, aggregation probabilities are often based on a nearest neighbor approximation scheme, whereby each non-representative state j takes the cost value of the "closest" representative state y, i.e.,

$$\phi_{jy_j} = 1 \qquad \text{if } y_j \text{ is the closest representative state to } j.$$

Then all states j for which a given representative state y is the closest to j (the footprint of y) are assigned equal approximate cost $\tilde{J}(j) = r^*_y$.

Methods for Solving the Aggregate Problem

The most straightforward way to solve the aggregate problem is to compute the aggregate problem transition probabilities $\hat{p}_{xy}(u)$ [cf. Eq. (3.79)] and transition costs $\hat{g}(x,u)$ [cf. Eq. (3.80)] by either an algebraic calculation or by simulation. The aggregate problem may then be solved by any one of the standard methods, such as VI or PI. This exact calculation is plausible if the number of representative states is relatively small. An alternative possibility is to use a VI or PI method that is based on simulation. We refer to a discussion of these methods in the author's books [Ber12], Section 6.5.2, and [Ber19a], Section 6.3. The idea is that a simulator for the original problem can be used to construct a simulator for the aggregate problem; cf. Fig. 3.6.3.

An important observation is that if the original problem is deterministic and hard aggregation is used, the aggregate problem is also deterministic, and can be solved by shortest-path like methods. This is true for both discounted problems and for undiscounted shortest path-type problems. In the latter case, the termination state of the original problem must be included as a representative state in the aggregate problem. However, if hard aggregation is not used, the aggregate problem will be stochastic, because of the introduction of the aggregation probabilities. Of course, once the aggregate problem is solved and the lookahead approximation $\tilde{J}$ is obtained, a deterministic structure in the original problem can be exploited to facilitate the on-line lookahead minimizations.

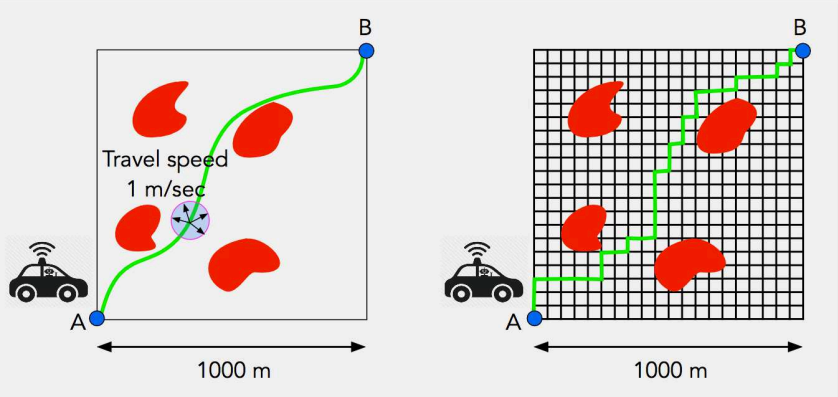

Figure 3.6.5 Illustration of discretization issues for problems with infinite state and control spaces.

3.6.2 Continuous Control Space Discretization

Aggregation with representative states extends without difficulty to problems with a continuous state space, as long as the control space is finite. Then once the representative states and the aggregation probabilities have been defined, the corresponding aggregate problem is a discounted problem with finite state and control spaces, which can be solved with the standard methods. The only potential difficulty arises when the disturbance space is also infinite, in which case the calculation of the transition probabilities and expected stage costs of the aggregate problem must be obtained by some form of integration process.

The case where both the state and the control spaces are continuous is somewhat more complicated, because both of these spaces must be discretized using representative state-control pairs, instead of just representative states. The following example illustrates what may happen if we use representative state discretization only.

Example 3.6.2 (Continuous Shortest Path Discretization)

Suppose that we want to find the fastest route for a car to travel between two points A and B located at the opposite ends of a square with side 1000 meters, while avoiding some known obstacles. We assume a constant car speed of 1 meter per second and that the car can drive in any direction; cf. Fig. 3.6.5.

Let us consider discretizing the space with a square grid (a set of representative states), and restrict the directions of motion to horizontal and vertical, so that at each stage the car moves from a grid point to one of the four closest grid points. Thus in the discretized version of the problem the car travels with a sequence of horizontal and vertical moves as indicated in the right side of Fig. 3.6.5. Is it possible to approximate the fastest route arbi-

trarily closely with the optimal solution of the discretized problem, assuming a sufficiently fine grid?

The answer is no! To see this note that in the discretized problem the optimal travel time is 2000 secs, regardless of how fine the discretization is. On the other hand, in the continuous space/nondiscretized problem the optimal travel time can be as little as $\sqrt{2} \cdot 1000$ secs (this corresponds to the favorable case where the straight line from A to B does not meet an obstacle).

The difficulty in the preceding example is that *the state space is discretized finely but the control space is not*. What is needed is to introduce a fine discretization of the control space as well, through some set of "representative controls." We can deal with this situation with a suitable form of discretized aggregate problem, which when solved provides an appropriate form of cost function approximation for use with one-step lookahead. The discretized problem is a stochastic infinite horizon problem, even if the original problem is deterministic. Further discussion of this approach is outside our scope, and we refer to the sources cited at the end of the chapter. Under reasonable assumptions we can show consistency, i.e., that the optimal cost function of the discretized problem converges to the optimal cost function of the original continuous spaces problem as the discretization of both the state and the control spaces becomes increasingly fine.

3.6.3 Continuous State Space - POMDP Discretization

Aggregation with representative states is very well suited to problems with continuous state space and a finite control space because it results in an aggregate problem with finite state and control spaces, so the control discretization issue discussed in the preceding section does not arise. This situation arises prominently in POMDP problems, as we will explain in this section.

Let us consider an α-discounted DP problem, where the state space is a bounded convex subset B of a Euclidean space, such as the unit simplex, but the control space U is finite. We use b to denote the states, to emphasize the connection with belief states in POMDP and to distinguish them from x, which we will use to denote representative states. Bellman's equation has the form $J = TJ$, with the Bellman operator T defined by

$$(TJ)(b) = \min_{u \in U} \mathop{E}_{w} \Big\{ g(b,u,w) + \alpha J\big(f(b,u,w)\big) \Big\}, \qquad b \in B.$$

We introduce a set of representative states $\{x_1, \ldots, x_m\} \subset B$. We assume that the convex hull of $\{x_1, \ldots, x_m\}$ is equal to B, so each state $b \in B$ can be expressed as

$$b = \sum_{i=1}^{m} \phi_{bx_i} x_i,$$

where $\{\phi_{bx_i} \mid i = 1, \ldots, m\}$ is a probability distribution:

$$\phi_{bx_i} \geq 0, \ i = 1, \ldots, m, \quad \sum_{i=1}^{m} \phi_{bx_i} = 1, \qquad \text{for all } b \in B.$$

We view ϕ_{bx_i} as aggregation probabilities.

Consider the operator $\hat{T}$ that transforms a vector $r = (r_{x_1}, \ldots, r_{x_m})$ into the vector $\hat{T}r$ with components $(\hat{T}r)(x_1), \ldots, (\hat{T}r)(x_m)$ defined by

$$(\hat{T}r)(x_i) = \min_{u \in U} \mathop{E}_{w} \left\{ g(x_i, u, w) + \alpha \sum_{j=1}^{m} \phi_{f(x_i,u,w)\, x_j} r_{x_j} \right\}, \qquad i = 1, \ldots, m,$$

where $\phi_{f(x_i,u,w)\, x_j}$ are the aggregation probabilities of the state $f(x_i, u, w)$. It can then be shown that $\hat{T}$ is a contraction mapping with respect to the maximum norm (we give the proof for a similar result in the next section). Bellman's equation for an aggregate finite-state discounted DP problem whose states are $x_1, \ldots, x_m$ has the form

$$r_{x_i} = (\hat{T}r)(x_i), \qquad i = 1, \ldots, m,$$

and has a unique solution.

The transitions in this problem occur as follows: from state x_i under control u, we first move to $f(x_i, u, w)$ at cost $g(x_i, u, w)$, and then we move to a state x_j, $j = 1, \ldots, m$, according to the probabilities

$$\phi_{f(x_i,u,w)\, x_j}.$$

The optimal costs $r^*_{x_i}$, $i = 1, \ldots, m$, of this problem can often be obtained by standard VI and PI methods that may or may not use simulation. We can then approximate the optimal cost function of the original problem by

$$\tilde{J}(b) = \sum_{i=1}^{m} \phi_{bx_i} r^*_{x_i}, \qquad \text{for all } b \in B,$$

and reasonably expect that the optimal discretized solution converges to the optimal as the number of representative states increases.

In the case where B is the belief space of an α-discounted POMDP, the representative states/beliefs and the aggregation probabilities define an aggregate problem, which is a finite-state α-discounted problem with a perfect state information structure. This problem can be solved with exact DP methods if either the aggregate transition probabilities and transition costs can be obtained analytically (in favorable cases) or if the number of representative states is small enough to allow their calculation by simulation. The aggregate problem can also be addressed with the approximate DP methods that we have discussed earlier, such as problem approximation/certainty equivalence approaches. Another possibility is the use of a rollout method, which is well-suited for an on-line implementation. See also the paper by Li, Hammar, and Bertsekas [LHB25], which develops a sophisticated POMDP aggregation methodology, based on belief features, and provides extensive computational results.

3.6.4 General Aggregation

We will now discuss a more general aggregation framework for the infinite horizon n-state α-discounted problem. We essentially replace the representative states x with *subsets* $I_x \subset \{1, \ldots, n\}$ of the original state space.

General Aggregation Framework

We introduce a finite subset $\mathcal{A}$ of aggregate states, which we denote by symbols such as x and y. We define:

(a) A collection of subsets $I_x \subset \{1, \ldots, n\}$, $x \in \mathcal{A}$.

(b) A probability distribution over $\{1, \ldots, n\}$ for each $x \in \mathcal{A}$, denoted by $\{d_{xi} \mid i = 1, \ldots, n\}$, and referred to the *disaggregation probabilities of* x. We require that the distribution corresponding to x is concentrated on the subset I_x:

$$d_{xi} = 0, \qquad \text{for all } i \notin I_x,\ x \in \mathcal{A}. \tag{3.82}$$

(c) For each original system state $j \in \{1, \ldots, n\}$, a probability distribution over $\mathcal{A}$, denoted by $\{\phi_{jy} \mid y \in \mathcal{A}\}$, and referred to as the *aggregation probabilities of* j.

The aggregation and disaggregation probabilities specify a dynamic system involving both aggregate and original system states; cf. Fig. 3.6.6. In this system:

(i) From aggregate state x, we generate an original system state $i \in I_x$ according to d_{xi}.

(ii) We generate transitions between original system states i and j according to $p_{ij}(u)$, with cost $g(i, u, j)$.

(iii) From original system state j, we generate aggregate state y according to ϕ_{jy}.

The optimal costs of the aggregate states $y \in \mathcal{A}$ in the aggregate problem are denoted by r^*_y, and they define approximate costs for the original problem through the interpolation formula

$$\tilde{J}(j) = \sum_{y \in \mathcal{A}} \phi_{jy} r^*_y, \qquad j = 1, \ldots, n. \tag{3.83}$$

Our general aggregation framework is illustrated in Fig. 3.6.6. While the sets I_x are often constructed by using features, we will formulate our aggregation framework in a general form, and introduce features later. Note

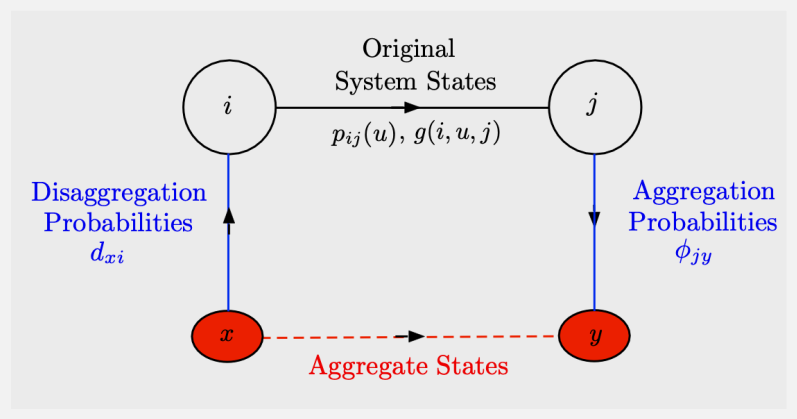

Figure 3.6.6 Illustration of the aggregate system, and the transition mechanism and the costs per stage of the aggregate problem.

that if each set I_x consists of a single state, we obtain the representative states framework of the preceding section. In this case the disaggregation distribution $\{d_{xi} \mid i \in I_x\}$ is just the atomic distribution that assigns probability 1 to the unique state in I_x. Consistent with the special case of representative states, the disaggregation probability d_{xi} may be interpreted as a "measure of the relation of x and i."

The aggregate problem is a DP problem with an enlarged state space that consists of two copies of the original state space $\{1, \ldots, n\}$ plus the set of aggregate states $\mathcal{A}$. We introduce the corresponding optimal vectors $\tilde{J}_0$, $\tilde{J}_1$, and $r^* = \{r_x^* \mid x \in \mathcal{A}\}$ where:

r_x^* is the optimal cost-to-go from aggregate state x.

$\tilde{J}_0(i)$ is the optimal cost-to-go from original system state i that has just been generated from an aggregate state (left side of Fig. 3.6.6).

$\tilde{J}_1(j)$ is the optimal cost-to-go from original system state j that has just been generated from an original system state (right side of Fig. 3.6.6).

Note that because of the intermediate transitions to aggregate states, $\tilde{J}_0$ and $\tilde{J}_1$ are different.

These three vectors satisfy the following three Bellman equations:

$$r_x^* = \sum_{i \in I_x} d_{xi} \tilde{J}_0(i), \qquad x \in \mathcal{A}, \tag{3.84}$$

$$\tilde{J}_0(i) = \min_{u \in U(i)} \sum_{j=1}^{n} p_{ij}(u)\big(g(i,u,j) + \alpha \tilde{J}_1(j)\big), \qquad i = 1, \ldots, n, \tag{3.85}$$

$$\tilde{J}_1(j) = \sum_{y \in \mathcal{A}} \phi_{jy} r_y^*, \qquad j = 1, \ldots, n. \tag{3.86}$$

The objective is to solve for the optimal costs r_x^* of the aggregate states in order to obtain approximate costs for the original problem through the interpolation formula

$$\tilde{J}(j) = \sum_{y \in \mathcal{A}} \phi_{jy} r_y^*, \qquad j = 1, \ldots, n;$$

cf. Eq. (3.83).

By combining the three Bellman equations (3.84)-(3.86), we see that r^* satisfies

$$r_x^* = \sum_{i \in I_x} d_{xi} \min_{u \in U(i)} \sum_{j=1}^{n} p_{ij}(u) \left(g(i,u,j) + \alpha \sum_{y \in \mathcal{A}} \phi_{jy}\, r_y^* \right), \qquad x \in \mathcal{A}, \tag{3.87}$$

or equivalently $r^* = Hr^*$, where H is the operator that maps the vector r to the vector Hr with components

$$(Hr)(x) = \sum_{i \in I_x} d_{xi} \min_{u \in U(i)} \sum_{j=1}^{n} p_{ij}(u) \left(g(i,u,j) + \alpha \sum_{y \in \mathcal{A}} \phi_{jy}\, r_y \right), \quad x \in \mathcal{A}. \tag{3.88}$$

It can be shown that H *is a contraction mapping with respect to the maximum norm*, and thus the composite Bellman equation (3.87) has r^* as its unique solution. To see this, we note that for any vectors r and r', we have

$$\begin{aligned}
(Hr)(x) &= \sum_{i \in I_x} d_{xi} \min_{u \in U(i)} \sum_{j=1}^{n} p_{ij}(u) \left(g(i,u,j) + \alpha \sum_{y \in \mathcal{A}} \phi_{jy}\, r_y \right) \\
&\leq \sum_{i \in I_x} d_{xi} \min_{u \in U(i)} \sum_{j=1}^{n} p_{ij}(u) \Big(g(i,u,j) + \alpha \sum_{y \in \mathcal{A}} \phi_{jy} \left(r'_y + \|r - r'\| \right) \Big) \\
&= (Hr')(x) + \alpha \|r - r'\|,
\end{aligned}$$

where $\|\cdot\|$ is the maximum norm, and the equality follows from the definition of $(Hr')(x)$, and the fact that d_{xi}, $p_{ij}(u)$, and ϕ_{jy} are probabilities. It follows that

$$(Hr)(x) - (Hr')(x) \leq \alpha \|r - r'\|, \qquad x \in \mathcal{A}.$$

By reversing the roles of r and r', we also have

$$(Hr')(x) - (Hr)(x) \leq \alpha \|r - r'\|, \qquad x \in \mathcal{A},$$

so that

$$\big|(Hr')(x) - (Hr)(x)\big| \leq \alpha \|r - r'\|, \qquad \text{for all } x \in \mathcal{A}.$$

By taking the maximum over $x \in \mathcal{A}$, we obtain that

$$\|Hr - Hr'\| \leq \alpha \|r - r'\|, \tag{3.89}$$

and we conclude that H is a maximum norm contraction.

Note that the composite Bellman equation (3.87) has dimension equal to the number of aggregate states, which is potentially much smaller than n. To apply the aggregation framework of this section, we may solve exactly this equation for the optimal aggregate costs r^*_x, $x \in \mathcal{A}$, by simulation-based analogs of the VI and PI methods, and obtain a cost function approximation for the original problem through the interpolation formula (3.83); see [Ber12], Section 6.5.2 and [Ber19a], Section 6.3. These methods have strong convergence properties thanks to the contraction property (3.89).

3.6.5 Types of Aggregation and Error Bounds

Let us consider the set

$$S_y = \{j \mid \phi_{jy} > 0\}, \qquad y \in \mathcal{A}, \tag{3.90}$$

called the *footprint* of aggregate state y (extending our earlier terminology of Section 3.6.1). It consists of all states j that we view as "related" to aggregate state y. Since we have viewed I_x as the set of states i that are "related" to aggregate state x, it makes sense to assume that for every aggregate state x we have

$$I_x \subset S_x, \qquad \text{for all } x \in \mathcal{A}. \tag{3.91}$$

Intuitively, this means that if a state i is "related" to aggregate state x in the disaggregation process, then i is also "related" to x in the aggregation process.

The case of *hard aggregation* is of special interest. Here, for each state j, we have $\phi_{jy} = 0$ for all aggregate states y, except a single one, denoted y_j, for which $\phi_{jy_j} = 1$. In this case, the footprints of all the aggregate states are disjoint and form a partition of the state space, i.e.,

$$\cup_{x \in \mathcal{A}} S_x = \{1, \ldots, n\};$$

cf. our discussion of Section 3.6.1. Then, the one-step lookahead approximation

$$\tilde{J}(j) = \sum_{y \in \mathcal{A}} \phi_{jy} r^*_y, \qquad j = 1, \ldots, n,$$

is piecewise constant; it is constant and equal to r_y^* for all j in the footprint set S_y of y. The condition (3.91) states that $\phi_{ix} = 1$ if $d_{xi} > 0$, or in words, that the set I_x into which the aggregate state x disaggregates is a subset of the footprint set of x.

Aside from hard aggregation and aggregation with representative states, there are several other special cases of aggregation that have received attention in the literature:

(a) *Soft aggregation*: This is an extension of hard aggregation, where there is a "soft" boundary between the sets of the state space partition, i.e., the footprint sets overlap partially. The aggregation probabilities are chosen to be positive for the states of overlap, so that the cost approximation $\tilde{J}$ is piecewise constant, except along the states of footprint overlap, where $\tilde{J}$ changes "smoothly;" see Singh, Jaakkola, and Jordan [SJJ95].

(b) *Aggregation with representative features*: Here the aggregate states are characterized by nonempty subsets of original system states, which, however, may not form a partition of the original state space. In an important example of this scheme, we choose a collection of distinct representative feature vectors, and we associate each one of them with an aggregate state consisting of the subset of original system states that share the corresponding feature value (see [Ber12], Section 6.5, or [Ber19a], Section 6.2). The author's paper [Ber18a] provides an overview of feature-based aggregation, and discusses ways to combine the methodology with the use of deep neural networks.

We can show the following error bound, first given by Tsitsiklis and Van Roy [Van95], [TsV96], for the case of hard aggregation, and extended to the more general case, where just the condition (3.91) holds, by Li and Bertsekas [LiB25b].

Proposition 3.6.1: (Error Bound for General Aggregation) Let the condition (3.91) hold. Then, we have

$$\big|J^*(j) - \tilde{J}(j)\big| \leq \frac{\epsilon}{1-\alpha}, \qquad \text{for all } j \text{ such that } j \in S_y,\ y \in \mathcal{A},$$

where ϵ is the maximum variation of the optimal cost function J^* over the footprint sets S_y, $y \in \mathcal{A}$:

$$\epsilon = \max_{y \in \mathcal{A}} \max_{i,j \in S_y} \big|J^*(i) - J^*(j)\big|.$$

The meaning of the preceding proposition is that if the optimal cost

function J^* varies by at most ϵ within each footprint set S_y, the aggregation scheme yields a piecewise constant approximation to the optimal cost function that is within $\epsilon/(1-\alpha)$ of the optimal. The paper [LiB25b] also provides an example where the bound fails to hold because the condition (3.91) is violated; see Exercise 3.3.

Selecting the Aggregate States

Generally, the method to select the aggregate states is an important issue, for which there is no mathematical theory at present. In practice, intuition and problem-specific insights often suggest reasonable choices, which can then be fine-tuned through experimentation. For example, suppose that the optimal cost function J^* is piecewise constant over a partition $\{S_y \mid y \in \mathcal{A}\}$ of the state space $\{1, \ldots, n\}$. By this we mean that for some vector

$$r^* = \big\{r_y^* \mid y \in \mathcal{A}\big\},$$

we have

$$J^*(j) = r_y^* \qquad \text{for all } j \in S_y,\ y \in \mathcal{A}.$$

Then from Prop. 3.6.1 it follows that the hard aggregation scheme with $I_x = S_x$ for all $x \in \mathcal{A}$ is exact, meaning that r_x^* are the optimal costs of the aggregate states x in the aggregate problem. This suggests that *the states in the footprint set S_y corresponding to an aggregate state y should have roughly equal optimal cost*, in line with the error bound of Prop. 3.6.1.

Expanding on this idea, suppose that through some special insight into the problem's structure or preliminary calculations, we know some features of the system's state that can "predict well" its optimal cost when combined through some approximation architecture, e.g., one that is linear. Then it seems reasonable to form the set of aggregate states $\mathcal{A}$ of a hard aggregation scheme so that the sets I_y and S_y consist of states with "similar features" for every $y \in \mathcal{A}$. This approach, suggested in the book [BeT96], Section 3.1.2, is known as *feature-based aggregation*. The next section considers this possibility, and provides a way to introduce features and nonlinearities into the aggregation architecture, without compromising its other favorable aspects.

3.6.6 Aggregation Using Features

Let us focus on the guideline for hard aggregation discussed above: *states i that belong to the same footprint set S_y should have nearly equal optimal costs*, i.e.,

$$\max_{i,j \in S_y} \big|J^*(i) - J^*(j)\big| \approx 0, \qquad \text{for all } y \in \mathcal{A}.$$

This raises the question of how to select the sets S_y according to this guideline.

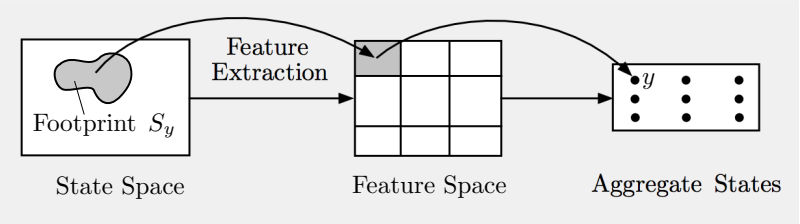

Figure 3.6.7 Feature-based hard aggregation using a partition of the space of features. Each aggregate state y has a footprint S_y that consists of states with "similar" features, i.e., states that map into the same subset of a partition in the space of features.

One approach is to use a *feature mapping*, i.e., a function F that maps a state i into an m-dimensional feature vector $F(i)$; cf. Example 3.1.7. In particular, suppose that F has the property that states i with nearly equal feature vector have nearly equal optimal cost $J^*(i)$. Then we can form the sets S_y by grouping together states with nearly equal feature vectors. Specifically, given F, we introduce a more or less regular partition of the feature space [the subset of $\Re^m$ that consists of all possible feature vectors $F(i)$]. This partition induces a possibly irregular collection of subsets of the original state space. Each of these subsets can then be used as the footprint of a distinct aggregate state; see Fig. 3.6.7.

Note that in the resulting aggregation scheme the number of aggregate states may become very large. However, this approach offers a significant advantage over the linear feature-based architectures of Section 3.1, where each feature is assigned a single weight: in feature-based hard aggregation, a weight is assigned to each subset of the feature space partition (possibly a weight to every possible feature value, in the extreme case where each feature value is viewed by itself as a distinct set of the partition). In effect we use aggregation to construct a *nonlinear* (piecewise constant) feature-based architecture, which may be much more powerful than the corresponding linear architecture of Section 3.1.

A question that now arises is how to obtain a suitable feature vector when there is no obvious choice, based on problem-specific considerations. One option, proposed in the author's paper [Ber18a] and also discussed in the book [Ber19a] (Section 6.4), is to obtain "good" features by using a neural network. More generally, any method that automatically generates features from data may be used.

3.6.7 Biased Aggregation

In this section we will introduce an extension of the preceding aggregation frameworks, called *biased aggregation*.† It involves a vector $V =$

† The aggregation framework of this section was proposed in the author's paper [Ber18b], which contains much additional material. It is related to a clas-

$\big(V(1),\ldots,V(n)\big)$, called the *bias vector* or *bias function*, which affects the cost structure of the aggregate problem, and biases the values of its optimal cost function towards their correct levels. For practical purposes the values $V(i)$ at various states should be readily accessible through simple computation or precomputation. Generally, the bias function V is obtained with some method that approximates J^*, such as for example neural network-based approximate PI, rollout, or problem approximation.

When $V = 0$, biased aggregation is identical to the aggregation scheme of Section 3.6.4. When $V \neq 0$, biased aggregation yields an approximation to J^* that is equal to V plus a local correction $\hat{J}$; see Fig. 3.6.8. The method relies on a simple cost function change, as indicated below and explained later.

Biased Aggregation Method

We modify the original problem by replacing the cost per stage,

$$g(i,u,j),$$

with

$$g(i,u,j) - V(i) + \alpha V(j). \tag{3.92}$$

We then apply the aggregation method of Section 3.6.4 to the modified cost problem to obtain a cost approximation $\hat{J}$. The function

$$\tilde{J}(i) = \hat{J}(i) + V(i)$$

forms an approximation to the optimal cost function $J^*(i)$ of the original problem.

To justify the biased aggregation method, consider the optimal cost function $\hat{J}$ of the modified cost problem, i.e., the one with cost per stage given by Eq. (3.92). It satisfies the corresponding Bellman equation:

$$\hat{J}(i) = \min_{u\in U(i)} \sum_{j=1}^{n} p_{ij}(u)\big(g(i,u,j) - V(i) + \alpha V(j) + \alpha\hat{J}(j)\big), \quad i = 1,\ldots,n,$$

or equivalently

$$\hat{J}(i) + V(i) = \min_{u\in U(i)} \sum_{j=1}^{n} p_{ij}(u)\Big(g(i,u,j) + \alpha\big(\hat{J}(j) + V(j)\big)\Big), \quad i = 1,\ldots,n.$$

sical DP scheme, known as *cost shaping* in the RL literature; see e.g., [Ber19a], the references quoted there.

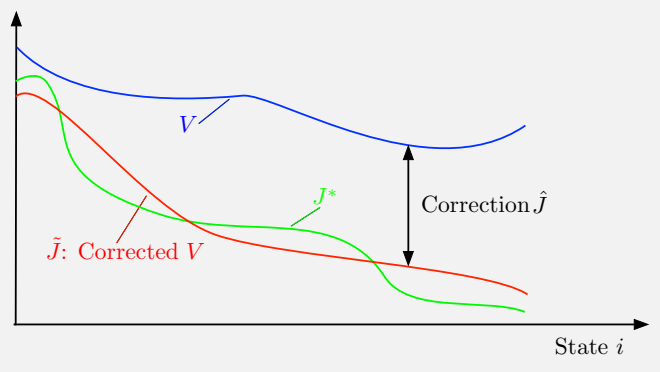

Figure 3.6.8 Schematic illustration of biased aggregation. It provides an approximation $\tilde{J}$ to J^* that is equal to the bias function V plus a local correction $\hat{J}$, which is obtained by solving a modified cost problem with the aggregation method of Section 3.6.4.

By comparing this equation with the Bellman equation for the original problem, we see that the optimal cost functions of the modified and the original problems are related by

$$J^*(i) = \hat{J}(i) + V(i), \qquad i = 1, \ldots, n,$$

and that the two problems have the same optimal policies. This of course assumes that the modified cost problem is solved exactly. If instead it is solved approximately using (unbiased) aggregation, the choice of V and the approximation architecture may affect substantially the character of the resulting approximation in value space scheme and the quality of suboptimal policies obtained.

Figure 3.6.9 provides an interpretation of biased aggregation, which is consistent to the one of the (unbiased) aggregation scheme of Section 3.6.4. It involves three sets of states: two copies of the original state space, as shown in the figure, as well as a finite set $\mathcal{A}$ of aggregate states. The state transitions go from a state in $\mathcal{A}$ to a state in the left state space copy, according to disaggregation probabilities, then to a state in the right state space copy, and then back to a state in $\mathcal{A}$, according to aggregation probabilities, and the process is repeated. At a state i in the left state space copy we must choose a control $u \in U(i)$, and then transition to a state j in the right state space copy at a cost $g(i, u, j)$ according to the original system transition probabilities $p_{ij}(u)$.

The key insight here is that *biased aggregation can be viewed as unbiased aggregation applied to a modified DP problem*, with cost per stage given by Eq. (3.92), which is equivalent to the original DP problem in the sense that it has the same optimal policies. Thus any unbiased aggregation scheme and algorithm, when applied to the modified DP problem, yields a biased aggregation scheme and algorithm for the original DP problem. As

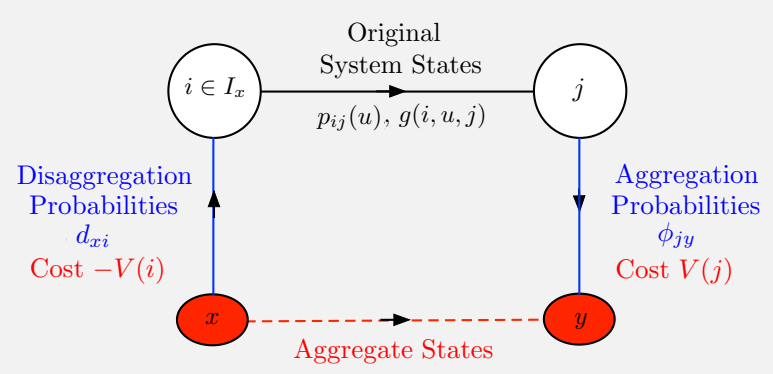

Figure 3.6.9 Illustration of the transition mechanism and the costs per stage of the aggregate problem in biased aggregation. When the bias function V is identically zero, we obtain the aggregation framework of Section 3.6.4.

a result, we can straightforwardly transfer results, algorithms, and intuition from our earlier unbiased aggregation analysis to the biased aggregation framework, by applying them to the unbiased aggregation framework that corresponds to the modified stage cost (3.92). Moreover, we may use simulation-based algorithms for policy evaluation, policy improvement, and Q-learning for the aggregate problem, with the only requirement that the value $V(i)$ for any state i is available when needed.

Regarding the choice of V, consistent with our earlier analysis, the general principle is that V should capture a fair amount of the nonlinearity or "shape" of J^*. In the case of hard aggregation, based on the error bound of Prop. 3.6.1, the function V should be chosen so that the variation of $J^*(i) - V(i)$ is as small as possible within the corresponding footprint sets. This suggests that V should be chosen as a good approximation to J^* (within a constant value).

3.6.8 Asynchronous Distributed Aggregation

Let us now discuss the distributed solution of large-scale discounted DP problems using cost function approximation, multiple agents/processors, and hard aggregation. Here we partition the original system states into aggregate states/subsets $x \in \mathcal{A} = \{x_1, \ldots, x_m\}$, and we employ a network of processors/agents. Each processor updates asynchronously a detailed/exact local cost function, defined on a single aggregate state/subset. Each processor also maintains an aggregate cost for its aggregate state, which is a weighted average of the detailed cost of the (original system) states in the processor's subset, weighted by the corresponding disaggregation probabilities. These aggregate costs are communicated between processors and are used to perform the local updates.

In a synchronous VI method of this type, each processor $\ell = 1, \ldots, m$, maintains and updates a (local) cost $J(i)$ for every original system state $i \in x_\ell$, and an aggregate cost

$$R(\ell) = \sum_{i \in x_\ell} d_{x_\ell i} J(i),$$

where $d_{x_\ell i}$ are the corresponding disaggregation probabilities. We generically denote by J and R the vectors with components $J(i)$, $i = 1, \ldots, n$, and $R(\ell)$, $\ell = 1, \ldots, m$, respectively. These components are updated according to

$$J_{k+1}(i) = \min_{u \in U(i)} H_\ell(i, u, J_k, R_k), \qquad \text{for all } i \in x_\ell, \tag{3.93}$$

with

$$R_k(\ell) = \sum_{i \in x_\ell} d_{x_\ell i} J_k(i), \qquad \ell = 1, \ldots, m, \tag{3.94}$$

where the mapping H_ℓ is defined for all $\ell = 1, \ldots, m$, $i \in x_\ell$, $u \in U(i)$, and $J \in \Re^n$, $R \in \Re^m$, by

$$H_\ell(i, u, J, R) = \sum_{j=1}^{n} p_{ij}(u) g(i, u, j) + \alpha \sum_{j \in x_\ell} p_{ij}(u) J(j) + \alpha \sum_{j \notin x_\ell} p_{ij}(u) R\big(x(j)\big), \tag{3.95}$$

and where for each original system state j, we denote by $x(j)$ the subset to which j belongs [i.e., $j \in x(j)$]. Thus the iteration (3.93) is the same as ordinary VI, except that instead of $J(j)$, we use the aggregate costs $R\big(x(j)\big)$ for the states j whose costs are updated by other processors.

It is possible to show that the iteration (3.93)-(3.94) involves a sup-norm contraction mapping of modulus α, so it converges to the unique solution of the system of equations in (J, R)

$$J(i) = \min_{u \in U(i)} H_\ell(i, u, J, R), \qquad R(\ell) = \sum_{i \in x_\ell} d_{x_\ell i} J(i), \qquad \text{for all } i \in x_\ell,\ \ell = 1, \ldots, m. \tag{3.96}$$

This follows from the fact that $\{d_{x_\ell i} \mid i = 1, \ldots, n\}$ is a probability distribution. We may view the equations (3.96) as a set of Bellman equations for an "aggregate" DP problem, which similar to our earlier discussion, involves both the original and the aggregate system states. The difference from the Bellman equations (3.84)-(3.86) is that the mapping (3.95) involves $J(j)$ rather than $R\big(x(j)\big)$ for $j \in x_\ell$.

In the algorithm (3.93)-(3.94), all processors ℓ must be updating their local costs $J(i)$ and aggregate costs $R(\ell)$ synchronously, and communicate the aggregate costs to the other processors before a new iteration may begin. This is often impractical and time-wasting. In a more practical asynchronous version of the method, the aggregate costs $R(\ell)$ may be outdated

to account for communication "delays" between processors. Moreover, the costs $J(i)$ need not be updated for all i; it is sufficient that they are updated by each processor ℓ only for a (possibly empty) subset of $I_{\ell,k}$ of the aggregate state/set x_ℓ. In this case, the iteration (3.93)-(3.94) is modified to take the form

$$J_{k+1}(i) = \min_{u \in U(i)} H_\ell\big(i, u, J_k, R_{\tau_{1,k}}(1), \ldots, R_{\tau_{m,k}}(m)\big), \qquad \text{for all } i \in I_{\ell,k}, \tag{3.97}$$

with $0 \leq \tau_{\ell,k} \leq k$ for $\ell = 1, \ldots, m$, and

$$R_\tau(\ell) = \sum_{i \in x_\ell} d_{x_\ell i} J_\tau(i), \qquad \text{for all } \ell = 1, \ldots, m.$$

The differences $k - \tau_{\ell,k}$, $\ell = 1, \ldots, m$, in Eq. (3.97) may be viewed as "delays" between the current time k and the times $\tau_{\ell,k}$ when the corresponding aggregate costs were computed at other processors. For convergence, it is of course essential that every $i \in x_\ell$ belongs to $I_{\ell,k}$ for infinitely many k (so each cost component is updated infinitely often), and $\lim_{k\to\infty} \tau_{\ell,k} = \infty$ for all $\ell = 1, \ldots, m$ (so that processors eventually communicate more recently computed aggregate costs to other processors).

Convergence of this type of method can be established using the asynchronous convergence theory for DP developed by the author in the paper [Ber82a] (see the books [BeT89], [Ber12], [Ber22b] for a more detailed treatment). The proof is based on the sup-norm contraction property of the mapping underlying Eq. (3.96). The monotonicity property is also sufficient to establish asynchronous convergence, and this is useful in the convergence analysis of related aggregation algorithms for undiscounted DP models (see the paper by Bertsekas and Yu [BeY10]).

3.7 NOTES, SOURCES, AND EXERCISES

Section 3.1: Our discussion of approximation architectures, neural networks, and training has been limited, and aimed just to provide the connection with approximate DP. The literature on the subject is vast, and some of the textbooks mentioned in the references to Chapter 1 provide detailed accounts and many sources, in addition to the ones given in Sections 3.1 and 3.2.

There are two broad directions of inquiry in parametric architectures:

(1) The design of architectures, either in a general or a problem-specific context.

(2) The training of neural networks, as well as other linear and nonlinear architectures.

Research along both of these directions has been extensive and is continuing.

Methods for selection of basis functions have received much attention, particularly in the context of neural network research and deep reinforcement learning (see e.g., the book by Goodfellow, Bengio, and Courville [GBC16]). For discussions that are focused outside the neural network area, see Bertsekas and Tsitsiklis [BeT96], Keller, Mannor, and Precup [KMP06], Jung and Polani [JuP07], Bertsekas and Yu [BeY09], and Bhatnagar, Borkar, and Prashanth [BBP13]. Moreover, there has been considerable research on optimal feature selection within given parametric classes (see Menache, Mannor, and Shimkin [MMS05], Yu and Bertsekas [YuB09], Busoniu et al. [BBD10a], and Di Castro and Mannor [DiM10]).

Incremental algorithms are the principal methods for training approximation architectures. They are supported by substantial theoretical analysis, which addresses issues of convergence, rate of convergence, stepsize selection, and component order selection. Moreover, incremental algorithms have been extended to constrained optimization settings, where the constraints are also treated incrementally, first by Nedić [Ned11], and then by several other authors: Bertsekas [Ber11a], Wang and Bertsekas [WaB15], [WaB16], Bianchi [Bia16], Iusem, Jofre, and Thompson [IJT18]. It is beyond our scope to cover this analysis. The author's surveys [Ber10a] and [Ber15b], and convex optimization and nonlinear programming textbooks [Ber15a], [Ber16], collectively contain an extensive account of incremental methods, including the Kaczmarz, incremental gradient, subgradient, aggregated gradient, Newton, Gauss-Newton, and extended Kalman filtering methods, and give many references. The book [BeT96] and paper [BeT00] by Bertsekas and Tsitsiklis, and the survey by Bottou, Curtis, and Nocedal [BCN18] provide theoretically oriented treatments.

Linear feature-based architectures can also be trained with temporal difference methods, such as TD(λ), LSTD(λ), and LSPE(λ), which are not discussed in this book; see [BeT96], [Ber12], [Ber19a], [SuB18]. These methods can be viewed as algorithms for solving general linear systems of equations by Monte Carlo simulation, a subject that is of interest beyond approximate DP and entails much subtlety. We refer to the work of H. Yu, M. Wang, and the author for an in-depth analysis and discussion [BeY07], [BeY09], [YuB10], [Ber11c], [WaB13a], [WaB13b]. Section 7.3 of the book [Ber12] provides an extensive textbook treatment and additional references.

Section 3.2: The publicly and commercially available neural network training programs incorporate heuristics for scaling and preprocessing data, stepsize selection, initialization, etc, which can be very effective in specialized problem domains. We refer to books on neural networks such as Bishop [Bis95], Goodfellow, Bengio, and Courville [GBC16], Haykin [Hay08]. The recent book by Bishop and Bishop [BiB24] includes discussions of deep neural networks and transformers.

Deep neural networks have created a lot of excitement in the machine learning field, in view of some high profile successes in image and speech recognition, and in RL with the AlphaGo and AlphaZero programs. One question is whether and for what classes of target functions we can enhance approximation power by increasing the number of layers while keeping the number of weights constant. For discussion, analysis, and speculation around this question, see Bengio [Ben09], Liang and Srikant [LiS16], Yarotsky [Yar17], and Daubechies et al. [DDF19].

Another important research question relates to the role of overparametrization in the success of deep neural networks. With more weights than training data, the training problem has infinitely many solutions, each providing an architecture that fits the training data perfectly. The question then is how to select a solution that works well on test data (i.e., data outside the training set); see Zhang et al. [ZBH16], [ZBH21], Belkin, Ma, and Mandal [BMM18], Belkin, Rakhlin, and Tsybakov [BRT18], Soltanolkotabi, Javanmard, and Lee [SJL18], Bartlett et al. [BLL19], Hastie et al. [HMR19], Muthukumar, Vodrahalli, and Sahai [MVS19], Su and Yang [SuY19], Sun [Sun19], Vaswani et al. [VLK21], Zhang et al. [ZBH21], and the discussions in the machine learning books by Hardt and Recht [HaR21], and Bishop and Bishop [BiB24].

Section 3.3: Fitted value iteration has a long history; it was mentioned by Bellman among others. It has interesting properties, and at times exhibits pathological/unstable behavior due to accumulation of errors over a long horizon (see [Ber19a], Section 5.2).

The approximate policy iteration method of Section 3.3.3 has been proposed by Fern, Yoon, and Givan [FYG06], and variants have also been discussed and analyzed by several other authors. The method (with some variations) has been used to train a tetris playing computer program that performs impressively better than programs that are based on other variants of approximate policy iteration; see Scherrer [Sch13], Scherrer et al. [SGG15], and also Gabillon, Ghavamzadeh, and Scherrer [GGS13], who also provide an analysis of the method. The RL and approximate DP books collectively describe several alternative simulation-based methods for policy evaluation; see e.g., [BeT96], [SuB18], [Ber12], Chapters 6 and 7. These include temporal difference methods, which enjoyed much popularity in the early days of RL. They are stochastic iterative algorithms that are closely related to Galerkin approximation, a major computational approach for solving large scale equations, as first observed by Yu and Bertsekas [YuB10], and Bertsekas [Ber11c]; see also Szepesvari [Sze11].

Simulation-based methods for approximate DP can benefit significantly from the use of parallel and distributed computation. A simple possibility is to parallelize the collection of Monte Carlo simulation samples. The book [Ber20a] describes distributed versions of approximate policy iteration, which are based on partitioning of the state space.

The original proposal of SARSA (Section 3.3.4) is attributed to Rummery and Niranjan [RuN94], with related work presented in the papers by Peng and Williams [PeW96], and Wiering and Schmidhuber [WiS98]. The ideas of the DQN algorithm attracted much attention following the paper by Mnih et al. [MKS15], which reported impressive test results on a suite of 49 classic Atari 2600 games.

The rollout and approximate PI methodology for POMDP of Section 3.3.5 was described in the author's RL book [Ber19a]. It was extended and tested in the paper by Bhattacharya et al. [BBW20] in the context of a challenging pipeline repair problem. A skillful application of approximate PI in combination with deep neural networks was given by Rybicki and Nelson [RyN25].

Advantage updating (Section 3.3.6) was proposed by Baird [Bai93], [Bai94], and is discussed further in Section 6.6 of the neuro-dynamic programming book [BeT96]. The differential training methodology (Section 3.3.7) was proposed by the author in the paper [Ber97b], and followup work was presented by Weaver and Baxter [WeB99].

Generally, the challenges of implementing successfully approximate value and policy iteration schemes are quite formidable, and tend to be underestimated, because the literature naturally tends to place emphasis on success stories, and tends to underreport failures. In practice, the training difficulties, particularly exploration, must often be addressed on a case-by-case basis, and may require long and tricky parameter tuning, with little guarantee of ultimate success or even a diagnosis of the causes of failure. By contrast, approximation in value space with long multistep lookahead and simple terminal cost function approximation, and rollout (a single policy iteration starting from a base policy), while less ambitious, are typically much easier to implement, and often attain considerable success reliably. An intermediate approach that often works well is to use truncated rollout with a terminal cost function approximation that is trained with data.

Section 3.4: Classification (sometimes called "pattern classification" or "pattern recognition") is a major subject in machine learning, for which there are many approaches, an extensive literature, and an abundance of public domain and commercial software; see e.g. the textbooks by Bishop [Bis95], [Bis06], Duda, Hart, and Stork [DHS12], and Hardt and Recht [HaR21]. Approximation in policy space was formulated as a classification problem in the context of DP by Lagoudakis and Parr [LaP03], and was followed up by several other authors (see e.g., Dimitrakakis and Lagoudakis [DiL08], Lazaric, Ghavamzadeh, and Munos [LGM10], Gabillon et al. [GLG11], Liu and Wei [LiW14], Farahmand et al. [FPB15], and the references quoted there). While we have focused on a classification approach that makes use of least squares regression and a parametric architecture, other classification methods may also be used. For example the paper [LaP03] discusses the use of nearest neighbor schemes, support

vector machines, as well as neural networks.

Section 3.5: Our coverage of policy gradient and random search methods has aimed to provide an entry point into the field, and has been restricted to the off-line training of policies. For a detailed discussion and references on policy gradient methods, we refer to the book by Sutton and Barto [SuB18], the monograph by Deisenroth, Neumann, and Peters [DNP11], and the survey by Grondman et al. [GBL12]. An influential paper in this context by Williams [Wil92] proposed among others the likelihood-ratio policy gradient method given here. The methods of [Wil92] are commonly referred to as REINFORCE in the literature (see e.g., [SuB18], Ch. 13). For recent work on these and related methods, which give many additional references, see Zhang et al. [ZKZ20], Bhatnagar [Bha23], Bhandari and Russo [BhR24], Maniyar et al. [MPM24], and Muller and Montufar [MuM24]. For general references on stochastic optimization, including stochastic gradient methods, see the books by Amari [Ama16], and Bertsekas and Tsitsiklis [BeT96], and the lecture notes by Duchi [Duc18].

There are several early works on search along randomly chosen directions (Rastrigin [Ras63], Matyas [Mat65], Aleksandrov, Sysoyev, and Shemeneva [ASS68], Rubinstein [Rub69]). For some more modern works, see Spall [Spa92], [Spa03], Duchi et al. [DJW12], [DJW15], and Nesterov and Spokoiny [NeS17]. For early works on simulation-based policy gradient schemes for various DP problems, see Glynn [Gly87], [Gly90], L'Ecuyer [L'Ec91], Fu and Hu [FuH94], Jaakkola, Singh, and Jordan [JSJ95], Cao and Chen [CaC97], Cao and Wan [CaW98]. More recent works have focused on the use of natural gradient scaling and a trust region; see the discussion and the references in Section 3.5.2.

Policy gradient-like methods: The main challenge in the successful implementation of policy gradient methods is twofold:

(a) The difficulties with slow convergence. The detrimental effects of simulation noise contribute further to slow convergence. Much work has been directed towards variations that address these difficulties, including the use of a baseline and variance reduction methods (Greensmith, Bartlett, and Baxter [GBB04], Greensmith [Gre05]), or second order information (Wang and Paschalidis [WaP17], and the references quoted there); cf. Section 3.5.2.

(b) The presence of local minima. In RL applications, the cost function is typically nonconvex, particularly when neural networks are involved, giving rise to many local minima of dubious quality (see Exercise 3.2). This is a serious concern, which has been mitigated in practice by trying many starting points and other more or less heuristic randomization devices.

A further concern is that the method may yield a randomized policy [see Exercise 3.2(c)], which has to be converted to a nonrandomized policy for

on-line implementation. Finally, a general issue to contend with arises in practical contexts that call for on-line replanning and possibly on-line policy retraining, e.g., in adaptive control.

The natural gradient approach has been used extensively, and has been effective in improving the convergence rate of unscaled policy gradient methods; see the references given in Section 3.5.2. Its origins lie with concepts of information geometry, developed principally by Amari and described in his book [Ama16]. The constrained version of the natural gradient method, described in Section 3.5.2, is presented here for the first time, and has not been tested extensively. An alternative possibility to treat constraints directly, without a softmax reparametrization is the mirror descent algorithm, described in the context of policy gradient methods by Xiao [Xia22].

We have not covered actor-critic methods within the policy gradient context. These methods were introduced in the paper by Barto, Sutton, and Anderson [BSA83]. The more recent works of Sutton et al. [SMS99], Baxter and Bartlett [BaB01], Konda and Tsitsiklis [KoT99], [KoT03], Marbach and Tsitsiklis [MaT01], [MaT03], Peters and Schaal [PeS08], and Bhatnagar et al. [BSG09] have been influential; see also the survey by Grondman et al. [GBL12]. Actor-critic algorithms that are suitable for POMDP and involve gradient estimation have been given by H. Yu [Yu05], and Estanjini, Li, and Paschalidis [ELP12].

Random search methods: The cross-entropy method was initially developed in the context of rare event simulation and was later adapted for use in optimization. For textbook accounts, see Rubinstein and Kroese [RuK04], [RuK13], [RuK16], and Busoniu et al. [BBD10a], and for surveys see de Boer et al. [BKM05], and Kroese et al. [KRC13]. The method was proposed for policy search in an approximate DP context by Mannor, Rubinstein, and Gat [MRG03]. For recent analysis, see Joseph and Bhatnagar [JoB16], [JoB18].

It is generally thought that the cross entropy method, while applicable to problems with unrestricted parameter dimension, is effective primarily for a low-dimensional parameter space. One such context where the method was successfully applied is the game of tetris; see the papers by Szita and Lorincz [SzL06], and Thiery and Scherrer [ThS09]. These papers report much superior results to the ones obtained earlier by Bertsekas and Ioffe [BeI96], using approximate PI methods, and by Kakade [Kak02] using policy gradient methods (comparable results were obtained later with approximate PI methods by Gabillon, Ghavamzadeh, and Scherrer [GGS13]).

Are the policy gradient and random search approaches related? At first glance the answer seems negative, in view of the fundamental conceptual differences between these two types of methods: while gradient-like methods are guided by the gradient at the current iterate to move towards an improved iterate, random search methods involve stochastic exploration

of the parameter space, with no strict requirement for cost improvement. In the context of RL applications, however, there is substantial commonality of ideas between the two types of methods. Policy gradient methods rely on randomization of the optimization variables (e.g., randomized policies) and on randomization of the starting parameter to escape from local minima (cf. the 2-state problem of Exercise 3.2). Conversely, random search methods often bias their random exploration towards lower cost solutions (cf. the cross entropy method and Fig. 3.5.4). Thus, while their mechanisms differ, policy gradient and random search methods are quite similar in RL practice, through shared principles of stochastic exploration and cost-driven iterative improvement. In fact, methods that combine random search for global exploration with gradient-based local optimization have been proposed since the 1990s. Application contexts, have included hybrid systems (e.g., processes with discrete/continuous dynamics), and integer programming (e.g., combinations of branch-and-bound with gradient methods).

Section 3.6: The aggregation approach has a long history in scientific computation and operations research (see for example Bean, Birge, and Smith [BBS87], Chatelin and Miranker [ChM82], Douglas and Douglas [DoD93], and Rogers et al. [RPW91]). It was introduced in the simulation-based approximate DP context, mostly in the form of VI; see Singh, Jaakkola, and Jordan [SJJ95], Gordon [Gor95], and Tsitsiklis and Van Roy [Van95], [TsV96]. It was further discussed in the neuro-dynamic programming book [BeT96], Sections 3.1.2 and 6.7.

In the RL literature, aggregation, as described here, is sometimes referred to as "state abstraction." Another scheme, called *options*, has been introduced by Sutton, Precup, and Singh [SPS99], and can be viewed as a formalization of temporal abstraction. It introduces additional multi-step "macro-actions" that are available at some states. Macro-actions transfer the state of the system to one of a selected set of states through a sequence of actions at a given cost. Options can be viewed as a problem approximation approach: the set of actions available at a state is augmented with the additional macro-actions. The possibility of a synergistic combination of the options formalism with state aggregation has been discussed by Ciosek and Silver [Cio15], [CiS15].

The material on POMDP discretization (cf. Section 3.6.3) is based on the paper by Yu and Bertsekas [YuB04]. This paper provides also a similar discretization scheme for the average cost case where the Bellman equation need not have a solution, but the scheme nevertheless provides lower bounds to optimal average cost functions which are difficult to compute. Here we have focused on the simpler discounted case. The paper by Li, Hammar, and Bertsekas [LHB25], which develops a more sophisticated POMDP aggregation methodology, and provides extensive computational results.

The aggregation framework with representative features was introduced in the author's DP book [Ber12], was discussed in detail in the RL textbook [Ber19a] (Chapter 6), and was further developed in the survey paper [Ber18a], which provides an expanded view of the methodology. These sources also provide several VI and PI simulation-based methods for solving the aggregate problem. The paper by Yu and Bertsekas [YuB12] and the book [Ber12] discuss a view of aggregated equations as projected equations that involve a Euclidean norm or seminorm projection. This view provides a connection with the temporal difference and Galerkin approximation methodologies, both of which also involve Euclidean projections.

Biased aggregation (Section 3.6.7) was introduced in the author's paper [Ber18b], which contains further analysis, discusses connections with rollout algorithms, and suggests additional methods. A noteworthy result given in that paper is that the bound of Prop. 3.6.1 admits an extension whereby the optimal cost function J^* is replaced by $J^* - V$. This shows that if the variation of $J^* - V$ is small, the performance of the aggregation method improves accordingly.

Distributed asynchronous aggregation (Section 3.6.8) was first proposed in the paper by Bertsekas and Yu [BeY10] (Example 2.5); see also the discussions in the author's DP books [Ber12] (Section 6.5.4) and [Ber22b] (Example 1.2.11). A recent computational study, related to distributed traffic routing, was given by Vertovec and Margellos [VeM23].

Aggregation may also be used as a policy evaluation method in the context of policy iteration with linear feature-based cost function approximations. Within this context, the aggregation approach provides an alternative to the temporal difference approach. These two approaches are described and compared in the author's approximate policy iteration survey paper [Ber11b]. Generally speaking, aggregation methods are characterized by stronger theoretical properties, such as Bellman operator monotonicity, resilience to policy oscillations, and better error bounds. On the other hand, they are more restrictive in their use of linear approximation architectures, compared with temporal difference methods (see [Ber11b], [Ber18a]).

EXERCISES

3.1 (Proof of Prop. 3.4.1)

Complete the details of the following proof of Prop. 3.4.1. Fix c, and for any scalar y, consider for a given x the conditional expected value $E\{(z(c,c')-y)^2 \mid x\}$. Here the random variable $z(c,c')$ takes the value 1 with probability $p(c\,|\,x)$ and the value 0 with probability $1-p(c\,|\,x)$, so we have

$$E\Big\{\big(z(c,c')-y\big)^2 \;\Big|\; x\Big\} = p(c\,|\,x)(y-1)^2 + \big(1-p(c\,|\,x)\big)y^2.$$

We minimize this expression with respect to y, by setting to 0 its derivative, i.e.,

$$0 = 2p(c\,|\,x)(y-1) + 2\big(1-p(c\,|\,x)\big)y = 2\big(-p(c\,|\,x)+y\big).$$

We thus obtain the minimizing value of y, namely $y^* = p(c\,|\,x)$, so that

$$E\Big\{\big(z(c,c')-p(c\,|\,x)\big)^2 \mid x\Big\} \le E\Big\{\big(z(c,c')-y\big)^2 \mid x\Big\}, \qquad \text{for all scalars } y.$$

We set $y = h(c,x)$ in the above expression and obtain

$$E\Big\{\big(z(c,c')-p(c\,|\,x)\big)^2 \mid x\Big\} \le E\Big\{\big(z(c,c')-h(c,x)\big)^2 \mid x\Big\}.$$

Since this is true for all x, we also have

$$\sum_x \xi(x) E\Big\{\big(z(c,c')-p(c\,|\,x)\big)^2 \;\Big|\; x\Big\} \le \sum_x \xi(x) E\Big\{\big(z(c,c')-h(c,x)\big)^2 \;\Big|\; x\Big\},$$

showing that Eq. (3.35) holds for all functions h and all classes c.

3.2 (Local Minima in Policy Gradient Optimization)

This exercise explores some of the pitfalls of policy gradient optimization. Consider a deterministic α-discounted DP problem with two states, Left and Right, denoted by L and R, respectively, and the two controls Move-to-the-Left and Move-to-the-Right, denoted by u_L and u_R, respectively. The transition probabilities are

$$p_{LL}(u_L) = 1, \quad p_{LR}(u_L) = 0, \qquad p_{LL}(u_R) = 0, \quad p_{LR}(u_R) = 1,$$

$$p_{RR}(u_L) = 0, \quad p_{RL}(u_L) = 1, \qquad p_{RR}(u_R) = 1, \quad p_{RL}(u_R) = 0.$$

The transition cost for $L \to R$ is 2, and for $R \to L$ it is 2. The transition cost for $L \to L$ is 1, and for $R \to R$ it is 0. Starting from R, the optimal policy is to

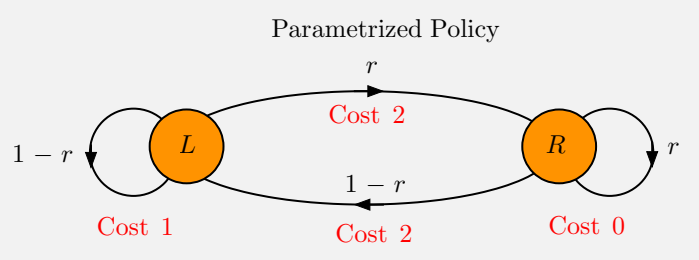

Figure 3.7.1 Markov chain and parametrized policy in Exercise 3.2.

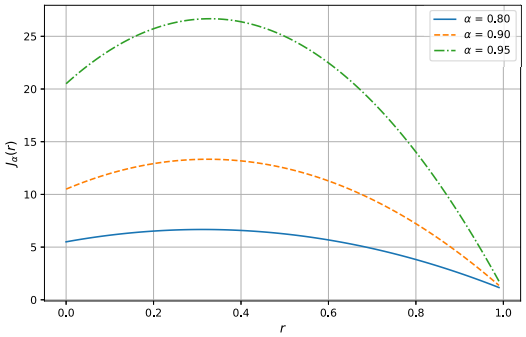

Figure 3.7.2 Plot of $J_\alpha(r)$ for various values of α; cf. Exercise 3.2(a).

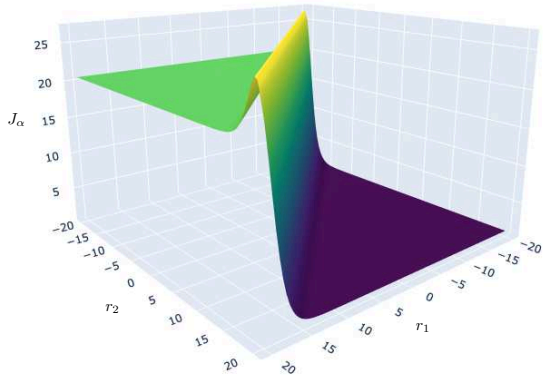

Figure 3.7.3 Two-dimensional plot of $J_\alpha(r)$ for $\alpha = .95$, where $r = (r_1, r_2)$ are the parameters of the soft-max distribution that specifies the randomized policy; cf. Exercise 3.2(b).

stay in R. Starting from L, the optimal policy is to stay in L if $\alpha < 1/2$ and to move to R if $\alpha > 1/2$.

(a) Consider policies that move to R with probability r and to L with probability $1 - r$, starting from each of the two states; cf. Fig. 3.7.1. The

costs $J_\alpha(L;r)$ and $J_\alpha(R;r)$ of such a policy starting from L and from R, respectively, can be computed from the corresponding Bellman equation:

$$J_\alpha(L;r) = (1-r)\big(1+\alpha J_\alpha(L;r)\big) + r\big(2+\alpha J_\alpha(R;r)\big),$$

$$J_\alpha(R;r) = (1-r)\big(2+\alpha J_\alpha(L;r)\big) + r\alpha J_\alpha(R;r).$$

Consider also the expected cost of the policy corresponding to r, assuming the initial states are L or R with equal probability $1/2$:

$$J_\alpha(r) = \frac{1}{2}\big(J_\alpha(L;r) + J_\alpha(R;r)\big). \tag{3.98}$$

Compute and plot $J_\alpha(r)$ for $r \in [0,1]$ and several values of $\alpha \in (0,1)$. Verify that $J_\alpha(r)$ is concave as a function of r, and that the values $r=0$ and $r=1$ are both local minima of $J_\alpha(r)$. Moreover, a policy gradient method converges to either one of these local minima if started close enough to it; see Fig. 3.7.2. (This exercise is adapted from an example given in the paper [BhR24].)

(b) Consider an equivalent soft-max parametrization of policies. Here, we replace the single parameter r of part (a), which is constrained by $0 \le r \le 1$, with two parameters r_1 and r_2, which are unconstrained. In particular, we move to state R with probability

$$\frac{e^{r_2}}{e^{r_1}+e^{r_2}},$$

starting from each of the two states, and we move to L with probability

$$\frac{e^{r_1}}{e^{r_1}+e^{r_2}},$$

starting from each of the two states. Compute and plot $J_\alpha(r_1,r_2)$ for several values of $\alpha \in (0,1)$. Verify that $J_\alpha(r_1,r_2)$ has no local minima if started close enough to it; see Fig. 3.7.3. Explain this observation in light of Fig. 3.7.2.

(c) Suppose that we change the problem from minimization to maximization, i.e., we view the "costs" in Fig. 3.7.1 as "rewards" to be maximized over an infinite horizon. Argue that the plots of Fig. 3.7.2, give the reward $J_\alpha(r)$ of Eq. (3.98) correctly, so that for the values of α used in Fig. 3.7.2, there is a unique value of r that maximizes $J_\alpha(r)$, and it lies strictly between 0 and 1. Thus a policy gradient method will tend to produce a randomized policy.

3.3 (Counterexample to the Aggregation Error Bound [LiB25b])

This exercise provides an example where the error bound of Prop. 3.6.1 fails to hold when the condition (3.91) is violated. Consider a system involving two absorbing states, 1 and 2, i.e.,

$$p_{11} = 1, \qquad p_{22} = 1, \qquad p_{12} = 0, \qquad p_{21} = 0,$$

with self transition costs

$$g(1,1) = 0, \qquad g(2,2) = 1.$$

Thus the infinite horizon costs (without aggregation) are

$$J^*(1) = 0, \qquad J^*(2) = \frac{1}{1-\alpha}.$$

Assume that there are two aggregate states x_1 and x_2 that disaggregate into states 1 and 2, respectively, but aggregate states 2 and 1, respectively, i.e.,

$$d_{x_1 1} = 1, \qquad d_{x_2 2} = 1, \qquad \phi_{1 x_2} = 1, \qquad \phi_{2 x_1} = 1.$$

Then $\epsilon = 0$ since the footprint sets S_{x_1} and S_{x_2} consist of a single state. Show that the true aggregation error is positive, i.e., the aggregation process is not exact. *Hint*: Verify that the sequence of generated costs starting from aggregate state x_1 is

$$\{0, \alpha, 0, \alpha^3, 0, \alpha^5, \ldots\},$$

while the sequence of generated costs starting from aggregate state x_2 is

$$\{1, 0, \alpha^2, 0, \alpha^4, 0, \alpha^6, \ldots\},$$

so we have $\tilde{J}(i) \neq J^*(i)$ for both states $i = 1, 2$.

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Neuro-Dynamic Programming
Dimitri P. Bertsekas and John N. Tsitsiklis

Athena Scientific, 1996
512 pp., hardcover, ISBN 1-886529-10-8

This is the first textbook that fully explains the neuro-dynamic programming/reinforcement learning methodology, a breakthrough in the practical application of neural networks and dynamic programming to complex problems of planning, optimal decision making, and intelligent control.

From the review by George Cybenko for IEEE Computational Science and Engineering, May 1998:

"Neuro-Dynamic Programming is a remarkable monograph that integrates a sweeping mathematical and computational landscape into a coherent body of rigorous knowledge. The topics are current, the writing is clear and to the point, the examples are comprehensive and the historical notes and comments are scholarly."

"In this monograph, Bertsekas and Tsitsiklis have performed a Herculean task that will be studied and appreciated by generations to come. I strongly recommend it to scientists and engineers eager to seriously understand the mathematics and computations behind modern behavioral machine learning."

Among its special features, the book:

- Describes and unifies a large number of NDP methods, including several that are new
- Describes new approaches to formulation and solution of important problems in stochastic optimal control, sequential decision making, and discrete optimization
- Rigorously explains the mathematical principles behind NDP
- Illustrates through examples and case studies the practical application of NDP to complex problems from optimal resource allocation, optimal feedback control, data communications, game playing, and combinatorial optimization
- Presents extensive background and new research material on dynamic programming and neural network training

Neuro-Dynamic Programming is the winner of the 1997 INFORMS CSTS prize for research excellence in the interface between Operations Research and Computer Science

Reinforcement Learning and Optimal Control
Dimitri P. Bertsekas

Athena Scientific, 2019
388 pp., hardcover, ISBN 978-1-886529-39-7

This book explores the common boundary between optimal control and artificial intelligence, as it relates to reinforcement learning and simulation-based neural network methods. These are popular fields with many applications, which can provide approximate solutions to challenging sequential decision problems and large-scale dynamic programming (DP). The aim of the book is to organize coherently the broad mosaic of methods in these fields, which have a solid analytical and logical foundation, and have also proved successful in practice.

The book discusses both approximation in value space and approximation in policy space. It adopts a gradual expository approach, which proceeds along four directions:

- From exact DP to approximate DP: We first discuss exact DP algorithms, explain why they may be difficult to implement, and then use them as the basis for approximations.
- From finite horizon to infinite horizon problems: We first discuss finite horizon exact and approximate DP methodologies, which are intuitive and mathematically simple, and then progress to infinite horizon problems.
- From model-based to model-free implementations: We first discuss model-based implementations, and then we identify schemes that can be appropriately modified to work with a simulator.

The mathematical style of this book is somewhat different from the one of the author's DP books, and the 1996 neuro-dynamic programming (NDP) research monograph, written jointly with John Tsitsiklis. While we provide a rigorous, albeit short, mathematical account of the theory of finite and infinite horizon DP, and some fundamental approximation methods, we rely more on intuitive explanations and less on proof-based insights. Moreover, our mathematical requirements are quite modest: calculus, a minimal use of matrix-vector algebra, and elementary probability (mathematically complicated arguments involving laws of large numbers and stochastic convergence are bypassed in favor of intuitive explanations).

The book is supported by on-line video lectures and slides, as well as new research material, some of which has been covered in the present monograph.

Rollout, Policy Iteration, and Distributed Reinforcement Learning

Dimitri P. Bertsekas

Athena Scientific, 2020
480 pp., hardcover, ISBN 978-1-886529-07-6

This book develops in greater depth some of the methods from the author's Reinforcement Learning and Optimal Control textbook (Athena Scientific, 2019). It presents new research, relating to rollout algorithms, policy iteration, multiagent systems, partitioned architectures, and distributed asynchronous computation.

The application of the methodology to challenging discrete optimization problems, such as routing, scheduling, assignment, and mixed integer programming, including the use of neural network approximations within these contexts, is also discussed.

Much of the new research is inspired by the remarkable AlphaZero chess program, where policy iteration, value and policy networks, approximate lookahead minimization, and parallel computation all play an important role.

Among its special features, the book:

- Presents new research relating to distributed asynchronous computation, partitioned architectures, and multiagent systems, with application to challenging large scale optimization problems, such as combinatorial/discrete optimization, as well as partially observed Markov decision problems.
- Describes variants of rollout and policy iteration for problems with a multiagent structure, which allow the dramatic reduction of the computational requirements for lookahead minimization.
- Establishes connections of rollout algorithms and model predictive control, one of the most prominent control system design methodology.
- Expands the coverage of some research areas discussed in the author's 2019 textbook Reinforcement Learning and Optimal Control.
- Provides the mathematical analysis that supports the Newton step interpretations and the conclusions of the present book.

The book is supported by on-line video lectures and slides, as well as new research material, some of which has been covered in the present monograph.